

APPENDIX D

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May 26, 2026

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OF:**

APPLICATION NUMBER: 17/259,395

FILING DATE: *July 16, 2021*

PATENT NUMBER: 11749866

ISSUE DATE: *September 05, 2023*



Certified by

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE

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TRANSMITTAL LETTER TO THE UNITED STATES DESIGNATED/ELECTED OFFICE (DO/EO/US) CONCERNING A SUBMISSION UNDER 35 U.S.C. 371		Attorney Docket No. LGCHEM 3.3F-3059
		U.S. Application No. (if known, see 37 CFR 1.5) Not Yet Assigned
International Application No. PCT/KR2019/015396	International Filing Date 13 November 2019	Priority Date Claimed 28 November 2018
Title of Invention Secondary Battery		
First Named Inventor Geon Woo Min		
Applicant herewith submits to the United States Designated/Elected Office (DO/EO/US) the following items and other information. 1. ☒ This is an express request to begin national examination procedures (35 U.S.C. 371(f)). NOTE: The express request under 35 U.S.C. 371(f) will not be effective unless the requirements under 35 U.S.C. 371(c)(1), (2), and (4) for payment of the basic national fee, copy of the International Application and English translation thereof (if required), and the oath or declaration of the inventor(s) have been received. 2. ☒ A copy of the International Application (35 U.S.C. 371(c)(2)) is attached hereto (not required if the International Application was previously communicated by the International Bureau or was filed in the United States Receiving Office (RO/US)). 3. An English language translation of the International Application (35 U.S.C. 371(c)(2)) a. ☒ is attached hereto. b. ☐ has been previously submitted under 35 U.S.C. 154(d)(4). 4. An oath or declaration of the inventor(s) (35 U.S.C. 371(c)(4)) a. ☒ is attached. b. ☐ was previously filed in the international phase under PCT Rule 4.17(iv). Items 5 to 8 below concern amendments made in the international phase. <u>PCT Article 19 and 34 amendments</u> 5. ☐ Amendments to the claims under PCT Article 19 are attached (not required if communicated by the International Bureau) (35 U.S.C. 371(c)(3)). 6. ☐ English translation of the PCT Article 19 amendment is attached (35 U.S.C. 371(c)(3)). 7. ☐ English translation of annexes (Article 19 and/or 34 amendments only) of the International Preliminary Examination Report is attached (35 U.S.C. 371(c)(5)). <u>Cancellation of amendments made in the international phase</u> 8a. ☐ Do not enter the amendment made in the international phase under PCT Article 19. 8b. ☐ Do not enter the amendment made in the international phase under PCT Article 34. NOTE: A proper amendment made in English under Article 19 or 34 will be entered in the U.S. national phase application absent a clear instruction from applicant not to enter the amendment(s). The following items 9 to 17 concern a document(s) or information included. 9. ☒ An Information Disclosure Statement under 37 CFR 1.97 and 1.98. 10. ☒ A preliminary amendment. 11. ☒ An Application Data Sheet under 37 CFR 1.76. 12. ☒ A substitute specification. NOTE: A substitute specification cannot include claims. See 37 CFR 1.125(b). 13. ☒ A power of attorney and/or change of address letter. 14. ☐ A computer-readable form of the sequence listing in accordance with PCT Rule 13ter.3 and 37 CFR 1.821-1.825 (not required if sequence listing in text format was indicated on the PCT Request as part of the International Application and the sequence listing was published as part of the International application). 15. ☐ Assignment papers (<i>cover sheet and document(s)</i>). Name of Assignee: _____ 16. ☐ 37 CFR 3.73(c) Statement (<i>when there is an Assignee</i>). _____		

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U.S. APPLN. No. (if known – see 37 CFR 1.5) Not Yet Assigned	INTERNATIONAL APPLICATION No. PCT/KR2019/015396	ATTORNEY DOCKET No. LGCHEM 3.3F-3059
17. ☒ Other items or information: Marked Up Specification 35 Pages		
The following fees have been submitted.		CALCULATIONS
18. ☒ Basic national fee (37 CFR 1.492(a)).....\$320		\$ 320.00
19. ☒ Examination fee (37 CFR 1.492(c))		
• If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4) \$0 • All other situations\$800		\$ 800.00
20. ☒ Search fee (37 CFR 1.492(b))		
• If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4) \$0 • Search fee (37 CFR 1.445(a)(2)) has been paid on the international application to the USPTO as an International Searching Authority \$140 • International Search Report prepared by an ISA other than the US and provided to the Office or previously communicated to the US by the IB \$540 • All other situations \$700		\$ 540.00
TOTAL OF 18, 19, and 20 =		\$ 1,660.00
☐ Additional fee for specification and drawings filed in paper over 100 sheets (excluding sequence listing in compliance with 37 CFR 1.821(c) or (e) in an electronic medium or computer program listing in an electronic medium) (37 CFR 1.492(j)). Fee for each additional 50 sheets of paper or fraction thereof \$420		
Total Sheets	Extra Sheets	Number of each additional 50 or fraction thereof (round up to a whole number)
18 - 100 =	/ 50 =	x \$420
Surcharge for furnishing any of the search fee, examination fee, or the oath or declaration after the date of commencement of the national stage (37 CFR 1.492(h)).....\$160		\$
CLAIMS	NUMBER FILED	NUMBER EXTRA
Total claims	12 - 20 =	0
Independent claims	1 - 3 =	0
		x \$100
		x \$480
MULTIPLE DEPENDENT CLAIM(S) (if applicable)		+ \$860
Fee for submission of Sequence Listing text file of 300 MB to 800 MB (37 CFR 1.21(o)(1)).....\$1,060		\$
Fee for submission of Sequence Listing text file of more than 800 MB (37 CFR 1.21(o)(2)).....\$10,500		\$
Processing fee for furnishing the English translation later than 30 months from the earliest claimed priority date (37 CFR 1.492(i)).....\$140 +		\$
TOTAL OF ABOVE CALCULATIONS =		\$ 1,660.00
☐ Applicant asserts small entity status. See 37 CFR 1.27. Fees above are reduced by 1/2.		
☐ Applicant certifies micro entity status. See 37 CFR 1.29. Fees above are reduced by 3/4. Applicant must attach form PTO/SB/15A or B or equivalent.		
TOTAL NATIONAL FEE =		\$ 1,660.00
Fee for recording the enclosed assignment (37 CFR 1.21(h)). The assignment must be accompanied by an appropriate cover sheet (37 CFR 3.28, 3.31).....\$50.00 per property +		\$
TOTAL FEES ENCLOSED =		\$ 1,660.00

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- a. ☐ A check in the amount of \$ _____ to cover the above fees is enclosed.
- b. ☐ Please charge my Deposit Account No. _____ in the amount of \$ _____ to cover the above fees.
- c. ☒ The Director is hereby authorized to charge additional fees which may be required, or credit any overpayment, to Deposit Account No. 12-1095 as follows:
- i. ☒ any required fee.
- ii. ☐ any required fee except for excess claims fees required under 37 CFR 1.492(d) and (e) and multiple dependent claim fee required under 37 CFR 1.492(f).
- d. ☐ Fees are to be charged to a credit card. **WARNING:** Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. The PTO-2038 should only be mailed or faxed to the USPTO. However, when paying the basic national fee, the PTO-2038 may NOT be faxed to the USPTO.

ADVISORY: If filing by EFS-Web, do **NOT** attach the PTO-2038 form as a PDF along with your EFS-Web submission. Please be advised that this is **not** recommended and by doing so your **credit card information may be displayed via PAIR**. To protect your information, it is recommended to pay fees online by using the electronic payment method.

NOTE: Where an appropriate time limit under 37 CFR 1.495 has not been met, a petition to revive (37 CFR 1.137(a) or (b)) must be filed and granted to restore the International Application to pending status.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

- ☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013, and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE 1: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

NOTE 2: A U.S. national stage application may not claim priority to the international application of which it is the national phase. The filing date of a U.S. national stage application is the international filing date. See 35 U.S.C. 363.

Correspondence Address

- ☒ The address associated with Customer Number: 86765 **OR** ☐ Correspondence address below

Name					
Address					
City		State		Zip Code	
Country				Telephone	
Email					

Signature	/Jordan T. Riviello/	Date	January 11, 2021
Name (Print/Type)	Jordan T. Riviello	Registration No. (Attorney/Agent)	72,515

[Page 3 of 3]

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: January 11, 2021

Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

(12) 특허협력조약에 의하여 공개된 국제출원

(19) 세계지식재산권기구
국제사무국

(43) 국제공개일
2020년 6월 4일 (04.06.2020)

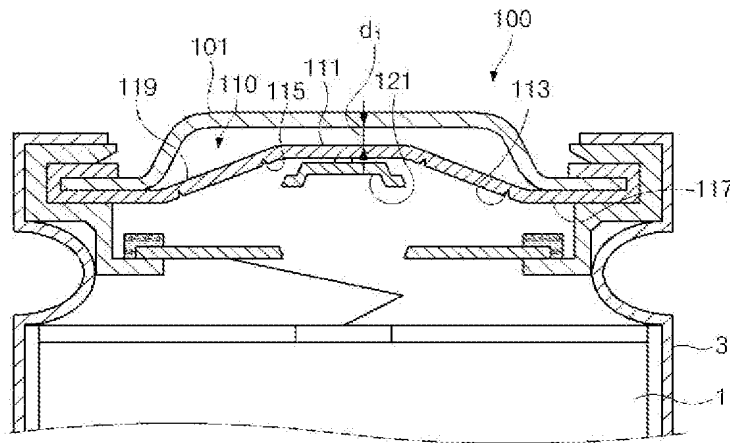


(10) 국제공개번호
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- (51) 국제특허분류: *H01M 2/12* (2006.01) *H01M 2/02* (2006.01)
H01M 2/34 (2006.01)
- (21) 국제출원번호: PCT/KR2019/015396
- (22) 국제출원일: 2019년 11월 13일 (13.11.2019)
- (25) 출원언어: 한국어
- (26) 공개언어: 한국어
- (30) 우선권정보: 10-2018-0149369 2018년 11월 28일 (28.11.2018)KR
- (71) 출원인: 주식회사 엘지화학 (LG CHEM, LTD.) [KR/KR]; 07336 서울시 영등포구 여의대로 128, Seoul (KR).
- (72) 발명자: 민건우 (MIN, Geon Woo); 34122 대전시 유성구 문지로 188 LG화학 기술연구원, Daejeon (KR). 이재준 (LEE, Je Jun); 34122 대전시 유성구 문지로 188 LG화학 기술연구원, Daejeon (KR). 정상석 (JUNG, Sang Suk); 34122 대전시 유성구 문지로 188 LG화학 기술연구원, Daejeon (KR).
- (74) 대리인: 특허법인 태평양 (BAE, KIM & LEE IP GROUP); 06626 서울시 서초구 강남대로 343, 11층, Seoul (KR).
- (81) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.
- (84) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), 유라시아 (AM, AZ, BY, KG, KZ, RU, TJ, TM), 유럽 (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI

(54) Title: SECONDARY BATTERY

(54) 발명의 명칭: 이차전지



(57) Abstract: The present invention relates to a secondary battery capable of improving a venting pressure distribution. In addition, the present invention relates to a secondary battery comprising: an electrode assembly; a can member for accommodating the electrode assembly therein; and a top cap assembly for covering the opening of the can member, wherein the top cap assembly comprises a top cap positioned on a top part and a safety vent part positioned below the top cap, and the safety vent part comprises a breaking part which breaks when the internal pressure in the can member increases above a predetermined level; a breaking notch part for forming a boundary of the breaking part; and a bending notch part formed at a position spaced apart from the breaking notch part toward the center of the breaking part, on the bottom surface of the breaking part facing the inside of the can member, and the bending notch part is bent when the internal pressure in the can member increases.

[다음 쪽 계속]

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(BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

공개:

— 국제조사보고서와 함께 (조약 제21조(3))

(57) 요약서: 본 발명은 벤딩(venting)압 산포를 개선할 수 있는 이차전지에 관한 것이다. 또한, 본 발명은 전극 조립체, 상기 전극 조립체를 내부에 수용하는 캔부재 및 상기 캔부재의 개구부를 덮는 탭캡 조립체를 포함하는 이차전지에 있어서, 상기 탭캡 조립체는 최상부에 위치되는 탭캡과 상기 탭캡의 하부에 위치되는 안전벤트부를 포함하고, 상기 안전벤트부는, 상기 캔부재 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부, 상기 파단부의 경계를 형성하는 파단 노치부 및 상기 캔부재의 내측을 따라보는 상기 파단부의 저면에서, 상기 파단 노치부로부터 상기 파단부의 중앙측으로 이격된 위치에 형성된 벤딩 노치부를 포함하고, 상기 캔부재 내부의 내압 증가 시 상기 벤딩 노치부는 벤딩(bending)되는 것을 특징으로 한다.

명세서

발명의 명칭: 이차전지

기술분야

- [1] 관련출원과의 상호인용
- [2] 본 출원은 2018년 11월 28일자 한국 특허 출원 제10-2018-0149369호에 기초한 우선권의 이익을 주장하며, 해당 한국 특허 출원의 문헌에 개시된 모든 내용은 본 명세서의 일부로서 포함된다.
- [3] 기술분야
- [4] 본 발명은 이차전지에 관한 것으로서, 보다 자세하게는 벤팅(venting)압 산포를 개선할 수 있는 이차전지에 관한 것이다.

배경기술

- [5] 물질의 물리적 반응이나 화학적 반응을 통해 전기에너지를 생성시켜 외부로 전원을 공급하게 되는 전지(cell, battery)는 각종 전기전자 기기로 둘러싸여 있는 생활환경에 따라, 진공로 공급되는 교류 전원을 획득하지 못할 경우나 직류전원이 필요할 경우 사용하게 된다.
- [6] 이와 같은 전지 중에서 화학적 반응을 이용하는 화학전지인 일차전지와 이차전지가 일반적으로 많이 사용되고 있는데, 일차전지는 건전지로 통칭되는 것으로 소모성 전지이다. 또한, 이차전지는 전류와 물질 사이의 산화환원과정에서 다수 반복 가능한 소재를 사용하여 제조되는 재충전식 전지로서, 전류에 의해 소재에 대한 환원반응이 수행되면 전원이 충전되고, 소재에 대한 산화반응이 수행되면 전원이 방전되는데, 이와 같은 충전-방전이 반복적으로 수행되면서 전기가 생성되게 된다.
- [7] 한편, 이차 전지 중 리튬 이온 전지는 양극 도전 포일과 음극 도전 포일에 각각 활물질을 일정한 두께로 코팅하고, 상기 양 도전 포일 사이에는 분리막이 개재되도록 하여 대략 젤리 롤(jelly roll) 내지는 원통 형태로 다수 회 권취하여 제작한 전극 조립체를 원통형 또는 각형 캔, 파우치 등에 수납하고 이를 밀봉 처리하여 제작된다.
- [8] 대한민국 공개특허공보 제10-2011-0095118호에는 종래의 이차 전지가 개시되어 있다.
- [9] 이차 전지는 내부 압력 증가 시에 탭캡 조립체를 통해 내부 가스를 외부로 배출하며, 탭캡 조립체는 내부 가스의 배출을 위한 안전벤트(vent)가 형성된다.
- [10] 이러한 안전벤트는 탭캡 조립체와 캔의 사이에 마련된 벤트 플랜지부에 형성된 노치가 캔 내부 압력 증가 시 절개되어 가스 배출을 위한 통로를 형성하는 구성이다.
- [11] 특히, 원형 전지는 과충전 등 이상 상황 발생 시 전지 폭발 및 발화를 방지하기 위하여 안전벤트와 같은 특정 부품에서 벤팅(venting)을 유도한다.

[12] 이러한 안전벤트는 기 설계된 벤딩압에서 벤딩되지 않으면 안전문제가 발생할 수 있다.

[13] 그런데 종래의 안전벤트는 벤딩압 측정 시 안전벤트의 노치(notch)가 절개되기 전에 탑캡(top cap)에 의해 간섭을 받아 벤딩이 지연되는 문제점이 있었다.

발명의 상세한 설명

기술적 과제

[14] 따라서 본 발명은 위와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 과제는 탑캡과의 간섭을 방지하여 벤딩압 산포를 개선할 수 있는 이차전지를 제공하는 것이다.

과제 해결 수단

[15] 본 발명의 일 실시예에 따른 이차전지는 전극 조립체, 상기 전극 조립체를 내부에 수용하는 캔부재 및 상기 캔부재의 개구부를 덮는 탑캡 조립체를 포함하는 이차전지에 있어서, 상기 탑캡 조립체는 최상부에 위치되는 탑캡과 상기 탑캡의 하부에 위치되는 안전벤트부를 포함하고, 상기 안전벤트부는, 상기 캔부재 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부, 상기 파단부의 경계를 형성하는 파단 노치부 및 상기 캔부재의 내측을 바라보는 상기 파단부의 저면에서, 상기 파단 노치부로부터 상기 파단부의 중앙측으로 이격된 위치에 형성된 벤딩 노치부를 포함하고, 상기 캔부재 내부의 내압 증가 시 상기 벤딩 노치부는 벤딩(bending)되는 것을 특징으로 한다.

[16] 상기 파단 노치부는 상기 안전벤트부의 하부 표면으로부터 오목하게 형성되고, 상기 벤딩 노치부 또한 상기 안전벤트부의 하부 표면으로부터 오목하게 형성되며, 상기 벤딩 노치부의 오목한 깊이는 상기 파단 노치부의 오목한 깊이보다 얕을 수 있다.

[17] 상기 파단 노치부는 상기 파단부의 상기 저면에 형성될 수 있다.

[18] 상기 탑캡 조립체는 상기 파단부에 부착되며 상기 캔부재의 내부 내압 증가 시 파단되어 전류를 차단하는 차단부를 구비하는 전류차단장치를 더 포함할 수 있다.

[19] 상기 전류차단장치의 차단부는 상기 파단부에서 상기 벤딩 노치부의 둘레 내측에 부착될 수 있다.

[20] 상기 벤딩 노치부는 연속되는 원형 또는 타원형일 수 있다.

[21] 상기 벤딩 노치부는 불연속되는 원형 또는 타원형일 수 있다.

[22] 상기 벤딩 노치부는 서로 길이가 다르게 복수 개로 이루어질 수 있다.

[23] 상기 파단 노치부는 불연속되는 원형 또는 타원형일 수 있다.

[24] 상기 안전벤트부의 테두리부와 상기 파단부 사이의 연결부는 상측으로 벤딩(bending)될 수 있다.

[25] 상기 파단부는 상기 연결부에서 상기 벤딩 노치부 측방향으로 높이가 낮아지게 기울어지는 경사부분 및 상기 경사부분의 단부에서 상기 파단부의 중앙측으로

연장되고 수평한 형태의 수평부분을 포함하고, 상기 벤딩 노치부는 상기 경사부분과 상기 수평부분의 경계에 위치할 수 있다.

- [26] 상기 안전벤트부의 테두리 단부와 상기 파단 노치부 사이의 이격 거리와 상기 파단 노치부와 상기 벤딩 노치부 사이의 이격 거리는 동일할 수 있다.

발명의 효과

- [27] 본 발명에 따르면, 안전벤트부의 지면에 파단 노치부 외에 벤딩 노치부를 추가로 형성하여 캔부재 내부 내압 상승 시 안전벤트부가 탭캡에 간섭되지 않게 하는 효과가 있다.
- [28] 또한, 안전벤트부가 탭캡에 간섭되지 않아 파단 노치부가 기 설정된 벤딩압에서 안정되게 벤딩되게 함에 따라 벤딩압 산포를 개선하는 효과가 있다.

도면의 간단한 설명

- [29] 도 1은 본 발명의 제1 실시예에 따른 이차전지의 사시도이다.
- [30] 도 2는 도 1을 A-A 방향으로 절단하여 이차전지의 상부 내부의 주요부를 도시한 부분 단면도이다.
- [31] 도 3은 도 2에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.
- [32] 도 4는 도 3에서 안전벤트부의 파단 노치부가 절단되어 벤딩되는 것을 도시한 사용상태도이다.
- [33] 도 5는 본 발명의 제1 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [34] 도 6은 도 5를 B-B 방향으로 절단하여 도시한 단면도이다.
- [35] 도 7은 본 발명의 제2 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [36] 도 8은 본 발명의 제3 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [37] 도 9는 도 8을 C-C 방향으로 절단하여 도시한 단면도이다.
- [38] 도 10은 종래의 안전벤트부가 설치된 이차전지를 도 3과 같은 단면에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.
- [39] 도 11은 종래의 안전벤트부가 설치된 이차전지와 본 발명의 안전벤트부가 설치된 이차전지의 벤딩압 산포를 비교하여 나타낸 산포도이다.

발명의 실시를 위한 형태

- [40] 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예에 따른 이차전지에 대하여 상세히 설명하기로 한다.

[41]

- [42] 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다. 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의

가장 바람직한 일 실시예에 불과하고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물이 있을 수 있음을 이해하여야 한다.

- [43] 도면에서 각 구성요소 또는 그 구성요소를 이루는 특정 부분의 크기는 설명의 편의 및 명확성을 위하여 과장되거나 생략되거나 또는 개략적으로 도시되었다. 따라서, 각 구성요소의 크기는 실제크기를 전적으로 반영하는 것은 아니다. 관련된 공지기능 혹은 구성에 대한 구체적인 설명이 본 발명의 요지를 불필요하게 흐릴 수 있다고 판단되는 경우, 그러한 설명은 생략하도록 한다.

[44]

- [45] 도 1은 본 발명의 제1 실시예에 따른 이차전지의 사시도이고, 도 2는 도 1을 A-A 방향으로 절단하여 이차전지의 상부 내부의 주요부를 도시한 부분 단면도이다.

- [46] 도 1 및 도 2에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 전극 조립체(1), 상기 전극 조립체(1)를 내부에 수용하는 캔부재(3) 및 상기 캔부재(3)의 개구부를 덮는 탭캡 조립체(100)를 포함하는 이차전지에 있어서, 상기 탭캡 조립체(100)는 최상부에 위치되는 탭캡(101)과 상기 탭캡(101)의 하부에 위치되는 안전벤트부(110)를 포함하고, 상기 안전벤트부(110)는 상기 캔부재(3) 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부(111), 상기 파단부(111)의 경계를 형성하는 파단 노치부(113) 및 상기 캔부재(3)의 내측을 바라보는 상기 파단부(111)의 저면에서, 상기 파단 노치부(113)로부터 상기 파단부(111)의 중앙측으로 이격된 위치에 형성된 벤딩 노치부(115)를 포함하고, 상기 캔부재(3) 내부의 내압 증가 시 상기 벤딩 노치부(115)는 벤딩(bending)되는 것을 특징으로 한다.

- [47] 전극 조립체(1)는, 예를 들어, 양극 활물질이 도포된 양극과 음극 활물질이 도포된 음극 및 양극과 음극 사이에 개재된 분리막을 복수 회 적층하여 제작할 수 있다.

- [48] 그리고 전극 조립체(1)는 양극과 분리막 및 음극을 적층한 적층체를 젤리 롤 형태로 권취하여 제작할 수도 있다.

- [49] 캔부재(3)는 원형 리튬이온 2차 전지에서 대략 원통형의 위쪽이 개방된 형상을 가진 금속재질의 용기이거나 각형 리튬이온 2차 전지에 대략 각형의 위쪽이 개방된 형상을 가진 금속재질의 용기일 수 있으며, 일반적으로 가볍고 부식에 대처가 용이한 알루미늄 또는 알루미늄 합금이 사용된다.

- [50] 캔부재(3)는 전극 조립체(1)와 전해액(미도시)의 용기가 되고, 전극 조립체(1)가 캔부재(3)의 개방된 상단, 즉, 상단개구부를 통해 캔부재(3)에 삽입된 뒤 캔부재(3)의 상단개구부는 탭캡 조립체(100)에 의해 봉해진다.

- [51] 탭캡 조립체(100)는 캔부재(3)의 내부 내압 증가 시 파단되어 전류를 차단하는 차단부(121)를 구비하는 전류차단장치(120)를 더 포함할 수 있다.

- [52] 도 3은 도 2에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가

파단된 것을 도시한 사용상태도이다.

- [53] 도 3을 참조하면, 진류차단장치(120)는 차단부(121)를 안전벤트부(110)의 파단부(111)에 부착하는데 캔부재(3)의 내부 내압 증가 시 차단부(121)가 진류차단장치(120)로부터 파단되면서 안전벤트부(110)의 파단부(111)와 함께 탐캡 조립체(100)의 탐캡(101) 방향인 상측으로 이동하게 된다.
- [54] 본 발명에서는 진류차단장치(120)의 차단부(121)가 파단부(111)에서 벤딩 노치부(115)의 둘레 내측의 정중앙에 부착될 수 있다.
- [55] 안전벤트부(110)의 파단 노치부(113)는 캔부재(3)의 내부 내압 증가 시 절개되기 위해 안전벤트부(110)의 표면으로부터 오목하게 형성된다.
- [56] 파단 노치부(113)는 안전벤트부(110)의 상면과 저면 중 어느 하나에 형성될 수 있는데, 안전벤트부(110)가 캔부재(3)의 내부측을 바라보는 안전벤트부(110)의 저면에 형성되면 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 상승했을 때 탐캡(101)과의 이격 거리(d)를 확보하는데 유리 할 수 있다.
- [57] 벤딩 노치부(115)는 파단 노치부(113)와 마찬가지로 안전벤트부(110)의 표면으로부터 오목하게 만입된 형태(또는 파인 형태)로 형성되는데 벤딩 노치부(115)의 오목한 깊이는 파단 노치부(113)의 오목한 깊이보다 얕을 수 있다.
- [58] 이러한 이유는 벤딩 노치부(115)가 캔부재(3)의 내부 내압에 의해 절개되지 않고 벤딩(bending)되게 하기 위해 노치(notch)의 두께를 절개되지 않을 정도의 두께로 유지하기 위함 일 수 있다.
- [59] 그리고 벤딩 노치부(115)는 안전벤트부(110)의 저면에 형성되어 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 이동하였을 때에 탐캡(101)과의 이격거리(d)를 확보하게 할 수 있다.
- [60] 도 4는 도 3에서 안전벤트부의 파단 노치부가 절단되어 벤딩되는 것을 도시한 사용상태도이다.
- [61] 도 4에 도시된 바와 같이, 안전벤트부(110)가 캔부재(3)의 내부 내압의 의해 상측으로 이동하였을 때 안전벤트부(110)의 저면에 형성된 벤딩 노치부(115)의 벤딩에 의해 탐캡(101)과의 이격거리(d)를 확보하게 되면 파단 노치부(113)가 기 설정된 벤팅(venting)압에 의해 절개되어 안전하게 캔부재(3)의 내부 가스를 외부로 배출(venting)하게 할 수 있다.
- [62] 도 5는 본 발명의 제1 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [63] 도 5에 도시된 바와 같이, 본 발명의 일 실시예에 따르면 파단 노치부(113)는 안전벤트부(110)의 저면에 불연속되는 원형 또는 타원형으로 형성될 수 있다.
- [64] 이와 같이 파단 노치부(113)는 안전벤트부(110)의 저면의 형상에 대응되게 형성될 수 있다. 즉, 안전벤트부(110)의 저면의 형상이 원형이면 파단 노치부(113)의 형상도 원형을 형성할 수 있고, 안전벤트부(110)의 저면의 형상이 타원형이면 파단 노치부(113)의 형상도 타원형을 형성할 수 있다.
- [65] 벤딩 노치부(115)는 안전벤트부(110)의 저면에 연속되는 원형 또는 타원형을 형성할 수 있다.

- [66] 벤딩 노치부(115)도 파단 노치부(113)와 마찬가지로 안전벤트부(110)의 저면의 형상에 대응되게 형성될 수 있다.
- [67] 또한, 벤딩 노치부(115)는 파단 노치부(113)보다 둘레의 크기가 작게 형성되어 안전벤트부(110)의 저면에서 파단 노치부(113)의 둘레 안쪽에 형성될 수 있다.
- [68] 한편, 안전벤트부(110)의 저면에서 보아 안전벤트부(110)의 저면 테두리 단부와 파단 노치부(113) 사이의 이격된 거리와 파단 노치부(113)와 벤딩 노치부(115) 사이의 이격된 거리를 어느 일측으로 치우치지 않고 동일하게 한다면 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 이동하였을 때 안전벤트부(110)와 탭캡(101)과의 이격 거리를 가장 효과적으로 확보하게 할 수 있다.
- [69] 도 6은 도 5를 B-B 방향으로 절단하여 도시한 단면도이다.
- [70] 도 6에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 안전벤트부(110)의 테두리부(117)와 파단부(111) 사이의 연결부(119)는 상측으로 벤딩(bending)될 수 있다.
- [71] 즉, 연결부(119)가 상측으로 벤딩되었기 때문에 캔부재(3) 내부의 내압에 의해 안전벤트부(110)가 상측으로 이동하여도 파단 노치부(113)가 쉽게 절개되지 않고 안전벤트부(110)가 상측으로 이동하게 할 수 있다.
- [72] 파단부(111)는 연결부(119)에서 벤딩 노치부(115) 방향으로 높이가 낮아지게 기울어지는 경사부분(111a)을 포함할 수 있다.
- [73] 그리고 경사부분(111a)의 단부에서 파단부(111)의 중앙측으로 연장되고 수평한 형태의 수평부분(111b)을 포함할 수 있다.
- [74] 그리고 벤딩 노치부(115)는 경사부분(111a)과 수평부분(111b)의 경계에 위치할 수 있다.
- [75] 즉, 경사부분(111a)이 수평부분(111b) 방향으로 높이가 낮아지게 기울어짐에 따라 연결부(119)가 상측으로 벤딩되어 높이가 높아진 것을 보상해 주어 파단부(111)의 수평부분(111b)이 가장 하측에 위치하면서 수평을 유지하게 할 수 있다.
- [76] 이와 같이 파단부(111)는 하측에 설치된 전류차단장치(120)의 차단부(121)와 결합하기 위해 수평부분(111b)이 안전벤트부(110)에서 가장 하측에 위치할 수 있다.
- [77] 또한, 벤딩 노치부(115)가 경사부분(111a)과 수평부분(111b)의 경계에 위치하게 되면, 탭캡(101)과의 최적의 이격거리(d)를 확보하도록 벤딩 노치부(115)가 벤딩 되도록 하는데 현저히 효과적일 수 있다.
- [78] 도 7은 본 발명의 제2 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [79] 도 7에 도시된 바와 같이, 본 발명의 제2 실시예에 따른 이차전지는 안전벤트부(110)의 벤딩 노치부(115)가 불연속되는 원형 또는 타원형을 형성할 수 있다.
- [80] 벤딩 노치부(115)가 불연속되는 원형 또는 타원형을 형성한다면 벤딩

노치부(115)에서 오목한 부분을 최소화할 수 있어 캔부재(3)의 내압 상승 시 벤딩 노치부(115)가 절개되는 것을 효과적으로 방지할 수 있다.

[81] 도 8은 본 발명의 제3 실시예에 따른 안전벤트부만을 도시한 저면도이고, 도 9는 도 8을 C-C 방향으로 절단하여 도시한 단면도이다.

[82] 도 8 및 도 9에 도시된 바와 같이, 본 발명의 제3 실시예에 따른 이차전지는 안전벤트부(110)의 벤딩 노치부(115)가 서로 길이가 다르게 복수 개로 형성될 수 있다.

[83] 벤딩 노치부(115)가 서로 길이가 다르게 복수 개로 형성되면 캔부재(3)의 순간 내압 상승에 의해 안전벤트부(110)가 급속한 속도로 상승하여도 상승 충격을 복수 개의 벤딩 노치부(115)가 벤딩되면서 완화시킬 수 있다.

[84] 즉, 순간 급격한 상승 충격에 의해 파단 노치부(113)가 기 설정된 벤딩압 이전의 압력에서 파단되는 것을 방지할 수 있다.

[85] 도 10은 비교예로서 제시된다. 종래의 안전벤트부가 설치된 이차전지를 도 3과 같은 단면에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.

[86] 도 10에 도시된 바와 같이, 종래의 이차전지는 안전벤트부(110')에 파단 노치부(113')만 형성하며, 본원발명에서와 같은 벤딩 노치부를 형성하지 않고 만들어진단.

[87] 이러한 종래의 이차전지는 캔부재(3) 내부의 내압 상승 시 안전벤트부(110')가 탭캡(101') 방향으로 이동하는데, 가스의 내압에 의해 안전벤트부(110')에서 캔부재(3)와 결합된 테두리로부터 가장 거리가 먼 안전벤트부(110')의 중앙 부위가 가장 상측으로 솟아올라 수 있다.

[88] 따라서 안전벤트부(110')의 중앙 부위가 탭캡(101')의 인근까지 솟아올라 탭캡(101')과의 이격 거리(d_2)가 대략 123 μ m 정도까지 근접할 수 있다. 또 더 나아가 안전벤트부(110')의 중앙 부위가 탭캡(101')에 닿을 수 있다.

[89] 이와 같이 안전벤트부(110')와 탭캡(101')과의 이격 거리(d_2)가 너무 가깝거나 닿게 되면, 파단 노치가 절개되어 내부 가스를 벤딩해야 하는 기 설정된 벤딩압에서 절개되지 않아 벤딩이 안정적으로 이루어지지 않는 경우가 발생할 수 있다.

[90] 이러한 경우 이차전지의 폭발, 발열, 발화 등의 안전 문제가 발생할 수 있다.

[91] 도 3에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 캔부재(3) 내부의 내압 상승 시 파단부(111)의 경사부분(111a)이 상측으로 꺾이면서 안전벤트부(110)가 탭캡(101) 방향으로 이동하는데, 안전벤트부(110)가 탭캡(101)을 향하여 이동할 때 벤딩 노치부(115)가 꺾이면서(bending) 캔부재(3) 내부의 내압을 일부 흡수하여 파단부(111)의 수평부분(111b)이 탭캡(101) 방향으로 솟구치는 것을 방지할 수 있다.

[92] 따라서 안전벤트부(110)와 탭캡(101) 사이의 이격 거리(d_1)를 대략 186 μ m 정도로 충분히 확보할 수 있다.

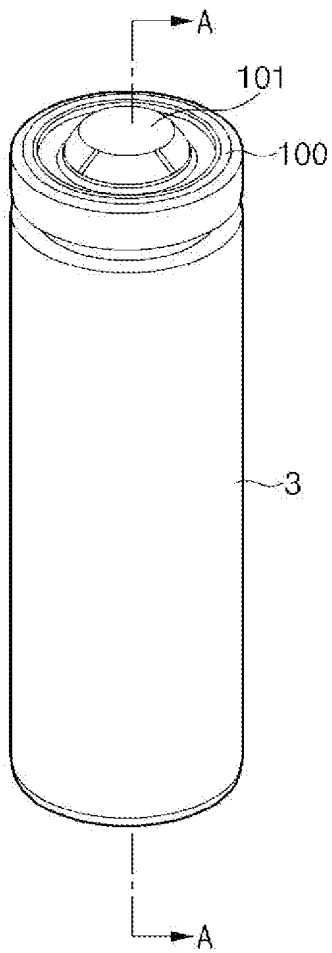
- [93] 이와 같이 안전벤트부(110)와 탐캡(101) 사이의 이격 거리(d_1)를 확보함에 따라, 기 설정된 벤팅압에서 파단 노치부(113)가 절개되는 것이 탐캡(101)에 의해 간섭되는 것을 방지할 수 있어 안정적으로 켄부재(3) 내부의 가스를 벤팅할 수 있다.
- [94] 도 11은 종래의 안전벤트부가 설치된 이차전지와 본 발명의 안전벤트부가 설치된 이차전지의 벤팅압 산포를 비교하여 나타낸 산포도이다.
- [95] 도 11에서 V_0 는 벤팅압의 산포 집합이고, M_1 는 벤팅압의 최대치이고 M_n 은 벤팅압의 최소치이며, V_1 , V_2 는 각각 벤팅압 산포의 일부분이다.
- [96] 도 11을 참조하여, 종래의 벤딩 노치부가 형성되지 않은 안전벤트부의 벤팅압과 개선된 본 발명의 벤딩 노치부가 형성된 안전벤트부의 벤팅압을 비교해보면, 종래의 안전벤트부의 벤팅압의 산포에 비해 본 발명의 안전벤트부의 벤팅압의 산포가 개선된 것을 알 수 있는데 그 차이는 대략 1.5배 정도의 차이가 날 수 있다.
- [97] 특히, 종래의 안전벤트부는 켄부재 내부 가스의 벤팅 과정에서 파단 노치부가 절개되지 않은 상태에서 안전벤트부의 중앙부가 탐캡에 닿게되어 기 설정된 벤팅압에서 정상적으로 벤팅이 되지 않는 것과 같은 비정상적인 벤팅 과정에서 벤팅압의 변화가 크게 발생하는 산포를 나타내는 것이 V_1 및 V_2 일 수 있다.
- [98]
- [99] 상술한 바와 같이 본 발명에 따르면, 안전벤트의 저면에 파단 노치부 외에 벤딩 노치부를 추가로 형성하여 켄부재 내부 내압 상승 시 안전벤트가 탐캡에 간섭되지 않게 하는 효과가 있다.
- [100] 또한, 안전벤트가 탐캡에 간섭되지 않아 파단 노치부가 기 설정된 벤팅압에서 벤팅되게 함에 따라 벤팅압 산포를 개선하는 효과가 있다.
- [101]
- [102] 이상과 같이 본 발명에 따른 이차전지를 예시된 도면을 참고하여 설명하였으나, 본 발명은 이상에서 설명된 실시예와 도면에 의해 한정되지 않으며, 특허청구범위 내에서 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자들에 의해 다양한 실시가 가능하다.

청구범위

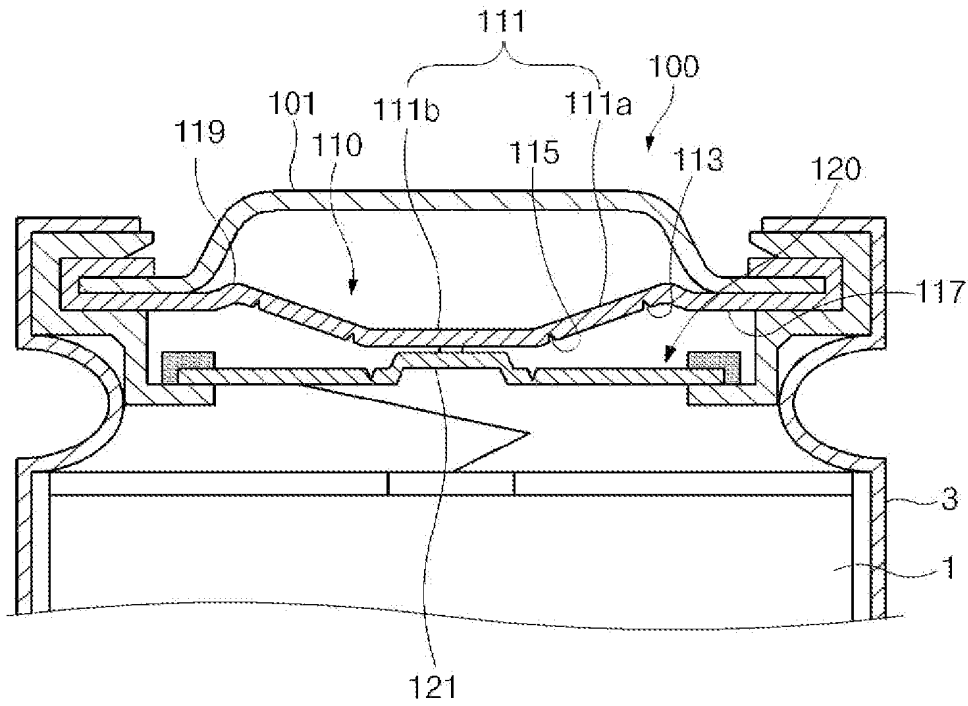
- [청구항 1] 전극 조립체(1);
 상기 전극 조립체(1)를 내부에 수용하는 캔부재(3); 및
 상기 캔부재(3)의 개구부를 덮는 탐캡 조립체(100);를 포함하는
 이차전지에 있어서,
 상기 탐캡 조립체(100)는 최상부에 위치되는 탐캡(101)과 상기
 탐캡(101)의 하부에 위치되는 안전벤트부(110)를 포함하고,
 상기 안전벤트부(110)는,
 상기 캔부재(3) 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는
 파단부(111);
 상기 파단부(111)의 경계를 형성하는 파단 노치부(113); 및
 상기 캔부재(3)의 내측을 바라보는 상기 파단부(111)의 저면에서, 상기
 파단 노치부(113)로부터 상기 파단부(111)의 중앙측으로 이격된 위치에
 형성된 벤딩 노치부(115)를 포함하고,
 상기 캔부재(3) 내부의 내압 증가 시 상기 벤딩 노치부(115)는
 벤딩(bending)되는 것을 특징으로 하는 이차전지.
- [청구항 2] 청구항 1에 있어서,
 상기 파단 노치부(113)는 상기 안전벤트부(110)의 하부 표면으로부터
 오목하게 형성되고,
 상기 벤딩 노치부(115) 또한 상기 안전벤트부(110)의 하부 표면으로부터
 오목하게 형성되며,
 상기 벤딩 노치부(115)의 오목한 깊이는 상기 파단 노치부(113)의 오목한
 깊이보다 얕은 것을 특징으로 하는 이차전지.
- [청구항 3] 청구항 1에 있어서,
 상기 파단 노치부(113)는 상기 파단부(111)의 상기 저면에 형성되는 것을
 특징으로 하는 이차전지.
- [청구항 4] 청구항 1에 있어서,
 상기 탐캡 조립체(100)는,
 상기 파단부(111)에 부착되며 상기 캔부재(3)의 내부 내압 증가 시
 파단되어 전류를 차단하는 차단부(121)를 구비하는 전류차단장치(120)를
 더 포함하는 것을 특징으로 하는 이차전지.
- [청구항 5] 청구항 4에 있어서,
 상기 전류차단장치(120)의 차단부(121)는 상기 파단부(111)에서 상기
 벤딩 노치부(115)의 둘레 내측에 부착되는 것을 특징으로 하는 이차전지.
- [청구항 6] 청구항 1에 있어서,
 상기 벤딩 노치부(115)는 연속되는 원형 또는 타원형인 것을 특징으로
 하는 이차전지.

- [청구항 7] 청구항 1에 있어서,
상기 벤딩 노치부(115)는 불연속되는 원형 또는 다원형인 것을 특징으로 하는 이차전지.
- [청구항 8] 청구항 1에 있어서,
상기 벤딩 노치부(115)는 서로 길이가 다르게 복수 개로 이루어지는 것을 특징으로 하는 이차전지.
- [청구항 9] 청구항 1에 있어서,
상기 파단 노치부(113)는 불연속되는 원형 또는 다원형인 것을 특징으로 하는 이차전지.
- [청구항 10] 청구항 1에 있어서,
상기 안전벤트부(110)의 테두리부(117)와 상기 파단부(111) 사이의 연결부(119)는 상측으로 벤딩(bending)된 것을 특징으로 하는 이차전지.
- [청구항 11] 청구항 10에 있어서,
상기 파단부(111)는,
상기 연결부(119)에서 상기 벤딩 노치부(115) 측방향으로 높이가 낮아지게 기울어지는 경사부분(111a) 및
상기 경사부분의 단부에서 상기 파단부(111)의 중앙측으로 연장되고 수평한 형태의 수평부분(111b)을 포함하고,
상기 벤딩 노치부(115)는 상기 경사부분(111a)과 상기 수평부분(111b)의 경계에 위치하는 것을 특징으로 하는 이차전지.
- [청구항 12] 청구항 1에 있어서,
상기 안전벤트부(110)의 테두리 단부와 상기 파단 노치부(113) 사이의 이격 거리와 상기 파단 노치부(113)와 상기 벤딩 노치부(115) 사이의 이격 거리는 동일한 것을 특징으로 하는 이차전지.

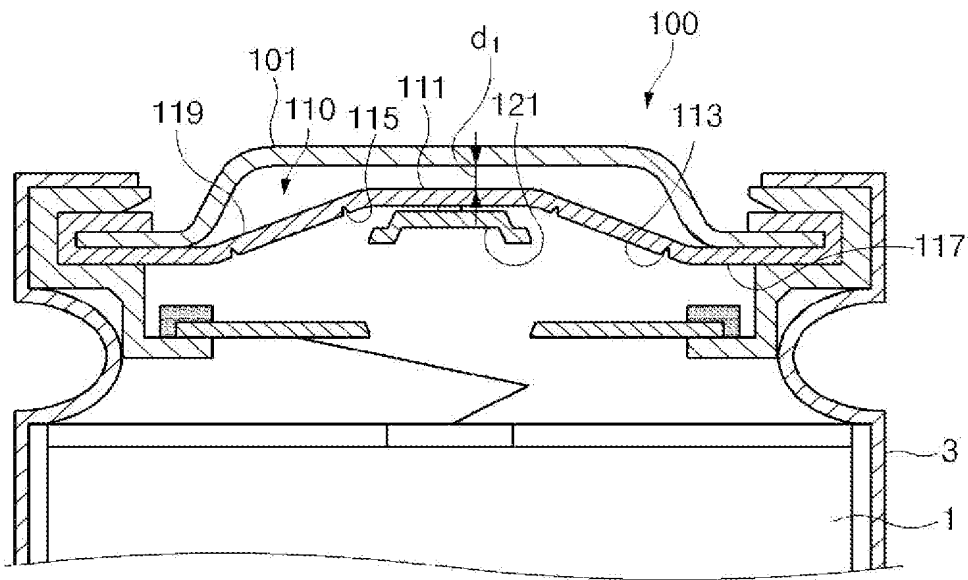
[도 1]



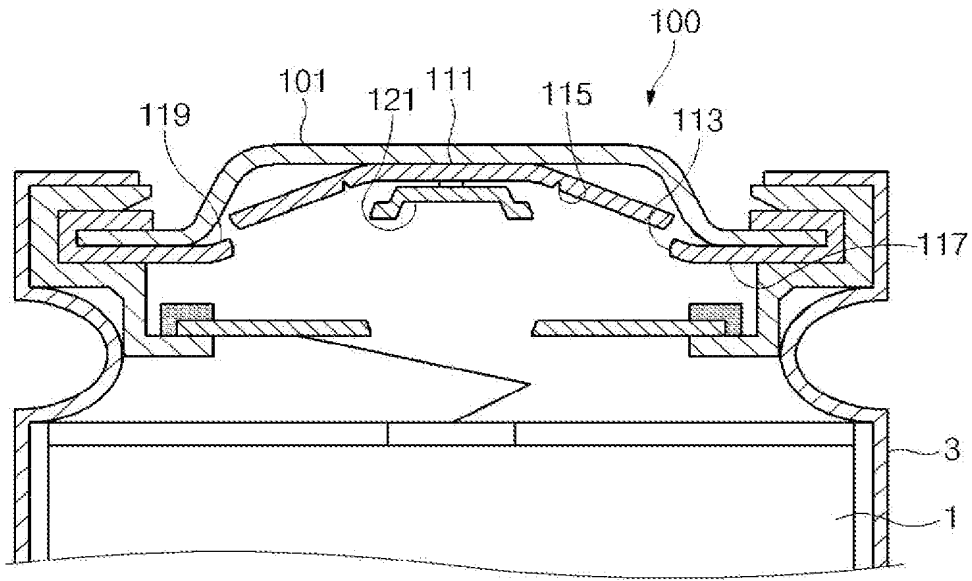
[도2]



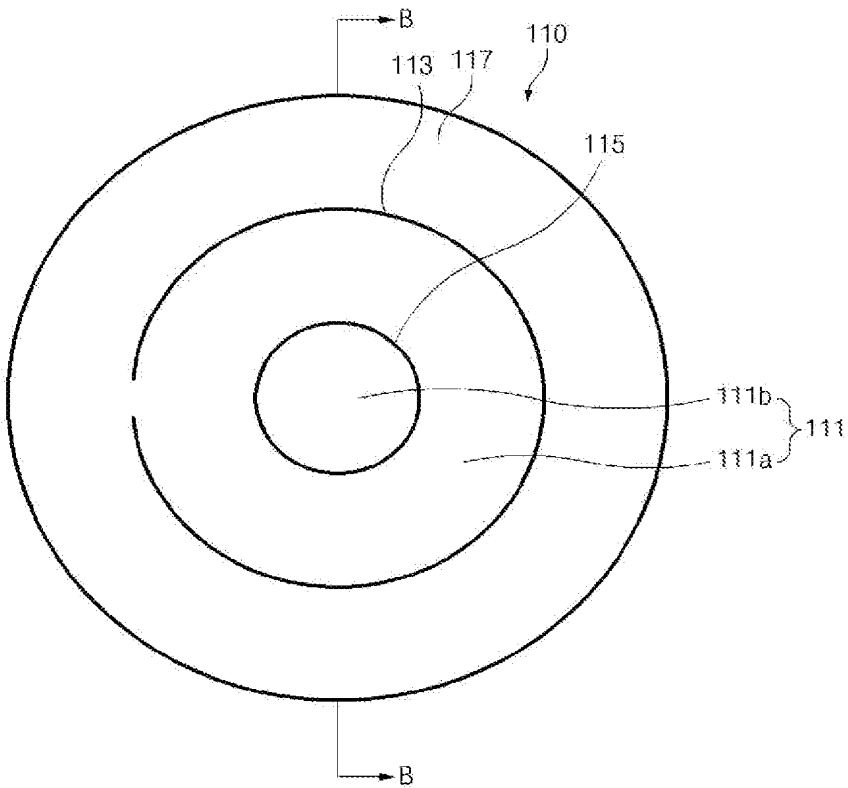
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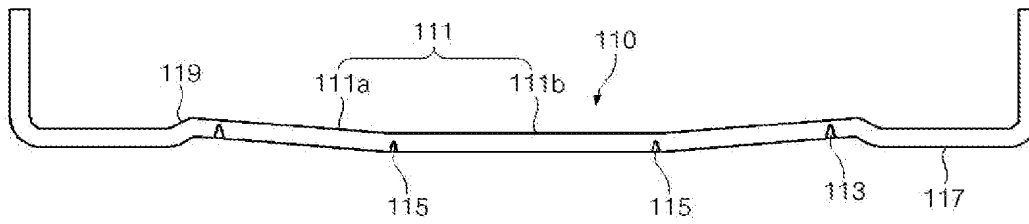
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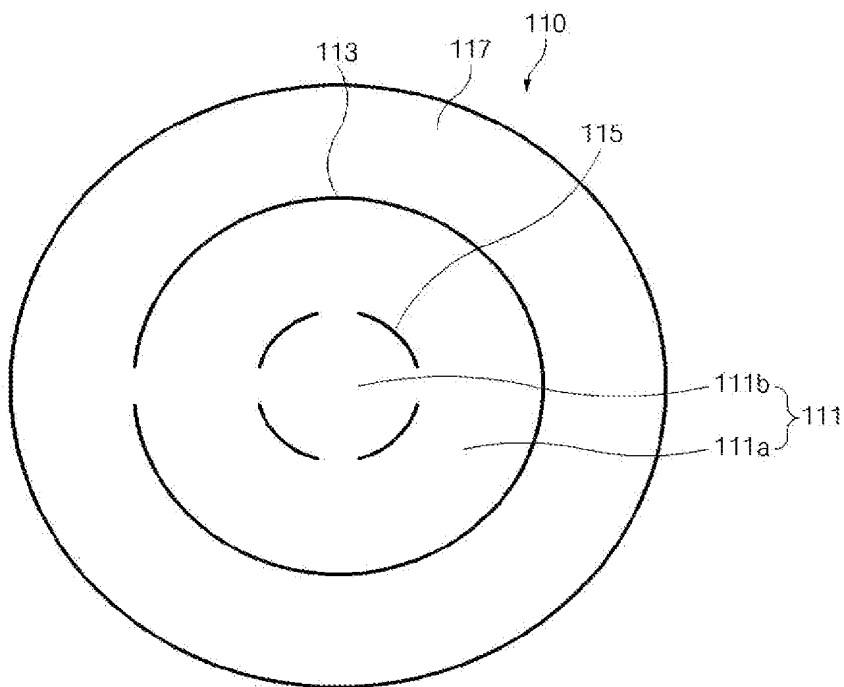
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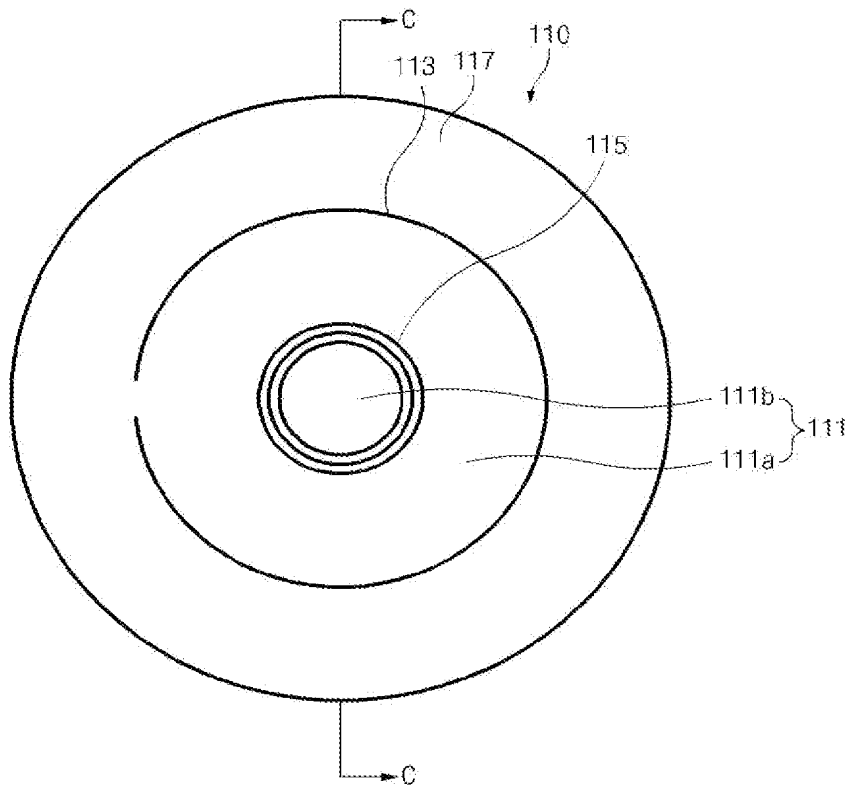
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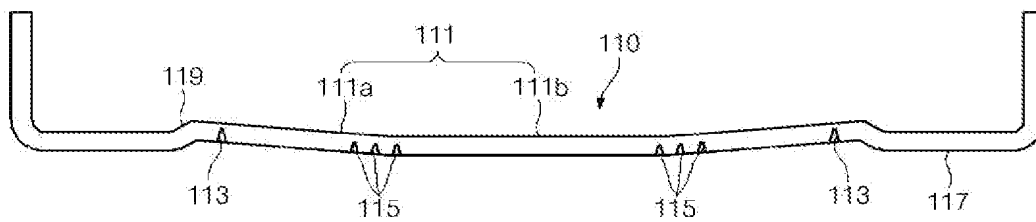
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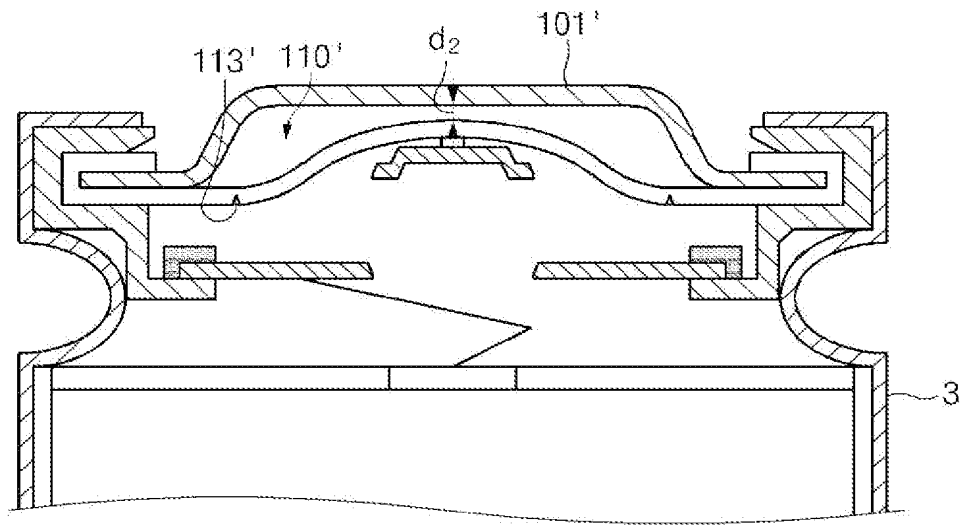
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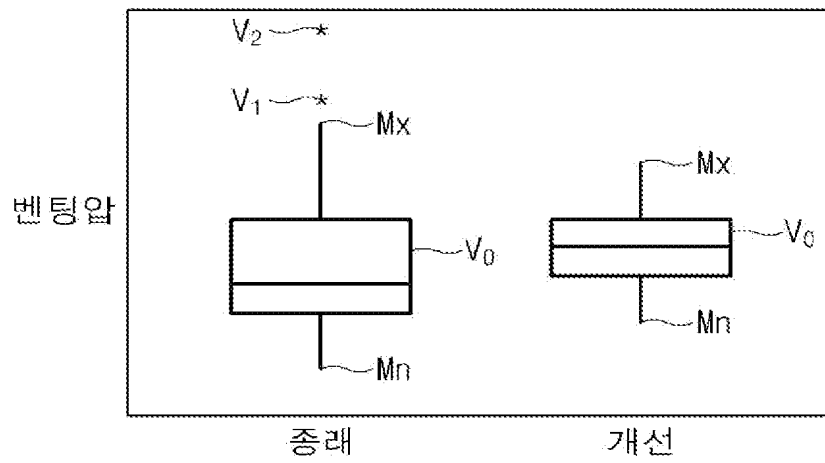
[도9]



[도10]



[도11]



INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2019/015396

A. CLASSIFICATION OF SUBJECT MATTER

H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models: IPC as above

Japanese utility models and applications for utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: gas, notch, bending, fracture

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 05 August 1994 See paragraphs [0019]-[0022], claim 1 and figures 1-4.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 24 December 1996 See paragraph [0043] and figure 11.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 13 April 2001 See paragraphs [0016], [0017] and figures 1-5.	4-9
A	KR 10-2015-0035267 A (LG CHEM, LTD.) 06 April 2015 See the entire document.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 02 October 2018 See the entire document.	1-12



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"B" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"J" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"G" document member of the same patent family

Date of the actual completion of the international search

19 FEBRUARY 2020 (19.02.2020)

Date of mailing of the international search report

20 FEBRUARY 2020 (20.02.2020)

Name and mailing address of the ISA/KR



Korean Intellectual Property Office
Government Complex Daejeon Building 4, 189, Cheongro-ro, Seo-gu,
Daejeon, 35288, Republic of Korea

Facsimile No. +82-42-481-8578

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.

PCT/KR2019/015396

Patent document cited in search report	Publication date	Patent family member	Publication date
JP 06-215746 A	05/08/1994	None	
JP 08-339793 A	24/12/1996	None	
JP 2001-102038 A	13/04/2001	None	
KR 10-2015-0035267 A	06/04/2015	KR 10-1667966 B1	20/10/2016
US 10090497 B2	02/10/2018	CN 107293687 A	24/10/2017
		KR 10-2017-0116370 A	19/10/2017
		US 2017-0294635 A1	12/10/2017

A. 발명이 속하는 기술분류(국제특허분류(IPC))

H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i

B. 조사된 분야

조사된 최소문헌(국제특허분류를 기재)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

조사된 기술분야에 속하는 최소문헌 이외의 문헌

한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우))

eKOMPASS(특허청 내부 검색시스템) & 키워드: 가스(gas), 노치(notch), 벤딩(bending), 파단(fracture)

C. 관련 문헌

카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 1994.08.05 단락 [0019]~[0022], 청구항 1 및 도면 1~4 참조.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 1996.12.24 단락 [0043] 및 도면 11 참조.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 2001.04.13 단락 [0016], [0017] 및 도면 1~5 참조.	4-9
A	KR 10-2015-0035267 A (주식회사 엘지화학) 2015.04.06 전체 문헌 참조.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 2018.10.02 전체 문헌 참조.	1-12

☐ 추가 문헌이 C(제속)에 기재되어 있습니다.☒ 대응특허에 관한 별지를 참조하십시오.

* 인용된 문헌의 특별 카테고리:

- "A" 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌
- "D" 본 국제출원에서 출원인이 인용한 문헌
- "E" 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후 "X" 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 진균성 또는 진보성이 없는 것으로 본다.
- "L" 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌
- "O" 아무 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌
- "P" 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌
- "I" 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌
- "Y" 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명의 진보성이 없는 것으로 본다.
- "&" 동일한 대응특허문헌에 속하는 문헌

국제조사의 실제 완료일

2020년 02월 19일 (19.02.2020)

국제조사보고서 발송일

2020년 02월 20일 (20.02.2020)

ISA/KR의 명칭 및 우편주소

대한민국 특허청
(35208) 대전광역시 서구 청사로 189,
4동 (문산동, 정부대전청사)

팩스 번호 +82-42-481-8578

심사관

김연경

전화번호 +82-42-481-3325



국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
JP 06-215746 A	1994/08/05	없음	
JP 08-339793 A	1996/12/24	없음	
JP 2001-102038 A	2001/04/13	없음	
KR 10-2015-0035267 A	2015/04/06	KR 10-1667966 B1	2016/10/20
US 10090497 B2	2018/10/02	CN 107293687 A	2017/10/24
		KR 10-2017-0116370 A	2017/10/19
		US 2017-0294635 A1	2017/10/12

DESCRIPTION**SECONDARY BATTERY****CROSS-REFERENCE TO RELATED APPLICATION**

[0001] The present application is a national phase entry under 35 U.S.C. § 371 of PCT/KR2019/015396, filed on November 13, 2019 and claims the benefit of the priority of Korean Patent Application No. 10-2018-0149369, filed on November 28, 2018, which ~~is~~ are hereby incorporated by reference in their~~its~~ entirety.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery, and more particularly, to a secondary battery capable of improving venting pressure distribution.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction of materials to supply power to the outside are used when alternate current (AC) power to be supplied to the building is not obtained, or direct current (DC) power is required according to the living environments surrounded by various electrical/electronic devices.

[0004] Among such batteries, primary batteries and secondary batteries, which are chemical cells using chemical reaction, are being generally used. The primary batteries are consumable cells which are collectively referred to as dry cells. Also, the secondary batteries are rechargeable batteries manufactured by using a material in which oxidation and reduction processes between current and a material are capable

of being repeated several times. That is, power is recharged when the reduction reaction to the material is performed by the current, and power is discharged when the oxidation reaction to the material is performed. The recharging and discharging are repeatedly performed to generate electricity.

[0005] A lithium ion battery among the secondary batteries is manufactured in such a manner in which: an active material is applied, with a predetermined thickness, to each of a positive electrode conductive foil and a negative electrode conductive foil; a separator is disposed between both the conductive foils so as to be wound several times in an approximate jelly roll or cylindrical shape and thereby to manufacture an electrode assembly; and the electrode assembly is accommodated in a cylindrical or prismatic can, a pouch, or the like and is sealed.

[0006] A secondary battery according to the related art is disclosed in Korean Patent Publication No. 10-2011-0095118.

[0007] In the secondary battery, an internal gas is vented to the outside through a top cap assembly when an internal pressure increases, and the top cap assembly includes a safety vent for venting the internal gas.

[0008] Such a safety vent has a structure in which a notch defined in a vent flange part disposed between the top cap assembly and a can is cut when the internal pressure within the can increases, and thus a passage for venting the gas is provided.

[0009] Particularly, a certain component, such as the safety vent, in a circular battery induces venting to prevent the battery from being exploded and ignited when an abnormal accident such as overcharging occurs.

[0010] Such a safety vent may cause a safety problem when the venting is not induced at a pre-designed venting pressure.

[0011] Also, a top cap interferes with the safety vent according to the related art during measurement of a venting pressure before the notch of the safety vent is cut, and thus the venting may be delayed.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0012] Accordingly, the present invention has been made to solve the above problem, and an object of the present invention is to provide a secondary battery in which the interference with a top cap is prevented to be able to improve the venting pressure distribution.

TECHNICAL SOLUTION

[0013] A secondary battery according to an embodiment of the present invention includes: an electrode assembly; a can member that accommodates the electrode assembly therein; and a top cap assembly that covers an opening of the can member, wherein the top cap assembly includes a top cap positioned at the uppermost portion and a safety vent part positioned below the top cap, wherein the safety vent part includes: a rupture portion ruptured when an internal pressure within the can member increases to a predetermined pressure or more; a rupture notch portion that defines a boundary of the rupture portion; and a bending notch portion which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing the inside of the can member, wherein the bending notch portion is bent when the internal pressure within the can member increases.

[0014] The rupture notch portion may be recessed from a lower

surface of the safety vent part, and the bending notch portion may be recessed from the lower surface of the safety vent part, wherein a recessed depth of the bending notch portion is less than that of the rupture notch portion.

[0015] The rupture notch portion may be provided in the bottom surface of the rupture portion.

[0016] The top cap assembly may further include a current interrupt device provided with an interrupt portion which is attached to the rupture portion and ruptured to interrupt current when the internal pressure within the can member increases.

[0017] The interrupt portion of the current interrupt device may be attached to the rupture portion inside a circumference of the bending notch portion.

[0018] The bending notch portion may have a continuous circular or elliptical shape.

[0019] The bending notch portion may have a discontinuous circular or elliptical shape.

[0020] The bending notch portion may be provided in plurality, and the plurality of bending notch portions have different lengths.

[0021] The rupture notch portion may have a discontinuous circular or elliptical shape.

[0022] A connection portion between an edge portion and the rupture portion of the safety vent part may be bent upward.

[0023] The rupture portion may include: an inclined section inclined so that a height thereof gradually decreases from the connection portion toward the bending notch portion; and a horizontal section having a horizontal shape and extending from an end of the inclined section toward the center of the

rupture portion, wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section.

[0024] A spaced distance between an edge end of the safety vent part and the rupture notch portion may be equal to a spaced distance between the rupture notch portion and the bending notch portion.

ADVANTAGEOUS EFFECTS

[0025] According to the present invention, the bending notch portion as well as the rupture notch portion is further provided in the bottom surface of the safety vent part, and thus the safety vent part is not interfered with the top cap when the internal pressure within the can member increases.

[0026] Also, since the safety vent part is not interfered with the top cap, the rupture notch portion stably performs the venting at the preset venting pressure, and thus the venting pressure distribution may be improved.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a perspective view of a secondary battery according to a first embodiment of the present invention.

[0028] FIG. 2 is a partial cross-sectional view taken along line A-A and illustrating main parts in an upper inner portion of the secondary battery.

[0029] FIG. 3 is a view illustrating a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in the internal pressure of a can member of FIG. 2.

[0030] FIG. 4 is a view illustrating a state of use in which a rupture notch portion of a safety vent part of FIG. 3 is cut to provide venting.

[0031] FIG. 5 is a bottom view illustrating only the safety vent part according to the first embodiment of the present invention.

[0032] FIG. 6 is a cross-sectional view taken along line B-B of FIG. 5.

[0033] FIG. 7 is a bottom view illustrating only a safety vent part according to a second embodiment of the present invention.

[0034] FIG. 8 is a bottom view illustrating only a safety vent part according to a third embodiment of the present invention.

[0035] FIG. 9 is a cross-sectional view taken along line C-C of FIG. 8.

[0036] FIG. 10 is a view of a secondary battery in which a safety vent part according to the related art is installed, illustrating, on the same cross-section of FIG. 3, a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in an internal pressure of a can member.

[0037] FIG. 11 shows the degrees of distribution for comparing the venting pressure distribution in the secondary battery in which the safety vent part according to the related art is installed and the venting pressure distribution in the secondary battery in which the safety vent part according to the present invention is installed.

MODE FOR CARRYING OUT THE INVENTION

[0038] Hereinafter, preferred embodiments of the present invention will now be described in detail with reference to the accompanying drawings.

[0039] Terms or words used in this specification and claims

should not be restrictively interpreted as ordinary meanings or dictionary-based meanings, but should be interpreted as meanings and concepts conforming to the scope of the present invention on the basis of the principle that an inventor can properly define the concept of a term to describe and explain his or her invention in the best ways. Therefore, the embodiments described in this specification and the configurations illustrated in the drawings are only preferred embodiments of the present invention, and may not describe the technical idea thoroughly. Accordingly, it should be understood that various equivalents and modifications that can be substituted for the embodiments might be provided at the filing date of the present application.

[0040] In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience and clarity of illustration. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

[0041] FIG. 1 is a perspective view of a secondary battery according to a first embodiment of the present invention, and FIG. 2 is a partial cross-sectional view taken along line A-A and illustrating main parts in an upper inner portion of the secondary battery.

[0042] As illustrated in FIGS. 1 and 2, a secondary battery according to the embodiment includes an electrode assembly 1, a can member 3 for accommodating the electrode assembly 1 therein, and a top cap assembly 100 for covering an opening of

the can member 3. The top cap assembly 100 includes a top cap 101 positioned at the uppermost portion and a safety vent part 1010 positioned below the top cap 101. The safety vent part 110 includes a rupture portion 111 which ruptures when an internal pressure within the can member 3 increases to a predetermined pressure or more, a rupture notch portion 113 that defines a boundary of the rupture portion 111, and a bending notch portion 115 which is disposed at a position spaced apart from the rupture notch portion 113 toward a center of the rupture portion 111 on a bottom surface of the rupture portion 111 facing the inside of the can member 3. The bending notch portion 115 is bent when the internal pressure within the can member 3 increases.

[0043] The electrode assembly 1 may be manufactured by stacking, multiple times, a positive electrode coated with a positive electrode active material, a negative electrode coated with a negative electrode active material, and a separator interposed between the negative electrode and the positive electrode.

[0044] Also, the electrode assembly 1 may be manufactured by winding, in a jelly roll structure, a stack body in which the positive electrodes, the separators, and the negative electrodes are stacked on each other.

[0045] The can member 3 is a metal container having an open top surface of an approximately cylindrical structure in a cylindrical lithium ion secondary battery or a metal container having an open top surface of an approximately prismatic structure in a prismatic lithium ion secondary battery. The can member 3 is made of aluminum or an aluminum alloy that is generally lightweight and easy to manage corrosion.

[0046] The can member 3 serves as a container for the electrode assembly 1 and an electrolyte (not shown). The electrode assembly 1 is inserted into the can member 3 via an upper open end, i.e., an upper end that is open in the can member 3, and then the upper open end of the can member 3 is sealed by the top cap assembly 100.

[0047] The top cap assembly 100 may further include a current interrupt device 120 provided with an interrupt portion 121 which is configured to be ruptured to interrupt current when an internal pressure within the can member 3 increases.

[0048] FIG. 3 is a view illustrating a state of use in which the interrupt portion is ruptured from the current interrupt device due to an increase in the internal pressure of the can member of FIG. 2.

[0049] Referring to FIG. 3, the interrupt portion 121 of the current interrupt device 120 is attached to the rupture portion 111 of the safety vent part 110. The rupture portion 121 is ruptured from the current interrupt device 120 and moves upward toward the top cap 101 of the top cap assembly 100, together with the rupture portion 111 of the safety vent part 110.

[0050] According to the present invention, the rupture portion 121 of the current interrupt device 120 may be attached to the center in the rupture portion 111 inside a circumference of the bending notch portion 115.

[0051] The rupture notch portion 113 of the safety vent part 110 is recessed from the surface of the safety vent part 110 so as to be cut when the internal pressure within the can member 3 increases.

[0052] The rupture notch portion 113 may be defined in one of a top surface and a bottom surface of the safety vent part 110.

In a case in which the rupture notch portion 113 is provided in the bottom surface of the safety vent part 110 facing the inside of the can member 3, it may be advantageous to secure a spaced distance d to the top cap 101 when the safety vent part 110 rises upward due to the internal pressure within the can member 3.

[0053] Similar to the rupture notch portion 113, the bending notch portion 115 is concavely indented (or grooved) from the surface of the safety vent part 110. The recessed depth of the bending notch portion 115 is less than the recessed depth of the rupture notch portion 113.

[0054] This may be done for maintaining a thickness of a notch to a thickness not allowing the notch to be cut, and thus the bending notch portion 115 is not cut but bent due to the internal pressure within the can member 3.

[0055] Also, the bending notch portion 115 is provided in the bottom surface of the safety vent part 110, and thus the spaced distance d to the top cap 101 may be secured when the safety vent part 110 is moved upward due to the internal pressure within the can member 3.

[0056] FIG. 4 is a view illustrating a state of use in which the rupture notch portion of the safety vent part of FIG. 3 is cut to provide venting.

[0057] As illustrated in FIG. 4, in a case in which the spaced distance d to the top cap 101 is secured by the bending of the bending notch portion 115 provided in the bottom surface of the safety vent part 110 when the safety vent part 110 is moved upward due to the internal pressure inside the can member 3, the rupture notch portion 113 is cut at a preset venting pressure, and thus the gas within the can member 3 may be

safely vented to the outside.

[0058] FIG. 5 is a bottom view illustrating only the safety vent part according to the first embodiment of the present invention.

[0059] According to the embodiment of the present invention as illustrated in FIG. 5, the rupture notch portion 113 may have a discontinuous circular or elliptical shape in the bottom surface of the safety vent part 110.

[0060] As described above, the rupture notch portion 113 may be provided corresponding to the bottom surface of the safety vent part 110. That is, when the bottom surface of the safety vent part 110 has a circular shape, the rupture notch portion 113 may also have the circular shape, and when the bottom surface of the safety vent part 110 has an elliptical shape, the rupture notch portion 113 may also have the elliptical shape.

[0061] The bending notch portion 115 may have a continuous circular or elliptical shape in the bottom surface of the safety vent part 110.

[0062] Similar to the rupture notch portion 113, the bending notch portion 115 may also be provided corresponding to the bottom surface of the safety vent part 110.

[0063] Also, the bending notch portion 115 has a smaller circumferential size than the rupture notch portion 113, and thus may be provided in the bottom surface of the safety vent part 110 inside a circumference of the rupture notch portion 113.

[0064] Here, if a spaced distance between an edge end on the bottom surface of the safety vent part 110 and the rupture notch portion 113 is equal, without being biased to the one

side, to a spaced distance between the rupture notch portion 113 and the bending notch portion 115 when viewed from the bottom surface of the safety vent part 110, the spaced gap between the safety vent part 110 and the top cap 101 may be most effectively secured when the safety vent part 110 is moved upward due to the internal pressure within the can member 3.

[0065] FIG. 6 is a cross-sectional view taken along line B-B of FIG. 5.

[0066] As illustrated in FIG. 6, in the secondary battery according to the embodiment of the present invention, a connection portion 119 between an edge portion 117 and the rupture portion 111 of the safety vent part 110 may be bent upward.

[0067] That is, since the connection portion 119 is bent upward, although the safety vent part 110 is moved upward due to the internal pressure within the can member 3, the rupture notch portion 113 is not easily cut, and the safety vent part 110 may be moved upward.

[0068] The rupture portion 111 may include an inclined section 111a which is inclined so that a height thereof gradually decreases from the connection portion 119 toward the bending notch portion 115.

[0069] Also, the rupture portion 111 may include a horizontal section 111b having a horizontal shape and extending from an end of the inclined section 111a toward the center of the rupture portion 111.

[0070] Also, the bending notch portion 115 may be positioned on a boundary between the inclined section 111a and the horizontal section 111b.

[0071] That is, as the inclined section 111a is inclined so

that a height thereof gradually decreases toward the horizontal section 111b, an increase in the height of the connection portion 119 by being bent upward is compensated. Thus, the horizontal section 111b of the rupture portion 111 is allowed to be in a lowermost position so as to maintain the horizontal level.

[0072] As described above, the horizontal section 111b of the rupture portion 111 may be positioned at the lowermost portion of the safety vent part 110 so as to be coupled to the interrupt portion 121 of the current interrupt device 120 installed at the lower side.

[0073] Also, when the bending notch portion 115 is positioned on the boundary between the inclined section 111a and the horizontal section 111b, it may be significantly effective in bending the bending notch portion 115 to secure an optimal spaced distance d to the top cap 101.

[0074] FIG. 7 is a bottom view illustrating only a safety vent part according to a second embodiment of the present invention.

[0075] As illustrated in FIG. 7, in a secondary battery according to the second embodiment of the present invention, a bending notch portion 115 of a safety vent part 110 may have a discontinuous circular or elliptical shape.

[0076] When the bending notch portion 115 has the discontinuous circular or elliptical shape, a recessed region of the bending notch portion 115 may be minimized, and thus the bending notch portion 115 may be effectively prevented from being cut when an internal pressure inside a can member 3 increases.

[0077] FIG. 8 is a bottom view illustrating only a safety

vent part according to a third embodiment of the present invention, and FIG. 9 is a cross-sectional view taken along line C-C of FIG. 8.

[0078] As illustrated in FIGS. 8 and 9, in a secondary battery according to a third embodiment of the present invention, a plurality of bending notch portions 115 of the safety vent part 110 are provided having different lengths.

[0079] In a case in which the plurality of bending notch portions 115 are provided having different lengths, even when the safety vent part 110 rapidly rises due to an instantaneous increase in an internal pressure within the can member 3, the plurality of bending notch portions 115 may be bent to mitigate a rising impact.

[0080] That is, the rupture notch portions 113 may be prevented from being cut, at a pressure before reaching a preset venting pressure, due to the instant and sudden rising impact.

[0081] FIG. 10 is provided as Comparative example. FIG. 10 is a view of a secondary battery in which a safety vent part according to the related art is installed, illustrating, on the same cross-section of FIG. 3, a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in an internal pressure of a can member.

[0082] In the secondary battery according to the related art as illustrated in FIG. 10, only a rupture notch portion 113' is provided in a safety vent part 110' without the bending notch portion of the present invention.

[0083] In such a secondary battery according to the related art, the safety vent part 110' moves toward a top cap 101' when an internal pressure within a can member 3 increases.

Here, a central portion of the safety vent part 110' farthest away from an edge of the safety vent part 110' coupled to the can member 3 may rise to the uppermost side due to the internal pressure of a gas.

[0084] Thus, the central portion of the safety vent part 110' soars to the vicinity of the top cap 101', and approaches the top cap 101' until a spaced distance d_2 therebetween is close to approximately 123 μm . Furthermore, the central portion of the safety vent part 110' may come into contact with the top cap 101'.

[0085] A rupture notch is not cut when the spaced distance d_2 between the safety vent part 110' and the top cap 101' is very small, or the safety vent part 110' comes into contact with the top cap 101' as described above. That is, since the rupture notch is not cut at a preset venting pressure at which an internal gas should be vented, the venting may not be performed stably.

[0086] In this case, safety problems such as explosion, heat generation, and ignition of a secondary battery may occur.

[0087] In the secondary battery according to the embodiment of the present invention as illustrated in FIG. 3, when the internal pressure within the can member 3 increases, the safety vent part 110 moves toward the top cap 101 while the inclined section 111a of the rupture portion 111 is bent upward. When the safety vent part 110 moves toward top cap 101, the bending notch portion 115 is bent and absorbs a portion of the internal pressure within the can member 3, and thus the horizontal section 111b of the rupture portion 111 may be prevented from soaring toward top cap 101.

[0088] Thus, a spaced distance d_1 between the safety vent part

110 and the top cap 101 may be sufficiently secured to about 186 μm .

[0089] As the spaced distance d_1 between the safety vent part 110 and the top cap 101 is secured as described above, the rupture notch portion 113 is cut at a preset vent pressure, which may prevent interference with the top cap 101 and stably vent the gas within the can member 3.

[0090] FIG. 11 shows the degrees of distribution for comparing the venting pressure distribution in the secondary battery in which the safety vent part according to the related art is installed and the venting pressure distribution in the secondary battery in which the safety vent part according to the present invention is installed.

[0091] In FIG. 11, V_0 is a distribution group of venting pressures, M_x is the maximum value of the venting pressures, M_n is the minimum value of the venting pressures, and each of V_1 and V_2 is a portion of the venting pressure distribution.

[0092] When comparing the venting pressures of the safety vent part according to the related art in which the bending notch portion is not provided and the venting pressures of the improved safety vent part according to the present invention in which the bending notch portion is provided with reference to FIG. 11, it may be found that the venting pressure distribution of the safety vent part according to the present invention is improved as compared to the venting pressure distribution of the safety vent part according to the related art. The difference therebetween may be about 1.5 times.

[0093] Particularly, during the venting of gas within the can member in the safety vent part according to the related art, the central portion of the safety vent part comes into contact

with the top cap in a state in which the rupture notch portion is not cut, and thus the venting is not properly performed at the preset venting pressure. The distribution in which the venting pressures greatly change during abnormal venting described above may be indicated as V_1 and V_2 .

[0094] According to the present invention as described above, the bending notch portion as well as the rupture notch portion is further provided in the bottom surface of the safety vent, and thus the safety vent is not interfered with the top cap when the internal pressure within the can member increases.

[0095] Also, since the safety vent is not interfered with the top cap, the rupture notch portion performs the venting at the preset venting pressure, and thus the venting pressure distribution may be improved.

[0096] Although the secondary battery according to the present invention has been described with reference to the exemplary drawings as described above, the present invention is not limited to the embodiments and drawings described and illustrated above, and various embodiments may be made within the scope of the appended claims by those skilled in the art pertaining to the present invention.

DESCRIPTION**SECONDARY BATTERY****CROSS-REFERENCE TO RELATED APPLICATION**

[0001] The present application is a national phase entry under 35 U.S.C. § 371 of PCT/KR2019/015396, filed on November 13, 2019 and claims the benefit of the priority of Korean Patent Application No. 10-2018-0149369, filed on November 28, 2018, which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery, and more particularly, to a secondary battery capable of improving venting pressure distribution.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction of materials to supply power to the outside are used when alternate current (AC) power to be supplied to the building is not obtained, or direct current (DC) power is required according to the living environments surrounded by various electrical/electronic devices.

[0004] Among such batteries, primary batteries and secondary batteries, which are chemical cells using chemical reaction, are being generally used. The primary batteries are consumable cells which are collectively referred to as dry cells. Also, the secondary batteries are rechargeable batteries manufactured by using a material in which oxidation and reduction processes between current and a material are capable of being repeated several times. That is, power is recharged

when the reduction reaction to the material is performed by the current, and power is discharged when the oxidation reaction to the material is performed. The recharging and discharging are repeatedly performed to generate electricity.

[0005] A lithium ion battery among the secondary batteries is manufactured in such a manner in which: an active material is applied, with a predetermined thickness, to each of a positive electrode conductive foil and a negative electrode conductive foil; a separator is disposed between both the conductive foils so as to be wound several times in an approximate jelly roll or cylindrical shape and thereby to manufacture an electrode assembly; and the electrode assembly is accommodated in a cylindrical or prismatic can, a pouch, or the like and is sealed.

[0006] A secondary battery according to the related art is disclosed in Korean Patent Publication No. 10-2011-0095118.

[0007] In the secondary battery, an internal gas is vented to the outside through a top cap assembly when an internal pressure increases, and the top cap assembly includes a safety vent for venting the internal gas.

[0008] Such a safety vent has a structure in which a notch defined in a vent flange part disposed between the top cap assembly and a can is cut when the internal pressure within the can increases, and thus a passage for venting the gas is provided.

[0009] Particularly, a certain component, such as the safety vent, in a circular battery induces venting to prevent the battery from being exploded and ignited when an abnormal accident such as overcharging occurs.

[0010] Such a safety vent may cause a safety problem when the venting is not induced at a pre-designed venting pressure.

[0011] Also, a top cap interferes with the safety vent according to the related art during measurement of a venting pressure before the notch of the safety vent is cut, and thus the venting may be delayed.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0012] Accordingly, the present invention has been made to solve the above problem, and an object of the present invention is to provide a secondary battery in which the interference with a top cap is prevented to be able to improve the venting pressure distribution.

TECHNICAL SOLUTION

[0013] A secondary battery according to an embodiment of the present invention includes: an electrode assembly; a can member that accommodates the electrode assembly therein; and a top cap assembly that covers an opening of the can member, wherein the top cap assembly includes a top cap positioned at the uppermost portion and a safety vent part positioned below the top cap, wherein the safety vent part includes: a rupture portion ruptured when an internal pressure within the can member increases to a predetermined pressure or more; a rupture notch portion that defines a boundary of the rupture portion; and a bending notch portion which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing the inside of the can member, wherein the bending notch portion is bent when the internal pressure within the can member increases.

[0014] The rupture notch portion may be recessed from a lower surface of the safety vent part, and the bending notch portion may be recessed from the lower surface of the safety vent part, wherein a recessed depth of the bending notch portion is less than that of the rupture notch portion.

[0015] The rupture notch portion may be provided in the bottom surface of the rupture portion.

[0016] The top cap assembly may further include a current interrupt device provided with an interrupt portion which is attached to the rupture portion and ruptured to interrupt current when the internal pressure within the can member increases.

[0017] The interrupt portion of the current interrupt device may be attached to the rupture portion inside a circumference of the bending notch portion.

[0018] The bending notch portion may have a continuous circular or elliptical shape.

[0019] The bending notch portion may have a discontinuous circular or elliptical shape.

[0020] The bending notch portion may be provided in plurality, and the plurality of bending notch portions have different lengths.

[0021] The rupture notch portion may have a discontinuous circular or elliptical shape.

[0022] A connection portion between an edge portion and the rupture portion of the safety vent part may be bent upward.

[0023] The rupture portion may include: an inclined section inclined so that a height thereof gradually decreases from the connection portion toward the bending notch portion; and a horizontal section having a horizontal shape and extending

from an end of the inclined section toward the center of the rupture portion, wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section.

[0024] A spaced distance between an edge end of the safety vent part and the rupture notch portion may be equal to a spaced distance between the rupture notch portion and the bending notch portion.

ADVANTAGEOUS EFFECTS

[0025] According to the present invention, the bending notch portion as well as the rupture notch portion is further provided in the bottom surface of the safety vent part, and thus the safety vent part is not interfered with the top cap when the internal pressure within the can member increases.

[0026] Also, since the safety vent part is not interfered with the top cap, the rupture notch portion stably performs the venting at the preset venting pressure, and thus the venting pressure distribution may be improved.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a perspective view of a secondary battery according to a first embodiment of the present invention.

[0028] FIG. 2 is a partial cross-sectional view taken along line A-A and illustrating main parts in an upper inner portion of the secondary battery.

[0029] FIG. 3 is a view illustrating a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in the internal pressure of a can member of FIG. 2.

[0030] FIG. 4 is a view illustrating a state of use in which a rupture notch portion of a safety vent part of FIG. 3 is cut to provide venting.

[0031] FIG. 5 is a bottom view illustrating only the safety vent part according to the first embodiment of the present invention.

[0032] FIG. 6 is a cross-sectional view taken along line B-B of FIG. 5.

[0033] FIG. 7 is a bottom view illustrating only a safety vent part according to a second embodiment of the present invention.

[0034] FIG. 8 is a bottom view illustrating only a safety vent part according to a third embodiment of the present invention.

[0035] FIG. 9 is a cross-sectional view taken along line C-C of FIG. 8.

[0036] FIG. 10 is a view of a secondary battery in which a safety vent part according to the related art is installed, illustrating, on the same cross-section of FIG. 3, a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in an internal pressure of a can member.

[0037] FIG. 11 shows the degrees of distribution for comparing the venting pressure distribution in the secondary battery in which the safety vent part according to the related art is installed and the venting pressure distribution in the secondary battery in which the safety vent part according to the present invention is installed.

MODE FOR CARRYING OUT THE INVENTION

[0038] Hereinafter, preferred embodiments of the present invention will now be described in detail with reference to the accompanying drawings.

[0039] Terms or words used in this specification and claims should not be restrictively interpreted as ordinary meanings or dictionary-based meanings, but should be interpreted as meanings and concepts conforming to the scope of the present invention on the basis of the principle that an inventor can properly define the concept of a term to describe and explain his or her invention in the best ways. Therefore, the embodiments described in this specification and the configurations illustrated in the drawings are only preferred embodiments of the present invention, and may not describe the technical idea thoroughly. Accordingly, it should be understood that various equivalents and modifications that can be substituted for the embodiments might be provided at the filing date of the present application.

[0040] In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience and clarity of illustration. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

[0041] FIG. 1 is a perspective view of a secondary battery according to a first embodiment of the present invention, and FIG. 2 is a partial cross-sectional view taken along line A-A

and illustrating main parts in an upper inner portion of the secondary battery.

[0042] As illustrated in FIGS. 1 and 2, a secondary battery according to the embodiment includes an electrode assembly 1, a can member 3 for accommodating the electrode assembly 1 therein, and a top cap assembly 100 for covering an opening of the can member 3. The top cap assembly 100 includes a top cap 101 positioned at the uppermost portion and a safety vent part 110 positioned below the top cap 101. The safety vent part 110 includes a rupture portion 111 which ruptures when an internal pressure within the can member 3 increases to a predetermined pressure or more, a rupture notch portion 113 that defines a boundary of the rupture portion 111, and a bending notch portion 115 which is disposed at a position spaced apart from the rupture notch portion 113 toward a center of the rupture portion 111 on a bottom surface of the rupture portion 111 facing the inside of the can member 3. The bending notch portion 115 is bent when the internal pressure within the can member 3 increases.

[0043] The electrode assembly 1 may be manufactured by stacking, multiple times, a positive electrode coated with a positive electrode active material, a negative electrode coated with a negative electrode active material, and a separator interposed between the negative electrode and the positive electrode.

[0044] Also, the electrode assembly 1 may be manufactured by winding, in a jelly roll structure, a stack body in which the positive electrodes, the separators, and the negative electrodes are stacked on each other.

[0045] The can member 3 is a metal container having an open top surface of an approximately cylindrical structure in a cylindrical lithium ion secondary battery or a metal container having an open top surface of an approximately prismatic structure in a prismatic lithium ion secondary battery. The can member 3 is made of aluminum or an aluminum alloy that is generally lightweight and easy to manage corrosion.

[0046] The can member 3 serves as a container for the electrode assembly 1 and an electrolyte (not shown). The electrode assembly 1 is inserted into the can member 3 via an upper open end, i.e., an upper end that is open in the can member 3, and then the upper open end of the can member 3 is sealed by the top cap assembly 100.

[0047] The top cap assembly 100 may further include a current interrupt device 120 provided with an interrupt portion 121 which is configured to be ruptured to interrupt current when an internal pressure within the can member 3 increases.

[0048] FIG. 3 is a view illustrating a state of use in which the interrupt portion is ruptured from the current interrupt device due to an increase in the internal pressure of the can member of FIG. 2.

[0049] Referring to FIG. 3, the interrupt portion 121 of the current interrupt device 120 is attached to the rupture portion 111 of the safety vent part 110. The rupture portion 121 is ruptured from the current interrupt device 120 and moves upward toward the top cap 101 of the top cap assembly 100, together with the rupture portion 111 of the safety vent part 110.

[0050] According to the present invention, the rupture portion 121 of the current interrupt device 120 may be attached

to the center in the rupture portion 111 inside a circumference of the bending notch portion 115.

[0051] The rupture notch portion 113 of the safety vent part 110 is recessed from the surface of the safety vent part 110 so as to be cut when the internal pressure within the can member 3 increases.

[0052] The rupture notch portion 113 may be defined in one of a top surface and a bottom surface of the safety vent part 110. In a case in which the rupture notch portion 113 is provided in the bottom surface of the safety vent part 110 facing the inside of the can member 3, it may be advantageous to secure a spaced distance d to the top cap 101 when the safety vent part 110 rises upward due to the internal pressure within the can member 3.

[0053] Similar to the rupture notch portion 113, the bending notch portion 115 is concavely indented (or grooved) from the surface of the safety vent part 110. The recessed depth of the bending notch portion 115 is less than the recessed depth of the rupture notch portion 113.

[0054] This may be done for maintaining a thickness of a notch to a thickness not allowing the notch to be cut, and thus the bending notch portion 115 is not cut but bent due to the internal pressure within the can member 3.

[0055] Also, the bending notch portion 115 is provided in the bottom surface of the safety vent part 110, and thus the spaced distance d to the top cap 101 may be secured when the safety vent part 110 is moved upward due to the internal pressure within the can member 3.

[0056] FIG. 4 is a view illustrating a state of use in which the rupture notch portion of the safety vent part of FIG. 3 is cut to provide venting.

[0057] As illustrated in FIG. 4, in a case in which the spaced distance d to the top cap 101 is secured by the bending of the bending notch portion 115 provided in the bottom surface of the safety vent part 110 when the safety vent part 110 is moved upward due to the internal pressure inside the can member 3, the rupture notch portion 113 is cut at a preset venting pressure, and thus the gas within the can member 3 may be safely vented to the outside.

[0058] FIG. 5 is a bottom view illustrating only the safety vent part according to the first embodiment of the present invention.

[0059] According to the embodiment of the present invention as illustrated in FIG. 5, the rupture notch portion 113 may have a discontinuous circular or elliptical shape in the bottom surface of the safety vent part 110.

[0060] As described above, the rupture notch portion 113 may be provided corresponding to the bottom surface of the safety vent part 110. That is, when the bottom surface of the safety vent part 110 has a circular shape, the rupture notch portion 113 may also have the circular shape, and when the bottom surface of the safety vent part 110 has an elliptical shape, the rupture notch portion 113 may also have the elliptical shape.

[0061] The bending notch portion 115 may have a continuous circular or elliptical shape in the bottom surface of the safety vent part 110.

[0062] Similar to the rupture notch portion 113, the bending notch portion 115 may also be provided corresponding to the bottom surface of the safety vent part 110.

[0063] Also, the bending notch portion 115 has a smaller circumferential size than the rupture notch portion 113, and thus may be provided in the bottom surface of the safety vent part 110 inside a circumference of the rupture notch portion 113.

[0064] Here, if a spaced distance between an edge end on the bottom surface of the safety vent part 110 and the rupture notch portion 113 is equal, without being biased to the one side, to a spaced distance between the rupture notch portion 113 and the bending notch portion 115 when viewed from the bottom surface of the safety vent part 110, the spaced gap between the safety vent part 110 and the top cap 101 may be most effectively secured when the safety vent part 110 is moved upward due to the internal pressure within the can member 3.

[0065] FIG. 6 is a cross-sectional view taken along line B-B of FIG. 5.

[0066] As illustrated in FIG. 6, in the secondary battery according to the embodiment of the present invention, a connection portion 119 between an edge portion 117 and the rupture portion 111 of the safety vent part 110 may be bent upward.

[0067] That is, since the connection portion 119 is bent upward, although the safety vent part 110 is moved upward due to the internal pressure within the can member 3, the rupture notch portion 113 is not easily cut, and the safety vent part 110 may be moved upward.

[0068] The rupture portion 111 may include an inclined section 111a which is inclined so that a height thereof gradually decreases from the connection portion 119 toward the bending notch portion 115.

[0069] Also, the rupture portion 111 may include a horizontal section 111b having a horizontal shape and extending from an end of the inclined section 111a toward the center of the rupture portion 111.

[0070] Also, the bending notch portion 115 may be positioned on a boundary between the inclined section 111a and the horizontal section 111b.

[0071] That is, as the inclined section 111a is inclined so that a height thereof gradually decreases toward the horizontal section 111b, an increase in the height of the connection portion 119 by being bent upward is compensated. Thus, the horizontal section 111b of the rupture portion 111 is allowed to be in a lowermost position so as to maintain the horizontal level.

[0072] As described above, the horizontal section 111b of the rupture portion 111 may be positioned at the lowermost portion of the safety vent part 110 so as to be coupled to the interrupt portion 121 of the current interrupt device 120 installed at the lower side.

[0073] Also, when the bending notch portion 115 is positioned on the boundary between the inclined section 111a and the horizontal section 111b, it may be significantly effective in bending the bending notch portion 115 to secure an optimal spaced distance d to the top cap 101.

[0074] FIG. 7 is a bottom view illustrating only a safety vent part according to a second embodiment of the present invention.

[0075] As illustrated in FIG. 7, in a secondary battery according to the second embodiment of the present invention, a bending notch portion 115 of a safety vent part 110 may have a discontinuous circular or elliptical shape.

[0076] When the bending notch portion 115 has the discontinuous circular or elliptical shape, a recessed region of the bending notch portion 115 may be minimized, and thus the bending notch portion 115 may be effectively prevented from being cut when an internal pressure inside a can member 3 increases.

[0077] FIG. 8 is a bottom view illustrating only a safety vent part according to a third embodiment of the present invention, and FIG. 9 is a cross-sectional view taken along line C-C of FIG. 8.

[0078] As illustrated in FIGS. 8 and 9, in a secondary battery according to a third embodiment of the present invention, a plurality of bending notch portions 115 of the safety vent part 110 are provided having different lengths.

[0079] In a case in which the plurality of bending notch portions 115 are provided having different lengths, even when the safety vent part 110 rapidly rises due to an instantaneous increase in an internal pressure within the can member 3, the plurality of bending notch portions 115 may be bent to mitigate a rising impact.

[0080] That is, the rupture notch portions 113 may be prevented from being cut, at a pressure before reaching a

preset venting pressure, due to the instant and sudden rising impact.

[0081] FIG. 10 is provided as Comparative example. FIG. 10 is a view of a secondary battery in which a safety vent part according to the related art is installed, illustrating, on the same cross-section of FIG. 3, a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in an internal pressure of a can member.

[0082] In the secondary battery according to the related art as illustrated in FIG. 10, only a rupture notch portion 113' is provided in a safety vent part 110' without the bending notch portion of the present invention.

[0083] In such a secondary battery according to the related art, the safety vent part 110' moves toward a top cap 101' when an internal pressure within a can member 3 increases. Here, a central portion of the safety vent part 110' farthest away from an edge of the safety vent part 110' coupled to the can member 3 may rise to the uppermost side due to the internal pressure of a gas.

[0084] Thus, the central portion of the safety vent part 110' soars to the vicinity of the top cap 101', and approaches the top cap 101' until a spaced distance d_2 therebetween is close to approximately 123 μm . Furthermore, the central portion of the safety vent part 110' may come into contact with the top cap 101'.

[0085] A rupture notch is not cut when the spaced distance d_2 between the safety vent part 110' and the top cap 101' is very small, or the safety vent part 110' comes into contact with the top cap 101' as described above. That is, since the rupture notch is not cut at a preset venting pressure at which

an internal gas should be vented, the venting may not be performed stably.

[0086] In this case, safety problems such as explosion, heat generation, and ignition of a secondary battery may occur.

[0087] In the secondary battery according to the embodiment of the present invention as illustrated in FIG. 3, when the internal pressure within the can member 3 increases, the safety vent part 110 moves toward the top cap 101 while the inclined section 111a of the rupture portion 111 is bent upward. When the safety vent part 110 moves toward top cap 101, the bending notch portion 115 is bent and absorbs a portion of the internal pressure within the can member 3, and thus the horizontal section 111b of the rupture portion 111 may be prevented from soaring toward top cap 101.

[0088] Thus, a spaced distance d_1 between the safety vent part 110 and the top cap 101 may be sufficiently secured to about 186 μm .

[0089] As the spaced distance d_1 between the safety vent part 110 and the top cap 101 is secured as described above, the rupture notch portion 113 is cut at a preset vent pressure, which may prevent interference with the top cap 101 and stably vent the gas within the can member 3.

[0090] FIG. 11 shows the degrees of distribution for comparing the venting pressure distribution in the secondary battery in which the safety vent part according to the related art is installed and the venting pressure distribution in the secondary battery in which the safety vent part according to the present invention is installed.

[0091] In FIG. 11, V_0 is a distribution group of venting pressures, M_x is the maximum value of the venting pressures,

M_n is the minimum value of the venting pressures, and each of V_1 and V_2 is a portion of the venting pressure distribution.

[0092] When comparing the venting pressures of the safety vent part according to the related art in which the bending notch portion is not provided and the venting pressures of the improved safety vent part according to the present invention in which the bending notch portion is provided with reference to FIG. 11, it may be found that the venting pressure distribution of the safety vent part according to the present invention is improved as compared to the venting pressure distribution of the safety vent part according to the related art. The difference therebetween may be about 1.5 times.

[0093] Particularly, during the venting of gas within the can member in the safety vent part according to the related art, the central portion of the safety vent part comes into contact with the top cap in a state in which the rupture notch portion is not cut, and thus the venting is not properly performed at the preset venting pressure. The distribution in which the venting pressures greatly change during abnormal venting described above may be indicated as V_1 and V_2 .

[0094] According to the present invention as described above, the bending notch portion as well as the rupture notch portion is further provided in the bottom surface of the safety vent, and thus the safety vent is not interfered with the top cap when the internal pressure within the can member increases.

[0095] Also, since the safety vent is not interfered with the top cap, the rupture notch portion performs the venting at the preset venting pressure, and thus the venting pressure distribution may be improved.

[0096] Although the secondary battery according to the present invention has been described with reference to the exemplary drawings as described above, the present invention is not limited to the embodiments and drawings described and illustrated above, and various embodiments may be made within the scope of the appended claims by those skilled in the art pertaining to the present invention.

**DECLARATION AND ASSIGNMENT FOR
UTILITY OR DESIGN APPLICATIONS**

Attorney Docket Number: _____

Title of Invention **SECONDARY BATTERY**

As the below named inventor(s) (hereinafter designated as the undersigned), I/We hereby declare that:

This declaration
is directed to:

The attached application, or



United States application or PCT international application number _____

filed on _____

The above-identified application was made or authorized to be made by me/us.

I/We believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

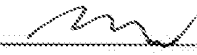
I/We hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WHEREAS, the undersigned as a sole inventor (if only one name is listed below) or as a joint inventor (if plural names are listed below) has invented certain new and useful improvements in the application identified above.

AND, WHEREAS **LG CHEM, LTD.**, (hereinafter referred to as assignee), a corporation of Korea whose mailing address is 128, Yeoui-daero, Yeongdeungpo-gu, Seoul 07336, Republic of Korea is desirous of acquiring the entire right, title and interest in the same.

NOW, THEREFORE, in consideration of the sum of one dollar, the receipt whereof is acknowledged, and other good and valuable consideration, the undersigned hereby sell(s), assign(s) and set(s) over to said assignee my/our entire right, title and interest for the United States of America and all other countries in and to said invention and the aforesaid utility patent application and all original, divisional, continuation, substitute or reissue applications and patents applied for or granted therefor in the United States of America and all other countries, for said invention, including without limitation all applications and patents for said invention claiming priority or benefit of the aforesaid utility application pursuant to any law or treaty, and including all claims for damages by reason of past infringement with the right to sue for and collect the same for its own use and the use of its successors, assigns and legal representatives, and including the right to claim such priority or benefit and the Commissioner of Patents and Trademarks is hereby authorized and requested to issue all patents on said improvements or resulting therefrom to said assignee herein, as assignee of the entire interest therein; and the undersigned for me/us and my/our legal representatives, heirs and assigns do hereby agree and covenant without further remuneration, to execute and deliver all original, divisional, continuation, reissue and other applications for Letters Patent on said improvements and all assignments thereof to said assignee or its assigns, to communicate to said assignee or its representatives all facts known to the undersigned respecting said improvements, whenever requested, to testify in any interferences or other legal proceedings in which any of said applications or patents may become involved, to sign all lawful papers, make all rightful oaths, and to do generally everything necessary to aid assignee, its successors, assigns and nominees to obtain patent protection for said improvements in all countries, the expenses incident to said applications to be borne and paid by said assignee.

And I do hereby authorize my attorneys to insert on this deed the filing date and application number of said application when known.

FULL NAME OF INVENTOR(S)Inventor one: MIN, Geon WooDate: 20.09.22Signature: Additional inventors or a legal representative are being named on 1 additional form(s) attached hereto.**PLEASE SEE THE WARNINGS ON THE LAST PAGE OF THIS FORM.**

DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS	ADDITIONAL INVENTOR(S) Supplemental Sheet
Inventor two: <u>LEE, Je Jun</u> Signature: <u>Lee Je Jun</u>	Date: <u>20. 08. 22</u>
Inventor three: <u>JUNG, Sang Suk</u> Signature: _____	Date: _____
Inventor four: _____ Signature: _____	Date: _____
Inventor five: _____ Signature: _____	Date: _____
Inventor six: _____ Signature: _____	Date: _____
Inventor seven: _____ Signature: _____	Date: _____
Inventor eight: _____ Signature: _____	Date: _____
Inventor nine: _____ Signature: _____	Date: _____
Inventor ten: _____ Signature: _____	Date: _____
WARNING: According to 37 C.F.R. 1.63 (c), "[a] person may not execute an oath or declaration unless the person has reviewed and understands the contents of the application, including the claims and is aware of the duty to disclose to the Office all information known to be material to patentability as defined in 1.56." Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.	



DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS	Attorney Docket Number:
Title of Invention	SECONDARY BATTERY

As the below named inventor(s) (hereinafter designated as the undersigned), I/We hereby declare that:
This declaration
is directed to:

☒ The attached application, or
☐ United States application or PCT international application number _____
 filed on _____

The above-identified application was made or authorized to be made by me/us.

I/We believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I/We hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WHEREAS, the undersigned as a sole inventor (if only one name is listed below) or as a joint inventor (if plural names are listed below) has invented certain new and useful improvements in the application identified above.

AND, WHEREAS **LG CHEM, LTD.,** (hereinafter referred to as assignee), a corporation of Korea whose mailing address is 128, Yeoui-daero, Yeongdeungpo-gu, Seoul 07336, Republic of Korea is desirous of acquiring the entire right, title and interest in the same.

NOW, THEREFORE, in consideration of the sum of one dollar, the receipt wherof is acknowledged, and other good and valuable consideration, the undersigned hereby sell(s), assign(s) and set(s) over to said assignee my/our entire right, title and interest for the United States of America and all other countries in and to said invention and the aforesaid utility patent application and all original, divisional, continuation, substitute or reissue applications and patents applied for or granted therefor in the United States of America and all other countries, for said invention, including without limitation all applications and patents for said invention claiming priority or benefit of the aforesaid utility application pursuant to any law or treaty, and including all claims for damages by reason of past infringement with the right to sue for and collect the same for its own use and the use of its successors, assigns and legal representatives, and including the right to claim such priority or benefit and the Commissioner of Patents and Trademarks is hereby authorized and requested to issue all patents on said improvements or resulting therefrom to said assignee herein, as assignee of the entire interest therein; and the undersigned for me/us and my/our legal representatives, heirs and assigns do hereby agree and covenant without further remuneration, to execute and deliver all original, divisional, continuation, reissue and other applications for Letters Patent on said improvements and all assignments thereof to said assignee or its assigns, to communicate to said assignee or its representatives all facts known to the undersigned respecting said improvements, whenever requested, to testify in any interferences or other legal proceedings in which any of said applications or patents may become involved, to sign all lawful papers, make all rightful oaths, and to do generally everything necessary to aid assignee, its successors, assigns and nominees to obtain patent protection for said improvements in all countries, the expenses incident to said applications to be borne and paid by said assignee.

And I do hereby authorize my attorneys to insert on this deed the filing date and application number of said application when known.

FULL NAME OF INVENTOR(S)

Inventor one: **MIN, Geon Woo** Date:_____

Signature: _____

☒ Additional inventors or a legal representative are being named on 1 additional form(s) attached hereto.

PLEASE SEE THE WARNINGS ON THE LAST PAGE OF THIS FORM.

DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS	ADDITIONAL INVENTOR(S) Supplemental Sheet
Inventor two: <u>LEE, Je Jun</u> Signature: _____	Date: _____
Inventor three: <u>JUNG, Sang Suk</u> Signature: <u>[Signature]</u>	Date: <u>2020. 6. 20</u>
Inventor four: _____ Signature: _____	Date: _____
Inventor five: _____ Signature: _____	Date: _____
Inventor six: _____ Signature: _____	Date: _____
Inventor seven: _____ Signature: _____	Date: _____
Inventor eight: _____ Signature: _____	Date: _____
Inventor nine: _____ Signature: _____	Date: _____
Inventor ten: _____ Signature: _____	Date: _____
WARNING: According to 37 C.F.R. 1.63 (c), "[a] person may not execute an oath or declaration unless the person has reviewed and understands the contents of the application, including the claims and is aware of the duty to disclose to the Office all information known to be material to patentability as defined in 1.56." Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Geon Woo Min	
	Art Unit	N/A	
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGCHEM 3.3F-3059	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	10090497	B2	2018-10-02	Kim et al.	

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20050214641	A1	2005-09-29	Kim	
	2	20100086835	A1	2010-04-08	KIM	
	3	20110200854	A1	2011-08-18	Bak et al.	
	4	20120114979	A1	2012-05-10	KIM et al.	

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²ⁱ	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Geon Woo Min	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

1	2001102038	JP	A	2001-04-13	Toyo Kohan Co Ltd	Machine generated English translation attached.	×
2	H06215746	JP	A	1994-08-05	Matsushita Electric Ind Co Ltd	Machine generated English translation attached.	☒
3	H08339793	JP	A	1996-12-24	Mitsubishi Cable Ind Ltd	Machine generated English translation attached.	☒
4	100578805	KR	B1	2006-05-11	Samsung Sdi Co., Ltd	U.S. Publication No. 20050214641 A1 is an English language counterpart.	☒
5	101058386	KR	B1	2011-08-22	Lg Chem. Ltd	U.S. Publication No. 20120114979 A1 is an English language counterpart.	☒
6	20100038054	KR	A	2010-04-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20100086835 A1 is an English language counterpart.	☒
7	20110095118	KR	A	2011-08-24	Samsung Sdi Co., Ltd	U.S. Publication No. 20110200854 A1 is an English language counterpart.	☒
8	20150035267	KR	A	2015-04-06	Lg Chem. Ltd	Machine generated English translation attached.	☒

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NON-PATENT LITERATURE DOCUMENTS

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report for Application No. PCT/KR2019/015396, mailed February 20, 2020, pages 1-2.	×

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Geon Woo Min
Art Unit	N/A
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Geon Woo Min
Art Unit	N/A
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2021-01-11
Name/Print	Jordan T. Riviello	Registration Number	72,515

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

- ☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above: 86765
- OR
- ☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

- ☒ The address associated with the above-mentioned Customer Number

OR

- ☐ The address associated with Customer Number:

OR

Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

LG Chem, Ltd.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature	<i>Kim Min Ho</i>	Date (Optional)	<i>June 28, 2016</i>
Name	Minho Kim		
Title	General Manager		

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ Total of 1 forms are submitted.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: January 11, 2021
Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

International Application No.: PCT/KR2019/015396

Confirmation No.: N/A

International Filing Date: November 13, 2019

Art Unit: N/A

For: Secondary Battery

Examiner: Not Yet Assigned

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

It is respectfully requested that the references listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited references are legally available prior art or that the same are pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-3059, from which the undersigned is authorized to draw.

Dated: January 11, 2021

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant(s)

LD-458

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

Secondary Battery

First Named Inventor/Applicant Name:

Geon Woo Min

Filer:

James A. Vezeris/Jessica Ortiz

Attorney Docket Number:

LGCHEM 3.3F-3059

Filed as Large Entity

Filing Fees for U.S. National Stage under 35 USC 371

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
NATIONAL STAGE FEE	1631	1	320	320
NATL STAGE SEARCH FEE - REPORT PROVIDED	1642	1	540	540
NATIONAL STAGE EXAM - ALL OTHER CASES	1633	1	800	800

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1660

Electronic Acknowledgement Receipt

EFS ID:	41610426
Application Number:	17259395
International Application Number:	PCT/KR2019/015396
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	James A. Vezeris/Jessica Ortiz
Filer Authorized By:	James A. Vezeris
Attorney Docket Number:	LGCHEM 3.3F-3059
Receipt Date:	11-JAN-2021
Filing Date:	
Time Stamp:	15:49:45
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1660
RAM confirmation Number	E20211AF50340444
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	LGCHEM3430_- _Application_Data_Sheet_Fillable_PDF.pdf	1255863	no	9
			0ccca811703e41021744d62fd650bfe33c226e8e		
Warnings:					
Information:					
2	Foreign Reference	LGCHEM3430_- _JP2001102038A.pdf	8797071	no	9
			976b7271f69e37bb592c21a42bb6cc43f0b36229		
Warnings:					
Information:					
3	Foreign Reference	LGCHEM3430_- _Translation_of_JP2001102038A.pdf	319863	no	10
			8a17a083a23114d1832c19adbb2825dbe9c7c352		
Warnings:					
Information:					
4	Preliminary Amendment	Preliminary_Amendment.pdf	33029	no	7
			deeb832bfa1fbfd79349bd5d7b655cb410e845a		
Warnings:					
Information:					
5	Foreign Reference	LGCHEM3430_- _JPH06215746A.pdf	4986964	no	6
			74f57e83670e8b4dcb756a3278b896b03dfffa2c1		
Warnings:					
Information:					
6	Foreign Reference	LGCHEM3430_- _Translation_of_JPH06215746A.pdf	290245	no	7
			1393293c359f6785dc1c609bc8b8c232ce312f52		
Warnings:					
Information:					

7	Foreign Reference	LGCHEM3430_- _JPH08339793A.pdf	24163828	no	19
			c9db6832842b3478646cffc0a531bf20ea902d87		
Warnings:					
Information:					
8	Foreign Reference	LGCHEM3430_- _Translation_of_JPH08339793 A.pdf	476226	no	31
			ca112419855ea846511bcd157ba179b972d3ee21		
Warnings:					
Information:					
9	Foreign Reference	LGCHEM3430_- _KR100578805B1.pdf	5458180	no	13
			ac69417f4f7e781a9fe9ceaf2ef3a60e942932e		
Warnings:					
Information:					
10	Foreign Reference	LGCHEM3430_- _KR101058386B1.pdf	4755651	no	26
			1f04aebb42f338145caffba4ba83e5a472ba4a46		
Warnings:					
Information:					
11	Foreign Reference	LGCHEM3430_- _KR20100038054A.pdf	3558048	no	20
			d1602fa78954e9c9cf9744120307656efa149ba0		
Warnings:					
Information:					
12	Foreign Reference	LGCHEM2286_- _KR1020110095118A.pdf	1603911	no	14
			4924c0cb111f1751bd0cec552b60e5fa74820d59		
Warnings:					
Information:					
13	Foreign Reference	LGCHEM3430_- _KR20150035267A.pdf	2253263	no	12
			36ed5260d4e2d9880ae1b0dfb66f9ae7fdd0096a		
Warnings:					
Information:					

14	Foreign Reference	LGCHEM3430_- _Translation_of_KR2015003526 7A.pdf	328936	no	14
			e04caf1563066832926804012f1d168b4075ee34		
Warnings:					
Information:					
15	Non Patent Literature	LGCHEM3430_- _ISR.pdf	68483	no	2
			23fac9b2d1a2df680b2cf41e2b85d36162cb2eaf5		
Warnings:					
Information:					
16	Non Patent Literature	LGCHEM3430_- _ISR_ENG.pdf	111950	no	2
			5ba58ced3cf0d140cbbce25c118a24bf3df057d2		
Warnings:					
Information:					
17	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM3430_- _Information_Disclosure_State ment_Fillable_PDF.pdf	1034880	no	5
			198c8d9a9c716178a9db50ef2d8e1ec6d45b7fda		
Warnings:					
Information:					
18	Transmittal of New Application	Transmittal_Letter_to_the_Uni ted_States_Designated- Elected_Office.pdf	48907	no	3
			6c3da2f01addfcc22f4699aac348c53f56e13706		
Warnings:					
Information:					
19	Documents submitted with 371 Applications	Published_PCT_Publication. PDF	7294176	no	22
			1e4c39b673597f41878876555123fe1f3be295d4		
Warnings:					
Information:					
20	Specification	English_Translation_of_PCT_A pplication.pdf	91595	no	21
			aacafad68f2a91c6a6fb945b82c188ddc7b9061e		
Warnings:					
Information:					

21	Specification	Substitute_Spec_-_01_06_2021.pdf	84340	no	17
			4212dcc0da10ed2dc066a848fce4f86abb2b971e		
Warnings:					
Information:					
22	Specification	Clean_Spec_-_01_06_2021.pdf	92690	no	18
			f91e98c4bdf9716241e0900d3109b204a439f941		
Warnings:					
Information:					
23	Oath or Declaration filed	Executed_Declaration-Assignment_Document.PDF	290243	no	4
			02828df2867bbde5d871063013e281aa6a39bea2		
Warnings:					
Information:					
24	Power of Attorney	LGCHEM_Power_of_Attorney.PDF	512304	no	1
			eac1f0267196dbfb0ea5fab5de5bb2dcd21f2a9		
Warnings:					
Information:					
25	Transmittal Letter	IDS_-_Information_Disclosure_State-ment_-_Reference_Enclosed_LD-458.pdf	21104	no	1
			ce2542a3530b02f17fcd197842077133d032d3d3		
Warnings:					
Information:					
26	Fee Worksheet (SB06)	fee-info.pdf	35220	no	2
			147e5d55cde35f77fcd8fe5ae3de8bd64e461684		
Warnings:					
Information:					
Total Files Size (in bytes):			67966970		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Geon Woo		Min		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Daejeon		Country of Residence ⁱ	KR	
Mailing Address of Inventor:					
Address 1		LG Chem Research Park, 188, Munji-ro			
Address 2		Yuseong-gu			
City	Daejeon		State/Province		
Postal Code	34122		Country ⁱ	KR	
Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Je Jun		Lee		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City			Country of Residence ⁱ		
Mailing Address of Inventor:					
Address 1		LG Chem Research Park, 188, Munji-ro			
Address 2		Yuseong-gu			
City	Daejeon		State/Province		
Postal Code	34122		Country ⁱ	KR	
Inventor	3				Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Sang Suk		Jung	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Daejeon	Country of Residence ⁱ	KR	

Mailing Address of Inventor:				
Address 1	LG Chem Research Park, 188, Munji-ro			
Address 2	Yuseong-gu			
City	Daejeon	State/Province		
Postal Code	34122	Country ⁱ	KR	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	86765		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	Secondary Battery		
Attorney Docket Number	LGCHEM 3.3F-3059	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)	3

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country ⁱ

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	86765		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78. When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	a 371 of international	PCT/KR2019/015396	2019-11-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		

			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ^j (if applicable)
10-2018-0149369	KR	2018-11-28	
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	LG Chem, Ltd.	
Mailing Address Information For Applicant:		
Address 1	128, Yeoui-daero	
Address 2	Yeongdeungpo-gu	
City	Seoul	State/Province
Country	KR	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		

Assignee	1		
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.			
Remove			
If the Assignee or Non-Applicant Assignee is an Organization check here. ☒			
Organization Name	LG Chem, Ltd.		
Mailing Address Information For Assignee including Non-Applicant Assignee:			
Address 1	128, Yeoui-daero		
Address 2	Yeongdeungpo-gu		
City	Seoul	State/Province	
Country ⁱ	KR	Postal Code	07336
Phone Number		Fax Number	
Email Address			
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add			

Signature:[Remove](#)

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is **not** checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Jordan T. Riviello/		Date (YYYY-MM-DD)	2021-01-11
First Name	Jordan	Last Name	Riviello	Registration Number
				72,515
Additional Signature may be generated within this form by selecting the Add button. Add				

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-3059
		Application Number	
Title of Invention	Secondary Battery		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



Espacenet

Bibliographic data: KR100578805 (B1) — 2006-05-11

CAP ASSEMBLY AND SECONDARY BATTERY USING THE SAME

Inventor(s): KIM, YONG SAM

Applicant(s): SAMSUNG SDI CO., LTD

Classification: - **international:** H01M2/00; H01M2/02; H01M2/04; H01M2/08; H01M2/12; H01M2/34; H01M6/50
- **cooperative:** A01K69/06 (KR); H01M2/0413 (EP); H01M2/1241 (EP); H01M2200/20 (EP); Y02E60/10 (EP)

Application number: KR20040019953 20040324 Global Dossier

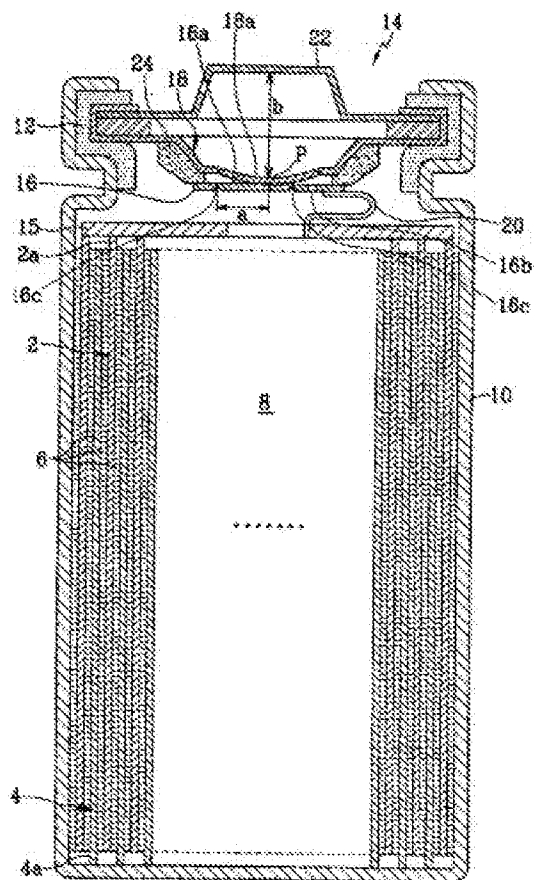
Priority number (s): KR20040019953 20040324

Also published as: CN100359717 (C) CN1697209 (A) JP2005276838 (A) KR20050094664 (A) US2005214641 (A1)

Abstract not available for KR100578805 (B1)

Abstract of corresponding document: US2005214641 (A1)

A cap assembly for a secondary battery may be constructed with a cap cover, a tab plate arranged along one side of the cap cover, and a vent plate arranged between the cap cover and the tab plate. The tab plate has a current breaking portion configured to be torn by a predetermined force, and the vent plate has a vent dedicated to be altered, or broken, by a predetermined force. The current breaking portion is attachedly fixed to the vent, and the center point of the current breaking portion and the point where the current breaking portion is attached to the vent are not coincident.



(19)대한민국특허청(KR)
(12) 등록특허공보(B1)

(51) . Int. Cl.	(45) 공고일자	2006년05월11일
H01M 2/04 (2006.01)	(11) 등록번호	10-0578805
	(24) 등록일자	2006년05월04일

(21) 출원번호	10-2004-0019953	(65) 공개번호	10-2005-0094664
(22) 출원일자	2004년03월24일	(43) 공개일자	2005년09월28일

(73) 특허권자 삼성에스디아이 주식회사
 경기 수원시 영통구 신동 575

(72) 발명자 김용삼
 서울특별시금천구독산4동188-34

(74) 대리인 유미특허법인

참사관 : 이선희

(54) 캡 조립체와 이를 구비한 이차 전지

요약

본 발명은 내압 상승으로 인한 전지의 폭발을 방지하는 안전 장치를 고출력 성능을 목적으로 개선한 캡 조립체와 이를 구비한 이차 전지에 관한 것으로서,

상기 캡 조립체는, 전극 조립체가 내장된 케이스와 결합하는 캡 커버와; 캡 커버 내측에 위치하며 전극 조립체의 전극탭이 고정되고, 전극탭이 고정된 부위로부터 이격된 곳에 가변부가 형성되는 탭 플레이트와; 캡 커버와 탭 플레이트 사이에 위치하며 가변부를 향해 돌출되어 가변부에 고정되는 밴트를 갖는 밴트 플레이트를 포함하며, 가변부는 평면에서 보았을 때 밴트와 가변부의 고정점이 가변부의 중심점으로부터 편심된 형상으로 이루어진다.

대표도

도 1

색인어

전지, 이차전지, 캡조립체, 캡커버, 밴트, 밴트플레이트, 탭플레이트, 노치

명세서

도면의 간단한 설명

도 1은 본 발명의 실시예에 따른 이차 전지의 부분 단면도이다.

도 2는 도 1에 도시한 탭 플레이트의 저면도이다.

도 3은 본 발명의 실시예에 따른 이차 전지에서 가변부의 다른 실시예를 설명하기 위해 도시한 탭 플레이트의 저면도이다.

도 4와 도 5는 본 발명의 실시예에 따른 이차 전지에서 내압 상승시 댄트 플레이트와 탭 플레이트의 작동 과정을 보여주는 부분 단면도이다.

도 6은 종래 기술에 의한 이차 전지의 부분 단면도이다.

도 7은 도 6에 도시한 댄트 플레이트의 평면도이다.

도 8은 종래 기술에 의한 이차 전지에서 내압 상승시 댄트 플레이트의 작동 과정을 보여주는 부분 단면도이다.

발명의 상세한 설명

발명의 목적

발명이 속하는 기술 및 그 분야의 종래기술

본 발명은 이차 전지에 관한 것으로서, 보다 상세하게는 내압 상승으로 인한 전지의 폭발을 방지하는 안전 장치를 고출력 성능을 목적으로 개선한 캡 조립체와 이를 구비한 이차 전지에 관한 것이다.

일반적으로 이차 전지는 충전이 불가능한 일차 전지와 달리 충전과 방전이 가능한 전지로서, 대표적으로 니켈수소(Ni-MH) 전지와 리튬(Li) 전지 및 리튬이온(Li-ion) 전지 등이 있다. 특히 리튬 이차 전지는 작동 전압과 단위 중량당 에너지 밀도가 높아 휴대용 전자기기 분야에 더욱 적합하며, 최근 들어서는 하이브리드 전기 자동차(Hybrid Electric Vehicle; HEV)의 모터 구동원으로 적용이 활발하게 연구되고 있다.

상기 이차 전지는 충전조건 이상으로 과충전되거나 전극 조립체의 양극판과 음극판이 쇼트되는 경우 전지 내부에 가스가 발생하고, 이 가스에 의해 내부 압력이 높아져 폭발 및 발화의 위험을 안고 있다.

따라서 이차 전지는 과충전과 쇼트에 따른 문제점을 해결하기 위하여 내부에 안전 장치를 구비하고 있는데, 상기 안전 장치로는 첫-다운 세퍼레이터, 온도 상승시 전류를 차단하는 피티씨(PTC; positive temperature coefficient) 소자 및 압력 상승시 가스를 배출하여 압력을 낮추는 안전 댄트(safety vent) 등이 잘 알려져 있다.

도 6은 종래 기술에 의한 이차 전지의 부분 단면도이고, 도 7은 도 6에 도시한 댄트 플레이트의 평면도이다.

도면을 참고하면, 이차 전지는 격리판(1)을 사이로 양극판(3)과 음극판(5)이 권취된 전극 조립체(7)와, 전해액과 함께 전극 조립체(7)를 수용하는 케이스(9)와, 가스켓(11)을 매개로 케이스(9)의 개구부 상단에 위치하여 케이스(9)를 밀봉하는 캡 조립체(13)와, 안전 장치로서 캡 조립체(13) 내부에 위치하는 댄트 플레이트(15)를 포함한다.

상기 캡 조립체(13)는 그 위에서부터 아래로 캡 커버(17)와 댄트 플레이트(15)와 고리 모양의 절연체(19) 및 탭 플레이트(21)가 순서대로 위치한다. 탭 플레이트(21)는 양극탭(23)과 용접으로 고정되며, 가스 통과를 위한 다수의 홀들(21a)을 형성한다.

상기 댄트 플레이트(15)는 중앙에 오목한 형상의 댄트(15a)를 형성하며 이 댄트(15a) 중심에서 탭 플레이트(21)와 용접으로 고정된다. 그리고 댄트(15)의 둘레를 따라서는 기계적 방법 혹은 예칭법에 의해 일정한 깊이를 지니는 노치(15b)가 형성되는데, 이 노치(15b)에 의해 댄트(15a)가 댄트 플레이트(15)의 다른 부위에 비해 상대적으로 취약해져 변형을 일으킬 수 있다. 이 때, 댄트(15a)는 그 중심점이 탭 플레이트(21)와의 용접점(A)과 일치하는 원형 혹은 타원형 구조로 이루어진다.

이로서 전지 내부에 가스가 발생하여 내압이 높아지면, 도 8에 도시한 바와 같이 압력에 의해 댄트(15a)가 위로 상승하면서 탭 플레이트(21)와 분리되어 양극탭(23)에서 캡 커버(17)를 향한 전류 흐름이 차단되며, 내압이 심한 경우에는 노치(15b) 부분을 기준으로 댄트(15a)가 파열되면서 가스를 배출하게 된다.

이와 같이 종래의 이차 전지는 밴트(15a)가 용접점(A)을 기준으로 탭 플레이트(21)로부터 분리되면서 전류를 차단하는 구조로서, 밴트(15a)의 용이한 작동을 위해 밴트(15a)와 탭 플레이트(21)의 용접 강도를 그다지 크게 설정하지 않는 것이 일반적이다. 즉, 종래에는 탭 플레이트(21)와 밴트 플레이트(15) 사이에 전류 통로만이 형성될 수 있도록 최소한의 용접 강도로 두 부재를 고정시키고 있다.

이에 따라 종래의 이차 전지에 있어서는 밴트(15a)와 탭 플레이트(21) 사이의 충분하지 못한 용접 상태로 인해 이 부분에 대한 접촉 저항을 높이고 있다. 이러한 구조는 전극 조립체(7)의 전류를 캡 조립체(13)로 충분하게 유도하지 못하는 결과를 초래하여 해당 전지의 특성을 극대화하지 못할 수 있으며, 이는 특히 HEV의 모터 구동용과 같은 고출력 전지에 적합하지 않은 구조라 할 수 있다.

발명이 이루고자 하는 기술적 과제

따라서 본 발명은 상기한 문제점을 해소하기 위한 것으로서, 본 발명의 목적은 전극 조립체에서 캡 조립체로 향하는 전류 흐름 상에서 발생할 수 있는 저항 인자들을 최소화하면서 밴트의 원활한 작동을 유도할 수 있는 캡 조립체 및 이를 구비한 이차 전지를 제공하는데 있다.

발명의 구성 및 작용

상기의 목적을 달성하기 위하여 본 발명은,

전극 조립체가 내장된 케이스와 결합하는 캡 커버와, 캡 커버 내측에 위치하며 전극 조립체의 전극탭이 고정되고, 전극탭이 고정된 부위로부터 이격된 곳에 가변부가 형성되는 탭 플레이트와, 캡 커버와 탭 플레이트 사이에 위치하며 가변부를 향해 돌출되어 가변부에 고정되는 밴트를 갖는 밴트 플레이트를 포함하며, 가변부는 평면에서 보았을 때 밴트와 가변부의 고정점이 가변부의 중심점으로부터 편심된 형상으로 이루어지는 이차 전지의 캡 조립체를 제공한다.

상기 탭 플레이트는 가변부의 둘레를 따라 일정한 깊이를 지닌 노치를 형성하며, 가변부 외측에 적어도 1의 통기공을 형성한다.

상기 고정점을 관통하여 이 고정점과 가까운 측의 가변부 가장자리 부위와 고정점과 먼 측의 가변부 가장자리 부위를 양분하는 선을 가정할 때, 가변부는 고정점과 먼 측의 가장자리 부위 길이가 고정점과 가까운 측의 가장자리 부위 길이보다 큰 비대칭 형상으로 이루어진다. 이러한 가변부는 사다리꼴과 삼각형 중 어느 하나의 형상으로 이루어질 수 있다.

또한, 상기의 목적을 달성하기 위하여 본 발명은,

격리판을 사이에 두고 위치하는 양,음극판을 포함하는 전극 조립체와; 전극 조립체가 내장되는 공간부를 구비하는 케이스와; 케이스와 결합하는 캡 커버와, 캡 커버 내측에 위치하며 상기 양극판과 연결된 양극탭이 고정되고 양극탭이 고정된 부위로부터 이격된 곳에 가변부가 형성되는 탭 플레이트와, 캡 커버와 탭 플레이트 사이에 위치하며 가변부를 향해 돌출되어 가변부에 고정되는 밴트를 갖는 밴트 플레이트를 포함하는 캡 조립체를 포함하며, 가변부는 평면에서 보았을 때 밴트와 가변부의 고정점이 가변부의 중심점으로부터 편심된 형상으로 이루어지는 이차 전지를 제공한다.

상기 탭 플레이트는 가변부의 둘레를 따라 일정한 깊이를 지닌 노치를 형성하며, 가변부 외측에 적어도 1의 통기공을 형성한다.

이하, 첨부한 도면을 참고하여 본 발명의 바람직한 실시예를 보다 상세하게 설명하면 다음과 같다.

도 1은 본 발명의 실시예에 따른 이차 전지의 부분 단면도이다.

도면을 참고하면, 이차 전지는 양극판(2)과 음극판(4)이 격리판(6)을 사이에 두고 위치하는 전극 조립체(8)와, 전해액과 함께 전극 조립체(8)를 수용하는 케이스(10)와, 가스켓(12)을 매개로 케이스(10)의 개구부 상단에 위치하여 케이스(10)를 밀봉하는 캡 조립체(14)와, 캡 조립체(14) 내부에 구비되며 내압 상승시 파단되어 전류 흐름을 차단하는 탭 플레이트(16)와 밴트 플레이트(18)를 포함한다.

상기 케이스(10)는 알루미늄, 알루미늄 합금 또는 니켈이 도금된 스틸과 같은 도전성 금속 재료로 형성되고, 그 형상은 전극 조립체(8)가 위치하는 내부 공간부를 가진 원통형 또는 육면체 또는 그 이외의 형상으로 이루어질 수 있다.

상기 전극 조립체(8)는 양극판(2)과 음극판(4)이 격리판(6)을 사이에 두고 적층된 구성 또는 양극판(2)과 격리판(6) 및 음극판(4)이 적층된 상태에서 이들이 감겨 형성된 젤리롤(jelly-roll) 타입으로 이루어질 수 있다. 도면에서는 일례로 원통형 케이스(10)에 젤리롤 타입의 전극 조립체(8)가 위치하는 경우를 도시하였다.

상기와 같이 구성된 전극 조립체(8)에서, 음극판(4)에 활물질이 도포되지 않은 부지부(4a)가 케이스(10)와 접촉하여 케이스(10)가 음극 단자로 기능할 수 있고, 양극탭(20)이 양극판(2)과 캡 조립체(14) 사이에 설치되어 이들을 전기적으로 연결함으로써 캡 조립체(14)가 양극 단자로 기능할 수 있다.

본 실시예에서 캡 조립체(14)는 그 위에서부터 아래로 캡 커버(22), 밴트 플레이트(18), 절연체(24) 및 탭 플레이트(16)가 순서대로 위치한다. 탭 플레이트(16)는 그 상부면에서 밴트 플레이트(18)의 밴트(18a)와 용접을 통해 고정되고, 하부면에서 양극탭(20)과 고정된다. 탭 플레이트(16)는 또한 가스 통과를 위한 다수의 통기공(16b)을 구비한다.

여기서, 본 실시예에 의한 이차 전지는 양극탭(20)에서 탭 플레이트(16)와 밴트 플레이트(18)를 거쳐 캡 커버(22)로 향하는 전류 흐름 상에서 발생할 수 있는 저항 인자들을 최소화하도록 탭 플레이트(16)의 구조를 개선하여 전지의 출력 성능을 높인 구성을 제공한다.

도 2는 도 1에 도시한 탭 플레이트의 저면도이다.

도 1과 도 2를 참고하면, 탭 플레이트(16)는 그 중앙에 가변부(16a)를 형성하고, 가변부(16a)가 밴트 플레이트(18)의 밴트(18a)와 용접을 통해 고정되는 구조를 이룬다. 가변부(16a)는 탭 플레이트(16)의 다른 부위에 비해 상대적으로 취약해져 쉽게 변형을 일으킬 수 있는 부분으로서, 이를 위해 가변부(16a)의 둘레를 따라 기계적 방법 혹은 에칭법으로 일정한 깊이를 지닌 노치(16c)를 형성한다. 상기 탭 플레이트(16)는 가변부(16a) 외측에 다수의 통기공(16b)을 형성하며, 가변부(16a) 외측에서 양극탭(20)과 용접을 통해 고정된다.

특히 본 실시예에서 탭 플레이트(16)는 밴트 플레이트(18)와 용접을 통해 고정될 때 용접 강도를 높여 밴트 플레이트(18)에 단단하게 고정된다. 이로써 탭 플레이트(16)는 밴트 플레이트(18)와의 고정점, 즉 용접점(A)에서의 접촉 저항을 낮추어 고전류 흐름을 용이하게 한다.

따라서 본 실시예의 이차 전지는 내압 상승시 탭 플레이트(16)와 밴트 플레이트(18)의 용접점(A)이 분리되지 않고, 전지 내부에서 발생한 가스가 밴트(18a)를 밀어올리는 힘에 의해 탭 플레이트(16)의 가변부(16a) 전체가 탭 플레이트(16)로부터 파단되어 전류를 차단시키는 구성으로 이루어진다.

이를 위하여 상기 가변부(16a)는 밴트(18a)와의 용접점(A)이 가변부(16a)의 형상 중심점과 일치하지 않고 편심된 형상으로 이루어져 내압 상승시 용접점(A)과 가까운 부분의 가변부(16a) 가장자리 부위로부터 먼저 파단이 일어나도록 하고 있다.

이와 동시에 상기 용접점(A)을 관통하는 가상의 선을 가정하여 이 선이 용접점(A)과 가까운 측의 가변부(16a) 가장자리 부위와 먼 측의 가변부(16a) 가장자리 부위를 양분할 때, 가변부(16a)는 용접점(A)과 먼 측의 가장자리 부위 길이가 용접점(A)과 가까운 측의 가장자리 부위 길이보다 큰 비대칭 형상으로 이루어진다. 이로서 가변부(16a)는 탭 플레이트(16)와 접촉하는 가장자리 측단면의 면적을 확대시킴으로써 탭 플레이트(16)와 가변부(16a) 사이의 저항을 낮추도록 하고 있다.

도 2를 참고하면, 상기 가변부(16a)는 사다리꼴 모양으로서 밴트와의 용접점(A)이 가변부(16a)의 중심점과 일치하지 않고 편심되어 있으며, 용접점(A)을 관통하는 가상의 선(B선)이 용접점(A)과 가까운 측의 가변부(16a) 가장자리 부위(도면을 기준으로 B선 우측에 위치하는 가변부 가장자리 부위)와 용접점(A)과 먼 측의 가장자리 부위(도면을 기준으로 B선 좌측에 위치하는 가변부 가장자리 부위)를 양분할 때, 가변부(16a)는 용접점(A)과 먼 측의 가장자리 길이가 용접점(A)과 가까운 측의 가장자리 길이보다 큰 비대칭 형상으로 이루어진다.

도 3은 가변부의 다른 실시예를 설명하기 위해 도시한 탭 플레이트의 저면도이다.

도면을 참고하면, 탭 플레이트(16')의 가변부(16d)는 대략적인 삼각형 모양으로서 용접점(A)이 가변부(16d)의 중심점과 일치하지 않고 편심되어 있으며, 용접점(A)을 관통하는 가상의 선(B선)이 용접점(A)과 가까운 측의 가변부(16d) 가장자리 부위(도면을 기준으로 B선의 우측에 위치하는 가변부 가장자리 부위)와 용접점(A)과 먼 측의 가장자리 부위(도면을 기준으로 B선의 좌측에 위치하는 가변부 가장자리 부위)를 양분할 때, 가변부(16d)는 용접점(A)과 먼 측의 가장자리 길이가 용접점(A)과 가까운 측의 가장자리 길이보다 큰 비대칭 형상으로 이루어진다.

상기에서는 가변부(16a, 16d)가 사다리꼴 또는 삼각형인 경우를 설명하였으나, 가변부의 형상은 이에 한정되지 않고 전술한 형상 조건을 만족하는 형상이면 모두 무방하다.

도 4와 도 5는 내압 상승시 밴트 플레이트와 탭 플레이트의 작동 과정을 보여주는 이차 전지의 부분 단면도이다.

도 4를 참고하면, 이차 전지가 충전조건 이상으로 과충전되거나 전극 조립체의 양극판과 음극판이 쇼트되어 전지 내부에 가스가 발생하고 내부 압력이 높아질 때, 상기 가스가 탭 플레이트(16)의 통기공(16b)을 통해 밴트 플레이트(18)의 밴트(18a)를 위로 밀어 올리게 된다.

이 때, 본 실시예의 이차 전지는 가변부(16a)와 밴트(18a)가 용접에 의해 공고하게 고정되어 있으므로 용접점(A)에서 분리가 일어나지 않고, 밴트(18a)가 위로 상승하는 변형이 일어남과 동시에 용접점(A)과 가까운 측의 가변부(16a) 가장자리 부위가 탭 플레이트(16)로부터 분리되면서 파단이 일어나고, 이 파단에 의해 가변부(16a) 전체가 탭 플레이트(16)로부터 분리되어 전류 차단이 이루어진다.

그리고 도 5에 도시한 바와 같이 전류 차단이 일어난 후에도 전지 내압이 상승하는 경우에는 밴트(18a)가 밴트 플레이트(18)로부터 파단되어 가스 배출이 이루어짐으로써 전지의 폭발과 발화의 위험을 낮추게 된다.

이와 같이 본 실시예에서는 탭 플레이트(16)와 밴트 플레이트(18)를 고정시키는 용접점(A)에서의 접촉 저항을 낮추고, 가변부(16a)가 탭 플레이트(16)와 접촉하는 가장자리 측단면의 면적을 확대시킴으로써 가변부(16a)와 탭 플레이트(16) 사이의 저항을 낮추고 있다. 따라서 본 실시예의 이차 전지는 전극 조립체(8)에서 캡 조립체(14)로 향하는 전류 흐름 상에서 발생할 수 있는 저항 인자들을 최소화하여 고효율 성능을 구현할 수 있다.

상기에서는 본 발명의 바람직한 실시예에 대하여 설명하였지만, 본 발명은 이에 한정되는 것이 아니고 특허청구범위와 발명의 상세한 설명 및 첨부한 도면의 범위 안에서 여러 가지로 변형하여 실시하는 것이 가능하고 이 또한 본 발명의 범위에 속하는 것은 당연하다.

발명의 효과

이와 같이 본 실시예에 의한 이차 전지는 전술한 탭 플레이트 형상에 의해 탭 플레이트와 밴트 플레이트간 접촉 저항을 낮추고, 가변부와 탭 플레이트의 접촉 면적을 확대시킨다. 따라서 본 실시예의 이차 전지는 양극탭에서 탭 플레이트와 밴트 플레이트를 거쳐 캡 커버를 향하는 전류 흐름 상에서 발생할 수 있는 저항 인자들을 최소화하여 고효율 흐름을 용이하게 하고, 그 결과 전지의 출력 성능을 높이는 효과를 갖는다.

(57) 청구의 범위

청구항 1.

전극 조립체가 내장된 케이스와 결합하는 캡 커버와;

상기 캡 커버 내측에 위치하며, 상기 전극 조립체의 전극탭이 고정되고, 전극탭이 고정된 부위로부터 이격된 곳에 가변부가 형성되는 탭 플레이트; 및

상기 캡 커버와 탭 플레이트 사이에 위치하며, 상기 가변부를 향해 돌출되어 가변부에 고정되는 밴트를 갖는 밴트 플레이트

를 포함하며,

상기 가변부는 평면에서 보았을 때 상기 뱀트와 가변부의 고정 점이 가변부의 중심점으로부터 편심된 형상으로 이루어지는 이차 전지의 캡 조립체.

청구항 2.

제1항에 있어서,

상기 탭 플레이트는 상기 가변부의 둘레를 따라 일정한 깊이를 지닌 노치를 형성하는 이차 전지의 캡 조립체.

청구항 3.

제1항에 있어서,

상기 탭 플레이트는 상기 가변부 외측에 적어도 1의 통기공을 형성하는 이차 전지의 캡 조립체.

청구항 4.

제1항에 있어서,

상기 고정점을 관통하여 이 고정점과 가까운 측의 가변부 가장자리 부위와 고정점과 먼 측의 가변부 가장자리 부위를 양분하는 선을 가정할 때, 상기 가변부는 고정점과 먼 측의 가장자리 부위 길이가 고정점과 가까운 측의 가장자리 부위 길이보다 큰 비대칭 형상으로 이루어지는 이차 전지의 캡 조립체.

청구항 5.

제4항에 있어서,

상기 가변부는 사다리꼴과 삼각형 중 어느 하나의 형상으로 이루어지는 이차 전지의 캡 조립체.

청구항 6.

제1항에 있어서,

상기 가변부와 뱀트가 용접으로 고정되는 이차 전지의 캡 조립체.

청구항 7.

격리판을 사이에 두고 위치하는 양,음극판을 포함하는 전극 조립체와;

상기 전극 조립체가 내장되는 공간부를 구비하는 케이스; 및

상기 케이스와 결합하는 캡 커버와, 캡 커버 내측에 위치하며 상기 양극판과 연결된 양극탭이 고정되고 양극탭이 고정된 부위로부터 이격된 곳에 가변부가 형성되는 탭 플레이트와, 캡 커버와 탭 플레이트 사이에 위치하며 가변부를 향해 돌출되어 가변부에 고정되는 뱀트를 갖는 뱀트 플레이트를 포함하는 캡 조립체

를 포함하며,

상기 가변부는 평면에서 보았을 때 상기 램프와 가변부의 고정 점이 가변부의 중심점으로부터 편심된 형상으로 이루어지는 이차 전지.

청구항 8.

제7항에 있어서,

상기 탭 플레이트는 상기 가변부의 둘레를 따라 일정한 깊이를 지닌 노치를 형성하는 이차 전지.

청구항 9.

제7항에 있어서,

상기 탭 플레이트는 상기 가변부 외측에 적어도 1의 통기공을 형성하는 이차 전지.

청구항 10.

제7항에 있어서,

상기 고정점을 관통하여 이 고정점과 가까운 측의 가변부 가장자리 부위와 고정점과 먼 측의 가변부 가장자리 부위를 양분하는 선을 가정할 때, 상기 가변부는 고정점과 먼 측의 가장자리 부위 길이가 고정점과 가까운 측의 가장자리 부위 길이보다 큰 비대칭 형상으로 이루어지는 이차 전지.

청구항 11.

제10항에 있어서,

상기 가변부는 사다리꼴과 삼각형 중 어느 하나의 형상으로 이루어지는 이차 전지.

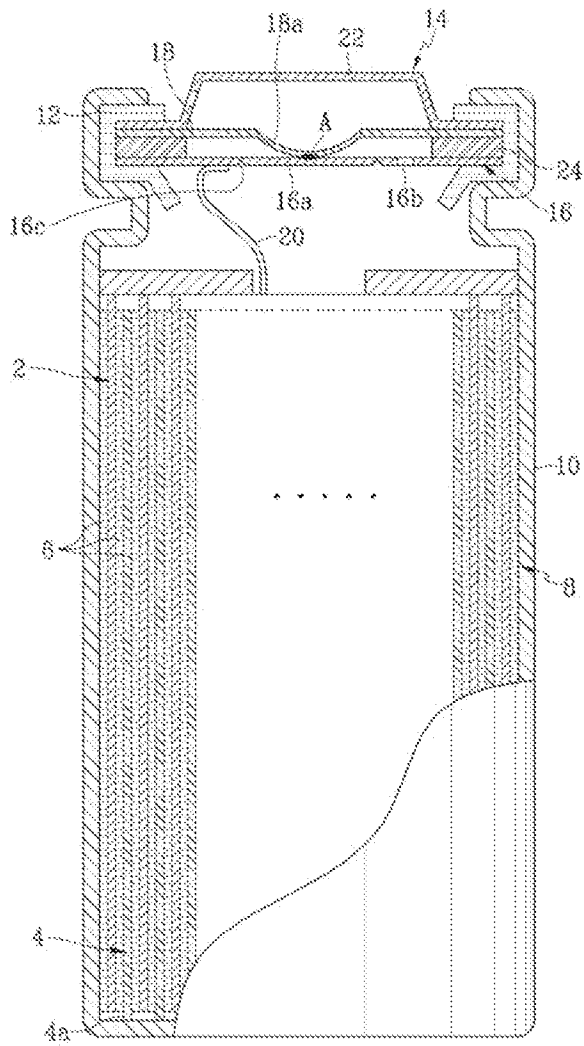
청구항 12.

제7항에 있어서,

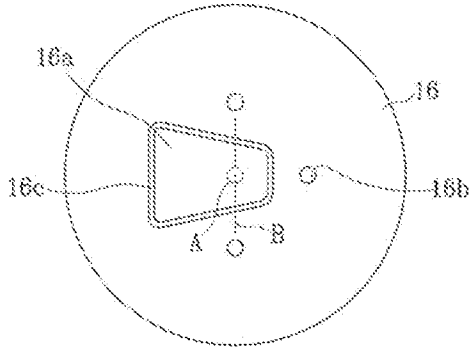
상기 가변부와 램프가 용접으로 고정되는 이차 전지.

도면

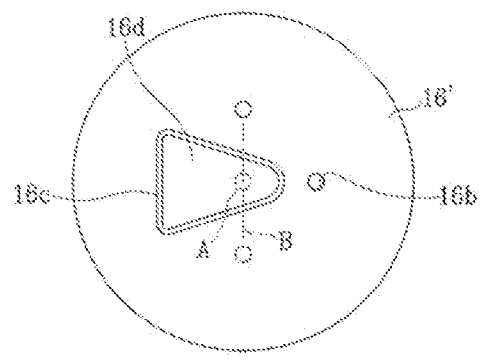
도면1



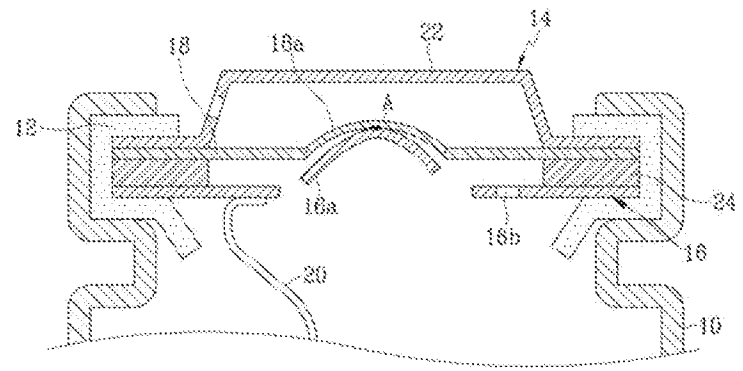
도면2



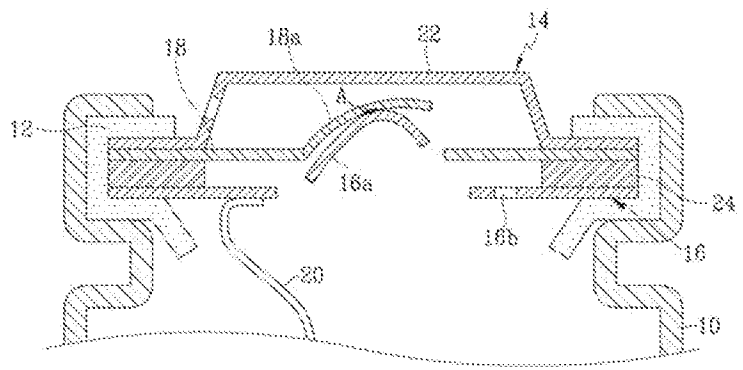
도면3



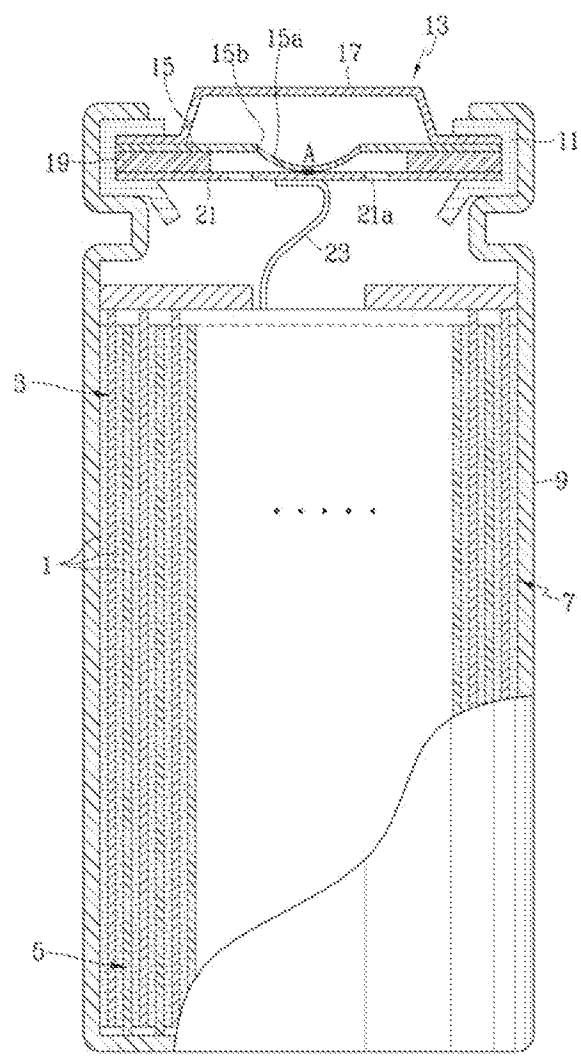
도면4



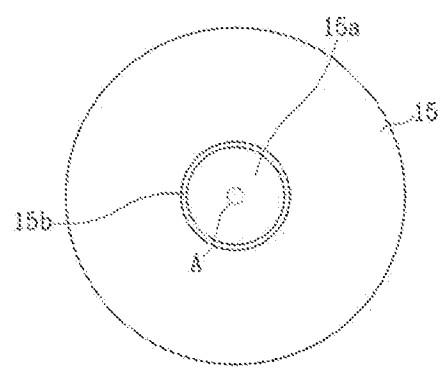
도면5



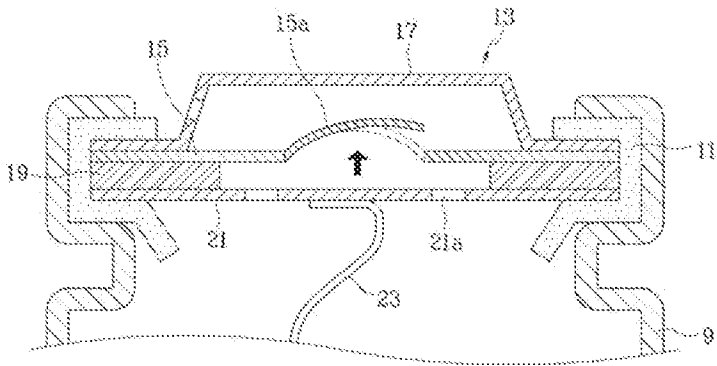
도면6



도면7



도면3





Espacenet

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CYLINDRICAL SECONDARY BATTERY OF IMPROVED SAFETY

Inventor(s): KIM, JI YOUNG, ; KIM, JUNG JIN, ; LEE, KWAN SOO, ; KIM, SUNG JONG, ; KIM, SOO RYOUNG, ; KU, CHA HUN, ; RYU, DUK HYUN, ; JUNG, BYUNG KYU

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Abstract of KR20110017835 (A)

PURPOSE: A cylindrical rechargeable battery is provided to effectively discharge high pressure gas inside a battery by comprising a current interruption member having through-holes with relatively large area and to induce immediate short circuits.

CONSTITUTION: A cylindrical rechargeable battery houses an electrode assembly produced by the winding of an anode/separating film/cathode and an electrolytic solution inside a cylindrical can. The cap assembly mounted on the open upper part of the cylindrical can comprises: a safety vent formed with a predetermined notch so as to be ruptured by high-pressure gas in the battery; a current-blocking member which is coupled by welding to the lower edge of the safety vent and blocks current when the internal pressure of the battery rises; and a gasket for the current-blocking member surrounding the outer circumferential surface of the current-blocking member, and wherein the current-blocking member comprises two or more through holes for discharging gas, and the through holes are formed with a size such that their overall surface area is from 20 to 50% of the overall surface area of the current-blocking member.



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(73) 특허권자

주식회사 엘지화학

서울특별시 영등포구 여의도동 20

(72) 발명자

김지영

서울특별시 광진구 구의3동 현대프라임아파트 1동 808호

김정진

대전광역시 서구 둔산동 아너스빌 901호

(뒷면에 계속)

(74) 대리인

손창규

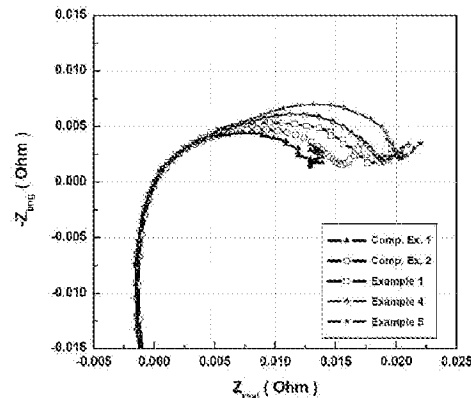
심사관 : 이창희

(54) 안전성이 향상된 원통형 이차전지

(57) 요약

본 발명은 양극/분리막/음극을 권취하여 제조된 전극조립체 및 전해액이 원통형 캔에 내장되어 있는 원통형 전지로서, 상기 원통형 캔의 개방 상단부에 탑재되는 캡 어셈블리에는, 전지의 고압 가스에 의해 파열되도록 소정의 노치가 형성되어 있는 안전벤트; 상기 안전벤트의 하단에 용접에 의해 결합되어 있고, 전지 내압 상승시 전류를 차단하는 전류차단 부재; 및 상기 전류차단 부재의 외주면을 감싸는 전류차단 부재용 가스켓;을 포함하고, 상기 전류차단 부재는 가스의 배출을 위한 둘 또는 그 이상의 관통구들을 포함하고 있으며, 상기 관통구들은 그것의 전체 면적이 전류차단 부재의 전체 면적대비 20 ~ 50%의 크기로 형성되어 있는 원통형 전지를 제공한다.

도면도 - 도18



(72) 발명자

이관수

대전광역시 서구 만년동 상록수아파트 105동 701호

김성종

대전광역시 대덕구 송촌동 선비마을3단지아파트
309동 1402호

김수령

충청북도 청원군 오창읍 양청리 724-14번지 스카이
빌 303호

구자훈

대전광역시 유성구 도룡동 385-11번지 더포엠 1동
803호

류덕현

대전광역시 유성구 도룡동 LG화학사원아파트 8동
510호

정병규

대전광역시 유성구 도룡동 LG화학사원아파트 3동
203호

특허청구의 범위

청구항 1

양극/분리막/음극을 권취하여 제조된 전극조립체('젤리-롤') 및 전해액이 원통형 캔에 내장되어 있는 원통형 전지로서, 상기 원통형 캔의 개방 상단부에 탑재되는 캡 어셈블리에는,

전지의 고압 가스에 의해 파열되도록 소정의 노치가 형성되어 있는 안전벤트;

상기 안전벤트의 하단에 용접에 의해 결합되어 있고, 전지 내압 상승시 전류를 차단하는 전류차단 부재; 및

상기 전류차단 부재의 외주면을 감싸는 전류차단 부재용 가스켓;

을 포함하고,

상기 전류차단 부재는 가스의 배출을 위한 둘 또는 그 이상의 관통구들을 포함하고 있으며, 상기 관통구들은 그것의 전체 면적이 전류차단 부재의 전체 면적대비 20 ~ 50%의 크기로 형성되어 있는 것을 특징으로 하는 원통형 전지.

청구항 2

제 1 항에 있어서, 상기 전류차단 부재는 안전벤트의 만입부의 하단에 용접 부착되어 있는 상향 돌출된 돌출부가 중앙에 형성되어 있으며, 젤리-롤의 양극에 연결되어 있는 양극 리드가 상기 돌출부를 제외한 부위의 하단면을 통해 전기적으로 연결되어 있고, 상기 돌출부를 중심으로 동심원상으로 3 ~ 5개의 관통홀들과 상기 관통홀들을 연결하며 노치가 형성되어 있는 브릿지가 형성되어 있는 구조로 이루어진 것을 특징으로 하는 원통형 전지.

청구항 3

제 1 항에 있어서, 상기 관통구들은 그것의 전체 면적이 전류차단 부재의 전체 면적대비 30 ~ 40%의 크기로 형성되어 있는 것을 특징으로 하는 원통형 전지.

청구항 4

제 1 항에 있어서, 상기 관통구들은 전류차단 부재의 중심을 기준으로 방사상 대칭 위치에서 원호 형상으로 형성되어 있는 것을 특징으로 하는 원통형 전지.

청구항 5

제 2 항에 있어서, 3개의 관통구들이 전류차단 부재의 중심을 기준으로 50 ~ 80도의 각도 범위에 대응하는 길이의 원호 형상을 이루며 동일한 간격으로 배열되어 있는 것을 특징으로 하는 원통형 전지.

청구항 6

제 2 항에 있어서, 상기 관통홀들은 관통구들에 대응하는 위치에서 전류차단 부재의 중심을 기준으로 40 ~ 55도의 각도 범위에 대응하는 길이의 원호 형상을 이루며 동일한 간격으로 배열되어 있는 것을 특징으로 하는 원통형 전지.

청구항 7

제 1 항에 있어서, 상기 안전벤트는 중앙이 하향 만입된 형상으로 되어 있고, 상기 하향 만입된 형상의 만입부를 형성하는 상절곡 부위 및 하절곡 부위에는 각각 제 1 노치 및 제 2 노치가 형성되어 있으며, 상부에 형성되는 제 1 노치는 일측이 개방된 개곡선의 구조로 되어 있는 것을 특징으로 하는 원통형 전지.

청구항 8

제 7 항에 있어서, 상기 제 2 노치는 서로 대향하는 양단이 개방된 개곡선의 구조를 이루어진 것을 특징으로 하는 원통형 전지.

청구항 9

제 7 항에 있어서, 상기 제 1 노치의 크기는 전류차단 부재에 형성된 관통구들의 중심선을 연결한 가상원의 크기에 대응하는 크기를 가지는 것을 특징으로 하는 원통형 전지.

청구항 10

제 7 항에 있어서, 상기 제 1 노치의 개방 부위는 제 1 노치부의 절취시 만입부가 완전히 절취되는 것을 방지할 수 있도록, 안전벤트의 중심에 대해 20 ~ 50도의 각도에 대응하는 원호 길이로 형성되어 있는 것을 특징으로 하는 원통형 전지.

청구항 11

제 8 항에 있어서, 상기 제 2 노치에서 개방된 양단은 제 1 노치의 개방 부위와 일치하지 않는 위치에 형성되어 있는 것을 특징으로 하는 원통형 전지.

청구항 12

제 1 항에 있어서, 상기 캡 어셈블리는 전류차단 부재, 전류차단 부재용 가스켓, 안전벤트, PTC 소자, 및 하나 이상의 가스 배출구가 형성되어 있는 탭 캡의 적층 구조로 이루어져 있고, 상기 적층 구조의 외주면에 가스켓이 추가로 장착되어 있는 것을 특징으로 하는 원통형 전지.

청구항 13

제 1 항에 있어서, 상기 전류차단 부재용 가스켓은 중앙에 개구가 형성되어 있고, 상기 중앙 개구는 전류차단 부재의 상기 관통구와 연통된 형상으로 이루어져 있는 것을 특징으로 하는 원통형 전지.

청구항 14

제 13 항에 있어서, 상기 전류차단 부재의 관통구, 전류차단 부재용 가스켓의 개구, 및 안전벤트의 노치는 일직선 상으로 연통되는 위치로 형성되어 있는 것을 특징으로 하는 원통형 전지.

청구항 15

제 1 항에 있어서, 상기 원통형 캔의 내부에는 안전성을 향상시키기 위한 첨가제('전해액 첨가제')를 포함하는 전해액이 내장되어 있는 것을 특징으로 하는 원통형 전지.

청구항 16

제 15 항에 있어서, 상기 전해액 첨가제는 150℃ 이상에서 온도 상승에 따라 순차적으로 전해액 분해 반응에 의해 가스를 발생시키는 것을 특징으로 하는 원통형 전지.

청구항 17

제 16 항에 있어서, 상기 전해액 첨가제는 150 내지 250℃에서 제 1 차 전해액 분해 반응을 일으키고 250 내지 300℃에서 제 2 차 전해액 분해 반응을 일으키는 것을 특징으로 하는 원통형 전지.

청구항 18

제 15 항에 있어서, 상기 전해액 첨가제는 플루오르 에틸렌 카보네이트(fluoroethylene carbonate)인 것을 특징으로 하는 원통형 전지.

청구항 19

제 15 항에 있어서, 상기 전해액 첨가제는 전해액 중량을 기준으로 11 ~ 40 중량%로 첨가되는 것을 특징으로 하는 원통형 전지.

청구항 20

제 15 항에 있어서, 상기 전해액 첨가제는 전해액 중량을 기준으로 15 ~ 35 중량%로 첨가되는 것을 특징으로 하는 원통형 전지.

청구항 21

전극조립체 및 전해액이 원통형 캔에 내장되어 있는 구조의 전지에서 캔의 개방 상단부에 탑재되는 캡 어셈블리로서,

상기 전지의 전해액에는 150℃ 이상에서 온도 상승에 따라 순차적으로 전해액 분해 반응에 의해 가스를 발생시키는 첨가제가 전해액 중량을 기준으로 11 ~ 40 중량%로 포함되어 있고,

상기 캡 어셈블리에는 가스 배출 및 전극단자의 연결을 위해 중앙에 개구가 천공되어 있고 그것의 주변에 하나 또는 그 이상의 관통구가 천공되어 있는 관상형 절연부재가 젤리-롤의 상단에 탑재되어 있으며, 상기 관통구는 관상형 절연부재의 면적대비 20 ~ 50%의 크기로 형성되어 있는 구조로 이루어진 것을 특징으로 하는 캡 어셈블리.

청구항 22

제 21 항에 있어서, 상기 관상형 절연부재의 관통구는 전류차단 부재용 가스켓의 중앙 개구 및 전류차단 부재의 관통구와 일직선 상으로 연통되는 위치로 형성되어 있는 것을 특징으로 하는 캡 어셈블리.

명세서

기술분야

[0001] 본 발명은 안전성이 향상된 원통형 이차전지에 관한 것으로, 더욱 상세하게는, 양극/분리막/음극을 권취하여 제조된 전극조립체 및 전해액이 원통형 캔에 내장되어 있는 원통형 전지로서, 상기 원통형 캔의 개방 상단부에 탑재되는 캡 어셈블리에는, 전지의 고압 가스에 의해 파괴되도록 소정의 노치가 형성되어 있는 안전벤트, 상기 안전벤트의 하단에 용접에 의해 결합되어 있고, 전지 내압 상승시 전류를 차단하는 전류차단 부재, 및 상기 전류차단 부재의 외주면을 감싸는 전류차단 부재용 가스켓을 포함하고, 상기 전류차단 부재는 가스의 배출을 위한 둘 또는 그 이상의 관통구들을 포함하고 있으며, 상기 관통구들은 그것의 전체 면적이 전류차단 부재의 전체 면적대비 20 ~ 50%의 크기로 형성되어 있는 원통형 전지에 관한 것이다.

배경기술

[0002] 모바일 기기에 대한 기술 개발과 수요가 증가함에 따라 에너지원으로서의 이차전지의 수요가 급격히 증가하고 있고, 그러한 이차전지 중 높은 에너지 밀도와 방전 전압의 리튬 이차전지에 대해 많은 연구가 행해졌고 또한 상용화되어 널리 사용되고 있다.

[0003] 이차전지는 전지케이스의 형상에 따라, 전극조립체가 원통형 또는 각형의 금속 캔에 내장되어 있는 원통형 전지 및 각형 전지와, 전극조립체가 알루미늄 라미네이트 시트의 파우치형 케이스에 내장되어 있는 파우치형 전지로 분류된다. 그 중 원통형 전지는 상대적으로 용량이 크고 구조적으로 안정하다는 장점을 가진다.

[0004] 전지케이스에 내장되는 상기 전극조립체는 양극/분리막/음극의 적층 구조로 이루어진 총방전이 가능한 발전소자로서, 활물질이 도포된 긴 시트형의 양극과 음극 사이의 분리막을 개재하여 권취한 젤리-롤형과, 소정 크기의 다수의 양극과 음극을 분리막이 개재된 상태에서 순차적으로 적층한 스택형으로 분류된다. 그 중 젤리-롤형 전극조립체는 제조가 용이하고 중량당 에너지 밀도가 높은 장점을 가지고 있다.

[0005] 이와 관련하여, 도 1에는 일반적인 원통형 전지의 수직 단면 사시도가 모식적으로 도시되어 있다.

[0006] 도 1을 참조하면, 원통형 이차전지(100)는 젤리-롤형(권취형) 전극조립체(120)를 원통형 케이스(130)에 수납하고, 원통형 케이스(130) 내에 전해액을 주입한 후에, 케이스(130)의 개방 상단에 전극 단자(예를 들어, 양극 단자; 도시하지 않음)가 형성되어 있는 탑 캡(140)을 결합하여 제작한다.

[0007] 전극조립체(120)는 양극(121)과 음극(122), 및 이들 사이에 분리막(123)을 개제한 후 등근 형태로 감은 구조로서, 그것의 권심(젤리-롤의 중심부)에는 원통형의 센터 핀(150)이 삽입되어 있다. 센터 핀(150)은 일반적으로 소정의 강도를 부여하기 위해 금속 소재로 이루어져 있으며, 판재를 등글게 절곡한 중공형의 원통형 구조로 이루어져 있다. 이러한 센터 핀(150)은 전극조립체를 고정 및 지지하는 작용과 총방전 및 작동시 내부 반응에 의해 발생하는 가스를 방출하는 통로로서 작용한다.

[0008] 한편, 리튬 이차전지는 안전성이 낮다는 단점을 가지고 있다. 예를 들어, 전지가 대략 4.5V 이상으로 과충전되

는 경우에는 양극 활물질의 분해반응이 일어나고, 음극에서 리튬 금속의 수지상(dendrite) 성장과, 전해액의 분해반응 등이 일어난다. 이러한 과정에서 열이 수반되어 상기와 같은 분해반응과 다수의 부반응들이 급속히 진행되며, 급기야는 전지의 발화 및 폭발이 유발되기도 한다.

[0009] 따라서, 이러한 문제점을 해소하기 위하여, 일반적인 원통형 이차전지에는 전지의 비정상적인 작동시 전류를 차단하고 내압을 해소하기 위한 전류차단 부재(Current Interruptive Device; CID)와 안전벤트(vent)가 전극조립체와 상단 캡 사이의 공간에 장착되어 있다.

[0010] 이를 도 2 내지 4를 참조하여 이하에서 설명한다.

[0011] 이들 도면을 참조하면, 탑 캡(10)은 돌출된 형태로 양극 단자를 형성하고 배기구가 천공되어 있으며, 그것의 하부에 전지 내부의 온도 상승시 전지 저항이 크게 증가하여 전류를 차단하는 PTC 소자(positive temperature coefficient element: 20), 정상적인 상태에서는 하향 돌출된 형상으로 되어 있고 전지 내부의 압력 상승시 돌출되면서 과열되어 가스를 배기하는 안전벤트(30), 및 상단 일측 부위가 안전벤트(30)에 결합되어 있고 하단 일측이 전극조립체(40)의 양극에 연결되어 있는 접속 플레이트(50)가 순차적으로 위치되어 있다.

[0012] 따라서, 정상적인 작동조건에서 전극조립체(40)의 양극은 리드(42), 접속 플레이트(50), 안전벤트(30) 및 PTC 소자(20)를 경유하여 탑 캡(10)에 연결되어 통전을 이룬다.

[0013] 그러나, 과충전 등과 같은 원인에 의해 전극조립체(40) 쪽으로부터 가스가 발생하여 내압이 증가하면, 도 3에서와 같이, 안전벤트(30)는 그것의 형상이 역전되면서 상향 돌출되게 되고, 이때, 안전벤트(30)가 접속 플레이트(50)로부터 분리되어 전류가 차단되게 된다. 따라서, 과충전이 더 이상 진행되지 않도록 하여 안전성을 확보한다. 그럼에도 불구하고, 계속적으로 내압이 증가하면, 도 4에서와 같이, 안전벤트(30)가 과열되고 가압 가스는 그러한 과열 부위를 경유하여 탑 캡(10)의 배기구를 통해 배기됨으로써, 전지의 폭발을 방지하게 된다.

[0014] 이러한 작동 과정은 전극조립체 부위에서 발생하는 가스의 양, 그러한 가스가 얼마나 효과적으로 안전벤트에 전달되느냐 등에 따라 좌우된다. 예를 들어, 많은 양의 가스가 발생하더라도, 그것이 안전벤트에 효과적으로 전달되지 못하면, 소망하는 안전화 작동 과정이 진행되지 못하게 된다. 더욱이, 많은 양의 가스가 단시간 내에 발생하고, 그러한 가스가 안전벤트에 도달하여 소정의 작동 과정을 진행시키지 못하면, 전지의 내압이 순간적으로 증가하여 폭발을 유발할 수 있다.

[0015] 또한, 가스의 발생은 고온 조건에서 열에 의한 전해액 분해 반응으로 유발되는데, 단시간 내에 전지에서의 발열량이 급격히 증가하면, 열폭주 현상이 일어날 수 있다. 열폭주 현상은 전지가 계속적인 통전 상태에 놓여 있을 때 발생하거나 또는 더욱 가속화됨으로써, 전지가 발화 또는 폭발할 위험이 매우 커지므로 안전성에 심각한 문제가 있다.

[0016] 따라서, 전지의 비정상적인 작동 상태에서 발생하는 전지 내부의 가스를 보다 안정적으로 유발시키면서, 내압이 증가하는 경우, 전지 외부로 신속하게 가스를 배출할 수 있는 원통형 전지에 대한 개발의 필요성이 높은 실정이다.

발명의 내용

해결하려는 과제

[0017] 본 발명은 상기와 같은 종래기술의 문제점과 과거로부터 요청되어온 기술적 과제를 해결하는 것을 목적으로 한다.

[0018] 본 발명의 첫 번째 목적은, 발생된 가스가 특정한 구조의 캡 어셈블리 구조에 의해 안전부재들로 효과적으로 전달되어 소정의 안전화 과정이 진행될 수 있도록 하여, 궁극적으로 전지의 안전성을 향상시킬 수 있는 이차전지를 제공하는 것이다.

[0019] 본 발명의 두 번째 목적은, 온도 상승에 따라 순차적으로 전해액 분해 반응을 유발시켜, 전지 내부의 급격한 압력 상승을 억제하면서, 안전부재들이 효과적으로 작동할 수 있는 이차전지를 제공하는 것이다.

[0020] 본 발명의 세 번째 목적은, 전해액 분해 반응을 유발하되, 발열량을 최소화하여, 열폭주 현상이 초래되는 것을 방지할 수 있는 이차전지를 제공하는 것이다.

과제의 해결 수단

- [0021] 이러한 목적을 달성하기 위한 본 발명에 따른 원통형 전지는, 양극/분리막/음극을 권취하여 제조된 전극조립체('셀리-롤') 및 전해액이 원통형 캔에 내장되어 있는 원통형 전지로서,
- [0022] 상기 원통형 캔의 개방 상단부에 탑재되는 캡 어셈블리에는,
- [0023] 전지의 고압 가스에 의해 파열되도록 소정의 노치가 형성되어 있는 안전벤트;
- [0024] 상기 안전벤트의 하단에 용접에 의해 결합되어 있고, 전지 내압 상승시 전류를 차단하는 전류차단 부재; 및
- [0025] 상기 전류차단 부재의 외주면을 감싸는 전류차단 부재용 가스켓;
- [0026] 을 포함하고, 상기 전류차단 부재는 가스의 배출을 위한 둘 또는 그 이상의 관통구들을 포함하고 있으며, 상기 관통구들은 그것의 전체 면적이 전류차단 부재의 전체 면적대비 20 ~ 50%의 크기로 형성되어 있는 것으로 이루어져 있다.
- [0027] 따라서, 본 발명에 따른 원통형 전지는, 특정한 구조의 캡 어셈블리에 의해, 발생 가스가 안전부재들에 직접적으로 작용하도록 함으로써, 소망하는 단전을 유도하고 안전벤트의 파열에 의해 전지 외부로 신속하게 가스를 배출되어 전지의 안전성을 담보할 수 있다.
- [0028] 상기 전류차단 부재는 전지 내부의 압력 상승시 파열되어 전류를 차단하는 부재로서, 안전벤트의 만입부 하단에 용접 부착되어 있는 상향 돌출된 돌출부가 중앙에 형성되어 있고, 셀리-롤의 양극에 연결되어 있는 양극 리드가 상기 돌출부를 제외한 부위의 하단면을 통해 전기적으로 연결되어 있는 구조일 수 있다.
- [0029] 이와 같이, 전류차단 부재는 안전벤트에 결합되어 있어서 안전벤트의 형상 역전에 따라 전류차단 부재와 양극 사이의 연결 부위가 파열되면서 전류를 차단하게 된다. 구체적인 예에서, 상기 돌출부에는 외주면에 노치가 형성되어 있을 수 있고, 안전벤트의 형상 역전에 따라 노치 부위가 파열되면서 돌출부는 안전벤트와 결합된 상태로 전류차단 부재와 양극의 연결부위와 용이하게 분리될 수 있다.
- [0030] 또한, 상기 전류차단 부재에는 가스의 상향 이동을 위한 둘 또는 그 이상의 관통구가 형성되어 있고, 이를 통해 전지의 내부가스가 상승하여 안전벤트의 형상 역전을 유도한다.
- [0031] 그러나, 본 발명자들이 실험적으로 확인한 바에 따르면, 상기 전류차단 부재에 형성된 관통구의 면적이 20% 미만인 경우에는, 전지 내부의 고압 발생시 전지가 발화 내지 폭발에 이르는 비율이 높았다. 이는, 관통구를 통한 가스의 통과량이 적어 효과적인 단전이 이루어지지 못하거나, 또는 가스의 상향 이동 경로가 차단되거나 변경되어 전류차단에 걸리는 시간이 길어지기 때문인 것으로 추측된다. 한편, 상기 관통구의 크기가 지나치게 크면 소정의 강도를 확보하기 어렵고 가공시 부품의 뒤틀림 현상이 발생하여 바람직하지 않다.
- [0032] 이에, 본 발명에 따른 원통형 전지에서, 상기 전류차단 부재는 가스의 배출을 위한 둘 이상의 관통구를 포함하고 있고 그것의 전체 면적이 전류차단 부재의 면적대비 20 ~ 50%의 크기, 더욱 바람직하게는 30 ~ 40%의 크기로 형성되어 있어서, 가스의 상향 이동량(배출량)을 높이고 가스의 신속한 상향 이동(배출)을 유도할 수 있다.
- [0033] 상기 관통구들은 전류차단 부재의 중심을 기준으로 방사상 대칭 위치에서 원호 형상으로 형성되는 것이 바람직하다. 이와 같이 방사상 대칭 위치로 형성되는 경우에는 가스 배출이 일 방향으로 지나치게 편중되는 것을 방지할 수 있고 전류차단 부재 자체의 강도를 확보할 수 있다는 장점이 있다.
- [0034] 이 때, 상기 관통구들의 개수는 특별히 제한되는 것은 아니며 바람직하게는 2 ~ 4개일 수 있다.
- [0035] 하나의 바람직한 예에서, 3개의 관통구들이 전류차단 부재의 중심을 기준으로 50 ~ 80도의 각도 범위에 대응하는 길이의 원호 형상을 이루며 동일한 간격으로 배열되어 있는 구조일 수 있다.
- [0036] 한편, 상기 돌출부는 이를 중심으로 동심원상으로 3 ~ 5개의 관통홀들과 상기 관통홀들을 연결하며 노치가 형성되어 있는 브릿지가 형성되어 있는 구조일 수 있다. 이 때, 관통홀들은 상기 관통구들에 대응하는 위치에서, 바람직하게는 전류차단 부재의 중심을 기준으로 40 ~ 55도의 각도 범위에 대응하는 길이의 원호 형상을 이루며 동일한 간격으로 배열되어 있는 구조일 수 있다.
- [0037] 상기 안전벤트는 전지 내부 압력의 상승시 가스를 외부로 배출시켜 전지의 안전성을 담보하는 소자이다. 예를 들어, 전지의 내부에서 가스가 발생하여 임계치 이상으로 내압이 증가하였을 때, 안전벤트가 파열되고, 그러한 파열 부위로 배출되는 가스는 상단 캡에 형성되어 있는 하나 또는 둘 이상의 가스 배출구를 통해 외부로 배출될 수 있다.

- [0038] 상기 안전벤트는 바람직하게는 중앙이 하향 만입된 형상의 구조일 수 있다. 이러한 구조에서, 전지의 내압 상승시 안전벤트의 중앙에서 하향 만입된 형상이 상향 돌출된 형상으로 역전되면서, 전류차단 부재로부터 분리되어 전기적 연결을 차단하게 된다.
- [0039] 이러한 구조의 안전벤트의 작동 과정을 살펴보면, 전지 내부에서 발생한 가스는 전류차단 부재의 관통구들을 통과하여 안전벤트에 상향 압력을 가하게 되고, 일정한 압력(이하, '제 1 임계 압력'으로 약칭함) 이상에서 안전벤트의 하향 만입부의 형상이 역전되면서 상기 만입부에 부착되어 있는 돌출부가 전류차단 부재로부터 분리되어 전류차단 부재로부터 안전벤트로의 통전을 차단하게 된다.
- [0040] 상기 만입부는, 예를 들어, 상절곡 부위 및 하절곡 부위로 이루어지고, 상기 상절곡 부위에는 바람직하게는 제 1 노치가 형성되어 있을 수 있다. 종래기술 중에는 하절곡 부위만이 절취되는 구조가 제안되기도 하였으나, 하절곡 부위는 좁은 면적을 가지므로 효과적인 가스 방출에 적합하지 않았다. 그러나, 본 발명에서는 상절곡 부위에 파열을 위한 노치가 형성되어 있어서, 절취되는 면적이 상대적으로 크므로, 다량의 가스를 신속하게 배출할 수 있다.
- [0041] 상기 제 1 노치는, 안전벤트의 전류 차단에도 불구하고 계속적으로 압력이 상승하여 일정한 수준(이하, '제 2 임계 압력'으로 약칭함) 이상이 되면, 절취될 수 있으므로, 내부의 가압 가스가 탭 캡에 형성된 가스 배출구를 통해 원활하게 외부로 배출된다.
- [0042] 한편, 상기 상절곡 부위 전체가 안전벤트로부터 분리되는 것을 방지하기 위해 상기 제 1 노치는 바람직하게는 일측이 개방된 개곡선의 구조로 되어 있을 수 있다. 즉, 상기 일측의 개방 부위는 안전벤트에서 만입부와 이를 제외한 나머지 부위의 브릿지 역할을 한다.
- [0043] 상기 제 1 노치의 크기는 특별히 제한되지 않으나, 전류차단 부재로부터 상향 이동한 가스가 제 1 노치에 직접 압력을 가할 수 있도록, 바람직하게는 전류차단 부재에 형성된 관통구들의 중심선을 연결한 가상원의 크기에 대응하는 크기를 가질 수 있다.
- [0044] 또한, 상기 제 1 노치의 개방 부위는 앞서 설명한 바와 같이 제 1 노치부의 절취시 만입부가 완전히 절취되는 것을 방지할 수 있는 크기로서, 바람직하게는 안전벤트의 중심에 대해 20 ~ 50도의 각도, 더욱 바람직하게는 25 ~ 40도의 각도에 대응하는 원호 길이로 형성할 수 있다.
- [0045] 하나의 바람직한 예에서, 상기 안전벤트의 하절곡 부위에는 제 2 노치가 형성되어 있지만, 내압 상승시 상절곡 부위에서만 절취 현상이 발생할 수 있도록, 바람직하게는 서로 대향하는 양단이 개방된 개곡선의 구조로 이루어질 수 있다. 이와 같이 양단이 개방되어 있는 구조는 전지 내압이 상승하여도 하절곡 부위의 절취를 억제하고 상절곡 부위의 절취에 의한 개방을 더욱 효과적으로 유도한다.
- [0046] 상기 제 2 노치에서 개방된 양단은 상기 제 1 노치의 개방 부위와 일치하지 않는 위치에 형성되는 것이 바람직하고, 더욱 바람직한 예에서, 약 $\pm 90^\circ$ 의 위치에 각각 형성되는 것이 바람직하다.
- [0047] 바람직한 예에서, 상기 제 1 노치는 제 2 노치에 비해 우선적으로 파열(절취)될 수 있도록, 제 2 노치 보다 깊은 깊이로 형성될 수 있다.
- [0048] 상기 안전벤트는 특별히 제한되지 않으나, 임계치 이상의 내압 증가에 의해 파열(절취)될 수 있을 정도의 강성을 가지기 위하여, 바람직하게는 0.15 내지 0.4 mm의 두께를 가진 알루미늄 판재로 이루어진 것일 수 있다.
- [0049] 본 발명에 따른 원통형 전지의 캡 어셈블리는 경우에 따라, 전지 내부의 온도 상승시 전지저항이 크게 증가하여 전류를 차단하는 PTC 소자(positive temperature coefficient element) 등의 부재를 더 포함할 수 있다.
- [0050] 상기 캡 어셈블리는 전류차단 부재, 전류차단 부재용 가스켓, 안전벤트, PTC 소자, 및 하나 이상의 가스 배출구가 형성되어 있는 탭 캡의 적층 구조로 이루어져 있고, 상기 적층 구조의 외주면에 가스켓이 추가로 장착된 구조일 수 있다. 상기 PTC 소자는 상단 캡과 안전벤트 사이에 밀착된 상태로 개재될 수 있다.
- [0051] 또한, 상기 전류차단 부재용 가스켓은 전류차단 부재의 외주면을 감싸는 구조로서, 바람직한 예에서, 중앙에 개구가 형성되어 있는 구조일 수 있다. 이 때, 내부 가스의 상승 이동이 효율적으로 이루어지기 위해서는, 기압 가스의 이동 경로가 일직선이 되는 것이 바람직하므로, 상기 중앙 개구는 전류차단 부재의 상기 관통구를 가리지 않는 형상으로 이루어질 수 있다.
- [0052] 종래에는 가스켓이 전류차단 부재의 관통구를 일부 가리는 형상으로 되어 있어서, 관통구를 통과한 고압 가스가 가스켓에 의해 방해를 받아 배출량이 줄어들고 경로가 변경됨으로써 안전벤트의 즉각적인 작동을 유도하기 어려

있다. 그러나, 상기와 같이, 가스켓이 관통구를 가리지 않는 구조에서는 관통구를 통과한 고압 가스 전부가 직접적으로 안전벤트에 도달할 수 있어서 전지의 안전성을 더욱 담보할 수 있다.

[0053] 하나의 바람직한 예에서, 상기 전류차단 부재의 관통구, 전류차단 부재용 가스켓의 개구, 및 안전벤트의 노치는 일직선 상으로 연통되는 위치로 형성될 수 있다.

[0054] 그에 따라, 전지 내부로부터 발생한 고압 가스가 전류차단 부재의 관통구를 통해 가스켓의 중앙 개구를 거쳐 안전벤트의 노치로 이동하는 과정에서 가스의 경로가 일직선상으로 유지될 수 있으므로, 고압 가스가 집중적이고 신속하게 상향 이동될 수 있다. 따라서, 안전벤트의 즉각적인 작동을 유도할 수 있어서 전지의 안전성을 담보할 수 있다.

[0055] 하나의 바람직한 예에서, 상기 원통형 캔의 내부에는 열적 안전성을 향상시키기 위한 첨가제('전해액 첨가제')를 포함하는 전해액이 내장되어 있을 수 있다.

[0056] 상기 전해액 첨가제는 바람직하게는 온도 상승에 따라 순차적으로 전해액 분해 반응에 의해 가스를 발생시킬 수 있으며, 예를 들어, 150℃ 이상의 고온에서 반응하는 물질일 수 있다.

[0057] 따라서, 본 발명에 따른 원통형 전지는, 이후 실험예에서도 입증되는 바와 같이, 상기 전해액 첨가제에 의해, 발열량을 최소화하면서 온도 상승에 따른 순차적으로 전해액 분해 반응을 유도하여, 열폭주 현상과 전지 내압의 급격한 상승을 억제할 수 있다. 이러한 순차적인 전해액 분해 반응의 유도는 많은 양의 발생 가스가 안전부재들에 전달되는 과정에서 일종의 병목 현상이 초래되는 것을 방지하여 준다. 따라서, 상기 전해액 첨가제는 앞서 설명한 특징적인 캡 어셈블리 구조와의 상호 보완에 의해 상승적인 효과를 발휘할 수 있다.

[0058] 이를 위한 바람직한 예로서, 상기 전해액 첨가제는 150 내지 250℃에서 제 1 차 전해액 분해 반응을 일으키고 250 내지 300℃에서 제 2 차 전해액 분해 반응을 일으키는 물질일 수 있다. 예를 들어, 전해액 첨가제는 제 1 차 전해액 분해 반응에서 전체 분해 반응 대비 약 2/3의 분해 반응을 일으키고, 제 1 차 분해 반응보다 고온에서 전체 분해 반응 대비 약 1/3인 제 2 차 전해액 분해 반응을 일으킨다.

[0059] 따라서, 온도의 상승에 따라 단계적으로 가스를 발생시켜 전류차단 부재와 안전벤트가 작동되도록 함으로써, 단기간 내에 다량의 가스가 발생할 때 유발되는 문제점을 해결할 수 있고, 이러한 단계적인 가스 발생에도 불구하고, 앞서 설명한 특징적인 캡 어셈블리 구조에 의해, 발생 가스가 안전부재들에 직접적으로 작용하도록 함으로써, 소망하는 단전을 유도하고 안전벤트의 파열에 의해 전지 외부로 신속하게 가스를 배출할 수 있으므로, 궁극적으로 전지의 발화 및 폭발이 초래되는 것을 효과적으로 방지할 수 있다.

[0060] 상기 전해액 첨가제의 바람직한 예로는 플루오르 에틸렌 카보네이트(fluoroethylene carbonate; FEC)를 들 수 있지만, 그것만으로 한정되는 것은 아니다.

[0061] 본 출원의 발명자들이 행한 실험에 따르면, 전해액 첨가제는, 앞서 설명한 바와 같이 온도 상승에 따라 순차적으로 전해액 분해 반응을 유도할 뿐만 아니라, 양극의 표면에 보호막을 형성하여 양극 계면의 저항을 증가시키며, 이러한 계면 저항의 증가는 단락시 통전 전류량을 감소시켜 결과적으로 전지의 안정성을 향상시킬 수 있음을 확인하였다.

[0062] 이러한 전해액 첨가제는 전해액 전체 중량을 기준으로 바람직하게는 11 ~ 40 중량%를 포함시킬 수 있고, 더욱 바람직하게는 15 ~ 35 중량%로 포함시킬 수 있다.

[0063] 상기 함량이 너무 적은 경우에는 고온에서 순차적인 가스 발생 및 단락시 전류량 감소 효과를 기대하기 어렵고, 반대로 너무 많은 경우에는 전지용량의 감소 등을 초래하므로 바람직하지 않다.

[0064] 이러한 전해액 첨가제의 사용은 특히 고전압 셀, 예를 들어, 4.3V ~ 4.35V의 충방전 전위에서 작용하는 셀에 바람직하며, 이 경우, 전해액 첨가제의 바람직한 함량은 20 ~ 35 중량%일 수 있다.

[0065] 일반적으로 전지에 사용되는 전해액은 비수계 유기용매에 리튬염 등이 포함되어 있는 구성으로 이루어져 있으며, 이는 당업계에 공지되어 있다.

[0066] 상기 비수계 유기용매로는, 예를 들어, N-메틸-2-피롤리디논, 프로필렌 카보네이트, 에틸렌 카보네이트, 부틸렌 카보네이트, 디메틸 카보네이트, 디에틸 카보네이트, 감마-부티로 락톤, 1,2-디메톡시 에탄, 테트라하이드록시 프랑(franc), 2-메틸 테트라하이드로푸란, 디메틸술폰, 1,3-디옥소란, 포름아미드, 디메틸포름아미드, 디옥소란, 아세토니트릴, 니트로메탄, 포름산 메틸, 초산메틸, 인산 트리에스테르, 트리메톡시 메탄, 디옥소란 유도체, 설포란, 메틸 설포란, 1,3-디메틸-2-이미다졸리디논, 프로필렌 카보네이트 유도체, 테트라하이드로푸란

유도체, 에테르, 프로피온산 메틸, 프로피온산 에틸, 프로피온산 프로필 등의 비양자성 유기용매가 사용될 수 있다. 그 중에서도, 선형 카보네이트 화합물, 환형 카보네이트 화합물, 또는 이들의 혼합물로 이루어진 카보네이트계 용매가 특히 바람직하게 사용될 수 있다.

[0067] 상기 리튬염은 상기 비수계 전해질에 용해되기 좋은 물질로서, 예를 들어, LiCl, LiBr, LiI, LiClO₄, LiBF₄, LiB₁₀Cl₁₀, LiPF₆, LiCF₃SO₃, LiCF₃CO₂, LiAsF₆, LiSbF₆, LiAlCl₄, CH₃SO₃Li, CF₃SO₃Li, (CF₃SO₂)₂NLi, 클로로 보란 리튬, 저급 지방족 카르보산 리튬, 4 페닐 붕산 리튬, 이머드 등이 사용될 수 있다.

[0068] 경우에 따라서는, 상기 전해액에, 충방전 특성, 난연성 등의 개선을 목적에서, 예를 들어, 피리딘, 트리에틸포스파이트, 트리에탄올아민, 환상 에테르, 에틸렌 디아민, n-글라이머(glyme), 헥사 인산 트리 아마이드, 니트로벤젠 유도체, 유헤, 퀴논 이민 염료, N-치환 옥사졸리딘, N,N-치환 이미다졸리딘, 에틸렌 글리콜 디알킬 에테르, 암모늄염, 피롤, 2-메톡시 에탄올, 삼염화 알루미늄 등; 불연성을 부여하기 목적에서, 예를 들어, 사염화탄소, 삼불화에틸렌 등의 할로겐 함유 용매; 고온 보존 특성을 향상시키기 위한 목적에서, 예를 들어, 이산화탄소 가스, PRS(Propene sultone) 등이 더 포함될 수 있다.

[0069] 본 발명은 또한, 전극조립체 및 전해액이 원통형 캔에 내장되어 있는 구조의 전지에서 캔의 개방 상단부에 탑재되는 캡 어셈블리를 제공한다. 구체적으로, 상기 전지의 전해액에는 150℃ 이상에서 온도 상승에 따라 순차적으로 전해액 분해 반응에 의해 가스를 발생시키는 첨가제가 전해액 중량을 기준으로 11 ~ 40 중량%로 포함되어 있고, 상기 캡 어셈블리에는 가스 배출 및 전극단자의 연결을 위해 중앙에 개구가 천공되어 있고 그것의 주변에 하나 또는 그 이상의 관통구가 천공되어 있는 판상형 절연부재가 젤리-몰의 상단에 탑재될 수 있다.

[0070] 하나의 바람직한 예에서, 상기 판상형 절연부재의 관통구는 전류차단 부재용 가스켓의 중앙 개구 및 전류차단 부재의 관통구와 일직선 상으로 연통되는 크기와 위치로 형성될 수 있다. 따라서, 상기 판상형 절연부재의 관통구는 판상형 절연부재의 면적대비 20 ~ 50%의 크기로 형성될 수 있다.

[0071] 상기 절연부재는 절연성 소재라면 특별히 제한되지 않으며, 예를 들어, 폴리프로필렌 수지일 수 있다. 그 중에서도 고강도 폴리프로필렌 수지는 동일한 두께 대비로 우수한 기계적 강성을 발휘함으로써, 강한 외부 충격의 인가시에 젤리-몰의 이동과 단락을 방지할 수 있으므로, 특히 바람직하다.

발명의 효과

[0072] 본 발명에 따른 원통형 전지는, 상대적으로 큰 면적을 갖는 관통구가 형성된 전류차단 부재를 포함함으로써 전지 내부의 고압 가스를 효과적으로 배출할 수 있으며, 나아가 안전벤트에서 파열이 이루어지는 노치부를 관통구에 대응되도록 형성하는 경우 고압 가스의 배출이 일직선상으로 이루어질 수 있는 바, 고압 가스 발생시 안전벤트의 즉각적인 단전을 유도할 수 있고 가스 배출이 신속하게 이루어질 수 있어서 전지의 안전성을 크게 향상시킬 수 있다.

[0073] 더욱이, 특정한 전해액 첨가제를 전해액에 첨가하는 경우, 온도 상승에 따라 순차적으로 전해액 분해 반응을 일으켜 안정적으로 가스를 발생시킬 수 있으며, 단락 전류량을 감소시켜 고열 발생에 따른 발화 내지 폭발 가능성을 크게 낮출 수 있다.

도면의 간단한 설명

[0074] 도 1은 일반적인 원통형 이차전지의 수직 단면 사시도이다;

도 2 내지 4는 종래의 원통형 이차전지에서 안전소자의 작동에 의해 전류가 차단되고 고압 가스가 배출되는 일련의 과정에 대한 수직 단면도들이다;

도 5는 본 발명의 하나의 실시예에 따른 원통형 전지의 단면 모식도이다;

도 6 및 도 7은 도 5의 이차전지에 사용된 안전벤트의 사시도 및 평면도이다;

도 8 및 도 9는 도 5의 이차전지에 사용된 전류차단 부재의 사시도 및 평면도이다;

도 10 및 도 11은 도 5의 이차전지에 사용된 전류차단 부재용 가스켓의 사시도 및 평면도이다;

도 12 및 도 13은 도 5의 이차전지에 사용된 판상형 절연부재의 사시도 및 평면도이다;

도 14는 본 발명의 하나의 실시예에 따른 판상형 절연부재, 전류차단 부재 가스켓 및 전류차단 부재의 적층과정

의 모식도이다;

도 15 내지 17은 도 5의 이차전지에서 안전소자의 작동에 의해 전류가 차단되고 고압 가스가 배출되는 일련의 과정에 대한 수직 단면도들이다;

도 18은 실시예 1, 4 및 5와 비교예 1 및 2의 전지들에 대한 실험예 1의 계면저항 실험 결과를 보여주는 그래프이다.

발명을 실시하기 위한 구체적인 내용

- [0075] 이하 도면을 참조하여 본 발명을 더욱 상세하지만, 본 발명의 범주가 그것에 의해 한정되는 것은 아니다.
- [0076] 도 5에는 본 발명의 하나의 실시예에 따른 원통형 전지에서 캡 어셈블리 구조가 도시되어 있고, 도 6 및 도 7에는 도 5의 원통형 전지에 사용된 안전벤트의 사시도 및 평면도가 각각 도시되어 있다.
- [0077] 우선 도 5를 참조하면, 원통형 전지(100)는 원통형 캔(200)의 내부에 젤리-롤(110)을 삽입하고, 여기에 전해액을 주입한 뒤, 젤리-롤(110)의 상단에 판상형 절연부재(800)를 탑재하고, 원통형 캔(200)의 개방 상단에 캡 어셈블리(300)를 장착함으로써 제조된다.
- [0078] 상기 전해액에는 발열량을 최소화하면서 온도 상승에 따라 순차적으로 전해액 분해 반응을 유도하는 특성을 가진 첨가제가 포함되어 있다.
- [0079] 캡 어셈블리(300)는 원통형 캔(200)의 상부 비딩부(210)에 장착되는 기밀유지용 가스켓(400) 내부에 상단 캡(310), PTC 소자(350) 및 내부 압력 강하용 안전벤트(320)가 밀착되어 있는 구조로 이루어져 있다.
- [0080] 안전벤트(320)의 하단에는 전지 내부의 압력 상승시 파열되어 전류를 차단하는 전류차단 부재(600)가 용접에 의해 결합되어 있으며, 전류차단 부재(600)는 전류차단 부재용 가스켓(700)에 의하여 감싸여 있다.
- [0081] PTC 소자(350)는 상단 캡(310)과 안전벤트(320) 사이의 전류회로 상에 밀착된 상태로 개재되며, 전지 내부의 온도 상승시 전지저항이 크게 증가하여 전류를 차단한다. 상단 캡(310)은 중앙이 상향 돌출되어 있어서 외부 회로와의 접속에 의한 양극 단자로서의 역할을 수행하고, 돌출부 주변을 따라 원통형 캔(200) 내부의 압축 가스가 배출될 수 있는 가스 배출구(312)가 다수 개 형성되어 있다.
- [0082] 이러한 구조의 캡 어셈블리(300)에서, 안전벤트(320)의 구조를 도 6 및 7를 참조하여 더욱 구체적으로 살펴본다.
- [0083] 안전벤트(320)는 전류가 통하는 박막 구조물로서, 그것의 중앙에 하향 만입된 형상인 만입부(322)가 형성되어 있고, 상절곡 부위에는 제 1 노치(324)가 형성되어 있으며, 하절곡 부위에는 제 2 노치(326)가 형성되어 있다.
- [0084] 안전벤트(320)의 평면도를 도시한 도 7에 도시된 바와 같이, 상절곡 부위에 형성된 제 1 노치(324)는 일측이 개방된 개곡선의 구조로 되어 있다. 이러한 개방 부위는 미노치부(330)로서, 고압 가스에 의해 노치가 형성된 부위가 절취되어도 미노치부(330)에 의해 제 1 노치부 전체가 분리되는 것이 방지될 수 있다.
- [0085] 또한, 하절곡 부위에는 제 2 노치(326)가 형성되어 있으며, 서로 대향하는 양단이 개방된 개곡선의 구조로 이루어져 있다. 제 2 노치(326)의 개방된 양단 부위에는 2개의 미노치부들(340)이 위치한다. 이러한 미노치부들(340)에 의해 제 2 노치(326)가 절취에 의해 분리되는 것이 방지될 수 있다. 제 2 노치(326)의 분리를 보다 확실하게 방지하기 위해, 미노치부들(340)은 제 1 노치(324)에 형성된 미노치부(330)와 일치하지 않는 위치에 형성되는 것이 바람직하고, 더욱 바람직하게는 본 도면에서와 같이, 제 1 노치부(324)의 미노치부(330)의 중심선을 기준으로 약 $\pm 90^\circ$ 위치에 형성될 수 있고, 2개의 미노치부들(340)이 서로 대향하도록 형성될 수 있다.
- [0086] 도 8 및 도 9에는 도 5의 원통형 전지에 사용된 전류차단 부재(600)가 도시되어 있다.
- [0087] 도 5와 함께 도 8 및 도 9를 참조하면, 전류차단 부재(600)의 중앙에는 안전벤트(320)의 만입부 하단(322; 도 6 참조)에 용접 부착되는 상향 돌출된 돌출부(620)가 형성되어 있으며, 전류차단 부재(600)의 하단면 중 돌출부(620)를 제외한 부위에는 젤리-롤(110)의 양극에 연결되어 있는 양극 리드(420)가 전기적으로 연결된다.
- [0088] 돌출부(620)에는 이를 중심으로 동심원상으로 3개의 관통홀(630)과 관통홀(630)을 연결하는 노치(650)가 형성되어 있는 브릿지(640)가 형성되어 있다.
- [0089] 또한, 전류차단 부재(600)의 외주면에는 가스의 배출을 위한 3개의 관통구들(610)이 중심축을 중심으로 방사상 대칭 위치에서 원호 형상으로 형성되어 있다. 관통구들(610)은 그것의 전체 면적이 전류차단 부재(600)의 면적

대비 30%의 크기로 형성되어 있는 바, 전지 내부의 고압 가스의 배출량을 높일 수 있어서 신뢰성 있는 전류차단 효과를 발휘할 수 있다. 관통구들(610) 상호간에 대략 120도의 각도로 이격되어 있으며, 각각의 관통구의 형상 및 크기가 동일하게 형성되어 있고, 관통구들(610) 사이의 간격 역시 대략 동일하게 이루어져 있다. 이러한 구조에 의해, 전류차단 부재(600)는 가스 배출량을 최대화하면서도 높은 기계적 강도를 유지할 수 있다.

[0090] 이 때, 관통구들(610)의 중심선을 연결한 가상의 원(점선 참조)의 크기는 안전벤트(320; 도 6 및 7 참조)의 제 1 노치(324)의 크기와 대략 동일하게 형성되는 것이 바람직하다. 이에 따라, 소정 크기 이상의 내압 발생시, 관통구들(610)을 통과한 고압 가스가 직선 경로로 제 1 노치(324)를 가압하여 제 1 노치(324)의 절취를 촉진할 수 있으므로 신속한 가스 배출이 가능하다.

[0091] 도 10 및 도 11에는 전류차단 부재(600)를 감싸는 전류차단 부재용 가스켓(700)이 도시되어 있다. 설명의 편의를 위하여, 개구의 형상만을 표현하기 위하여 전류차단 부재용 가스켓은 간략히 표현하였다.

[0092] 도 5와 함께 도 10 및 도 11를 참조하면, 전류차단 부재용 가스켓(700)은 중앙에 개구(710)가 형성되어 있으며, 중앙 개구(710)는 전류차단 부재(600)의 관통구들(610)을 가리지 않는 형상으로 되어 있다. 즉, 전류차단 부재용 가스켓(700)의 형상은 전류차단 부재(600)에서 개구가 형성되어 있지 않은 부분의 외주면의 형상과 대략 일치하도록 이루어져 있어서, 전류차단 부재(600)의 외주면을 감싸면서 관통구(610)를 가리지 않으므로 관통구(610)가 가스 배출 통로로서 작용하는 것을 담보한다.

[0093] 한편, 도 12 및 도 13에는 도 5에 따른 원통형 전지(100)에서 젤리-롤(110)의 상단에 탑재되는 판상형 절연부재(800)의 일 예가 도시되어 있다. 도 2와 함께 이들 도면을 참조하면, 판상형 절연부재(800)는 가스 배출 및 전극단자의 연결을 위해 중앙에 개구(820)가 천공되어 있고 그것의 주변에 하나 또는 그 이상의 관통구(810)가 천공되어 있다. 판상형 절연부재(800)의 관통구(810)는 전류차단 부재(600)에 형성된 관통구(610)에 대응하는 형상, 크기 및 위치로 형성될 수 있으며, 판상형 절연부재(800)의 면적대비 대략 30%의 크기로 형성되어 있다.

[0094] 도 14에는 원통형 전지에 장착되는 순서대로 판상형 절연부재(800), 전류차단 부재용 가스켓(700), 전류차단 부재(600) 및 안전벤트(320)의 적층형태가 모식적으로 도시되어 있다.

[0095] 상기 설명한 도 2 내지 도 7과 함께 도 14를 참조하면, 전지 내부의 가압 가스가 효과적으로 외부로 배출될 수 있도록 판상형 절연부재(800)의 관통구(810)와 전류차단 부재 가스켓(700)의 중앙 개구(710), 전류차단 부재(600)의 관통구(610), 및 안전벤트(320)의 제 1 노치(324)는 일직선 상으로 연통되는 위치로 형성되어 있다.

[0096] 도 15 내지 17에는 전지 내부의 압력이 비정상적으로 상승하는 경우 가압된 가스가 배출되는 과정이 모식적으로 도시되어 있다.

[0097] 이들 도면을 참조하면, 전지 내부의 압력이 비정상적으로 상승할 때 가압된 가스는 전류차단 부재(600)의 관통구(610) 및 관통홀(630)을 통과하여 안전벤트(320)에 상향 압력을 가하게 된다.

[0098] 그러한 압력에 의해 도 16에서와 같이 안전벤트의 하향 만입부(322)가 들러 올려지면서 상기 만입부(322)에 용접 부착되어 있는 돌출부(620)가 제 1 임계 압력 이상에서 전류차단 부재(600)로부터 용이하게 분리되어 전류차단 부재(600)로부터 안전벤트(320)로의 통전을 차단하게 된다.

[0099] 이러한 전류의 차단에도 불구하고 계속적으로 압력이 상승하여 제 2 임계 압력 이상이 되면, 도 17에서와 같이 안전벤트(320)의 제 1 노치(324)가 절취되면서 내부의 가압 가스가 상단 캡의 가스 배출구(312)를 통해 외부로 배출되게 된다. 본 발명에서는 직경이 큰 제 1 노치(324)가 절취됨에 따라 직경이 작은 제 2 노치(326)가 절취되는 경우에 비해 가스 배출량이 커지므로, 고압 가스의 신속한 배출을 유도할 수 있다.

[0100] 이하에서는 실시예를 통해 본 발명의 내용을 상술하지만, 본 발명의 범주가 그것에 의해 한정되는 것은 아니다.

[0101] [실시예 1]

[0102] 원통형 이차전지를 다음과 같이 제조하였다.

[0103] 1-1. 양극 및 음극의 제조

[0104] LiCoO_2 가 함유된 양극 활물질을 알루미늄 집전체에 도포한 후, 집전체의 단부에 상향 돌출되도록 양극 탭을 부착

하는 것으로 양극을 제조하고, 인조흑연이 함유된 음극 활물질을 구리 집전체에 도포한 후, 집전체의 단부에 상향 돌출되도록 음극 탭을 부착하는 것으로 음극을 제조하였다.

[0105] 1-2. 전해액의 제조

[0106] 전해액으로서 1.0M LiPF₆을 포함하는 EC(ethylene carbonate)/PC(propylene carbonate)/DMC(dimethyl carbonate) = 30/5/65 (체적비) 조성의 비수계 용매에, 전해액 첨가제로서 플루오르 에틸렌 카보네이트(fluoroethylene carbonate)가 전해액 전체 조성을 기준으로 11 중량%로 포함되도록 리튬 이차전지용 전해액을 제조하였다.

[0107] 1-3. 센터 핀을 이용한 적층체의 권취

[0108] 상기 1-1의 양극과 음극 사이에 분리막을 개재한 시트형 적층체를 센터 핀에 끼워 350 g/cm²의 응력이 가해지도록 권취함으로써 젤리-롤을 제조하였다.

[0109] 제조한 젤리-롤을 원통형 케이스에 넣고 리드선을 연결한 후, 상기 1-2의 전해액을 주입하였다.

[0110] 1-4. 원통형 이차전지의 제조

[0111] 외경 16 mm 및 두께 0.3 mm 알루미늄 판재에, 도 7에서와 같이, 30도의 원호 크기를 제외한 부위에 8.0 mm 직경과 0.06 mm 두께의 제 1 노치를 형성하고 4.0 mm 직경과 0.1 mm 두께의 제 2 노치를 형성한 후, 중앙을 하향으로 0.65 mm 깊이만큼 돌출된 만입부를 가지도록 안전벤트를 제조하였다.

[0112] 또한, 외경 11 mm 및 두께 0.5 mm의 알루미늄 판재에, 도 9에서와 같이, 3개의 관통구들을 방사상으로 형성하고 관통구들의 총 면적이 상기 판재의 전체면적 대비 약 30%가 되도록 하였다. 그런 다음 중앙에 직경 1.53 mm 및 돌출 높이 0.20 mm의 돌출부를 형성하고, 폭 0.6 mm 및 원주 길이 2.61 mm의 관통홀을 중앙 돌출부의 중심으로 부터 1.5 mm의 거리에 3개 천공한 뒤, 관통홀을 잇는 각각의 브릿지에 두께 약 70 μ m의 노치를 형성하여 전류차단 부재를 제조하였다.

[0113] 또한, 외경 12 mm 및 두께 0.5 mm의 폴리프로필렌 판재에, 도 11에서와 같이, 전류차단 부재의 관통구를 가리지 않는 구조의 전류차단 부재용 가스켓을 제조하였다.

[0114] 상기 전류차단 부재의 외주면을 전류차단 부재용 가스켓에 삽입하고, 상기 안전벤트의 만입부 밑면을 전류차단 부재의 돌출부 윗면에 레이저 용접하여 부착시켰다.

[0115] 상기 1-3의 원통형 케이스 개방 상단에 판상형 절연부재를 장착한 뒤, 캔의 윗면을 비딩하여 고정시키고, 그러한 비딩부에 전류차단 부재가 부착되어 있는 상기 안전벤트와, PTC 소자 및 상단 캡이 삽입되어 있는 가스켓을 끼워넣고 캔의 상단을 안쪽으로 프레스하여 가스켓을 클립핑함으로써 전지를 제조하였다.

[0116] [실시예 2]

[0117] 안전벤트의 제조시 제 1 노치의 두께를 0.1 mm, 제 2 노치의 두께를 0.06 mm로 하였다는 점을 제외하고는 실시예 1과 동일한 방법으로 전지를 제조하였다.

[0118] [실시예 3]

[0119] 안전벤트의 제조시 제 1 노치를 7 mm 직경과 0.06 mm 두께로 형성하였다는 점을 제외하고는 실시예 1과 동일한 방법으로 전지를 제조하였다.

[0120] [실시예 4]

[0121] 전해액의 제조시 플루오르 에틸렌 카보네이트가 전해액 전체 조성을 기준으로 20 중량%로 포함되도록 한 점을 제외하고는 실시예 1과 동일한 방법으로 전지를 제조하였다.

[0122] [실시예 5]

- [0123] 전해액의 제조시 플루오르 에틸렌 카보네이트를 전해액 전체 조성을 기준으로 30 중량%로 포함되도록 한 점을 제외하고는 실시예 1과 동일한 방법으로 전지를 제조하였다.
- [0124] [실시예 6]
- [0125] 전해액의 제조시 PC(propylene carbonate)/MP(methyl propionate)=8:92 (체적비) 조성의 비수계 용매를 사용한 점을 제외하고는 실시예 4와 동일한 방법으로 전지를 제조하였다.
- [0126] [실시예 7]
- [0127] 전해액의 제조시 PC/MP=8:92 (체적비) 조성의 비수계 용매를 사용한 점을 제외하고는 실시예 7과 동일한 방법으로 전지를 제조하였다.
- [0128] [비교예 1]
- [0129] 전해액의 제조시 전해액 첨가제를 첨가하지 않았다는 점을 제외하고는 실시예 1과 동일한 방법으로 전지를 제조하였다.
- [0130] [비교예 2]
- [0131] 전해액의 제조시 플루오르 에틸렌 카보네이트를 전해액 전체 조성을 기준으로 5 중량%로 포함되도록 한 점을 제외하고는 실시예 1과 동일한 방법으로 전지를 제조하였다.
- [0132] [비교예 3]
- [0133] 상기 전류차단 부재의 관통구의 총 면적이 상기 전류차단 부재의 전체면적 대비 18%가 되도록 하였다는 점을 제외하고는 실시예 1과 동일한 방법으로 전지를 제조하였다.
- [0134] [실험예 1]
- [0135] 상기 실시예 1, 4 및 5에서 제작된 전지들과 비교예 1 및 2에서 제작된 전지들을 각각 준비하여 양극 계면 저항을 측정하였고, 그 결과를 도 18에 나타내었다.
- [0136] 도 18을 참조하면, EIS(Electrochemical impedance spectroscopy) 장비를 이용하여 진동수(frequency)당 저항값을 측정한 뒤, 그 성향을 semi-circle로 변환하여 나타낸 그래프로써, 플루오르 에틸렌 카보네이트를 전해액에 첨가한 실시예 1의 전지는 상기 전해액 첨가제를 포함하지 않은 전지(비교예 1)과 상기 전해액 첨가제를 5 중량%로 포함하는 전지(비교예 2)보다 양극 계면의 저항값이 상대적으로 증가함을 알 수 있다. 특히, 전해액 첨가제를 20 중량%로 포함하는 전지(실시예 4)와 30 중량%로 포함하는 전지(실시예 5)에서 양극 계면의 저항값이 크게 증가함을 확인할 수 있다.
- [0137] [실험예 2]
- [0138] 상기 실시예 1 및 4 내지 7에서 제작된 전지들과 비교예 1 및 2에서 제작된 전지들을 각각 2개씩 준비하여, 만충전 상태에서 강제로 외부 단락을 유발시킨 후 단락 전류량을 측정하였고, 그 결과를 하기 표 1에 나타내었다.

[0139] <표 1> (단위: current(A))

	1	2	Avg.
실시예 1	46.8	50	48.4
실시예 4	44.5	45.8	45.2
실시예 5	40.9	43.2	42.0
실시예 6	45.3	45.6	45.4
실시예 7	43.9	42.1	43.0
비교예 1	51.2	53.6	52.4
비교예 2	51	52	51.5

[0140]

[0141] 상기 표 1에서 보는 바와 같이, 플루오르 에틸렌 카보네이트를 전해액에 첨가한 실시예 1의 전지는 상기 전해액 첨가제를 포함하지 않은 전지(비교예 1)과 상기 전해액 첨가제를 5 중량%로 포함하는 전지(비교예 2)보다 단전시의 전류량이 상대적으로 적음을 알 수 있다.

[0142] 더욱이, 전해액 첨가제를 20 중량%로 포함하는 전지(실시예 4)와 30 중량%로 포함하는 전지(실시예 5)는 11 중량%를 포함하는 전지(실시예 1)보다도 현저히 낮은 단전 전류량을 나타냄을 확인할 수 있다. 이러한 현상은 실시예 4 및 5와 비수계 전해액의 조성을 달리하는 실시예 6 및 7에서도 유사하게 나타난다.

[0143] 따라서, 단락시의 발열량이 전류량에 비례함을 고려할 때, 본 발명에 따른 전지들은 낮은 발열량에 의해 상대적으로 안전함을 알 수 있다.

[0144] [실험예 3]

[0145] 상기 실시예 1 및 4 내지 7에서 제작된 전지들과 비교예 1 및 2에서 제작된 전지들의 열적 안전성을 비교하였고, 그 결과를 하기 표 2에 나타내었다.

[0146] <표 2>

	Onset temp. (°C)	발열량 (J/g)			
		1 peak	2 peak	3 peak	Total
실시예 1	206 ± 1	71 ± 4	47 ± 4	-	118 ± 6
실시예 4	202 ± 1	55 ± 3	45 ± 5	-	100 ± 6
실시예 5	204 ± 1	50 ± 4	40 ± 4	-	90 ± 4
실시예 6	203 ± 1	53 ± 2	50 ± 4	-	103 ± 4
실시예 7	205 ± 1	48 ± 5	43 ± 5	-	91 ± 6
비교예 1	198 ± 1	223 ± 6			223 ± 6
비교예 2	204 ± 1	98 ± 11	56 ± 5	49 ± 6	203 ± 6

[0147]

[0148] 상기 표 2에서 'Onset temp.'는 열분해 반응의 시작 온도를 나타내며, '발열량'은 열 분석시 peak 면적을 나타낸 상대값으로, 상기 두 값을 비교하여 열적 안전성 정도를 평가하였다.

[0149] 실험적으로 확인한 바로는, 플루오르 에틸렌 카보네이트를 포함하지 않는 전지인 비교예 1의 전지는 대략 210℃에서 한 차례 전해액 분해 반응이 유발됨에 반하여, 플루오르 에틸렌 카보네이트가 첨가된 실시예 1의 전지에서는 대략 225℃와 275℃에서 각각 전해액 분해 반응이 유발됨이 확인되었다. 따라서, 일 순간에 가스 발생이 집중되는 현상이 억제될 수 있음을 알 수 있다.

[0150] 또한, 상기 표 2에서 보는 바와 같이, 실시예 1의 전지에서는 발열량이 비교예 1의 전지에 대해 대략 50% 수준으로서 고열 발생에 따른 안전성 문제가 덜 심각함을 알 수 있다. 특히, 전해액 첨가제의 함량이 20 중량%와 30 중량%인 실시예 4 및 5의 전지들은 실시예 1의 전지보다도 현저히 낮은 발열량을 나타냄으로써, 안전성이 더

속 우수함을 확인할 수 있다. 이와 같이 현저한 발열량 감소 현상은 PC/MP 조성의 비수계 전해액을 사용하는 실시예 6 및 7에서도 유사하게 나타난다.

[0151] 한편, 전해액 첨가제를 5 중량%로 첨가한 비교예 2의 전지는 11 중량%로 첨가한 실시예 1의 전지와 비교하여 발열량이 70% 이상 증가하였는 바, 이는 플루오르 에틸렌 카보네이트를 첨가하더라도 그것의 함량이 대략 10 중량% 이하인 경우에는 첨가에 따른 효과가 매우 낮음을 의미한다.

[0152]

[0153] [실험예 4]

[0154] 상기 실시예 1 ~ 3에서 제작된 전지와 비교예 3에서 각각 제작된 전지들에 대해 nail test를 실시하였고, 그 결과를 하기 표 3에 나타내었다.

[0155] <표 3>

	관통구의 면적비	안전벤트의 절취부위	폭발여부	전지 캔의 변형여부
실시예 1	30 %	제 1 노치	X	X
실시예 2	30 %	제 2 노치	X	Δ
실시예 3	30 %	내부노치(직경확대)	X	Δ
비교예 3	18 %	제 1 노치	O	O

[0156]

[0157] 상기 표 3을 참조하면, 본 발명에 따라 관통구의 면적비가 30%인 전류차단 부재를 사용한 경우 전지의 폭발 또는 전지 캔의 변형이 발생하지 않았다. 반면에, 관통구의 면적비가 18%인 전류차단 부재를 사용한 비교예 3의 전지는 nail test시 전지가 폭발하였다. 또한, 실시예 2 및 3에서와 같이 제 2 노치부가 절취되어 안전벤트에서 가스 분출이 원활히 일어나지 못하는 경우는 전지의 폭발은 일어나지 않았으나, 전지 캔의 변형이 발생하였다.

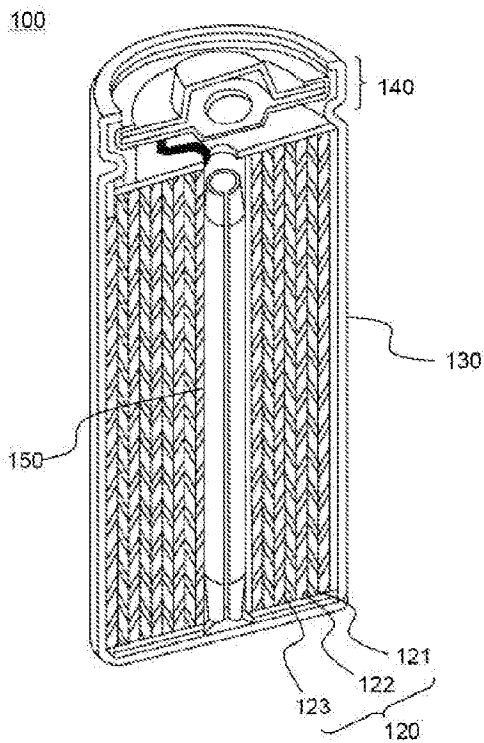
[0158] 고용량의 전지의 경우 안전벤트의 절취부 및 전류차단 부재의 관통구의 크기가 충분치 못한 경우, 충분히 가스를 통과시킬 수 없고, 안전벤트의 절취부가 내부에 위치하는 제 2 노치인 경우, 노치의 직경을 확대하여도 절취 후 안전벤트의 경사진 내부 노치부에 의해 가스 분출에 대한 저항이 발생하는 문제점이 있음을 확인할 수 있다.

[0159] 이상 검토한 바와 같이, 전지 내부의 가압 가스를 효과적으로 배출하기 위해서는 전류차단 부재의 관통구의 면적의 확대와 함께, 안전벤트의 절취부를 제 1 노치부로 하는 구조가 전지 안전성의 상승효과를 발휘할 수 있음을 알 수 있다.

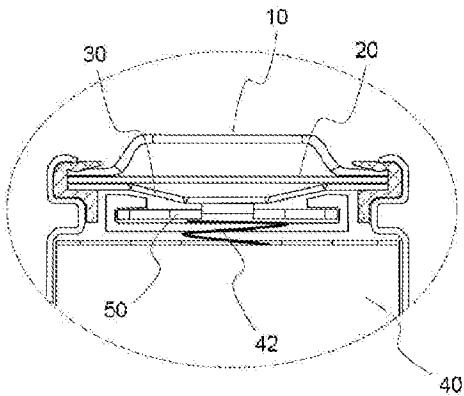
[0160] 본 발명이 속한 분야에서 통상의 지식을 가진 자라면, 상기 내용을 바탕으로 본 발명의 범주내에서 다양한 응용 및 변형을 행하는 것이 가능할 것이다.

도면

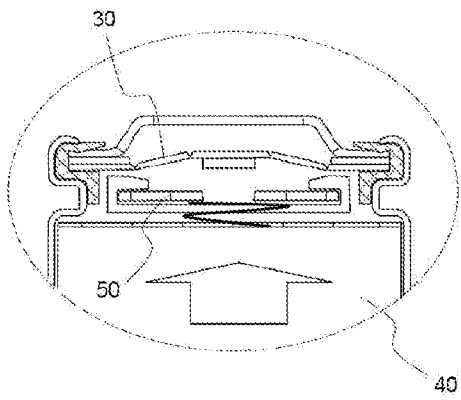
도면1



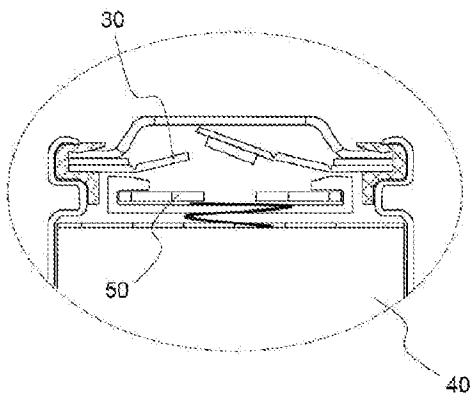
도면2



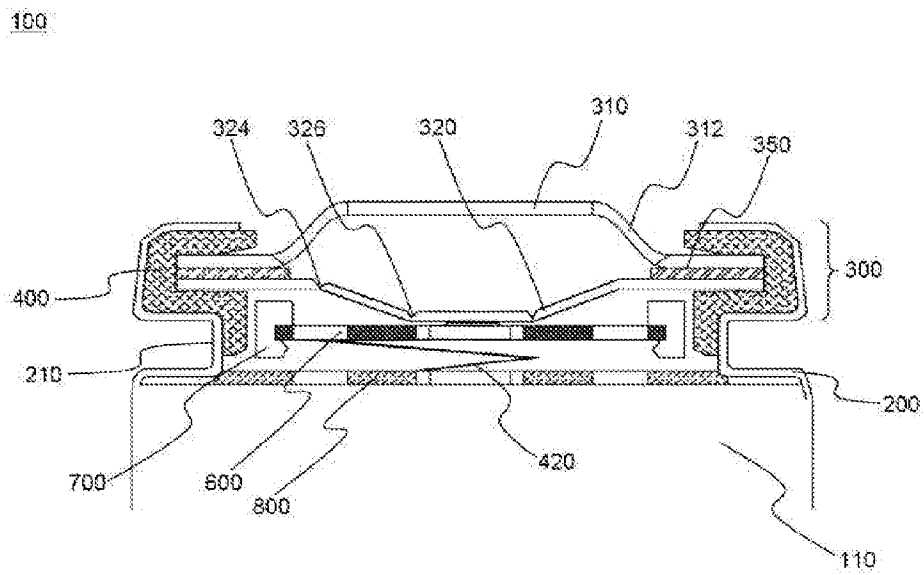
도면3



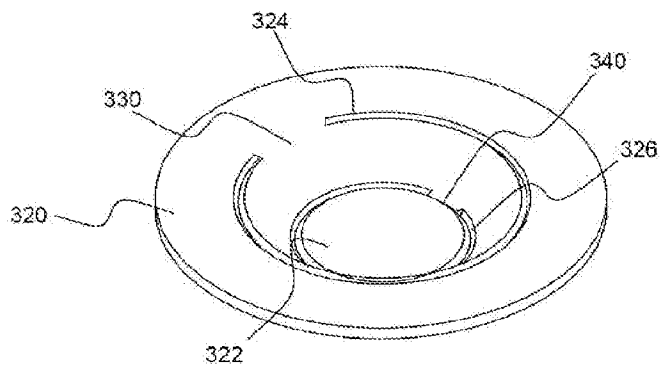
도면4



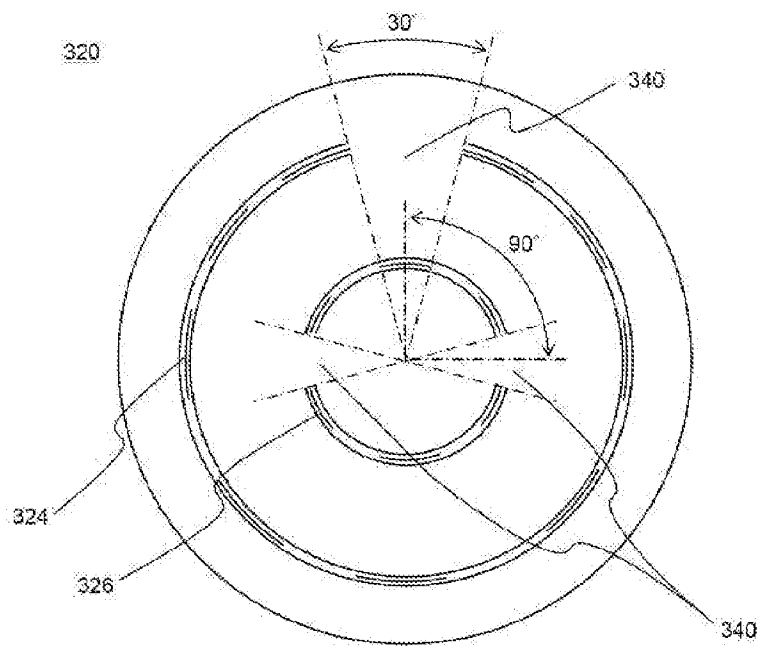
도면5



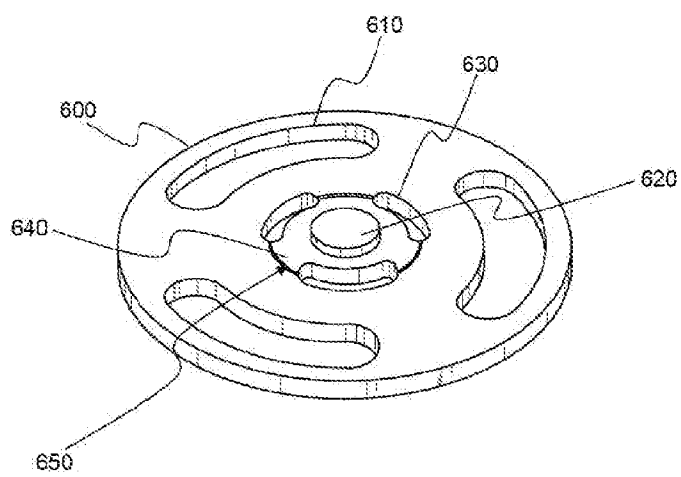
도면6



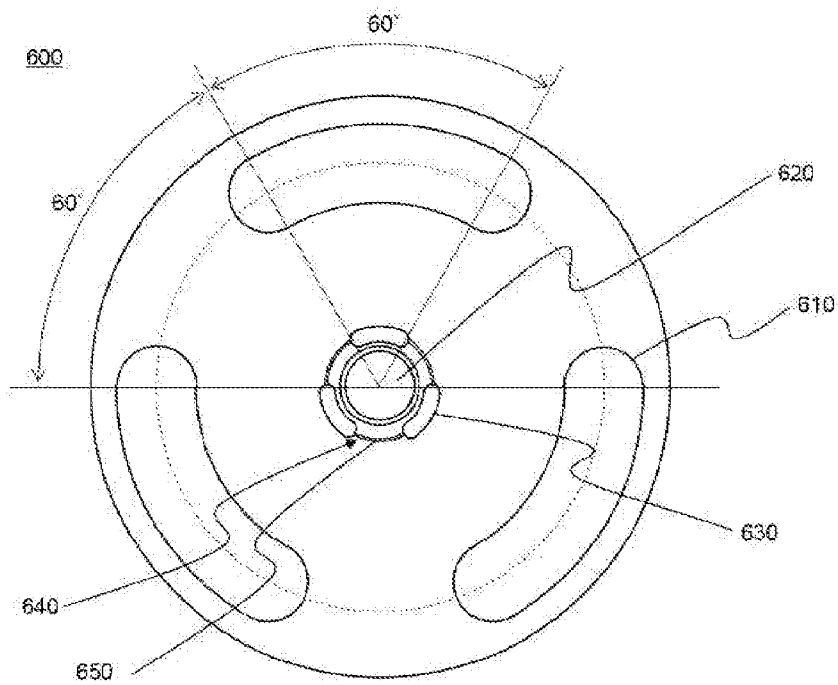
도면7



도면8

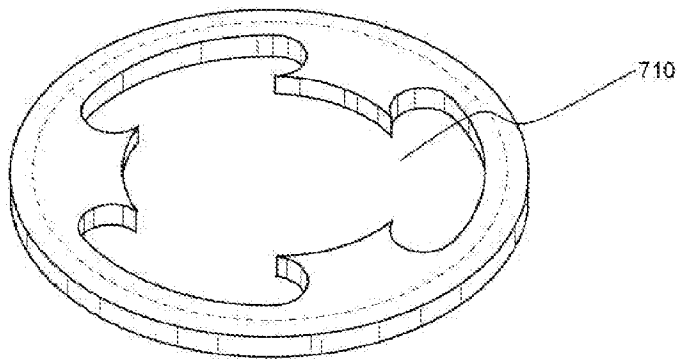


도면9



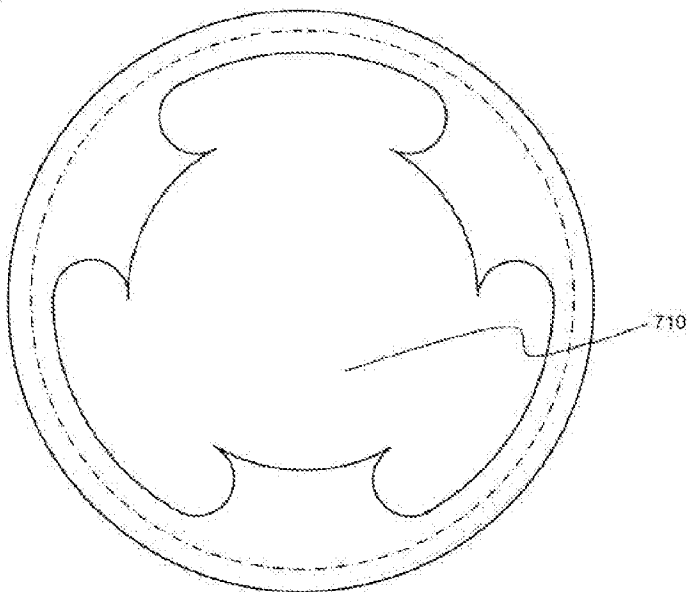
도면10

700



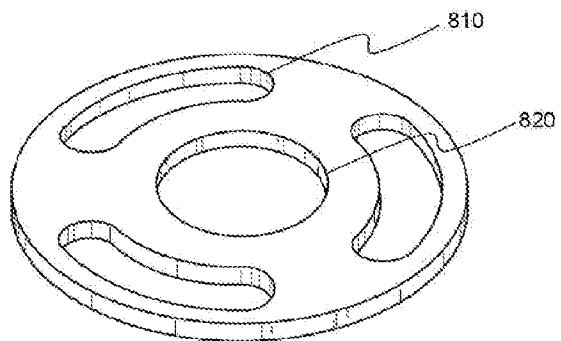
도면11

700



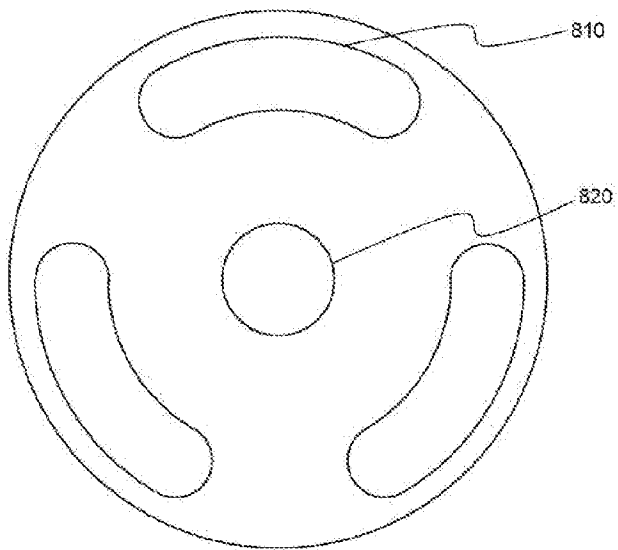
도면12

800

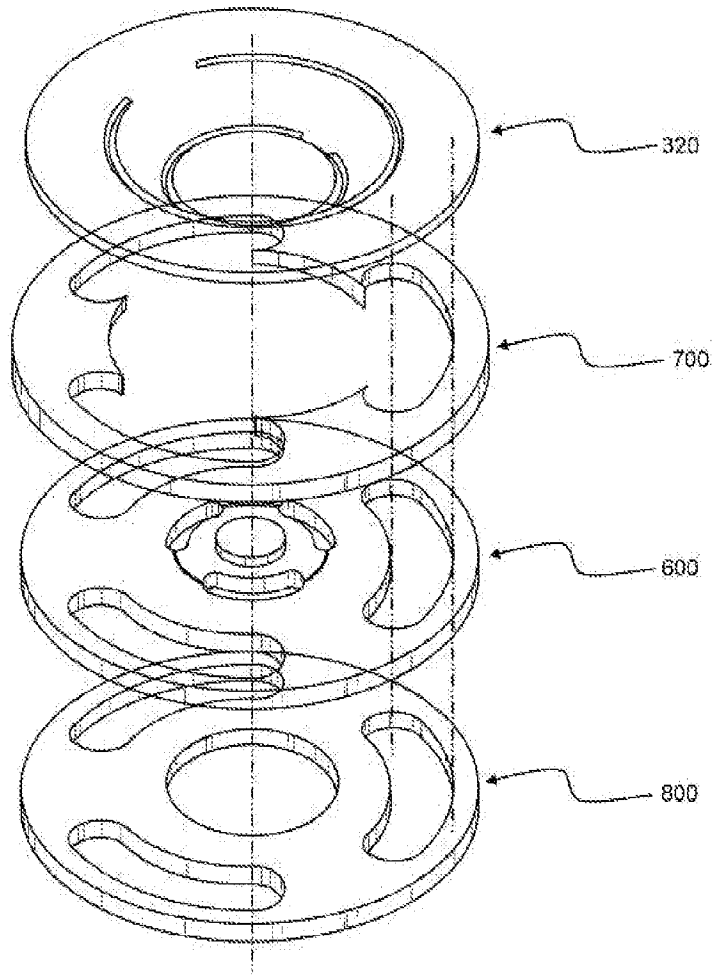


도면13

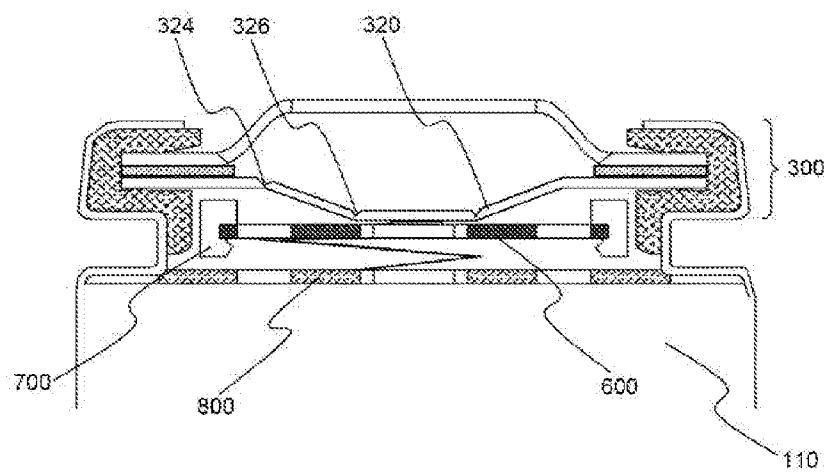
800



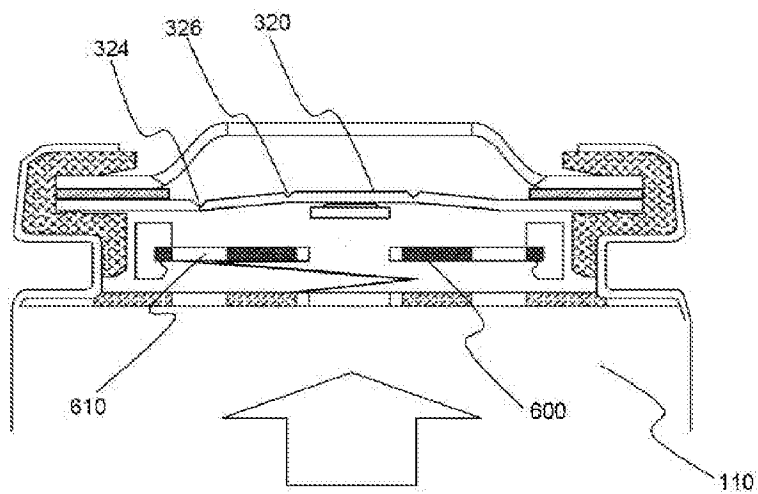
도면14



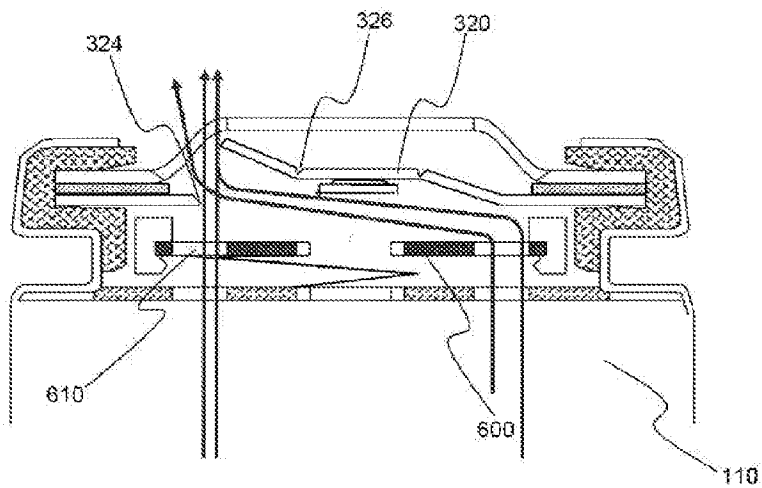
도면15



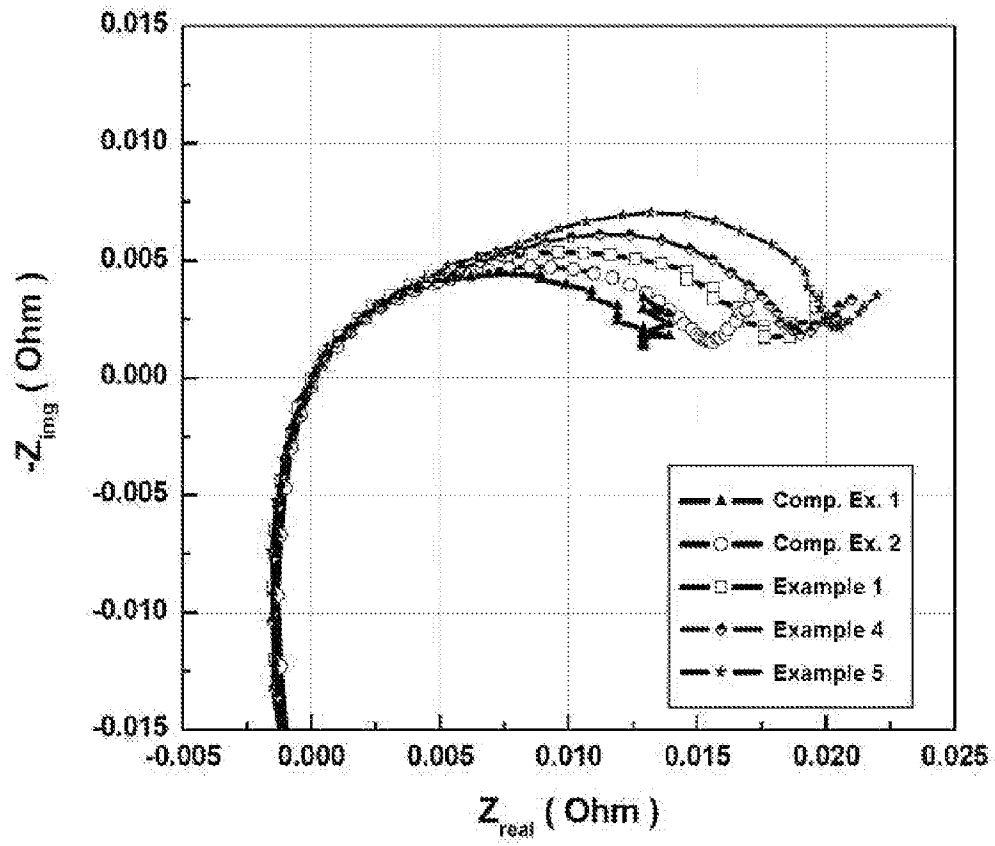
도면16



도면17



도면18





Espacenet

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RECHARGEABLE BATTERY

Inventor(s): KIM JIN HEE [KR] ± (KIM, JIN HEE)

Applicant(s): SAMSUNG SDI CO LTD [KR] ± (SAMSUNG SDI CO., LTD)

Classification: - international: H01M2/10; H01M2/12; H01M4/58; H01M4/66
 - cooperative: H01M10/052 (EP, US); H01M10/058 (EP, US);
H01M10/4235 (EP, US); H01M2/1241 (EP, US);
H01M2/34 (EP, US); H01M2/345 (EP, US);
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US8993138 (B2) less

Abstract of KR20100038054 (A)

PURPOSE: A rechargeable battery is provided to prevent the cut off of current and the breaking of vent, thereby improving reliability and safety. CONSTITUTION: A rechargeable battery(100) comprises: an electrode assembly(110) comprising a positive electrode, a negative electrode, and a separator between the positive electrode and the negative electrode, the positive electrode comprising a current collector and a positive active material coupled to the current collector, wherein the positive active material is a lithium-nickel-cobalt-manganese oxide(NCM)-based positive active material; a case(120) containing the electrode assembly; and a cap assembly(140) coupled to the case, the cap assembly comprising a vent plate(160) having a notch (163) configured to fracture the vent plate due to a predefined vent fracture pressure B inside the case and a current interrupt portion configured to interrupt electrical connection due to a predetermined current interrupt pressure A inside the case, characterized in that a ratio of the current interrupt pressure A and the vent fracture pressure B satisfies the following equation: $A/B = 1.22 - 0.39X + D$ ($-0.08 \leq D \leq 0.08$), wherein A represents the current interrupt pressure measured in kgf/cm² of the current interrupt portion; B represents the vent fracture pressure measured in kgf/cm² of the notch, X represents a capacity of the rechargeable battery measured in Ah, and D is a factor within the range of $-0.08 \leq D \leq 0.08$.

FIG. 2A

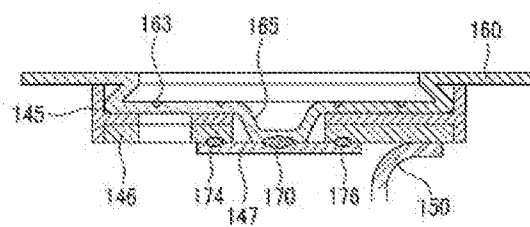


FIG. 2B

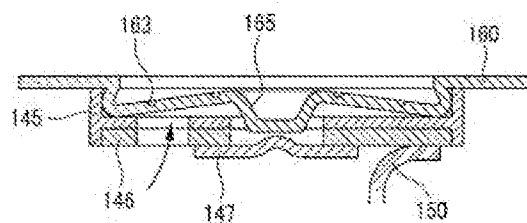
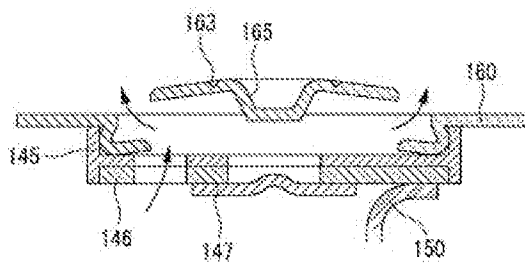


FIG. 2C





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(71) 출원인

삼성에스디아이 주식회사

경기도 수원시 영통구 매탄동 673-7

(72) 발명자

김진희

서울특별시 강북구 미아6동 1265-241

(74) 대리인

팬코리아특허법인

전체 청구항 수 : 총 23 항

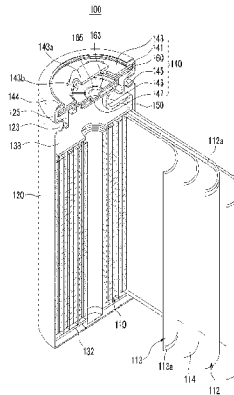
(54) 이차 전지

(57) 요약

본 발명에 따른 이차 전지는 신뢰성과 안전성을 확보할 수 있도록, 양극, 음극 및 상기 양극과 상기 음극 사이에 개재되는 세퍼레이터를 포함하는 전극 조립체와, 상기 전극 조립체가 내장되는 케이스와, 상기 케이스와 결합되는 캡 조립체를 포함하는 이차 전지에 있어서, 상기 양극은 집전체와 상기 집전체 상에 형성된 3원계 양극 활물질로 이루어진 양극 활물질층을 포함하고, 상기 캡 조립체는 노치가 형성된 벤트 플레이트와 케이스 내부의 압력 상승으로 전기적인 연결을 차단하는 전류 차단부를 포함하며, 상기 전류 차단부의 전류 차단압을 A, 상기 벤트 플레이트의 노치가 파단되는 벤트 파단압을 B, 이차 전지의 용량을 X라 할 때,

$A/B = 1.22 - 0.39X + D$ (단, $-0.08 \leq D \leq 0.08$)의 조건을 만족한다.

배터리 - 도1



특허청구의 범위

청구항 1

양극, 음극 및, 상기 양극과 음극 사이에 개재되는 세퍼레이터를 포함하는 전극 조립체;

상기 전극 조립체를 포함하는 케이스;

상기 케이스에 체결되는 캡 조립체

를 포함하고,

상기 양극은 집전체와 상기 집전체에 형성되는 양극 활물질을 포함하고,

상기 캡 조립체는 벤트 플레이트와 상기 케이스 내에서의 전류 차단압으로 인해 전기적 연결을 차단하는 전류 차단부를 포함하고,

상기 벤트 플레이트는 상기 케이스 내에서 벤트 파단압으로 인해 벤트 플레이트가 파단되도록 형성되는 노치를 포함하며,

하기 식을 만족하는 이차 전지.

$$A/B = 1.22 - 0.39X + D \quad (-0.08 \leq D \leq 0.08),$$

여기서, A는 상기 전류 차단부의 상기 전류 차단압 (kgf/cm^2)을 나타내고, B는 상기 노치의 상기 벤트 파단압 (kgf/cm^2)을 나타내며, X는 상기 이차 전지의 용량 (Ah)을 나타낸다.

청구항 2

제 1 항에 있어서,

상기 캡 조립체는,

상기 케이스에 체결되는 캡 엽;

서브 플레이트;

상기 벤트 플레이트와 상기 서브 플레이트 사이에 구비되는 캡 다운; 및

상기 캡 다운과 상기 서브 플레이트 사이에 구비되는 돌기

를 포함하고,

상기 벤트 플레이트는 상기 캡 엽과 상기 서브 플레이트 사이에 위치하며 상기 케이스의 바닥을 향해 돌출되는 블록부를 포함하고,

상기 서브 플레이트는 상기 블록부에 용접되고, 상기 전극 조립체에 전기적으로 연결되는 이차 전지.

청구항 3

제 2 항에 있어서,

상기 돌기는 상기 서브 플레이트와 상기 캡 다운에 함께 용접되는 용접부를 포함하는 이차 전지.

청구항 4

제 2 항에 있어서,

상기 전극 조립체에 전기적으로 연결되는 리드 부재를 더 포함하고, 상기 캡 다운은 상기 리드 부재를 통해 상기 전극 조립체에 전기적으로 연결되는 이차 전지.

청구항 5

제 2 항에 있어서,

상기 서브 플레이트는 상기 캡 다운을 통해 상기 전극 조립체에 전기적으로 연결되는 이차 전지.

청구항 6

제 2 항에 있어서,

상기 캡 다운과 상기 벤트 플레이트 사이에 절연판을 더 포함하는 이차 전지.

청구항 7

제 2 항에 있어서,

상기 전류 차단부는 상기 볼록부와 상기 서브 플레이트에 함께 용접되는 용접부를 포함하는 이차 전지.

청구항 8

제 1 항에 있어서,

상기 케이스가 원통 형상으로 이루어지는 이차 전지.

청구항 9

제 1 항에 있어서,

상기 활물질은 과충전 첨가제를 상기 양극 활물질의 0.5 내지 2.0wt% 범위 내에 속하는 양으로 포함하는 이차 전지.

청구항 10

제 9 항에 있어서,

상기 과충전 첨가제는 탄소 화합물을 포함하는 이차 전지.

청구항 11

제 10 항에 있어서,

상기 탄소 화합물은 탄산리튬을 포함하는 이차 전지.

청구항 12

제 1 항에 있어서,

상기 전류 차단부의 전류 차단압은 8 kgf/cm^2 과 9 kgf/cm^2 사이 범위에 속하는 이차 전지.

청구항 13

제 1 항에 있어서,

상기 노치의 벤트 파단압은 20 kgf/cm^2 와 25 kgf/cm^2 사이 범위에 속하는 이차 전지.

청구항 14

제 1 항에 있어서,

상기 전류 차단부의 상기 전류 차단압은 8 kgf/cm^2 과 9 kgf/cm^2 사이 범위에 속하고,

상기 노치의 벤트 파단압은 20 kgf/cm^2 와 25 kgf/cm^2 사이 범위에 속하는 이차 전지.

청구항 15

제 1 항에 있어서,

상기 양극 활물질은 상기 집전체에 도포되는 이차 전지.

청구항 16

제 15 항에 있어서,
상기 집전체는 알루미늄을 포함하는 이차 전지.

청구항 17

제 1 항에 있어서,
상기 양극 활물질은 3원계 (NCM; Lithium Nickel-Cobalt-Manganese Oxide, LiNiCoMnO_2) 양극 활물질인 이차 전지.

청구항 18

양극, 음극, 및 상기 양극과 음극의 사이에 개재되는 세퍼레이터를 포함하는 전극 조립체;
상기 전극 조립체가 내장되는 케이스; 및
상기 케이스에 체결되는 캡 조립체
를 포함하고,
상기 양극은 집전체와 상기 집전체에 형성되는 양극 활물질을 포함하고,
상기 캡 조립체는 벤트 플레이트와 상기 케이스 내에서의 전류 차단압으로 인해 전기적 연결이 차단되도록 하는 전류 차단부를 포함하며,
상기 벤트 플레이트는 상기 케이스 내에서의 벤트 파단압으로 인해 상기 벤트 플레이트를 포함하도록 구성되는 노치를 포함하고,
상기 전류 차단부의 상기 전류 차단압은 8 kgf/cm^2 과 9 kgf/cm^2 사이 범위에 속하며,
상기 벤트 플레이트의 상기 노치의 상기 벤트 파단압은 20 kgf/cm^2 와 25 kgf/cm^2 사이 범위에 속하는 이차 전지.

청구항 19

제 18 항에 있어서,
양극 활물질은 3원계 (NCM; Lithium Nickel-Cobalt-Manganese Oxide, LiNiCoMnO_2) 양극 활물질인 이차 전지.

청구항 20

제 19 항에 있어서,
상기 3원계 양극 활물질은 양극 활물질의 0.5 내지 2.0wt% 사이 범위 내에 속하는 양의 과충전 첨가제를 포함하는 이차 전지.

청구항 21

제 20 항에 있어서,
상기 과충전 첨가제는 탄산리튬을 포함하는 이차 전지.

청구항 22

양극, 음극 및 상기 양극과 음극의 사이에 개재되는 세퍼레이터를 포함하는 전극 조립체, 상기 전극 조립체가 내장되는 케이스, 및 상기 케이스에 체결되는 캡 조립체를 포함하고, 상기 양극은 집전체와 상기 집전체에 형성되는 양극 활물질을 포함하고, 상기 캡 조립체는 벤트 플레이트와 전류 차단부를 포함하며, 상기 벤트 플레이트는 노치를 포함하는 이차 전지 형성방법에 있어서,
상기 케이스 내에서의 전류 차단압 A (kgf/cm^2)로 인해 전기적 연결을 차단하는 상기 전류 차단부를 구성하고,

상기 케이스 내에서의 벤트 파단압 B (kgf/cm^2)로 인해 벤트 플레이트를 파단하는 노치를 구성하고,
 상기 이차 전지가 용량 X (Ah)를 갖도록 구성되고,
 상기 이차 전지가 식 $A/B = 1.22 - 0.39X + D$ ($-0.08 \leq D \leq 0.08$)를 만족하도록 구성하는 이차 전지 형성방법.

청구항 23

양극, 음극 및 상기 양극과 음극의 사이에 개재되는 세퍼레이터를 포함하는 전극 조립체, 상기 전극 조립체가 내장되는 케이스, 및 상기 케이스에 체결되는 캡 조립체를 포함하고, 상기 양극은 집전체와 상기 집전체에 형성되는 양극 활물질을 포함하고, 상기 캡 조립체는 벤트 플레이트와 상기 케이스 내에서의 전류 차단압으로 인해 전기적 연결을 차단하는 전류 차단부를 포함하며, 상기 벤트 플레이트는 상기 케이스 내에서의 벤트 파단압으로 인해 상기 벤트 플레이트를 파단하도록 구성되는 노치를 포함하는 이차 전지 형성방법에 있어서,

상기 전류 차단부의 상기 전류 차단압을 8 kgf/cm^2 과 9 kgf/cm^2 사이 범위에 속하도록 설정하고,

상기 벤트 플레이트의 상기 노치의 상기 벤트 파단압을 20 kgf/cm^2 와 25 kgf/cm^2 사이 범위에 속하도록 설정하는 이차 전지 형성방법.

발명서

발명의 상세한 설명

기술분야

[0001] 본 발명은 이차 전지에 관한 것으로서, 보다 상세하게는 캡 조립체의 구조를 갖는 이차 전지에 관한 것이다.

배경기술

[0002] 이차 전지(rechargeable battery)는 충전이 불가능한 일차 전지와는 달리 충전 및 방전이 가능한 전지이다. 저용량 이차 전지(예를 들어, 하나의 셀로 이루어진 이차 전지)의 경우, 휴대폰이나 노트북 컴퓨터 및 캠코더와 같이 휴대가 가능한 소형 전자기기에 사용된다. 대용량 이차 전지(예를 들어, 복수개의 셀이 팩 형태로 연결된 이차 전지)는 하이브리드 전기 자동차 등의 모터 구동용 전원으로 널리 사용되고 있다.

[0003] 이러한 이차 전지는 여러 가지 형상으로 제조되고 있는데, 대표적인 형상으로는 원통형, 각형을 들 수 있다.

[0004] 그리고 이러한 복수 개의 이차 전지는 대전력을 필요로 하는 전기 자동차 등의 모터 구동에 사용될 수 있도록 직렬로 연결되어 대용량의 이차 전지 모듈을 구성할 수 있다.

[0005] 이차 전지는 양극과 음극이 세퍼레이터(separator)를 사이에 두고 위치하는 전극 조립체와 전극 조립체가 내장되는 공간을 구비하는 케이스, 케이스를 밀폐하는 캡 조립체를 포함한다.

[0006] 이차 전지가 원통형으로 형성되는 경우, 전극 조립체의 양극 및 음극에는 활물질이 도포되지 않은 무지부가 형성되고, 양극 무지부와 음극 무지부는 서로 다른 방향을 향하도록 배치된다.

[0007] 음극 무지부에는 음극 집전판이 부착되고, 양극 무지부에는 양극 집전판이 부착된다. 음극 집전판은 케이스와 전기적으로 연결되고, 양극 집전판은 캡 조립체와 전기적으로 연결되어 외부로 전류를 유도하게 된다. 따라서 케이스는 음극 단자 역할을 하고, 캡 조립체에 설치된 캡 엽이 양극 단자 역할을 한다.

[0008] 이차 전지가 충전과 방전을 되풀이하는 동안, 이차 전지의 내부에는 가스가 발생하여 내부 압력이 상승하며, 이차 전지 내부의 압력 상승을 방지하면, 이차 전지가 폭발할 위험이 있다. 이를 방지하기 위해서 캡 엽의 아래에는 소정 압력에서 파단될 수 있도록 노치가 형성된 벤트 플레이트가 설치된다.

[0009] 벤트 플레이트는 아래로 돌출된 볼록부를 구비하며 볼록부는 전극 조립체와 전기적으로 연결된 서브 플레이트와 용접으로 접합된다. 이차 전지 내부의 압력이 상승하면, 먼저 볼록부가 서브 플레이트에서 이탈되어 벤트 플레이트와 서브 플레이트 사이의 전류를 차단한다. 이후에 계속적으로 압력이 상승하면 벤트 플레이트에 형성된 노치가 파단되어 외부로 가스를 방출한다.

[0010] 전류가 차단되는 전류 차단압과 벤트 플레이트의 노치가 파단되는 벤트 파단압은 이차 전지의 신뢰성과 안전성

측면에서 중요한 요소이다.

- [0011] 즉, 즉 전류 차단압과 벤트 파단압을 낮게 설정하면, 안전성은 증가하나, 내부압력이 조금만 상승하더라도 전지가 작동하지 못하는 문제가 발생하게 된다.
- [0012] 특히 3원계(NCM; Lithium Nickel-Cobalt-Manganese Oxide, LiNiCoMnO_2) 양극 활물질이 적용된 전지의 경우, 코발트계(LCO, LiCoO_2) 양극 활물질이 적용된 전지와 다른 압력 거동을 하는 바, 3원 양극 활물질이 적용된 전지의 전류 차단압과 벤트 파단압은 코발트계 양극 활물질이 적용된 전지와 상이하게 설정된다.
- [0013] 코발트계 양극 활물질에 적용된 전류 차단압과 벤트 파단압을 3원계 양극 활물질이 적용된 전지에 그대로 적용할 경우에는 이차 전지의 신뢰성과 안전성이 저하되는 문제가 발생한다.

발명의 내용

해결 하고자 하는 과제

- [0014] 본 발명의 일 측면은 신뢰성과 안전성이 향상된 이차 전지를 제공함에 있다.

과제 해결수단

- [0015] 본 발명의 일 실시예에 따른 이차 전지는 양극, 음극 및 상기 양극과 상기 음극 사이에 개재되는 세퍼레이터를 포함하는 전극 조립체와, 상기 전극 조립체가 내장되는 케이스와 케이스와 결합되는 캡 조립체를 포함하는 이차 전지에 있어서, 상기 양극은 집전체와 상기 집전체 상에 형성된 3원계 양극 활물질로 이루어진 양극 활물질층을 포함하고, 상기 캡 조립체는 노치가 형성된 벤트 플레이트와 케이스 내부의 압력 상승으로 전기적인 연결을 차단하는 전류 차단부를 포함하며, 상기 전류 차단부의 전류 차단압(kgf/cm^2)을 A, 상기 벤트 플레이트의 노치가 파단되는 벤트 파단압(kgf/cm^2)을 B, 이차 전지의 용량(Ah)을 X라 할 때, $A/B = 1.22 - 0.39X + D$ (단, $-0.08 \leq D \leq 0.08$)의 조건을 만족한다.
- [0016] 상기 캡 조립체는 상기 케이스와 결합된 캡 업과 상기 캡 업의 아래에 설치되며 상기 케이스의 바닥을 향하여 돌출된 볼록부를 갖는 벤트 플레이트와, 상기 벤트 플레이트의 아래에 설치되어 상기 볼록부와 용접되며 상기 전극 조립체와 전기적으로 연결된 서브 플레이트와 상기 벤트 플레이트와 상기 서브 플레이트 사이에 배치된 캡 다운, 및 상기 캡 다운과 상기 서브 플레이트 사이에 설치된 돌기를 포함한다.
- [0017] 상기 전류 차단부는 상기 볼록부와 상기 서브 플레이트가 용접으로 접합된 부분일 수 있으며, 상기 이차 전지는 원통형으로 이루어질 수 있다.
- [0018] 상기 양극 활물질에는 0.5wt% 내지 2.0wt% 사이의 과충전 첨가제가 포함될 수 있으며, 상기 과충전 첨가제는 탄소 화합물을 포함할 수 있으며, 상기 탄소 화합물은 LiCO_3 로 이루어질 수 있다.
- [0019] 상기 캡 다운은 상기 전극 조립체와 전기적으로 연결된 리드부재를 매개로 상기 전극 조립체와 전기적으로 연결되고, 상기 서브 플레이트는 상기 캡 다운을 매개로 상기 전극 조립체와 전기적으로 연결될 수 있다.
- [0020] 그리고 상기 캡 다운과 상기 벤트 플레이트 사이에는 절연판이 설치될 수 있다.
- [0021] 상기 전류 차단부의 전류 차단압은 8 kgf/cm^2 과 9 kgf/cm^2 사이 범위에 속한다.
- [0022] 상기 노치의 벤트 파단압은 20 kgf/cm^2 와 25 kgf/cm^2 사이 범위에 속한다.
- [0023] 상기 양극 활물질은 상기 집전체에 도포되며, 상기 집전체는 알루미늄을 포함할 수 있다.
- [0024] 상기 양극 활물질은 3원계 (NCM; Lithium Nickel-Cobalt-Manganese Oxide, LiNiCoMnO_2) 양극 활물질이 될 수 있다.

효 과

- [0025] 이와 같이 본 발명의 실시예에 따르면 3원계 양극 활물질을 적용한 이차 전지에 있어서 적절한 시기에 전류가 차단되고, 벤트가 파단되어 안전성이 향상된다.

[0026] 또한, 본 발명의 실시예에 따르면, 전류의 차단 및 벤트의 과단이 조기에 발생하는 것을 방지하여 신뢰성이 향상된다.

[0027] 또한, 본 발명의 실시예에 따르면, 양극 활물질에 과충전 첨가제가 포함되어 신뢰성 및 안전성이 더욱 향상된다.

발명의 실시를 위한 구체적인 내용

[0028] 이하, 첨부한 도면을 참조로 본 발명의 실시예에 대해 당업자가 용이하게 실시할 수 있도록 상세히 설명한다. 그러나, 본 발명은 여러가지 상이한 형태로 구현될 수 있으며 여기에서 설명하는 실시예에 한정되는 것은 아니다.

[0029] 도 1은 본 발명의 제1 실시예에 따른 이차 전지를 도시한 절개 사시도이다.

[0030] 도 1을 참조하여 설명하면, 본 실시예에 따른 이차 전지(100)는 양극(112)과 음극(113)이 세퍼레이터(114)를 사이에 두고 위치하는 전극 조립체(110)와, 전해액과 함께 전극 조립체(110)를 수용할 수 있도록 일측 선단에 개구부가 형성된 케이스(120)를 포함한다. 그리고 케이스(120)의 개구부에는 케이스(120)를 밀봉하는 캡 조립체(140)가 개스킷(144)을 매개로 설치된다.

[0031] 보다 구체적으로 설명하면, 상기 케이스(120)는 알루미늄, 알루미늄 합금 또는 니켈이 도금된 스틸과 같은 도전성 금속으로 이루어진다.

[0032] 그리고 본 실시예에 따른 케이스(120)의 형상은 전극 조립체(110)가 위치하는 내부 공간을 가진 원통형으로 이루어진다. 캡 조립체(140)를 케이스(120)에 끼운 후 클램핑하여 고정시키게 되는데, 이 과정에서 케이스(120)에는 비딩부(123)와 크림핑부(125)가 형성된다.

[0033] 본 실시예에 따른 전극 조립체(110)는 양극(112)과 세퍼레이터(114) 및 음극(113)이 적층된 후, 와류상으로 감겨진 원통형(또는 쉘리 롤) 타입으로 이루어지나, 전극 조립체(110)의 구조가 반드시 이것으로 한정되는 것은 아니고 다른 구조로도 이루어질 수 있다.

[0034] 그리고 양극(112)의 상단에는 양극 활물질이 도포되지 않은 양극 무지부(112a)가 형성되어 양극 집전판(138)과 전기적으로 연결된다. 또한, 음극(113)의 하단에는 음극 활물질이 도포되지 않은 음극 무지부(113a)가 형성되어 음극 집전판(132)과 전기적으로 연결된다.

[0035] 음극(113)은 구리 또는 다른 적합한 금속으로 이루어진 집전체에 탄소계 활물질이 도포된 구조로 이루어지고, 양극(112)은 알루미늄 또는 다른 적합한 금속으로 이루어진 집전체에 3원계(NCM; Lithium Nickel-Cobalt-Manganese Oxide, LiNiCoMnO₂) 양극 활물질이 도포된 구조로 이루어진다.

[0036] 캡 조립체(140)는 돌출된 외부 단자(143a)와 배기구(143b)가 형성된 캡 엽(143)과, 캡 엽(143)의 아래에 설치되며 설정된 압력 조건에서 파손되어 가스를 방출할 수 있도록 노치(163)가 형성된 벤트 플레이트(160)를 포함한다. 벤트 플레이트(160)는 설정된 압력 조건에서 전극 조립체(110)와 캡 엽(143)의 전기적 연결을 차단하는 역할을 한다.

[0037] 캡 엽(143)과 벤트 플레이트(160)의 사이에는 양성온도소자(positive temperature coefficient element)(141)가 설치되는 바, 양성온도소자(141)는 일정 온도를 넘으면 전기저항이 거의 무한대까지 커지는 장치로서, 이차 전지(100)가 정해진 값 이상의 온도가 되었을 때, 전류의 흐름을 차단시키는 역할을 한다.

[0038] 벤트 플레이트(160)의 중앙에는 아래로 돌출된 볼록부(165)가 형성되며, 볼록부(165)의 하면에는 서브 플레이트(147)가 용접으로 부착된다.

[0039] 벤트 플레이트(160)와 서브 플레이트(147) 사이에는 캡 다운(146)이 설치되는데, 캡 다운(146)은 원판 형태로 이루어지며 중앙에는 볼록부(165)를 끼움할 수 있도록 홀이 형성된다.

[0040] 캡 다운(146)과 벤트 플레이트(160) 사이에는 절연부재(145)가 설치되어 캡 다운(146)과 벤트 플레이트(160)를 절연하는데, 절연부재(145)에도 캡 엽(143)의 볼록부(165)가 삽입되는 홀이 형성된다.

[0041] 이에 따라 벤트 플레이트(160)의 볼록부(165)가 홀들을 통과하여 서브 플레이트(147)와 용이하게 접합될 수 있다.

[0042] 서브 플레이트(147)는 볼록부(165)와 캡 다운(146)에 각각 용접되며, 캡 다운(146)은 리드부재(150)를 통해서

전극 조립체(110)과 전기적으로 연결된다. 이에 따라, 벤트 플레이트(160)로 용이하게 전류를 전달할 수 있으며, 벤트 플레이트(160)는 캡 업(143)과 접합되어 캡 업(143)의 외부 단자(143a)로 전류를 전달한다.

- [0043] 도 2a 내지 도 2c는 벤트 플레이트가 분리되어 파단되는 과정을 도시한 공정도이다.
- [0044] 도 2a 내지 도 2c를 참조하여 설명하면, 서브 플레이트(147)와 볼록부(165)는 초음파 용접으로 접합되어 서브 플레이트(147)와 볼록부(165) 사이에 전류 차단부(170)가 형성된다. 또한, 서브 플레이트(147)와 캡 다운(146)은 레이저 용접으로 접합되어 서브 플레이트(147)와 캡 다운(146) 사이에 용접부(174, 176)가 형성된다.
- [0045] 충전과 방전을 반복하는 동안 스웰링 현상이 발생하여 이차 전지의 내부 압력이 증가하면, 도 2b에 도시된 바와 같이, 전류 차단부(170)에서 볼록부(165)가 압력으로 서브 플레이트(147)로부터 이탈한다. 볼록부(165)가 서브 플레이트(147)에서 이탈하면 벤트 플레이트(160)와 서브 플레이트(147) 사이의 전류가 차단되는데, 볼록부(165)가 서브 플레이트(147)에서 이탈하는 압력을 전류 차단압이라고 한다.
- [0046] 한편, 도 2c에 도시한 바와 같이 전지의 내부 압력이 더욱 증가하면 벤트 플레이트(160)에 형성된 노치(163)가 파단되어 전지 내부의 가스가 외부로 배출되는데, 노치(163)가 파단되는 압력을 벤트 파단압이라고 한다.
- [0047] 양극 활물질은 코발트계(LCO, LiCoO_2) 양극 활물질과 3원계(NCM; Lithium Nickel-Cobalt-Manganese Oxide, LiNiCoMnO_2) 양극 활물질로 나눌 수 있다. 코발트계 양극 활물질은 안전성이 우수하지만, 코발트 가격의 상승으로 원자재 공급의 어려움과 수익성에 문제가 발생하여 3원계 양극 활물질에 대한 연구가 진행 중이다.
- [0048] 한편, 3원계 양극 활물질의 경우 전해액과의 반응으로 인한 가스 발생 프로파일이 코발트계 활물질과 상이하기 때문에 안전성을 확보하기 위해서는 3원계 양극 활물질이 적용된 전지에 대한 적절한 전류 차단압과 적절한 벤트 파단압을 설정하는 것은 중요하다.
- [0049] 도 3은 만충전 전지를 90℃에 방치했을 때 시간의 경과로 발생한 가스에 따라 코발트계 양극 활물질을 적용한 전지와 3원계 양극 활물질을 적용한 전지의 내부 압력 변화를 측정한 그래프이다. 도 3에 도시된 바와 같이, 코발트계 양극 활물질의 경우, 처음부터 일정 기간 동안은 가스의 발생량이 적으나, 일정 시간이 지난 이후에는 가스가 급격하게 발생하는 것을 알 수 있으며, 3원계 양극 활물질의 경우, 처음에는 가스의 발생량이 많으나, 시간이 지남에 따라 가스가 거의 발생하지 않는 것을 알 수 있다.
- [0050] 이차 전지에서는 신뢰성과 안전성이 모두 확보되어야 하는데, 신뢰성과 안전성은 서로 상반된 관계이다.
- [0051] 안전성을 높이면 신뢰성이 하락하게 되고 신뢰성을 높이면 안전성이 하락할 수 밖에 없다. 예를 들어 신뢰성 평가항목인 만충전 90도 방치 시, 8시간 동안 CID (전류차단장치) 작동 없을 것이라는 스펙을 맞추기 위해서는 전류 차단압을 높게 설정하는 것이 유리하다. 90℃ 방치 항목은 극한 고온 환경에서도 전원이 차단되지 않고 모바일 기기들을 사용하기 위한 신뢰성 항목이다. 하지만 안전성 평가항목인 2C(C-rate)로 단일 과충전에서 발화, 폭발 없을 것이라는 스펙을 맞추기 위해서는 발화, 폭발 전에 과충전 전류를 차단해줘야 하기 때문에 과충전 되면서 발생한 가스에 의해서 작동되는 전류 차단압을 낮게 설정하는 것이 유리하다.
- [0052] 전지 내에서 가스가 발생하는 가장 큰 이유는 전해액과 양극 활물질의 반응 때문에 전해액이 분해되기 때문이다.
- [0053] 본 발명은 하기 실시예를 참조하여 보다 상세하게 설명될 수 있다. 그러나 이들 실시예들은 예시의 목적으로만 이해되어야 하며 본 발명의 범위를 제한하고자 하는 것이 아니다.

예

[0054]

	양극 활물질	LiCoO_2 (wt%)	전류 차단압 (kgf/cm^2)	벤트 파단압 (kgf/cm^2)	제1항목	제2항목	제3항목
					90℃ 방치 (전류차단 작동시간, hr)	2C단열 과충전	150℃ 열노출 (발화시간, min)
비교예1	LCO	0	6	26	OK(15.0)	OK	OK(15)
비교예2	LCO	0	7	30	OK(16.1)	OK	OK(16)
비교예3	NCM	0	6	26	NG(6.3)	NG	NG(7)
비교예4	NCM	0	7	30	NG(7.2)	NG	NG(8)
비교예5	NCM	0	6	20	NG(6.4)	NG	OK(20)
비교예6	NCM	0	7	25	NG(6.5)	NG	OK(22)

비교예7	NCM	0	8	26	OK(10.2)	NG	NG(5)
비교예8	NCM	0	9	30	OK(9.3)	NG	NG(6)
비교예9	NCM	0.2	8	20	OK(8.9)	NG	OK(21)
비교예10	NCM	0.2	9	25	OK(9.5)	NG	OK(19)
비교예11	NCM	0.4	8	20	OK(10.5)	NG	OK(17)
비교예12	NCM	0.4	9	25	OK(11)	NG	OK(18)
비교예13	NCM	2.2	8	20	NG(5.6)	OK	OK(22)
비교예14	NCM	2.2	9	25	NG(5.8)	OK	OK(23)
비교예15	NCM	1.0	11	20	OK(10.2)	NG	OK(23)
비교예16	NCM	1.0	9	18	NG(4.6)	OK	OK(20)
실시예1	NCM	0.5	8	20	OK(12.5)	OK	OK(30)
실시예2	NCM	0.5	9	25	OK(13.6)	OK	OK(31)
실시예3	NCM	1.0	8	20	OK(15.7)	OK	OK(33)
실시예4	NCM	1.0	9	25	OK(16.5)	OK	OK(29)
실시예5	NCM	2.0	8	20	OK(17.1)	OK	OK(30)
실시예6	NCM	2.0	9	25	OK(19.9)	OK	OK(32)

- [0055] 표 1은 양극 활물질과 전류 차단압 및 벤트 파단압에 따른 세가지 항목의 실험 결과를 나타낸다.
- [0056] 제1항목은 1C(2.4Ah), 4.2V로 만충전된 10개의 전지를 90℃로 가열된 오븐에 넣은 후, 전압 변화를 관찰한 것이다. 이때, 8시간 동안 전류가 차단되지 않고, 전압이 4.0V 이상 유지되면 OK로 그렇지 않으면 NG로 분류하였다.
- [0057] 제2항목은 1C(2.4Ah), 4.2V로 만충전된 10개의 전지 외부를 단열재로 감싼 후, 2C, 18V로 3시간 동안 과충전하여 관찰한 것이다. 이때, 하여 전류가 차단되어 발화 또는 파열이 없을 때, OK로 그렇지 않으면 NG로 분류하였다.
- [0058] 제3항목은 온도가 1분에 5℃씩 상승하여 30분 후 150℃에 도달하는 오븐에 1C(2.4Ah), 4.2V로 만충전된 10개의 전지를 넣어 전압 및 온도 변화를 관찰한 것이다. 전지 온도가 150℃에 도달한 후, 10분 내에 벤트가 파단되어 발화 및 파열이 없을 때 OK로 그렇지 않으면 NG로 분류하였다.
- [0059] 제1항목은 신뢰성에 대한 항목이며, 제2, 3항목은 안전성에 대한 항목이다.
- [0060] <비교예>
- [0061] 비교예 1은 코발트계 양극 활물질을 사용한 전지에서 전류 차단압을 6 kgf/cm²으로 설정하고, 벤트 파단압을 26 kgf/cm²으로 설정하였다. 비교예 2는 코발트계 양극 활물질을 사용한 전지에서 전류 차단압을 7 kgf/cm²으로 설정하고, 벤트 파단압을 30 kgf/cm²으로 설정하였다. 표 1을 살펴보면 비교예 1 및 비교예 2는 제1, 2, 3 항목을 모두 만족시키는 것을 알 수 있다.
- [0062] 코발트계 양극 활물질의 경우 가스 발생량이 적어서 상기한 항목들을 모두 만족하였다.
- [0063] 비교예 3은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 6 kgf/cm²으로 설정하고, 벤트 파단압을 26 kgf/cm²으로 설정하였다. 비교예 4는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 7 kgf/cm²으로 설정하고, 벤트 파단압을 30 kgf/cm²으로 설정하였다. 표 1을 살펴보면 비교예 3 및 비교예 4는 모든 항목을 만족시키지 못하는 것을 알 수 있다.
- [0064] 3원계 양극 활물질의 경우 가스 발생 변화가 코발트계 양극 활물질과 달라서 상기한 조건에서 신뢰성 및 안전성을 만족하지 못한다.
- [0065] 비교예 5는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 6 kgf/cm²으로 설정하고, 벤트 파단압을 20 kgf/cm²으로 설정하였다. 비교예 6은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 6 kgf/cm²으로 설정하고, 벤트 파단압을 25 kgf/cm²으로 설정하였다. 3원계 양극 활물질의 경우 방치 특성에 있어서 후반으로 갈수록 가스의 발생량이 적으므로 열노출 시에도 벤트를 파단시킬 정도로 충분한 가스가 발생하지 않기 때문에,

비교예 3, 4에서 안전성 조건을 만족시키지 못한 것으로 판단되어 벤트 파단압을 낮추었다.

- [0066] 비교예 5와 비교예 6의 결과를 표 1에서 살펴보면 벤트 파단압을 낮추어 제3 항목은 만족하였지만, 제1, 2 항목은 만족하지 못한 것을 알 수 있다.
- [0067] 3원계 양극 활물질은 초기에 가스 발생량이 많기 때문에 90℃에 방치했을 때, 코발트계 양극 활물질에 비하여 빨리 전류 차단압에 도달하게 된다. 따라서 3원계 양극 활물질을 사용하는 전지의 경우, 코발트계 양극 활물질에 비하여 전류 차단압을 더 크게 설정하여야만 90℃에서 방치했을 때, 8시간 동안 전류 차단 장치의 작동이 없을 것이라는 제1 항목을 만족시킬 수 있다.
- [0068] 비교예 7은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 8 kgf/cm²으로 설정하고, 벤트 파단압을 26 kgf/cm²으로 설정하였다. 비교예 8은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm²으로 설정하고, 벤트 파단압을 30 kgf/cm²으로 설정하였다.
- [0069] 표 1에 나타난 바와 같이, 비교예 7 및 비교예 8의 경우 전류 차단압을 높여주어 제1 항목은 만족하지만, 제2 항목 및 제3 항목은 만족하지 못하였다.
- [0070] 상기한 비교예들을 바탕으로 3원계 양극 활물질을 적용한 전지에서 제1 항목과 제3 항목은 전류 차단압과 벤트 파단압의 조절을 통해서 만족시킬 수 있음을 알 수 있었지만, 제2 항목을 전류 차단압과 벤트 파단압의 조절로 쉽게 만족시킬 수 없었다.
- [0071] 제2 항목을 만족시키기 위하여 과충전 첨가제를 도입하였다. 과충전 첨가제인 LiCO₃는 과충전 시 전해액과 반응하여 가스를 발생시킬 수 있는 첨가제인 바, LiCO₃의 함량을 조절하여 과충전 시에만 전류 차단압에 도달할 정도의 가스를 발생시킬 수 있다.
- [0072] 비교예 9는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 8 kgf/cm²으로 설정하고, 벤트 파단압을 20 kgf/cm²으로 설정하였으며, 양극 활물질에 LiCO₃를 0.2wt% 첨가하였다. 비교예 10은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm²으로 설정하고, 벤트 파단압을 25 kgf/cm²으로 설정하였으며, 양극 활물질에 LiCO₃를 0.2wt% 첨가하였다.
- [0073] 표 1에 나타난 바와 같이, 비교예 9 및 비교예 10의 경우 차단압 및 벤트 파단압을 높여주어 제1 항목 및 제3 항목은 만족하지만, LiCO₃의 함량이 적어서 제2 항목은 만족하지 못하였다.
- [0074] 비교예 11은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 8 kgf/cm²으로 설정하고, 벤트 파단압을 20 kgf/cm²으로 설정하였으며, 양극 활물질에 LiCO₃를 0.4wt% 첨가하였다. 비교예 12는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm²으로 설정하고, 벤트 파단압을 25 kgf/cm²으로 설정하였으며, 양극 활물질에 LiCO₃를 0.4wt% 첨가하였다.
- [0075] 표 1에 나타난 바와 같이, 비교예 비교예 11 및 비교예 12의 경우 전류 차단압 및 벤트 파단압을 높여주어 제1 항목 및 제3 항목은 만족하지만, LiCO₃의 함량이 적어서 제2 항목은 만족하지 못하였다.
- [0076] 비교예 13은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 8 kgf/cm²으로 설정하고, 벤트 파단압을 20 kgf/cm²으로 설정하였으며, 양극 활물질에 LiCO₃를 2.2wt% 첨가하였다. 비교예 14는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm²으로 설정하고, 벤트 파단압을 25 kgf/cm²으로 설정하였으며, 양극 활물질에 LiCO₃를 2.2wt% 첨가하였다.
- [0077] 표 1에 나타난 바와 같이, 비교예 비교예 13 및 비교예 14의 경우 벤트 파단압을 높이고, LiCO₃의 함량을 증가시켜서 제2 항목 및 제3 항목은 만족하지만, LiCO₃의 함량이 너무 많아서 초기에 전류가 차단되므로 제1 항목은 만족하지 못하였다.

- [0078] 비교예 15는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 11 kgf/cm^2 으로 설정하고, 벤트 파단압을 20 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 1.0wt% 첨가하였다. 표 1에 나타난 바와 같이, 비교예 15는 전류 차단압이 9 kgf/cm^2 보다 더 높은 11 kgf/cm^2 로 설정되어 전류가 제때에 차단되지 못함으로써 발화되는 문제가 발생하여 제2 항목을 만족하지 못하였다.
- [0079] 비교예 16은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm^2 으로 설정하고, 벤트 파단압을 18 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 1.0wt% 첨가하였다. 표 1에 나타난 바와 같이 비교예 16은 벤트 파단압이 20 kgf/cm^2 보다 더 낮은 18 kgf/cm^2 로 설정되어 벤트가 조기에 파단되는 문제가 발생하였다.
- [0080] <실시에>
- [0081] 실시예 1은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 8 kgf/cm^2 으로 설정하고, 벤트 파단압을 20 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 0.5wt% 첨가하였다.
- [0082] 표 1에 나타난 바와 같이, 실시예 1은 전류 파단압 및 벤트 파단압을 높이고, LiCO_3 를 적정하게 포함시켜서 제 1, 2, 3 항목을 만족하였다.
- [0083] 실시예 2은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm^2 으로 설정하고, 벤트 파단압을 25 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 0.5wt% 첨가하였다.
- [0084] 표 1에 나타난 바와 같이, 실시예 2는 전류 파단압 및 벤트 파단압을 높이고, LiCO_3 를 적정하게 포함시켜서 제 1, 2, 3 항목을 만족하였다.
- [0085] 실시예 3은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 8 kgf/cm^2 으로 설정하고, 벤트 파단압을 20 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 1.0wt% 첨가하였다.
- [0086] 표 1에 나타난 바와 같이, 실시예 3은 전류 파단압 및 벤트 파단압을 높이고, LiCO_3 를 적정하게 포함시켜서 제 1, 2, 3 항목을 만족하였다.
- [0087] 실시예 4는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm^2 으로 설정하고, 벤트 파단압을 25 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 1.0wt% 첨가하였다.
- [0088] 표 1에 나타난 바와 같이, 실시예 4는 전류 파단압 및 벤트 파단압을 높이고, LiCO_3 를 적정하게 포함시켜서 제 1, 2, 3 항목을 만족하였다.
- [0089] 실시예 5는 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 8 kgf/cm^2 으로 설정하고, 벤트 파단압을 20 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 2.0wt% 첨가하였다.
- [0090] 표 1에 나타난 바와 같이, 실시예 5는 전류 파단압 및 벤트 파단압을 높이고, LiCO_3 를 적정하게 포함시켜서 제 1, 2, 3 항목을 만족하였다.
- [0091] 실시예 6은 3원계 양극 활물질을 사용한 전지에서 전류 차단압을 9 kgf/cm^2 으로 설정하고, 벤트 파단압을 25 kgf/cm^2 으로 설정하였으며, 양극 활물질에 LiCO_3 를 2.0wt% 첨가하였다.
- [0092] 표 1에 나타난 바와 같이, 실시예 6은 전류 파단압 및 벤트 파단압을 높이고, LiCO_3 를 적정하게 포함시켜서 제 1, 2, 3 항목을 만족하였다.
- [0093] 상기에서 살펴본 바와 같이, 3원계 양극 활물질을 적용한 전지에 있어서, 양극 활물질에 첨가되는 과충전 첨가제인 LiCO_3 의 함량은 0.5wt% 내지 2.0wt%일 때, 가장 효과가 우수한 것을 알 수 있다. 즉, 비교예 11 및 비교

에 12에 나타난 바와 같이 과충전 첨가제가 0.5wt% 보다 더 적게 첨가된 경우에는 가스의 발생량이 적어서 전류가 적당한 시기에 차단되지 못하고 발화되는 문제가 있었으며, 비교예 13 및 비교예 14에 나타난 바와 같이, 과충전 첨가제가 2.0wt% 보다 더 많이 첨가된 경우에는 가스의 발생량이 많아서 신뢰성이 저하되는 문제가 발생하였다.

[0094] 또한, 3원계 양극 활물질을 적용한 전지에 있어서 전류 차단압은 8 kgf/cm^2 내지 9 kgf/cm^2 로 설정될 때, 가장 우수한 효과를 나타내었으며, 벤트 파단압은 20 kgf/cm^2 내지 25 kgf/cm^2 로 설정될 때, 가장 우수한 효과를 나타내었다.

[0095] 즉, 비교예 6에 나타난 바와 같이 전류 차단압이 8 kgf/cm^2 보다 더 작으면 발화 위험이 없음에도 전류가 조기에 차단되어 신뢰성이 저하되는 문제가 있었으며, 비교예 15에 나타난 바와 같이 전류 차단압이 9 kgf/cm^2 보다 더 크면 전류가 적시에 차단되지 못하여 발화 또는 폭발하는 문제가 발생하였다.

[0096] 또한, 비교예 16에 나타난 바와 같이 벤트 파단압이 20 kgf/cm^2 보다 더 작으면 벤트가 조기에 파단되는 문제가 있었으며, 비교예 7에 나타난 바와 같이 벤트 파단압이 25 kgf/cm^2 보다 더 크면 벤트가 적시에 파단되지 못하여 전지가 발화 또는 폭발하는 문제가 발생하였다.

[0097] 다만, 전류 차단압 및 벤트 파단압은 전지의 용량에 따라 달라지게 되는 바, 전지의 용량에 따른 전류 차단압과 벤트 파단압의 관계를 도 4 및 도 5에 나타내었다.

[0098] 도 4는 코발트계 양극 활물질을 적용한 전지의 용량에 따른 전류 차단압과 벤트 파단압의 관계를 나타낸 그래프이다.

표 2

[0099]	용량(Ah)=X	2.2	2.4	2.6	2.8	3.0
	전류 차단압(kgf/cm^2)=A	8	8	9.5	15	15
	벤트 파단압(kgf/cm^2)=B	20	20	20	30	30
	A/B=Y	0.4	0.4	0.48	0.5	0.5

[0100] 표 2는 용량에 따른 전류 차단압과 벤트 파단압의 관계를 나타낸 것으로서, 표 2를 바탕으로 도 4의 그래프를 작성하였다.

[0101] 도 4에 도시된 바와 같이, 코발트계 양극 활물질을 적용한 전지의 경우, 용량이 증가할 수록 전류 차단압을 벤트 파단압으로 나눈 전류 차단압과 벤트 파단압의 비율인 Y 값이 점차 증가하는 것을 알 수 있다.

[0102] 상기한 자료를 바탕으로 용량(X)과 전류 차단압과 벤트 파단압의 비율(Y)의 관계를 정형화하면 아래의 식 1을 도출할 수 있으며, 수학식 1을 도 4에 나타내었다.

수학식 1

[0103]
$$Y = 0.06500 + 0.1500X$$

[0104] 상기한 수학식 1에 나타난 바와 같이 코발트계 양극 활물질의 경우 용량(X)이 증가할 수록 전류 차단압과 벤트 파단압의 비율(Y)도 증가하는 것을 알 수 있다.

[0105] 한편, 도 5는 3원계 양극 활물질을 적용한 전지의 용량에 따른 전류 차단압과 벤트 파단압의 관계를 나타낸 그래프이다.

표 3

[0106]	용량(Ah)=X	2.0	2.0	2.0	2.2	2.2	2.2	2.4	2.4	2.4
	전류 차단압(kgf/cm^2)=A	9.0	9.5	9.5	8	8.5	9.0	8	8	8
	벤트 파단압(kgf/cm^2)=B	21.2	18.1	20	25	18	24.3	21.6	25	30
	A/B=Y	0.425	0.525	0.475	0.32	0.47	0.37	0.37	0.32	0.27

- [0107] 표 3은 용량에 따른 전류 차단압과 벤트 파단압의 관계를 나타낸 것으로서, 표 3을 바탕으로 도 5의 그래프를 작성하였다.
- [0108] 표 3에 기재된 자료를 도 5에서 원형으로 점으로 표시하였다.
- [0109] 도 5에 도시된 바와 같이, 3원계 양극 활물질을 적용한 전지의 경우, 용량이 증가할 수록 전류 차단압을 벤트 파단압으로 나눈 전류 차단압과 벤트 파단압의 비율인 Y 값이 점차 감소하는 것을 알 수 있다.
- [0110] 상기한 자료를 바탕으로 용량(X)과 전류 차단압과 벤트 파단압의 비율(Y)의 관계를 정형화하면 아래의 수학적 2를 도출할 수 있으며, 수학적 2를 도 5에 도식화하여 나타내었다.

수학적 2

- [0111] $Y = 1.22 - 0.39X + D$ (단, $-0.08 \leq D \leq 0.08$)
- [0112] 또는,
- [0113] $Y = 1.224 - 0.3875X + D$ (단, $-0.08 \leq D \leq 0.08$)
- [0114] 도 5에서 실선은 $Y = 1.22 - 0.39X$ 을 나타내며, 절선들은 각각 $Y = 1.22 - 0.39X + 0.08$ 과 $Y = 1.22 - 0.39X - 0.08$ 을 나타낸다.
- [0115] 상기한 수학적 2에 나타난 바와 같이 코발트계 양극 활물질의 경우 용량(X)이 증가할수록 전류 차단압과 벤트 파단압의 비율(Y)은 감소하는 것을 알 수 있다.

표 4

[0116]

용량(Ah)=X	2.0	2.2	2.4
전류 차단압(kgf/cm ²)=A	9.0	7.0	8
벤트 파단압(kgf/cm ²)=B	26	14	19
A/B=Y	0.346	0.5	0.42

- [0117] 표 4는 비교예를 나타낸 것인데, 도 5에서 삼각형의 점으로 표시하였다. 용량이 2.0Ah인 점에서는 벤트가 파단되지 못하여 전지가 폭발하는 문제가 발생하였으며, 용량이 2.2Ah인 점에서는 전류가 조기에 차단되고 벤트가 조기에 파단되어 신뢰성을 확보하지 못하는 문제가 발생하였으며, 용량이 2.4Ah인 점에서는 벤트가 조기에 파단되어 신뢰성을 확보하지 못하는 문제가 발생하였다.
- [0118] 3원계 양극 활물질을 적용한 전지에 있어서 용량에 따른 전류 차단압과 벤트 파단압의 관계가 상기한 수학적 2를 따르는 경우에는 이차 전지의 신뢰성과 안전성을 충분히 확보할 수 있다.
- [0119] 바꿔 말하면, 이차 전지는 신뢰성과 안전성을 모두 확보할 필요가 있다. 그러나 전지의 신뢰성은 안전성과는 반비례 관계에 있다. 또한, 코발트계 양극 활물질은 초기 구동시간 동안에 소량의 가스를 발생시키지만, 소정 시간 이후에는 다량의 가스를 발생시킨다. 반대로, 3원계 양극 활물질은 초기 구동시간 동안에 다량의 가스를 발생시키지만, 시간이 지나면서 가스 발생량은 더 이상 증가하지 않는다.
- [0120] 따라서, 상기한 바와 같이, 본 발명의 실시예는 양극을 포함하는 전극 조립체, 상기 전극 조립체를 내장하는 케이스, 및 상기 케이스에 체결되는 캡 조립체를 구비한 이차 전지를 포함한다. 여기서, 양극은 집전체와 이에 결합된 3원계 양극 활물질을 포함하고, 캡 조립체는 전류 차단부, 및 노치를 갖는 벤트 플레이트를 포함한다. 상기 전류 차단부는 상기 케이스 내에서 전류 차단압 A (kgf/cm²)로 인해 전기적 연결이 차단되도록 구성된다. 노치는 케이스 내에서의 벤트 파단압 B (kgf/cm²)로 인해 벤트 플레이트가 파단되도록 구성된다. 상기 이차 전지는 X (Ah)의 용량을 가지도록 구성되며, 상기 이차 전지는 $A/B = 1.22 - 0.39X + D$ ($-0.88 \leq D \leq 0.08$). 일 실시예에서, 전류 차단부의 전류 차단압은 약 8 kgf/cm²과 약 9 kgf/cm² 사이에 속하며, 벤트 플레이트의 노치의 벤트 파단압은 약 20 kgf/cm²와 약 25 kgf/cm² 사이에 속한다. 또한, 상기 3원계 양극 활물질은 과충전 첨가제를 약 0.5 내지 2.0wt%의 사이 범위의 양을 포함할 수 있다. 상기 과충전 첨가제는 LiCo₃를 포함한다. 이와 같이, 3원

계 양극 활물질이 적용된 이차 전지는 적절한 순간에 전류가 차단되고 벤트가 파단되기 때문에 향상된 안정성과 신뢰성을 가진다. 또한, 신뢰성과 안전성은 양극 활물질에 존재하는 파충진 첨가제로 인해 더욱 향상된다.

[0121] 상기에서는 본 발명의 바람직한 실시예에 대하여 설명하였지만, 본 발명은 이에 한정되는 것이 아니고 특허청구 범위와 발명의 상세한 설명 및 첨부한 도면의 범위 안에서 여러 가지로 변형하여 실시하는 것이 가능하고 이 또한 본 발명의 범위에 속하는 것은 당연하다.

도면의 간단한 설명

[0122] 도 1은 본 발명의 제1 실시예에 따른 이차 전지를 도시한 절개 사시도이다.

[0123] 도 2a 내지 도 2c는 본 발명의 제1 실시예에 이차 전지의 전류가 차단되고 벤트가 파단되는 과정을 도시한 도면이다.

[0124] 도 3은 만충전 전지를 90℃에 방치했을 때 시간의 경과로 발생한 가스에 따라 코발트계 양극 활물질을 적용한 전지와 3원계 양극 활물질을 적용한 전지의 내부 압력 변화를 측정한 그래프이다.

[0125] 도 4는 코발트계 양극 활물질을 적용한 전지의 용량에 따른 전류 차단압과 벤트 판단압의 관계를 나타낸 그래프이다.

[0126] 도 5는 3원계 양극 활물질질을 적용한 전지의 용량에 따른 전류 차단압과 벤트 판단압의 관계를 나타낸 그래프이다

[0127] <도면의 주요부분에 대한 참조 부호의 설명>

[0128] 100: 이차 전지 110: 전극 조립체

[0129] 112: 양극 113: 음극

[0130] 114: 세퍼레이터 120: 케이스

[0131] 140: 캡 조립체 141: 양성온도소자

[0132] 143: 캡 업 145: 절연부재

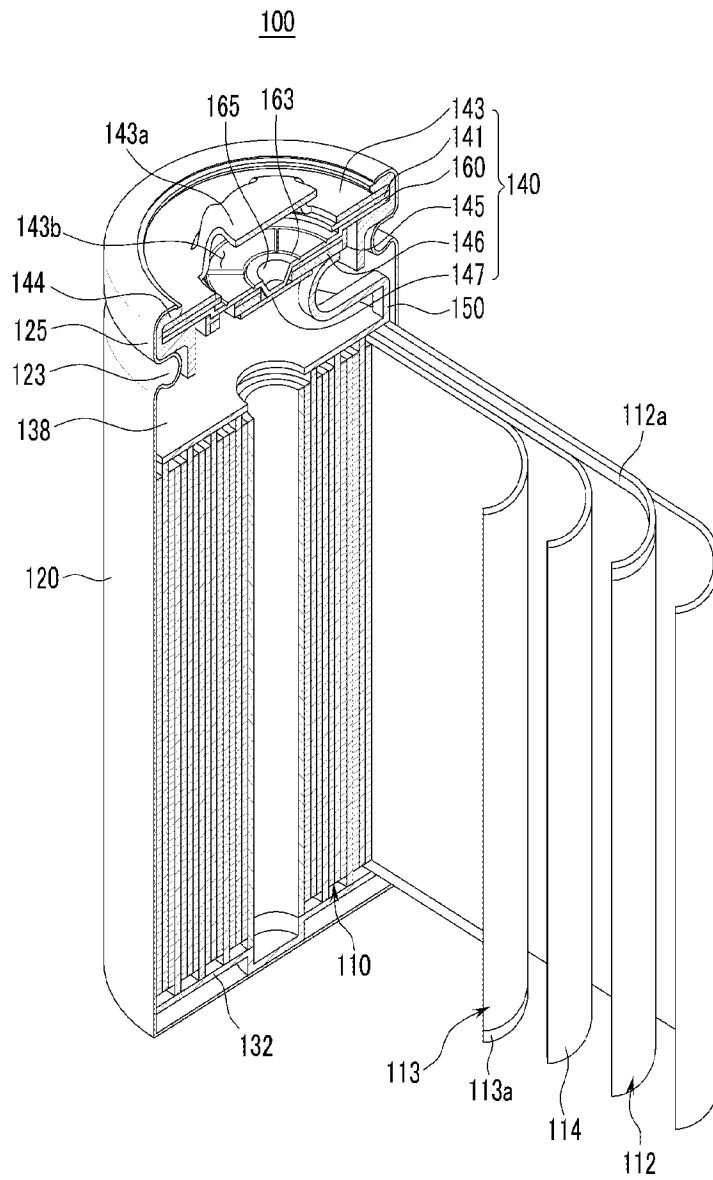
[0133] 146: 컵 다운 147: 서브 플레이트

[0134] 160: 벤트 플레이트 163: 노치

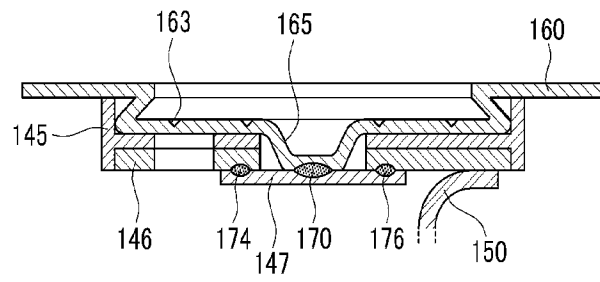
[0135] 165: 불록부 170: 전류 차단부

도면

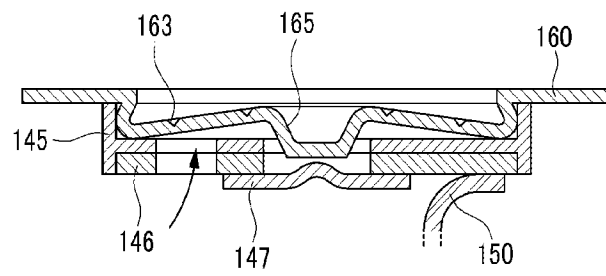
도면



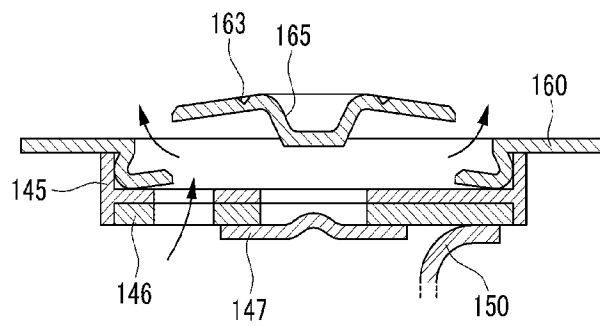
도면2a



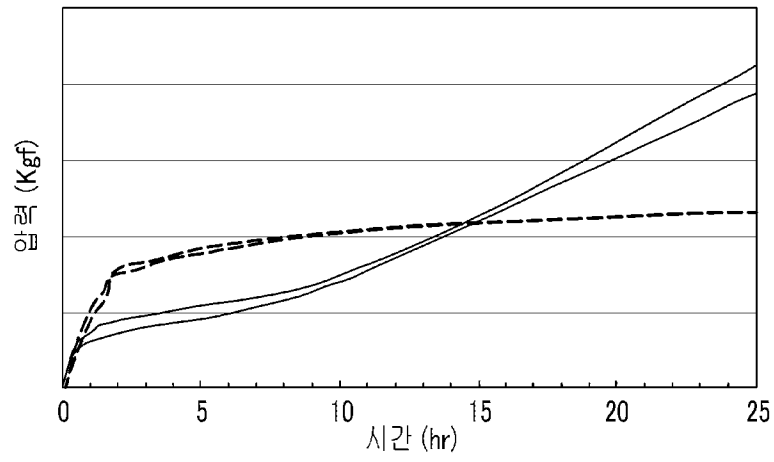
도면2b



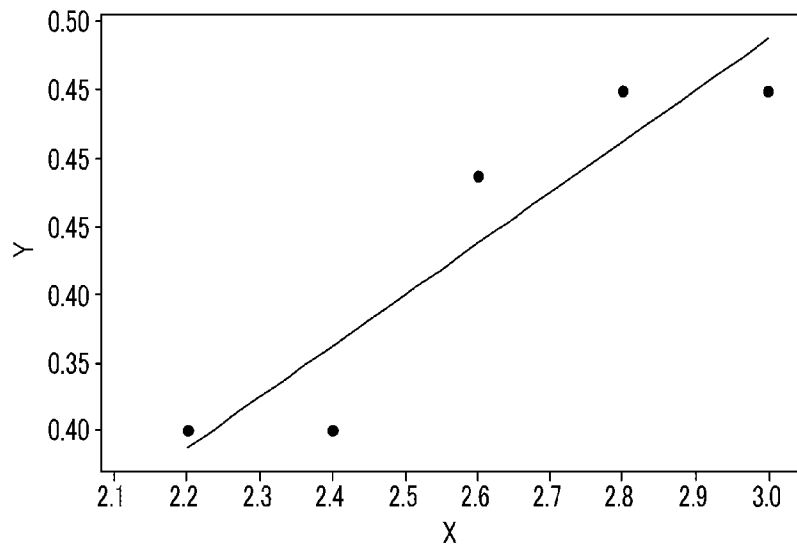
도면2c



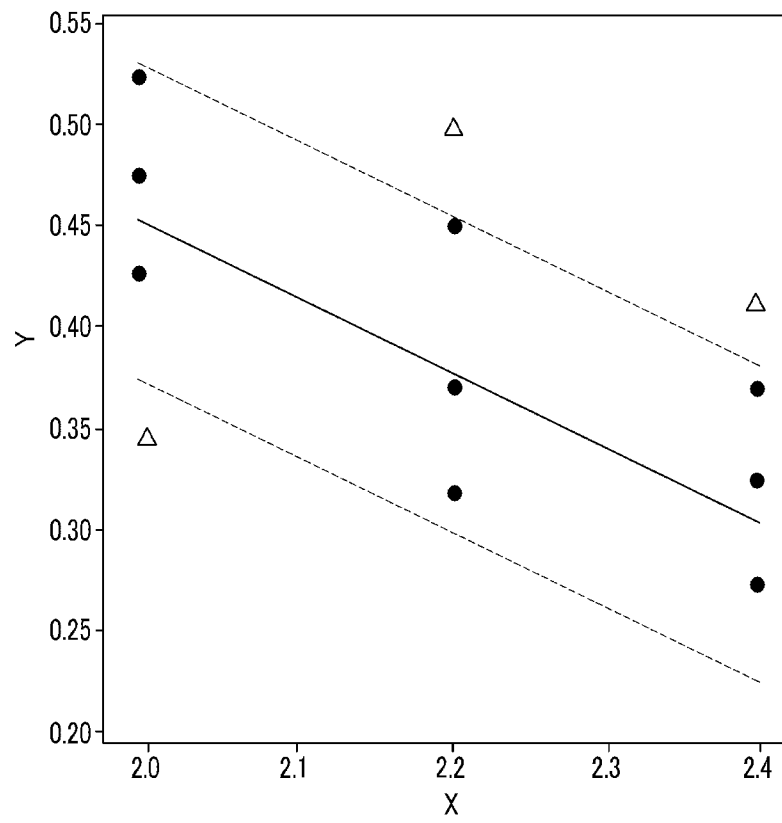
도면3

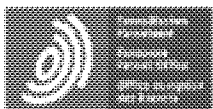


도면4



도 25





Espacenet

Bibliographic data: KR20110095118 (A) — 2011-08-24

RECHARGEABLE BATTERY

Inventor(s): BAK HYO RIM [KR]; PARK JUNE HYOUNG [KR] ± (BAK, HYO RIM, ; PARK, JUNE HYOUNG)

Applicant(s): SAMSUNG SDI CO LTD [KR] ± (SAMSUNG SDI CO., LTD)

Classification: - international: H01M2/04; H01M2/12
- cooperative: H01M2/1241 (EP); H01M10/02 (EP)

Application number: KR20100122706 20101203 Global Dossier

Priority number(s): US20100957314 20101130 ; US20100305906P 20100218

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EP2362465 (A3) JP2011171299 (A) JP5398756 (B2)
KR101201812 (B1) US2011200854 (A1) US8597809 (B2) less

Abstract of KR20110095118 (A)

PURPOSE: A rechargeable battery is provided to improve an aperture ratio in a vent member, to stably secure the exhaust path of gas, and to enhance the safety of the rechargeable battery. **CONSTITUTION:** A rechargeable battery(100) comprises: an electrode assembly(110); a case(120) for housing the electrode assembly; and a cap assembly(140) including a head, a cap plate vent members(160), and vent portions having two or more notches and support portions. The vent member is bent along two or more lines adjacent to the support portion by being broken in two or more notches.



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(71) 출원인

삼성에스디아이 주식회사

경기 용인시 기흥구 공세동 428-5

(72) 발명자

박효림

경기 용인시 기흥구 공세동 428-5

박준형

경기 과천시 별양동 주공아파트 501동 1201호

(74) 대리인

팬코리아특허법인

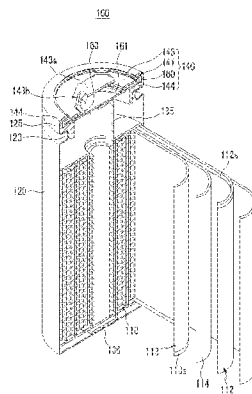
전체 청구항 수 : 총 20 항

(54) 이차 전지

(57) 요약

본 발명은 기스를 신속하게 배출함으로써 안전성이 향상된 이차 전지에 관한 것이다. 본 발명의 일 실시예에 따른 이차 전지는 전극 조립체, 상기 전극 조립체를 수용하는 케이스, 및 헤드(head)와 적어도 하나의 배기구를 갖는 캡 플레이트 및 두 개 이상의 노치(notch)와 지지부를 갖는 벤트 부재를 포함하는 캡 조립체를 포함한다. 상기 벤트 부재는 상기 두 개 이상의 노치에서 파단되어 상기 지지부에 연결한 두 개 이상의 라인을 따라 절곡된다.

도 1



특허청구의 범위

청구항 1

선극 조립체;

상기 선극 조립체를 수용하는 케이스; 및

헤드(head)와 적어도 하나의 배기구를 갖는 캡 플레이트 및 두 개 이상의 노치(notch)와 지지부를 갖는 벤트 부재를 포함하는 캡 조립체;

를 포함하고,

상기 벤트 부재는 상기 두 개 이상의 노치에서 파단되어 상기 지지부에 인접한 두 개 이상의 라인을 따라 절곡되는, 이차 전지.

청구항 2

제1항에 있어서,

상기 지지부는 상기 노치와 상기 라인(line) 사이에 형성되는, 이차 전지.

청구항 3

제1항에 있어서,

상기 지지부의 적어도 일부분은 상기 캡 플레이트의 상기 헤드의 아래에 위치하는, 이차 전지.

청구항 4

제1항에 있어서,

적어도 하나의 상기 노치가 파단되어 적어도 하나의 상기 라인을 따라 절곡될 때 상기 벤트 부재는 적어도 하나의 플랩(flap)을 형성하는, 이차 전지.

청구항 5

제4항에 있어서,

상기 적어도 하나의 플랩은 상기 캡 플레이트의 상기 헤드의 가장자리부와 접촉하는 에지(edge)를 포함하는, 이차 전지.

청구항 6

제4항에 있어서,

상기 적어도 하나의 플랩은 상기 캡 플레이트의 헤드의 바닥면과 접촉하는 에지(edge)를 포함하는, 이차 전지.

청구항 7

제4항에 있어서,

상기 적어도 하나의 플랩에서, 적어도 하나의 상기 라인과 적어도 하나의 상기 노치 사이의 길이는 적어도 하나의 상기 라인과 상기 캡 플레이트의 상기 헤드의 가장자리부 사이의 거리와 동일한, 이차 전지.

청구항 8

제4항에 있어서,

상기 적어도 하나의 플랩에서, 적어도 하나의 상기 라인과 적어도 하나의 상기 노치 사이의 길이는 적어도 하나의 상기 라인과 상기 캡 플레이트의 상기 헤드의 가장자리부 사이의 거리보다 작은, 이차 전지.

청구항 9

제8항에 있어서,

적어도 하나의 상기 라인과 적어도 하나의 상기 노치 사이의 상기 길이는 상기 벤트 부재와 상기 캡 플레이트의 상기 헤드 사이의 높이와 같거나 그보다 큰, 이차 전지.

청구항 10

제4항에 있어서,

상기 적어도 하나의 플랩은 상기 벤트 부재에 대하여 30° 내지 90° 의 각도로 절곡되도록 형성되는, 이차 전지.

청구항 11

제1항에 있어서,

상기 두 개 이상의 노치는 상기 벤트 부재의 원주 방향을 따라 곡선으로 형성되는, 이차 전지.

청구항 12

제11항에 있어서,

상기 지지부는 상기 두 개 이상의 노치 사이에서 상기 벤트 부재의 중앙부에 배치되는 연장부로 형성되는, 이차 전지.

청구항 13

제11항에 있어서,

상기 노치는 세 개 이상으로 형성되어 상기 벤트 부재의 원주 방향을 따라 곡선으로 형성되고,
상기 지지부는 상기 세 개 이상의 노치 사이에서 상기 벤트 부재의 중앙부에 배치되는, 이차 전지.

청구항 14

제13항에 있어서,

상기 세 개 이상의 노치와 상기 지지부 사이의 각각의 거리는 동일한, 이차 전지.

청구항 15

제3항에 있어서,

상기 두 개 이상의 라인은 상기 캡 플레이트의 상기 헤드의 길이와 같거나 그보다 작은 길이를 갖는, 이차 전지.

청구항 16

제1항에 있어서,

상기 벤트 부재는 상기 적어도 하나의 배기구통을 통해 상기 전극 조립체로부터 가스를 방출하도록 형성되는, 이차 전지.

청구항 17

제1항에 있어서,

상기 전극 조립체는 양극 및 음극을 포함하고,

상기 양극 및 상기 음극 중 어느 하나는 상기 벤트 부재와 전기적으로 연결되고, 상기 양극 및 상기 음극 중 더 큰 하나는 상기 케이스의 바닥과 전기적으로 연결되는, 이차 전지.

청구항 18

전극 조립체;

상기 전극 조립체를 수용하는 케이스; 및

복수의 노치를 갖는 하부 플레이트를 포함하는 캡 조립체;

를 포함하고,

상기 하부 플레이트는 상기 노치에서 파단되어 상기 노치 사이의 중심부에 위치하는 라인을 따라 절곡되는, 이차 전지.

청구항 19

제18항에 있어서,

상기 노치는 상기 하부 플레이트의 원주 방향을 따라 곡선으로 형성되는, 이차 전지.

청구항 20

제18항에 있어서,

상기 하부 플레이트는 상기 노치에서 파단되고 상기 라인을 따라 절곡되어 복수의 플랩을 형성하고, 상기 플랩은 인접하는 플랩이 절곡되는 각도를 제한하는, 이차 전지.

발명 상세

기술 분야

[0001] 본 발명은 이차 전지에 관한 것으로서, 보다 상세하게는 펌프 부재의 구조를 개선한 이차 전지에 관한 것이다.

배경 기술

[0002] 이차 전지(rechargeable battery)는 충전이 불가능한 일차전지와는 달리 충전 및 방전이 가능한 전지이다. 저용량의 이차 전지는 휴대폰이나 노트북 컴퓨터 및 캠코더와 같이 휴대가 가능한 소형 전자기기에 사용되고, 대용량 전지는 하이브리드 자동차 등의 모터 구동용 전원 또는 대용량 전력 저장 장치로 사용되고 있다.

[0003] 최근 들어 고에너지 밀도의 비수전해액을 이용한 고출력 이차 전지가 개발되고 있으며, 이러한 고출력 이차 전지는 대전력을 필요로 하는 기기 예컨대, 전기 자동차 등의 모터 구동에 사용될 수 있도록 복수 개의 이차 전지를 직렬로 연결하여 대용량의 전지 모듈로 구성된다. 이러한 이차 전지는 원통형과 각형 등으로 이루어질 수 있다.

[0004] 이차 전지가 충전과 방전을 되풀이하는 동안, 이차 전지의 내부에는 가스가 발생하여 내부 압력이 상승한다. 이차 전지 내부의 압력 상승을 그대로 방치하면, 이차 전지가 폭발할 위험이 있다.

발명의 내용

해결하려는 과제

[0005] 따라서 본 발명은 상기한 바와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 목적은 가스를 신속하게 배출함으로써 안전성이 향상된 이차 전지를 제공함에 있다.

과제의 해결 수단

[0006] 본 발명의 일 실시예에 따른 이차 전지는 전극 조립체, 상기 전극 조립체를 수용하는 케이스, 및 헤드(head)와 적어도 하나의 배기구를 갖는 캡 플레이트 및 두 개 이상의 노치(notch)와 지지부를 갖는 펌프 부재를 포함하는 캡 조립체를 포함한다. 상기 펌프 부재는 상기 두 개 이상의 노치에서 파단되어 상기 지지부에 인접한 두 개 이상의 라인을 따라 절곡된다.

[0007] 상기 지지부는 상기 노치와 상기 라인(line) 사이에 형성될 수 있다.

- [0008] 상기 지지부의 적어도 일부는 상기 캡 플레이트의 상기 헤드의 아래에 위치할 수 있다.
- [0009] 적어도 하나의 상기 노치가 파단되어 적어도 하나의 상기 라인을 따라 절곡될 때 상기 벤트 부재는 적어도 하나의 플랩(flap)을 형성할 수 있다.
- [0010] 상기 적어도 하나의 플랩은 상기 캡 플레이트의 상기 헤드의 가장자리부와 접촉하는 에지(edge)를 포함할 수 있다.
- [0011] 상기 적어도 하나의 플랩은 상기 캡 플레이트의 헤드의 바닥면과 접촉하는 에지(edge)를 포함할 수 있다.
- [0012] 상기 적어도 하나의 플랩에서, 적어도 하나의 상기 라인과 적어도 하나의 상기 노치 사이의 길이는 적어도 하나의 상기 라인과 상기 캡 플레이트의 상기 헤드의 가장자리부 사이의 거리와 동일할 수 있다.
- [0013] 상기 적어도 하나의 플랩에서, 적어도 하나의 상기 라인과 적어도 하나의 상기 노치 사이의 길이는 적어도 하나의 상기 라인과 상기 캡 플레이트의 상기 헤드의 가장자리부 사이의 거리보다 작을 수 있다.
- [0014] 적어도 하나의 상기 라인과 적어도 하나의 상기 노치 사이의 상기 길이는 상기 벤트 부재와 상기 캡 플레이트의 상기 헤드 사이의 높이와 같거나 그보다 클 수 있다.
- [0015] 상기 적어도 하나의 플랩은 상기 벤트 부재에 대하여 30° 내지 90° 의 각도로 절곡되도록 형성될 수 있다.
- [0016] 상기 두 개 이상의 노치는 상기 벤트 부재의 원주 방향을 따라 곡선으로 형성될 수 있다.
- [0017] 상기 지지부는 상기 두 개 이상의 노치 사이에서 상기 벤트 부재의 중앙부에 배치되는 연장부로 형성될 수 있다.
- [0018] 상기 노치는 세 개 이상으로 형성되어 상기 벤트 부재의 원주 방향을 따라 곡선으로 형성될 수 있고, 상기 지지부는 상기 세 개 이상의 노치 사이에서 상기 벤트 부재의 중앙부에 배치될 수 있다.
- [0019] 상기 세 개 이상의 노치와 상기 지지부 사이의 각각의 거리는 동일할 수 있다.
- [0020] 상기 두 개 이상의 라인은 상기 캡 플레이트의 상기 헤드의 길이와 같거나 그보다 작은 길이를 가질 수 있다.
- [0021] 상기 벤트 부재는 상기 적어도 하나의 배기구를 통해 상기 전극 조립체로부터 가스를 방출하도록 형성될 수 있다.
- [0022] 상기 전극 조립체는 양극 및 음극을 포함할 수 있고, 상기 양극 및 상기 음극 중 어느 하나는 상기 벤트 부재와 전기적으로 연결될 수 있고, 상기 양극 및 상기 음극 중 다른 하나는 상기 케이스의 바닥과 전기적으로 연결될 수 있다.
- [0023] 본 발명의 다른 실시예에 따른 이차 전지는 전극 조립체, 상기 전극 조립체를 수용하는 케이스 및 복수의 노치를 갖는 하부 플레이트를 포함하는 캡 조립체를 포함한다. 상기 하부 플레이트는 상기 노치에서 파단되어 상기 노치 사이의 중심부에 위치하는 라인을 따라 절곡된다.
- [0024] 상기 노치는 상기 하부 플레이트의 원주 방향을 따라 곡선으로 형성될 수 있다.
- [0025] 상기 하부 플레이트는 상기 노치에서 파단되고 상기 라인을 따라 절곡되어 복수의 플랩을 형성할 수 있고, 상기 플랩은 인접하는 플랩이 절곡되는 각도를 제한할 수 있다.

발명의 효과

- [0026] 본 발명의 일 실시예에 따르면, 벤트 부재에서 개구율이 향상되고 가스의 배출통로를 안정적으로 확보하여 이차 전지의 안전성이 향상된다.

도면의 간단한 설명

- [0027] 도 1은 본 발명의 제1 실시예에 따른 이차 전지를 도시한 절개 사시도이다.
- 도 2a는 본 발명의 제1 실시예에 따른 이차 전지의 벤트 부재를 도시한 사시도이다.
- 도 2b는 도 2a에 도시된 벤트 부재의 노치가 찢어져 개방된 상태를 도시한 사시도이다.
- 도 2c는 본 발명의 제1 실시예에 따른 이차 전지의 벤트 부재와 캡 플레이트의 관계를 나타낸 평면도이다.

도 3는 본 발명의 제1 실시예에 따른 이차 전지의 부분 단면도이다.

도 4는 도 3의 밴드 부재의 노치가 찢어져 개방된 상태를 도시한 부분 단면도이다.

도 5a는 본 발명의 제2 실시예에 따른 이차 전지를 도시한 부분 단면도이다.

도 5b는 도 5a의 밴드 부재의 노치가 찢어져 개방된 상태를 도시한 부분 단면도이다.

도 6a은 본 발명의 제3 실시예에 따른 이차 전지의 밴드 부재를 도시한 사시도이다.

도 6b는 도 6a에 도시된 밴드 부재의 노치가 찢어져 개방된 상태를 도시한 사시도이다.

도 7은 본 발명의 제3 실시예에 따른 이차 전지를 도시한 부분 단면도이다.

발명을 실시하기 위한 구체적인 내용

- [0028] 이하, 첨부한 도면을 참고로 하여 본 발명이 속하는 기술 분야에서 통상의 지식을 가진 자가 용이하게 실시할 수 있도록 본 발명의 실시예를 상세히 설명한다. 그러나, 본 발명은 여러 가지 상이한 형태로 구현될 수 있으며 이하에서 설명하는 실시예에 한정되지 않는다. 그리고, 본 명세서 및 도면에서 동일한 부호는 동일한 구성 요소를 나타낸다.
- [0029] 도 1은 본 발명의 일 실시예에 따른 이차 전지를 도시한 절개 사시도이다.
- [0030] 도 1을 참조하여 설명하면, 본 실시예에 따른 이차 전지(100)는 양극(112)과 음극(113)이 세퍼레이터(114)를 사이에 두고 위치하는 전극 조립체(110)와, 전해액과 전극 조립체(110)를 수용할 수 있도록 일측 선단이 개구부가 형성된 케이스(120)를 포함한다. 그리고, 케이스(120)의 개구부에는 케이스(120)를 밀봉하는 캡 조립체(140)가 설치된다.
- [0031] 보다 구체적으로 설명하면, 케이스(120)는 알루미늄, 알루미늄 합금 또는 니켈이 도금된 스텔과 같은 도전성 금속으로 이루어질 수 있다. 그리고, 본 실시예에 따른 케이스(120)의 형상은 전극 조립체(110)이 위치하는 내부 공간을 가진 원통형으로 이루어질 수 있다. 캡 조립체(140)를 케이스(120)에 끼운 후, 클램핑하여 캡 조립체(140)를 케이스(120)에 고정시킬 수 있는데, 이 과정에서 케이스(120)에는 비딩부(123)와 클램핑부(125)가 형성될 수 있다.
- [0032] 본 실시예에 따른 전극 조립체(110)는 양극(112)과 세퍼레이터(114) 및 음극(113)이 적층된 후, 와류상으로 감겨진 원통형 타입으로 이루어질 수 있으나, 전극 조립체(110)의 구조가 반드시 이것으로 한정되는 것은 아니고 다른 구조로도 이루어질 수 있다. 양극(112), 음극(113), 및 세퍼레이터(114)는 일방향으로 이어진 띠 형상으로 이루어질 수 있다.
- [0033] 양극(112)은 양극 집전체와 양극 집전체의 양면에 형성된 양극 활물질층을 포함할 수 있고, 음극(113)은 음극 집전체와 음극 집전체의 양면에 형성된 음극 활물질층을 포함할 수 있다. 세퍼레이터(114)는 양극(112)과 음극(113) 사이에 개재되어 양극(112)과 음극(113)을 절연하며 이온이 이동하는 통로를 제공할 수 있다.
- [0034] 그리고 양극(112)의 상단에는 양극 활물질층이 형성되지 않은 양극 무지부(112a)가 형성될 수 있고, 음극(113)의 하단에는 음극 활물질층이 형성되지 않은 음극 무지부(113a)가 형성될 수 있다. 본 실시예에서는 양극 무지부(112a) 및 음극 무지부(113a)가 전극 조립체(110)의 상단과 하단에 각각 형성된 것으로 예시하고 있으나, 본 발명이 이에 제한되는 것은 아니다. 양극 무지부(112a)는 양극(112)의 길이 방향 일측 단부에 형성되고 음극 무지부(113a)는 음극(113)의 길이방향 타측 단부에 형성될 수도 있다.
- [0035] 캡 조립체(140)는 캡 플레이트(143)와, 캡 플레이트(143)의 아래에 설치된 밴드 부재(160)와 캡 플레이트(143)를 감싸면서 케이스(120)와 접하도록 배치된 가스켓(144), 및 캡 조립체(140)와 밴드 부재(160) 사이에 배치된 안전 플레이트(141)를 포함할 수 있다.
- [0036] 캡 플레이트(143)에는 상부로 돌출된 헤드(143a)와 관통된 배기구(143b)가 형성될 수 있고, 밴드 부재(160)에는 설정된 압력 조건에서 찢어지는 노치(161, 163)가 형성될 수 있다. 캡 플레이트(143)와 밴드 부재(160) 사이에는 캡 플레이트(143)와 밴드 부재(160)를 전기적으로 연결하는 안전 플레이트(141)가 설치될 수 있다. 안전 플레이트(141)는 양성온도소자(positive temperature coefficient element)로 이루어질 수 있는데, 양성온도소자는 일정 온도를 넘으면 전기저항이 거의 무한대까지 커지는 소자로서, 이차 전지(100)가 정해진 값 이상의 온도를 초과할 때 충전 및 방전 전류의 흐름을 차단시키는 역할을 한다.

- [0037] 가스켓(144)은 캡 플레이트(143), 안전 플레이트(141), 밴드 부재(160)의 둘레를 감싸도록 설치되어, 캡 조립체(140)를 케이스(120)와 절연할 수 있다.
- [0038] 양극(112)은 양극 집전 탭(135)을 매개로 밴드 부재(160)와 전기적으로 연결될 수 있고, 음극(113)은 음극 집전 탭(136)을 매개로 케이스(120)의 바닥과 전기적으로 연결될 수 있다. 본 실시예에서는 양극 집전 탭(135)이 밴드 부재(160)에 직접 연결된 것으로 예시하고 있으나, 본 발명이 이에 제한되는 것은 아니며, 밴드 부재(160)에 중간부재가 설치되어 밴드 부재(160)와 양극 집전 탭(135)이 중간부재를 매개로 연결될 수도 있다.
- [0039] 도 2a는 본 발명의 제1 실시예에 따른 밴드 부재를 도시한 사시도이고, 도 2b는 도 2a에 도시된 밴드 부재의 노치가 찢어져 개방된 상태를 도시한 사시도이며, 도 2c는 본 발명의 제1 실시예에 따른 아차 전지의 밴드 부재와 캡 플레이트의 관계를 나타내는 평면도이다.
- [0040] 도 2a, 도 2b 및 도 2c를 참조하여 설명하면, 본 실시예에 따른 밴드 부재(160)에는 제1 노치(161)와 제2 노치(163)가 형성되는데, 노치들(161, 163)은 각각 호형의 선분으로 이루어질 수 있다. 본 실시예에서는 노치(161, 163)가 호형상으로 이루어진 것으로 예시하고 있으나 본 발명이 이에 제한되는 것은 아니며, 노치는 삼각형, 사각형 등 다양한 형태로 이루어질 수 있다. 본 실시예에 따른 제1 노치(161) 및 제2 노치(163)는 만원형으로 이루어지며, 제1 노치(161)와 제2 노치(163)는 서로 이격 배치된다. 각 노치(161, 163)의 양단을 연결하는 절곡 라인(164) 사이에는 지지부(165)가 형성될 수 있는데, 지지부(165)는 노치들(161, 163)이 찢어져 형성된 개구(162)를 분리하는 역할을 한다. 제1 노치(161)와 제2 노치(163)는 지지부(165)를 사이에 두고 대칭된 형태로 이루어질 수 있다.
- [0041] 노치(161, 163)는 절곡 라인(164)에서 밴드 부재(160)의 외측을 향하여 볼록하게 돌출된 선 형태로 이루어질 수 있고, 제1 노치(161)와 지지부(165) 사이에는 제1 노치(161)의 파단 시에 절곡되어 상승하는 제1 파단 조각(167) 위치할 수 있고, 제2 노치(163)와 지지부(165) 사이에는 제2 노치(163)의 파단 시에 절곡되어 상승하는 제2 파단 조각(168)이 위치할 수 있다. 한편, 본 실시예에서 이들 파단 조각(167, 168)은 플랩(flap)으로 형성될 수 있다.
- [0042] 제2 노치는 제1 노치와 동일한 구조로 이루어지므로, 제1 노치에 대한 설명으로 제2 노치에 대한 설명을 대체한다.
- [0043] 도 2c를 참조하면, 지지부(165)는 캡 플레이트(143)의 헤드(143a)의 아래에 위치하고, 이를 초과하여 연장되지 않는다. 즉, 상부에서 바라볼 때 절곡 라인(164)과 지지부(165)는 헤드(143a)의 안쪽에 위치한다.
- [0044] 만약 절곡 라인(164)이 헤드(143a)의 안쪽에 위치하지 않으면, 밴드 부재(160)가 개방될 때 헤드(143a)의 외곽에 위치한 지지부(165)의 일부분이 배기구(143b)로 향하는 통로를 차단하게 된다. 하지만 본 실시예의 절곡 라인(164)은 헤드(143a)의 안쪽에 위치하기 때문에 지지부(165)는 배기구(143b)로 향하는 통로를 차단하지 않게 되고, 제1 및 제2 파단 조각(167, 168)은 가스가 개구(162)에서 배기구(143b)로 배출될 수 있도록 안내한다.
- [0045] 도 3에 도시된 바와 같이, 절곡 라인(164)에서 제1 노치(161)까지의 최대 거리를 장폭(L2)이라 할 때, 본 실시예에 따른 제1 노치(161)의 장폭(L2)은 절곡 라인(164)에서 캡 플레이트(143)에 형성된 배기구(143b)의 상단, 즉 헤드(143a)의 가장자리부까지의 거리(L3)와 동일한 거리로 이루어질 수 있다. 이에 따라, 도 4에 도시된 바와 같이 아차 전지의 내부 압력이 상승하여 제1 노치(161)가 찢어질 경우, 제1 파단 조각(167)의 상단이 배기구(143b)의 내측단과 맞닿아 가스가 이동하는 통로를 충분히 확보할 수 있다. 다만, 본 발명이 이에 제한되는 것은 아니며, 장폭(L2)은 절곡 라인(164)에서 배기구(143b)의 상단까지의 거리(L3)보다 더 작게 형성될 수 있다.
- [0046] 또한, 노치(161, 163)가 지지부(165)를 사이에 두고 이격 형성되므로 노치(161, 163)가 파단될 때, 각 노치(161, 163)가 서로 간섭하지 않고 용이하게 파단될 수 있으며, 개구(162)가 2곳 이상 형성될 수 있다.
- [0047] 또한, 제1 노치(161)가 절곡 라인(164)에 대하여 밴드 부재(160)의 외측을 향하여 볼록하도록 형성되므로 제1 노치(161)가 파단되어 제1 파단 조각(167)이 상승하면, 제1 파단 조각(167)은 제1 노치(161)의 파단으로 형성된 개구(162)와 배기구(143b) 사이에 위치하지 않고, 배기구(143b)와 개구(162)의 옆쪽에 위치하게 된다. 즉, 도면을 살펴보면, 제1 파단 조각(167)의 하단은 개구(162)의 내측단에 접할 수 있고, 제1 파단 조각(167)의 상단, 즉 제1 파단 조각(167)의 에지(edge)도 배기구(143b)의 내측단에 접할 수 있음을 알 수 있다. 이 때, 내측이라 함은 밴드 부재(160)의 중앙을 기준으로 안쪽에 위치하는 것을 의미한다. 이에 따라, 개구(162)를 통해서 방출된 가스는 제1 파단 조각(167)의 안내를 받아 배기구(143b)로 이동할 수 있어, 보다 신속하게 가스가 배출될 수 있다. 그러나, 제1 노치(161)가 절곡 라인(164)보다 내측에 위치하고, 배기구(143b)가 캡 플레이트(143)의 중

심에서 이격되어 원주 방향으로 배치된 경우에는, 개구(162)와 배기구(143b) 사이에 제1 파단 조각(167)이 위치하게 되므로 제1 파단 조각(167)이 가스의 진행을 방해하는 문제가 발생한다.

- [0048] 내압의 상승으로 제1 노치(161)가 파단되면, 제1 파단 조각(167)은 절곡 라인(164)에서 지지부(165)에 대하여 30° 내지 90° 의 경사로 절곡될 수 있다. 제1 파단 조각(167)의 절곡 각도가 30° 보다 더 작으면 이차 전지 내부의 가스가 신속하게 배출되지 못하는 문제점이 있으며, 제1 파단 조각(167)의 절곡 각도가 90° 보다 더 크면 제1 파단 조각(167)의 상단과 인접한 영역에서 와류가 발생하여 가스의 배출 속도가 저하되는 문제가 있다. 즉, 제1 파단 조각(167)의 굽힘 경사가 90° 보다 더 크면 가스가 배출되는 방향과 역방향으로 진입하여 역류하는 현상이 발생할 수 있고, 이에 따라 역류하는 가스로 인하여 가스의 배출이 지연될 수 있다.
- [0049] 가스의 배출 속도는 이차 전지의 안전성 측면에서 매우 중요한데, 가스의 배출 속도가 지연되면 조건에 따라서 이차 전지가 폭발하는 심각한 문제가 발생할 수 있다. 가스의 배출은 급박한 상황에서 짧은 시간 동안 이루어지는 바, 더욱 짧은 시간 동안 신속하게 가스가 배출되어야 안전성을 향상시킬 수 있다.
- [0050] 10개의 이차 전지가 포함된 전지 모듈을 화염의 상황에 두고 폭발하는지 여부를 실험하였다. 반원보다 큰 하나의 노치가 형성된 비교예의 이차 전지는 4개의 전지가 폭발하고, 6개의 전지가 발화하였으나, 본 실시예에 따른 이차 전지는 10개 모두 발화만하고 폭발이 발생하지 않았다. 상기한 실험을 통해서, 화재 현장이나 매우 온도가 높은 이상 상황에서도 본 실시예에 따른 이차 전지는 폭발하지 않음을 확인할 수 있었다. 이는 이차 전지가 폭발하기 이전에 밸트 부재(160)가 신속하게 개방되어 내부의 가스를 외부로 배출할 수 있었기 때문이다.
- [0051] 도 5a는 본 발명의 제2 실시예에 따른 이차 전지를 도시한 부분 단면도이고 도 5b는 도 5에서 노치가 찢어져 밸트 부재가 개방된 상태를 도시한 부분 단면도이다. 도 5를 참조하여 설명하면, 본 실시예에 따른 이차 전지는 밸트 부재(170)의 구조를 제외하고는 상기한 제1 실시예에 따른 이차 전지와 동일한 구조로 이루어지므로 동일한 구조에 대한 중복 설명은 생략한다.
- [0052] 밸트 부재(170)에는 제1 실시예와 동일하게, 호형으로 이루어진 제1 노치(171)와 제2 노치(172)가 지지부(175)를 사이에 두고 이격 형성될 수 있다. 제1 노치(171)와 제2 노치(172)는 지지부(175)를 기준으로 대칭되도록 형성될 수 있으며, 제1 노치(171)의 양단을 이은 선분과 제2 노치(172)의 양단을 이은 선분은 절곡 라인(174)이 될 수 있다. 이차 전지의 내압이 상승하면 제1 노치(171)가 파단되어 절곡 라인(174)에서 제1 파단 조각(177)이 절곡되어 상승할 수 있고, 제2 노치(172)도 파단되어 절곡 라인(174)에서 제2 파단 조각(178)이 절곡되어 상승할 수 있다.
- [0053] 본 실시예에 따른 밸트 부재(170)는 절곡 라인(174)에서 제1 노치까지의 최대 거리인 상폭(L4)은 절곡 라인(174)에서 배기구(143b)의 내측단까지의 거리(L5) 보다 더 작게 형성될 수 있고, 절곡 라인(174)에서 캡 플레이트(143)의 상단, 즉 헤드(143a)의 바닥면까지의 거리(h1)보다 더 크게 형성될 수 있다. 이에 의하면, 파단 조각들(177, 178)의 상단이 배기구(143b)의 내측단과 절곡 라인(174)의 직상방 사이에 위치할 수 있어, 개구(173)를 통해서 배출되는 가스를 배기구(143b)로 신속하게 안내할 수 있다.
- [0054] 도 6a는 본 발명의 제3 실시예에 따른 이차 전지의 밸트 부재를 도시한 사시도이고, 도 6b는 도 6a의 밸트 부재에서 노치가 찢어져 개방된 상태를 도시한 사시도이다.
- [0055] 도 6a 및 도 6b를 참조하여 설명하면, 본 실시예에 따른 이차 전지는 밸트 부재(180)를 제외하고는 상기한 제1 실시예에 따른 이차 전지와 동일한 구조로 이루어지므로 동일한 구조에 대한 중복 설명은 생략한다.
- [0056] 밸트 부재(180)에는 제1 노치(181)와 제2 노치(182), 및 제3 노치(183)가 형성된다. 노치들(181, 182, 183)은 밸트 부재(180)의 원주 방향을 따라 등간격으로 배치될 수 있고, 밸트 부재(180)의 외측을 향하여 볼록하게 만곡된 곡선으로 이루어질 수 있다. 이 때, 노치들(181, 182, 183)의 양단을 이은 선분이 절곡 라인(184)이 될 수 있다. 캡 플레이트(143)에는 3개의 배기구(143b)가 형성될 수 있으며, 배기구(143b)의 수에 대응되도록 3개의 노치(181, 182, 183)가 형성될 수 있다. 이 때, 노치들(181, 182, 183)은 각 배기구의 아래에 대응되는 위치에 배치될 수 있다. 이차 전지의 내압 상승으로 노치들(181, 182, 183)이 찢어지면 제1 파단 조각(186), 제2 파단 조각(187), 및 제3 파단 조각(188)이 절곡 라인(184)에서 굽혀져 상승할 수 있다. 노치들(181, 182, 183) 사이의 밸트 부재(180)의 중앙에는 파단 조각들(186, 187, 188)을 지지하는 지지부(185)가 형성될 수 있다.
- [0057] 파단 조각들(186, 187, 188)이 상승하면 밸트 부재(180)에 개구(189)가 형성되는데, 개구는 배기구의 아래에 대응되는 위치에 형성될 수 있다. 이에 따라, 각 개구(189)를 통해서 배출되는 가스는 대응되는 배기구(143b)를

동해서 신속하게 배출될 수 있다.

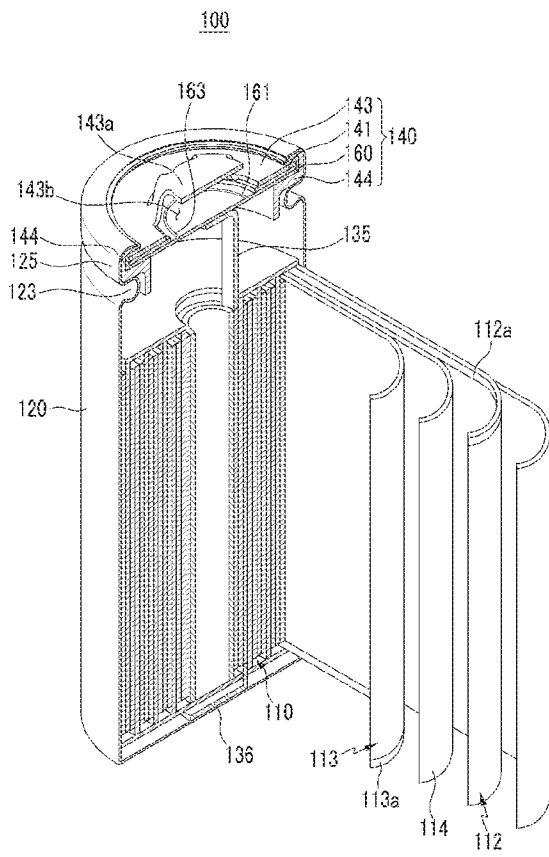
- [0058] 제2 노치(182)와 제3 노치(183)는 제1 노치(181)와 동일한 구조로 이루어지므로, 제1 노치(181)에 대한 설명으로 제2 노치(182)와 제3 노치(183)에 대한 설명을 대체한다.
- [0059] 절곡 라인(184)에서 제1 노치(181)까지의 최대 거리인 장폭(L2)은 절곡 라인(184)에서 캡 플레이트(143)의 상단, 즉 헤드(143a)의 바닥면까지의 거리(h1)보다 더 작게 형성될 수 있다. 이에, 파단 조각들(186, 187, 188)은 파단 압력에 따라서 임의의 각도로 자유롭게 절곡될 수 있으나, 이웃하는 파단 조각들(186, 187, 188)과 간섭이 발생하여 절곡 각도가 제한될 수 있어, 파단 조각(186, 187, 188)이 지나치게 절곡되는 것은 방지할 수 있다.
- [0060] 상기에서는 본 발명의 바람직한 실시예들에 대하여 설명하였지만, 본 발명은 이에 한정되는 것이 아니고 특허청구범위와 발명의 상세한 설명 및 첨부한 도면의 범위 안에서 여러 가지로 변형하여 실시하는 것이 가능하고 이 또한 본 발명의 범위에 속하는 것은 당연하다.

부호의 설명

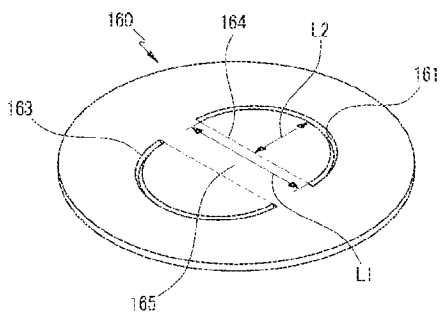
- [0061]
- | | |
|-------------------------|-------------------------|
| 100: 이차 전지 | 110: 전극 조립체 |
| 112: 양극 | 113: 음극 |
| 114: 새퍼레이터 | 120: 케이스 |
| 135: 양극 집전 탭 | 136: 음극 집전 탭 |
| 140: 캡 조립체 | 141: 안전 플레이트 |
| 143: 캡 플레이트 | 143a: 헤드 |
| 143b: 배기구 | 144: 가스켓 |
| 160, 170, 180: 벤트 부재 | 161, 171, 181: 제1 노치 |
| 163, 172, 182: 제2 노치 | 162, 173, 189: 개구 |
| 164, 174, 184: 절곡 라인 | 165, 175, 185: 지지부 |
| 167, 177, 186: 제1 파단 조각 | 168, 178, 187: 제2 파단 조각 |
| 183: 제3 노치 | 188: 제3 파단 조각 |

도면

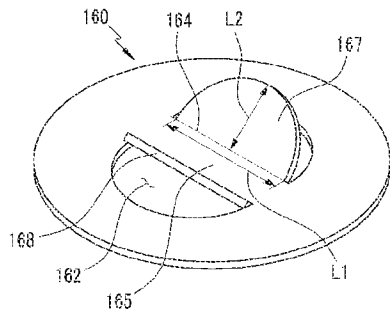
도면1



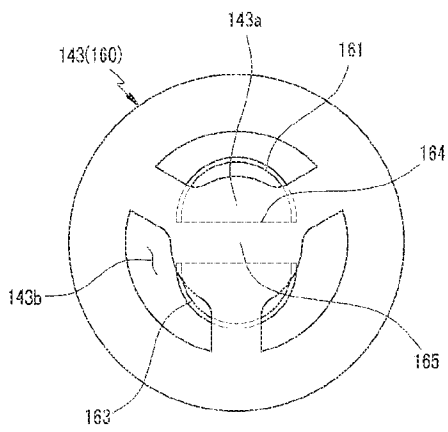
도면2a



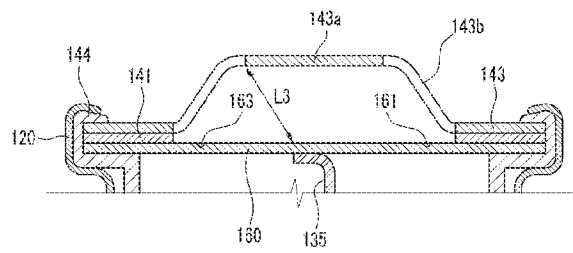
도면2b



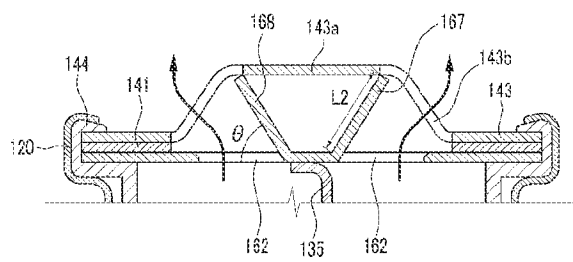
도면2c



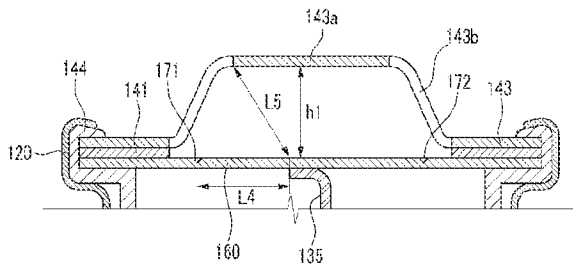
도면3



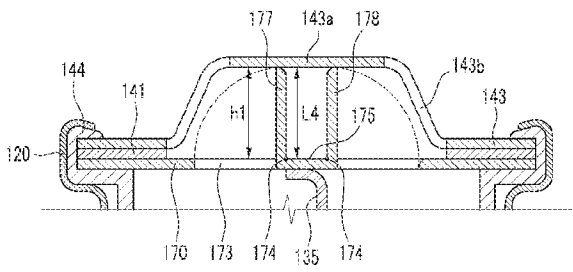
도면4



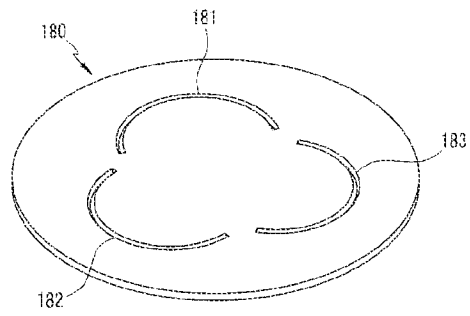
도면5a



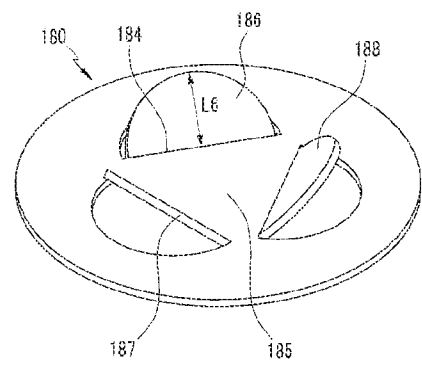
도면5b



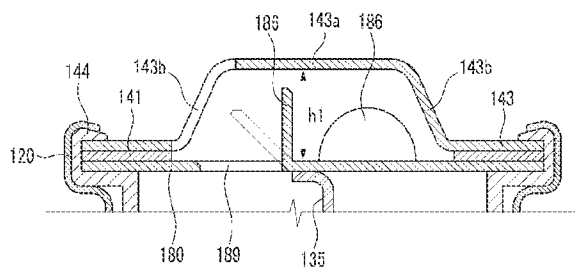
도면6a



도면6b



도면7



특허협력조약

발신: 국제조사기관

수신:

특허법인 태평양

대한민국 06626 서울시 서초구 강남대로 343, 11층

PCT

국제조사기관의 견해서

(PCT규칙 43의2.1)

발송일
(일/월/년)

2020년 02월 20일 (20.02.2020)

출원인 또는 대리인의 서류참조기호

2019OPLC3194

추가적인 조치

아래 2를 참조

국제출원번호

PCT/KR2019/015396

국제출원일 (일/월/년)

2019년 11월 13일 (13.11.2019)

우선일 (일/월/년)

2018년 11월 28일 (28.11.2018)

국제특허분류(IPC)

H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i

출원인

주식회사 엘지화학

1. 본 견해서는 다음 기재란에 관한 내용을 포함합니다.



제1기재란 견해서의 기초



제2기재란 우선권



제3기재란 신규성, 진보성 및 산업상이용가능성에 관한 견해 부작성



제4기재란 발명의 단일성 결여



제5기재란 신규성, 진보성 또는 산업상이용가능성에 관한 견해(PCT규칙 43의2.1(a)(i)), 이를 뒷받침하는 인용문헌 및 설명



제6기재란 특이 인용문헌



제7기재란 국제출원의 흠결



제8기재란 국제출원에 관한 의견

2. 추가적인 조치

국제예비심사가 청구되면, 본 견해서는 국제예비심사기관("IPEA")의 견해서로 간주될 것입니다. 다만, 출원인이 본 기관 이외의 기관을 IPEA로 선택하고, 그 선택된 IPEA가 PCT규칙 66.1의2(b)에 따라 본 국제조사기관의 견해서가 위와 같이 간주되지 않을 것임을 국제사무국에 통보한 경우에는 그러하지 않습니다.

본 견해서가 상기와 같이 IPEA의 견해서로 간주되는 경우, 출원인은 서식 PCT/ISA/220의 발송일로부터 3월 또는 우선일 부터 22월 중 늦게 만료되는 날 이전에 의견서 및 보정서(해당하는 경우)를 IPEA에 제출할 수 있습니다.

다른 선택사항에 대하여는 서식 PCT/ISA/220에 대한 안내문을 참조하십시오.

ISA/KR의 명칭 및 우편주소



대한민국 특허청
(35208) 대전광역시 서구 청사로 189,
4동 (둔산동, 정부대전청사)

팩스번호 +82-42-481-8578

본 견해서의 완료일

2020년 02월 19일 (19.02.2020)

심사관

김연경

전화번호 +82-42-481-3325



제1기재란 본 견해서의 기초

1. 언어와 관련하여, 본 견해서는 아래에 기초하여 작성되었습니다.

☒ 출원시의 언어로 된 국제출원

☐ 국제조사를 위하여 _____ 로 번역되어 제출된 국제출원의 번역문
(PCT규칙 12.3(a) 및 23.1(b))

2. ☐ 본 견해서는 PCT규칙 91에 따라 당해 기관이 허가하였거나 당해 기관에 통보된 **명백한 잘못의 정정**을 고려하여 작성되었습니다(PCT규칙 43의2.1(b)).

3. ☐ 국제출원에 개시된 핵산염기 및/또는 아미노산 서열과 관련하여, 본 견해서는 아래에 기초하여 작성되었습니다.

a. ☐ 아래의 형태로 출원시 국제출원의 일부를 구성하는 서열목록

☐ 부록 C/ST.25 텍스트 파일

☐ 서면 혹은 이미지 파일

b. ☐ PCT 규칙 13의3.1(a)에 따라 국제출원과 함께 국제조사만을 목적으로 부록 C/ST.25 텍스트 파일의 형태로 제출된 서열목록

c. ☐ 국제조사만을 목적으로 국제출원일 이후에 아래 형태로 제출된 서열목록

☐ 부록 C/ST.25 텍스트 파일 (규칙 13의3.1(a))

☐ 서면 혹은 이미지 파일 (규칙 13의3.1(b) 및 시행세칙 713)

4. ☐ 추가로 서열목록에 대하여 하나 이상의 버전이나 사본이 제출된 경우, 후속 버전 또는 추가된 사본에 기재되어 있는 정보가 출원시 출원의 일부를 구성하는 정보와 동일하거나 또는 출원시의 개시범위를 벗어나지 않는다는 진술서가 제출되었습니다.

5. 추가 의견:

제5기제란 신규성, 진보성 또는 산업상이용가능성에 관한 견해(PCT규칙 43의2.1(a)(i)), 이를 뒷받침하는 인용문헌 및 설명

1. 견해

신규성 (N)	청구항	1-12	있음
	청구항	없음	없음
진보성 (IS)	청구항	없음	있음
	청구항	1-12	없음
산업상 이용가능성 (IA)	청구항	1-12	있음
	청구항	없음	없음

2. 인용문헌 및 설명:

참고한 인용문헌은 다음과 같습니다.

D1: JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 1994.08.05

D2: JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 1996.12.24

D3: JP 2001-102038 A (TOYO KOHAN CO., LTD.) 2001.04.13

I. 신규성 및 진보성(PCT 제33조(2) 및 (3))

D1에는 전지; 전지 케이스(8); 외부 단자판(5); 금속제 안전밸브(1); 전지 내부 압력으로 인해 파단되는 제 2 반전부; 꺾어짐부(1a); 및 중심부가 윗쪽으로 들어올려질 수 있는 꺾어 구부러짐부(1b)가 형성된 제 1 반전부를 포함하는 장치 (단락 [0019]-[0021], 청구항 1 및 도면 1-4 참조)가 개시되어 있습니다. 다만, 청구항 1은 파단 노치부(113) 및 벤딩 노치부(115)를 포함하는 점에서 D1과 차이가 있으나, 이는 통상의 기술자가 D2의 취약부(8c)의 형태 (단락 [0043] 및 도면 11 참조)로부터 쉽게 설계 변경하여 도출할 수 있습니다.

청구항 2는 D1의 제 1 반전부가 비교적 낮은 설정 전지 내부 압력으로 반전 변형하고, 제 2 반전부가, 비교적 높은 설정 전지 내부 압력으로 반전 변형하는 것 (청구항 1 참조) 으로부터 쉽게 설계 변경하여 도출할 수 있습니다.

청구항 3은 D2의 취약부(8c)의 형태 (단락 [0043] 및 도면 11 참조)로부터 쉽게 설계 변경하여 도출할 수 있습니다.

청구항 4는 D3의 얇은 부분(26)이 파단하여 부상하게 되는 리드 부착판(19)의 중앙 용접 부(20) (단락 [0016],[0017] 및 도면 1,4 참조)로부터 쉽게 설계 변경하여 도출할 수 있습니다.

청구항 5-9는 D3의 C자 모양 긴 구멍(32)과 긴 구멍(33)으로 대응하는 박재(21b)의 부분에 따라 각각 판막(35,36)이 형성되어 파단 압력 이상이 되면 파단되는 것 (단락 [0017] 및 도면 3,5 참조)으로부터 쉽게 설계 변경하여 도출할 수 있습니다.

청구항 10은 D1의 꺾어짐부(1a) (단락 [0022] 및 도면 4 참조)로부터 쉽게 설계 변경하여 도출할 수 있습니다.

청구항 11은 D1의 금속제 안전밸브(1) (단락 [0019] 및 도면 1-4 참조)로부터 쉽게 설계 변경하여 도출할 수 있습니다.

추가 기재란에 계속

추가 기재란

이전 기재란의 공간이 충분하지 아니한 경우.

제5 기재란의 연속

청구항 12는 통상의 기술자라면 일반적으로 채용할 수 있는 정도의 기술적 사항에 해당합니다.

따라서 청구항 1-3, 10-12는 D1 및 D2의 결합에 의해, 청구항 4-9는 D1-D3의 결합에 의해 진보성이 없습니다.

II. 산업상 이용가능성(PCT 제33조(4))

청구항 1-12는 산업상 이용가능합니다.

PCT 출원서

(전자적 형태가 원본)

0	수리관청 전용	
0-1	국제출원번호	PCT/KR2019/015396
0-2	국제출원일자	2019년 11월 13일 (13.11.2019)
0-3	수리관청 명칭 및 "PCT 국제출원"	RO/KR
0-4	서식 PCT/RO/101 - PCT 출원서	
0-4-1	우측에 기재된 바와 같이 작성되었다.	PCT-SAFE 버전 3.51.087.263 MT/FOP 20190401/0.20.5.24
0-5	신청 아래 서명인은 본 국제 출원서가 특허협력조약에 의해 처리될 것을 청구합니다.	
0-6	출원인이 지정한 수리관청	대한민국 특허청 (RO/KR)
0-7	출원인 또는 대리인의 서류참조기호	2019OPLC3194
I	발명의 명칭	이차전지
II	출원인	
II-1	이 사람은	오직 출원인 (applicant only)
II-2	우측 지정국에 관한 출원인	모든 지정국 (all designated States)
II-4ko	성명	주식회사 엘지화학
II-4en	Name:	LG CHEM, LTD.
II-5ko	주소	대한민국 07336 서울시 영등포구 여의대로 128
II-5en	Address:	128, Yeoui-daero Yeongdeungpo-gu Seoul 07336 Republic of Korea
II-6	국적	대한민국 KR
II-7	거주국	대한민국 KR
II-11	출원인 코드	1-2001-013456-3
III-1	출원인 및/또는 발명자	
III-1-1	이 사람은	오직 발명자 (inventor only)
III-1-3	우측 지정국에 관한 발명자	모든 지정국 (all designated States)
III-1-4ko	성명	민건우
III-1-4en	Name (LAST, First):	MIN, Geon Woo
III-1-5ko	주소	대한민국 34122 대전시 유성구 문지로 188 LG화학 기술연구원
III-1-5en	Address:	LG Chem Research Park, 188, Munji-ro Yuseong-gu Daejeon 34122 Republic of Korea

PCT 출원서

(전자적 형태가 원본)

III-2 III-2-1 III-2-3 III-2-4ko III-2-4en III-2-5ko III-2-5en	출원인 및/또는 발명자 이 사람은 우측 지정국에 관한 발명자 성명 Name (LAST, First): 주소 Address:	오직 발명자 (inventor only) 모든 지정국 (all designated States) 이제준 LEE, Je Jun 대한민국 34122 대전시 유성구 문지로 188 LG화학 기술연구원 LG Chem Research Park, 188, Munji-ro Yuseong-gu Daejeon 34122 Republic of Korea
III-3 III-3-1 III-3-3 III-3-4ko III-3-4en III-3-5ko III-3-5en	출원인 및/또는 발명자 이 사람은 우측 지정국에 관한 발명자 성명 Name (LAST, First): 주소 Address:	오직 발명자 (inventor only) 모든 지정국 (all designated States) 정상석 JUNG, Sang Suk 대한민국 34122 대전시 유성구 문지로 188 LG화학 기술연구원 LG Chem Research Park, 188, Munji-ro Yuseong-gu Daejeon 34122 Republic of Korea
IV-1 IV-1-1ko IV-1-1en IV-1-2ko IV-1-2en IV-1-3 IV-1-4 IV-1-5 IV-1-5(a) IV-1-6	대리인 또는 대표자 아래에 기재된 자는 관할 국제기관에 대하여 우측에 표시된 자격으로 출원인을 대리하는 것으로 선임되었다. 성명 Name: 주소 Address: 전화번호 팩스번호 이메일 주소 이메일 사용동의 수리관청, 국제조사기관, 국제사무국, 국제예비심사기관이 필요 시 이 이메일 주소를 사용하여 이 국제 출원과 관련하여 발행된 통지서를 송부할 것에 동의한다. 대리인 코드	대리인 특허법인 태평양 BAE, KIM & LEE IP GROUP 대한민국 06626 서울시 서초구 강남대로 343, 11층 11th Floor, 343, Gangnam-daero Seocho-gu Seoul 06626 Republic of Korea 82-2-2188-5454 82-2-2188-5455 ip@bkl.co.kr 오직 전자적 형태의 통지서만 송부 (서면 통지서는 미발송) 9-2006-100042-1

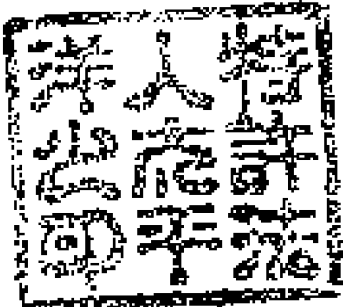
PCT 출원서

(전자적 형태가 원본)

V	지정국		
V-1	본 출원서의 제출로, 규칙 4.9(a)에 따라, 부여될 수 있는 모든 종류의 권리 보호를 위하여, 그리고 해당하는 경우 지역특허 및 국내특허 모두를 위하여 당해 국제출원일에 PCT에 기속되는 모든 계약국이 지정된다.		
V-2	V-2란은 출원서 제출시 또는 규칙 26의2.1에 의해 그 이후 출원서 제6기제관에 위 특정 관련 계약국의 국내 선출원에 대한 우선권주장이 포함되어 있을 경우 당해 계약국의 국내법에 의해 해당 국내 선출원의 효력이 상실되는 것을 방지하기 위한 목적으로 당해 계약국의 지정을 제외하는 데에만 사용될 수 있다 (지정 제외시 이의 취소 불가능).	KR	
VI-1	선국내출원에 대한 우선권 주장	2018년 11월 28일 (28.11.2018) 10-2018-0149369 KR	
VI-1-1	출원일		
VI-1-2	출원번호		
VI-1-3	파리협약 당사국명 또는 WTO 회원국명		
VI-2	우선권서류 신청	VI-1	
	수리관청에 대하여 위에 명시된 선출원의 인증증본을 준비하여 국제사무국에 송부하여 줄 것을 신청한다.		
VI-3	인용에 의한 보완 조약 제11조(1)(iii)(d) 또는 (e)에서 규정하는 국제출원의 요소, 또는 규칙 20.5(a)에서 규정하는 명세서, 청구 범위 또는 도면의 일부가 본 국제출원에는 포함되어 있지 않지만 조약 제11조(1)(iii) 규정의 요소 중 하나 이상이 수리관청에 최초로 접수된 날에 우선권주장의 기초가 된 선출원에 완전히 포함되어 있는 경우, 그 요소 또는 부분은 규칙 20.6 규정에 의한 확인을 조건으로, 규칙 20.6의 규정과 관련하여 본 국제출원에 있어서 인용에 의해 보완된다.		
VII-1	국제조사기관(ISA) 선택	대한민국 특허청 (ISA/KR)	
VIII	선언서	선언서 개수	
VIII-1	발명자의 신원에 관한 선언	-	
VIII-2	국제출원일에 특허출원 및 특허를 받을 수 있는 출원인의 자격에 관한 선언	-	
VIII-3	국제출원일에 선출원의 우선권을 주장할 수 있는 출원인의 자격에 관한 선언	-	
VIII-4	발명자 선언(미국에 대한 지정의 경우에 한함)	-	
VIII-5	신규성을 해치지 아니하는 개시 또는 신규성 상실의 예외에 관한 선언	-	
IX	체크 리스트	용지 수	전자적 파일 첨부
IX-1	출원서(선언서 포함)	4	✓
IX-2	명세서	8	✓
IX-3	청구범위	2	✓
IX-4	요약서	1	✓
IX-5	도면	6	✓
IX-7	용지매수 소개	21	
	첨부 항목	서면 첨부	전자적 파일 첨부
IX-8	수수료 계산 용지	-	✓
IX-11	포괄위임장 사본	-	포괄위임장 번호 2019OPLC3194
IX-20	요약서에 수반되어야 할 도면 번호	3	
IX-21	국제출원의 출원 언어	한국어	

PCT 출원서

(전자적 형태가 원본)

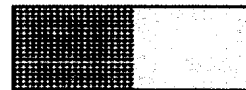
X-1	출원인, 대리인 또는 대표자의 서명 또는 날인	
X-1-1	성명	특허법인 태평양
X-1-2	서명인의 성명	이은경
X-1-3	권한 (출원서를 통해 서명자의 자격이 명백하지 않은 경우에는 그 자격도 표시)	대표

수리관청 전용

10-1	국제출원으로 제출된 서류의 실제 접수일	2019년 11월 13일 (13.11.2019)
10-2	도면	
10-2-1	접수	
10-2-2	미접수	
10-3	국제출원으로 제출된 서류를 완성하는 서류 또는 도면의 추후 기간내 제출에 따른 정정된 실제 접수일	
10-4	PCT 제11조(2)에 따라 제출이 요구된 보완서로서 기간내 제출된 보완서의 접수일	
10-5	국제조사기관(ISA)	ISA/KR
10-6	조사료 납부시까지 지연된 조사용 사본의 송부	

국제 사무국 전용

11-1	국제 사무국의 기록원본 접수일	
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별첨 사본은 아래 출원의 원본과 동일함을 증명함.

This is to certify that the following application annexed hereto
is a true copy from the records of the Korean Intellectual
Property Office.

출 원 번 호 : 10-2018-0149369
Application Number

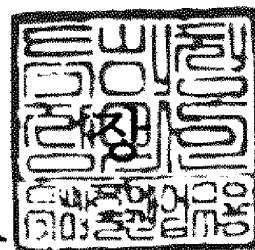
출 원 년 월 일 : 2018년 11월 28일
Filing Date NOV 28, 2018

출 원 인 : 주식회사 엘지화학
Applicant(s) LG CHEM, LTD.

2019 년 11 월 21 일

특 허 청

COMMISSIONER



【서지사항】

【서류명】	특허출원서
【참조번호】	5285
【출원구분】	특허출원
【출원인】	
【명칭】	주식회사 엘지화학
【특허고객번호】	1-2001-013456-3
【대리인】	
【명칭】	특허법인 태평양
【대리인번호】	9-2006-100042-1
【지정된변리사】	이은경, 이종욱, 양성욱, 남정우
【포괄위임등록번호】	2011-015093-1
【발명의 국문명칭】	이차전지
【발명의 영문명칭】	SECONDARY BATTERY
【발명자】	
【성명】	민건우
【성명의 영문표기】	MIN, Geon Woo
【주민등록번호】	000000-0XXXXXX
【우편번호】	34122
【주소】	대전광역시 유성구 문지로 188 LG화학 기술연구원 내
【발명자】	
【성명】	이제준

【성명의 영문표기】 LEE, Je Jun

【주민등록번호】 000000-0XXXXXX

【우편번호】 34122

【주소】 대전광역시 유성구 문지로 188 LG화학 기술연구원 내

【발명자】

【성명】 정상석

【성명의 영문표기】 JUNG, Sang Suk

【주민등록번호】 000000-0XXXXXX

【우편번호】 34122

【주소】 대전광역시 유성구 문지로 188 LG화학 기술연구원 내

【출원언어】 국어

【취지】 위와 같이 특허청장에게 제출합니다.

대리인 특허법인 태평양

(서명 또는 인)

【수수료】

【출원료】 0 면 46,000 원

【가산출원료】 28 면 0 원

【우선권주장료】 0 건 0 원

【심사청구료】 0 항 0 원

【합계】 46,000 원

【발명의 설명】

【발명의 명칭】

이차전지{SECONDARY BATTERY}

【기술분야】

【0001】 본 발명은 이차전지에 관한 것으로서, 보다 자세하게는 벤팅(venting)압 산포를 개선할 수 있는 이차전지에 관한 것이다.

【발명의 배경이 되는 기술】

【0002】 물질의 물리적 반응이나 화학적 반응을 통해 전기에너지를 생성시켜 외부로 전원을 공급하게 되는 전지(cell, battery)는 각종 전기전자 기기로 둘러싸여 있는 생활환경에 따라, 건물로 공급되는 교류 전원을 획득하지 못할 경우나 직류전원이 필요할 경우 사용하게 된다.

【0003】 이와 같은 전지 중에서 화학적 반응을 이용하는 화학전지인 일차전지와 이차전지가 일반적으로 많이 사용되고 있는데, 일차전지는 건전지로 통칭되는 것으로 소모성 전지이다. 또한, 이차전지는 전류와 물질 사이의 산화환원과정이 다수 반복 가능한 소재를 사용하여 제조되는 재충전식 전지로서, 전류에 의해 소재에 대한 환원반응이 수행되면 전원이 충전되고, 소재에 대한 산화반응이 수행되면 전원이 방전되는데, 이와 같은 충전-방전이 반복적으로 수행되면서 전기가 생성되게 된다.

【0004】한편, 이차 전지 중 리튬 이온 전지는 양극 도전 포일과 음극 도전 포일에 각각 활물질을 일정한 두께로 코팅하고, 상기 양 도전 포일 사이에는 분리막이 개재되도록 하여 대략 젤리 롤(jelly roll) 내지는 원통 형태로 다수 회 권취하여 제작한 전극 조립체를 원통형 또는 각형 캔, 파우치 등에 수납하고 이를 밀봉 처리하여 제작된다.

【0005】대한민국 공개특허공보 제10-2011-0095118호에는 종래의 이차 전지가 개시되어 있다.

【0006】이차 전지는 내부 압력 증가 시에 탭캡 조립체를 통해 내부 가스를 외부로 배출하며, 탭캡 조립체는 내부 가스의 배출을 위한 안전벤트(vent)가 형성된다.

【0007】이러한 안전벤트는 탭캡 조립체와 캔의 사이에 마련된 벤트 플랜지부에 형성된 노치가 캔 내부 압력 증가 시 절개되어 가스 배출을 위한 통로를 형성하는 구성이다.

【0008】특히, 원형 전지는 과충전 등 이상 상황 발생 시 전지 폭발 및 발화를 방지하기 위하여 안전벤트와 같은 특정 부품에서 벤팅(venting)을 유도한다.

【0009】이러한 안전벤트는 기 설계된 벤팅압에서 벤팅되지 않으면 안전문제가 발생할 수 있다.

【0010】그런데 종래의 안전벤트는 벤팅압 측정 시 안전벤트의 노치(notch)가 절개되기 전에 탭캡(top cap)에 의해 간섭을 받아 벤팅이 지연되는 문제점이 있

었다.

【발명의 내용】

【해결하고자 하는 과제】

【0011】 따라서 본 발명은 위와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 과제는 탭캡과의 간섭을 방지하여 벤딩압 산포를 개선할 수 있는 이차전지를 제공하는 것이다.

【과제의 해결 수단】

【0012】 본 발명의 일 실시예에 따른 이차전지는 전극 조립체, 상기 전극 조립체를 내부에 수용하는 캔부재 및 상기 캔부재의 개구부를 덮는 탭캡 조립체를 포함하는 이차전지에 있어서, 상기 탭캡 조립체는 최상부에 위치되는 탭캡과 상기 탭캡의 하부에 위치되는 안전벤트부를 포함하고, 상기 안전벤트부는, 상기 캔부재 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부, 상기 파단부의 경계를 형성하는 파단 노치부 및 상기 캔부재의 내측을 바라보는 상기 파단부의 저면에, 상기 파단 노치부로부터 상기 파단부의 중앙측으로 이격된 위치에 형성된 벤딩 노치부를 포함하고, 상기 캔부재 내부의 내압 증가 시 상기 벤딩 노치부는 벤딩(bending)되는 것을 특징으로 한다.

【0013】 상기 파단 노치부는 상기 안전벤트부의 하부 표면으로부터 오목하게 형성되고, 상기 벤딩 노치부 또한 상기 안전벤트부의 하부 표면으로부터 오목하게 형성되며, 상기 벤딩 노치부의 오목한 깊이는 상기 파단 노치부의 오목한 깊이보다

알 수 있다.

【0014】 상기 파단 노치부는 상기 파단부의 상기 저면에 형성될 수 있다.

【0015】 상기 탑캡 조립체는 상기 파단부에 부착되며 상기 캔부재의 내부 내압 증가 시 파단되어 전류를 차단하는 차단부를 구비하는 전류차단장치를 더 포함할 수 있다.

【0016】 상기 전류차단장치의 차단부는 상기 파단부에서 상기 벤딩 노치부의 둘레 내측에 부착될 수 있다.

【0017】 상기 벤딩 노치부는 연속되는 원형 또는 타원형일 수 있다.

【0018】 상기 벤딩 노치부는 불연속되는 원형 또는 타원형일 수 있다.

【0019】 상기 벤딩 노치부는 서로 길이가 다르게 복수 개로 이루어질 수 있다.

【0020】 상기 파단 노치부는 불연속되는 원형 또는 타원형일 수 있다.

【0021】 상기 안전벤트부의 테두리부와 상기 파단부 사이의 연결부는 상측으로 벤딩(bending)될 수 있다.

【0022】 상기 파단부는 상기 연결부에서 상기 벤딩 노치부 측방향으로 높이가 낮아지게 기울어지는 경사부분 및 상기 경사부분의 단부에서 상기 파단부의 중앙측으로 연장되고 수평한 형태의 수평부분을 포함하고, 상기 벤딩 노치부는 상기 경사부분과 상기 수평부분의 경계에 위치할 수 있다.

【0023】 상기 안전벤트부의 테두리 단부와 상기 파단 노치부 사이의 이격 거리와 상기 파단 노치부와 상기 벤딩 노치부 사이의 이격 거리는 동일할 수 있다.

【발명의 효과】

【0024】 본 발명에 따르면, 안전벤트부의 저면에 파단 노치부 외에 벤딩 노치부를 추가로 형성하여 캔부재 내부 내압 상승 시 안전벤트부가 탭캡에 간섭되지 않게 하는 효과가 있다.

【0025】 또한, 안전벤트부가 탭캡에 간섭되지 않아 파단 노치부가 기 설정된 벤팅압에서 안정되게 벤팅되게 함에 따라 벤팅압 산포를 개선하는 효과가 있다.

【도면의 간단한 설명】

【0026】 도 1은 본 발명의 제1 실시예에 따른 이차전지의 사시도이다.

도 2는 도 1을 A-A 방향으로 절단하여 이차전지의 상부 내부의 주요부를 도시한 부분 단면도이다.

도 3은 도 2에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.

도 4는 도 3에서 안전벤트부의 파단 노치부가 절단되어 벤팅되는 것을 도시한 사용상태도이다.

도 5는 본 발명의 제1 실시예에 따른 안전벤트부만을 도시한 저면도이다.

도 6은 도 5를 B-B 방향으로 절단하여 도시한 단면도이다.

도 7은 본 발명의 제2 실시예에 따른 안전벤트부만을 도시한 저면도이다.

도 8은 본 발명의 제3 실시예에 따른 안전벤트부만을 도시한 저면도이다.

도 9는 도 8을 C-C 방향으로 절단하여 도시한 단면도이다.

도 10은 종래의 안전벤트부가 설치된 이차전지를 도 3과 같은 단면에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.

도 11은 종래의 안전벤트부가 설치된 이차전지와 본 발명의 안전벤트부가 설치된 이차전지의 벤팅압 산포를 비교하여 나타낸 산포도이다.

【발명을 실시하기 위한 구체적인 내용】

【0027】 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예에 따른 이차전지에 대하여 상세히 설명하기로 한다.

【0028】 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다. 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의 가장 바람직한 일 실시예에 불과하고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물이 있을 수 있음을 이해하여야 한다.

【0029】도면에서 각 구성요소 또는 그 구성요소를 이루는 특정 부분의 크기는 설명의 편의 및 명확성을 위하여 과장되거나 생략되거나 또는 개략적으로 도시되었다. 따라서, 각 구성요소의 크기는 실제크기를 전적으로 반영하는 것은 아니다. 관련된 공지기능 혹은 구성에 대한 구체적인 설명이 본 발명의 요지를 불필요하게 흐릴 수 있다고 판단되는 경우, 그러한 설명은 생략하도록 한다.

【0031】도 1은 본 발명의 제1 실시예에 따른 이차전지의 사시도이고, 도 2는 도 1을 A-A 방향으로 절단하여 이차전지의 상부 내부의 주요부를 도시한 부분 단면도이다.

【0032】도 1 및 도 2에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 전극 조립체(1), 상기 전극 조립체(1)를 내부에 수용하는 캔부재(3) 및 상기 캔부재(3)의 개구부를 덮는 탑캡 조립체(100)를 포함하는 이차전지에 있어서, 상기 탑캡 조립체(100)는 최상부에 위치되는 탑캡(101)과 상기 탑캡(101)의 하부에 위치되는 안전벤트부(110)를 포함하고, 상기 안전벤트부(110)는 상기 캔부재(3) 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부(111), 상기 파단부(111)의 경계를 형성하는 파단 노치부(113) 및 상기 캔부재(3)의 내측을 바라보는 상기 파단부(111)의 저면에서, 상기 파단 노치부(113)로부터 상기 파단부(111)의 중앙측으로 이격된 위치에 형성된 벤딩 노치부(115)를 포함하고, 상기 캔부재(3) 내부의 내압 증가 시 상기 벤딩 노치부(115)는 벤딩(bending)되는 것을 특징으로 한다.

【0033】전극 조립체(1)는, 예를 들어, 양극 활물질이 도포된 양극과 음극 활물질이 도포된 음극 및 양극과 음극 사이에 개재된 분리막을 복수 회 적층하여 제작할 수 있다.

【0034】그리고 전극 조립체(1)는 양극과 분리막 및 음극을 적층한 적층체를 젤리 롤 형태로 권취하여 제작할 수도 있다.

【0035】캔부재(3)는 원형 리튬이온 2차 전지에서 대략 원통형의 위쪽이 개방된 형상을 가진 금속재질의 용기이거나 각형 리튬이온 2차 전지에 대략 각형의 위쪽이 개방된 형상을 가진 금속재질의 용기일 수 있으며, 일반적으로 가볍고 부식에 대처가 용이한 알루미늄 또는 알루미늄 합금이 사용된다.

【0036】캔부재(3)는 전극 조립체(1)와 전해액(미도시)의 용기가 되고, 전극 조립체(1)가 캔부재(3)의 개방된 상단, 즉, 상단개구부를 통해 캔부재(3)에 삽입된 뒤 캔부재(3)의 상단개구부는 탭캡 조립체(100)에 의해 봉해진다.

【0037】탭캡 조립체(100)는 캔부재(3)의 내부 내압 증가 시 파단되어 전류를 차단하는 차단부(121)를 구비하는 전류차단장치(120)를 더 포함할 수 있다.

【0038】도 3은 도 2에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.

【0039】도 3을 참조하면, 전류차단장치(120)는 차단부(121)를 안전벤트부(110)의 파단부(111)에 부착하는데 캔부재(3)의 내부 내압 증가 시 차단부(121)가 전류차단장치(120)로부터 파단되면서 안전벤트부(110)의 파단부(111)와 함께 탭캡

조립체(100)의 탭캡(101) 방향인 상측으로 이동하게 된다.

【0040】 본 발명에서는 전류차단장치(120)의 차단부(121)가 파단부(111)에서 벤딩 노치부(115)의 둘레 내측의 정중앙에 부착될 수 있다.

【0041】 안전벤트부(110)의 파단 노치부(113)는 캔부재(3)의 내부 내압 증가 시 절개되기 위해 안전벤트부(110)의 표면으로부터 오목하게 형성된다.

【0042】 파단 노치부(113)는 안전벤트부(110)의 상면과 저면 중 어느 하나에 형성될 수 있는데, 안전벤트부(110)가 캔부재(3)의 내부측을 바라보는 안전벤트부(110)의 저면에 형성되면 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 상승했을 때 탭캡(101)과의 이격 거리(d)를 확보하는데 유리 할 수 있다.

【0043】 벤딩 노치부(115)는 파단 노치부(113)와 마찬가지로 안전벤트부(110)의 표면으로부터 오목하게 만입된 형태(또는 파인 형태)로 형성되는데 벤딩 노치부(115)의 오목한 깊이는 파단 노치부(113)의 오목한 깊이보다 얇을 수 있다.

【0044】 이러한 이유는 벤딩 노치부(115)가 캔부재(3)의 내부 내압에 의해 절개되지 않고 벤딩(bending)되게 하기 위해 노치(notch)의 두께를 절개되지 않을 정도의 두께로 유지하기 위함 일 수 있다.

【0045】 그리고 벤딩 노치부(115)는 안전벤트부(110)의 저면에 형성되어 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 이동하였을 때에 탭캡(101)과의 이격거리(d)를 확보하게 할 수 있다.

【0046】 도 4는 도 3에서 안전벤트부의 파단 노치부가 절단되어 벤팅되는 것을 도시한 사용상태도이다.

【0047】 도 4에 도시된 바와 같이, 안전벤트부(110)가 캔부재(3)의 내부 내압의 의해 상측으로 이동하였을 때 안전벤트부(110)의 저면에 형성된 벤딩 노치부(115)의 벤딩에 의해 탑캡(101)과의 이격거리(d)를 확보하게 되면 파단 노치부(113)가 기 설정된 벤팅(venting)압에 의해 절개되어 안전하게 캔부재(3)의 내부 가스를 외부로 배출(venting)하게 할 수 있다.

【0048】 도 5는 본 발명의 제1 실시예에 따른 안전벤트부만을 도시한 저면도이다.

【0049】 도 5에 도시된 바와 같이, 본 발명의 일 실시예에 따르면 파단 노치부(113)는 안전벤트부(110)의 저면에 불연속되는 원형 또는 타원형으로 형성될 수 있다.

【0050】 이와 같이 파단 노치부(113)는 안전벤트부(110)의 저면의 형상에 대응되게 형성될 수 있다. 즉, 안전벤트부(110)의 저면의 형상이 원형이면 파단 노치부(113)의 형상도 원형을 형성할 수 있고, 안전벤트부(110)의 저면의 형상이 타원형이면 파단 노치부(113)의 형상도 타원형을 형성할 수 있다.

【0051】 벤딩 노치부(115)는 안전벤트부(110)의 저면에 연속되는 원형 또는 타원형을 형성할 수 있다.

【0052】 벤딩 노치부(115)도 파단 노치부(113)와 마찬가지로 안전벤트부(110)의 저면의 형상에 대응되게 형성될 수 있다.

【0053】 또한, 벤딩 노치부(115)는 파단 노치부(113)보다 둘레의 크기가 작게 형성되어 안전벤트부(110)의 저면에서 파단 노치부(113)의 둘레 안쪽에 형성될 수 있다.

【0054】 한편, 안전벤트부(110)의 저면에서 보아 안전벤트부(110)의 저면 테두리 단부와 파단 노치부(113) 사이의 이격된 거리와 파단 노치부(113)와 벤딩 노치부(115) 사이의 이격된 거리를 어느 일측으로 치우치지 않고 동일하게 한다면 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 이동하였을 때 안전벤트부(110)와 탑캡(101)과의 이격 거리를 가장 효과적으로 확보하게 할 수 있다.

【0055】 도 6은 도 5를 B-B 방향으로 절단하여 도시한 단면도이다.

【0056】 도 6에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 안전벤트부(110)의 테두리부(117)와 파단부(111) 사이의 연결부(119)는 상측으로 벤딩(bending)될 수 있다.

【0057】 즉, 연결부(119)가 상측으로 벤딩되었기 때문에 캔부재(3) 내부의 내압에 의해 안전벤트부(110)가 상측으로 이동하여도 파단 노치부(113)가 쉽게 절개되지 않고 안전벤트부(110)가 상측으로 이동하게 할 수 있다.

【0058】 파단부(111)는 연결부(119)에서 벤딩 노치부(115) 방향으로 높이가 낮아지게 기울어지는 경사부분(111a)을 포함할 수 있다.

【0059】 그리고 경사부분(111a)의 단부에서 파단부(111)의 중앙측으로 연장되고 수평한 형태의 수평부분(111b)을 포함할 수 있다.

【0060】 그리고 벤딩 노치부(115)는 경사부분(111a)과 수평부분(111b)의 경계에 위치할 수 있다.

【0061】 즉, 경사부분(111a)이 수평부분(111b) 방향으로 높이가 낮아지게 기울어짐에 따라 연결부(119)가 상측으로 벤딩되어 높이가 높아진 것을 보상해 주어 파단부(111)의 수평부분(111b)이 가장 하측에 위치하면서 수평을 유지하게 할 수 있다.

【0062】 이와 같이 파단부(111)는 하측에 설치된 전류차단장치(120)의 차단부(121)와 결합하기 위해 수평부분(111b)이 안전벤트부(110)에서 가장 하측에 위치할 수 있다.

【0063】 또한, 벤딩 노치부(115)가 경사부분(111a)과 수평부분(111b)의 경계에 위치하게 되면, 탐캡(101)과의 최적의 이격거리(d)를 확보하도록 벤딩 노치부(115)가 벤딩 되도록 하는데 현저히 효과적일 수 있다.

【0064】 도 7은 본 발명의 제2 실시예에 따른 안전벤트부만을 도시한 저면도이다.

【0065】 도 7에 도시된 바와 같이, 본 발명의 제2 실시예에 따른 이차전지는 안전벤트부(110)의 벤딩 노치부(115)가 불연속되는 원형 또는 타원형을 형성할 수 있다.

【0066】벤딩 노치부(115)가 불연속되는 원형 또는 타원형을 형성한다면 벤딩 노치부(115)에서 오목한 부분을 최소화할 수 있어 캔부재(3)의 내압 상승 시 벤딩 노치부(115)가 절개되는 것을 효과적으로 방지할 수 있다.

【0067】도 8은 본 발명의 제3 실시예에 따른 안전벤트부만을 도시한 저면도이고, 도 9는 도 8을 C-C 방향으로 절단하여 도시한 단면도이다.

【0068】도 8 및 도 9에 도시된 바와 같이, 본 발명의 제3 실시예에 따른 이차전지는 안전벤트부(110)의 벤딩 노치부(115)가 서로 길이가 다르게 복수 개로 형성될 수 있다.

【0069】벤딩 노치부(115)가 서로 길이가 다르게 복수 개로 형성되면 캔부재(3)의 순간 내압 상승에 의해 안전벤트부(110)가 급속한 속도로 상승하여도 상승충격을 복수 개의 벤딩 노치부(115)가 벤딩되면서 완화시킬 수 있다.

【0070】즉, 순간 급격한 상승 충격에 의해 파단 노치부(113)가 기 설정된 벤팅압 이전의 압력에서 파단되는 것을 방지할 수 있다.

【0071】도 10은 비교예로서 제시된다. 종래의 안전벤트부가 설치된 이차전지를 도 3과 같은 단면에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.

【0072】도 10에 도시된 바와 같이, 종래의 이차전지는 안전벤트부(110')에 파단 노치부(113')만 형성하며, 본원발명에서와 같은 벤딩 노치부를 형성하지 않고 만들어진다는 점에 차이가 있다.

【0073】 이러한 종래의 이차전지는 캔부재(3) 내부의 내압 상승 시 안전벤트부(110')가 탭캡(101') 방향으로 이동하는데, 가스의 내압에 의해 안전벤트부(110')에서 캔부재(3)와 결합된 테두리로부터 가장 거리가 먼 안전벤트부(110')의 중앙 부위가 가장 상측으로 솟아오를 수 있다.

【0074】 따라서 안전벤트부(110')의 중앙 부위가 탭캡(101')의 인근까지 솟아올라 탭캡(101')과의 이격 거리(d_2)가 대략 123um 정도까지 근접할 수 있다. 또 더 나아가 안전벤트부(110')의 중앙 부위가 탭캡(101')에 닿을 수 있다.

【0075】 이와 같이 안전벤트부(110')와 탭캡(101')과의 이격 거리(d_2)가 너무 가깝거나 닿게 되면, 파단 노치가 절개되어 내부 가스를 벤팅해야 하는 기 설정된 벤팅압에서 절개되지 않아 벤팅이 안정적으로 이루어지지 않는 경우가 발생할 수 있다.

【0076】 이러한 경우 이차전지의 폭발, 발열, 발화 등의 안전 문제가 발생할 수 있다.

【0077】 도 3에 도시된 바와 같이, 본 발명의 일실시예에 따른 이차전지는 캔부재(3) 내부의 내압 상승 시 파단부(111)의 경사부분(111a)이 상측으로 꺾이면서 안전벤트부(110)가 탭캡(101) 방향으로 이동하는데, 안전벤트부(110)가 탭캡(101)을 향하여 이동할 때 벤딩 노치부(115)가 꺾이면서(bending) 캔부재(3) 내부의 내압을 일부 흡수하여 파단부(111)의 수평부분(111b)이 탭캡(101) 방향으로 솟구치는 것을 방지할 수 있다.

【0078】 따라서 안전벤트부(110)와 탑캡(101) 사이의 이격 거리(d_1)를 대략 186 μ m 정도로 충분히 확보할 수 있다.

【0079】 이와 같이 안전벤트부(110)와 탑캡(101) 사이의 이격 거리(d_1)를 확보함에 따라, 기 설정된 벤팅압에서 파단 노치부(113)가 절개되는 것이 탑캡(101)에 의해 간섭되는 것을 방지할 수 있어 안정적으로 캔부재(3) 내부의 가스를 벤팅할 수 있다.

【0080】 도 11은 종래의 안전벤트부가 설치된 이차전지와 본 발명의 안전벤트부가 설치된 이차전지의 벤팅압 산포를 비교하여 나타낸 산포도이다.

【0081】 도 11에서 V_0 는 벤팅압의 산포 집합이고, M_x 는 벤팅압의 최대치이고 M_n 은 벤팅압의 최소치이며, V_1 , V_2 는 각각 벤팅압 산포의 일부분이다.

【0082】 도 11을 참조하여, 종래의 벤딩 노치부가 형성되지 않은 안전벤트부의 벤팅압과 개선된 본 발명의 벤딩 노치부가 형성된 안전벤트부의 벤팅압을 비교해보면, 종래의 안전벤트부의 벤팅압의 산포에 비해 본 발명의 안전벤트부의 벤팅압의 산포가 개선된 것을 알 수 있는데 그 차이는 대략 1.5배 정도의 차이가 날 수 있다.

【0083】 특히, 종래의 안전벤트부는 캔부재 내부 가스의 벤팅 과정에서 파단 노치부가 절개되지 않은 상태에서 안전벤트부의 중앙부가 탑캡에 닿게되어 기 설정된 벤팅압에서 정상적으로 벤팅이 되지 않는 것과 같은 비정상적인 벤팅 과정에서 벤팅압의 변화가 크게 발생하는 산포를 나타내는 것이 V_1 및 V_2 일 수 있다.

【0085】 상술한 바와 같이 본 발명에 따르면, 안전벤트의 저면에 파단 노치부 외에 벤딩 노치부를 추가로 형성하여 캔부재 내부 내압 상승 시 안전벤트가 탭캡에 간섭되지 않게 하는 효과가 있다.

【0086】 또한, 안전벤트가 탭캡에 간섭되지 않아 파단 노치부가 기 설정된 벤딩압에서 벤딩되게 함에 따라 벤딩압 산포를 개선하는 효과가 있다.

【0088】 이상과 같이 본 발명에 따른 이차전지를 예시된 도면을 참고하여 설명하였으나, 본 발명은 이상에서 설명된 실시예와 도면에 의해 한정되지 않으며, 특허청구범위 내에서 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자들에 의해 다양한 실시가 가능하다.

【부호의 설명】

【0089】 1: 전극 조립체

3: 캔부재

100: 탭캡 조립체

110: 안전벤트부

111: 파단부

111a: 경사부분

111b: 수평부분

113: 파단 노치부

115: 벤딩 노치부

117: 파단부

119: 연결부

120: 전류차단장치

121: 차단부

【청구범위】

【청구항 1】

전극 조립체(1);

상기 전극 조립체(1)를 내부에 수용하는 캔부재(3); 및

상기 캔부재(3)의 개구부를 덮는 탭캡 조립체(100); 를 포함하는 이차전지에
있어서,

상기 탭캡 조립체(100)는 최상부에 위치되는 탭캡(101)과 상기 탭캡(101)의
하부에 위치되는 안전벤트부(110)를 포함하고,

상기 안전벤트부(110)는,

상기 캔부재(3) 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단
부(111);

상기 파단부(111)의 경계를 형성하는 파단 노치부(113); 및

상기 캔부재(3)의 내측을 바라보는 상기 파단부(111)의 저면에서, 상기 파단
노치부(113)로부터 상기 파단부(111)의 중앙측으로 이격된 위치에 형성된 벤딩 노
치부(115)를 포함하고,

상기 캔부재(3) 내부의 내압 증가 시 상기 벤딩 노치부(115)는 벤딩
(bending)되는 것을 특징으로 하는 이차전지.

【청구항 2】

청구항 1에 있어서,

상기 파단 노치부(113)는 상기 안전벤트부(110)의 하부 표면으로부터 오목하게 형성되고,

상기 벤딩 노치부(115) 또한 상기 안전벤트부(110)의 하부 표면으로부터 오목하게 형성되며,

상기 벤딩 노치부(115)의 오목한 깊이는 상기 파단 노치부(113)의 오목한 깊이보다 얕은 것을 특징으로 하는 이차전지.

【청구항 3】

청구항 1에 있어서,

상기 파단 노치부(113)는 상기 파단부(111)의 상기 저면에 형성되는 것을 특징으로 하는 이차전지.

【청구항 4】

청구항 1에 있어서,

상기 탭캡 조립체(100)는,

상기 파단부(111)에 부착되며 상기 캔부재(3)의 내부 내압 증가 시 파단되어 전류를 차단하는 차단부(121)를 구비하는 전류차단장치(120)를 더 포함하는 것을 특징으로 하는 이차전지.

【청구항 5】

청구항 4에 있어서,

상기 전류차단장치(120)의 차단부(121)는 상기 파단부(111)에서 상기 벤딩

노치부(115)의 둘레 내측에 부착되는 것을 특징으로 하는 이차전지.

【청구항 6】

청구항 1에 있어서,

상기 벤딩 노치부(115)는 연속되는 원형 또는 타원형인 것을 특징으로 하는 이차전지.

【청구항 7】

청구항 1에 있어서,

상기 벤딩 노치부(115)는 불연속되는 원형 또는 타원형인 것을 특징으로 하는 이차전지.

【청구항 8】

청구항 1에 있어서,

상기 벤딩 노치부(115)는 서로 길이가 다르게 복수 개로 이루어지는 것을 특징으로 하는 이차전지.

【청구항 9】

청구항 1에 있어서,

상기 파단 노치부(113)는 불연속되는 원형 또는 타원형인 것을 특징으로 하는 이차전지.

【청구항 10】

청구항 1에 있어서,

상기 안전벤트부(110)의 테두리부(117)와 상기 파단부(111) 사이의 연결부(119)는 상측으로 벤딩(bending)된 것을 특징으로 하는 이차전지.

【청구항 11】

청구항 10에 있어서,

상기 파단부(111)는,

상기 연결부(119)에서 상기 벤딩 노치부(115) 측방향으로 높이가 낮아지게 기울어지는 경사부분(111a) 및

상기 경사부분의 단부에서 상기 파단부(111)의 중앙측으로 연장되고 수평한 형태의 수평부분(111b)을 포함하고,

상기 벤딩 노치부(115)는 상기 경사부분(111a)과 상기 수평부분(111b)의 경계에 위치하는 것을 특징으로 하는 이차전지.

【청구항 12】

청구항 1에 있어서,

상기 안전벤트부(110)의 테두리 단부와 상기 파단 노치부(113) 사이의 이격 거리와 상기 파단 노치부(113)와 상기 벤딩 노치부(115) 사이의 이격 거리는 동일한 것을 특징으로 하는 이차전지.

【요약서】**【요약】**

본 발명은 벤팅(venting)압 산포를 개선할 수 있는 이차전지에 관한 것이다.

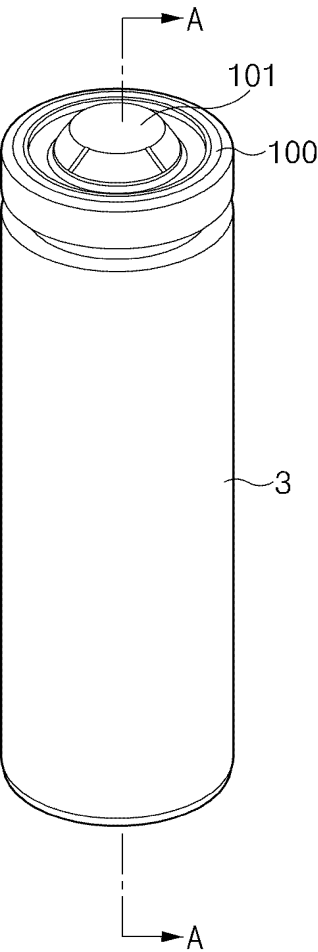
또한, 본 발명은 전극 조립체, 상기 전극 조립체를 내부에 수용하는 캔부재 및 상기 캔부재의 개구부를 덮는 탑캡 조립체를 포함하는 이차전지에 있어서, 상기 탑캡 조립체는 최상부에 위치되는 탑캡과 상기 탑캡의 하부에 위치되는 안전벤트부를 포함하고, 상기 안전벤트부는, 상기 캔부재 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부, 상기 파단부의 경계를 형성하는 파단 노치부 및 상기 캔부재의 내측을 바라보는 상기 파단부의 저면에서, 상기 파단 노치부로부터 상기 파단부의 중앙측으로 이격된 위치에 형성된 벤딩 노치부를 포함하고, 상기 캔부재 내부의 내압 증가 시 상기 벤딩 노치부는 벤딩(bending)되는 것을 특징으로 한다.

【대표도】

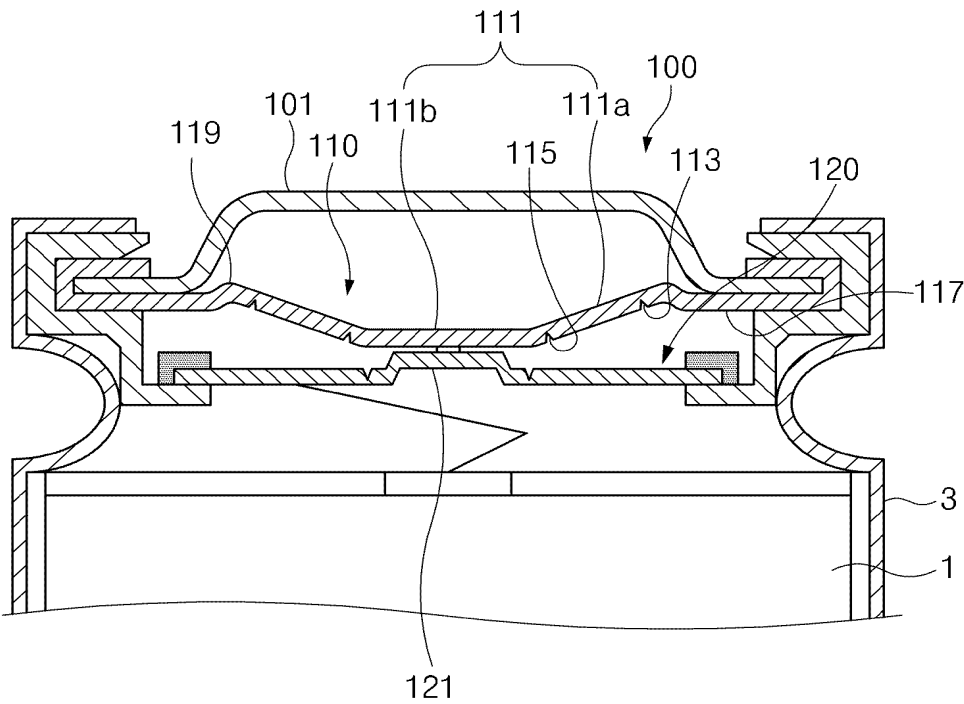
도 3

【도면】

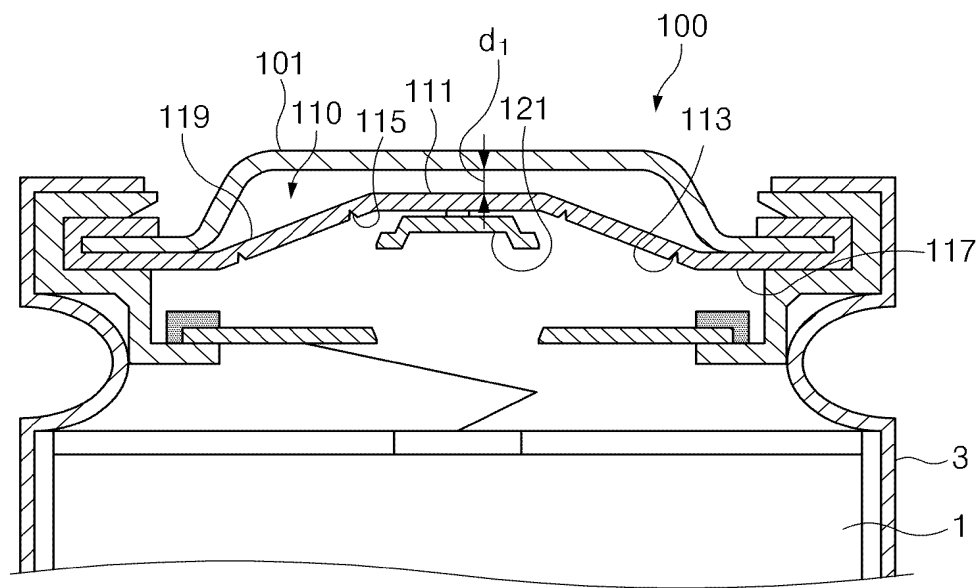
【도 1】



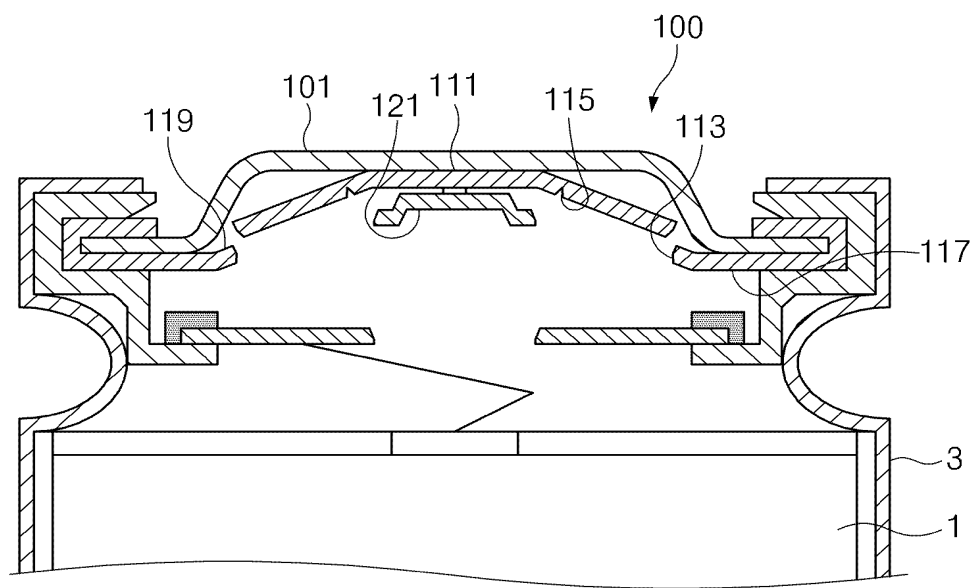
【도 2】



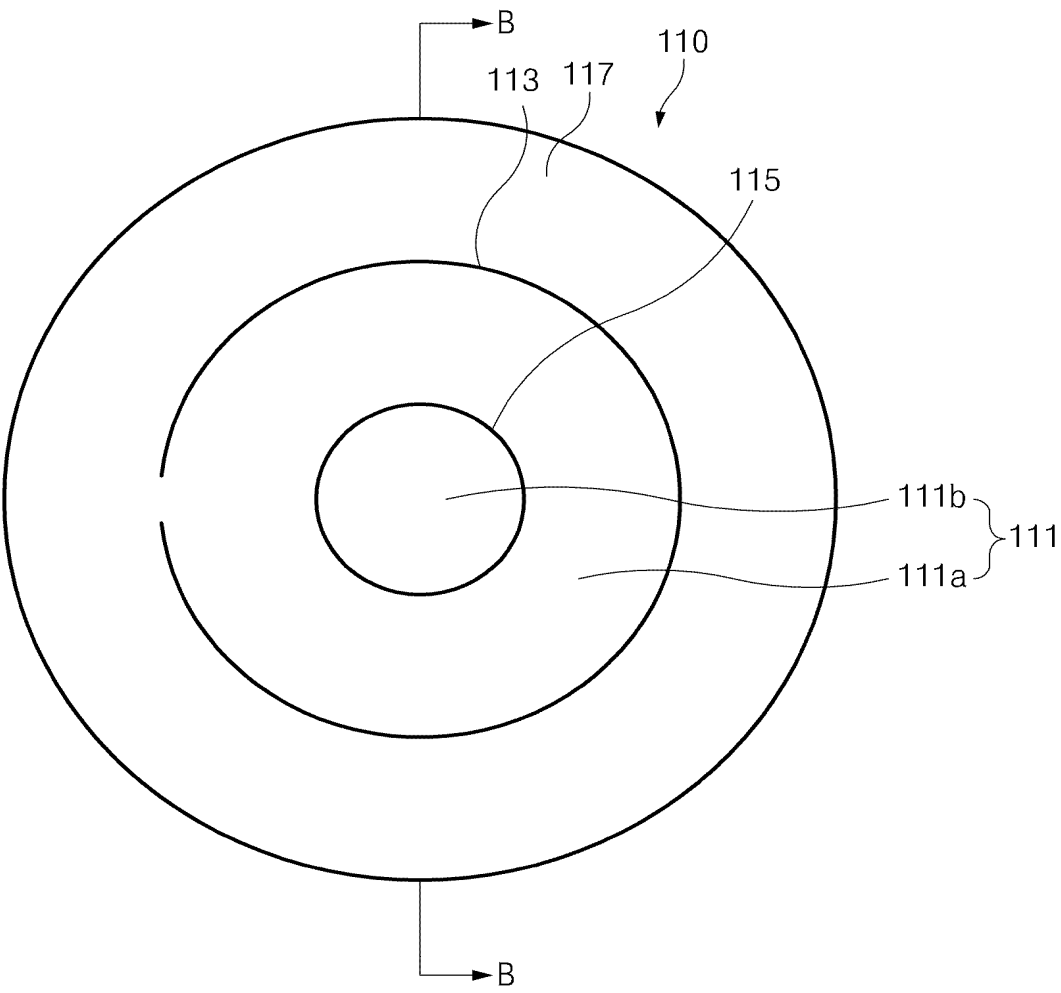
【도 3】



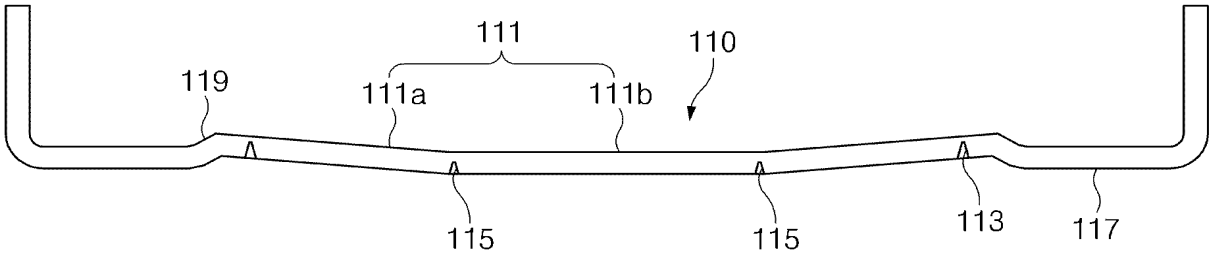
【도 4】



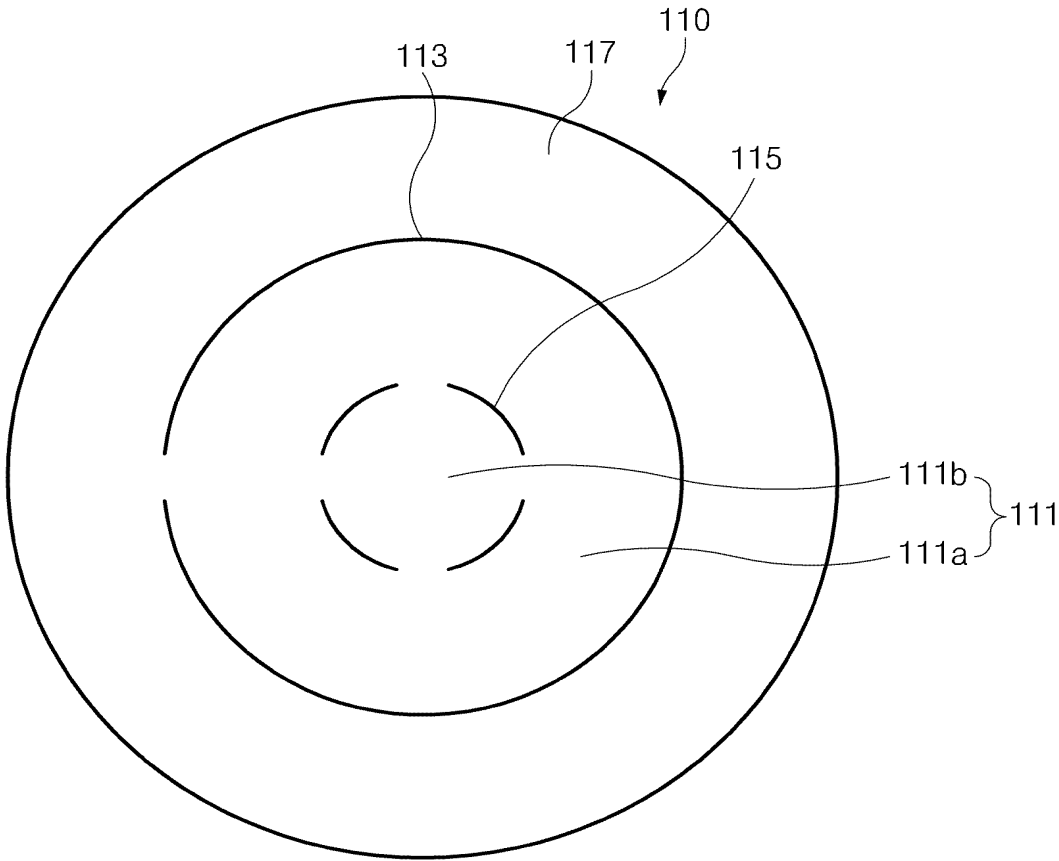
【도 5】



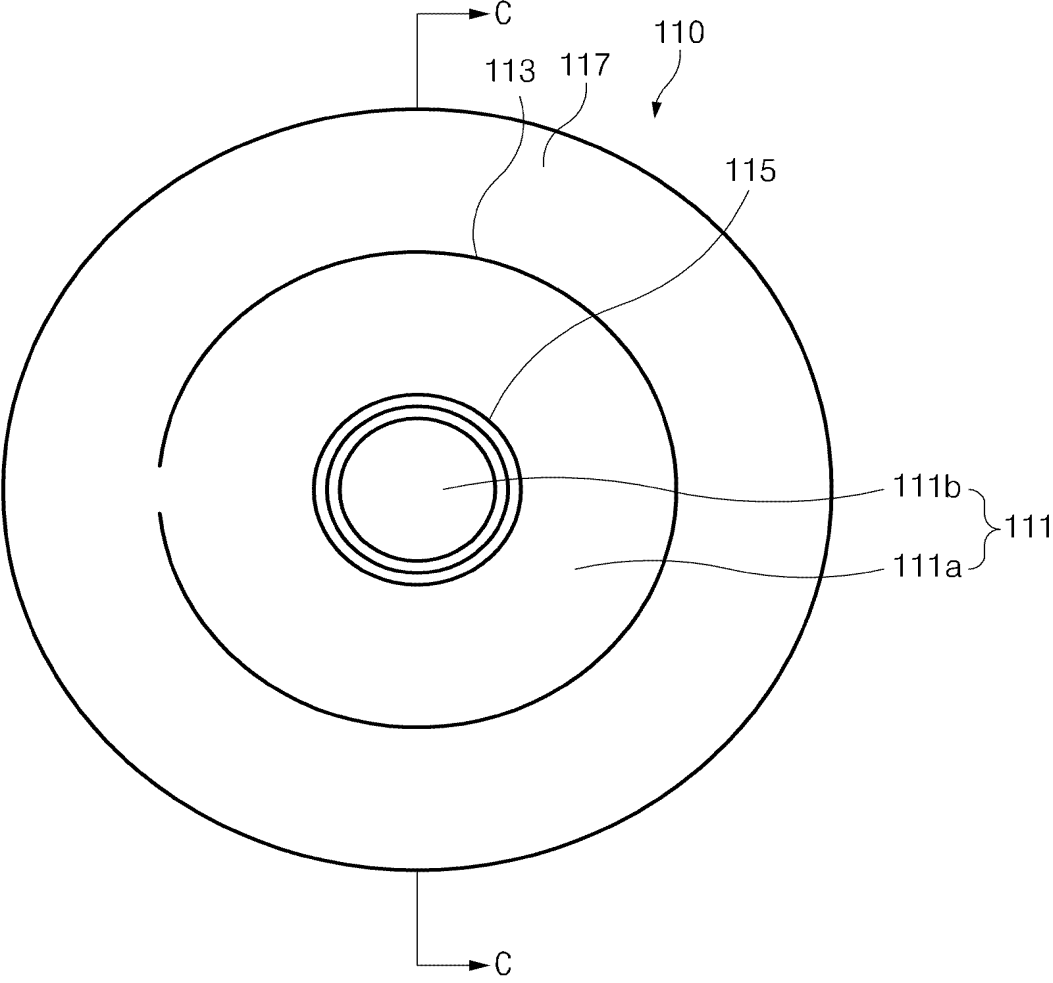
【도 6】



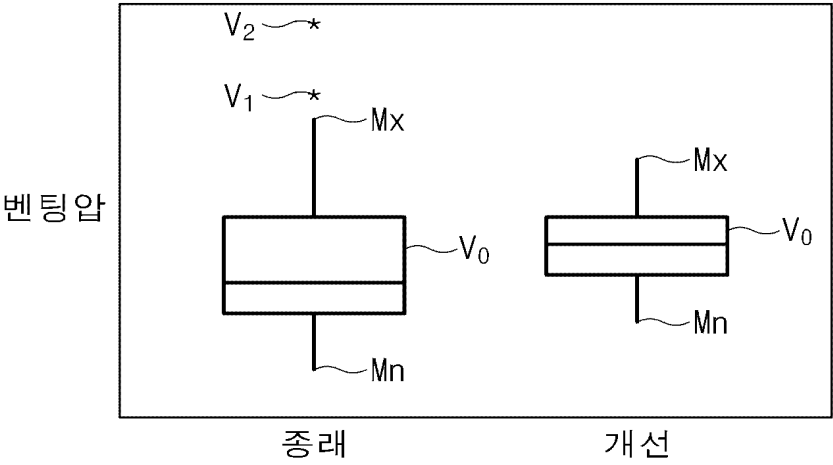
【도 7】



【도 8】



【도 11】



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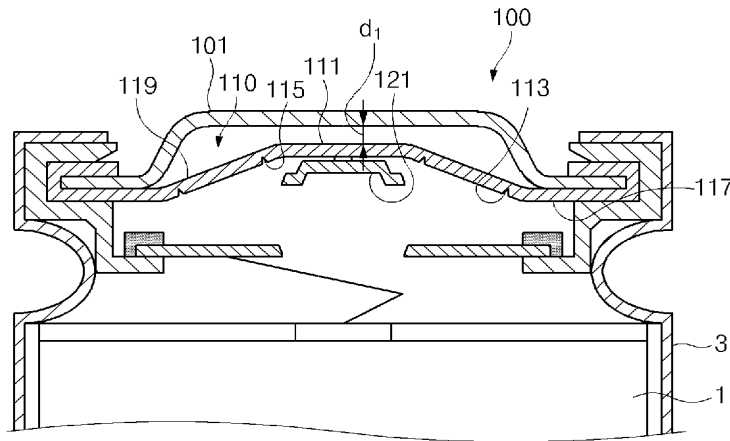
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- (72) 발명자: 민건우 (MIN, Geon Woo); 34122 대전시 유성구 문지로 188 LG화학 기술연구원, Daejeon (KR). 이제준 (LEE, Je Jun); 34122 대전시 유성구 문지로 188 LG화학 기술연구원, Daejeon (KR). 정상석 (JUNG, Sang Suk); 34122 대전시 유성구 문지로 188 LG화학 기술연구원, Daejeon (KR).
- (74) 대리인: 특허법인 태평양 (BAE, KIM & LEE IP GROUP); 06626 서울시 서초구 강남대로 343, 11층, Seoul (KR).
- (81) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.
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(54) Title: SECONDARY BATTERY

(54) 발명의 명칭: 이차전지



(57) **Abstract:** The present invention relates to a secondary battery capable of improving a venting pressure distribution. In addition, the present invention relates to a secondary battery comprising: an electrode assembly; a can member for accommodating the electrode assembly therein; and a top cap assembly for covering the opening of the can member, wherein the top cap assembly comprises a top cap positioned on a top part and a safety vent part positioned below the top cap, and the safety vent part comprises a breaking part which breaks when the internal pressure in the can member increases above a predetermined level; a breaking notch part for forming a boundary of the breaking part; and a bending notch part formed at a position spaced apart from the breaking notch part toward the center of the breaking part, on the bottom surface of the breaking part facing the inside of the can member, and the bending notch part is bent when the internal pressure in the can member increases.

(BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML,
MR, NE, SN, TD, TG).

공개:

— 국제조사보고서와 함께 (조약 제21조(3))

(57) 요약서: 본 발명은 벤팅(venting)압 산포를 개선할 수 있는 이차전지에 관한 것이다. 또한, 본 발명은 전극 조립체, 상기 전극 조립체를 내부에 수용하는 캔부재 및 상기 캔부재의 개구부를 덮는 탭캡 조립체를 포함하는 이차전지에 있어서, 상기 탭캡 조립체는 최상부에 위치되는 탭캡과 상기 탭캡의 하부에 위치되는 안전벤트부를 포함하고, 상기 안전벤트부는, 상기 캔부재 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부, 상기 파단부의 경계를 형성하는 파단 노치부 및 상기 캔부재의 내측을 바라보는 상기 파단부의 저면에서, 상기 파단 노치부로부터 상기 파단부의 중앙측으로 이격된 위치에 형성된 벤딩 노치부를 포함하고, 상기 캔부재 내부의 내압 증가 시 상기 벤딩 노치부는 벤딩(bending)되는 것을 특징으로 한다.

명세서

발명의 명칭: 이차전지

기술분야

[1] 관련출원과의 상호인용

[2] 본 출원은 2018년 11월 28일자 한국 특허 출원 제10-2018-0149369호에 기초한 우선권의 이익을 주장하며, 해당 한국 특허 출원의 문헌에 개시된 모든 내용은 본 명세서의 일부로서 포함된다.

[3] 기술분야

[4] 본 발명은 이차전지에 관한 것으로서, 보다 자세하게는 벤팅(venting)압 산포를 개선할 수 있는 이차전지에 관한 것이다.

배경기술

[5] 물질의 물리적 반응이나 화학적 반응을 통해 전기에너지를 생성시켜 외부로 전원을 공급하게 되는 전지(cell, battery)는 각종 전기전자 기기로 둘러싸여 있는 생활환경에 따라, 건물로 공급되는 교류 전원을 획득하지 못할 경우나 직류전원이 필요할 경우 사용하게 된다.

[6] 이와 같은 전지 중에서 화학적 반응을 이용하는 화학전지인 일차전지와 이차전지가 일반적으로 많이 사용되고 있는데, 일차전지는 건전지로 통칭되는 것으로 소모성 전지이다. 또한, 이차전지는 전류와 물질 사이의 산화환원과정이 다수 반복 가능한 소재를 사용하여 제조되는 재충전식 전지로서, 전류에 의해 소재에 대한 환원반응이 수행되면 전원이 충전되고, 소재에 대한 산화반응이 수행되면 전원이 방전되는데, 이와 같은 충전-방전이 반복적으로 수행되면서 전기가 생성되게 된다.

[7] 한편, 이차 전지 중 리튬 이온 전지는 양극 도전 포일과 음극 도전 포일에 각각 활물질을 일정한 두께로 코팅하고, 상기 양 도전 포일 사이에는 분리막이 개재되도록 하여 대략 젤리 롤(jelly roll) 내지는 원통 형태로 다수 회 권취하여 제작한 전극 조립체를 원통형 또는 각형 캔, 파우치 등에 수납하고 이를 밀봉 처리하여 제작된다.

[8] 대한민국 공개특허공보 제10-2011-0095118호에는 종래의 이차 전지가 개시되어 있다.

[9] 이차 전지는 내부 압력 증가 시에 탭캡 조립체를 통해 내부 가스를 외부로 배출하며, 탭캡 조립체는 내부 가스의 배출을 위한 안전벤트(vent)가 형성된다.

[10] 이러한 안전벤트는 탭캡 조립체와 캔의 사이에 마련된 벤트 플렌지부에 형성된 노치가 캔 내부 압력 증가 시 절개되어 가스 배출을 위한 통로를 형성하는 구성이다.

[11] 특히, 원형 전지는 과충전 등 이상 상황 발생 시 전지 폭발 및 발화를 방지하기 위하여 안전벤트와 같은 특정 부품에서 벤팅(venting)을 유도한다.

[12] 이러한 안전벤트는 기 설계 된 벤팅압에서 벤팅되지 않으면 안전문제가 발생할 수 있다.

[13] 그런데 종래의 안전벤트는 벤팅압 측정 시 안전벤트의 노치(notch)가 절개되기 전에 탑캡(top cap)에 의해 간섭을 받아 벤팅이 지연되는 문제점이 있었다.

발명의 상세한 설명

기술적 과제

[14] 따라서 본 발명은 위와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 과제는 탑캡과의 간섭을 방지하여 벤팅압 산포를 개선할 수 있는 이차전지를 제공하는 것이다.

과제 해결 수단

[15] 본 발명의 일 실시예에 따른 이차전지는 전극 조립체, 상기 전극 조립체를 내부에 수용하는 캔부재 및 상기 캔부재의 개구부를 덮는 탑캡 조립체를 포함하는 이차전지에 있어서, 상기 탑캡 조립체는 최상부에 위치되는 탑캡과 상기 탑캡의 하부에 위치되는 안전벤트부를 포함하고, 상기 안전벤트부는, 상기 캔부재 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부, 상기 파단부의 경계를 형성하는 파단 노치부 및 상기 캔부재의 내측을 바라보는 상기 파단부의 저면에서, 상기 파단 노치부로부터 상기 파단부의 중앙측으로 이격된 위치에 형성된 벤딩 노치부를 포함하고, 상기 캔부재 내부의 내압 증가 시 상기 벤딩 노치부는 벤딩(bending)되는 것을 특징으로 한다.

[16] 상기 파단 노치부는 상기 안전벤트부의 하부 표면으로부터 오목하게 형성되고, 상기 벤딩 노치부 또한 상기 안전벤트부의 하부 표면으로부터 오목하게 형성되며, 상기 벤딩 노치부의 오목한 깊이는 상기 파단 노치부의 오목한 깊이보다 얕을 수 있다.

[17] 상기 파단 노치부는 상기 파단부의 상기 저면에 형성될 수 있다.

[18] 상기 탑캡 조립체는 상기 파단부에 부착되며 상기 캔부재의 내부 내압 증가 시 파단되어 전류를 차단하는 차단부를 구비하는 전류차단장치를 더 포함할 수 있다.

[19] 상기 전류차단장치의 차단부는 상기 파단부에서 상기 벤딩 노치부의 둘레 내측에 부착될 수 있다.

[20] 상기 벤딩 노치부는 연속되는 원형 또는 타원형일 수 있다.

[21] 상기 벤딩 노치부는 불연속되는 원형 또는 타원형일 수 있다.

[22] 상기 벤딩 노치부는 서로 길이가 다르게 복수 개로 이루어질 수 있다.

[23] 상기 파단 노치부는 불연속되는 원형 또는 타원형일 수 있다.

[24] 상기 안전벤트부의 테두리부와 상기 파단부 사이의 연결부는 상측으로 벤딩(bending)될 수 있다.

[25] 상기 파단부는 상기 연결부에서 상기 벤딩 노치부 측방향으로 높이가 낮아지게 기울어지는 경사부분 및 상기 경사부분의 단부에서 상기 파단부의 중앙측으로

연장되고 수평한 형태의 수평부분을 포함하고, 상기 벤딩 노치부는 상기 경사부분과 상기 수평부분의 경계에 위치할 수 있다.

- [26] 상기 안전벤트부의 테두리 단부와 상기 파단 노치부 사이의 이격 거리와 상기 파단 노치부와 상기 벤딩 노치부 사이의 이격 거리는 동일할 수 있다.

발명의 효과

- [27] 본 발명에 따르면, 안전벤트부의 저면에 파단 노치부 외에 벤딩 노치부를 추가로 형성하여 캔부재 내부 내압 상승 시 안전벤트부가 탭캡에 간섭되지 않게 하는 효과가 있다.
- [28] 또한, 안전벤트부가 탭캡에 간섭되지 않아 파단 노치부가 기 설정된 벤팅압에서 안정되게 벤팅되게 함에 따라 벤팅압 산포를 개선하는 효과가 있다.

도면의 간단한 설명

- [29] 도 1은 본 발명의 제1 실시예에 따른 이차전지의 사시도이다.
- [30] 도 2는 도 1을 A-A 방향으로 절단하여 이차전지의 상부 내부의 주요부를 도시한 부분 단면도이다.
- [31] 도 3은 도 2에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.
- [32] 도 4는 도 3에서 안전벤트부의 파단 노치부가 절단되어 벤팅되는 것을 도시한 사용상태도이다.
- [33] 도 5는 본 발명의 제1 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [34] 도 6은 도 5를 B-B 방향으로 절단하여 도시한 단면도이다.
- [35] 도 7은 본 발명의 제2 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [36] 도 8은 본 발명의 제3 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [37] 도 9는 도 8을 C-C 방향으로 절단하여 도시한 단면도이다.
- [38] 도 10은 종래의 안전벤트부가 설치된 이차전지를 도 3과 같은 단면에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.
- [39] 도 11은 종래의 안전벤트부가 설치된 이차전지와 본 발명의 안전벤트부가 설치된 이차전지의 벤팅압 산포를 비교하여 나타낸 산포도이다.

발명의 실시를 위한 형태

- [40] 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예에 따른 이차전지에 대하여 상세히 설명하기로 한다.

[41]

- [42] 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다. 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의

가장 바람직한 일 실시예에 불과하고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물이 있을 수 있음을 이해하여야 한다.

[43] 도면에서 각 구성요소 또는 그 구성요소를 이루는 특정 부분의 크기는 설명의 편의 및 명확성을 위하여 과장되거나 생략되거나 또는 개략적으로 도시되었다. 따라서, 각 구성요소의 크기는 실제크기를 전적으로 반영하는 것은 아니다. 관련된 공지기능 혹은 구성에 대한 구체적인 설명이 본 발명의 요지를 불필요하게 흐릴 수 있다고 판단되는 경우, 그러한 설명은 생략하도록 한다.

[44]

[45] 도 1은 본 발명의 제1 실시예에 따른 이차전지의 사시도이고, 도 2는 도 1을 A-A 방향으로 절단하여 이차전지의 상부 내부의 주요부를 도시한 부분 단면도이다.

[46] 도 1 및 도 2에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 전극 조립체(1), 상기 전극 조립체(1)를 내부에 수용하는 캔부재(3) 및 상기 캔부재(3)의 개구부를 덮는 탭캡 조립체(100)를 포함하는 이차전지에 있어서, 상기 탭캡 조립체(100)는 최상부에 위치되는 탭캡(101)과 상기 탭캡(101)의 하부에 위치되는 안전벤트부(110)를 포함하고, 상기 안전벤트부(110)는 상기 캔부재(3) 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는 파단부(111), 상기 파단부(111)의 경계를 형성하는 파단 노치부(113) 및 상기 캔부재(3)의 내측을 바라보는 상기 파단부(111)의 저면에서, 상기 파단 노치부(113)로부터 상기 파단부(111)의 중앙측으로 이격된 위치에 형성된 벤딩 노치부(115)를 포함하고, 상기 캔부재(3) 내부의 내압 증가 시 상기 벤딩 노치부(115)는 벤딩(bending)되는 것을 특징으로 한다.

[47] 전극 조립체(1)는, 예를 들어, 양극 활물질이 도포된 양극과 음극 활물질이 도포된 음극 및 양극과 음극 사이에 개재된 분리막을 복수 회 적층하여 제작할 수 있다.

[48] 그리고 전극 조립체(1)는 양극과 분리막 및 음극을 적층한 적층체를 젤리 롤 형태로 권취하여 제작할 수도 있다.

[49] 캔부재(3)는 원형 리튬이온 2차 전지에서 대략 원통형의 위쪽이 개방된 형상을 가진 금속재질의 용기이거나 각형 리튬이온 2차 전지에 대략 각형의 위쪽이 개방된 형상을 가진 금속재질의 용기일 수 있으며, 일반적으로 가볍고 부식에 대처가 용이한 알루미늄 또는 알루미늄 합금이 사용된다.

[50] 캔부재(3)는 전극 조립체(1)와 전해액(미도시)의 용기가 되고, 전극 조립체(1)가 캔부재(3)의 개방된 상단, 즉, 상단개구부를 통해 캔부재(3)에 삽입된 뒤 캔부재(3)의 상단개구부는 탭캡 조립체(100)에 의해 봉해진다.

[51] 탭캡 조립체(100)는 캔부재(3)의 내부 내압 증가 시 파단되어 전류를 차단하는 차단부(121)를 구비하는 전류차단장치(120)를 더 포함할 수 있다.

[52] 도 3은 도 2에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가

파단된 것을 도시한 사용상태도이다.

- [53] 도 3을 참조하면, 전류차단장치(120)는 차단부(121)를 안전벤트부(110)의 파단부(111)에 부착하는데 캔부재(3)의 내부 내압 증가 시 차단부(121)가 전류차단장치(120)로부터 파단되면서 안전벤트부(110)의 파단부(111)와 함께 탭캡 조립체(100)의 탭캡(101) 방향인 상측으로 이동하게 된다.
- [54] 본 발명에서는 전류차단장치(120)의 차단부(121)가 파단부(111)에서 벤딩 노치부(115)의 둘레 내측의 정중앙에 부착될 수 있다.
- [55] 안전벤트부(110)의 파단 노치부(113)는 캔부재(3)의 내부 내압 증가 시 절개되기 위해 안전벤트부(110)의 표면으로부터 오목하게 형성된다.
- [56] 파단 노치부(113)는 안전벤트부(110)의 상면과 저면 중 어느 하나에 형성될 수 있는데, 안전벤트부(110)가 캔부재(3)의 내부측을 바라보는 안전벤트부(110)의 저면에 형성되면 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 상승했을 때 탭캡(101)과의 이격 거리(d)를 확보하는데 유리 할 수 있다.
- [57] 벤딩 노치부(115)는 파단 노치부(113)와 마찬가지로 안전벤트부(110)의 표면으로부터 오목하게 만입된 형태(또는 파인 형태)로 형성되는데 벤딩 노치부(115)의 오목한 깊이는 파단 노치부(113)의 오목한 깊이보다 얕을 수 있다.
- [58] 이러한 이유는 벤딩 노치부(115)가 캔부재(3)의 내부 내압에 의해 절개되지 않고 벤딩(bending)되게 하기 위해 노치(notch)의 두께를 절개되지 않을 정도의 두께로 유지하기 위함 일 수 있다.
- [59] 그리고 벤딩 노치부(115)는 안전벤트부(110)의 저면에 형성되어 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 이동하였을 때에 탭캡(101)과의 이격거리(d)를 확보하게 할 수 있다.
- [60] 도 4는 도 3에서 안전벤트부의 파단 노치부가 절단되어 벤딩되는 것을 도시한 사용상태도이다.
- [61] 도 4에 도시된 바와 같이, 안전벤트부(110)가 캔부재(3)의 내부 내압의 의해 상측으로 이동하였을 때 안전벤트부(110)의 저면에 형성된 벤딩 노치부(115)의 벤딩에 의해 탭캡(101)과의 이격거리(d)를 확보하게 되면 파단 노치부(113)가 기 설정된 벤팅(venting)압에 의해 절개되어 안전하게 캔부재(3)의 내부 가스를 외부로 배출(venting)하게 할 수 있다.
- [62] 도 5는 본 발명의 제1 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [63] 도 5에 도시된 바와 같이, 본 발명의 일 실시예에 따르면 파단 노치부(113)는 안전벤트부(110)의 저면에 불연속되는 원형 또는 타원형으로 형성될 수 있다.
- [64] 이와 같이 파단 노치부(113)는 안전벤트부(110)의 저면의 형상에 대응되게 형성될 수 있다. 즉, 안전벤트부(110)의 저면의 형상이 원형이면 파단 노치부(113)의 형상도 원형을 형성할 수 있고, 안전벤트부(110)의 저면의 형상이 타원형이면 파단 노치부(113)의 형상도 타원형을 형성할 수 있다.
- [65] 벤딩 노치부(115)는 안전벤트부(110)의 저면에 연속되는 원형 또는 타원형을 형성할 수 있다.

- [66] 벤딩 노치부(115)도 파단 노치부(113)와 마찬가지로 안전벤트부(110)의 저면의 형상에 대응되게 형성될 수 있다.
- [67] 또한, 벤딩 노치부(115)는 파단 노치부(113)보다 둘레의 크기가 작게 형성되어 안전벤트부(110)의 저면에서 파단 노치부(113)의 둘레 안쪽에 형성될 수 있다.
- [68] 한편, 안전벤트부(110)의 저면에서 보아 안전벤트부(110)의 저면 테두리 단부와 파단 노치부(113) 사이의 이격된 거리와 파단 노치부(113)와 벤딩 노치부(115) 사이의 이격된 거리를 어느 일측으로 치우치지 않고 동일하게 한다면 안전벤트부(110)가 캔부재(3)의 내부 내압에 의해 상측으로 이동하였을 때 안전벤트부(110)와 탭캡(101)과의 이격 거리를 가장 효과적으로 확보하게 할 수 있다.
- [69] 도 6은 도 5를 B-B 방향으로 절단하여 도시한 단면도이다.
- [70] 도 6에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 안전벤트부(110)의 테두리부(117)와 파단부(111) 사이의 연결부(119)는 상측으로 벤딩(bending)될 수 있다.
- [71] 즉, 연결부(119)가 상측으로 벤딩되었기 때문에 캔부재(3) 내부의 내압에 의해 안전벤트부(110)가 상측으로 이동하여도 파단 노치부(113)가 쉽게 절개되지 않고 안전벤트부(110)가 상측으로 이동하게 할 수 있다.
- [72] 파단부(111)는 연결부(119)에서 벤딩 노치부(115) 방향으로 높이가 낮아지게 기울어지는 경사부분(111a)을 포함할 수 있다.
- [73] 그리고 경사부분(111a)의 단부에서 파단부(111)의 중앙측으로 연장되고 수평한 형태의 수평부분(111b)을 포함할 수 있다.
- [74] 그리고 벤딩 노치부(115)는 경사부분(111a)과 수평부분(111b)의 경계에 위치할 수 있다.
- [75] 즉, 경사부분(111a)이 수평부분(111b) 방향으로 높이가 낮아지게 기울어짐에 따라 연결부(119)가 상측으로 벤딩되어 높이가 높아진 것을 보상해 주어 파단부(111)의 수평부분(111b)이 가장 하측에 위치하면서 수평을 유지하게 할 수 있다.
- [76] 이와 같이 파단부(111)는 하측에 설치된 전류차단장치(120)의 차단부(121)와 결합하기 위해 수평부분(111b)이 안전벤트부(110)에서 가장 하측에 위치할 수 있다.
- [77] 또한, 벤딩 노치부(115)가 경사부분(111a)과 수평부분(111b)의 경계에 위치하게 되면, 탭캡(101)과의 최적의 이격거리(d)를 확보하도록 벤딩 노치부(115)가 벤딩 되도록 하는데 현저히 효과적일 수 있다.
- [78] 도 7은 본 발명의 제2 실시예에 따른 안전벤트부만을 도시한 저면도이다.
- [79] 도 7에 도시된 바와 같이, 본 발명의 제2 실시예에 따른 이차전지는 안전벤트부(110)의 벤딩 노치부(115)가 불연속되는 원형 또는 타원형을 형성할 수 있다.
- [80] 벤딩 노치부(115)가 불연속되는 원형 또는 타원형을 형성한다면 벤딩

노치부(115)에서 오목한 부분을 최소화할 수 있어 캔부재(3)의 내압 상승 시 벤딩 노치부(115)가 절개되는 것을 효과적으로 방지할 수 있다.

[81] 도 8은 본 발명의 제3 실시예에 따른 안전벤트부만을 도시한 저면도이고, 도 9는 도 8을 C-C 방향으로 절단하여 도시한 단면도이다.

[82] 도 8 및 도 9에 도시된 바와 같이, 본 발명의 제3 실시예에 따른 이차전지는 안전벤트부(110)의 벤딩 노치부(115)가 서로 길이가 다르게 복수 개로 형성될 수 있다.

[83] 벤딩 노치부(115)가 서로 길이가 다르게 복수 개로 형성되면 캔부재(3)의 순간 내압 상승에 의해 안전벤트부(110)가 급속한 속도로 상승하여도 상승 충격을 복수 개의 벤딩 노치부(115)가 벤딩되면서 완화시킬 수 있다.

[84] 즉, 순간 급격한 상승 충격에 의해 파단 노치부(113)가 기 설정된 벤팅압 이전의 압력에서 파단되는 것을 방지할 수 있다.

[85] 도 10은 비교예로서 제시된다. 종래의 안전벤트부가 설치된 이차전지를 도 3과 같은 단면에서 캔부재의 내압이 상승하여 전류차단장치로부터 차단부가 파단된 것을 도시한 사용상태도이다.

[86] 도 10에 도시된 바와 같이, 종래의 이차전지는 안전벤트부(110')에 파단 노치부(113')만 형성하며, 본원발명에서와 같은 벤딩 노치부를 형성하지 않고 만들어진다.

[87] 이러한 종래의 이차전지는 캔부재(3) 내부의 내압 상승 시 안전벤트부(110')가 탭캡(101') 방향으로 이동하는데, 가스의 내압에 의해 안전벤트부(110')에서 캔부재(3)와 결합된 테두리로부터 가장 거리가 먼 안전벤트부(110')의 중앙 부위가 가장 상측으로 솟아올라 수 있다.

[88] 따라서 안전벤트부(110')의 중앙 부위가 탭캡(101')의 인근까지 솟아올라 탭캡(101')과의 이격 거리(d_2)가 대략 123 μm 정도까지 근접할 수 있다. 또 더 나아가 안전벤트부(110')의 중앙 부위가 탭캡(101')에 닿을 수 있다.

[89] 이와 같이 안전벤트부(110')와 탭캡(101')과의 이격 거리(d_2)가 너무 가깝거나 닿게 되면, 파단 노치가 절개되어 내부 가스를 벤팅해야 하는 기 설정된 벤팅압에서 절개되지 않아 벤팅이 안정적으로 이루어지지 않는 경우가 발생할 수 있다.

[90] 이러한 경우 이차전지의 폭발, 발열, 발화 등의 안전 문제가 발생할 수 있다.

[91] 도 3에 도시된 바와 같이, 본 발명의 일실시예에 따른 이차전지는 캔부재(3) 내부의 내압 상승 시 파단부(111)의 경사부분(111a)이 상측으로 꺾이면서 안전벤트부(110)가 탭캡(101) 방향으로 이동하는데, 안전벤트부(110)가 탭캡(101)을 향하여 이동할 때 벤딩 노치부(115)가 꺾이면서(bending) 캔부재(3) 내부의 내압을 일부 흡수하여 파단부(111)의 수평부분(111b)이 탭캡(101) 방향으로 솟구치는 것을 방지할 수 있다.

[92] 따라서 안전벤트부(110)와 탭캡(101) 사이의 이격 거리(d_1)를 대략 186 μm 정도로 충분히 확보할 수 있다.

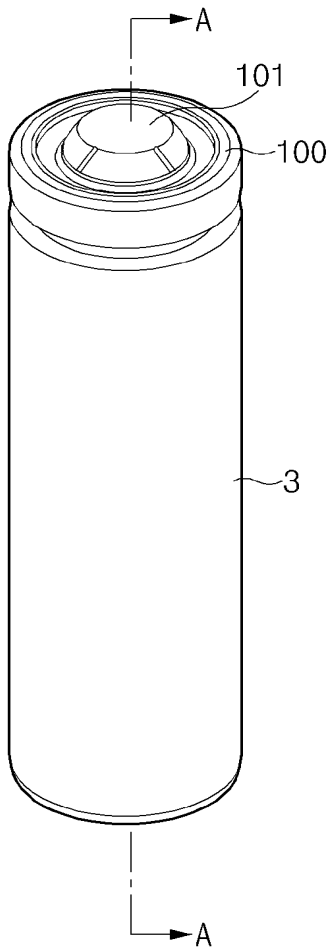
- [93] 이와 같이 안전벤트부(110)와 탐캡(101) 사이의 이격 거리(d_1)를 확보함에 따라, 기 설정된 벤팅압에서 파단 노치부(113)가 절개되는 것이 탐캡(101)에 의해 간섭되는 것을 방지할 수 있어 안정적으로 캔부재(3) 내부의 가스를 벤팅할 수 있다.
- [94] 도 11은 종래의 안전벤트부가 설치된 이차전지와 본 발명의 안전벤트부가 설치된 이차전지의 벤팅압 산포를 비교하여 나타낸 산포도이다.
- [95] 도 11에서 V_0 는 벤팅압의 산포 집합이고, M_x 는 벤팅압의 최대치이고 M_n 은 벤팅압의 최소치이며, V_1 , V_2 는 각각 벤팅압 산포의 일부분이다.
- [96] 도 11을 참조하여, 종래의 벤딩 노치부가 형성되지 않은 안전벤트부의 벤팅압과 개선된 본 발명의 벤딩 노치부가 형성된 안전벤트부의 벤팅압을 비교해보면, 종래의 안전벤트부의 벤팅압의 산포에 비해 본 발명의 안전벤트부의 벤팅압의 산포가 개선된 것을 알 수 있는데 그 차이는 대략 1.5배 정도의 차이가 날 수 있다.
- [97] 특히, 종래의 안전벤트부는 캔부재 내부 가스의 벤팅 과정에서 파단 노치부가 절개되지 않은 상태에서 안전벤트부의 중앙부가 탐캡에 닿게되어 기 설정된 벤팅압에서 정상적으로 벤팅이 되지 않는 것과 같은 비정상적인 벤팅 과정에서 벤팅압의 변화가 크게 발생하는 산포를 나타내는 것이 V_1 및 V_2 일 수 있다.
- [98]
- [99] 상술한 바와 같이 본 발명에 따르면, 안전벤트의 저면에 파단 노치부 외에 벤딩 노치부를 추가로 형성하여 캔부재 내부 내압 상승 시 안전벤트가 탐캡에 간섭되지 않게 하는 효과가 있다.
- [100] 또한, 안전벤트가 탐캡에 간섭되지 않아 파단 노치부가 기 설정된 벤팅압에서 벤팅되게 함에 따라 벤팅압 산포를 개선하는 효과가 있다.
- [101]
- [102] 이상과 같이 본 발명에 따른 이차전지를 예시된 도면을 참고하여 설명하였으나, 본 발명은 이상에서 설명된 실시예와 도면에 의해 한정되지 않으며, 특허청구범위 내에서 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자들에 의해 다양한 실시가 가능하다.

청구범위

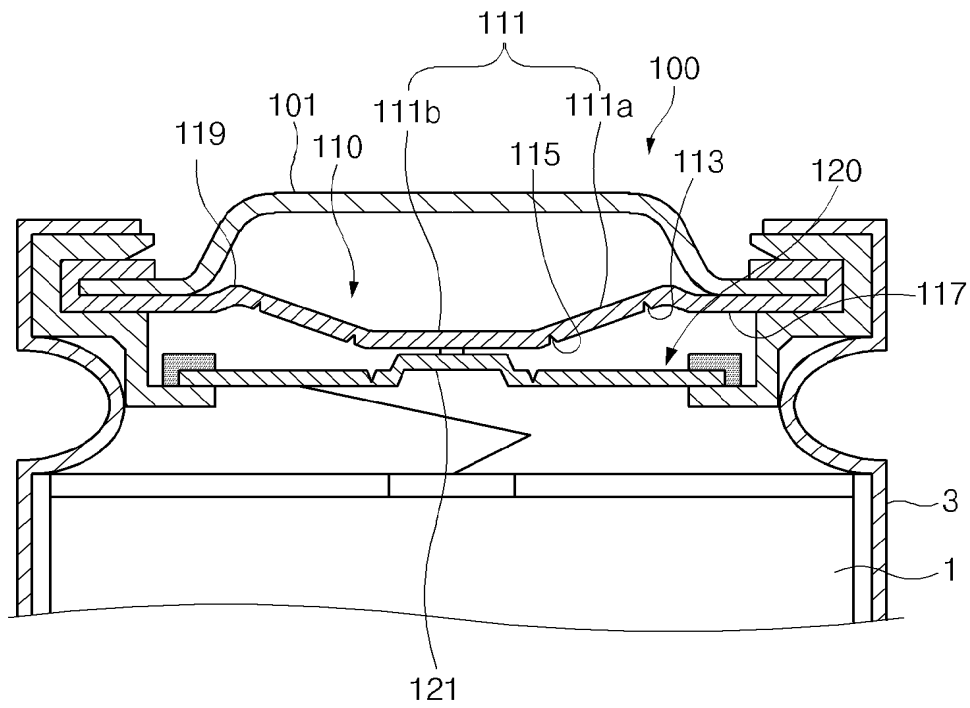
- [청구항 1] 전극 조립체(1);
 상기 전극 조립체(1)를 내부에 수용하는 캔부재(3); 및
 상기 캔부재(3)의 개구부를 덮는 탭캡 조립체(100); 를 포함하는
 이차전지에 있어서,
 상기 탭캡 조립체(100)는 최상부에 위치되는 탭캡(101)과 상기
 탭캡(101)의 하부에 위치되는 안전벤트부(110)를 포함하고,
 상기 안전벤트부(110)는,
 상기 캔부재(3) 내부의 내압이 소정의 압력 이상으로 증가 시 파단되는
 파단부(111);
 상기 파단부(111)의 경계를 형성하는 파단 노치부(113); 및
 상기 캔부재(3)의 내측을 바라보는 상기 파단부(111)의 저면에서, 상기
 파단 노치부(113)로부터 상기 파단부(111)의 중앙측으로 이격된 위치에
 형성된 벤딩 노치부(115)를 포함하고,
 상기 캔부재(3) 내부의 내압 증가 시 상기 벤딩 노치부(115)는
 벤딩(bending)되는 것을 특징으로 하는 이차전지.
- [청구항 2] 청구항 1에 있어서,
 상기 파단 노치부(113)는 상기 안전벤트부(110)의 하부 표면으로부터
 오목하게 형성되고,
 상기 벤딩 노치부(115) 또한 상기 안전벤트부(110)의 하부 표면으로부터
 오목하게 형성되며,
 상기 벤딩 노치부(115)의 오목한 깊이는 상기 파단 노치부(113)의 오목한
 깊이보다 얕은 것을 특징으로 하는 이차전지.
- [청구항 3] 청구항 1에 있어서,
 상기 파단 노치부(113)는 상기 파단부(111)의 상기 저면에 형성되는 것을
 특징으로 하는 이차전지.
- [청구항 4] 청구항 1에 있어서,
 상기 탭캡 조립체(100)는,
 상기 파단부(111)에 부착되며 상기 캔부재(3)의 내부 내압 증가 시
 파단되어 전류를 차단하는 차단부(121)를 구비하는 전류차단장치(120)를
 더 포함하는 것을 특징으로 하는 이차전지.
- [청구항 5] 청구항 4에 있어서,
 상기 전류차단장치(120)의 차단부(121)는 상기 파단부(111)에서 상기
 벤딩 노치부(115)의 둘레 내측에 부착되는 것을 특징으로 하는 이차전지.
- [청구항 6] 청구항 1에 있어서,
 상기 벤딩 노치부(115)는 연속되는 원형 또는 타원형인 것을 특징으로
 하는 이차전지.

- [청구항 7] 청구항 1에 있어서,
상기 벤딩 노치부(115)는 불연속되는 원형 또는 타원형인 것을 특징으로 하는 이차전지.
- [청구항 8] 청구항 1에 있어서,
상기 벤딩 노치부(115)는 서로 길이가 다르게 복수 개로 이루어지는 것을 특징으로 하는 이차전지.
- [청구항 9] 청구항 1에 있어서,
상기 파단 노치부(113)는 불연속되는 원형 또는 타원형인 것을 특징으로 하는 이차전지.
- [청구항 10] 청구항 1에 있어서,
상기 안전벤트부(110)의 테두리부(117)와 상기 파단부(111) 사이의 연결부(119)는 상측으로 벤딩(bending)된 것을 특징으로 하는 이차전지.
- [청구항 11] 청구항 10에 있어서,
상기 파단부(111)는,
상기 연결부(119)에서 상기 벤딩 노치부(115) 측방향으로 높이가 낮아지게 기울어지는 경사부분(111a) 및
상기 경사부분의 단부에서 상기 파단부(111)의 중앙측으로 연장되고 수평한 형태의 수평부분(111b)을 포함하고,
상기 벤딩 노치부(115)는 상기 경사부분(111a)과 상기 수평부분(111b)의 경계에 위치하는 것을 특징으로 하는 이차전지.
- [청구항 12] 청구항 1에 있어서,
상기 안전벤트부(110)의 테두리 단부와 상기 파단 노치부(113) 사이의 이격 거리와 상기 파단 노치부(113)와 상기 벤딩 노치부(115) 사이의 이격 거리는 동일한 것을 특징으로 하는 이차전지.

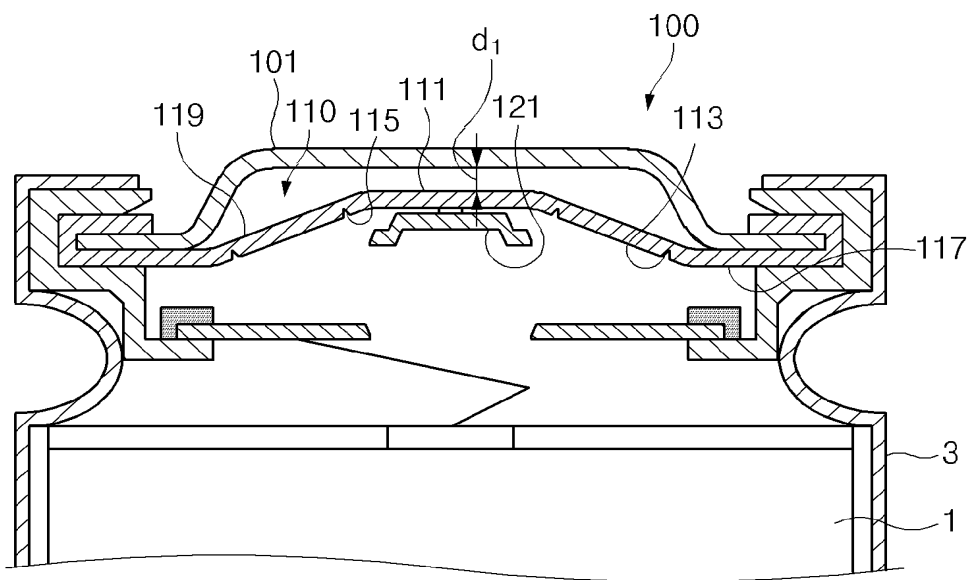
[도 1]



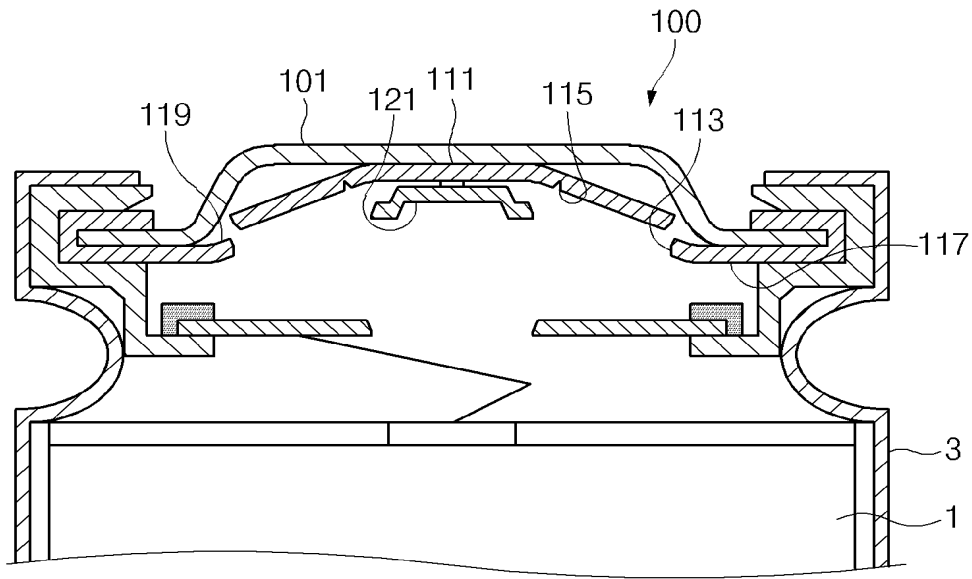
[도2]



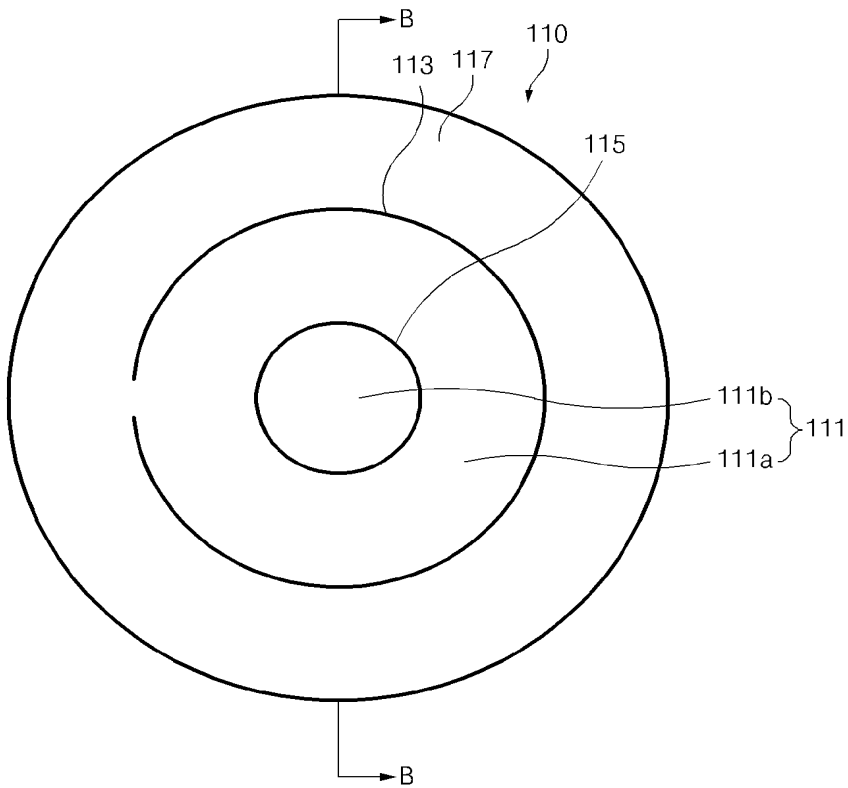
[도3]



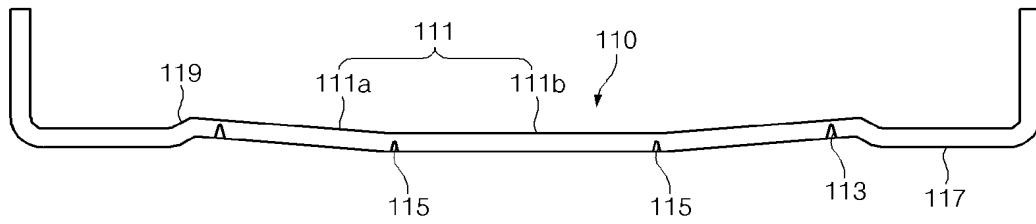
[도4]



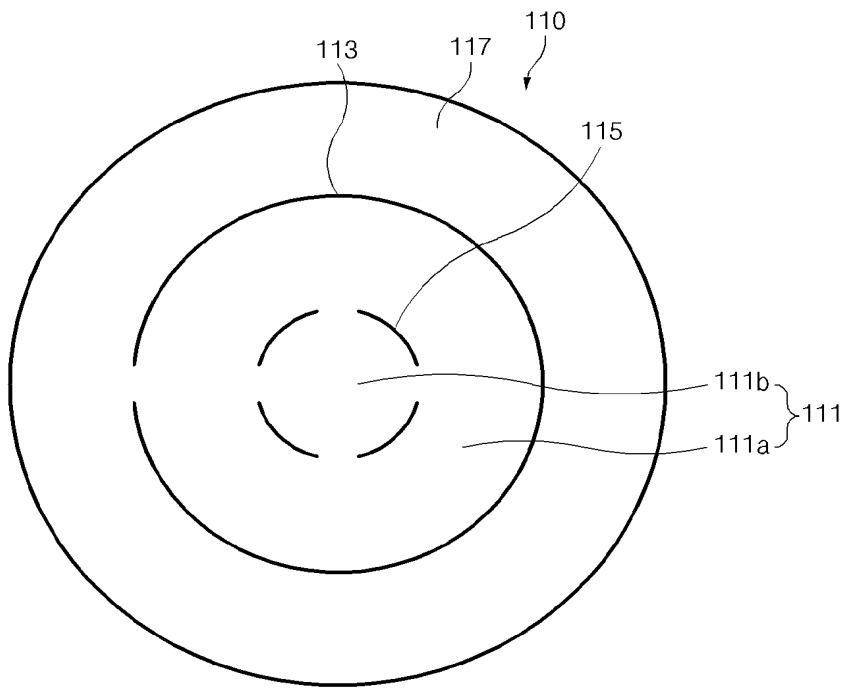
[도5]



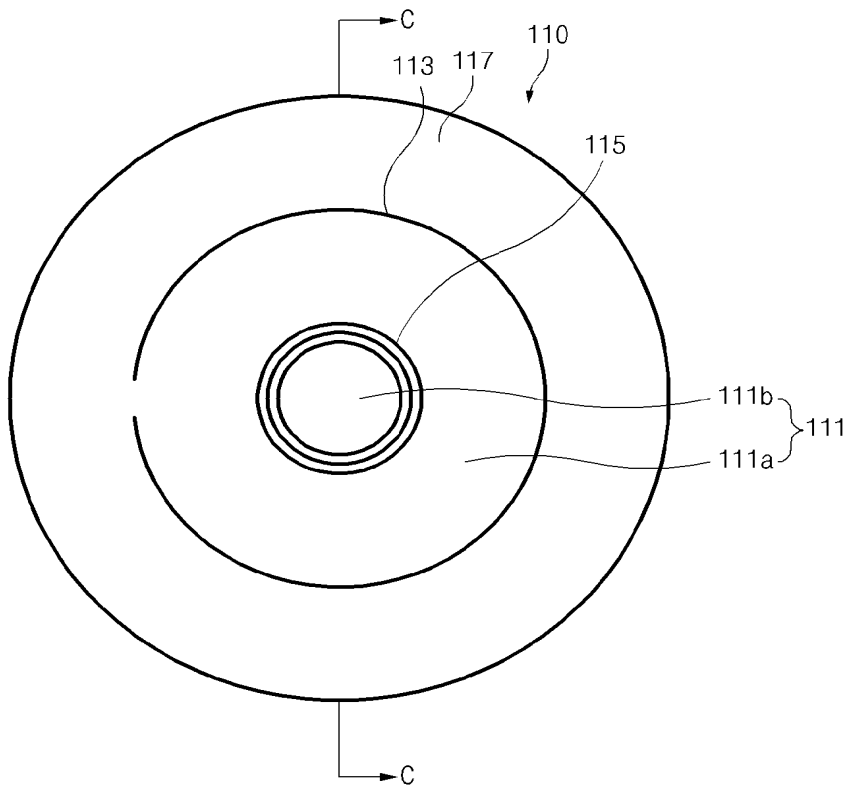
[도6]



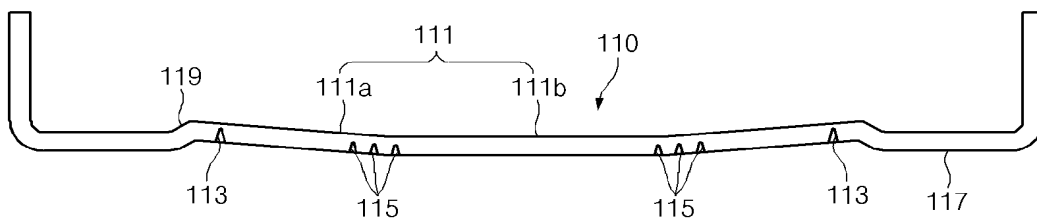
[도7]



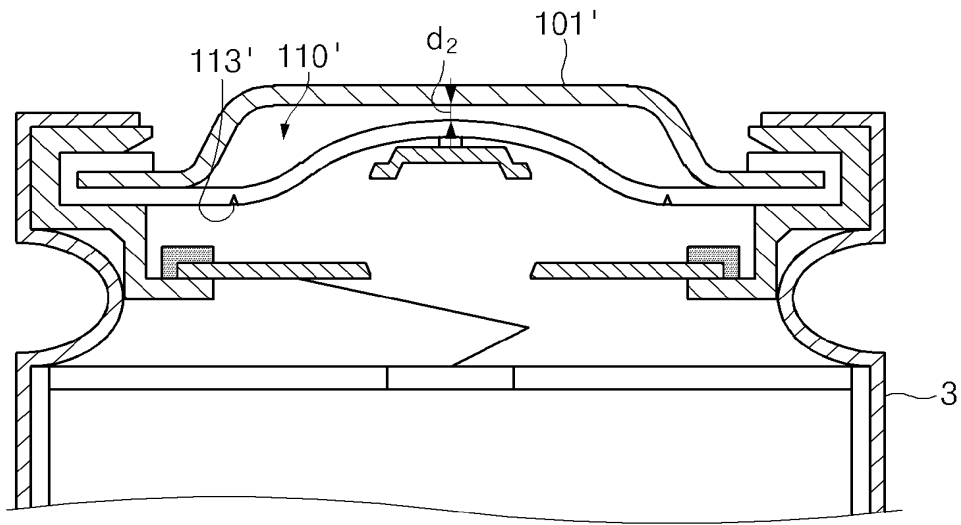
[도8]



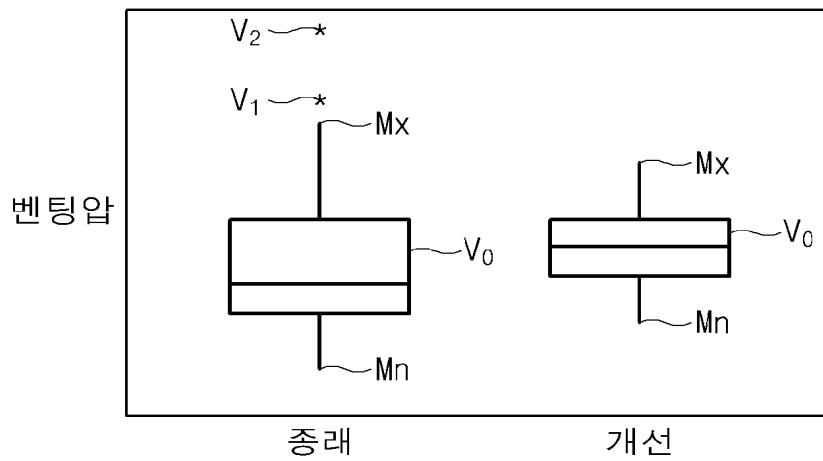
[도9]



[도10]



[도11]



INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2019/015396**A. CLASSIFICATION OF SUBJECT MATTER*****H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i***

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models: IPC as above

Japanese utility models and applications for utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: gas, notch, bending, fracture

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 05 August 1994 See paragraphs [0019]-[0022], claim 1 and figures 1-4.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 24 December 1996 See paragraph [0043] and figure 11.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 13 April 2001 See paragraphs [0016], [0017] and figures 1-5.	4-9
A	KR 10-2015-0035267 A (LG CHEM, LTD.) 06 April 2015 See the entire document.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 02 October 2018 See the entire document.	1-12



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

“A” document defining the general state of the art which is not considered to be of particular relevance

“E” earlier application or patent but published on or after the international filing date

“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

“O” document referring to an oral disclosure, use, exhibition or other means

“P” document published prior to the international filing date but later than the priority date claimed

“T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

“X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

“Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

“&” document member of the same patent family

Date of the actual completion of the international search

19 FEBRUARY 2020 (19.02.2020)

Date of mailing of the international search report

20 FEBRUARY 2020 (20.02.2020)

Name and mailing address of the ISA/KR

Korean Intellectual Property Office
Government Complex Daejeon Building 4, 189, Cheongsu-ro, Seo-gu,
Daejeon, 35208, Republic of Korea

Facsimile No. +82-42-481-8578

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.

PCT/KR2019/015396

Patent document cited in search report	Publication date	Patent family member	Publication date
JP 06-215746 A	05/08/1994	None	
JP 08-339793 A	24/12/1996	None	
JP 2001-102038 A	13/04/2001	None	
KR 10-2015-0035267 A	06/04/2015	KR 10-1667966 B1	20/10/2016
US 10090497 B2	02/10/2018	CN 107293687 A	24/10/2017
		KR 10-2017-0116370 A	19/10/2017
		US 2017-0294635 A1	12/10/2017

A. 발명이 속하는 기술분류(국제특허분류(IPC))

H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i

B. 조사된 분야

조사된 최소문헌(국제특허분류를 기재)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

조사된 기술분야에 속하는 최소문헌 이외의 문헌

한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우))

eKOMPASS(특허청 내부 검색시스템) & 키워드: 가스(gas), 노치(notch), 벤딩(bending), 파단(fracture)

C. 관련 문헌

카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 1994.08.05 단락 [0019]-[0022], 청구항 1 및 도면 1-4 참조.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 1996.12.24 단락 [0043] 및 도면 11 참조.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 2001.04.13 단락 [0016],[0017] 및 도면 1-5 참조.	4-9
A	KR 10-2015-0035267 A (주식회사 엘지화학) 2015.04.06 전체 문헌 참조.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 2018.10.02 전체 문헌 참조.	1-12

☐ 추가 문헌이 C(계속)에 기재되어 있습니다.

☒ 대응특허에 관한 별지를 참조하십시오.

* 인용된 문헌의 특별 카테고리:

“A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌

“D” 본 국제출원에서 출원인이 인용한 문헌

“E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후 “X”에 공개된 선출원 또는 특허 문헌

“L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌

“O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌

“P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌

“T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌

“X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다.

“Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다.

“&” 동일한 대응특허문헌에 속하는 문헌

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대한민국 특허청
(35208) 대전광역시 서구 청사로 189,
4동 (둔산동, 정부대전청사)

팩스 번호 +82-42-481-8578

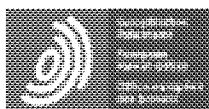
심사관

김연경

전화번호 +82-42-481-3325



국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
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JP 08-339793 A	1996/12/24	없음	
JP 2001-102038 A	2001/04/13	없음	
KR 10-2015-0035267 A	2015/04/06	KR 10-1667966 B1	2016/10/20
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		US 2017-0294635 A1	2017/10/12



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SAFETY DEVICE IN ENCLOSED CELL AND LEAD MOUNTING PLATE IN THE SAME

Inventor(s): SAIJO KINJI; OKAMOTO HIROAKI; KAWAMURA HIROAKI; KANEKO AKIRA ± (SAIJO KINJI, ; OKAMOTO HIROAKI, ; KAWAMURA HIROAKI, ; KANEKO AKIRA)

Applicant(s): TOYO KOHAN CO LTD ± (TOYO KOHAN CO LTD)

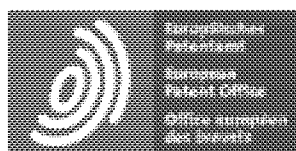
Classification: - **international:** *H01M2/12; H01M2/26; H01M2/34;*
(IPC1-7): H01M2/34
- **cooperative:** Y02E60/10 (EP)

Application number: JP19990281142 19991001

Priority number (s): JP19990281142 19991001

Abstract of JP2001102038 (A)

PROBLEM TO BE SOLVED: To provide an enclosed cell with a safety device for preventing explosion caused by a sudden rising of internal pressure that may occur in overcharging and short-circuiting, and method for fabricating a lead mounting plate.
SOLUTION: The positive pole cover 16 is installed in an external vessel 11, and comprises a lead mounting plate 19 forming an inmost cover and being connected to the positive pole 13 of an electrode body 12 through a positive pole lead 18, a shield plate 21 forming a middle cover and being connected electrically the lead mounting plate 19 through a central welding part 20, the shield plate being deformed in the applying direction of an internal force dependent upon the rising of the internal pressure in a inner space of battery, and sealing plate 22 forming an outmost cover and being connected electrically to the shield plate 21. A gas passage and a thin wall 26 are formed at peripherals of the central welding part 20 of the lead mounting plate 19, if the shield plate 19 is deformed by the rising of internal pressure, the current is interrupted as the thin wall 26 is broken, the lead mounting plate 19 includes a thin base member 19a and a thin foil member 19b, the thin wall 26 may be formed by only the thin foil member 19b as the thin base member 19a is cut and removed from it.



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DESCRIPTION JP2001102038A

[0001]

INDUSTRIAL APPLICABILITY The present invention relates to a safety device for a sealed battery having an explosion-proof function and a lead mounting plate for the safety device for a sealed battery using the same.

[0002]

PROBLEM TO BE SOLVED: In recent years, a secondary battery using a non-aqueous electrolyte solution for a lithium battery or a lithium ion battery using a non-aqueous electrolyte solution is widely used in a portable electronic device or the like.

While such a secondary battery is characterized by having a high electromotive force, the electrode body including the positive electrode and the negative electrode housed in the outer can undergoes a chemical change to increase the internal pressure, which may cause bursting. For example, when a non-aqueous electrolyte battery such as a lithium secondary battery is overcharged or a short circuit is caused by misuse and a large current flows, the non-aqueous electrolyte in the electrode body is decomposed and gas is generated. May occur. When such gas gradually fills the outer can and the internal pressure inside the outer can rises, the battery finally bursts.

[0003]

In order to prevent such a battery from exploding, various forms of sealed batteries have been conventionally developed, and as one form thereof, for example, a safety device described in JP-A-7-105933 is provided. There is a sealed battery.

As shown in FIGS. 10 and 11, the explosion-proof valve 60, which deforms in the internal pressure application direction as the battery internal pressure rises, and the lead body mounting member 61 are welded together with the explosion-proof valve 60 of the lead mounting member 61. A pair of pressure introduction ports 63 and a pair of thin-walled portions 64 are provided on the outer peripheral side portion of the welded portion 62 of the above, and the lead body 65 is attached to the lead attachment member 61. Note that FIG. 10 shows a state in which the thin portion 64 is broken.

[0004]

With the above configuration, in a normal state, the lead mounting member 61 and the explosion-proof valve 60 can be electrically connected to each other and function as a battery.

On the other hand, when the internal pressure of the battery rises due to overcharging or a short circuit and reaches a predetermined pressure, the explosion-proof valve 60 receives the internal pressure of the battery and is deformed in the internal pressure application direction, and the thin portion 64 provided on the lead mounting member 61 is formed. It breaks, the electrical connection between the lead body mounting member 61 and the explosion-proof valve 60 is lost, the current is cut off, further generation of gas inside the battery can be prevented, and the battery can be prevented from bursting. ...

[0005]

However, since the thin-walled portion 64 is formed on the lead mounting member 61 by press working or etching, the thickness of the thin-walled portion 64 is not constant, and as a result, the lead mounting is performed. The internal battery pressure that cuts off the electrical connection between the member 61 and the explosion-proof valve 60 becomes inconsistent, and depending on the sealed battery, the lead mounting member 61 and the explosion-proof valve 60 even though the battery internal pressure has reached the set breaking pressure. The electrical connection of the battery will not be interrupted, and the reliability of the safety of the sealed battery will be significantly impaired.

The present invention is intended to solve such a problem, and when the internal battery pressure rises above a certain level, the electrical connection between the lead mounting plate and the shielding plate can be reliably cut off, and the safety of the sealed battery. It is an object of the present invention to provide a safety device for a sealed battery and a lead mounting plate in the safety device that can sufficiently secure the above.

[0006]

[Means for Solving the Problems] In a sealed battery safety device according to the present invention for achieving the above object, a positive electrode lid attached to one end of an outer can is formed with an innermost lid and via a positive electrode lead. A lead mounting plate connected to the positive electrode of the electrode body and an intermediate lid are formed and electrically connected to the lead mounting plate via a central weld, and the internal pressure is applied in conjunction with an increase in internal pressure in the internal space of the battery. It is composed of a shielding plate that transforms into a shield plate and a sealing plate that forms the outermost lid and is electrically connected to the shielding plate, and a gas flow hole and a thin wall portion are provided around the central welded portion of the lead mounting plate to provide internal pressure. In a sealed battery safety device in which the thin-walled part breaks and cuts off the current when the shielding plate is deformed due to the rise of the lead mounting plate, the lead mounting plate is laminated with a thick-walled base material and a thin-walled foil material bonded to the lower surface thereof. It has a structure, and the gas flow hole is formed by integrally penetrating the base material and the foil material, and the thin-walled portion is formed by cutting off the base material and forming only from the foil material.

[0007]

Therefore, it is always ensured by welding the lead mounting plate to the central welded portion of the shielding plate in the closed space.

On the other hand, when the internal pressure inside the battery rises sharply and exceeds the set current cutoff pressure, the shield plate is deformed by the pressure of the decomposition gas and the thin part of the lead mounting plate made of foil material having a certain thickness is surely broken. Then, the electrical conduction between the lead mounting plate and the shielding plate is cut off to prevent further generation of the decomposed gas and prevent the decomposed gas from flowing out to the outside.

[0008]

The sealed battery safety device according to the first invention described above also has the following features.

The lead mounting plate is a shielding plate made of a clad metal plate in which a foil material made of nickel or the like is bonded to the lower surface of a base material made of aluminum or the like by clad bonding.

Here, the combinations of the base material and the foil material to be the clad metal plate are as follows. Either may be used as a base material or a foil material. That is, aluminum / aluminum, aluminum / stainless steel, stainless steel / stainless steel, stainless steel / nickel, nickel / nickel, copper / copper, copper / stainless steel, and the like. The thickness of the base material is preferably about 20 μm to 200 μm , and the thickness of the foil material is preferably about 10 to 100 μm . Further, the clad metal plate used as the lead mounting plate is, for example, an ultra-low pressure inert gas of 1×10^{-1} to 1×10^{-4} Torr, as disclosed earlier in Japanese Patent Application Laid-Open No. 1-224184 by the applicant. In an atmosphere, a metal substrate having a joint surface and a foil material are each grounded to one electrode A, and an AC of 1 to 50 MHz is applied between the other electrode B which is insulated and supported to perform glow discharge. It can be produced by sputter etching treatment so that the area of the electrode exposed in the plasma generated by the glow discharge is 1/3 or less of the area of the electrode B.

[0009]

The shielding plate is made of a clad metal plate in which a thin foil material made of nickel or the like is clad-bonded to the lower surface of a thick base material such as aluminum, and has an arc angle of 180° or more around the central welded portion surface of the base material. C-shaped elongated holes are concentrically provided, and strength-reducing elongated holes are formed at locations inside the C-shaped elongated holes of the base material, and both ends of the strength-reducing elongated holes and the C-shaped elongated holes are formed. A narrow bent portion is formed between the portion recessed from both ends by a predetermined distance, and a valve film is formed by the C-shaped elongated hole and the foil material portion corresponding to the strength reducing elongated hole, respectively, and inside the outer can. When the pressure of the lead mounting plate exceeds the set current breaking pressure, the shielding plate is bent in an upwardly convex state along the planned bending portion, breaks the thin portion of the lead mounting plate, and the electricity between the lead mounting plate and the shielding plate is broken. When the target connection is cut off and the pressure inside the current cutoff outer can exceeds the set film breaking pressure, the valve membrane is broken.

Here, preferably, the set current cutoff pressure is set to 4 to 5 kg / cm², and the set film breaking pressure is set to 20 kg / cm².

[0010]

By using such a cutoff plate, when the pressure inside the outer can exceeds the set current cutoff pressure, the cutoff plate is easily deformed and the thin portion of the lead mounting plate is easily broken to break the lead mounting plate and the cutoff plate. In spite of such electrical cutoff, the chemical reaction inside the outer can progresses and decomposition gas is generated, and the internal pressure rises further and is set. When the film breaking pressure is exceeded, the valve membrane provided on the shielding plate breaks, so that the decomposed gas is released to the outside through the gas flow hole, the operating space, the valve membrane, and the gas vent hole provided on the lead mounting plate. It is possible to prevent the sealed battery from exploding.

[0011]

Similar to the lead mounting plate, the clad metal plate used as the shielding plate is also, for example, 1×10^{-1} to 1×10^{-4} Torr, as disclosed by the applicant in Japanese Patent Application Laid-Open No. 1-224164. In an ultra-low pressure inert gas atmosphere, a metal substrate having a joint surface and a foil material are each grounded to one electrode A, and an AC of 1 to 50 MHz is applied between the other electrode B which is insulated and supported. The electrode can be manufactured by sputter etching so that the area of the electrode exposed to the plasma generated by the glow discharge is 1/3 or less of the area of the electrode B.

[0012]

A PTC thermistor element made of an annular plate is interposed between the shielding plate and the sealing plate, and the PTC thermistor element raises the temperature of the safety device of the sealed battery and makes it difficult for current to flow from this surface. Also tries to prevent explosion due to overcurrent.

The lead mounting plate in the safety device for the sealed battery according to the present invention for achieving the above object is formed by welding a pair of elongated holes for forming a thin wall portion in the base material at intervals by using a press. A pair of foil materials having the same shape as the base material are joined to one surface of the base material to form a laminated plate, and both ends of the elongated holes for forming a thin wall portion of the laminated plate are connected to each other by using a press. The gas flow holes are drilled at parallel intervals, and the thin wall portion is formed by the foil material portion corresponding to the long hole for forming the thin wall portion, and the lead of the central weld is formed in the island-shaped portion surrounded by the gas flow hole and the thin wall portion. A welded surface on the mounting plate side is formed.

[0013]

BEST MODE FOR CARRYING OUT THE INVENTION Hereinafter, the present invention will be specifically described with reference to one embodiment shown in the attached figure.

First, the configuration of the safety device for the sealed battery according to the embodiment of the present invention will be described with reference to FIGS. 1 to 5.

As shown in FIGS. 1 to 5, the electrode body 12 is housed in the outer can 11 that also serves as the negative electrode terminal.

The electrode body 12 has a structure in which a laminate of a positive electrode 13, a separator 14, and a negative electrode 15 is spirally wound. A safety device for a sealed battery that also has an explosion-proof function and a terminal is provided at the upper end opening of the outer can 11, and the safety device substantially covers the positive electrode lid 16 having the following configuration with an insulating gasket. It is configured by caulking and fixing to the upper end opening of the outer can 11 via the 17.

[0014]

As shown in FIGS. 1 to 4, the positive electrode lid 16 substantially forms the innermost lid and is connected to the positive electrode 13 of the electrode body 12 via the positive electrode lead 18 and the lead mounting plate 19 and the intermediate lid. A shielding plate 21 that forms the outermost lid and is electrically connected to the lead mounting plate 19 via the central welded portion 20, and a sealing plate 22 that forms the outermost lid and is electrically connected to the shielding plate 21. ...

Further, the positive electrode lid 16 has an insulating plate 23 annularly interposed between the lead mounting plate 19 and the shielding plate 21, and a PTC thermistor element 24 composed of an annular plate between the shielding plate 21 and the sealing plate 22. It is installed.

[0015]

Next, the configuration of each part of the positive electrode lid 16 having the above configuration will be described.

As shown in FIGS. 1 to 3 and 5, the lead mounting plate 19 has a laminated structure composed of a thick base material 19a and a thin foil material 19b bonded to the lower surface thereof. Then, as shown in FIGS. 1 and 2, a pair of gas flow holes 25 and a pair of thin-walled portions 26 are provided around the central welded portion 20 in the lead mounting plate 19. Here, the gas flow hole 25 is formed by integrally penetrating the base material 19a and the foil material 19b, and the thin-walled portion 26 is formed only from the foil material 19b by cutting the base material 19a. As described above, in the present embodiment, since the thin portion 26 is made of the foil material 19b having a certain thickness, it can be reliably broken at the set current breaking pressure as described later.

[0016]

Through the gas flow hole 25, the internal space 27 of the outer can and the operating space 28 formed between the lead mounting plate 19 and the shielding plate 21 are communicated with each other.

As shown in FIGS. 1 to 4, the central welded portion 20 that electrically connects the lead mounting plate 19 and the shielding plate 21 is the lead mounting plate side welding surface 29 forming the central portion of the lead mounting plate 19 and the shielding plate. It is formed by the shielding plate side welded surface 30 forming the central portion of 21. As shown in FIGS. 1, 4 and 5, the shielding plate 21 has a laminated structure composed of a thick base material 21a and a thin foil material 21b bonded to the lower surface thereof. Then, as shown in FIGS. 1 and 3, around the shield plate side welded surface 30 of the shield plate 21, the C-shaped elongated hole 32 has an arc angle θ of 180° or more, except for the connecting tab portion 31. It is formed concentrically. Further, inside the C-shaped elongated hole 32, an arc-shaped elongated hole 33 for reducing strength that bends in a direction facing the C-shaped elongated hole 32 is formed. Both ends of the strength-reducing elongated hole 33 are arranged in a polymerized state at both ends of the C-shaped elongated hole 32. A narrow bent portion 34 is formed between both ends 33a of the strength reducing elongated hole 33 and a portion 32a recessed from both ends of the C-shaped elongated hole 32 by a predetermined distance. In FIG. 2, 34a indicates a planned bending line.

[0017]

Further, in the present embodiment, as shown in FIG. 1, the valve membranes 35 and 36 are formed by the portion of the foil material 21b corresponding to the C-shaped elongated hole 32 and the strength reducing elongated hole 33, respectively.

The thicknesses of these valve membranes 35 and 36 are set so as to break when the set breaking pressure (for example, 20 kg / cm²) or more is reached. Specifically, when the shielding plate 21 is made of a clad metal plate in which a foil material 21b is clad on a thick (for example, 50 μ m) base material 21a made of aluminum, for example, a copper foil of 10 μ m is used as the foil material 21b. be able to. By forming the central welded portion 20 and the shielding plate 21 as described above, when the

pressure inside the outer can 11 increases and exceeds the set current breaking pressure, the valve membranes 35 and 36 are plastically deformed and as shown in FIG. The shielding plate 21 is bent in an upwardly convex state along the planned bending portion 34, which is the weakest part in terms of strength. As a result, the central welded portion 20 is lifted, and the thin portion 26 of the lead mounting plate 19 receives a tensile force and breaks. In this way, when the pressure of the decomposition gas reaches the set current cutoff pressure, the shielding plate 21 is rapidly deformed by providing the C-shaped elongated hole 32 and the elongated hole 33 for reducing the strength, and the thin wall of the lead mounting plate 19 is formed. The thin portion 26 of the lead mounting plate 19 can be lifted, and the thin portion 26 of the lead mounting plate 19 is formed from the foil material. Therefore, the thin portion 26 of the lead mounting plate 19 is reliably and quickly broken at the set current cutoff pressure to lead the lead. The electrical continuity between the mounting plate 19 and the shielding plate 21 can be cut off.

[0018]

Next, the operation of the safety device for the sealed battery having the above configuration will be described with reference to FIGS. 1 to 3.

In the above-mentioned sealed battery, for example, when a large current flows due to an overcharged state, the large current generates highly corrosive decomposition gas in the outer can 11, the pressure in the outer can 11 increases, and the battery is left as it is. Then, the sealed battery will explode. However, in the present embodiment, when the pressure of the decomposition gas in the outer can 11 exceeds the set current cutoff pressure, the central welded portion 20 is quickly pulled up as shown in FIG. 3, and the thin portion 26 of the lead mounting plate 19 is lifted. It breaks and the electrical continuity between the lead mounting plate 19 and the shielding plate 21 is quickly cut off. Therefore, it is possible to prevent further generation of the decomposed gas and surely prevent the internal pressure of the outer can 11 from further increasing and exploding, and the valve membranes 35 and 36 have not been broken yet. It is possible to prevent the decomposed gas harmful to the human body from flowing out to the outside, and it is possible to protect the environment.

[0019]

At this time, since the thickness of the thin-walled portion 26 is the thickness of the foil material 19b, the thickness of the thin-walled portion 26 can be easily and accurately set to the thickness corresponding to the set current cutoff pressure, and the inside of the outer can 11 is disassembled. When the gas pressure reaches the set current cutoff pressure, the thin-walled portion 26 can be reliably broken, and the electrical continuity between the lead mounting plate 19 and the shield plate 21 can be quickly cut off.

Further, the set current cutoff pressure can be easily changed by adjusting the thickness of the thin portion 26.

[0020]

Further, in the unlikely event that the chemical reaction proceeds in the outer can 11 despite the above-mentioned interruption of electrical conduction, decomposition gas is further generated, the internal pressure rises, and the set film breaking pressure is exceeded, as shown in FIG. As described above, either or both of the valve membranes 35 and 36 are broken, and the decomposed gas passes through the space between the operating space 28 and the sealing plate 21 and the sealing plate 22 and the degassing hole 37 provided in the sealing plate 22. Since it is quickly released to the outside, it is possible to reliably prevent the explosion of the sealed battery.

[0021]

As described above, by using the sealed battery safety device according to the present embodiment, it is possible to reliably cut off the current and release the decomposed gas to the outside of the battery, and prevent the sealed battery from exploding. be able to.

Furthermore, by releasing the decomposed gas to the outside of the battery only in the unlikely event, adverse effects on the human body and the environment can be suppressed as much as possible.

Further, as shown in FIGS. 1 to 4, in the present embodiment, since the PTC thermistor element 24 made of an annular plate is interposed between the shielding plate 21 and the sealing plate 22, due to the generation of decomposition gas. When the temperature of the safety device of the sealed battery rises, it becomes difficult for the current to flow, and from this aspect as well, it is possible to prevent an explosion due to overcurrent.

[0022]

Next, a method of manufacturing the lead mounting plate in the safety device for the sealed battery according to the present invention will be described with reference to FIGS. 6 to 8.

The sealed battery 40 shown in FIGS. 6 to 8 has a square structure in which the square lead mounting plate 42 is mounted on the square container 41, but the circular sealed battery 40 described with reference to FIGS. 1 to 5 is shown. Needless to say, the method for manufacturing a lead mounting plate according to the present invention can be applied to a type battery. As shown in FIG. 7A, a pair of elongated holes 44 for forming a thin wall portion are formed in a thick base material 43 having a rectangular shape at parallel intervals by using a press. As shown in FIG. 7B, a foil material 45 having the same shape as the base material 43 is joined to one surface of the base material 43 to form a laminated plate 46 made of a clad metal plate or the like. As shown in FIG. 7 (c), a pair of gas flow holes 47 are bored at parallel intervals in a state where both ends of the elongated holes 44 for forming the thin wall portion of the laminated plate 46 are in contact with each other using a press. At the same time, the thin-walled portion 48 is formed by the foil material portion corresponding to the elongated hole 44 for forming the thin-walled portion. Further, the lead mounting plate side welded surface 49 of the island-shaped portion can be formed in the central welded portion of the shielding plate (not shown) surrounded by the pair of gas flow holes 47 and the pair of thin-walled portions 48. As described above, by using the method for manufacturing the lead mounting plate in the safety device for the sealed battery according to the present invention, the lead mounting plate having a good quality laminated structure can be manufactured at low cost.

[0023]

INDUSTRIAL APPLICABILITY As described above, in the safety device for a sealed battery according to claim 1, a laminated structure in which a lead mounting plate is made of a thick base material and a thin foil material bonded to the lower surface thereof is used. The gas flow hole is formed by integrally penetrating the base material and the foil material, and the thin-walled portion is formed by cutting off the base material and forming the thin-walled portion only from the foil material.

Therefore, the thickness of the thin-walled portion is determined by the thickness of the foil material, and the thickness of the thin-walled portion can be easily and accurately set to the thickness corresponding to the set current cutoff pressure. When the pressure of the decomposition gas reaches the set current cutoff pressure, the thin-walled portion can be surely broken, and the electrical conduction between the lead mounting plate and the shielding plate can be cut off stably and quickly. Further, the set current cutoff pressure can be easily changed by adjusting the thickness of the thin portion.

[0024]

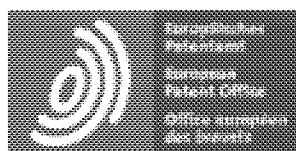
In the sealed battery safety device according to claim 2, the lead mounting plate can be manufactured at low cost by using a clad metal plate in which a foil material is clad bonded to the lower surface of the base material.

In the sealed battery safety device according to claim 3, when the pressure of the decomposition gas reaches the set current cutoff pressure, the shielding plate is rapidly deformed by providing the C-shaped elongated hole and the elongated hole for reducing the strength. The thin part of the lead mounting plate can be lifted, and coupled with the fact that the thin part of the lead mounting plate is formed from the foil material, the thin part of the lead mounting plate is reliably and quickly broken at the set current cutoff pressure. The electrical continuity between the lead mounting plate and the shielding plate can be cut off.

[0025]

In the safety device for the sealed battery according to claim 4, by interposing a PTC thermistor element made of an annular plate between the shielding plate and the sealing plate, the PTC element makes it difficult for current to flow, and from this aspect. Can prevent explosion due to overcurrent.

The method for manufacturing a lead mounting plate in a sealed battery safety device according to claim 5 is to create a pair of gas flow holes at intervals in a state where both ends of the elongated holes for forming thin-walled portions of the laminated plate are in contact with each other. At the same time, the thin-walled portion was formed by the foil material portion corresponding to the elongated hole for forming the thin-walled portion, and the lead mounting plate welded surface of the central welded portion was formed in the island-shaped portion surrounded by the gas flow hole and the thin-walled portion. , The lead mounting plate can be manufactured at low cost and with good yield.



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CLAIMS JP2001102038A

1.

The positive electrode lid attached to one end of the outer can forms the innermost lid and the lead attachment plate connected to the positive electrode of the electrode body via the positive electrode lead, and the intermediate lid is formed and the lead is formed through the central welded portion. A shielding plate that is electrically connected to the mounting plate and deforms in the direction of applying internal pressure in conjunction with an increase in internal pressure in the internal space of the battery, and a sealing plate that forms the outermost lid and is electrically connected to the shielding plate. A closed type in which a gas flow hole and a thin-walled portion are provided around the central welded portion of the lead mounting plate, and when the shielding plate is deformed due to an increase in the internal pressure, the thin-walled portion breaks to cut off the current. In a battery safety device, the lead mounting plate has a laminated structure composed of a thick base material and a thin foil material joined to the lower surface thereof, and integrally penetrates the base material and the foil material through the gas flow hole. A safety device for a sealed battery, characterized in that the thin portion is formed by cutting off the base material and formed only from the foil material.

2.

The safety device for a sealed battery according to claim 1, wherein the lead mounting plate is a clad metal plate in which the foil material is bonded to the lower surface of the base material by clad bonding.

3.

The shielding plate is made of a clad metal plate made of a thick base material and a thin foil material bonded to the lower surface thereof, and is concentrically C at an arc angle of 180 ° C. or higher around the central welded portion surface of the base material. A long hole for strength reduction is provided at a position inside the C-shaped long hole of the base material, and both ends of the long hole for strength reduction and both ends of the C-shaped long hole are formed. A narrow bent portion is formed between the portion retracted by a predetermined distance, and a valve film is formed by the portion of the foil material corresponding to the C-shaped elongated hole and the strength reducing elongated hole, respectively. When the pressure inside the outer can exceeds the set current cutoff pressure, the shielding plate is bent in an upwardly convex state along the planned bending portion, and the thin portion of the lead mounting plate is broken to the lead mounting plate. The sealing according to claim 1 or 2, wherein the electrical connection with the shielding plate is cut off, and the valve membrane is broken when the pressure inside the current blocking outer can exceeds the set film breaking pressure. Type battery safety device.

4.

The safety device for a sealed battery according to any one of claims 1 to 3, wherein a PTC thermistor element made of an annular plate is interposed between the shielding plate and the sealing plate.

5.

Using a press, a pair of elongated holes for forming thin-walled portions are formed in the base material at parallel intervals, and a foil material having the same shape as the base material is joined to one surface of the base material to form a laminated plate. Then, using a press, a pair of gas flow holes are formed at intervals in a state where both ends of the thin-walled portion-forming elongated holes of the laminated plate are in contact with each other, and the thin-walled portion-forming elongated holes are supported. A thin-walled portion is formed by the foil material portion to be formed, and a lead mounting plate-side welded surface of the central welded portion is formed in an island-shaped portion surrounded by the gas flow hole and the thin-walled portion. Lead mounting plate in the safety device of the type battery.



Espacenet

Bibliographic data: JP2001102038 (A) — 2001-04-13

SAFETY DEVICE IN ENCLOSED CELL AND LEAD MOUNTING PLATE IN THE SAME

Inventor(s): SAIJO KINJI; OKAMOTO HIROAKI; KAWAMURA HIROAKI;
KANEKO AKIRA ± (SAIJO KINJI, ; OKAMOTO HIROAKI, ;
KAWAMURA HIROAKI, ; KANEKO AKIRA)

Applicant(s): TOYO KOHAN CO LTD ± (TOYO KOHAN CO LTD)

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Abstract of JP2001102038 (A)

PROBLEM TO BE SOLVED: To provide an enclosed cell with a safety device for preventing explosion caused by a sudden rising of internal pressure that may occur in overcharging and short-circuiting, and method for fabricating a lead mounting plate.
SOLUTION: The positive pole cover 16 is installed in an external vessel 11, and comprises a lead mounting plate 19 forming an inmost cover and being connected to the positive pole 13 of an electrode body 12 through a positive pole lead 18, a shield plate 21 forming a middle cover and being connected electrically the lead mounting plate 19 through a central welding part 20, the shield plate being deformed in the applying direction of an internal force dependent upon the rising of the internal pressure in a inner space of battery, and sealing plate 22 forming an outmost cover and being connected electrically to the shield plate 21. A gas passage and a thin wall 26 are formed at peripherals of the central welding part 20 of the lead mounting plate 19, if the shield plate 19 is deformed by the rising of internal pressure, the current is interrupted as the thin wall 26 is broken, the lead mounting plate 19 includes a thin base member 19a and a thin foil member 19b, the thin wall 26 may be formed by only the thin foil member 19b as the thin base member 19a is cut and removed from it.

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(71) 出願人 390003193

東洋鋼板株式会社
東京都千代田区四番町2番地12

(72) 発明者 西條 謙二

山口県下松市東豊井1296番地の1 東洋鋼
板株式会社技術研究所内

(72) 発明者 岡本 浩明

山口県下松市東豊井1296番地の1 東洋鋼
板株式会社技術研究所内

(74) 代理人 100100103

弁理士 太田 明男

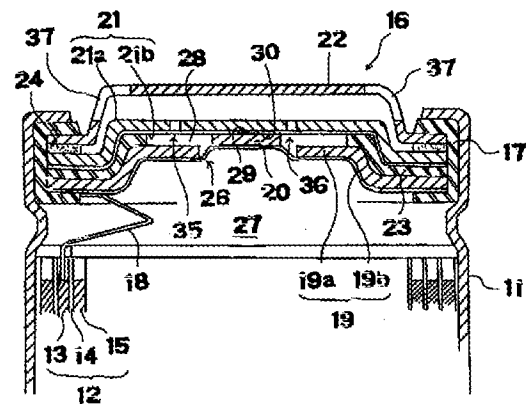
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(54) 【発明の名称】 密閉型電池の安全装置及び同装置におけるリード取付板

(57) 【要約】 (修正有)

【課題】 過充電時や短絡時に起きる急激な内圧上昇による破裂を確実に防止する密閉型電池の安全装置及びリード取付板の製造方法を提供する。

【解決手段】 外装缶11の一端に取り付けられる正極蓋16を、最内蓋を形成すると共に正極リード18を介して電極体12の正極13に接続されるリード取付板19と、中間蓋を形成すると共に中央溶接部20を介してリード取付板19に電気的に接続され、電池の内部空間における内圧上昇に連動して内圧付加方向に変形する遮蔽板21と、最外蓋を形成すると共に遮蔽板21に電気的に接続される封口板22とから構成し、リード取付板19の中央溶接部20の周囲にガス流通孔と薄肉部26を設け、内圧の上昇により遮蔽板21が変形すると薄肉部26が破断し電流を遮断し、リード取付板19が厚肉の基材19aと薄肉の箔材19bからなり、薄肉部26を基材19aを切除して箔材19bのみから形成している。



【特許請求の範囲】

【請求項1】 外装缶の一端に取り付けられる正極蓋を、最内蓋を形成すると共に正極リードを介して電極体の正極に接続されるリード取付板と、中間蓋を形成すると共に中央溶接部を介して前記リード取付板に電気的に接続され、電池の内部空間における内圧上昇に連動して内圧付加方向に変形する遮蔽板と、最外蓋を形成すると共に前記遮蔽板に電気的に接続される封口板とから構成し、前記リード取付板の前記中央溶接部の周囲にガス流通孔と薄肉部を設け、前記内圧の上昇により遮蔽板が変形すると前記薄肉部が破断し電流を遮断するようにした密閉型電池の安全装置において、

前記リード取付板が厚肉の基材とその下面に接合した薄肉の箔材からなる積層構造を具備し、前記ガス流通孔を前記基材と箔材を一体的に貫通することによって形成すると共に、前記薄肉部を前記基材を切除して前記箔材のみから形成したことを特徴とする密閉型電池の安全装置。

【請求項2】 前記リード取付板は、前記基材の下面に前記箔材をクラッド接合によって接合したクラッド金属板であることを特徴とする請求項1記載の密閉型電池の安全装置。

【請求項3】 前記遮蔽板が厚肉の基材とその下面に接合した薄肉の箔材からなるクラッド金属板からなり、前記基材の前記中央溶接部面の周りに180°以上の円弧角で同心円的にC字状長孔を設け、

前記基材の前記C字状長孔の内側をなす個所に強度低減用長孔を形成すると共に、該強度低減用長孔の両端と前記C字状長孔の両端より所定距離後退した部分との間に狭幅の折曲予定部を形成し、

前記C字状長孔と前記強度低減用長孔に対応する前記箔材の部分によってそれぞれ弁膜を形成し、

前記外装缶内の圧力が設定電流遮断圧力を超えると、前記遮蔽板が前記折曲予定部に沿って上方凸の状態では折り曲げられ、前記リード取付板の薄肉部を破断して前記リード取付板と前記遮蔽板との電気的接続が遮断され、前記電流遮断外装缶内の圧力が設定膜破断圧力を超えると前記弁膜が破断されるようにしたことを特徴とする請求項1又は2記載の密閉型電池の安全装置。

【請求項4】 前記遮蔽板と前記封口板との間に環状板からなるPTCサーミスタ素子が介設されていることを特徴とする請求項1～3のうちのいずれかの請求項記載の密閉型電池の安全装置。

【請求項5】 プレスを用いて、基材に一对の薄肉部形成用長孔を平行間隔を開けて穿設し、該基材の一面に該基材と同一形状を有する箔材を接合して積層板を形成し、プレスを用いて、前記積層板の前記薄肉部形成用長孔の両端部同士を連絡する状態で一对のガス流通孔を間隔を開けて穿設すると共に前記薄肉部形成用長孔に対応する前記箔材部分によって薄肉部を形成し、前記ガス流

通孔と前記薄肉部によって囲まれる島状部分に前記中央溶接部のリード取付板側溶接面を形成するようにしたことを特徴とする密閉型電池の安全装置におけるリード取付板。

【発明の詳細な説明】

【0001】

【発明の属する技術分野】本発明は、防爆機能を有する密閉型電池の安全装置及びそれを用いた密閉型電池の安全装置におけるリード取付板に関する。

【0002】

【従来の技術】近年、非水電解液を使用したリチウム電池やリチウムイオン電池等に非水電解液を用いた二次電池が携帯電子機器等に広く使用されつつある。このような二次電池は高い起電力を有するという特徴がある反面、外装缶内に収納された正極及び負極を備える電極体が化学変化を起こして内圧が高くなり、破裂が生じる場合がある。例えば、リチウム二次電池のような非水電解液電池を過充電状態にしたり、誤使用による短絡状態になって大電流が流れたりすると、電極体の中の非水電解液が分解されてガスが発生する場合がある。このようなガスが外装缶内に次第に充満し、外装缶内の内圧が上昇すると、最後には電池が破裂する。

【0003】このような電池の破裂を防止するため、従来においても、各種形態の密閉型電池が開発されており、その一形態として、例えば、特開平7-105933号公報に記載されている安全装置を具備する密閉型電池がある。これは、図10及び図11に示すように、電池内圧の上昇に伴い内圧付加方向に変形を生じる防爆弁60とリード体取付部材61とを溶接し、上記リード取付部材61の防爆弁60との溶接部分62の外周側部分に一对の圧力導入口63と一对の薄肉部64を設け、リード取付部材61にリード体65を取り付けることによって構成されている。なお、図10は薄肉部64が破断された状態を示している。

【0004】上記した構成によって、通常の状態では、リード取付部材61と防爆弁60との電気的接続を確保して電池として機能させることができる。一方、過充電や短絡などにより電池内圧が上昇して所定の圧力に達したときは、防爆弁60が電池内圧を受けて内圧付加方向に変形し、リード取付部材61に設けた薄肉部64が破断し、リード体取付部材61と防爆弁60との電気的接続を消失させ、電流を遮断し、電池内部におけるガスのそれ以上の発生を防止し、電池の破裂を未然に防止することができる。

【0005】

【発明が解決しようとする課題】しかし、リード取付部材61に薄肉部64を形成するに際しては、プレス加工やエッチングで行っているため、薄肉部64の厚みが一定せず、その結果、リード取付部材61と防爆弁60の電気的接続を遮断する電池内圧が一定しないことにな

り、密閉型電池によっては、電池内圧が設定遮断圧力に達しているにもかかわらずリード取付部材61と防爆弁60の電気的接続が遮断されない状態が生じることになり、密閉型電池の安全性への信頼性を著しく損なうことになる。本発明は、このような課題を解決しようとするものであり、電池内圧が一定以上に上昇すると確実にリード取付板と遮蔽板との電気的接続を遮断することができ、密閉型電池の安全を十分に確保できる密閉型電池の安全装置及び同安全装置におけるリード取付板を提供することを目的とする。

【0006】

【課題を解決するための手段】上記目的を達成するための本発明に係る密閉型電池の安全装置は、外装缶の一端に取り付けられる正極蓋を、最内蓋を形成すると共に正極リードを介して電極体の正極に接続されるリード取付板と、中間蓋を形成すると共に中央溶接部を介してリード取付板に電気的に接続され、電池の内部空間における内圧上昇に連動して内圧付加方向に変形する遮蔽板と、最外蓋を形成すると共に遮蔽板に電気的に接続される封口板とから構成し、リード取付板の中央溶接部の周囲にガス流通孔と薄肉部を設け、内圧の上昇により遮蔽板が変形すると薄肉部が破断し電流を遮断するようにした密閉型電池の安全装置において、リード取付板が厚肉の基材とその下面に接合した薄肉の箔材からなる積層構造を具備し、ガス流通孔を基材と箔材を一体的に貫通することによって形成すると共に、薄肉部を基材を切除して箔材のみから形成している。

【0007】従って、常時は、密閉空間内において、リード取付板を遮蔽板の中央溶接部に溶接することによって確実に確保されている。一方、電池内の内圧が急激に上昇して設定電流遮断圧力を超えると分解ガスの圧力によって遮蔽板が変形して一定の厚さを有する箔材からなるリード取付板の薄肉部を確実に破断し、リード取付板と遮蔽板の電気的導通を遮断して、分解ガスのそれ以上の発生を防止すると共に分解ガスが外部に流出するのを防止する。

【0008】また、上記した第1の発明に係る密閉型電池の安全装置は、以下の点にも特徴を有する。

① リード取付板は遮蔽板は、アルミニウム等からなる基材の下面にニッケル等からなる箔材をクラッド接合によって接合したクラッド金属板からなる。ここで、クラッド金属板となる基材及び箔材の組み合わせとしては下記のものがある。いずれを基材又は箔材としてもよい。すなわち、アルミニウム／アルミニウム、アルミニウム／ステンレススチール、ステンレススチール／ステンレススチール、ステンレススチール／ニッケル、ニッケル／ニッケル、銅／銅、銅／ステンレススチール、等である。また、基材の厚みは20 μ m～200 μ m程度とするのが好ましく、箔材の厚みは10～100 μ m程度とするのが好ましい。また、リード取付板として用いるク

ラッド金属板は、例えば、本出願人が先に特開平1-224184号公報で開示したように、 $1 \times 10^{-1} \sim 1 \times 10^{-4}$ Torr の極低圧不活性ガス雰囲気中で、接合面を有する金属基板と箔材をそれぞれアース接地した一方の電極Aとし、絶縁支持された他の電極Bとの間に1～50 MHzの交流を印加してグロー放電を行わせ、かつ、前記グロー放電によって生じたプラズマ中に露出される電極の面積が、電極Bの面積の1/3以下で、スパッタエッチング処理することによって製造することができる。

【0009】② 遮蔽板は、アルミニウム等の厚肉の基材の下面にニッケル等からなる薄肉の箔材をクラッド接合したクラッド金属板からなり、基材の前記中央溶接部面の周りに180℃以上の円弧角で同心円的にC字状長孔を設け、基材のC字状長孔の内側をなす個所に強度低減用長孔を形成すると共に、強度低減用長孔の両端とC字状長孔の両端より所定距離後退した部分との間に狭幅の折曲予定部を形成し、C字状長孔と強度低減用長孔に対応する箔材の部分によってそれぞれ弁膜を形成し、外装缶内の圧力が設定電流遮断圧力を超えると、遮蔽板が前記折曲予定部に沿って上方凸の状態で折り曲げられ、リード取付板の薄肉部を破断して前記リード取付板と遮蔽板との電気的接続が遮断され、電流遮断外装缶内の圧力が設定膜破断圧力を超えると弁膜が破断されるようにしている。ここで、好ましくは、設定電流遮断圧力は4～5 kg/cm² に、設定膜破断圧力は20 kg/cm² に設定する。

【0010】このような遮断板を用いることによって、外装缶内の圧力が設定電流遮断圧力を超えると、遮断板を容易に変形させてリード取付板の薄肉部を容易に破断してリード取付板と遮断板との電気的遮断を確実に行うことができると共に、このような電気的遮断にもかかわらず、万一、外装缶内の化学反応が進んで分解ガスが発生し、内部圧力がさらに上昇し設定膜破断圧力を超える場合には、遮蔽板に設けた弁膜が破断することによって、分解ガスはリード取付板に設けたガス流通孔、動作空間、弁膜、及び、ガス抜き穴を通して外部に放出され、密閉型電池が爆発するのを防止することができる。

【0011】遮蔽板として用いるクラッド金属板も、リード取付板と同様に、例えば、本出願人が先に特開平1-224184号公報で開示したように、 $1 \times 10^{-1} \sim 1 \times 10^{-4}$ Torr の極低圧不活性ガス雰囲気中で、接合面を有する金属基板と箔材をそれぞれアース接地した一方の電極Aとし、絶縁支持された他の電極Bとの間に1～50 MHzの交流を印加してグロー放電を行わせ、かつ、前記グロー放電によって生じたプラズマ中に露出される電極の面積が、電極Bの面積の1/3以下で、スパッタエッチング処理することによって製造することができる。

【0012】③ 遮蔽板と封口板との間に環状板からなる

PTCサーミスタ素子が介設されており、PTCサーミスタ素子によって、密閉型電池の安全装置の温度が上昇すると共に電流を流れにくくして、この面からも過電流による爆発を防止するようにしている。上記目的を達成するためのもう一つの本発明に係る密閉型電池の安全装置におけるリード取付板は、プレスを用いて、基材に一对の薄肉部形成用長孔を間隔を開けて穿設し、基材の一面に基材と同一形状を有する箔材を接合して積層板を形成し、プレスを用いて、積層板の薄肉部形成用長孔の両端部同士を連絡する状態で一对のガス流通孔を平行間隔を開けて穿設すると共に薄肉部形成用長孔に対応する箔材部分によって薄肉部を形成し、ガス流通孔と薄肉部によって囲まれる島状部分に中央溶接部のリード取付板側溶接面を形成する。

【0013】

【発明の実施の形態】以下、添付図に示す一実施の実施の形態を参照して、本発明を具体的に説明する。まず、本発明の一実施の形態に係る密閉型電池の安全装置の構成について、図1～図5を参照して説明する。図1～図5に示すように、負極端子を兼ねる外装缶11内には電極体12が収納されている。電極体12は、正極13、セパレータ14及び負極15の積層物を渦巻状に巻回した構成になっている。そして、外装缶11の上端開口部には、防爆機能と端子を兼ねる密閉型電池の安全装置が設けられており、安全装置は、実質的に、以下の構成を有する正極蓋16を、絶縁ガasket 17を介して外装缶11の上端開口部にカシメ固定することによって構成されている。

【0014】図1～図4に示すように、正極蓋16は、実質的に、最内蓋を形成すると共に正極リード18を介して電極体12の正極13に接続されるリード取付板19と、中間蓋を形成すると共に中央溶接部20を介してリード取付板19と電気的に接続される遮蔽板21と、最外蓋を形成すると共に遮蔽板21に電気的に接続される封口板22を具備する。また、正極蓋16は、リード取付板19と遮蔽板21との間に環状に絶縁板23を介設すると共に、遮蔽板21と封口板22との間に環状板からなるPTCサーミスタ素子24を介設している。

【0015】次に、上記した構成を有する正極蓋16の各部の構成について説明する。図1～3及び図5に示すように、リード取付板19は厚肉の基材19aとその下面に接合した薄肉の箔材19bからなる積層構造を具備している。そして、図1及び図2に示すように、リード取付板19において中央溶接部20の周りには一对のガス流通孔25と一对の薄肉部26が設けられている。ここに、ガス流通孔25は基材19aと箔材19bを一体的に貫通することによって形成されており、薄肉部26は基材19aを切除して箔材19bのみから形成されている。このように、本実施の形態では、薄肉部26は一定の厚さを有する箔材19bからなるので、後述するよ

うに、設定電流遮断圧力で確実に破断させることができる。

【0016】このガス流通孔25を通して外装缶の内部空間27と、リード取付板19と遮蔽板21との間に形成される動作空間28が連通されている。図1～図4に示すように、リード取付板19と遮蔽板21を電気的に接続する中央溶接部20は、リード取付板19の中央部をなすリード取付板側溶接面29と、遮蔽板21の中央部をなす遮蔽板側溶接面30によって形成される。図1、図4及び図5に示すように、遮蔽板21は厚肉の基材21aとその下面に接合した薄肉の箔材21bからなる積層構造を具備している。そして、図1及び図3に示すように、遮蔽板21の遮蔽板側溶接面30の周りには、連結タブ部31を除いて、C字状長孔32が180°以上の円弧角θで同心円的に形成されている。また、C字状長孔32の内側には、C字状長孔32と対向する方向に屈曲する弧状の強度低減用長孔33が形成されている。強度低減用長孔33の両端部はC字状長孔32の両端部に重合状態に配置されている。そして、強度低減用長孔33の両端33aとC字状長孔32の両端より所定距離後退した部分33aとの間には、狭幅の折曲予定部34が形成されている。なお、図2において、34aは折曲予定線を示す。

【0017】また、本実施の形態では、図1に示すように、C字状長孔32と強度低減用長孔33とに対応する箔材21bの部分によってそれぞれ弁膜35、36を形成している。そして、これらの弁膜35、36の厚みは、設定破断圧力（例えば、 20 kg/cm^2 ）以上になると破断するように設定されている。具体的には、遮蔽板21がアルミニウムからなる厚肉（例えば、 $50\mu\text{m}$ ）の基材21aに箔材21bをクラッドしたクラッド金属板からなる場合、箔材21bとしては、例えば $10\mu\text{m}$ の銅箔を用いることができる。中央溶接部20と遮蔽板21を上記した構成とすることによって、外装缶11内の圧力が増大し設定電流遮断圧力を超えると、図4に示すように、弁膜35、36が塑性変形すると共に、強度的に最も弱い部分である折曲予定部34に沿って遮蔽板21が上方凸の状態で折れ曲がることになる。その結果、中央溶接部20が持ち上がり、リード取付板19の薄肉部26が引張力を受け破断することになる。このように、分解ガスの圧力が設定電流遮断圧力になると、C字状長孔32と強度低減用長孔33を設けたことによって、遮蔽板21は速やかに変形してリード取付板19の薄肉部26を持ち上げることができ、リード取付板19の薄肉部26を箔材から形成したと相まって、設定電流遮断圧力で確実にかつ迅速にリード取付板19の薄肉部26を破断して、リード取付板19と遮蔽板21の電気的導通を遮断することができる。

【0018】次に、上記した構成を有する密閉型電池の安全装置の作動について、図1～図面3を参照して説明

する。上記した密閉型電池において、例えば、過充電状態により大電流が流れると、この大電流により外装缶11内に腐食性の高い分解ガスが発生し、外装缶11内の圧力が増大し、そのまま放置すると、密閉型電池が爆発することになる。しかし、本実施の形態では、外装缶11内の分解ガスの圧力が設定電流遮断圧力を超えると、図3に示すように中央溶接部20を速やかに引き上げてリード取付板19の薄肉部26を破断し、リード取付板19と遮蔽板21との電氣的導通が速やかに遮断する。従って、分解ガスのそれ以上の発生を防止して、外装缶11の内部圧がさらに上昇して爆発するのを確実に防止することができると共に、弁膜35、36は未だ破断されていないので、人体に有害な分解ガスが外部に流出するのを防止することができ、環境保護も図ることができる。

【0019】この際、薄肉部26の厚さが箔材19bの厚さなので、薄肉部26の厚さを設定電流遮断圧力に対応した厚みに容易かつ正確に設定することができ、外装缶11内の分解ガスの圧力が設定電流遮断圧力に達したとき、確実に薄肉部26を破断して、リード取付板19と遮蔽板21との電氣的導通が速やかに遮断することができる。また、薄肉部26の厚さを調整することによって、設定電流遮断圧力を容易に変更することができる。

【0020】さらに、万一、上記した電氣的導通の遮断にもかかわらず外装缶11内において化学反応が進み、分解ガスがさらに発生して内圧が上昇し、設定膜破断圧力を超えると、図4に示すように、弁膜35、36のいずれか又は両方が破断して、分解ガスが動作空間28から遮蔽板21と封口板22との間の空間及び封口板22に設けたガス抜き穴37を通過して外部に速やかに放出されるので、密閉型電池の爆発を確実に防止することができる。

【0021】このように、本実施の形態に係る密閉型電池の安全装置を用いることによって、電流遮断及び分解ガスの電池外部への放出を確実に行うことができ、密閉型電池の破裂を未然に防止することができる。さらに、分解ガスの電池外部への放出は万一の場合のみ行うことによって人体や環境への悪影響を可及的に抑制することができる。また、図1～図4に示すように、本実施の形態では、遮蔽板21と封口板22との間に環状板からなるPTCサーミスタ素子24が介設されているので、分解ガスの発生によって密閉型電池の安全装置の温度が上昇すると電流の流れにくくして、この面からも過電流による爆発を防止することができる。

【0022】次に、本発明に係る密閉型電池の安全装置におけるリード取付板の製造方法について、図6～図8を参照して説明する。なお、図6～図8に示す密閉型電池40は、角形容器41に角形のリード取付板42を装着する角形構造を示しているが、図1～図5を参照して説明した円形の密閉型電池についても、本発明に係るリ

ード取付板の製造方法を適用できることはいうまでもない。図7(a)に示すように、プレスを用いて、矩形形状を有する厚肉の基材43に一对の薄肉部形成用長孔44を平行間隔を開けて穿設する。図7(b)に示すように、基材43の一面に基材43と同一形状を有する箔材45を接合してクラッド金属板等からなる積層板46を形成する。図7(c)に示すように、プレスを用いて、積層板46の薄肉部形成用長孔44の両端部同士を連絡する状態で一对のガス流通孔47を平行間隔を開けて穿設すると共に薄肉部形成用長孔44に対応する箔材部分によって薄肉部48を形成する。また、一对のガス流通孔47と一对の薄肉部48によって囲まれる遮蔽板(図示しない)の中央溶接部に島状部分のリード取付板側溶接面49を形成することができる。このように、本発明に係る密閉型電池の安全装置におけるリード取付板の製造方法を用いることによって、良質の積層構造を有するリード取付板を安価に製造することができる。

【0023】

【発明の効果】以上説明してきたように、請求項1記載の密閉型電池の安全装置においては、リード取付板が厚肉の基材とその下面に接合した薄肉の箔材からなる積層構造を具備し、前記ガス流通孔を前記基材と箔材を一体的に貫通することによって形成すると共に、前記薄肉部を前記基材を切除して前記箔材のみから形成している。従って、薄肉部の厚さは箔材の厚さで決定されることになり、薄肉部の厚さを設定電流遮断圧力に対応した厚みに容易かつ正確に設定することができ、外装缶内の分解ガスの圧力が設定電流遮断圧力に達したとき、確実に薄肉部を破断して、リード取付板と遮蔽板との電氣的導通を安定してかつ速やかに遮断することができる。また、薄肉部の厚さを調整することによって、設定電流遮断圧力を容易に変更することができる。

【0024】請求項2記載の密閉型電池の安全装置においては、リード取付板を基材の下面に箔材をクラッド接合したクラッド金属板とすることによって、安価にリード取付板を製造することができる。請求項3記載の密閉型電池の安全装置においては、分解ガスの圧力が設定電流遮断圧力になると、C字状長孔と強度低減用長孔を設けたことによって、遮蔽板は速やかに変形してリード取付板の薄肉部を持ち上げることができ、リード取付板の薄肉部を箔材から形成したことと相まって、設定電流遮断圧力で確実にかつ迅速にリード取付板の薄肉部を破断して、リード取付板と遮蔽板の電氣的導通を遮断することができる。

【0025】請求項4記載の密閉型電池の安全装置においては、遮蔽板と封口板との間に環状板からなるPTCサーミスタ素子を介設することによって、PTC素子によって電流の流れにくくして、この面からも過電流による爆発を防止できる。請求項5記載の密閉型電池の安全装置におけるリード取付板の製造方法は、積層板の薄肉

部形成用長孔の両端部同士を連絡する状態で一对のガス流通孔を間隔を開けて穿設すると共に薄肉部形成用長孔に対応する箔材部分によって薄肉部を形成し、ガス流通孔と薄肉部によって囲まれる島状部分に中央溶接部のリード取付板溶接面を形成するようにしたので、リード取付板を安価に歩留り良く製造することができる。

【図面の簡単な説明】

【図1】通常使用状態における本発明の一実施の形態に係る密閉型電池の安全装置の構成説明図である。

【図2】同安全装置におけるリード取付板の平面図である。

【図3】同安全装置における遮蔽板の平面図である。

【図4】リード取付板と遮蔽板との電気的接続が遮断された状態の本発明の一実施の形態に係る密閉型電池の安全装置の構成説明図である。

【図5】弁膜が破断した状態における本発明の一実施の形態に係る密閉型電池の安全装置の構成説明図である。

【図6】密閉型電池が角形の場合の本発明の一実施の形態に係る密閉型電池の斜視図である。

【図7】本発明の一実施の形態に係る密閉型電池の安全装置におけるリード取付板の製造方法の工程説明図である。

【図8】図7のI-I線による断面図である。

【図9】図7のII-II線による断面図である。

【図10】従来の密閉型電池の安全装置の構造説明図である。

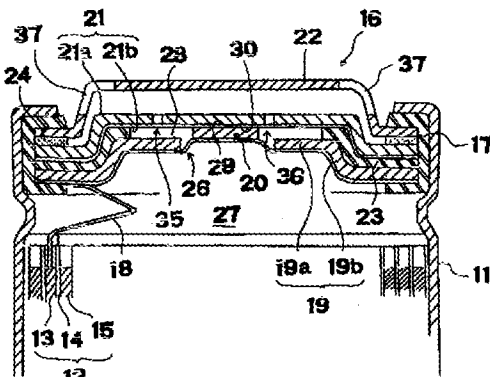
【図11】同装置におけるリード体取付部材の平面図(a)と横断面図(b)である。

【符号の説明】

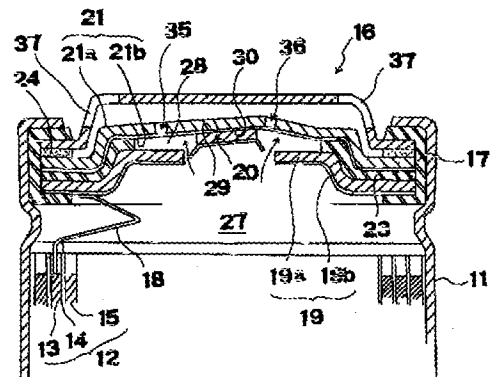
- 11 外装缶
- 12 電極体
- 13 正極
- 14 セパレータ
- 15 負極
- 16 正極蓋
- 17 絶縁ガasket

- 18 正極リード
- 19 リード取付板
- 19a 基材
- 19b 箔材
- 20 中央溶接部
- 21 遮蔽板
- 21a, 21b 箔材
- 22 封口板
- 23 絶縁板
- 24 PTCサーミスタ素子
- 25 ガス流通孔
- 26 薄肉部
- 27 内部空間
- 28 動作空間
- 29 リード取付板側溶接面
- 30 遮蔽板側溶接面
- 31 連結タブ部
- 32 C字状長孔
- 32a 部分
- 33 強度低減用長孔
- 33a 端
- 34 折曲予定部
- 34a 折曲予定線
- 35, 36 弁膜
- 37 ガス抜き穴
- 40 密閉型電池
- 41 角形容器
- 42 リード取付板
- 43 基材
- 44 薄肉部形成用長孔
- 45 箔材
- 46 積層板
- 47 ガス流通孔
- 48 薄肉部
- 49 リード取付板側溶接面

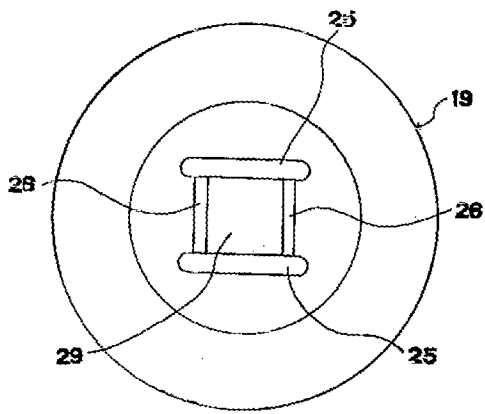
【図1】



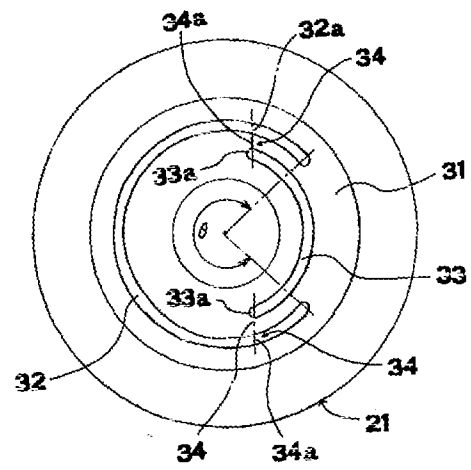
【図4】



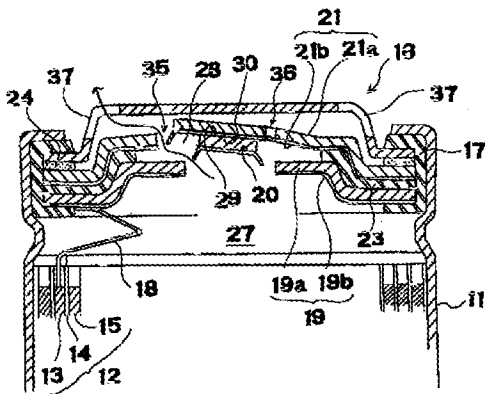
【図2】



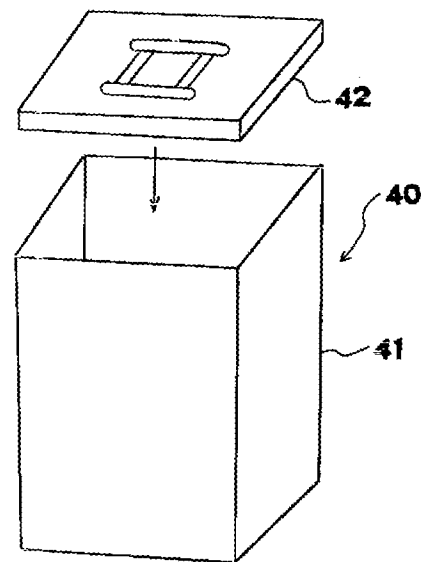
【図3】



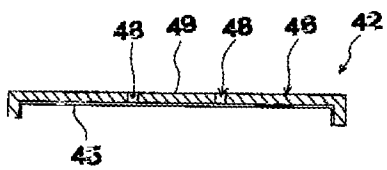
【図5】



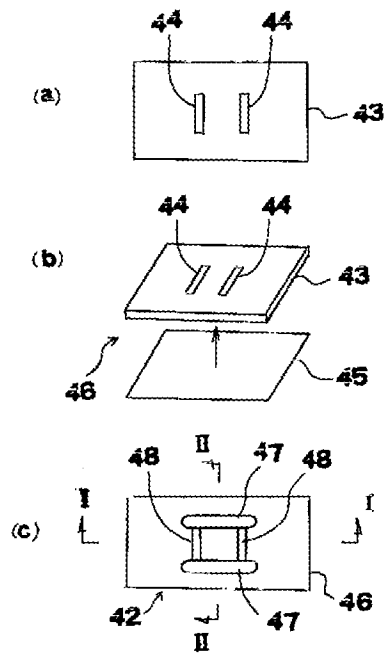
【図6】



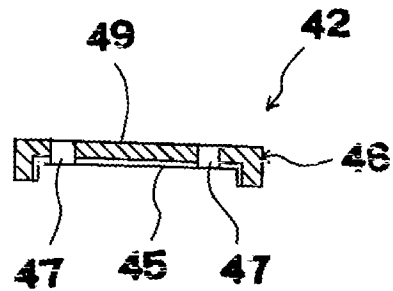
【図8】



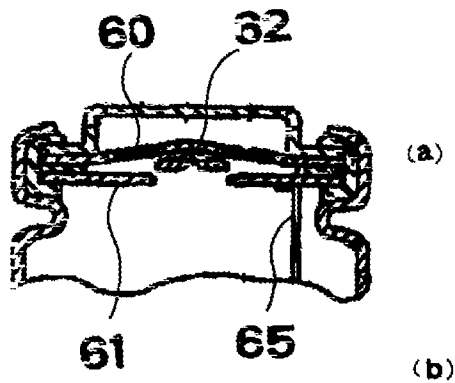
【図7】



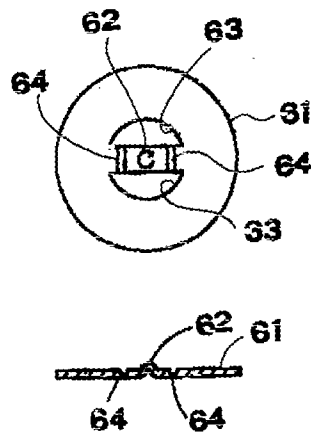
【図9】



【図10】



【図11】

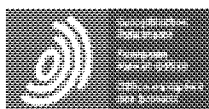


フロントページの続き

(72)発明者 河村 宏明
山口県下松市東豊井1302番地 東洋鋼鈑株
式会社下松工場内

(72)発明者 金子 彰
山口県下松市東豊井1302番地 東洋鋼鈑株
式会社下松工場内

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EXPLOSION-PROOF SAFETY DEVICE AND BATTERY HAVING IT

Inventor(s): MITAMURA TOMOKAZU; KONISHI AKIYA; TSURUTA KUNIO; MURAKAMI TETSUYA ± (MITAMURA TOMOKAZU, ; KONISHI AKIYA, ; TSURUTA KUNIO, ; MURAKAMI TETSUYA)

Applicant(s): MATSUSHITA ELECTRIC IND CO LTD ± (MATSUSHITA ELECTRIC IND CO LTD)

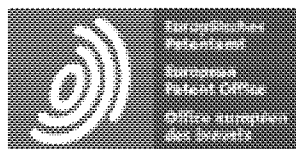
Classification: - **international:** H01M2/12; H01M2/34; (IPC1-7): H01M2/12
- **cooperative:** H01M50/3425 (EP); H01M50/572 (EP); H01M50/578 (EP); Y02E60/10 (EP)

Application number: JP19930005593 19930118

Priority number (s): JP19930005593 19930118

Abstract of JPH06215746 (A)

PURPOSE: To prevent the rupture of a battery, by interrupting an electric path between a terminal board and a power generating element, by a first inversion deformation when internal pressure in the battery is increased, and breaking a safety valve, concurrently with a second inversion deformation, to exhaust gas. **CONSTITUTION:** When battery internal pressure is increased by a short circuit, overcharge, and reverse charge, the battery internal pressure is transmitted to a metal-made safety valve 1 through an air vent hole provided on a metallic lead fitting plate 4, a metallic perforated terminal board 3, and an insulating ring 2. In the valve 1, since its outer peripheral part is calked by a battery case, when the battery internal pressure is reached to a given value, a center part is risen upward, a folded part 1b is reversely deformed, a fused part S is separated, and the connection with the terminal board 3 is released to interrupt current. Next, when battery internal pressure is suddenly increased, the valve 1 is ruptured concurrently with the reverse deformation of the folded part 1a of the valve 1 to exhaust gas to outside a battery through an air vent hole provided on a terminal board 5. Consequently, the rupture of the battery can be prevented.



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DESCRIPTION JPH06215746A

[0001]

[Industrial application field]

[0002]

The present invention relates to an explosion-proof safety device used for a battery.

[0003]

[Conventional technology]

[0004]

In recent years, portable and cordless electronic devices such as AV devices and personal computers have rapidly progressed, and there is a high demand for compact, lightweight, and high energy density sealed batteries as power sources for driving these devices.

In particular, lithium secondary batteries are highly expected as batteries having a particularly high voltage and high energy density.

[0005]

However, a sealed battery, for example, a lithium secondary battery that uses a lithium metal or carbon material as a negative electrode, generates gas in the battery when it is misused such as a short circuit, overcharge, or reverse charge, and the internal pressure rises. It may burst and lose its function or damage peripheral devices.

[0006]

In particular, if the overcharged state continues, there is a problem that the temperature of the battery may rise sharply or an explosion may occur due to the decomposition of the electrolytic solution and the decomposition of the active material.

[0007]

In order to prevent the battery from exploding in such a short circuit, overcharge, and reverse charge state, conventionally, as shown in, for example, Jitsukaisho 54-137734 and Jitsukosho 59-15398. , An explosion-proof safety device having a flexible thin plate (diaphragm) that breaks as the internal pressure of the battery rises and discharges the internal pressure to the outside of the battery has been proposed.

In this safety device, as shown in FIGS. 5A and 5B, as the internal pressure rises, the internal pressure causes the flexible thin plate 42 (diaphragm) to be deformed outward to break and destroy the gas. It is discharged to the outside of the battery to prevent the battery from exploding, but the short circuit, overcharge, and reverse charge state of the battery are continued.

Therefore, when this safety device is used for a battery having a high energy density (the amount of electric energy that can be extracted from the battery per unit volume or unit weight), for example, a lithium secondary battery, the internal pressure of the battery rises sharply, especially when the overcharged state continues. However, the discharge of gas due to the destruction of the flexible thin plate could not follow, resulting in an explosion.

[0008]

As an improvement plan, Japanese Patent Application Laid-Open No. 2-112151 and Japanese Patent Application Laid-Open No. 2-288603 propose an explosion-proof safety device that cuts off the current by peeling or cutting the lead and the explosion-proof valve as the battery internal pressure rises.

In this explosion-proof safety device, as shown in FIGS. 6A and 6B, the explosion-proof valve 21 that deforms outward as the internal pressure rises is brought into contact with the lead blocking stripper 22 by its protrusion 22a. It is attached in this way, and when a predetermined internal pressure is reached, the lead plate 23 is broken by the action of the explosion-proof valve 21 to cut off the current and prevent the explosion.

[0009]

However, in the case of the above-mentioned conventional explosion-proof safety device, as can be seen in FIG. 6, the protrusion 22a of the explosion-proof valve 21 made of a thin metal plate protruding through the hole of the lead blocking stripper 22. Since the lead plate 23 derived from one of the electrode plates is welded to the tip of the plate, the connection strength tends to be unstable.

If the connection strength varies, the internal battery pressure (operating pressure) will vary significantly when the current is cut off.

[0010]

Stabilizing the connection strength is possible by selecting appropriate welding conditions by laser welding, ultrasonic welding, or the like.

[0011]

However, among sealed batteries, lithium secondary batteries, which have a relatively high energy density, use a non-aqueous electrolytic solution consisting of an organic solvent with a low flash point as the electrolytic solution, so that the lead plate and explosion-proof battery that generate sparks are used. The connection with the valve must be made in advance before injecting the electrolyte into the battery.

That is, when assembling a battery, it is common to insert the electrode plate group into the battery case, connect the sealing plate to the lead plate, inject the electrolytic solution, and then seal the battery.

Therefore, even if the lead plate and the explosion-proof valve are connected with a constant connection strength, mechanical stress is applied to the connection part in the assembly process because the liquid injection process is performed before the sealing, and as a result, the connection strength varies. Occurred.

Reducing the mechanical stress applied to the connection between the lead plate and the explosion-proof valve as much as possible tends to be an obstacle in mass production with an automatic assembly machine or the like.

In addition, when the battery is subjected to mechanical vibration or an external impact such as dropping, the electrode plate group moves inside the battery case, the lead plate peels off from the explosion-proof valve, and there is a problem that the function as a battery is lost. ...

[0012]

Therefore, the present invention facilitates mass production of a battery and ensures the safety and reliability of the battery in a short-circuited, overcharged, or reverse-charged state even after an impact such as vibration or dropping is applied to the battery. The purpose is to provide an explosion-proof safety device that can be used.

[0013]

A means for solving the problem The explosion-proof safety device of the present invention is electrically connected to an external terminal plate at a peripheral portion and an internal perforated terminal plate at a central portion, respectively, and electricity from a power generation element to the external terminal plate. A safety valve that also acts as a target conduction path is provided, and this safety valve is arranged on the central side thereof, and is electrically connected to the internal perforated terminal plate in a normal state with a first reversing portion and the first reversing portion. It is provided with a second reversing part arranged outside the part, and the first reversing part is reversely deformed at a relatively low battery internal pressure to cut off the electrical connection between the internal perforated terminal plate and the safety valve. The second reversing portion is reversely deformed at a relatively high internal pressure of the battery, and at least a part of the second reversing portion is broken to discharge the gas in the battery. It is a thing.

[0014]

It is desirable that the reversing portion is formed by at least two bent portions, and a thin-walled portion is formed at the bent portion forming the second reversing portion.

[0015]

The battery of the present invention is characterized by including the above-mentioned explosion-proof safety device.

[0016]

[Action] With the above configuration, when the internal pressure of the battery rises due to a short circuit, overcharge, or reverse charge, when the internal pressure of the battery reaches the lower set value, the bent part corresponding to the lower setting Causes reverse deformation, which can block the electrical conduction path and prevent a sudden temperature rise of the battery, and when the internal pressure of the battery reaches the higher set value due to sudden gas generation. , The bent portion corresponding to the higher setting is destroyed at the same time as the reverse deformation occurs, and it is possible to dissipate the gas inside the battery and prevent the explosion.

[0017]

In addition, since the explosion-proof safety device itself has an electrical connection blocking function, it is possible to strengthen the electrical connection between the lead plate and the sealing plate, facilitating mass production of batteries, and further causing the batteries to vibrate, drop, etc. Even if the electrode plate group moves in the battery case due to the impact of the above, the lead plate does not peel off from the sealing plate, and a highly safe and reliable battery can be obtained.

[0018]

[Examples] Hereinafter, the present invention will be described with reference to specific examples together with drawings.

[0019]

1, FIG. 2 and FIG. 3 are views showing an assembly sealing plate including an explosion-proof safety device according to an embodiment of the present invention.

This assembled sealing plate is a metal internal perforated terminal plate 3 in which a flexible metal safety valve 1, an insulating ring 2, a protrusion 3a is provided in the center, and a ventilation hole 3b is formed around the protrusion 3a. It is composed of a metal lead mounting plate 4, an external terminal plate 5, and an insulating gasket 6.

The metal safety valve 1 is integrally formed with annular bent portions 1a and 1b, and the central portion of the metal safety valve 1 is a protrusion of a metal perforated terminal plate 3 projecting through an insulating ring 2. It is welded to the upper

surface of the portion 3a, and the metal safety valve 1 and the terminal plate 3 are electrically connected by forming the welded portion s.

It is appropriate that the metal safety valve 1 and the protrusion 3a of the metal perforated terminal plate 3 are welded by ultrasonic welding, resistance welding, laser welding, or the like.

As shown in FIG. 3, a thin-walled portion 1c may be provided on the bent portion 1a of the metal safety valve 1.

[0020]

Further, the metal reed mounting plate 4 is provided with a ventilation hole 4a, and the battery internal pressure is adjusted by the ventilation hole 4a, the inner opening of the insulating ring 2, the ventilation hole 3b of the metal perforated terminal plate 3, and the metal. It is configured to be sufficiently added to the safety valve 1.

[0021]

Next, the operation of the assembly sealing plate according to the present embodiment to cut off the current in response to an increase in the internal battery pressure will be described with reference to FIGS. 4A and 4B.

The assembly sealing plate is welded to the lead plate 7 attached to the electrode plate, and is attached by crimping the battery case 8 which also serves as the terminal plate after injecting the electrolytic solution to seal the battery.

When the internal pressure of the battery rises due to a short circuit, overcharging, reverse charging, etc., the internal pressure of the battery is increased by the metal safety valve 1 through the vent holes provided in the metal reed mounting plate 4, the metal perforated terminal plate 3, and the insulating ring 2. Is told to.

Since the outer peripheral portion of the metal safety valve 1 is crimped and fixed by the battery case 8, when the battery internal pressure reaches a predetermined value, the central portion is lifted upward and bent as shown in FIG. 4A. The portion 1b is inverted and deformed, and the welded portion s is peeled off by this stress, the connection with the metal perforated terminal plate 3 is broken, and the current is cut off.

[0022]

In addition, the current cutoff between the metal safety valve 1 and the metal perforated terminal plate 3 due to the increase in the battery internal pressure described above causes the metal safety valve 1 to be destroyed by the internal pressure in order to solve the short circuit, overcharge, or reverse charge state of the battery. Therefore, the welding strength of the metal safety valve 1 and the protrusion 3a of the metal perforated terminal plate 3 should be made smaller than the breaking strength of the metal safety valve 1. It is configured.

In this embodiment, the bending angle of the bent portion 1b of the metal safety valve 1 is set to an obtuse angle, and the bending angle of the bent portion 1a is set to an acute angle so that the bent portion 1b undergoes reverse deformation at a low pressure. ing.

[0023]

Further, since the metal safety valve 1 and the metal perforated terminal plate 3 are sufficiently separated by the reverse deformation, the metal safety valve 1 and the metal perforated terminal plate 3 come into contact with each other due to the fluctuation of the battery internal pressure after the current is cut off, causing a chattering phenomenon. It won't wake you up.

[0024]

Further, if the configuration is as shown in FIG. 3, that is, if the thin portion 1c is provided in the bent portion 1a of the metal safety valve 1, as described above, in the short-circuit, overcharge, and reverse charge states, some other factor Even when the internal pressure of the battery suddenly rises, the thin-walled portion 1c provided in the bent portion 1a of the metal safety valve 1 is destroyed at the same time as the inverted deformation of the bent portion 1a as shown in FIG. 4B. Therefore, the gas can be reliably discharged to the outside of the battery through the ventilation holes 5a provided in the terminal plate 5.

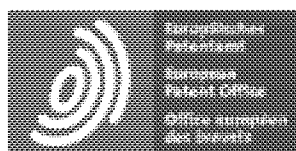
[0025]

By using the explosion-proof safety device of the present invention, since the explosion-proof safety device itself has a current cutoff mechanism, the welding strength with the lead plate can be relatively high, so that the trouble of the assembly sealing plate coming off in the battery assembly process occurs. This can be avoided and mass production of batteries can be facilitated.

Further, even if an impact such as vibration or dropping is applied to the battery, troubles such as battery malfunction due to the lead coming off from the assembly sealing plate can be avoided.

[0026]

According to the explosion-proof safety device of the present invention, it is possible to effectively prevent heat generation and explosion of a battery even at the time of short circuit, overcharging, and reverse charging, and to apply an impact such as vibration or dropping. However, it is possible to obtain a battery equipped with a highly reliable explosion-proof safety device that is less likely to cause malfunction.



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CLAIMS JPH06215746A

1.

A safety valve that is electrically connected to the external terminal plate at the peripheral portion and the internal perforated terminal plate at the central portion and also acts as an electrical conduction path from the power generation element to the external terminal plate is provided, and this safety valve is at the center thereof. A first reversing portion arranged on the side and electrically connected to the internal perforated terminal plate in a normal state, and a second reversing portion arranged outside the first reversing portion are provided. The first reversing part is inverted and deformed at a relatively low set battery internal pressure to cut off the electrical connection between the internal perforated terminal plate and the safety valve, and the second reversing part is compared. Explosion-proof safety device that reverses and deforms due to the internal pressure of the battery and at least partly breaks to discharge the gas inside the battery.

2.

The explosion-proof safety device according to claim 1, wherein the reversing portion is formed by at least two bent portions.

3.

The explosion-proof safety device according to claim 2, wherein a thin-walled portion is formed in a bent portion forming the second reversing portion.

4.

A battery provided with the explosion-proof safety device according to any one of claims 1 to 3.



Espacenet

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EXPLOSION-PROOF SAFETY DEVICE AND BATTERY HAVING IT

Inventor(s): MITAMURA TOMOKAZU; KONISHI AKIYA; TSURUTA KUNIO;
MURAKAMI TETSUYA ± (MITAMURA TOMOKAZU, ; KONISHI
AKIYA, ; TSURUTA KUNIO, ; MURAKAMI TETSUYA)

Applicant(s): MATSUSHITA ELECTRIC IND CO LTD ± (MATSUSHITA
ELECTRIC IND CO LTD)

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- cooperative: H01M2/1241 (EP); H01M2/34 (EP); H01M2/345
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PURPOSE: To prevent the rupture of a battery, by interrupting an electric path between a terminal board and a power generating element, by a first inversion deformation when internal pressure in the battery is increased, and breaking a safety valve, concurrently with a second inversion deformation, to exhaust gas. **CONSTITUTION:** When battery internal pressure is increased by a short circuit, overcharge, and reverse charge, the battery internal pressure is transmitted to a metal-made safety valve 1 through an air vent hole provided on a metallic lead fitting plate 4, a metallic perforated terminal board 3, and an insulating ring 2. In the valve 1, since its outer peripheral part is calked by a battery case, when the battery internal pressure is reached to a given value, a center part is risen upward, a folded part 1b is reversely deformed, a fused part S is separated, and the connection with the terminal board 3 is released to interrupt current. Next, when battery internal pressure is suddenly increased, the valve 1 is ruptured concurrently with the reverse deformation of the folded part 1a of the valve 1 to exhaust gas to outside a battery through an air vent hole provided on a terminal board 5. Consequently, the rupture of the battery can be prevented.

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(71)出願人 000005321

松下電器産業株式会社

大阪府門真市大字門真1006番地

(72)発明者 三田村 知一

大阪府門真市大字門真1006番地 松下電器
産業株式会社内

(72)発明者 小西 昭治

大阪府門真市大字門真1006番地 松下電器
産業株式会社内

(72)発明者 鶴田 邦夫

大阪府門真市大字門真1006番地 松下電器
産業株式会社内

(74)代理人 弁理士 小鍛冶 明 (外2名)

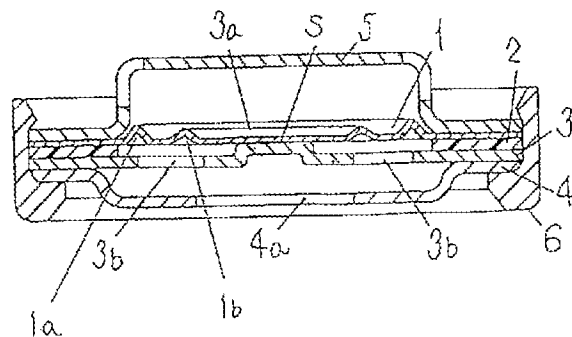
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(54)【発明の名称】 防爆安全装置および防爆安全装置を備えた電池

(57)【要約】

【目的】 電池が短絡、過充電、逆充電状態にあっても安全性と信頼性を確保できる防爆安全装置を提供することを目的とする。

【構成】 二重の反転変形構造を有し、電池内の内圧が上昇した時に、まず第一の反転変形によって端子板と発電要素間の電氣的導通経路を遮断し、次に第二の反転変形と同時に安全弁を破壊して電池内部の発生ガスを円滑に外部に排出し、電池の破裂を未然に防止する。



【特許請求の範囲】

【請求項1】 周縁部において外部端子板と、中央部において内部有孔端子板とそれぞれ電氣的に接続され、発電要素から外部端子板への電氣的導通経路としても作用する安全弁を備え、この安全弁が、その中央側に配置され、前記内部有孔端子板と通常状態においては電氣的に接続されている第1反転部と、該第1反転部の外方に配置された第2反転部とを備え、前記第1反転部が、比較的低い設定電池内部圧力で反転変形して、前記内部有孔端子板と安全弁との電氣的接続を遮断するようになっているとともに、前記第2反転部が、比較的高い設定電池内部圧力で反転変形するとともに、少なくとも一部が破断し、電池内ガスを排出するようになっている防爆安全装置。

【請求項2】 前記反転部が、少なくとも2重の折り曲がり部によって形成されている請求項1の防爆安全装置。

【請求項3】 前記第2反転部を形成する折れ曲がり部に、薄肉部が形成されている請求項2の防爆安全装置。

【請求項4】 請求項1ないし3のいずれかの防爆安全装置を備えた電池。

【発明の詳細な説明】

【0001】

【産業上の利用分野】

【0002】本発明は、電池に用いる防爆安全装置に関するものである。

【0003】

【従来の技術】

【0004】近年、AV機器あるいはパソコン等の電子機器のポータブル化、コードレス化が急速に進んでおり、これらの駆動用電源として小型、軽量で高エネルギー密度を有する密閉型電池への要望が高い。特にリチウム二次電池は、とりわけ高電圧、高エネルギー密度を有する電池として期待が大きい。

【0005】ところが、密閉型電池、例えばリチウム金属やカーボン材料を負極として使用するリチウム二次電池は、短絡、過充電、逆充電等の誤使用された時に、電池内にガスが発生し、内圧が上昇して破裂しその機能を失ったり、周辺機器に損傷を与えることがあった。

【0006】特に、過充電状態が継続すると、電解液の分解および活物質の分解により、電池が急激に温度上昇したり、爆発を招く恐れがあるという問題があった。

【0007】このような、短絡、過充電、逆充電状態における電池の爆発を未然に防止するため、従来は、例えば実開昭54-137734号公報、実公昭59-15398号公報に示されているように、電池内圧上昇に伴い破壊して内圧を電池外に排出する可撓性薄板（ダイヤフラム）を有する防爆安全装置が提案されている。この安全装置は、図5の（A）および（B）に示したように内圧の上昇に伴い、この内圧により可撓性薄板42（ダ

イアフラム）が外方に変形を生じて破壊してガスを電池外に排出し電池の爆発を未然に防止するものであるが、電池の短絡、過充電、逆充電状態は継続される。したがって、この安全装置をエネルギー密度（単位体積もしくは単位重量当りの電池から取り出せる電気エネルギー量）の高い電池、例えばリチウム二次電池に用いた場合、特に過充電状態が継続すると電池内圧が急激に上昇し、可撓性薄板の破壊によるガスの排出が追従せず、爆発にいたる場合があった。

【0008】その改良案として特開平2-112151号、特開平2-288063号には、電池内圧上昇に伴いリードと防爆弁を剥離もしくは切断して電流を遮断する防爆安全装置を提案している。この防爆安全装置は、図6の（A）および（B）に示したように内圧の上昇に伴い外方に変形を生じる防爆弁21を、その突起部22aがリード遮断用ストリッパ22に接触するようにして取り付け、所定の内圧に達したとき、防爆弁21の作用によりリード板23を破断するようにして電流を遮断し、爆発を防止するようにしたものである。

【0009】

【発明が解決しようとする課題】しかしながら、上記の従来の防爆安全装置の場合、図6で解るようにリード遮断用ストリッパ22の穴を介して突出した金属薄板製の防爆弁21の突起部22aの先端に一方の極板から導出したリード板23を溶接しているため、接続強度が不安定となりやすい。接続強度にバラツキが発生すると、電流遮断されときの電池内圧（作動圧）の著しいバラツキとなる。

【0010】接続強度を安定化することは、レーザー溶接、超音波溶着等で適切な溶接条件を選定することで可能である。

【0011】ところが、密閉型電池のなかでも比較的エネルギー密度の高いリチウム二次電池の場合、電解液として、低引火点の有機溶媒からなる非水電解液を用いるため、火花の発生を伴うリード板と防爆弁との接続は、電池に電解液を注液する前にあらかじめ実施する必要がある。つまり、電池を組み立てる場合、極板群を電池ケースに挿入した後に、封口板をリード板に接続し、電解液を注液し、しかる後に封口することが一般的である。そこで、一定の接続強度でリード板と防爆弁を接続しても、その後封口するまでに、注液工程を経るため組み立て工程において接続部に機械的ストレスが加わり、結果的に接続強度のバラツキが発生した。リード板と防爆弁との接続部に加わる機械的ストレスを可能な限り少なくすることは自動組立機などで大量生産する際に障害となりやすい。また、電池に機械的振動もしくは、落下等外部から衝撃を加えた場合、電池ケース内において極板群が移動し、リード板が防爆弁より剥離し、電池としての機能を失うという問題があった。

【0012】そこで、本発明は、電池の大量生産が容易

で、電池に振動、落下等の衝撃が加わった後においても、短絡、過充電、逆充電状態における電池の安全性と信頼性を確保することのできる防爆安全装置を提供することを目的とするものである。

【0013】

【課題を解決するための手段】本発明の防爆安全装置は、周縁部において外部端子板と、中央部において内部有孔端子板とそれぞれ電氣的に接続され、発電要素から外部端子板への電氣的導通経路としても作用する安全弁を備え、この安全弁が、その中央側に配置され、前記内部有孔端子板と通常状態においては電氣的に接続されている第1反転部と、該第1反転部の外方に配置された第2反転部とを備え、前記第1反転部が、比較的低い電池内部圧力で反転変形して、前記内部有孔端子板と安全弁との電氣的接続を遮断するようになっているとともに、前記第2反転部が、比較的高い電池内部圧力で反転変形するとともに、少なくとも一部が破断し、電池内ガスを排出するようになっていることを特徴とするものである。

【0014】上記反転部は、少なくとも2重の折れ曲がり部によって形成され、前記第2反転部を形成する折れ曲がり部に、薄肉部が形成されていることが望ましい。

【0015】本発明の電池は、上記の防爆安全装置を備えていることを特徴とするものである。

【0016】

【作用】以上の構成により、短絡、過充電、逆充電状態により、電池内圧が上昇した場合、まず低い方の設定値に電池の内部圧力が達した時に、低い方の設定に該当する折れ曲がり部が反転変形を生じ、このことにより電氣的導通経路を遮断し電池の急激な温度上昇を防止することができ、また、急激なガス発生によって高い方の設定値に電池の内部圧力が達した時には、高い方の設定に該当する折れ曲がり部が反転変形を生じると同時に破壊され、電池内部のガスを逸散させ爆発を未然に防ぐことが可能である。

【0017】また、防爆安全装置自体が電氣的接続遮断の機能を有するため、リード板と封口板との電氣的接続を強くすることが可能となり、電池の量産が容易であり、さらに電池に振動、落下等の衝撃が加わり極板群が電池ケース内を移動しても、リード板が封口板から剥離することもなく、安全性、信頼性の高い電池を得ることができる。

【0018】

【実施例】以下、図面とともに本発明を具体的な実施例によって説明する。

【0019】図1、図2および図3は、本発明の実施例における防爆安全装置を含む組立封口板を示す図である。この組立封口板は、可撓性の金属製安全弁1、絶縁リング2、中央に突起部3aが設けられ、この突起部3aの周囲に通気孔3bが形成された金属製内部有孔端子

板3、金属製リード取りつけ板4、外部端子板5および絶縁ガasket6より構成されている。金属製安全弁1は、環状の折り曲がり部1aおよび1bが一体に形成されており、また金属製安全弁1の中央部は、絶縁リング2を介して突出する金属製有孔端子板3の突起部3aの上面に溶着しており、溶着部sの形成により、金属製安全弁1と端子板3とは電氣的に接続されている。なお、金属製安全弁1と金属製有孔端子板3の突起部3aとの溶着は、超音波溶着もしくは抵抗溶接、レーザー溶接等により行うことが適切である。金属製安全弁1の折り曲がり部1aには図3に示したように薄肉部1cが設けられていてもよい。

【0020】また、金属製リード取りつけ板4には、通気孔4aが設けられており、電池内圧は、この通気孔4a、絶縁リング2の内開口および金属製有孔端子板3の通気孔3b、金属製安全弁1に充分加えられるよう構成されている。

【0021】次に、上記本実施例による組立封口板が、電池内圧の上昇に応じて電流を遮断する動作について、図4の(A)および(B)を参照して説明する。組立封口板は、極板に付属したリード板7と溶接され、電解液注液後に、端子板を兼ねる電池ケース8をかきしめることにより取り付けられ、電池を密閉している。短絡、過充電もしくは、逆充電等により電池内圧が上昇した場合、この電池内圧が金属製リード取りつけ板4、金属製有孔端子板3、絶縁リング2に設けられた通気孔を通して金属製安全弁1に伝えられる。金属製安全弁1はその外周部が電池ケース8によりかきしめられ固定されているため、電池内圧が所定の値に達すると、図4の(A)に示すように中心部が上方へ持ち上げられ折り曲がり部1bが反転変形し、この応力によって前記溶着部sが剥離され金属製有孔端子板3との接続が解かれて電流が遮断される。

【0022】なお、上述の電池内圧の上昇による金属製安全弁1と金属製有孔端子板3との電流遮断は、電池の短絡、過充電もしくは、逆充電状態を解くため内圧による金属製安全弁1の破壊される以前に確実に行う必要があるが、そのために、金属製安全弁1と金属製有孔端子板3の突起部3aとの溶着はその溶着強度を金属製安全弁1の破壊強度よりも小さくするよう構成されている。本実施例においては、金属製安全弁1の折り曲がり部1bの折り曲がり角度を鈍角に折れ曲がり部1aの折り曲がり角度を鋭角に構成し、折り曲がり部1bの方が低圧で反転変形を生じるようにしている。

【0023】さらに、反転変形によって金属製安全弁1と金属製有孔端子板3が充分隔離されるため、電流遮断後に電池内圧の変動によって金属製安全弁1と金属製有孔端子板3が接触しチャタリング現象を起こすこともない。

【0024】また、図3に示すように構成すれば、すな

わち、金属製安全弁1の折れ曲がり部1aに薄肉部1cを設ければ、上述のように、短絡、過充電、逆充電状態において、他のなんらかの要因で電池の内圧が急激に上昇した場合においても、図4の(B)に示すように前記金属製安全弁1の折れ曲がり部1aに設けた薄肉部1cが折れ曲がり部1aの反転変形と同時に破壊され、端子板5に設けた通気孔5aを通してガスを電池外部に確実に排出することができる。

【0025】本発明の防爆安全装置を用いれば、防爆安全装置自体が電流遮断機構を有するため、リード板との溶接強度を比較的高くすることができるので、電池組み立て工程における組立封口板はずれのトラブル発生を回避でき電池の量産化を容易にすることができる。また、電池に振動あるいは落下等の衝撃を加えても組立封口板からのリードはずれによる電池動作不良といったトラブルも回避できる。

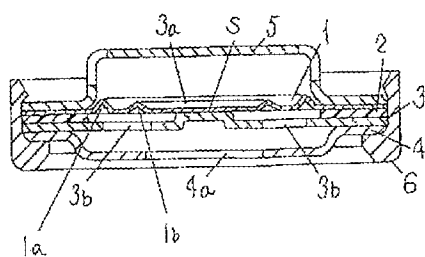
【0026】

【発明の効果】本発明における防爆安全装置によれば、短絡、過充電、逆充電時においても、電池の発熱、爆発を効果的に防止することができ、振動、落下等の衝撃を加えても動作不良が発生しにくい信頼性の高い防爆安全装置を備えた電池を得ることが可能である。

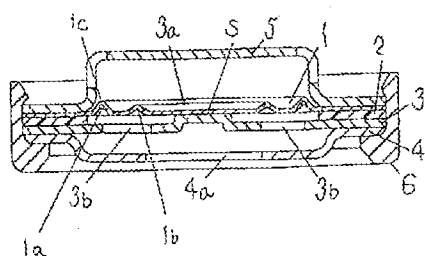
【図面の簡単な説明】

【図1】本発明の実施例の防爆安全装置を含む組立封口板の断面図である。

【図1】



【図3】



【図2】本発明の実施例の防爆安全装置を含む組立封口板の分解斜視図である。

【図3】本発明の実施例の防爆安全装置を含む組立封口板の断面図である。

【図4】上記防爆安全装置を備えた電池の作動状態を説明するための説明図である。

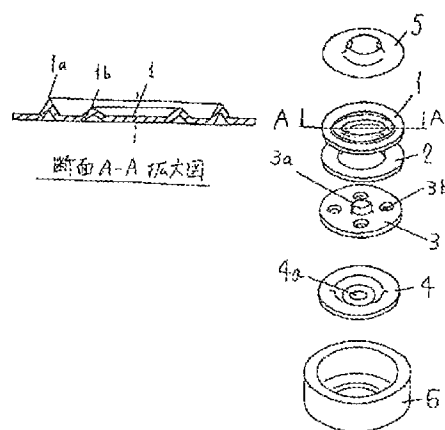
【図5】従来の密閉型電池断面図である。

【図6】従来の密閉型電池断面図である。

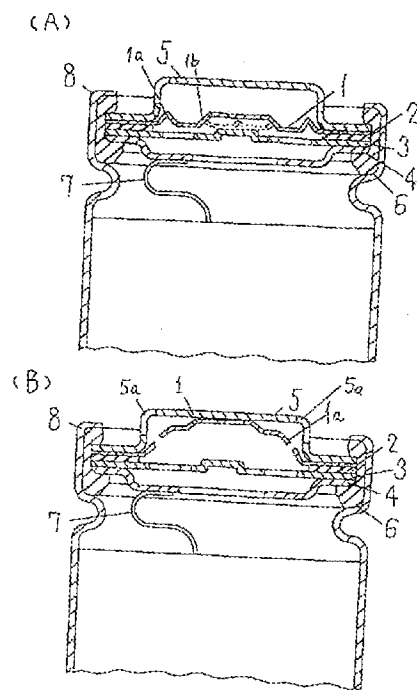
【符号の説明】

- 1 金属製安全弁
- 2 絶縁リング
- 3 金属製有孔端子板
- 4 金属製リード取り付け板
- 5 端子板
- 6 絶縁ガスカート
- 7 リード板
- 8 電池ケース
- 1a 折れ曲がり部
- 1b 折れ曲がり部
- 1c 薄肉部
- 3a 突起部
- 3b 通気孔
- 4a 通気孔
- 5a 通気孔

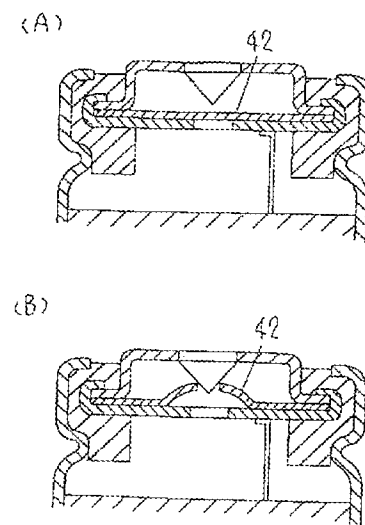
【図2】



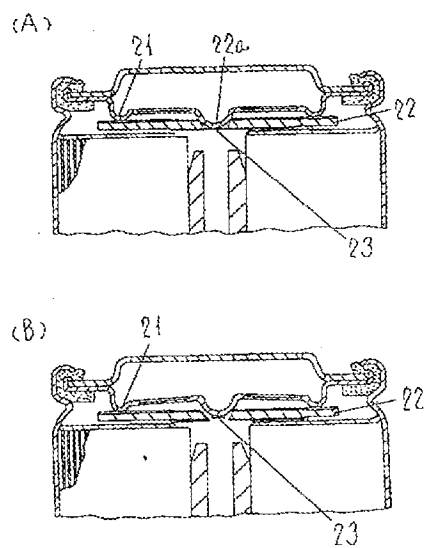
【図4】



【図5】



【図6】



フロントページの続き

(72)発明者 村上 哲哉
大阪府門真市大字門真1006番地 松下電器
産業株式会社内



Espacenet

Bibliographic data: JPH08339793 (A) — 1996-12-24**SAFETY DEVICE FOR SEALED BATTERY**

Inventor(s): KAMAUCHI MASA HARU; IJIRI YASUO; KASHIOKA TORU; OUCHI YUTAKA ± (KAMAUCHI MASA HARU, ; IJIRI YASUO, ; KASHIOKA TORU, ; OUCHI YUTAKA)

Applicant(s): MITSUBISHI CABLE IND LTD ± (MITSUBISHI CABLE IND LTD)

Classification: - **international:** H01M2/12; H01M2/34; (IPC1-7): H01M2/12; H01M2/34

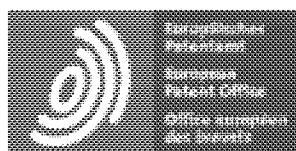
- **cooperative:** Y02E60/10 (EP)

Application number: JP19950256308 19951003

Priority number (s): JP19950256308 19951003 ; JP19950085409 19950411

Abstract of JPH08339793 (A)

PURPOSE: To eliminate a welding part from a current interrupting mechanism and to facilitate adjusting/setting for switching from connection to interruption of a current relating to a preset internal pressure value. CONSTITUTION: An opening part 1a of a battery can 1 is sealed by fixedly calking the periphery of a seal unit 2 through an insulating gasket 5. The seal unit is constituted by a rigid disk-shaped seal cover 6 connected with a lead wire 4 from a generating element 3, insulating ring 7, flexible conduction plate 8 with a circular annular protrusion 8b into contact to be electrically connected to the seal cover, rubber-made safety valve 30 tightly fitted to a through hole 31 with an upper end face adjacent to the conduction plate and a cap-shaped positive electrode terminal 9. Press fit resistance force in an intermediate step difference surface part of the safety valve against a small diametric hole part of the through hole is set so as to be smaller than pressing force to the safety valve based on a battery can internal pressure exceeding a preset internal pressure value, and when exceeding the preset internal pressure value, to relatively move upward to separate the protrusion by press fitting the safety valve, further when an internal pressure rises, to press fit a flange part of the safety valve, to spring out the through hole, so as to break a fragile part 8c. The safety valve is integrally vulcanized to be molded with the seal cover, to attain sureness of assembly.



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DESCRIPTION JPH08339793A

[0001]

The present invention relates to a safety device for a closed battery for dealing with an abnormal situation such as an abnormal increase in internal pressure due to overcharging or the like in a closed battery such as a secondary battery.

[0002]

Conventionally, as a safety device for this type of sealed battery, a lead plate to which a lead wire from a power generation element is connected is attached to the inner surface of the insulating plate, and the lead plate is used to form a through hole in the center of the insulating plate. A disc-shaped flexible explosion-proof valve formed of a conductive material is arranged so as to cover and cover the upper side of the insulating plate, and a protrusion at the center of the explosion-proof valve is fitted into the through hole to form the protrusion. The lead plate is known to be ultrasonically welded (see, for example, JP-A-2-112151 and JP-A-2-288603).

In these publications, the current is interrupted by the protrusion being lifted by an increase in the internal pressure of the battery can and the lead plate being broken at the peripheral edge of the through hole, or by the welded portion with the lead plate being peeled off. It is stated that it will be done.

[0003]

Further, as another safety device for a conventional sealed battery, a disc-shaped valve holder made of a conductive material to which a lead wire is connected and a protrusion in the center covered on the upper surface of the valve holder are formed on the upper surface of the valve holder. It is known that the valve body is made of a disc-shaped flexible valve body made of a conductive material that is spot-welded or ultrasonically welded, and the space between the valve body and the valve holder is communicated with the inside of the battery can. (For example, see JP-A-6-140011).

That is, in this case, the valve body is welded to the valve holder to which the lead wire is connected, thereby avoiding the inconvenience caused by directly welding the explosion-proof valve to the lead plate. Then, in this publication, the pressure in the space also rises due to the rise in the internal pressure of the battery can, the valve body is lifted upward, the welding between the central protrusion and the valve holder is disengaged, and the current is cut off. Is described.

[0004]

In both cases, when the internal pressure rises further, the flexible explosion-proof valve or the flexible valve body breaks at the notches provided in these and allows the gas inside to escape to the outside, thereby causing the battery itself to explode. It is designed to prevent it.

[0005]

However, in the above-mentioned conventional safety device for a sealed battery, since the welded portion is used in the current blocking mechanism, the current blocking is ensured depending on the strength of the welding itself. There is a risk of situations that cannot be done.

That is, if the welding strength is too high, the valve will not separate from the valve holder even if the internal pressure rises, and as a result, the current cannot be interrupted. Conversely, if the welding strength is too low, during normal use Due to slight internal pressure fluctuations and impacts, the valve peels off from the valve holder and the current is cut off, making it unusable.

[0006]

As described above, in the conventional one, it is necessary to surely set the welding strength to a predetermined value, but for that purpose, if the welding is performed in a dot shape by spot welding or the like, the current is passed through the point welded portion in normal use. Since the current is applied, the resistance may become too high and abnormal heat generation may occur.

On the contrary, if the welded portion is set to have a circumferential shape or the like, the welding strength increases in proportion to the welded area, so that the above-mentioned inconvenience occurs.

[0007]

Moreover, if the welding work that is just enough for the predetermined welding strength as described above is performed on the mass production line, the process control becomes extremely difficult, and the welding strength tends to vary among the products, and the reliability of the product is improved. It cannot be improved.

[0008]

The present invention has been made in view of such circumstances, and an object of the present invention is to eliminate a welded portion from a current cutoff mechanism of a safety device for a closed battery and to connect a current to a predetermined internal pressure value. The purpose is to facilitate adjustment and setting for switching from to the cutoff state.

[0009]

[Means for Solving the Problems] In order to achieve the above object, the invention according to claim 1 has a power generation element housed inside a battery can, and an opening facing the outside of the battery can is sealed by a sealing body. It is assumed that it has been done.

In this case, the sealing body is sealed with a flexible conductive plate formed to have flexibility by a conductive material and the inside of the battery can by the conductive material, and is not substantially deformed against the internal pressure thereof. It shall be provided with a sealing lid formed to have rigidity and a safety valve formed of an insulating material and relatively moving due to an increase in the internal pressure of the battery can.

Then, a through hole is provided in the substantially central portion of the sealing lid, the outer peripheral edge portion is airtightly attached to the opening of the battery can, and the lead wire from the power generation element is connected to the inner surface facing the inside of the battery can.

Further, the conductive plate is polymerized with the outer surface of the sealing lid at a position outside the sealing lid with the outer edge portion being the opening of the battery can via the insulating member, and with the sealing lid at a position near the through hole. Fix the position so that it comes into contact and is in an electrically connected state.

Further, the safety valve is positioned so that the tip surface is close to the conduction plate in a state where the safety valve is inserted in the inward and outward directions from the inside of the battery can to the conduction plate so as to seal the through hole in the through hole of the sealing lid. In addition, the inner surface of the hole constituting the through hole and the outer surface in the circumferential direction of the safety valve facing the inner surface of the hole resist the internal pressure until the internal pressure of the battery can rises and reaches the set internal pressure value. While maintaining the state in which the relative displacement of the safety valve with respect to the conduction plate is prevented, the safety valve is relatively displaced toward the conduction plate side by the action of the internal pressure exceeding the set internal pressure value to bring the

conduction plate into contact with the sealing lid. It is configured to be fitted to each other so as to exert a resistance force that causes them to separate.

[0010]

In the case of the above configuration, in the normal use state where proper discharge and proper charging are performed and the internal pressure inside the battery can is smaller than the set internal pressure value, the outer surface in the circumferential direction of the safety valve and the through hole of the sealing lid. Since the resistance force between the inner surface and the inner surface is larger than the internal pressure, the relative movement of the safety valve is prevented against the internal pressure, and the flexible conductive plate and the sealing lid to which the lead wire is connected come into contact with each other. Be maintained.

As a result, the electrical connection is maintained.

[0011]

On the other hand, when the internal pressure inside the battery can exceeds the set internal pressure value due to overcharging or the like, the pressing force for relatively moving the safety valve based on the internal pressure becomes larger than the resistance force, so that the safety valve becomes conductive. It moves relative to the plate side and pushes and bends the conductive plate outward, so that the contact with the sealing plate is separated.

As a result, the electrical connection state due to the contact is cut off and switched to the electrical cutoff state, so that the battery user can be made aware of the occurrence of an abnormality.

[0012]

In this case, since the sealing lid has rigidity that does not deform even when it receives the above internal pressure, the internal pressure acts directly and effectively on the safety valve, and the pressing force due to the relative movement of the safety valve is supported by the outer edge portion. It will act intensively in the central part of.

For this reason, when an internal pressure rise that exceeds the above-mentioned set internal pressure value occurs, the internal pressure is switched to the electrically cutoff state with better responsiveness as compared with the case where the internal pressure is indirectly applied to the entire disk-shaped explosion-proof valve. It becomes possible.

[0013]

Moreover, in such a current cutoff mechanism, the electrical connection state and the cutoff state are switched according to the internal pressure inside the battery can by setting the resistance force by fitting between the safety valve and the through hole. Compared with the case where the tensile breaking strength of the welded portion is set as in the conventional closed battery, it is easier to adjust and change the switching strength with respect to the internal pressure, and it is possible to surely set it to a predetermined value. Switching to the electrical cutoff state is ensured.

Further, the process control at the time of assembly can be facilitated, and the reliability of the product can be improved.

[0014]

In the invention according to claim 2, in the invention according to claim 1, the through hole is formed so as to have a uniform inner diameter in the inner and outer directions of the battery can, while the safety valve is formed in a piston shape.

Then, the inner peripheral surface of the through hole and the outer peripheral surface of the safety valve resist the internal pressure of the battery can by exerting a sliding friction resistance force between the two peripheral surfaces against the relative displacement of the safety valve to the conduction plate side. The structure is such that they are tightly fitted to each other.

[0015]

In the case of the above configuration, the safety valve is maintained in a state where its relative displacement is blocked up to the set internal pressure value, and the safety valve is allowed to be relatively displaced toward the conduction plate side for the battery can internal pressure exceeding the set internal pressure value. Since the resistance force between the safety valve and

the through hole is exerted by the sliding friction resistance force between the peripheral surfaces of the safety valve and the through hole, the resistance force can be easily adjusted or changed according to the internal pressure of the battery can.

That is, by setting and changing the hole diameter of the through hole, setting and changing the outer diameter of the safety valve, and setting and changing the material of the safety valve, sliding between the peripheral surfaces of the safety valve and the through hole. The dynamic friction resistance can be adjusted and changed.

[0016]

The invention according to claim 3 is the invention according to claim 2, wherein the safety valve has the same cross-sectional shape from the tip to the proximal end side corresponding to the through hole of the sealing lid, and the proximal end side is from the inner surface of the sealing lid to the inside of the battery can. It is composed of a columnar main body portion arranged so as to project toward the side, and a convex portion slightly protruding from the base end side of the main body portion to the periphery.

In addition, the internal pressure value when sliding and relative movement with the inner peripheral surface of the through hole under the internal pressure of the battery can is relatively larger at the convex portion forming position than at the main body portion. It is set to be.

[0017]

In the case of the above configuration, the safety valve is composed of a columnar main body portion and a convex portion formed on the base end side of the main body portion, and the main body portion corresponds to a through hole of the sealing lid from the tip end to the base end side. The battery has the same cross-sectional shape and is arranged so that the base end side protrudes from the inner surface of the sealing lid toward the inside of the battery can, and the convex portion slightly protrudes from the base end side of the main body to the periphery. In response to an increase in the internal pressure of the can that exceeds the set internal pressure value on the base end surface, the main body of the safety valve slides through the through hole of the sealing lid and moves relative to the convex part until it is stopped at the entrance of the through hole. The contact portion of the plate is separated from the sealing lid to cut off the current conduction.

Then, when the internal pressure further increases and exceeds the sliding frictional resistance force between the convex portion and the inner peripheral surface of the through hole, the convex portion slides through the through hole and jumps out, resulting in the inside of the battery can. The gas is discharged through the through hole, and the internal pressure of the battery can becomes atmospheric pressure. Therefore, in the event of an abnormality, the current is first cut off to make the battery user aware of the occurrence of the abnormality, and even if the user is left unattended, the internal pressure is released by venting the internal gas to prevent further increase in the internal pressure. Will be done.

[0018]

The invention according to claim 4 is the invention according to claim 2, wherein the safety valve has the same cross-sectional shape from the tip to the proximal end side corresponding to the through hole of the sealing lid, and the proximal end side is from the inner surface of the sealing lid to the inside of the battery can. It is composed of a columnar main body portion arranged so as to project to the side, and a flange-shaped pressure receiving portion expanded around the base end side of the main body portion.

In addition, the sealing lid is provided with a gas vent hole formed so as to communicate the inside and the outside of the battery can, and a closing body for sealing the gas vent hole. Then, the opening strength of the closing body is set so as to receive an internal pressure larger than the internal pressure of the battery can that brings the pressure receiving portion into contact with the inner surface of the sealing lid.

[0019]

In the case of the above configuration, the safety valve is composed of a columnar main body portion and a flange-shaped pressure receiving portion expanded on the base end side of the main body portion, while the sealing lid communicates the inside and the outside of the battery can with each other. Since the degassing hole to be degassed and the closing body that closes on the inner surface side of the sealing lid of the degassing hole and has a predetermined opening strength are provided, when the internal pressure of the battery can exceeds the set internal pressure value, the pressure received first The safety valve moves relative to each other until the part hits the inner surface of the sealing lid and is stopped, and the current is cut off.

Then, when the internal pressure of the battery can exceeds the set internal pressure value and further rises, the closing body is opened and the internal gas is released through the gas vent hole. In this case, since the pressure receiving portion of the safety valve is expanded more than the main body portion and the base end surface of the safety valve, which is the pressure receiving surface, is enlarged as compared with the case where only the main body portion is used, the conduction plate is more responsive to an increase in internal pressure. While it is possible to cut off the current by pushing up, even if the internal pressure exceeds the above set internal pressure value and further rises, the opening of the closing body prevents the internal pressure from rising further and destroys the sealed battery. Is avoided.

[0020]

The invention according to claim 5 is the invention according to claim 1, wherein the through hole is a large diameter hole portion on the inner side of the battery can, a small diameter hole portion on the conduction plate side, and these large diameter hole portion and the small diameter hole. It shall have a retaining surface formed between the portions and facing the inner side of the battery can.

On the other hand, the safety valve is inserted between the small-diameter portion through which the small-diameter hole portion is inserted to the outer side of the battery can, the large-diameter portion internally fitted into the large-diameter hole portion, and the stopper between the small-diameter portion and the large-diameter portion. It shall have a stepped surface formed so as to face the surface. Then, the stepped surface of the safety valve comes into contact with the contact surface of the through hole and the safety valve, and the portion from the stepped surface of the safety valve to the large diameter portion receives the internal pressure of the battery can and is a through hole. By exerting a press-fitting resistance force to be press-fitted into the small-diameter hole portion of the battery can, the battery cans are fitted to each other so as to resist the internal pressure of the battery can.

[0021]

In the case of the above configuration, the safety valve is maintained in a state where its relative displacement is blocked up to the set internal pressure value, and the safety valve is allowed to be relatively displaced toward the conduction plate side for the battery can internal pressure exceeding the set internal pressure value. Since the resistance force between the through hole and the through hole is exerted by the press-fitting resistance force of the portion from the stepped surface to the large diameter portion of the safety valve with respect to the small diameter hole portion of the through hole, the above resistance force is adjusted according to the internal pressure of the battery can. Alternatively, the change can be made more easily and reliably than in the case of the invention according to claim 2.

That is, the diameter of the small diameter hole portion of the through hole is set and changed, or the diameter of the large diameter portion of the safety valve is set and changed, that is, the diameter difference between the small diameter hole portion and the large diameter portion is adjusted. By adjusting the contact area between the contact surface of each through hole and the stepped surface of the safety valve, and by setting and changing the material of the safety valve, the press-fitting resistance to the through hole of the safety valve can be freely adjusted. .. Moreover, when assembling the safety valve to the sealing lid, the safety valve can be assembled at a predetermined inward / outward direction position with respect to the sealing lid and the conduction plate only by bringing the stepped surface of the safety valve into contact with the contact surface of the through hole. Therefore, the positioning of the safety valve with respect to the sealing lid is easy and reliable, and the manufacturing of the sealed battery is facilitated. In addition, the tip surface of the safety valve is securely positioned close to the conduction plate and the conduction plate is separated from the sealing lid. Can be reliably maintained in electrical connection with the sealing lid.

[0022]

The invention according to claim 6 is the invention according to claim 2 or 5, wherein the safety valve is formed of an elastic material.

Here, as the elastic material, for example, natural rubber, synthetic rubber such as EPDM (ethylene / propylene / diene copolymer), polypropylene, polyethylene or the like may be used.

[0023]

In the case of the above configuration, since the safety valve is made of an elastic material, the sliding friction resistance force or the press-fitting resistance force between the safety valve and the through hole of the sealing lid can be set more easily.

[0024]

The invention according to claim 7 is the invention according to claim 5, wherein the large diameter portion of the safety valve is projected inward of the battery can with the stepped surface in contact with the contact surface of the through hole. A base end side flange portion having a diameter larger than that of the large diameter hole portion of the through hole is integrally formed at the protruding end.

In addition, the outer diameter of the base end side flange portion is subjected to an internal pressure larger than the internal pressure of the battery can that abuts the base end side flange portion on the inner surface of the sealing lid so as to penetrate the through hole toward the conduction plate side. It is to be set.

[0025]

In the case of the above configuration, when the internal pressure of the battery can rises, the portion of the safety valve from the stepped surface to the large diameter portion is deformed by the internal pressure and press-fitted into the small diameter portion of the through hole.

Due to the relative displacement of the safety valve to the conduction plate side due to this press-fitting, the tip surface of the safety valve pushes up the conduction plate to switch to the current cutoff state, and the base end side flange portion comes into contact with the inner surface of the sealing lid.

As a result, the relative displacement of the safety valve is temporarily stopped due to the press-fitting resistance to the large-diameter hole portion of the proximal end side flange portion corresponding to the outer diameter of the proximal end side flange portion. Then, when the internal pressure of the battery can further increases and exceeds the press-fitting resistance of the base end side flange portion, the base end side flange portion is press-fitted into the large-diameter hole portion, and the entire safety valve passes through the through hole to the conduction plate side. As a result, the internal pressure of the battery can is released. Therefore, as the internal pressure of the battery can rises, a two-step safety operation is performed in which the current is cut off as the first step and the internal pressure is released as the second step.

[0026]

The invention according to claim 8 is the invention according to claim 5, wherein the inner peripheral surface of the small-diameter hole portion of the through hole is formed by a tapered hole surface whose diameter gradually increases toward the conduction plate side, and in addition, the safety valve. The outer peripheral surface of the small-diameter portion is formed by a tapered peripheral surface that abuts on the tapered hole surface in a state where the stepped surface abuts on the contact surface of the through hole.

[0027]

In the case of the above configuration, when the sealed battery is in a normal operating state, the tapered surface of the safety valve abuts on the tapered hole surface of the through hole, whereby the safety valve moves the inside of the through hole to the inside of the battery can. It is surely prevented from falling off.

Therefore, the safety valve is surely maintained in a state where the stepped surface is in contact with the contact surface of the through hole, and is maintained in a posture for surely performing a safe operation when the internal pressure of the battery can rises.

Therefore, from the product shipping stage to the use stage, the safety valve is surely prevented from moving and falling off, the safety valve is securely held against the sealing lid, and the safety operation when the internal pressure of the battery can rises is surely performed.

[0028]

The invention according to claim 9 is the invention according to claim 5, wherein the safety valve is integrally formed with the sealing lid by integral vulcanization molding using rubber for the sealing lid.

[0029]

In the case of the above configuration, since the safety valve is formed integrally with the sealing lid in advance, the work of assembling the safety valve to the sealing lid is omitted, and the production of the sealed battery can be facilitated and rationalized.

At the same time, the integrated vulcanization molding ensures that the safety valve is positioned in advance at a predetermined relative position with respect to the sealing lid and the conduction plate, and therefore, the electrical connection state between the sealing lid and the conduction plate during normal use is maintained. And, the safety operation of the safety valve when the internal pressure rises is surely performed.

[0030]

The invention according to claim 10 is the invention according to claim 1, wherein the tip surface of the safety valve and the inner surface of the flexible conductive plate are adhered to each other and bonded to each other.

[0031]

In the case of the above configuration, since the tip surface of the safety valve and the substantially central inner surface of the flexible conductive plate are adhered to each other, the flexible conductive plate itself is held by the sealing lid via the safety valve. ...

That is, the conduction plate supported by the outer edge portion is adhered to and supported by the safety valve at its substantially central portion, and the safety valve is held by the sealing lid by the sliding frictional resistance force between the inner peripheral surface of the through hole of the sealing lid. Therefore, the conductive plate is in contact with the sealing lid, that is, the electrically connected state is stably maintained.

Therefore, even if the conduction plate and the sealing lid are not welded, the above-mentioned electrical connection state is more reliably maintained, and the reliability of the product during normal use can be further improved.

[0032]

The invention according to claim 11 is the invention according to claim 1, wherein the safety valve receives the internal pressure of the battery can and protrudes from the through hole of the sealing lid to the outside of the battery can. A fragile portion that breaks due to bending deformation to the outside is provided.

[0033]

In the case of the above configuration, since the conductive plate is provided with a fragile portion that breaks when pushed and bent and deformed due to the jumping out of the safety valve, degassing from the inside of the battery can is performed more effectively.

[0034]

The invention according to claim 12 is the invention according to claim 1, wherein a PTC thermistor whose resistance value rapidly increases in a set temperature range is interposed in the middle of the current path between the lead wire and the sealing body. is there.

[0035]

In the case of the above configuration, since the PTC thermistor is interposed in the middle of the current path between the lead wire and the sealing body, the internal temperature of the battery can is properly discharged and charged normally. While the temperature is lower than the set temperature range in the state, the resistance value of the PTC thermistor is kept at a small value, and the current flows properly through the current path between the lead wire and the sealing body.

On the other hand, when the internal temperature rises above the set temperature range due to the occurrence of a short circuit or the like, the resistance value of the PTC thermistor rapidly increases in response to the rise, and the energization through the current path is substantially cut off. ..

[0036]

Further, in the invention of claim 13, in the invention of claim 1, the sealing body is arranged at an opening of the battery can and at an outer position of the conduction plate, and is electrically connected to the conduction plate. A positive electrode terminal is provided, and the positive electrode terminal, a conduction plate, an insulating member, and a sealing lid in which a safety valve is fitted are preliminarily coupled and integrated in a state of being overlapped with each other at their outer edges. It is something to do.

[0037]

In the case of the above configuration, as the sealing body, the sealing lid in which the safety valve is fitted in the through hole, the insulating member, the conduction plate in the state of being electrically connected to the sealing lid, and the positive electrode terminal, respectively. Since the outer edges of the batteries are preliminarily coupled and integrated in a state of being polymerized with each other, the work of assembling the battery can to the opening can be facilitated, and the safety valve is operated before the assembling. It will be possible to perform tests to confirm.

[0038]

BEST MODE FOR CARRYING OUT THE INVENTION Hereinafter, embodiments of the present invention will be described with reference to the drawings.

[0039]

<First Embodiment> FIG. 1 shows a sealing portion of a safety device for a sealed battery according to the first embodiment of the present invention.

The first embodiment is constructed based on the inventions according to claims 1 and 2.

In the figure, 1 is a battery can having an opening 1a, and 2 is a sealing body that closes the opening 1a to seal the battery can 1.

In FIG. 1, the inside and outside directions of the opening 1a of the battery can 1 are shown in the vertical direction.

That is, the inside of the battery can 1 is at the lower side of FIG. 1 and the outer side is at the upper side (the same applies to the drawings in the other embodiments below).

[0040]

The battery can 1 is a metal can body having a bottomed cylinder (cylindrical in the example, but may be a square cylinder), and a power generation element 3 is housed therein.

The power generation element 3 has a positive electrode plate, a negative electrode plate, an electrolytic solution, and the like, and the other end of the lead wire 4 to which one end is connected to the positive electrode plate constitutes the sealing body 2, which will be described later. It is connected to the lower surface 6a of the lid 6 by means such as welding.

A reduced diameter portion 1b is formed in the battery can 1 located above the power generation element 3, and the sealing body is formed between the reduced diameter portion 1b and the upper end caulking portion 1c in which the upper end opening edge is crimped. The outer peripheral edge of the battery can 1 is hermetically sandwiched via the insulating gasket 5 to seal the battery can 1.

[0041]

As shown in part in FIG. 2, the sealing body 2 includes a disk-shaped sealing lid 6, a donut plate-shaped insulating ring 7, a disk-shaped conducting plate 8, and a cap-shaped positive electrode terminal 9. And a piston-shaped safety valve 10.

Then, the sealing body 2 is sandwiched by the upper end caulking portion 1c, so that the sealing lid 6, the insulating ring 7, the conductive plate 8, and the positive electrode terminal 9 are integrated in this order in close contact with each other and polymerized. It is assembled.

[0042]

The sealing lid 6 is formed of a relatively thick wall made of a conductive material such as aluminum, titanium, an alloy thereof, or stainless steel, and has a rigidity that does not substantially deform against the internal pressure of the battery can 1. have.

A through hole 11 having a predetermined diameter is formed at a substantially central position of the sealing lid 6 in the vertical direction, and the safety valve 10 is tightly fitted into the through hole 11 in a state of being inserted from the bottom to the top.

[0043]

The conductive plate 8 is formed of the conductive material so as to have a predetermined flexibility, is insulated from the sealing lid 6 by an insulating ring 7 interposed therein, and has a thickness of the insulating ring 7. Minutes are arranged so as to be separated from the sealing lid 6 above.

Then, the lower surface 8a of the central portion of the conduction plate 8 and the upper end surface 10a which is the tip surface of the safety valve 10 are bonded to each other and connected to each other, and the conduction plate 8 has an annular protrusion surrounding the coupling portion. 8b is formed so as to project downward.

The protruding dimension of the protrusion 8b is set to be equal to or slightly larger than the thickness of the insulating ring 7 in the polymerized state, whereby the protruding peripheral surface of the protrusion 8b and the upper surface of the sealing lid 6 are densely set in the polymerized state. It is in contact with each other and is in an electrically connected state (state shown in FIG. 1).

In the drawing, 8c is a fragile portion formed in the vicinity of the formation range of the protrusion 8b, and the fragile portion 8c is formed by a notch groove formed so as to surround the binding site, and has a predetermined bending force. It is designed to break by acting.

[0044]

The positive electrode terminal 9 serves as a protective cap for protecting the conductive plate 8 and the like, and also serves as a positive electrode plate of the power generation element 3 via the lead wire 4, the sealing lid 6, the protrusion 8b, and the conductive plate 8. It is electrically connected and acts as a positive terminal for extracting power from the sealed battery.

Note that 9a in the drawing is a gas vent hole, and is formed through the space defined by the positive electrode terminal 9 and the conduction plate 8 so as to communicate with the outside.

[0045]

In the safety valve 10, a piston-shaped main body portion 10b having the same cross-sectional shape in the vertical direction and a convex portion 10c protruding in the outer peripheral direction from the lower end which is the base end of the main body portion 10b are the power generation elements. An insulating material that is not corroded by the electrolytic solution in 3 is preferably a material that is the insulating material and has elasticity, such as natural rubber, synthetic rubber such as EPDM (ethylene / propylene / diene copolymer), polyethylene, polypropylene. It is made of materials such as.

The upper end surface 10a of the main body 10b is adhered to the lower surface 8a of the central portion of the conductive plate 8 in the electrically connected state, and the lower end of the main body 10b is projected downward from the sealing lid 6 to form a through hole of the main body 10b. The convex portion 10c is set to be located at a position below a predetermined dimension from 11.

[0046]

The outer diameter of the main body 10b and the amount of protrusion of the convex portion 10c are related to the first and second set internal pressure values predetermined as the pressure for operating the safety valve 10, and each of the main body

10b or the convex portion 10c. It is determined that a predetermined sliding frictional resistance force is generated between the outer peripheral surface and the inner peripheral surface 11a of the through hole 11.

That is, until the internal pressure in the battery can 1 reaches the first set internal pressure value (for example, 10 to 15 kg / cm²), the sliding friction resistance force that prevents the relative movement of the safety valve 10 against the pressing force due to the internal pressure. When the internal pressure in the battery can 1 exceeds the first set internal pressure value, the pressing force due to the internal pressure becomes larger than the sliding friction force, and the safety valve 10 resists the sliding friction resistance force. The outer diameter of the main body 10b with respect to the inner diameter of the through hole 11 is determined so that a sliding frictional resistance force that causes the sliding friction resistance to move relative to the conduction plate 8 side, which is the outer side, acts.

Then, the outer diameter of the convex portion 10c with respect to the inner diameter of the through hole 11 is such that the internal pressure in the battery can 1 further increases beyond the first set internal pressure value and the second set internal pressure value (for example, about 20 kg / cm²).) Is exceeded, the pressing force due to the internal pressure becomes larger than the sliding frictional resistance between the outer peripheral surface of the convex portion 10c and the inner peripheral surface 11a of the through hole 11, and the convex portion 10c of the safety valve 10 penetrates. It is defined that a sliding frictional resistance force that jumps through the hole 11 acts.

[0047]

Further, the flexibility of the conduction plate 8 and the breaking strength of the fragile portion 8c are set according to the amount of relative movement of the safety valve 10 upward.

That is, the conduction plate 8 has an upward relative movement amount of the safety valve 10 until the convex portion 10c is stopped by the lower opening edge of the through hole 11 based on the internal pressure exceeding the first set internal pressure value. (See FIG. 3), and the fragile portion 8c is broken due to the bending corresponding to the relative movement amount of the safety valve 10 when the convex portion 10c jumps through the through hole 11 (See Fig. 4).

[0048]

In the sealed battery having the above configuration, when the internal pressure inside the battery can 1 is maintained at an appropriate internal pressure lower than the first set internal pressure value, the safety valve 10 has sliding friction between the through hole 11 and the inner peripheral surface 11a thereof. The relative movement is prevented by the resistance force, and the protrusion 8b of the conductive plate 8 whose vertical movement is blocked through the safety valve 10 is kept in contact with the sealing lid 6.

Therefore, the positive electrode plate of the power generation element is maintained in an electrically connected state in which it is electrically connected to the positive electrode terminal 9 via the lead wire 4, the sealing lid 6, the protrusion 8b, and the conductive plate 8. As a result, the positive electrode terminal 9 can be discharged and charged. Then, in this state, the outer peripheral surface of the safety valve 10 and the inner peripheral surface 11a of the through hole 11 are in close contact with each other, and the inside of the battery can 1 is kept in a sealed state.

[0049]

On the other hand, when the internal pressure inside the battery can 1 rises abnormally due to overcharging or the like while the sealed battery is being charged and the internal pressure exceeds the first set internal pressure value, it acts on the lower surface 10d of the safety valve 10 based on the internal pressure. The pressing force to be applied becomes larger than the sliding friction resistance force, whereby the safety valve 10 moves upward relative to the through hole 11 and the central portion of the conduction plate 8 is pushed upward.

As a result, the protrusion 8b of the conduction plate 8 is separated upward from the sealing lid 6, whereby the current from the power generation element 3 is cut off between the sealing lid 6 and the protrusion 8b, and the electric cutoff state (FIG. 3). It becomes the state shown). Since the charging current is cut off by this electrical cutoff and the decomposition of the electrolytic solution in the power generation element 3 is stopped, it is possible to suppress a further increase in the internal pressure. Moreover, since it is electrically cut off and does not function as a battery, it is possible to make the user aware of the occurrence of an abnormal state and the necessity of replacing the battery.

[0050]

Then, when the internal pressure in the battery can 1 rises to the extent that it exceeds the second set internal pressure value due to an increase in the internal pressure until the current is cut off, or due to some abnormality, the safety valve 10 described above The convex portion 10c is pushed into the through hole 11 and jumps upward through the through hole 11 (see FIG. 4).

The fragile portion 8c is broken by the excessive bending of the conduction plate 8 due to this jump-out, and the gas in the battery can 1 is discharged to the outside through the through hole 11, the broken portion, and the gas vent hole 9a of the positive electrode terminal 9. As a result, the internal pressure inside the battery can 1 drops to near the atmospheric pressure, and the destruction of the sealed battery due to an abnormal rise in the internal pressure is surely prevented.

[0051]

Therefore, when the internal pressure rises abnormally, as the first step, the current is cut off by setting the sliding friction resistance between the main body 10b of the piston-shaped safety valve 10 and the through hole 11, and the protrusion 8b and the sealing of the conduction plate 8. It is performed by a current cutoff mechanism composed of separation from the lid 6, and then, as a second step, the release of internal gas is performed to set the sliding frictional resistance force between the convex portion 10c of the safety valve 10 and the through hole 11. A two-step fail-safe operation is performed, which is performed by an internal pressure release mechanism composed of a breakage of the fragile portion 8c of the conduction plate 8.

[0052]

In such a sealed battery safety device, when the internal pressure is lower than the first set internal pressure value, the conductive plate 8 and the sealing lid 6 are polymerized via the insulating ring 7 and crimped by the caulking portion 1c. Therefore, the protrusion 8b of the conductive plate 8 and the upper surface of the sealing lid 6 are kept in contact with each other.

Moreover, since the upper surface 10a of the safety valve 10 is adhered to the lower surface 8a of the central portion of the conduction plate 8 and the safety valve 10 is held by the inner peripheral surface 11a of the through hole 11 so as not to move relative to the center of the conduction plate 8. The portion is held via the safety valve 10 so as not to be displaced relative to the sealing lid 6 in the vertical direction, and even if an impact or vibration acts on the battery can 1, the protrusion 8b and the sealing lid 6 are in contact with each other. The contact can be reliably maintained.

Therefore, even if the protrusion 8b of the conductive plate 8 and the sealing lid 6 are not welded, the above-mentioned electrical connection state is more reliably maintained, and the reliability of the product during normal use is further improved. ... That is, it is not necessary to perform welding for maintaining the electrical connection state at the time of normal use as in the conventional case, and the predetermined avoidance operation is not uncertain due to the welding. In addition, since it is not necessary to perform such delicate welding at the time of assembly, the process control in the production line becomes easy and the reliability of the product can be improved. Further, since the electrical connection between the conductive plate 8 and the sealing lid 6 can be made by contacting instead of welding, there is a restriction that the connecting surface of the electrical connection in the case of welding must be extremely small. Instead, an electrical connection can be made by a considerably large connecting surface of contact between the annular protrusion 8b and the sealing lid 6, and a stable current flow can be realized.

[0053]

Further, since the sealing lid 6 has a rigidity that does not deform even when it receives the internal pressure, the internal pressure acts directly and effectively on the lower surface 10d of the piston-shaped safety valve 10, and the relative movement of the safety valve 10 causes the outer peripheral edge. The pressing force acts intensively on the substantially central portion of the conduction plate 8 supported by the portion.

For this reason, when an internal pressure rise that exceeds the first set internal pressure value occurs, the internal pressure is indirectly cut off as compared with the case where the internal pressure is indirectly applied to the entire disk-shaped explosion-proof valve. Switching can be performed with good responsiveness to fluctuations in internal pressure.

[0054]

Moreover, since the switching between the electrical connection and the disconnection is mainly performed by setting the sliding frictional resistance force by setting the outer diameter of the safety valve 10 with respect to the inner diameter of the through

hole 11, the tensile breakage of the welded portion is performed as in the conventional closed battery. The setting and the change of the setting can be easily performed as compared with the case where the strength is set.

Therefore, it can be set to one that surely corresponds to the above-mentioned first or second set internal pressure value, and the switching timing from the electrical connection state to the cutoff state can be surely set according to the internal pressure in the battery can 1. Can be a thing. At this time, by forming the safety valve 10 from an elastic material, the sliding friction resistance force can be more reliably exerted, and the sliding friction resistance force to be exerted can be more easily set.

[0055]

<Second Embodiment> FIG. 5 shows a safety device for a sealed battery according to a second embodiment of the present invention.

This second embodiment is based on the structure of the first embodiment, and adds a current blocking mechanism due to an increase in internal temperature to the first embodiment to constitute the embodiment according to the invention according to claim 12. It is a thing.

[0056]

That is, in FIG. 5, reference numeral 12 denotes a PTC thermistor (Positive Temperature Coefficient thermistor) formed in the shape of a donut plate, and the PTC thermistor 12 is interposed between the conduction plate 8 and the positive electrode terminal 9. The conduction plate 8 and the positive electrode terminal 9 are connected to each other only via the PTC thermistor 12.

The interposition position of the PTC thermistor 12 may be any position as long as it is in the middle of the electrical conduction path between the power generation element 3 and the positive electrode terminal 9.

[0057]

The PTC thermistor 12 is a thermistor having a positive temperature coefficient whose resistance value increases as the temperature rises. In particular, the PTC thermistor 12 is a switching whose resistance value rapidly increases in a predetermined set temperature range (for example, 100 ° C. range). It is configured to have properties.

That is, in the PTC thermistor 12, the resistance value of the PTC thermistor 12 is extremely low in the normal temperature range up to the set temperature range, and therefore, a current flows between the conduction plate 8 and the positive electrode terminal 9 via the PTC thermistor 12. While flowing satisfactorily, in a high temperature range exceeding the set temperature range, the resistance value of the PTC thermistor 12 rapidly increases, so that the energization between the conduction plate 8 and the positive electrode terminal 9 is substantially cut off. It has become.

[0058]

In the case of the second embodiment having such a configuration, if the internal temperature inside the battery can 1 is within the normal temperature range up to the set temperature range, the lead wire 4, the sealing lid 6, and the protrusion from the positive electrode plate of the power generation element 3 The current path to the positive electrode terminal 9 via the 8b, the conduction plate 8, and the PTC thermistor 12 is in an electrically connected state, and discharge and charging can be performed from the positive electrode terminal 9.

On the other hand, if heat is generated in the battery can 1 due to overcharging or a short circuit and the internal temperature exceeds the set temperature range, the current flow is obstructed by the PTC thermistor 12 and the current path becomes electric. It becomes a target cutoff state. In particular, when the short circuit occurs, an excessive current flows in the path and the internal temperature rises sharply. Therefore, the PTC thermistor 12 immediately cuts off the excessive current flow between the power generation element 3 and the positive electrode terminal 9. be able to.

[0059]

Therefore, in addition to the current cutoff function based on the increase in internal pressure by separating the electrical connection between the protrusion 8b of the conduction plate 8 and the sealing lid 6 by the relative movement of the safety

valve 10, and the internal pressure release function by jumping out of the safety valve 10. The current cutoff function based on the increase in internal temperature as described above can also be added as a safety device.

[0060]

<Third Embodiment> FIG. 6 shows a safety device for a sealed battery according to a third embodiment of the present invention.

This third embodiment is based on the structure of the first embodiment, and the safety valve 13 is made to only switch from the connected state to the cutoff state of the current due to the increase in the internal pressure, and the internal pressure release after the current cutoff is closed. It is made to be performed by the elastic film 14 as described in the above, and relates to the invention according to claim 4.

[0061]

The safety valve 13 has a columnar main body portion 13a whose outer diameter is determined so as to exert a predetermined sliding friction resistance force in relation to the inner diameter of the through hole 11, and an outer diameter considerably larger than the inner diameter of the through hole 11. The flange-shaped pressure receiving portion 13b expanded so as to be integrally formed of the same material as the safety valve 10 of the first embodiment.

Then, the main body 13a of the safety valve 13 is tightly fitted to the inner peripheral surface 11a of the through hole 11 from below, and the sliding friction resistance force between the inner peripheral surface 11a of the through hole 11 causes the first embodiment. It is held so as not to move relative to the vertical direction against the internal pressure up to the first set internal pressure value described in the above.

The upper end surface 13c is adhered to the lower surface 8a of the central portion of the conduction plate 8 to maintain the contact between the protrusion 8b of the conduction plate 8 and the sealing lid 6. Further, the pressure receiving portion 13b is positioned slightly downward from the lower surface of the sealing lid 6.

[0062]

Further, gas vent holes 15a and 15b are formed through the sealing lid 6 and the conduction plate 8, respectively.

The lower surface side opening of the degassing hole 15a of the sealing lid 6 is covered with the elastic film 14, and the elastic film 14 is adhered to the lower surface of the sealing lid 6. As a result, the degassing hole 15a is sealed. Then, the elastic film 14 is formed of, for example, polyethylene or polypropylene so as to have a predetermined film thickness, thereby breaking under the internal pressure of the second set internal pressure value described in the first embodiment. The degassing hole 15a is opened. That is, the tensile breaking strength of the elastic film 14 is set as the opening strength as described above.

[0063]

Since the other configurations of the third embodiment are the same as those of the first embodiment, the same components are designated by the same reference numerals and detailed description thereof will be omitted.

[0064]

In the case of the third embodiment, in the range of the normal use state where the internal pressure in the battery can 1 is up to the first set internal pressure value, the power generation element 3 and the positive electrode terminal 9 are connected to the lead wire 4, the sealing lid 6, the protrusion 8b, and The inside of the battery can 1 is sealed by the sealing lid 6, the safety valve 13, and the elastic film 14 while being kept in an electrically connected state via the conduction plate 8.

[0065]

Then, when the internal pressure rises and exceeds the first set internal pressure value, the safety valve 13 moves upward relative to the protrusion 8b and the sealing until the pressure receiving portion 13b abuts on the lower surface of the sealing lid 6 and the relative movement is stopped. The contact with the lid 6 is separated.

As a result, the protrusion 8b and the sealing lid 6 are electrically cut off.

At this time, since the pressure receiving portion 13b of the safety valve 13 can have a considerably larger diameter than the main body portion 13a, it is possible to receive an internal pressure exceeding the first set internal pressure value in a wide area, and even if the internal pressure is the same, the first A larger pressing force than in the case of the first embodiment can be applied to the conductive plate 8 to separate the protrusion 8b from the sealing lid 6.

As a result, it is possible to switch to the current cutoff state with better responsiveness than in the case of the first embodiment.

[0066]

When the internal pressure rises and the internal pressure exceeds the second set internal pressure value, the elastic membrane 14 is broken by the internal pressure, and the internal gas is discharged to the outside through the degassing holes 15a, 15b and 9a. As a result, the internal pressure becomes close to the atmospheric pressure, and the internal pressure is released.

[0067]

Therefore, in the case of the third embodiment, the operation setting for interrupting the current at the first set internal pressure value can be performed by setting the sliding frictional resistance force between the main body portion 13a of the safety valve 13 and the through hole 11. On the other hand, it is possible to set the operation for releasing the internal pressure at the second set internal pressure value by setting the breaking strength of the elastic film 14, which is easier and more accurate than the case where the breaking strength of the welded portion is set. Can be done well.

[0068]

<Fourth Embodiment> FIG. 7 shows a safety device for a sealed battery according to a fourth embodiment of the present invention.

This fourth embodiment is constructed based on the invention according to claims 1 and 5.

In the figure, reference numeral 20 denotes a safety valve, and reference numeral 21 denotes a through hole formed by penetrating the substantially central portion of the sealing lid 6 in the vertical direction and in which the safety valve 20 is fitted.

[0069]

As shown in detail in FIG. 8, the through hole 21 is smaller than the large-diameter hole portion 21a formed in the lower half portion and having a predetermined inner diameter D2 and the large-diameter hole portion 21a formed in the upper half portion. It has a small-diameter hole portion 21b having an inner diameter D1 and a donut ring-shaped contact surface 21c formed between the large-diameter hole portion 21a and the small-diameter hole portion 21b so as to face downward, and is configured in a stepped shape. Has been done.

[0070]

Further, the safety valve 20 is formed on the upper end side, which is the tip side, and has an outer diameter smaller than the inner diameter D1 of the small diameter hole portion 21b, and is formed on the lower side of the small diameter portion 20a. It is formed in a stepped shape from a large diameter portion 20b having an outer diameter D2 that is substantially the same as the inner diameter of 21a, and a flange portion 20c protruding in the outer peripheral direction from the lower base end side of the large diameter portion 20b.

Then, the first stepped surface 20d that abuts the contact surface 21c between the small diameter portion 20a and the large diameter portion 20b, and the second stepped surface 20e between the large diameter portion 20b and the flange portion 20c are formed. It is formed.

Such a safety valve 20 is formed of an insulating material that is not corroded by the electrolytic solution in the power generation element 3 and that can exhibit predetermined deformation characteristics by receiving the internal pressure of the battery can 1.

Preferably, it is formed of the insulating material and a material elastically deformed by receiving the internal pressure, for example, natural rubber, synthetic rubber such as EPDM, polyethylene, polypropylene and the like.

[0071]

The small diameter portion 20a protrudes upward from the upper end opening of the through hole 21 in a state where the first stepped surface 20d is in contact with the contact surface 21c of the through hole 21, whereby the upper end surface thereof is upper. The end surface 20f is positioned close to the lower surface of the conductive plate 8.

Further, the large-diameter portion 20b is similarly projected downward from the lower end opening of the through hole 21 in a state where the first stepped surface 20d is in contact with the contact surface 21c of the through hole 21. The second stepped surface 20e is positioned below a predetermined dimension from the lower surface 6a of the sealing lid 6.

[0072]

In normal use, the safety valve 20 has a large diameter portion 20b internally fitted in the large diameter hole portion 21a, and a stepped surface 20d with respect to the sealing lid 6 is a contact surface of the through hole 21 due to sliding friction resistance between the two peripheral surfaces thereof. It is held in contact with 21b.

That is, the safety valve 20 is moved upward by the first stepped surface 20d coming into contact with the retaining surface 21b against the pressing force due to the internal pressure of the battery can 1 up to the first set internal pressure value described in the first embodiment. The relative displacement to the battery can 1 is prevented, whereby the inside of the battery can 1 is kept in a sealed state against the internal pressure of the battery can 1.

[0073]

The dimensions of the through hole 21 and the safety valve 20 are such that the safety valve 20 provides a predetermined press-fitting resistance to the through hole 21 in relation to the first and second set internal pressure values similar to those in the first embodiment. It is determined to exert.

That is, the inner diameter D1 of the small diameter hole portion 21b of the through hole 21 and the diameter D2 of the large diameter hole portion 21a are determined in consideration of the elastic deformation characteristics of the material for forming the safety valve 20. Specifically, first, the diameter D2 is set as a constant value, and by adjusting the value of the diameter D1 with respect to the diameter D2, an internal pressure exceeding the first set internal pressure value is received from the first step surface 20d of the safety valve 20. The portion extending to the large diameter portion 20b is elastically deformed so as to be press-fitted into the small diameter hole portion 21b having the diameter D1, and such a diameter value is defined as the diameter D1. Next, with respect to the protruding amount of the flange portion 20c of the safety valve 20, that is, the outer diameter of the flange portion 20c, the flange portion 20c is elastically deformed by an internal pressure exceeding the second set internal pressure value, and the through hole 21 is large. It is press-fitted into the diameter hole portion 21a and the small diameter hole portion 21b, and the value is set so that the through hole 21 jumps upward.

[0074]

In the case of the fourth embodiment having the above configuration, in normal use, the large diameter portion 20b of the safety valve 20 is fitted inside the large diameter hole portion 21a, and the safety valve 20 is closed by the sliding friction resistance force between both peripheral surfaces. The inside of the battery can 1 is kept airtight against the internal pressure up to the first set internal pressure value.

In this state, the protrusion 8b of the conduction plate 8 comes into contact with the sealing lid 6, and the sealing lid 6 and the conduction plate 8 are maintained in an electrically connected state (states shown by solid lines in FIGS. 7 and 8).

[0075]

When the internal pressure of the battery can 1 rises and exceeds the first set internal pressure value, the portion of the safety valve 20 from the stepped surface 20d to the large diameter portion 20b is elastically deformed and press-fitted into the small diameter hole portion 21b to move upward. It will be displaced relative to each other (see the safety valve 20 shown by the alternate long and short dash line in FIG. 8).

As a result, the upper end surface 20f of the safety valve 20 comes into contact with the lower surface 8a of the conductive plate 8 and pushes up the conductive plate 8 upward, and the protrusion 8b is separated upward from the sealing lid 6 to electrically shut off the sealed battery. Will be switched to. Under this internal pressurization, the second stepped surface 20e abuts on the lower surface 6a of the sealing lid 6 due to the upward relative displacement due to the press fitting, and the safety valve 20 is

further increased by the press fitting resistance to the large diameter hole portion 21a of the flange portion 20c. The relative displacement is suppressed, and the sealed battery is maintained in the above-mentioned electrically cut-off state.

[0076]

Further, when the internal pressure of the battery can 1 rises and exceeds the second set internal pressure value, the flange portion 20c is elastically deformed and press-fitted into the through hole 21 and jumps upward.

As a result, the fragile portion 8c of the conduction plate 8 is broken, and the internal pressure of the battery can 1 is released through the through hole 21, the above-mentioned broken portion, and the degassing hole 9a.

[0077]

In this way, as the internal pressure of the battery can 1 rises, the safety valve 20 can perform a two-step safety operation in which the internal pressure is first switched to the electrically cutoff state and then the internal pressure is released. It is possible to more easily and surely set the internal pressure for performing the conversion operation of the safety valve 20, that is, the setting of the press-fitting resistance force.

That is, in the case of the fourth embodiment, the press-fitting resistance is set in the small-diameter hole portion 21b having a diameter D1 from the first stepped surface 20d to the large-diameter portion 20b with respect to the internal pressure exceeding the first set internal pressure value. The value of the diameter D1 with respect to the diameter D2 (see FIG. 8) is set in consideration of the elastic deformation characteristic of the safety valve 20 so as to be press-fitted into, that is, the area of the contact surface 21c is adjusted and set. Be told. Therefore, the pressure for converting and operating the safety valve 20 from the electrically connected state to the electrically cutoff state can be reliably set based on the first set internal pressure value, and the setting is also the sliding friction resistance in the first embodiment and the like. This can be done more easily than when adjusting the force.

[0078]

Further, even when the conversion operating pressure is changed, it can be easily and surely performed while using the same safety valve 20 by changing the value of the diameter D1 of the small diameter hole portion 21b.

That is, the area of the contact surface 21c changes due to the change in the diameter D1, and the volume of the elastically deformed portion of the safety valve 20 when press-fitted into the small-diameter hole portion 21b changes with the change in the area of the contact surface 21c. The press-fitting resistance force into the small-diameter hole portion 21b can be changed.

[0079]

Moreover, in the work of assembling the safety valve 20 to the sealing lid 6, the large diameter portion 20b of the safety valve 20 is fitted into the large diameter hole portion 21a, and the first step surface 20d comes into contact with the contact surface 21c. If the inner fitting is stopped, the safety valve 20 can be reliably positioned at a predetermined position in the normal use state of the sealed battery, and therefore the safety valve 20 can be easily positioned with respect to the through hole 21.

[0080]

<Fifth Embodiment> FIG. 9 shows a safety device for a sealed battery according to a fifth embodiment of the present invention.

This fifth embodiment is based on the structure of the fourth embodiment, and the safety valve 30 and the through hole 31 are provided with tapered portions to prevent the safety valve 30 from falling off, and the safety valve 30 is connected to the sealing lid 6. It is formed integrally to save labor in assembly work.

[0081]

As shown in detail in FIG. 10, the through hole 31 includes a large diameter hole portion 31a in the lower half portion, a small diameter hole portion 31b in the upper half portion, and these large diameter hole portions, as in the fourth embodiment. It has a donut ring-shaped retaining surface 31c formed so as to face downward between the 31a and the small-diameter hole portion 31b.

The inner peripheral surface of the small-diameter hole portion 31b is formed of an upward tapered hole surface whose inner diameter gradually expands upward, whereby the small-diameter hole portion 31b is formed in a mortar shape.

[0082]

Further, the safety valve 30 has a small diameter portion 30a on the upper end side, a large diameter portion 30b formed under the small diameter portion 30a and having an outer diameter substantially the same as the inner diameter of the large diameter hole portion 31a, and the large diameter portion 30b. The flange portion 30c protruding from the lower end side in the outer peripheral direction is provided in the same manner as in the fourth embodiment, and the small diameter portion 30a and the large diameter portion 30b are in contact with the contact surface 31c. A second stepped surface 30e is formed between the large diameter portion 30b and the flange portion 30c so that the one stepped surface 30d abuts on the lower surface of the sealing lid 6.

The small diameter portion 30a has a tapered peripheral surface in which the outer peripheral surface of the lower half portion abuts on the tapered hole surface of the small diameter hole portion 31b of the through hole 31, and the upper half portion is from the upper end opening of the small diameter hole portion 31b. The upper end surface 30f projects upward to a position close to the lower surface 8a of the conductive plate 8 (see FIG. 9).

[0083]

Such a safety valve 30 is integrally formed with the sealing lid 6 by using the same material as in the fourth embodiment.

For example, when a rubber material such as EPDM is used, the safety valve 30 and the sealing lid 6 are integrally formed by vulcanization molding integrally with the sealing lid 6.

In this case, the sealing lid 6 may be arranged as an insert material in the molding mold of the safety valve 30, and an unvulcanized rubber material may be injected into the insert material for vulcanization. In addition to this, the safety valve 30 is manufactured separately from the sealing lid 6 using an elastic material such as the rubber material, and the manufactured safety valve 30 is fitted into the through hole 31, so that the conduction plate 8 and the like are formed. The safety valve 30 and the sealing lid 6 may be integrally formed in advance before assembling with other members.

[0084]

In the case of the fifth embodiment, first, since the safety valve 30 is integrally formed with the sealing lid 6, the sealing lid 6 is passed through the insulating ring 7 without considering the positional relationship between the safety valve 30 and the conduction plate 8. The sealing body 2 can be assembled only by assembling, which makes it possible to facilitate the assembling work.

In particular, when the safety valve 30 is formed by integral vulcanization molding with the sealing lid 6, the assembling work can be significantly reduced, and the safety valve 30 can be attached to the sealing lid 6 or the conduction plate 8. It can be reliably arranged in a predetermined position.

[0085]

Next, since the safety valve 30 is fitted in a state where the tapered peripheral surface of the small diameter portion 30a is in contact with the upward tapered hole surface of the through hole 31, the safety valve 30 is surely prevented from falling off downward. The safety valve 30 can be held against the sealing lid 6 in this state.

In addition, the second stepped surface 30e of the base end side flange portion 30c is in contact with the lower surface 6a of the sealing lid 6, and the sealing lid 6 is sandwiched between the second stepped surface 30c and the tapered peripheral surface from above and below. Therefore, the safety valve 30 can be more reliably held with respect to the sealing lid 6, and the reliability of the sealed battery from the manufacturing of the product to the time of shipment and the period of use is further improved. Can be made to.

[0086]

In the case of the fifth embodiment, the safety operation of the safety valve 30 due to the increase in the internal pressure of the battery can 1 is different from that of the fourth embodiment, and the second stepped surface 30e is in contact with the lower surface 6a of the sealing lid 6 in the normal use state. Since they are in contact with each other, it is performed as follows.

That is, when an internal pressure exceeding the first set internal pressure value is received, the portion of the safety valve 30 from the first stepped surface 30d to the large diameter portion 30b is elastically deformed and press-fitted into the small diameter hole portion 31b of the through hole 31, and at the same time. The portion from the second step surface 30e to the flange portion 30c is elastically deformed and press-fitted into the large-diameter hole portion 31a, whereby the safety valve 30 moves upward relative to the conduction plate 8 and the sealing lid via the protrusion 8b. The electrical connection state with 6 is switched to the cutoff state. Then, the internal pressure up to the second set internal pressure value is resisted by the press-fitting resistance force of the portion from the second step surface 30e to the flange portion 30c that is press-fitted into the large-diameter hole portion 31a to the small-diameter hole portion 31b, and the safety valve 30 is maintained in the above-mentioned electrically cut-off state. Further, when the internal pressure of the battery can 1 rises and the internal pressure exceeds the set internal pressure value, the portion from the second step surface 30e to the flange portion 30c is press-fitted into the small-diameter hole portion 31b and jumps upward. The fragile portion 8c is broken by the pressing force, and the internal pressure is released to the outside.

[0087]

<Sixth Embodiment> FIG. 11 shows a safety device for a sealed battery according to a sixth embodiment of the present invention.

This sixth embodiment is based on the structure of the fourth embodiment, and the safety valve 30 is prevented from falling off by the tapered hole surface and the tapered peripheral surface in the fifth embodiment by other means. ...

[0088]

As shown in detail in FIG. 12, the sealing lid 6' in the sixth embodiment has the same large-diameter hole portion 21a, the small-diameter hole portion 21b, and both hole portions 21a and 21b as in the fourth embodiment. A through hole 21 having a contact surface 21c is formed, and a safety valve 40 is internally fitted in the through hole 21.

[0089]

The safety valve 40 has an outer diameter substantially the same as the inner diameter of the small diameter hole portion 21b, and is fitted inside the inner peripheral surface of the small diameter hole portion 21b, and the small diameter portion 40a above the small diameter portion 40a. The upper end side flange portion 40b as the tip end side flange portion having an outer diameter larger than 21b and the large diameter hole portion 21a having an outer diameter substantially the same as the inner diameter of the large diameter hole portion 21a below the small diameter portion 40a. A large diameter portion 40c fitted inwardly on the inner peripheral surface of the above, and a lower end side flange portion 40d as a proximal end side flange portion having an outer diameter larger than the large diameter hole portion 21a on the lower side of the large diameter portion 40c. Have.

Then, in the safety valve 40, the first step surface 40e between the small diameter portion 40a and the large diameter portion 40c is on the contact surface 21c of the through hole 21, and between the small diameter portion 40a and the upper end side flange portion 40b. The second stepped surface 40f of the above corresponds to the upper surface 6c of the sealing lid 6', and the third stepped surface 40g between the large diameter portion 40c and the lower end side flange portion 40d corresponds to the lower surface 6a of the sealing lid 6', respectively. It is formed so as to touch.

The safety valve 40 is preliminarily formed integrally with the sealing lid 6' by the same method as in the fifth embodiment.

[0090]

The sealing lid 6' shown in the sixth embodiment is formed with a protrusion 6b' protruding upward from a position surrounding the safety valve 40, and the protrusion 6b' is formed on the lower surface 8a of the conduction plate 8'. The sealing lid 6' and the conductive plate 8' are brought into contact with each other to form an electrical connection.

At this time, the thickness of the insulating ring 7 is set to be slightly smaller than the protruding height of the protrusion 6b', and the sealing plate 6' and the conductive plate 8' or the like are assembled via the insulating ring 7 to form the conductive plate. The

central portion of 8' is bent in the vicinity of the fragile portion 8c position and is formed so as to slightly project upward. As a result, the conductive plate 8' and the protrusion 6b' are brought into close contact with each other in a state where the bending restoring force of the conductive plate 8' is applied, and the electrical connection between the conductive plate 8' and the protrusion 6b' is more reliable and good. In addition, the fragile portion 8c is more easily broken by jumping upward from the through hole 21 of the safety valve 40.

[0091]

Further, the fragile portion 8c may have a cut groove formed on only one side of the conductive plate 8 as in the above-described embodiment, but if the cut groove is formed on only one side, for example, when the sealing body 2 is automatically assembled. In some cases, it may be troublesome to distinguish the front and back sides of the conductive plate 8, and in consideration of this, as shown in FIG. 11, it is recommended to adopt a form in which cut grooves are formed from both sides of the conductive plate 8' as the fragile portion 8c, preferable.

[0092]

In the case of the sixth embodiment, the first step surface 40e and the third step surface 40g of the safety valve 40 abut on the contact surface 40e and the lower surface 6a which are the downward surfaces of the sealing lid 6', while the second step surface 40f. Since the safety valve 40 is in contact with the upper surface 6c, which is the upward surface of the sealing lid 6', the safety valve 40 is inserted into the through hole 21 with the sealing lid 6' sandwiched from above and below.

Therefore, as in the fifth embodiment, the safety valve 40 can be held against the sealing lid 6' in a state where the safety valve 40 is surely prevented from falling down, and is shipped after the product is manufactured and during the period of use. It is possible to further improve the reliability of the sealed battery until the above.

[0093]

The action / effect of forming the safety valve 40 integrally with the sealing lid 6' in advance and the action / effect of the safety operation of the safety valve 40 can be obtained in the same manner as in the fifth embodiment.

[0094]

<Other Embodiments> The present invention is not limited to the above-mentioned first to sixth embodiments, but includes various other embodiments.

That is, in the first to third embodiments, the upper end surfaces of the safety valves 10 and 13 and the lower surface of the conduction plate 8 are adhered to each other, but the present invention is not limited to this, and the safety valves 10 and 13 may be brought into contact with each other in a non-adhesive state. May be brought close to each other as in the fourth to sixth embodiments.

Even in these cases, by crimping the sealing lid 6, the insulating ring 7, the conductive plate 8, and the positive electrode terminal 9 with the crimped portion 1c, they are polymerized with each other and the above-mentioned contacted state is maintained.

[0095]

In the first to sixth embodiments, the flexible conductive plates 8, 8' are shown to be disk-shaped, but the present invention is not limited to this, and for example, a strip-shaped plate may be used.

In this case, even if the fragile portion 8c and the degassing hole 15b are not provided, the degassing can be performed by the safety valve 10 or the like jumping through the through hole 11 or the like.

[0096]

In the first to fifth embodiments, the electrical connection between the conductive plate 8 and the sealing lid 6 is performed via the protrusion 8b provided on the conductive plate 8 side, but the present invention is not limited to this, and instead of the protrusion 8b. In addition, as shown in FIG. 11 or 14, for example, a protrusion 6b' provided so as to surround the safety valves

10 and 40 on the sealing lid 6'side is brought into contact with the lower surface of the conductive plate 8, 8'. May be made to make an electrical connection.

[0097]

In the first to sixth embodiments, the protrusions 8b and 8b'are shown in an annular shape, but the present invention is not limited to this, and the shape may be any shape.

[0098]

In the first or second embodiment, the convex portion 10c of the safety valve 10 is shown to be continuous in the circumferential direction, but the present invention is not limited to this, and the convex portion may be formed in a part of the circumferential direction.

[0099]

In the first to fifth embodiments, a flat conductive plate 8 is used, but the present invention is not limited to this, and the conductive plate 8 is bent so that the central portion is slightly convex upward as in the sixth embodiment, for example. ' Or, as shown in FIG. 15, a conductive plate 8 " bent so that the central portion is slightly concave downward may be used.

In the case of FIG. 15, the conductive plate 8 " may be polymerized on the sealing lid 6 via a slightly thick insulating ring 7 '.

Further, in this case, the formation of the fragile portion 8c can be omitted by providing the gas vent holes 15b and 15b in the conduction plate 8 ".

[0100]

Further, as shown in FIG. 16, the configuration of the sealing body 2 in the first to sixth embodiments is a positive electrode terminal in a state where the sealing lid 6, the insulating ring 7 ", the conductive plate 8, and the PTC thermistor 12 are overlapped with each other. The sealing body 2 may be integrated by wrapping and sandwiching the outer peripheral edge of 9'with an extended outer peripheral portion 9b.

As a result, the work of assembling the battery can 1 to the opening 1a only needs to be done by assembling the integrated sealing body 2, and the efficiency of the assembling work can be improved.

Moreover, since the sealing body 2 can be assembled separately in advance, the operating state of the safety valve 10 can be tested before assembling the sealing body 2 to the battery can 1.

Further, although the one having the PTC thermistor 12 is shown in FIG. 16 above, the present invention is not limited to this, and the one having the structure of each embodiment without the above-mentioned PTC thermistor 12 may be integrated.

Further, this integration may be carried out by extending the outer peripheral portion of the sealing lid 6 to sandwich the positive electrode terminal 9, and further sandwiching the outer peripheral edge portion from the sealing lid 6 to the positive electrode terminal 9 by a separate sandwiching member. You may do so.

[0101]

Further, in the third embodiment, the elastic film 14 is shown as the closing body, but the closing body is not limited to this, and the closing body may be configured by, for example, a columnar closing valve tightly fitted in the gas vent hole 15a. ..

In this case, the shutoff valve is formed of an elastic material such as EPDM, and the sliding friction resistance between the outer peripheral surface of the shutoff valve and the inner peripheral surface of the gas vent hole 15a is the internal pressure up to the second set internal pressure value. Holds the shutoff valve so as not to move relative to each other and seals the gas vent hole 15a, while receiving an internal pressure exceeding the second set internal pressure value, jumps through the gas vent hole 15a upward and opens the gas vent hole 15a. It should be set to a value that will be used. That is, the opening strength of the shutoff valve is set by setting the sliding friction resistance force described above. By forming a convex portion such as the convex portion 10c in the safety valve 10 of the first embodiment at the lower end of the closing valve, the sliding friction resistance force can be easily set.

[0102]

[Example] In the sealed battery of the fourth embodiment described above, the plate thickness T (see FIG. 8) of the sealing lid 6 is set to 0.6 mm, and the large-diameter hole portion 21a and the small-diameter hole portion 21b of the through hole 21 are vertically oriented. The heights $T1$ and $T2$ are set to 0.3 mm, the diameter $D2$ of the large-diameter hole portion 21a is unified to 3 mm, and only the diameter $D1$ of the small-diameter hole portion 21b is changed in various ways within a range smaller than 3 mm. The lid 6 was manufactured as the first group.

In addition, among the above condition settings, six sealing lids 6 in which the diameter $D2$ of the large-diameter hole portion 21a was set to 4 mm and the diameter $D1$ of the small-diameter hole portion 21b was variously changed within a range smaller than 4 mm were manufactured as the second group. .

[0103]

On the other hand, as the safety valve 20, a required number of safety valves 20 having the same shape having the large diameter portion 20b having the diameter $D2$ were manufactured by EPDM so that the HS hardness was 60 degrees.

[0104]

Then, the safety valve 20 is fitted into the through hole 21 of each of the first and second groups of the sealing lids 6 so that the first stepped surface 20d abuts on the contact surface 21c, and the safety valve 20 in this state is fitted. A press-fitting test was performed by applying pressure P to the surface.

When this pressure P is gradually increased and the large diameter portion 20b of the safety valve 20 is press-fitted into the small diameter hole portion 21b and the second step surface 20e is displaced upward until it comes into contact with the lower surface 6a of the sealing lid 6. The pressure value of was calculated.

This pressure value is used as the working pressure at the time of press-fitting, and the obtained working pressure is shown in FIG. FIG. 13 shows the case where the working pressure is on the vertical axis and the diameter $D1$ of the small diameter hole portion 21b is on the horizontal axis, and the diameter $D2$ of the large diameter hole portion 21a is 3 mm in the first group (see the square mark in the figure).) And the case of the second group of 4 mm (see the triangular mark in the figure) are shown separately.

[0105]

According to this result, in order to set the first set internal pressure value for switching the electrical connection state to the cutoff state to, for example, 15 kgf / cm², the diameter $D2$ of the large diameter hole portion 21a is set to 3 mm (first group). In the case of), the diameter $D1$ of the small-diameter hole portion 2b may be set to approximately 2.6 mm, and when the diameter $D2$ is set to 4 mm (in the case of the second group), the diameter $D1$ of the small-diameter hole portion 2b may be set. It can be seen that the setting should be approximately 3.4 mm.

Further, in order to set the first set internal pressure value to 20 kgf / cm², when the diameter $D2$ of the large diameter hole portion 21a is set to 3 mm, the diameter $D1$ of the small diameter hole portion 2b may be set to approximately 2.5 mm. Further, it can be seen that when the diameter $D2$ is set to 4 mm, the diameter $D1$ of the small diameter hole portion 2b should be set to approximately 3.2 mm.

[0106]

Therefore, by measuring the press-fitting resistance characteristics as shown in FIG. 13, the safety valve 20 can be reliably operated at a predetermined internal pressure, and the dimensions $D1$ and $D2$ of the through holes 21 for the operation can be set. It can be done easily.

Moreover, even when the switching operating pressure to the electrically cutoff state is changed, the diameter $D1$ of the small diameter rear portion 21b can be immediately determined based on FIG. 9, and the safety valve 20 can be ensured by adjusting the change of the diameter $D1$. It can be operated with the switching operating pressure after the change to.

[0107]

[Effect of the invention] As described above, according to the safety device for the sealed battery according to the invention according to the first aspect, the sealing body is tightly fitted to the rigid sealing lid and the through hole of the sealing lid. Since it is composed of a safety valve sealed with a lid and a flexible conductive plate electrically connected to the sealing lid around the safety valve in an insulated state on the outer side of the sealing lid, it may be overcharged. When the internal pressure inside the battery can exceeds the set internal pressure value, the internal pressure is applied directly to the base end surface of the safety valve, and the relative movement of the safety valve makes the electrical connection to the electrical connection part of the conduction plate with the sealing lid. The pressing force can be applied intensively to the release side, and when the internal pressure rise occurs, the electrical cutoff state can be switched to with good responsiveness.

As a result, the battery user can quickly recognize the occurrence of an abnormality.

[0108]

Moreover, switching between electrical connection and disconnection according to the internal pressure inside the battery can is performed by setting the resistance force between the safety valve and the through hole against the internal pressure that attempts to relatively displace the safety valve toward the conduction plate side. Compared to the case where the breaking strength of the welded part is set like a conventional sealed battery, the setting and change can be done easily and surely, and the process control at the time of assembly becomes easy for the product. It is possible to improve reliability.

[0109]

According to the invention of claim 2, in addition to the effect of the invention of claim 1, the resistance force between the safety valve and the through hole against the internal pressure of the battery can is increased between both peripheral surfaces of the safety valve and the through hole. Since it is exerted by the sliding friction resistance force, the resistance force can be easily adjusted or changed according to the internal pressure of the battery can.

Then, the resistance force that keeps the safety valve in a state where the relative displacement is blocked up to the set internal pressure value and allows the relative displacement of the safety valve to the conduction plate side with respect to the battery can internal pressure exceeding the set internal pressure value is applied. It can be reliably exerted by the dynamic friction resistance.

According to the invention of claim 3, in addition to the effect of the invention of claim 2, the safety valve is composed of a columnar main body portion and a convex portion slightly protruding from the base end side of the main body portion. Therefore, when the internal pressure value of the battery can exceeds the set internal pressure value, the current is cut off due to the relative movement of the main body, and when the internal pressure further rises, the inside of the battery can is jumped out from the through hole of the convex portion. By releasing the internal pressure, it is possible to perform two-step safe operation. Therefore, when the internal pressure is abnormal, the current is first cut off to make the battery user aware of the occurrence of the abnormality, and even if the user is left unattended, the internal gas can be released to avoid an abnormal increase in the internal pressure. ...

[0110]

According to the invention of claim 4, in addition to the effect of the invention of claim 2, the safety valve is composed of a columnar main body portion and a flange-shaped pressure receiving portion expanded on the base end side of the main body portion. On the other hand, since the sealing lid is provided with a gas vent hole for communicating the inside and the outside of the battery can with each other and a closing body which is closed on the inner surface side of the sealing lid of the gas vent hole and has a predetermined opening strength, the set internal pressure is set. When the internal pressure rises beyond the value, it is possible to perform a two-step safety operation of shutting off the current by the relative movement of the main body and releasing the internal pressure by opening the closing body.

Moreover, when the current is cut off, the pressure receiving portion of the safety valve is expanded more than the main body portion, and the base end surface of the safety valve, which is the pressure receiving surface, can be expanded as compared with the case of the invention according to claim 4, so that the internal pressure rises. On the other hand, the conduction plate can be pushed up more responsively to cut off the current.

[0111]

According to the invention of claim 5, in addition to the effect of the invention of claim 1, the resistance force between the safety valve and the through hole against the internal pressure of the battery can is increased, and the stepped surface of the safety valve with respect to the small diameter hole portion of the through hole. Since it is exerted by the press-fitting resistance of the portion from the small diameter portion to the large diameter portion, the adjustment or change of the resistance according to the internal pressure of the battery can is made more easily and surely than in the case of the invention according to claim 2. Can be done.

Then, the safety valve is maintained in a state where its relative displacement is blocked up to the set internal pressure value, and the resistance force that allows the safety valve to be relatively displaced toward the conduction plate side with respect to the battery can internal pressure exceeding the set internal pressure value is provided. It can be surely exerted by the press-fitting resistance of.

[0112]

Moreover, when assembling the safety valve to the sealing lid, the safety valve can be securely assembled to the sealing lid and the conduction plate in the predetermined inward and outward directions simply by bringing the stepped surface of the safety valve into contact with the contact surface of the through hole. Therefore, the safety valve can be easily positioned with respect to the sealing lid.

As a result, it is possible to facilitate the manufacture of the sealed battery, and it is possible to reliably position the safety valve at a predetermined position and reliably maintain the state of electrical connection with the sealing lid.

[0113]

According to the invention of claim 6, in addition to the effect of the invention of claim 2 or 5, since the safety valve is made of an elastic material, sliding between the safety valve and the through hole of the sealing lid. The dynamic friction resistance force or the press-fitting resistance force can be set more easily, and the switching from the electrical connection state to the cutoff state due to the increase in the internal pressure can be performed more reliably.

[0114]

According to the invention of claim 7, in addition to the effect of the invention of claim 5, as the internal pressure of the battery can increases, first, the portion of the safety valve from the stepped surface to the large diameter portion has a small diameter of the through hole. The tip surface of the safety valve can be press-fitted into the hole to switch to the electrically shut-off state by pushing up the conduction plate, and then the base end side flange is press-fitted into the large-diameter hole in response to the further increased internal pressure, and the entire safety valve. Can penetrate the through hole toward the conduction plate side to release the internal pressure of the battery can.

Therefore, as the internal pressure of the battery can rises, the current is cut off as the first step, and the internal pressure is released as the second step, so that a two-step safe operation can be performed.

[0115]

According to the invention of claim 8, in addition to the effect of the invention of claim 5, when the sealed battery is in a normal use state, the tapered surface of the safety valve comes into contact with the tapered hole surface of the through hole. As a result, it is possible to reliably prevent the safety valve from falling out of the through hole to the inside of the battery can.

Therefore, it is possible to reliably maintain the stepped surface of the safety valve in contact with the contact surface of the through hole, and to maintain a posture for reliably performing safe operation when the internal pressure of the battery can rises. it can.

Therefore, the safety valve can be reliably held by the sealing lid from the shipping stage to the usage stage of the product, and the safety operation when the internal pressure of the battery can rises can be reliably performed.

[0116]

According to the invention of claim 9, in addition to the effect of the invention of claim 5 or 8, the safety valve is formed integrally with the sealing lid in advance, so that the safety valve is assembled to the sealing lid. It is possible to save labor and facilitate and rationalize the production of sealed batteries.

At the same time, the safety valve can be reliably positioned at a predetermined relative position with respect to the sealing lid and the conduction plate in advance, and the electrical connection state between the sealing lid and the conduction plate during normal use can be maintained and the safety valve when the internal pressure rises. It is possible to ensure safe operation.

[0117]

According to the invention of claim 10, in addition to the effect of the invention of claim 1, the tip surface of the safety valve and the inner surface of the substantially central portion of the flexible conductive plate are adhered to each other. The flexible conductive plate itself can be held by a rigid sealing lid by sliding frictional resistance between the safety valve and the through hole.

Therefore, the contact between the conductive plate and the sealing lid, that is, the electrical connection state can be stably maintained, and therefore, even if the conductive plate and the sealing lid are not welded, the electrical connection is described above. The connection state can be maintained more reliably, and the reliability of the product during normal use can be further improved.

[0118]

According to the invention of claim 11, in addition to the effect of the invention of claim 1, the conductive plate is provided with a fragile portion that breaks when pushed by the safety valve jumping out and bent and deformed. Degassing from the inside can be performed more effectively.

[0119]

Further, according to the invention of claim 12, in addition to the effect of the invention of claim 1, the resistance value rapidly increases in the set temperature range in the middle of the current path between the lead wire and the sealing body. Since the thermistor is interposed, the current can be properly passed through the current path while the internal temperature of the battery can is at an appropriate temperature lower than the set temperature, while the internal temperature causes a short circuit or the like. When the temperature rises above the set temperature range due to the above, the current through the current path can be substantially cut off by the rapid increase in the resistance value of the PTC thermistor in response to the rise, further improving safety. be able to.

[0120]

Further, according to the invention of claim 13, in addition to the effect of the invention of claim 1, the sealing body is provided with a sealing lid in which a safety valve is fitted, an insulating member, and the sealing lid and electricity. Since the outer edges of the conductive plate and the positive electrode terminal, which are connected to each other, are preliminarily coupled and integrated in a state of being overlapped with each other, they can be assembled to the opening of the battery can. In addition to facilitating the work, a test for confirming the operation of the safety valve can be performed before assembling the safety valve.

[0121]

A brief description of the drawing

[0122]

FIG. 1 is an upper sectional view showing a first embodiment of the present invention.

[0123]

FIG. 2 is an exploded perspective view in which a part of the sealing body is omitted.

[0124]

FIG. 3 is a diagram corresponding to FIG. 1 showing an electrically cutoff state.

[0125]

FIG. 4 is a diagram corresponding to FIG. 1 showing an internal pressure open state.

[0126]

5 is a diagram corresponding to FIG. 1 showing a second embodiment.

[0127]

6 is a view corresponding to FIG. 1 showing a third embodiment.

[0128]

FIG. 7 is a view corresponding to FIG. 1 showing a fourth embodiment.

[0129]

8 is a partially enlarged view of FIG. 7. FIG.

[0130]

9 is a view corresponding to FIG. 1 showing a fifth embodiment.

[0131]

10 is a partially enlarged view of FIG. 9. FIG.

[0132]

11 is a view corresponding to FIG. 1 showing a sixth embodiment.

[0133]

12 is a partially enlarged view of FIG. 11. FIG.

[0134]

13 is a relationship diagram between diameters D1 and D2 of the safety valve of FIG. 8 and operating pressure.

[0135]

FIG. 14 is a view corresponding to FIG. 1 showing another embodiment.

[0136]

15 is a view corresponding to FIG. 1 showing another embodiment different from FIG.

[0137]

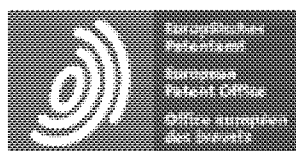
16 is a diagram corresponding to FIG. 1 showing another embodiment different from FIGS. 14 and 15.

[0138]

Code description

[0139]

1 Battery can 2 Sealing body 3 Power generation element 4 Lead wire 6,6' Sealing lid 7,7', 7 " Insulation ring (insulating member) 8,8', 8 " Conduction plate 9,9' Positive electrode terminal 10,13, 20, 30, 40 Safety valves 10a, 13c, 20f, 30f Upper end surface (tip surface) 10d, 13d Lower end surface (base end surface) 10b, 13a Main body 10c Convex 13b Pressure receiving portion 11, 21, 31 Through hole 12 PTC thermistor 14 Elastic film (closed body) 15a Degassing holes 20a, 30a, 40a Small diameter parts 20b, 30b, 40c Large diameter parts 20d, 30d, 40e First stepped surface (stepped surface) 20c, 30c, 40d Lower end side flange (base end) Side flange) 21a, 31a Large diameter hole 21b, 31b Small diameter hole 21c, 31c Stopping surface 30a Small diameter (tapered peripheral surface) 31b Small diameter hole (tapered hole surface)



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CLAIMS JPH08339793A

1.

In a sealed battery safety device in which the power generation element is housed inside the battery can and the outwardly facing opening of the battery can is sealed by a sealing body, the sealing body is made more flexible by a conductive material. A flexible conductive plate formed, a sealing lid formed of a conductive material to seal the inside of the battery can and having rigidity that does not substantially deform against the internal pressure thereof, and a battery can formed of an insulating material. The sealing lid is provided with a safety valve that moves relative to each other due to an increase in the internal pressure of the battery. The lead wire from the element is connected, and the conductive plate is overlapped with the outer surface of the sealing lid at a position outside the opening of the battery can at the outer edge portion via the insulating member, and the through hole is formed. The position is fixed so as to come into contact with the sealing lid and to be in an electrically connected state at a position in the vicinity of The tip surface is positioned close to the conduction plate in the state of being inserted in the inner and outer directions, and the inner surface of the hole forming the through hole and the outer surface in the circumferential direction of the safety valve facing the inner surface of the hole are the battery cans. Until the internal pressure rises and reaches the set internal pressure value, the safety valve is held in a state of being prevented from relative displacement of the safety valve with respect to the conduction plate, while the safety valve is made conductive by the action of the internal pressure exceeding the set internal pressure value. A safety device for a sealed battery, characterized in that the conductive battery is fitted to each other so as to exert a resistance force such that the conductive plate and the sealing lid are separated from each other by being relatively displaced toward the plate side.

2.

In claim 1, the through hole is formed so as to have a uniform inner diameter in the inner and outer directions of the battery can, while the safety valve is formed in a piston shape, and the inner peripheral surface of the through hole and the outer circumference of the safety valve. The surface is tightly fitted to each other so as to resist the internal pressure of the battery can by exerting the sliding friction resistance force between the two peripheral surfaces against the relative displacement of the safety valve to the conduction plate side. Sealed battery safety device.

3.

In claim 2, the safety valve has a columnar shape having the same cross-sectional shape corresponding to the through hole of the sealing lid from the tip end side to the base end side, and the base end side is arranged so as to project from the inner surface of the sealing lid toward the inside of the battery can. It is composed of a main body and a convex portion that slightly protrudes from the base end side of the main body to the periphery, and slides and moves relative to the inner peripheral surface of the through hole under the internal pressure of the battery can. A safety device for a sealed battery, characterized in that the internal pressure value at the time of the operation is set so that the convex portion forming position is relatively larger than the main body portion.

4.

In claim 2, the safety valve has a columnar body having the same cross-sectional shape corresponding to the through hole of the sealing lid from the tip end side to the base end side, and the base end side is arranged so as to protrude from the inner surface of the sealing lid toward the inside of the battery can. The sealing lid is composed of a portion and a flange-shaped pressure receiving portion expanded around the base end side of the main body portion, and the sealing lid is formed by a gas vent hole formed so as to communicate the inside and the outside of the battery can, and the gas vent hole. It is provided with a closing body that seals the degassing hole, and the opening strength of this closing body is set so as to receive an internal pressure larger than the internal pressure of the battery can that abuts the pressure receiving portion on the inner surface of the sealing lid. A safety device for sealed batteries featuring.

5.

In claim 1, the through hole is formed in the inside of the battery can formed between the large diameter hole portion on the inner side of the battery can, the small diameter hole portion on the conduction plate side, and the large diameter hole portion and the small diameter hole portion. The safety valve has a contact surface facing the side, and the safety valve has a small-diameter portion that inserts the small-diameter hole portion to the outer side of the battery can, a large-diameter portion that fits inside the large-diameter hole portion, and these small-diameter portions. It has a stepped surface formed so as to face the contact surface between the large diameter portion and the large diameter portion, and the through hole and the safety valve are the safety valve with respect to the contact surface of the through hole. When the stepped surface comes into contact with the battery can, the internal pressure of the battery can is received, and the portion of the safety valve from the stepped surface to the large diameter portion exerts a press-fitting resistance force to press-fit the small diameter portion of the through hole. A safety device for sealed batteries, characterized in that they are fitted together to resist the internal pressure of the battery.

6.

The safety device for a closed battery according to claim 2 or 5, wherein the safety valve is made of an elastic material.

7.

In claim 5, the large-diameter portion of the safety valve is projected inward of the battery can with the stepped surface in contact with the contact surface of the through hole, and the protruding end of the large-diameter portion is the large-diameter hole of the through hole. A base end side flange portion having a diameter larger than that of the portion is integrally formed, and the proximal end side flange portion exerts an internal pressure larger than the internal pressure of the battery can that abuts the proximal end side flange portion on the inner surface of the sealing lid. A safety device for a sealed battery, characterized in that its outer diameter is set so as to receive and penetrate the through hole toward the conduction plate side.

8.

In claim 5, the inner peripheral surface of the small-diameter hole portion of the through hole is formed of a tapered hole surface whose diameter gradually increases toward the conduction plate side, and the outer peripheral surface of the small-diameter portion of the safety valve has a stepped surface as the through hole. A safety device for a sealed battery, characterized in that it is composed of a tapered peripheral surface that abuts on the tapered hole surface in a state of being in contact with the retaining surface of the above.

9.

The safety device for a sealed battery according to claim 5 or 8, wherein the safety valve is integrally formed with the sealing lid by integral vulcanization molding using rubber for the sealing lid.

10.

A safety device for a closed battery according to claim 1, wherein the tip surface of the safety valve and the inner surface of the flexible conductive plate are bonded and bonded to each other.

11.

According to claim 1, the conduction plate is broken by bending deformation of the conduction plate to the outside caused by the safety valve protruding from the through hole of the sealing lid to the outside of the battery can by receiving the internal pressure of the battery can. A sealed battery safety device characterized by being provided with a fragile part.

12.

The safety device for a sealed battery according to claim 1, wherein a PTC thermistor whose resistance value rapidly increases in a set temperature range is interposed in the middle of the current path between the lead wire and the sealing body.

13.

According to claim 1, the sealing body includes a positive electrode terminal which is an opening of a battery can and is arranged at an outer position of a conductive plate and is electrically connected to the conductive plate. The positive electrode terminal and the conductive plate are provided. A safety device for a sealed battery, characterized in that an insulating member and a sealing lid in which a safety valve is fitted are previously coupled and integrated in a state of being overlapped with each other at their outer edges.



Espacenet

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SAFETY DEVICE FOR SEALED BATTERY

Inventor(s): KAMAUCHI MASA HARU; IJIRI YASUO; KASHIOKA TORU; OUCHI YUTAKA ± (KAMAUCHI MASA HARU, ; IJIRI YASUO, ; KASHIOKA TORU, ; OUCHI YUTAKA)

Applicant(s): MITSUBISHI CABLE IND LTD ± (MITSUBISHI CABLE IND LTD)

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H01M2/34
- cooperative: Y02E60/10 (EP)

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Abstract of JPH08339793 (A)

PURPOSE: To eliminate a welding part from a current interrupting mechanism and to facilitate adjusting/setting for switching from connection to interruption of a current relating to a preset internal pressure value. **CONSTITUTION:** An opening part 1a of a battery can 1 is sealed by fixedly calking the periphery of a seal unit 2 through an insulating gasket 5. The seal unit is constituted by a rigid disk-shaped seal cover 6 connected with a lead wire 4 from a generating element 3, insulating ring 7, flexible conduction plate 8 with a circular annular protrusion 8b into contact to be electrically connected to the seal cover, rubber-made safety valve 30 tightly fitted to a through hole 31 with an upper end face adjacent to the conduction plate and a cap-shaped positive electrode terminal 9. Press fit resistance force in an intermediate step difference surface part of the safety valve against a small diametric hole part of the through hole is set so as to be smaller than pressing force to the safety valve based on a battery can internal pressure exceeding a preset internal pressure value, and when exceeding the preset internal pressure value, to relatively move upward to separate the protrusion by press fitting the safety valve, further when an internal pressure rises, to press fit a flange part of the safety valve, to spring out the through hole, so as to break a fragile part 8c. The safety valve is integrally vulcanized to be molded with the seal cover, to attain sureness of assembly.

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(71) 出願人 000003263

三菱電線工業株式会社

兵庫県尼崎市東向島西之町8番地

(72) 発明者 鎌内 正治

兵庫県尼崎市東向島西之町8番地 三菱電線工業株式会社内

(72) 発明者 井尻 康夫

東京都千代田区丸の内3丁目4番1号(新国際ビル) 三菱電線工業株式会社東京事務所内

(74) 代理人 弁理士 前田 弘 (外2名)

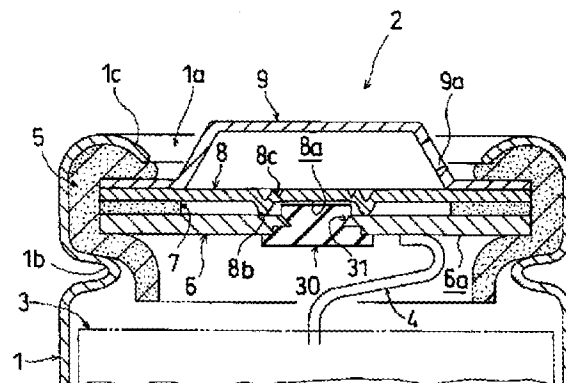
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(54) 【発明の名称】 密閉電池の安全装置

(57) 【要約】

【課題】 電流遮断機構から溶接部をなくし、かつ、設定内圧値に対する電流の接続から遮断への切換のための調整、設定の容易化を図る。

【解決手段】 封口体2の周囲を絶縁ガスケット5を介してかしめ固定することにより電池缶1の開口部1aを封止する。封口体を、発電要素3からのリード線4が接続される剛性円板状封口蓋6、絶縁リング7、円環状突起8bが当接して封口蓋と電気的に接続される可撓性導通板8、貫通孔31に密に嵌合されて上端面が導通板に近接されたゴム製安全弁30、及び、キャップ状正極端子9で構成する。貫通孔の小径孔部に対する安全弁の中間段差面部分の圧入抵抗力が設定内圧値を超える電池缶内圧に基づく安全弁への押圧力より小となるようにし、設定内圧値を超えると安全弁の圧入により上方に相対移動して突起を離し、さらに内圧が上昇すると安全弁の鍔部が圧入し貫通孔を飛び抜けて脆弱部8cを破断させるようにする。安全弁を封口蓋と一体加硫成形して組付けの確実化を図る。



【特許請求の範囲】

【請求項1】 発電要素が電池缶の内部に収容され、電池缶の外側に向いた開口部が封口体によって密閉されている密閉電池の安全装置において、

上記封口体は、導電材料により可撓性を有するように形成された可撓性導通板と、導電材料により電池缶の内部を密閉しその内圧に対抗して実質的に変形しない剛性を有するように形成された封口蓋と、絶縁材料により形成され電池缶の内圧上昇により相対移動する安全弁とを備え、

上記封口蓋は、略中央部に貫通孔を有し外周縁部が電池缶の開口部に気密に取付けられ、電池缶内に臨む内面に上記発電要素からのリード線が接続され、

上記導通板は、外縁部が電池缶の開口部であって上記封口蓋より外側位置で上記封口蓋の外周と絶縁部材を介して重合されて、上記貫通孔の近傍位置で上記封口蓋と当接して電氣的接続状態になるよう位置固定され、

上記安全弁は、上記封口蓋の貫通孔に対しこの貫通孔を封止するよう電池缶内部から上記導通板まで内外方向に挿通された状態で先端面が上記導通板に近接するよう位置付けられ、

上記貫通孔を構成する孔内面と、この孔内面に相対向する安全弁の周方向の外周面とは、電池缶の内圧が上昇して設定内圧値に達するまではその内圧に抵抗して上記導通板に対する安全弁の相対変位を阻止した状態に保持する一方、上記設定内圧値を超える内圧の作用により上記安全弁を上記導通板側に相対変位させて上記導通板と封口蓋との当接を離すような抵抗力を発揮するよう互いに嵌合されていることを特徴とする密閉電池の安全装置。

【請求項2】 請求項1において、貫通孔は電池缶の内外方向に一樣な内径を有するように形成される一方、安全弁はピストン状に形成されており、

上記貫通孔の内周面と、上記安全弁の外周面とが、上記安全弁の導通板側への相対変位に対する両周面間の摺動摩擦抵抗力を発揮することにより電池缶の内圧に抵抗するよう互いに密に嵌合されていることを特徴とする密閉電池の安全装置。

【請求項3】 請求項2において、安全弁は、先端から基端側まで封口蓋の貫通孔と対応する同一の断面形状を有し基端側が封口蓋内面から電池缶内方側に突出するよう配置された柱状の本体部と、この本体部の基端側から周囲に僅かに突出する凸部とから構成され、かつ、電池缶の内圧を受けて貫通孔の内周面との間で摺動して相対移動する際の内圧値が、上記凸部形成位置の方が上記本体部よりも相対的に大となるように設定されていることを特徴とする密閉電池の安全装置。

【請求項4】 請求項2において、安全弁は、先端から基端側まで封口蓋の貫通孔と対応する同一断面形状を有し基端側が封口蓋内面から電池缶内

方側に突出するよう配置された柱状の本体部と、この本体部の基端側の周囲に拡開されたフランジ状の受圧部とから構成され、

封口蓋は、電池缶内部と外部とを連通させるよう形成されたガス抜き孔と、このガス抜き孔を密閉する閉止体とを備え、この閉止体は上記受圧部を封口蓋内面に当接させる電池缶内圧より大の内圧を受けて開放するようにその開放強度が設定されていることを特徴とする密閉電池の安全装置。

【請求項5】 請求項1において、貫通孔は、電池缶の内方側の大径孔部と、導通板側の小径孔部と、これら大径孔部と小径孔部との間に形成された電池缶内方側に面する当止面とを有し、

安全弁は、上記小径孔部を電池缶外方側に挿通する小径部と、上記大径孔部に内嵌する大径部と、これら小径部と大径部との間に上記当止面と相対向するよう形成された階段状の段差面とを有し、

上記貫通孔と安全弁とは、上記貫通孔の当止面に対し上記安全弁の段差面が当接して、電池缶の内圧を受けてその安全弁の段差面から大径部にかけての部分が貫通孔の小径孔部に対して圧入しようとする圧入抵抗力を発揮することにより電池缶の内圧に抵抗するよう互いに嵌合されていることを特徴とする密閉電池の安全装置。

【請求項6】 請求項2または請求項5において、安全弁が、弾性材料により形成されていることを特徴とする密閉電池の安全装置。

【請求項7】 請求項5において、安全弁の大径部は段差面が貫通孔の当止面に当接した状態で電池缶内方に突出され、この大径部の突出端には上記貫通孔の大径孔部よりも大径の基端側鏝部が一体に形成されており、

上記基端側鏝部は、その基端側鏝部を封口蓋内面に当接させる電池缶内圧よりも大の内圧を受けて上記貫通孔を導通板側に突き抜けるようにその外径が設定されていることを特徴とする密閉電池の安全装置。

【請求項8】 請求項5において、貫通孔の小径孔部の内周面は、導通板側に向かって徐々に拡径するテーパ孔面により構成され、安全弁の小径部の外周面は、段差面が上記貫通孔の当止面に当接した状態で上記テーパ孔面に当接するテーパ周面により構成されていることを特徴とする密閉電池の安全装置。

【請求項9】 請求項5または請求項8において、安全弁は、封口蓋に対しゴムを用いた一体加硫成形により、その封口蓋と一体に形成されていることを特徴とする密閉電池の安全装置。

【請求項10】 請求項1において、安全弁の先端面と、可撓性導通板の内面とが互いに接合されて結合されていることを特徴とする密閉電池の安全装置。

【請求項11】 請求項1において、導通板には、安全弁が電池缶の内圧を受けて封口蓋の貫通孔から電池缶の外方側に飛び出すことにより生じた導通板の外方側への曲げ変形によって破断する脆弱部が設けられていることを特徴とする密閉電池の安全装置。

【請求項12】 請求項1において、リード線と封口体との間の電流経路の途中には、設定温度域で抵抗値が急増するPTCサーミスタが介装されていることを特徴とする密閉電池の安全装置。

【請求項13】 請求項1において、封口体は、電池缶の開口部であって導通板の外側位置に配設されてその導通板と電気的に接続された正極端子を備え、この正極端子と、導通板と、絶縁部材と、安全弁が嵌合された状態の封口蓋とがそれぞれの外縁部で互いに重合された状態に予め結合一体化されていることを特徴とする密閉電池の安全装置。

【発明の詳細な説明】

【0001】

【発明の属する技術分野】本発明は、例えば2次電池等の密閉電池において過充電等に起因する内圧の異常上昇等の異常事態に対処するための密閉電池の安全装置に関する。

【0002】

【従来の技術】従来より、この種の密閉電池の安全装置として、発電素子からのリード線が接続されたリード板を絶縁板の内面に取付けてそのリード板により絶縁板の中央の貫通孔を覆い、上記絶縁板の上側を覆うように導電材料により形成した円板状の可撓性防爆弁を配設し、この防爆弁の中央部の突起を上記貫通孔に内嵌させてその突起に上記リード板を超音波溶接したものが知られている（例えば、特開平2-112151号公報、特開平2-288063号公報参照）。これらの公報には、電池缶の内圧の上昇により上記突起が持ち上げられて上記リード板を貫通孔の周縁で破断することにより、または、上記リード板との溶着部が剥離することにより電流が遮断されることが記載されている。

【0003】また、他の従来の密閉電池の安全装置として、リード線が接続された導電材料よりなる円板状の弁保持体と、この上側に被せられて中央部の突起が上記弁保持体の上面にスポット溶接もしくは超音波溶接された導電材料よりなる円板状の可撓性弁体とからなり、この弁体と弁保持体との間の空間が電池缶内部と連通されたものが知られている（例えば、特開平6-140011号公報参照）。つまり、このものでは、リード線が接続された弁保持体に弁体を溶接することにより、上記の防爆弁をリード板に直接溶接することによる不都合を回避しようとするものである。そして、この公報には、電池缶の内圧の上昇により上記空間内の圧力も上昇し、弁体が上方に持ち上げられてその中央突起と弁保持体との溶接が外れて電流が遮断されることが記載されている。

【0004】そして、共に、さらに内圧が上昇すると、可撓性防爆弁もしくは可撓性弁体が、これらに設けられた切欠部分で破断して内部のガス等を外部に逃がすことにより、電池自体の爆発を防止するようになっている。

【0005】

【発明が解決しようとする課題】ところが、上記従来の密閉電池の安全装置においては、共に電流の遮断機構において、溶接部を利用するものであるため、その溶接自体の強度いかんによって電流遮断を確実にに行い得ない事態が生じるおそれがある。すなわち、上記溶接強度が余りに高いと、内圧が上昇しても弁が弁保持体と剥離せず、その結果、電流を遮断させることができず、逆に上記溶接強度が余りに低いと通常使用時のわずかな内圧変動や衝撃により弁が弁保持体から剥離して電流が遮断されて使用不能になってしまうことになる。

【0006】このように従来のものでは、溶接強度を確実に所定のものに設定する必要があるが、そのためにスポット溶接等により点状に溶接したのでは、通常使用時においてその点状の溶接部分を通して電流の通電が行われるため、抵抗が高くなり過ぎて異常な発熱を生じるおそれがある。逆に、溶接部を周状等に設定すると、溶接面積に比例して溶接強度が高くなるため、上記の如き不都合が生じることになる。

【0007】しかも、上記の如き所定の溶接強度に対し過不足のない溶接作業を大量生産ラインで行うことは、その工程管理が極めて困難となり、製品間で溶接強度のばらつきを招き易く、製品に対する信頼性の向上を図り得ない。

【0008】本発明は、このような事情に鑑みてなされたものであり、その目的とするところは、密閉電池の安全装置の電流遮断機構から溶接部をなくし、かつ、所定の内圧値に対する電流の接続状態から遮断状態への切換のための調整、設定の容易化を図ることにある。

【0009】

【課題を解決するための手段】上記目的を達成するために、請求項1記載の発明は、発電要素が電池缶の内部に收容され、電池缶の外側に向いた開口部が封口体によって密閉されているものを前提とする。このものにおいて、上記封口体を、導電材料により可撓性を有するように形成された可撓性導通板と、導電材料により電池缶の内部を密閉しその内圧に対抗して実質的に変形しない剛性を有するように形成された封口蓋と、絶縁材料により形成され電池缶の内圧上昇により相対移動する安全弁とを備えたものとする。そして、上記封口蓋の略中央部に貫通孔を有し、外周縁部を電池缶の開口部に気密に取付け、電池缶内に臨む内面に上記発電要素からのリード線を接続する。また、上記導通板を、外縁部が電池缶の開口部であって上記封口蓋より外側位置で上記封口蓋の外面と絶縁部材を介して重合させ、上記貫通孔の近傍位置で上記封口蓋と当接して電氣的接続状態になるよう位置

固定する。さらに、上記安全弁を、上記封口蓋の貫通孔に対しこの貫通孔を封止するよう電池缶内部から上記導通板まで内外方向に挿通された状態で先端面が上記導通板に近接するよう位置付ける。加えて、上記貫通孔を構成する孔内面と、この孔内面に相対向する安全弁の周方向の外面とを、電池缶の内圧が上昇して設定内圧値に達するまではその内圧に抵抗して上記導通板に対する安全弁の相対変位を阻止した状態に保持する一方、上記設定内圧値を超える内圧の作用により上記安全弁を上記導通板側に相対変位させて上記導通板と封口蓋との当接を離すような抵抗力を発揮するよう互いに嵌合する構成とするものである。

【0010】上記の構成の場合、適正放電、適正充電がなされている通常使用状態にあって電池缶内部の内圧が設定内圧値より小さい状態においては、安全弁の周方向の外面と封口蓋の貫通孔の孔内面との間の抵抗力の方が上記内圧よりも大きいためその内圧に対抗して上記安全弁の相対移動が阻止され、可撓性導通板とリード線が接続された封口蓋との当接が維持される。この結果、電気的接続状態に維持される。

【0011】一方、過充電等により電池缶内部の内圧が上記設定内圧値を超えると、上記抵抗力よりもその内圧に基づく安全弁を相対移動させようとする押圧力の方が大きくなるため、上記安全弁が導通板側に相対移動して導通板を外方に押し曲げ、封口板との当接が離れることになる。この結果、その当接による電気的接続状態が遮断されて電気的遮断状態に切換られ、電池の使用者に異常発生を認知させることができる。

【0012】この場合、封口蓋が上記内圧を受けても変形しない剛性を有しているため、その内圧が安全弁に直接かつ有効に作用しその安全弁の相対移動による押圧力が外縁部で支持された導通板の略中央部に集中的に作用することになる。このため、上記設定内圧値を超えるような内圧上昇が発生した場合に、従来の円板状の防爆弁の全体に内圧を間接的に作用させる場合と比べ、応答性よく電気的遮断状態に切換ることが可能となる。

【0013】しかも、このような電流遮断機構では電池缶内部の内圧に応じた電気的な接続状態と遮断状態との切換が、安全弁と貫通孔との間の嵌合による抵抗力の設定により行われるため、従来の密閉電池の如く溶接部の引張破断強度の設定により行われる場合と比べ、上記内圧に対する切換強度の調整及び変更が容易となる上、確実に所定のものに設定することが可能となって電気的な遮断状態への切換が確実になる。さらに、組付け時の工程管理の容易化が図られて製品に対する信頼性の向上が図り得る。

【0014】請求項2記載の発明は、請求項1記載の発明において、貫通孔を電池缶の内外方向に一樣な内径を有するように形成する一方、安全弁をピストン状に形成する。そして、上記貫通孔の内周面と、上記安全弁の外

周面とを、上記安全弁の導通板側への相対変位に対する両周面間の摺動摩擦抵抗力を発揮することにより電池缶の内圧に抵抗するように互いに密に嵌合させる構成とするものである。

【0015】上記の構成の場合、安全弁を設定内圧値まではその相対変位を阻止した状態に保持し、かつ、設定内圧値を超える電池缶内圧に対し安全弁の導通板側への相対変位を許容するという安全弁と貫通孔との間の抵抗力が、安全弁と貫通孔との両周面間の摺動摩擦抵抗力によって発揮されるため、電池缶内圧に応じた上記抵抗力の調整もしくは変更が容易になる。すなわち、貫通孔の孔径の設定及びその変更、または、安全弁の外径の設定及びその変更、さらには、上記安全弁の材質の設定及びその変更により、安全弁と貫通孔との両周面間の摺動摩擦抵抗力の調整、変更が行い得る。

【0016】請求項3記載の発明は、請求項2記載の発明において、安全弁を、先端から基端側まで封口蓋の貫通孔と対応する同一の断面形状を有し基端側が封口蓋内面から電池缶内方側に突出するよう配置された柱状の本体部と、この本体部の基端側から周囲に僅かに突出する凸部とから構成する。加えて、電池缶の内圧を受けて貫通孔の内周面との間で摺動して相対移動する際の内圧値が、上記凸部形成位置の方が上記本体部よりも相対的に大となるように設定するものである。

【0017】上記の構成の場合、安全弁が、柱状の本体部と、この本体部の基端側に形成された凸部とから構成され、上記本体部が先端から基端側まで封口蓋の貫通孔と対応する同一の断面形状を有して基端側が封口蓋内面から電池缶内方側に突出するよう配置され、上記凸部が本体部の基端側から周囲に僅かに突出されているため、電池缶の内圧の設定内圧値を超える上昇を基端面に受けて安全弁の本体部が封口蓋の貫通孔を摺動して凸部が貫通孔の入り口で止められるまで相対移動し、これにより、導通板の当接部が封口蓋から離れて電流の導通が遮断される。そして、内圧がさらに上昇して上記凸部と貫通孔の内周面との間の摺動摩擦抵抗力を上回ると、上記凸部が貫通孔を摺動して飛び抜け、この結果、電池缶内部のガスが貫通孔を通して排出され電池缶の内圧が大気圧状態となる。従って、異常時には、まず電流が遮断されて電池の使用者に異常発生を認知せしめ、それでも使用者が放置する場合であっても、内部のガス抜きによる内圧開放により内圧のそれ以上の上昇が回避される。

【0018】請求項4記載の発明は、請求項2記載の発明において、安全弁を、先端から基端側まで封口蓋の貫通孔と対応する同一断面形状を有し基端側が封口蓋内面から電池缶内方側に突出するよう配置された柱状の本体部と、この本体部の基端側の周囲に拡開されたフランジ状の受圧部とから構成する。加えて、封口蓋に、電池缶内部と外部とを連通させるよう形成されたガス抜き孔と、このガス抜き孔を密閉する閉止体とを備える。そし

て、この閉止体の開放強度を、上記受圧部を封口蓋内面に当接させる電池缶内圧より大の内圧を受けて開放するように設定するものである。

【0019】上記の構成の場合、安全弁が柱状の本体部と、その本体部の基端側で拡開されたフランジ状の受圧部とから構成される一方、封口蓋に電池缶内部と外部とを互いに連通させるガス抜き孔と、このガス抜き孔の封口蓋内面側で閉止し所定の開放強度を有する閉止体とが設けられているため、電池缶の内圧が設定内圧値を超えると、まず、上記受圧部が封口蓋内面に当たって止められるまで安全弁が相対移動し、電流が遮断される。そして、上記電池缶の内圧が設定内圧値を超えてさらに上昇すると、上記閉止体が開放し内部のガスがガス抜き孔を通して放出される。この場合、安全弁の受圧部が本体部よりも拡開されて受圧面である安全弁の基端面が本体部だけの場合と比べ拡大されているため、内圧の上昇に対し、より応答性よく導通板を押し上げて電流の遮断を行うことが可能になる一方、内圧が上記設定内圧値を超えてさらに上昇しても、閉止体の開放により、内圧のそれ以上の上昇が回避されて密閉電池の破壊が回避される。

【0020】請求項5記載の発明は、請求項1記載の発明において、貫通孔を、電池缶の内方側の大径孔部と、導通板側の小径孔部と、これら大径孔部と小径孔部との間に形成された電池缶内方側に面する当止面とを有するものとする。一方、安全弁を、上記小径孔部を電池缶外方側に挿通する小径部と、上記大径孔部に内嵌する大径部と、これら小径部と大径部との間に上記当止面と相対向するよう形成された階段状の段差面とを有するものとする。そして、上記貫通孔と安全弁とを、上記貫通孔の当止面に対し上記安全弁の段差面が当接し、電池缶の内圧を受けてその安全弁の段差面から大径部にかけての部分が貫通孔の小径孔部に対して圧入しようとする圧入抵抗力を発揮することにより電池缶の内圧に抵抗するように互いに嵌合させる構成とするものである。

【0021】上記の構成の場合、安全弁を設定内圧値まではその相対変位を阻止した状態に保持し、かつ、設定内圧値を超える電池缶内圧に対し安全弁の導通板側への相対変位を許容するという安全弁と貫通孔との間の抵抗力が、貫通孔の小径孔部に対する安全弁の段差面から大径部にかけての部分の圧入抵抗力的によって発揮されるため、電池缶内圧に応じた上記抵抗力の調整もしくは変更が請求項2記載の発明による場合と比べより容易にかつ確実にい行得る。すなわち、貫通孔の小径孔部の直径の設定及びその変更、または、安全弁の大径部の直径の設定及びその変更、つまり、上記小径孔部と大径部との互いの直径差を調整することによる各貫通孔の当止面と安全弁の段差面との当接面積の調整、さらには上記安全弁の材質の設定及びその変更により、安全弁の貫通孔に対する圧入抵抗力の調整が自由に行い得る。しかも、安全弁を封口蓋に対し組み付ける際に、その安全弁の段差面

を貫通孔の当止面に当接させるだけで、安全弁が封口蓋及び導通板に対し所定の内外方向位置に組み付けられる。このため、安全弁の封口蓋に対する位置決めが容易かつ確実となり、密閉電池の製造が容易になる上、その安全弁の先端面を導通板に近接する位置に確実に位置付けて導通板を封口蓋から離すことなく封口蓋と電気的接続状態に確実に維持し得る。

【0022】請求項6記載の発明は、請求項2または請求項5記載の発明において、安全弁を、弾性材料により形成するものである。ここで、弾性材料としては、例えば天然ゴム、EPDM（エチレン・プロピレン・ジエン共重合体）等の合成ゴム、ポリプロピレン、ポリエチレン等を用いればよい。

【0023】上記の構成の場合、安全弁が弾性材料により形成されているため、安全弁と封口蓋の貫通孔との間の摺動摩擦抵抗力または圧入抵抗力の設定がより一層容易に行い得る。

【0024】請求項7記載の発明は、請求項5記載の発明において、安全弁の大径部を段差面が貫通孔の当止面に当接した状態で電池缶内方に突出させ、この大径部の突出端に上記貫通孔の大径孔部よりも大径の基端側鏝部を一体に形成する。加えて、上記基端側鏝部の外径を、その基端側鏝部を封口蓋内面に当接させる電池缶内圧よりも大の内圧を受けて上記貫通孔を導通板側に突き抜けるように設定するものである。

【0025】上記の構成の場合、電池缶の内圧が上昇すると、その内圧を受けて安全弁の段差面から大径部にかけての部分が変形して貫通孔の小径孔部内に圧入することになる。この圧入による安全弁の導通板側への相対変位により安全弁の先端面が導通板を押し上げて電流の遮断状態に切換るとともに、上記基端側鏝部が封口蓋内面に当接させる。この結果、上記基端側鏝部の外径に対応して基端側鏝部の大径孔部に対する圧入抵抗を受けて上記安全弁の相対変位は一旦停止する。そして、上記電池缶の内圧がさらに上昇して上記基端側鏝部の圧入抵抗を超えると、その基端側鏝部が上記大径孔部内に圧入し、安全弁全体が貫通孔を導通板側に突き抜けることになり、この結果、電池缶の内圧が解放される。従って、電池缶の内圧の上昇に従って、第1段階として電流が遮断され、第2段階としてその内圧が解放されるという、2段階の安全動作が行われる。

【0026】請求項8記載の発明は、請求項5記載の発明において、貫通孔の小径孔部の内周面を導通板側に向かって徐々に拡径するテーパ孔面により構成し、加えて、安全弁の小径部の外周面を段差面が上記貫通孔の当止面に当接した状態で上記テーパ孔面に当接するテーパ孔面により構成するものである。

【0027】上記の構成の場合、密閉電池が通常の使用状態にある時には、安全弁のテーパ面が上記貫通孔のテーパ孔面に当接し、これにより、その安全弁が貫通孔内

を電池缶の内方側に脱落することが確実に防止される。このため、上記安全弁は、その段差面が貫通孔の当正面に当接した状態に確実に維持され、電池缶の内圧が上昇した場合に確実に安全動作を行う態勢に維持される。従って、製品の出荷段階から使用段階に至る間、安全弁の移動、脱落が確実に防止されて安全弁が封口蓋に対し確実に保持され、電池缶の内圧上昇時の安全動作が確実に行われる。

【0028】請求項9記載の発明は、請求項5記載の発明において、安全弁を、封口蓋に対しゴムを用いた一体加硫成形により、その封口蓋と一体に形成するものである。

【0029】上記の構成の場合、安全弁が予め封口蓋と一体に形成されているため、封口蓋に対する安全弁の組み付け作業を省略して密閉電池の製造の容易化、合理化が図られる。併せて、上記一体加硫成形により、予め安全弁が封口蓋及び導通板に対し所定の相対位置に確実に位置付けられ、このため、通常使用時における封口蓋と導通板との電気的接続状態の維持と、内圧上昇時の安全弁の安全動作とが確実に行われる。

【0030】請求項10記載の発明は、請求項1記載の発明において、安全弁の先端面と、可撓性導通板の内面とを互いに接着して結合するものである。

【0031】上記の構成の場合、安全弁の先端面と、可撓性導通板の略中央部内面とが接着されているため、可撓性導通板自身が安全弁を介して封口蓋により保持された状態になる。すなわち、外縁部により支持された導通板がその略中央部で安全弁と接着されて支持され、その安全弁が封口蓋の貫通孔の内周面との間の摺動摩擦抵抗力によって上記封口蓋に保持されるため、上記導通板は封口蓋との当接、つまり、電気的接続状態を安定的に保持されることになる。従って、導通板と封口蓋とが溶接されていなくても、上記の電気的接続状態がより確実に保持され、通常使用時の製品に対する信頼性についてより一層の向上が図られる。

【0032】請求項11記載の発明は、請求項1記載の発明において、導通板に、安全弁が電池缶の内圧を受けて封口蓋の貫通孔から電池缶の外方側に飛び出すことにより生じた導通板の外方側への曲げ変形によって破断する脆弱部を設けるようにするものである。

【0033】上記の構成の場合、安全弁の飛び抜けにより押されて曲げ変形した際に破断する脆弱部が導通板に設けられているため、電池缶内部からのガス抜きがより効果的に行われる。

【0034】また、請求項12記載の発明は、請求項1記載の発明において、リード線と封口体との間の電流経路の途中に、設定温度域で抵抗値が急増するPTCサーミスタを介装するものである。

【0035】上記の構成の場合、リード線と封口体との間の電流経路の途中に、PTCサーミスタが介装されて

いるため、電池缶の内部温度が適正放電、適正充電等の行われている通常使用状態における設定温度域より低い適正温度にある間は上記PTCサーミスタの抵抗値が小値に保たれ上記リード線と封口体との間の電流経路を通して適正に電流が流れる。一方、上記内部温度が短絡等の発生に起因して設定温度域以上に上昇した場合、その上昇に対応して上記PTCサーミスタの抵抗値が急増して上記電流経路を通しての通電が実質遮断される。

【0036】さらに、請求項13記載の発明は、請求項1記載の発明において、封口体を、電池缶の開口部であって導通板の外側位置に配設されてその導通板と電気的に接続された正極端子を備え、この正極端子と、導通板と、絶縁部材と、安全弁が嵌合された状態の封口蓋とがそれぞれの外縁部で互いに重合された状態に予め結合一体化されたものとして構成するものである。

【0037】上記の構成の場合、封口体として、貫通孔に安全弁が嵌合された状態の封口蓋と、絶縁部材と、上記封口蓋に電気的に接続された状態の導通板と、正極端子とのそれぞれの外縁部が互いに重合された状態に予め結合一体化されたものに構成しているため、電池缶の開口部への組み付け作業の容易化が図られる上、その組み付け前に上記安全弁の動作を確認するためのテストを行うことが可能になる。

【0038】

【発明の実施の形態】以下、本発明の実施形態を図面に基いて説明する。

【0039】＜第1実施形態＞図1は、本発明の第1実施形態に係る密閉電池の安全装置の封口部分を示す。本第1実施形態は、請求項1、2記載の発明を基本として構成されたものである。同図において、1は開口部1aを有する電池缶、2はその開口部1aを閉止して電池缶1を密封する封口体である。なお、図1では、電池缶1の開口部1aにおける内外方向が上下方向となるように図示されている。つまり、電池缶1の内方が図1の下方に、外方が上方になる（以下の他の実施形態における図において同じ）。

【0040】上記電池缶1は有底筒形（図例では円筒形であるが、角筒形であってもよい）の金属製缶体であり、その内部には発電要素3が収容されている。この発電要素3は、プラス電極板、マイナス電極板、及び、電解液等を有し、そのプラス電極板に一端が接続されたりード線4の他端が上記封口体2を構成する後述の封口蓋6の下面6aに溶接等の手段により接続されている。そして、上記発電要素3の上側位置の電池缶1には縮径部1bが形成されており、この縮径部1bと上端開口縁がかしめられた上端かしめ部1cとの間に上記封口体2の外周縁部が絶縁ガasket5を介して気密に挟み込まれて電池缶1が密封されている。

【0041】上記封口体2は、図2にも一部を示すように、円板状の封口蓋6と、ドーナツ板状の絶縁リング

7と、円板状の導通板8と、キャップ状の正極端子9と、ピストン状安全弁10とから構成されている。そして、上記封口蓋2が上記上端かしめ部1cにより挟み込まれることにより、下から封口蓋6、絶縁リング7、導通板8、及び、正極端子9の順に互いに密着して重合された状態に一体化されて組み付けられている。

【0042】上記封口蓋6は、アルミニウム、チタニウム、これらの合金、もしくは、ステンレス鋼等の導電材料により比較的厚肉に形成されて電池缶1の内圧に対抗して実質的に変形を生じない程度の剛性を有している。そして、この封口蓋6の略中央位置には所定径の貫通孔11が上下方向に貫通して形成され、この貫通孔11に上記安全弁10が下から上に挿通した状態で密に嵌合されている。

【0043】上記導通板8は、上記の導電材料により所定の可撓性を有するように形成されており、間に介在された絶縁リング7によって上記封口蓋6と絶縁されるとともに、その絶縁リング7の厚み分上記封口蓋6から上方に離されて配設されている。そして、上記導通板8の中央部下面8aと、上記安全弁10の先端面である上端面10aとが互いに接着されて結合されており、この導通板8にはこの結合部位を囲む円環状の突起8bが下方に突出して形成されている。この突起8bの突出寸法は上記の重合状態の絶縁リング7の厚みと等しいか僅かに大きく設定され、これにより、上記重合状態では上記突起8bの突出周面と封口蓋6の上面とが密に当接されて電氣的接続状態(図1に示す状態)とされている。なお、図面中8cは上記突起8bの形成範囲の近傍に形成された脆弱部であり、この脆弱部8cは、上記の結合部位を囲むように形成した切れ込み溝により構成され、所定の曲げ力が作用することにより破断するようになっている。

【0044】上記正極端子9は、上記の導通板8等を保護する保護キャップの役割を果たすとともに、リード線4、封口蓋6、突起8b、及び、導通板8を介して発電要素3のプラス電極板と電氣的に接続されて密閉電池から電力を取り出すためのプラス側の端子の役割を果たすようになっている。なお、図面中9aはガス抜き孔であり、正極端子9と導通板8とで区画される空間と外部とを連通するように貫通形成されている。

【0045】上記安全弁10は、上下方向に同一断面形状を有する所定径のピストン状の本体部10bと、この本体部10bの基端である下端から外周方向に膨出する凸部10cとが、上記発電要素3内の電解液により腐食されない絶縁材料により、好ましくは、その絶縁材料であってかつ弾性を有する材料、例えば天然ゴム、EPDM(エチレン・プロピレン・ジエン共重合体)等の合成ゴム、ポリエチレン、ポリプロピレン等の材料により形成されたものである。上記本体部10bは、その先端面である上端面10aが電氣接続状態の導通板8の中央部

下面8aと接着され、下端が封口蓋6から下方側に突出され、この本体部10bの貫通孔11から所定寸法下方位置に上記凸部10cが位置するように設定されている。

【0046】上記本体部10bの外径及び凸部10cの突出量は、上記安全弁10を動作させる圧力として予め定めた第1及び第2設定内圧値との関係で、上記本体部10bもしくは凸部10cの各外周面と貫通孔11の内周面11aとの間に所定の摺動摩擦抵抗力が生じるように定められる。すなわち、電池缶1内の内圧が上記第1設定内圧値(例えば10~15Kg/cm²)に到達するまでは、その内圧による押圧力に対抗して安全弁10の相対移動を阻止する摺動摩擦抵抗力が作用し、上記電池缶1内の内圧が上記第1設定内圧値を超えると、その内圧による押圧力の方が上記摺動摩擦力より大となって上記安全弁10をその摺動摩擦抵抗力に抗して外方側である導通板8側に相対移動させるような摺動摩擦抵抗力が作用するように、貫通孔11の内径に対する上記本体部10bの外径が定められている。そして、上記貫通孔11の内径に対する凸部10cの外径は、上記電池缶1内の内圧が上記第1設定内圧値を超えてさらに上昇して上記第2設定内圧値(例えば20Kg/cm²程度)を超えた場合に、その内圧による押圧力が上記凸部10cの外周面と貫通孔11の内周面11aと間の摺動摩擦抵抗力より大となって上記安全弁10の凸部10cが貫通孔11を飛び抜けるような摺動摩擦抵抗力が作用するように定められている。

【0047】また、この安全弁10の上方への相対移動量に対応して、上記導通板8の可撓性能及び脆弱部8cの破断強度の設定が行われている。すなわち、上記導通板8は、上記第1設定内圧値を超える内圧に基づき上記凸部10cが貫通孔11の下側開口縁により相対移動が停止されるまでの安全弁10の上方への相対移動量に対応する曲げが可能であり(図3参照)、上記凸部10cが貫通孔11を飛び抜けた場合の上記安全弁10の相対移動量に対応する曲げを受けて脆弱部8cが破断するようになっている(図4参照)。

【0048】上記構成の密閉電池において、電池缶1内の内圧が第1設定内圧値より低い適正内圧に保たれている状態では、安全弁10が貫通孔11に対しその内周面11aと間の摺動摩擦抵抗力によって相対移動を阻止された状態に保持されて、この安全弁10を介して上下方向の移動が阻止された導通板8の突起8bが封口蓋6と当接した状態に保たれる。このため、発電要素のプラス電極板がリード線4、封口蓋6、突起8b、及び、導通板8を介して正極端子9と電氣的に接続した電氣的接続状態に保たれる。これにより、上記正極端子9からの放電、充電が可能となる。そして、この状態では、安全弁10の外周面と、貫通孔11の内周面11aとが密着して電池缶1の内部が密封された状態に保たれる。

【0049】一方、密閉電池への充電中において、過充電等により電池缶1内の内圧が異常に上昇し、その内圧が上記第1設定内圧値を超えると、その内圧に基づき安全弁10の下面10dに作用する押圧力が上記摺動摩擦抵抗力より大となり、これにより、上記安全弁10が貫通孔11に対し上方に相対移動して導通板8の中央部が上方に押し上げられる。この結果、その導通板8の突起8bが封口蓋6から上方に離れ、これにより、発電要素3からの電流は封口蓋6と上記突起8bと間で遮断され、電氣的遮断状態(図3に示す状態)になる。この電氣的遮断により充電電流が遮断されて発電要素3内の電解液の分解が停止されるため、これ以上の内圧の上昇を抑制することができる。しかも、電氣的に遮断されて電池としての機能を果たさないことから、使用者に異常状態の発生、及び、その電池の交換の必要性を認知させることができる。

【0050】そして、上記の電流が遮断されるまでの間の内圧上昇、もしくは、何等かの異常によって電池缶1内の内圧が第2設定内圧値を超える程に上昇した場合には、上記の安全弁10の凸部10cが貫通孔11内に押し込まれ、その貫通孔11を上方に飛び抜ける(図4参照)。この飛び抜けによる導通板8の過大な曲げにより脆弱部8cが破断して、電池缶1内のガスが上記貫通孔11、上記破断箇所、及び、正極端子9のガス抜き孔9aを通して外部に放出され、これにより、電池缶1内の内圧が大気圧近くまで低下して、内圧の異常上昇による密閉電池の破壊が確実に防止される。

【0051】従って、内圧の異常上昇時には、まず、第1段階として、電流の遮断がピストン状安全弁10の本体部10bと貫通孔11との間の摺動摩擦抵抗力の設定及び導通板8の突起8bと封口蓋6との離反から構成される電流遮断機構により行われ、次に、第2段階として、内部ガスの放出が上記安全弁10の凸部10cと貫通孔11との間の摺動摩擦抵抗力の設定及び導通板8の脆弱部8cの破断から構成される内部圧力開放機構により行われるという、2段階のフェイルセーフ動作が行われる。

【0052】このような密閉電池の安全装置においては、内圧が第1設定内圧値より低い適正状態の時には、絶縁リング7を介して導通板8と封口蓋6とが重合した状態にかしめ部1cによってかしめられているため導通板8の突起8bと封口蓋6の上面とが当接した状態に保たれる。しかも、上記導通板8の中央部下面8aに対し安全弁10の上面10aが接着されその安全弁10が貫通孔11の内周面11aによって相対移動しないように保持されているため、導通板8の中央部位が安全弁10を介して封口蓋6に対し上下方向に相対変位しないように保持されることになり、電池缶1に衝撃、振動が作用しても上記の突起8bと封口蓋6との当接を確実に維持することができる。従って、導通板8の突起8bと封口

蓋6とが溶接されていなくても、上記の電氣的接続状態がより確実に保持され、通常使用時の製品に対する信頼性についてより一層の向上が図られる。つまり、従来の場合の如く通常使用時の電氣的接続状態を維持するための溶接を施す必要がなく、その溶接いかんによって所定の回避動作が不確実になることもない。その上、組み付け時にそのような微妙な溶接を施す必要がないために、その生産ラインにおける工程管理も容易となって製品に対する信頼性の向上を図ることができる。さらに、上記導通板8と封口蓋6との電氣的接続を溶接ではなく、当接により行うことができるため、溶接の場合における電氣的接続の接続面を極めて小さいものにしなければならないという制約がなく、円環状の突起8bと封口蓋6との当接というかなり大きな接続面により電氣的接続を行うことができ、安定した電流の流れを実現することができる。

【0053】また、封口蓋6が上記内圧を受けても変形しない剛性を有しているため、その内圧がピストン状安全弁10の下面10dに直接かつ有効に作用し、その安全弁10の相対移動によって、外周縁部で支持された導通板8の略中央部に押圧力が集中的に作用することになる。このため、上記第1設定内圧値を超えるような内圧上昇が発生した場合には、従来の円板状の防爆弁の全体に内圧を間接的に作用させる場合と比べ、電氣的遮断状態への切換を内圧の変動に対し応答性よく行うことができる。

【0054】しかも、電氣的な接続と遮断との切換が、主として貫通孔11の内径に対する安全弁10の外径の設定による摺動摩擦抵抗力の設定により行われるため、従来の密閉電池の如く溶接部の引張破断強度の設定により行われる場合と比べ、その設定及び設定の変更を容易に行うことができる。このため、上記の第1もしくは第2設定内圧値に確実に対応したものに設定することができ、電氣的な接続状態から遮断状態への切換タイミングを電池缶1内の内圧に応じて確実なものにすることができる。この際、上記安全弁10を弾性材料により形成することにより、上記摺動摩擦抵抗力をより確実に発揮させることができる上、その発揮する摺動摩擦抵抗力の設定をより容易に行うことができる。

【0055】<第2実施形態>図5は、本発明の第2実施形態に係る密閉電池の安全装置を示す。この第2実施形態は、第1実施形態の構造を基本として、この第1実施形態に対し内部温度上昇に伴う電流の遮断機構を付加して請求項12記載の発明に係る実施形態を構成したものである。

【0056】すなわち、図5において、12はドーナツ板状に形成されたPTCサーミスタ(Positive Temperature Coefficient サーミスタ; 正特性サーミスタ)であり、このPTCサーミスタ12は導通板8と正極端子9との間に介装されて、その導通板8と正極端子9と

が上記PTCサーミスタ12を介してのみ互いに接続されるようになっている。なお、このPTCサーミスタ12の介装位置は、発電要素3から正極端子9までの間の電気導通経路の途中であればいずれの位置でもよい。

【0057】上記PTCサーミスタ12は、温度上昇とともに抵抗値が増大する正の温度係数を有するサーミスタであり、特に上記PTCサーミスタ12は所定の設定温度域（例えば100℃域）で抵抗値が急激に増大するスイッチング特性を有するように構成されている。つまり、上記PTCサーミスタ12は、上記設定温度域までの通常温度範囲ではPTCサーミスタ12の抵抗値が極めて低く、従って、上記導通板8と正極端子9との間でPTCサーミスタ12を介して電流が良好に流れる一方、上記設定温度域を超える高温域では、上記PTCサーミスタ12の抵抗値が急増し、従って、上記導通板8と正極端子9との間での通電を実質的に遮断するようになっている。

【0058】このような構成の第2実施形態の場合、電池缶1内の内部温度が上記設定温度域までの通常温度範囲にあれば、発電要素3のプラス電極板からリード線4、封口蓋6、突起8b、導通板8、及び、PTCサーミスタ12を介して正極端子9までの電流経路が電氣的接続状態になり、その正極端子9から放電、充電を行うことができる。一方、過充電や短絡の発生に伴い電池缶1内で発熱し、その内部温度が上記設定温度域を超える事態になれば、上記PTCサーミスタ12で電流の流れが阻害されて上記電流経路が電氣的遮断状態になる。特に、上記短絡が発生すると、上記経路に過大電流が流れて急激に内部温度が上昇するため、上記PTCサーミスタ12により即座に発電要素3と正極端子9と間での過大電流の流れを遮断することができる。

【0059】従って、導通板8の突起8bと封口蓋6との電氣的接続を安全弁10の相対移動により離すことによる内圧上昇に基づく電流遮断機能、及び、上記安全弁10の飛び抜けによる内圧開放機能に加えて、上記の如き内部温度上昇に基づく電流遮断機能をも安全装置として付加することができる。

【0060】＜第3実施形態＞図6は本発明の第3実施形態に係る密閉電池の安全装置を示す。この第3実施形態は、第1実施形態の構造を基本としつつ、安全弁13には内圧上昇に伴う電流の接続状態から遮断状態への切換をのみ行わせ、電流遮断後の内圧開放を閉止体としての弾性膜14により行うようにしたものであり、請求項4記載の発明に係るものである。

【0061】上記安全弁13は、貫通孔11の内径との関係において所定の摺動摩擦抵抗力を発揮するように外径を定めた円柱状の本体部13aと、上記貫通孔11の内径よりもかなり大きい外径となるように拡開されたフランジ状の受圧部13bとが第1実施形態の安全弁10と同様の材料により一体に形成されている。そして、上

記安全弁13の本体部13aが貫通孔11の内周面11aに下から密に嵌合されて、その貫通孔11の内周面11aとの間の摺動摩擦抵抗力によって第1実施形態で説明した第1設定内圧値までの内圧に対抗して上下方向に相対移動しないように保持されている。その上端面13cが導通板8の中央部下面8aに接着されて、導通板8の突起8bと封口蓋6との当接を維持するようになっている。また、上記受圧部13bは封口蓋6の下面から下方に僅かに離れて位置するようになっている。

【0062】また、封口蓋6及び導通板8にはそれぞれガス抜き孔15a、15bが貫通して形成されている。上記封口蓋6のガス抜き孔15aの下面側開口は上記弾性膜14により覆われ、この弾性膜14は上記封口蓋6の下面に接着されている。これにより、上記ガス抜き孔15aが密閉されている。そして、上記弾性膜14は、例えばポリエチレンもしくはポリプロピレン等を材料として所定膜厚を有するように形成され、これにより、第1実施形態において説明した第2設定内圧値の内圧を受けて破断して上記ガス抜き孔15aを開放状態にするようになっている。つまり、開放強度として上記弾性膜14の引張破断強度が上記の如く設定されている。

【0063】なお、この第3実施形態のその他の構成は第1実施形態と同様であるため、同一構成部材には同一符号を付してその詳細な説明を省略する。

【0064】この第3実施形態の場合、電池缶1内の内圧が第1設定内圧値までの通常使用状態の範囲では、発電要素3と正極端子9とがリード線4、封口蓋6、突起8b、及び、導通板8を介して電氣的接続状態に保たれ、封口蓋6と安全弁13と弾性膜14とによって電池缶1の内部が密閉された状態になっている。

【0065】そして、上記内圧が上昇して第1設定内圧値を超えると、受圧部13bが封口蓋6の下面に当接して相対移動が止められるまで安全弁13が上方に相対移動して上記突起8bと封口蓋6との当接が離される。この結果、上記突起8bと封口蓋6との間で電氣的に遮断される。この際、上記安全弁13の受圧部13bは、本体部13aと比べかなり大径にすることができるため、上記第1設定内圧値を超える内圧を広い面積で受けることができ、同じ内圧でも、第1実施形態の場合と比べより大きい押圧力を導通板8に作用させて突起8bと封口蓋6との当接を離すことができる。これにより、第1実施形態の場合と比べ、より応答性よく電流の遮断状態への切換を行うことができる。

【0066】なお、上記内圧が上昇してその内圧が第2設定内圧値を超えると、その内圧を受けて弾性膜14が破断し、内部のガスがガス抜き孔15a、15b及び9aを通して外部に排出され、これにより、内圧が大気圧に近い状態になり、内圧の開放が行われる。

【0067】従って、本第3実施形態の場合、第1設定内圧値において電流を遮断させるための作動設定を、安

全弁13の本体部13aと貫通孔11との間の摺動摩擦抵抗力の設定により行うことができる一方、第2設定内圧値において内圧を開放させるための作動設定を弾性膜14の破断強度の設定により行うことができ、従来の溶接部の破断強度の設定により行う場合と比べ、容易かつ精度よく行うことができる。

【0068】<第4実施形態>図7は本発明の第4実施形態に係る密閉電池の安全装置を示す。この第4実施形態は、請求項1、5記載の発明を基本として構成されたものである。同図において、20は安全弁、21は封口蓋6の略中央部に上下方向に貫通して形成され上記安全弁20が内嵌された貫通孔である。

【0069】上記貫通孔21は、図8に詳細を示すように、下半部に形成され所定の内径D2を有する大径孔部21aと、上半部に形成され大径孔部21aよりも所定寸法小さい内径D1を有する小径孔部21bと、これら大径孔部21aと小径孔部21bとの間に下方に面するよう形成されたドーナツリング状の当止面21cとを有し、階段状に構成されている。

【0070】また、上記安全弁20は、先端側である上端側に形成され上記小径孔部21bの内径D1よりも小さい外径の小径部20aと、この小径部20aの下側に形成され上記大径孔部21aの内径とほぼ同じ外径D2の大径部20bと、この大径部20bの下側の基端側から外周方向に突出した鏝部20cとから階段状に形成されたものである。そして、上記小径部20aと大径部20bとの間に上記当止面21cに当接する第1段差面20dと、上記大径部20bと鏝部20cとの間に第2段差面20eとが形成されている。このような安全弁20は、発電要素3内の電解液により腐食されない絶縁材料であって電池缶1の内圧を受けて所定の变形特性を発揮し得る材料により形成されている。好ましくは、その絶縁材料であってかつ上記内圧を受けて弾性変形する材料、例えば天然ゴム、EPDM等の合成ゴム、ポリエチレン、ポリプロピレン等の材料により形成されている。

【0071】上記小径部20aは、第1段差面20dが貫通孔21の当止面21cに当接した状態で、貫通孔21の上端開口から上方に突出されており、これにより、その先端側である上端面20fが導通板8の下面に近接して位置付けられるようになっている。また、上記大径部20bは、同様に第1段差面20dが貫通孔21の当止面21cに当接した状態で、貫通孔21の下端開口から下方に突出されており、これにより、上記第2段差面20eが封口蓋6の下面6aから所定寸法下方に位置付けられるようになっている。

【0072】上記安全弁20は、通常使用時には、大径部20bが大径孔部21aに内嵌され、その両周面間の摺動摩擦抵抗によって封口蓋6に対し段差面20dが貫通孔21の当止面21bに当接した状態に保持されている。すなわち、第1実施形態において説明した第1設定

内圧値までの電池缶1の内圧による押圧力に対抗して、上記第1段差面20dが当止面21bに当接することにより、安全弁20は上方への相対変位が阻止されるようになっており、これにより、電池缶1の内圧に対抗して電池缶1内が密閉状態に保たれる。

【0073】そして、上記貫通孔21と安全弁20との各寸法は、第1実施形態と同様の第1及び第2設定内圧値との関係で、上記貫通孔21に対し安全弁20が所定の圧入抵抗力を発揮するように定められる。すなわち、安全弁20の形成材料の弾性変形特性を考慮して、上記貫通孔21の小径孔部21bの内径D1と、大径孔部21aの直径D2とが定められる。具体的には、まず、上記直径D2を一定値として定め、この直径D2に対する直径D1の値を調整することにより、第1設定内圧値を超える内圧を受けて安全弁20の第1段差面20dから大径部20bにかけての部分が弾性変形して上記直径D1の小径孔部21b内に圧入するようにし、このような直径値を上記直径D1として定める。次に、上記安全弁20の鏝部20cの突出量、すなわち、鏝部20cの外径について、上記第2設定内圧値を超える内圧を受けてその鏝部20cが弾性変形して貫通孔21の大径孔部21a及び小径孔部21bに圧入し、その貫通孔21を上方に飛び抜けるような値に設定する。

【0074】上記の構成の第4実施形態の場合、通常使用時には、安全弁20の大径部20bが大径孔部21aに内嵌して両周面間の摺動摩擦抵抗によって安全弁20が封口蓋6に保持されるとともに、第1設定内圧値までの内圧に対抗して電池缶1の内部を気密に維持する。この状態では、導通板8の突起8bは封口蓋6と当接して、封口蓋6と導通板8とは電氣的接続状態（図7及び図8に実線で示す状態）に保たれる。

【0075】そして、電池缶1の内圧が上昇して第1設定内圧値を超えると、安全弁20の段差面20dから大径部20bにかけての部分が弾性変形して小径孔部21b内に圧入して上方に相対変位することになる（図8の一点鎖線で示す安全弁20参照）。この結果、安全弁20の上端面20fが導通板8の下面8aに当接してその導通板8を上方に押し上げることになり、突起8bを封口蓋6から上方に離して密閉電池を電氣的遮断状態に切換ることになる。この内圧下では、上記圧入による上方への相対変位により第2段差面20eが封口蓋6の下面6aに当接し、鏝部20cの大径孔部21a等に対する圧入抵抗により安全弁20はそれ以上の相対変位が抑制され、密閉電池は上記電氣的遮断状態に維持される。

【0076】さらに、上記電池缶1の内圧が上昇して第2設定内圧値を超えると、上記鏝部20cが弾性変形して貫通孔21内に圧入していった上方に飛び抜けることになる。これにより、導通板8の脆弱部8cが破断し、電池缶1の内圧が貫通孔21、上記の破断箇所及びガス抜き孔9aを通して解放される。

【0077】このように電池缶1の内圧上昇に伴い、まず電氣的遮断状態への切換が行われ、次に内圧の解放が行われるという2段階の安全動作を安全弁20によって行なわせることができる上、その安全弁20の変換動作を行なわせる内圧設定、すなわち、圧入抵抗力の設定をより容易にかつ確実に行うことができる。すなわち、この第4実施形態の場合、上記圧入抵抗力の設定は、第1設定内圧値を超える内圧に対し第1段差面20dから大径部20bにかけての部分の直径D1の小径孔部21b内に圧入するように、安全弁20の弾性変形特性を考慮して直径D2(図8参照)に対する直径D1の値を設定する、つまり、当止面21cの面積を調整して設定することにより行われる。このため、安全弁20を電氣的接続状態から電氣的遮断状態へ変換動作させる圧力を確実に上記第1設定内圧値に基づいて設定することができ、その設定も第1実施形態等における摺動摩擦抵抗力を調整する場合と比べて容易に行うことができる。

【0078】また、その変換動作圧力を変更する場合においても、上記小径孔部21bの直径D1の値を変更することにより、同じ安全弁20を用いたままで容易かつ確実に行うことができる。すなわち、上記直径D1の変更により当止面21cの面積が変化し、この当止面21cの面積変化に伴い小径孔部21bに圧入する場合の安全弁20の弾性変形する部分の体積が変化して、上記小径孔部21b内への圧入抵抗力を変更させることができる。

【0079】しかも、上記安全弁20の封口蓋6に対する組み付け作業において、その安全弁20の大径部20bを大径孔部21aに内嵌させていき第1段差面20dが当止面21cに当接した段階で上記内嵌を止めれば、それだけで上記安全弁20は密閉電池の通常使用状態における所定位置に確実に位置付けることができ、従って、安全弁20の貫通孔21に対する位置決めを容易に行うことができる。

【0080】<第5実施形態>図9は、本発明の第5実施形態に係る密閉電池の安全装置を示す。この第5実施形態は、第4実施形態の構造を基本としつつ、安全弁30及び貫通孔31にそれぞれテーパ状の部分の設けて安全弁30の脱落防止を図るとともに、その安全弁30を封口蓋6と一体に形成して組み付け作業の省力化を図るようにしたものである。

【0081】上記貫通孔31は、図10に詳細を示すように、上記第4実施形態と同様に、下半部の大径孔部31aと、上半部の小径孔部31bと、これら大径孔部31aと小径孔部31bとの間に下方に面するよう形成されたドーナツリング状の当止面31cとを有している。そして、上記小径孔部31bの内周面は上方に向けて内径が徐々に拡大するよう上向きのテーパ孔面により構成され、これにより、上記小径孔部31bはすり鉢状に形成されている。

【0082】また、上記安全弁30は、上端側の小径部30aと、この小径部30aの下側に形成され上記大径孔部31aの内径とほぼ同じ外径の大径部30bと、この大径部30bの下端側から外周方向に突出した鍔部30cとを上記第4実施形態と同様に有するものであり、上記小径部30aと大径部30bとの間には上記当止面31cに当接する第1段差面30dが、上記大径部30bと鍔部30cとの間には封口蓋6の下面に当接する第2段差面30eがそれぞれ形成されている。そして、上記小径部30aは、下半部の外周面が上記貫通孔31の小径孔部31bのテーパ孔面に当接するテーパ周面とされ、上半部が上記小径孔部31bの上端開口から上端面30fが導通板8の下面8aに近接する位置まで上方に突出されている(図9参照)。

【0083】このような安全弁30は、上記第4実施形態と同様の材料を用いて封口蓋6と一体に形成される。例えばEPDM等のゴム材料を用い場合には、封口蓋6と一体に加硫成形することにより安全弁30と封口蓋6とを一体形成する。この場合、上記安全弁30の成形型の中に上記封口蓋6をインサート材として配設し、これに未加硫ゴム材料を注入して加硫すればよい。また、この他に上記安全弁30を上記ゴム材料等の弾性材料を用いて上記封口蓋6とは別に製造し、その製造した安全弁30を上記貫通孔31内に嵌め込むことにより、導通板8等の他の部材との組み付け前に予め安全弁30と封口蓋6とを一体に形成しておくようにしてもよい。

【0084】この第5実施形態の場合、まず、安全弁30が封口蓋6と一体に形成されているため、安全弁30と導通板8との位置関係を考慮することなく封口蓋6を絶縁リング7を介して組み付けるだけで封口体2の組み付け作業を行うことができ、これにより、その組み付け作業の容易化を図ることができる。特に、上記安全弁30を封口蓋6との一体加硫成形により形成する場合には、上記の組み付け作業の大幅な省力化を図ることができる上、安全弁30を封口蓋6もしくは導通板8に対して確実に所定の位置に配設することができる。

【0085】次に、上記安全弁30が、その小径部30aのテーパ周面が貫通孔31の上向きテーパ孔面に当接した状態に嵌合されているため、その安全弁30の下方への脱落を確実に防止した状態で安全弁30を封口蓋6に対し保持させることができる。そのうえ、基端側鍔部30cの第2段差面30eが封口蓋6の下面6aに当接されて、この第2段差面30cと上記テーパ周面とで封口蓋6を上下方向から挟んだ状態にされているため、上記封口蓋6に対する安全弁30の保持をより確実に行うことができ、製品の製造後、出荷されて使用期間中に至るまでの間の密閉電池に対する信頼性をより一層向上させることができる。

【0086】なお、この第5実施形態の場合における電池缶1の内圧上昇に伴う安全弁30の安全動作は、第4

実施形態と異なり、第2段差面30eが通常使用状態において封口蓋6の下面6aと当接しているため、以下のように行われる。すなわち、第1設定内圧値を超える内圧を受けると、安全弁30の第1段差面30dから大径部30bにかけての部分が弾性変形して貫通孔31の小径孔部31b内に圧入し、同時に、第2段差面30eから鏝部30cにかけての部分が弾性変形して大径孔部31a内に圧入し、これにより、安全弁30が上方に相対移動して突起8bを介した導通板8と封口蓋6との電気接続状態が遮断状態に切換えられる。そして、第2設定内圧値までの内圧に対し、上記大径孔部31a内に圧入した上記第2段差面30eから鏝部30cにかけての部分の小径孔部31bに対する圧入抵抗力によって抵抗し、安全弁30を上記の電気的遮断状態に維持する。さらに電池缶1の内圧が上昇してその内圧が設定内圧値を超えると、上記の第2段差面30eから鏝部30cにかけての部分が小径孔部31b内に圧入して上方に飛び抜ける結果、その押圧力を受けて脆弱部8cが破断して上記内圧を外部に解放する。

【0087】<第6実施形態>図11は、本発明の第6実施形態に係る密閉電池の安全装置を示す。この第6実施形態は、第4実施形態の構造を基本としつつ、第5実施形態におけるテーパ孔面とテーパ周面とによる安全弁30の脱落防止を、他の手段により図るようにしたものである。

【0088】この第6実施形態における封口蓋6'には、図12に詳細を示すように、第4実施形態と同様の大径孔部21aと、小径孔部21bと、両孔部21a、21b間の当止面21cとを有する貫通孔21が形成されており、この貫通孔21に対し安全弁40が内嵌されている。

【0089】上記安全弁40は、上記小径孔部21bの内径とほぼ同じ外径を有しその小径孔部21bの内周面に内嵌された小径部40aと、この小径部40aの上側で上記小径孔部21bより大きい外径を有する先端側鏝部としての先端側鏝部40bと、上記小径部40aの下側で上記大径孔部21aの内径とほぼ同じ外径を有しその大径孔部21aの内周面に内嵌された大径部40cと、この大径部40cの下側で上記大径孔部21aより大きい外径を有する基端側鏝部としての下端側鏝部40dとを有している。そして、上記安全弁40は、上記小径部40aと大径部40cとの間の第1段差面40eが上記貫通孔21の当止面21cに、上記小径部40aと先端側鏝部40bとの間の第2段差面40fが封口蓋6'の上面6cに、及び、上記大径部40cと下端側鏝部40dとの間の第3段差面40gが上記封口蓋6'の下面6aにそれぞれ当接するように形成されている。そして、上記安全弁40は、上記第5実施形態と同様の方法により、予め封口蓋6'と一体に形成されている。

【0090】なお、本第6実施形態において図示する封

口蓋6'には、上記安全弁40を囲む位置から上方に突出する突起6b'が形成されており、この突起6b'が導通板8'の下面8aに当接されて封口蓋6'と導通板8'とを電気的接続状態にするようになっている。この際、絶縁リング7の厚みが上記突起6b'の突出高さよりも若干小さく設定されており、この絶縁リング7を介して封口蓋6'と導通板8'等とを組み付けることにより上記導通板8'の中央部が脆弱部8c位置近傍で曲げられて上方にやや突出するように形成される。これにより、上記導通板8'の曲げ復元力が作用した状態で導通板8'と突起6b'とが密着され、その導通板8'と突起6b'との電気的接続をより一層確実かつ良好なものとするとともに、安全弁40の貫通孔21からの上方への飛び抜けにより上記脆弱部8cがより破断し易いようになっている。

【0091】また、脆弱部8cとしては前述の実施形態等の如く導通板8の片面にのみ切り溝を形成した形態にしてもよいが、片面のみの切り溝形態にすると封口体2を例えば自動組み立てする場合には導通板8の表裏判別が面倒になる場合があり、これを考慮すると、図11に示すように、脆弱部8cとして導通板8'の両面から切り溝を入れた形態を採用するのが好ましい。

【0092】この第6実施形態の場合、安全弁40の第1段差面40e及び第3段差面40gが封口蓋6'の下向き面である当止面40e及び下面6aに当接する一方、第2段差面40fが上記封口蓋6'の上向き面である上面6cに当接した状態にされているため、安全弁40が封口蓋6'を上下から挟み込んだ状態で貫通孔21に挿通されていることになる。従って、第5実施形態と同様に、安全弁40の下方への脱落を確実に防止した状態で安全弁40を封口蓋6'に対し保持させることができ、製品の製造後、出荷されて使用期間中に至るまでの間の密閉電池に対する信頼性をより一層向上させることができる。

【0093】なお、上記安全弁40を封口蓋6'と予め一体に形成することによる作用・効果、及び、その安全弁40の安全動作による作用・効果を、第5実施形態と同様に得ることができる。

【0094】<他の実施形態>なお、本発明は上記第1～第6実施形態に限定されるものではなく、その他種々の実施形態を包含するものである。すなわち、上記第1～第3実施形態では安全弁10、13の上端面と導通板8の下面とを互いに接合させているが、これに限らず、非接着状態で当接させてもよく、さらには第4～第6実施形態のごとく近接させてもよい。これらの場合であっても、封口蓋6、絶縁リング7、導通板8、及び、正極端子9をかしめ部1cでかしめることにより互いに重合されて上記の当接した状態が保持される。

【0095】上記第1～第6実施形態では、可撓性導通板8、8'として円板状のものを示したが、これに限ら

ず、例えば帯板状のものとしてもよい。この場合、脆弱部8cやガス抜き孔15bを設けなくても、安全弁10等が貫通孔11等を飛び抜けることによりガス抜きを行うことができる。

【0096】上記第1～第5実施形態では、導通板8と封口蓋6との電氣的接続を導通板8側に設けた突起8bを介して行っているが、これに限らず、上記突起8bの代わりに、例えば図11もしくは図14に示すように、封口蓋6'側に安全弁10、40を囲むように設けた円環状等の突起6b'を導通板8、8'の下面に当接させることにより電氣的接続を行うようにしてもよい。

【0097】上記第1～第6実施形態では突起8b、6b'として円環状のものを示したが、これに限らず、その形状はいずれのものでもよい。

【0098】上記第1もしくは第2実施形態では安全弁10の凸部10cとして周方向に連続するものを示したが、これに限らず、上記凸部は周方向の一部に形成するものでもよい。

【0099】上記第1～第5実施形態では平面状の導通板8を用いているが、これに限らず、例えば第6実施形態の如き中央部がやや上方に凸となるように屈曲させた導通板8'、または、図15に示すように、中央部がやや下方に凹となるように屈曲させた導通板8''を用いてもよい。図15の場合には、導通板8''をやや厚めの絶縁リング7'を介して封口蓋6に重合させればよい。さらに、この場合、その導通板8''にガス抜き孔15b、15bを設けることにより、脆弱部8cの形成を省略することもできる。

【0100】また、第1～第6実施形態における封口体2の構成を図16に示すように、封口蓋6、絶縁リング7''、導通板8、及び、PTCサーミスタ12を互いに重ねた状態で、正極端子9'の外周縁を延ばした外延部9bで包み込み挟持させるようにして、封口体2の一体化を図るようにしてもよい。これにより、電池缶1の開口部1aへの組み付け作業を上記一体化した封口体2を組み付けるだけで済み、組み付け作業の能率化を図ることができる。しかも、予め封口体2を別に組み付けることができるため、安全弁10の作動状態を上記封口体2の電池缶1への組み付け前にテストすることができる。また、上記の図16ではPTCサーミスタ12を有するものについて図示したが、これに限らず、上記のPTCサーミスタ12を設けない各実施形態の構造のものを一体化するようにしてもよい。さらに、この一体化を封口蓋6の外周部を延長して正極端子9を挟持するようにしてもよく、さらに別体の挟持部材により封口蓋6から正極端子9までの外周縁部を挟持するようにしてもよい。

【0101】さらに、上記第3実施形態では、閉止体として弾性膜14を示したが、これに限らず、例えばガス抜き孔15aに密に内嵌させた柱状の閉止弁により閉止体を構成してもよい。この場合、上記閉止弁をEPDM

等の弾性材料により形成し、この閉止弁の外周面とガス抜き孔15aの内周面との摺動摩擦抵抗力として、第2設定内圧値までの内圧に対しては閉止弁を相対移動しないように保持してガス抜き孔15aを密閉する一方、第2設定内圧値を超える内圧を受けて上記ガス抜き孔15aを上方に飛び抜けてガス抜き孔15aを開放状態にすることになるような値に設定すればよい。つまり、上記の摺動摩擦抵抗力の設定により閉止弁の開放強度の設定を行うものである。なお、上記の閉止弁の下端に第1実施形態の安全弁10における凸部10cの如き凸部を形成することにより、上記の摺動摩擦抵抗力の設定を容易に行い得る。

【0102】

【実施例】上記の第4実施形態の密閉電池において、封口蓋6の板厚T（図8参照）を0.6mmに、貫通孔21の大径孔部21a及び小径孔部21bの上下方向の高さT1、T2をそれぞれ0.3mmに、その大径孔部21aの直径D2を3mmにそれぞれ統一し、小径孔部21bの直径D1のみ3mmより小さい範囲で種々に変化させた4つの封口蓋6を第1グループとして製作した。また、上記条件設定の内、大径孔部21aの直径D2を4mmにし、小径孔部21bの直径D1を4mmより小さい範囲で種々に変化させた6つの封口蓋6を第2グループとして製作した。

【0103】一方、安全弁20として、上記直径D2の大径部20bを有する同一形状のものをEPDMによりHS硬度が60度になるように所要数製作した。

【0104】そして、上記第1及び第2のグループの各封口蓋6の貫通孔21に対し上記安全弁20を第1段差面20dが当止面21cに当接するように内嵌させ、この状態の安全弁20に対し圧力Pを作用させて圧入試験を行った。この圧力Pを徐々に増大させていき、上記安全弁20の大径部20bが小径孔部21bに圧入して第2段差面20eが封口蓋6の下面6aに当接するまで上方に相対変位する時の圧力値を求めた。この圧力値を圧入時の作動圧力とし、得た作動圧力を図13に示す。この図13は、上記作動圧力を縦軸に、上記小径孔部21bの直径D1を横軸にとり、大径孔部21aの直径D2が3mmの第1グループの場合（同図に四角形のマーク参照）と、4mmの第2グループの場合（同図に三角形のマーク参照）とに分けて示したものである。

【0105】この結果によれば、電氣的接続状態を遮断状態に切換るための第1設定内圧値を例えば15kgf/cm²とするには、大径孔部21aの直径D2を3mmにした場合（第1グループの場合）には小径孔部21bの直径D1としてほぼ2.6mmを設定すればよく、また、上記直径D2を4mmにした場合（第2グループの場合）には小径孔部21bの直径D1としてほぼ3.4mmを設定すればよいことが分かる。また、上記第1設定内圧値を20kgf/cm²とするには、大径孔部21

aの直径D2を3mmにした場合には小径孔部2bの直径D1としてほぼ2.5mmを設定すればよく、また、上記直径D2を4mmにした場合には小径孔部2bの直径D1としてほぼ3.2mmを設定すればよいことが分かる。

【0106】従って、図13のごとき圧入抵抗特性を測定しておくことにより、安全弁20を所定の内圧で確実に作動させることができ、その作動を行なわせるための貫通孔21の寸法D1、D2の設定を容易に行うことができる。しかも、その電氣的遮断状態への切換作動圧力を変更する場合にも、小径部21bの直径D1を図9に基づいて即座に決定することができ、その直径D1の変更調整により安全弁20を確実に変更後の切換作動圧力で作動させることができる。

【0107】

【発明の効果】以上説明したように、請求項1記載の発明における密閉電池の安全装置によれば、封口体を、剛性を有する封口蓋と、この封口蓋の貫通孔に密に嵌合させて封止した安全弁と、上記封口蓋の外方側に絶縁状態で重ね上記安全弁の周囲で封口蓋と電氣的に接続させた可撓性導通板とから構成しているため、過充電等により電池缶内部の内圧が設定内圧値を超えると、その内圧を安全弁の基端面に直接的に作用させてその安全弁の相対移動により導通板の封口蓋との電氣的接続部位にその電氣的接続を離す側に集中的に押圧力を作用させることができ、上記内圧上昇の発生に際し応答性よく電氣的遮断状態に切換えることができる。これにより、電池の使用者に異常発生を素早く認知させることができる。

【0108】しかも、電池缶内部の内圧に応じた電氣的な接続と遮断との切換を、安全弁を導通板側に相対変位させようとする内圧に対する安全弁と貫通孔との間の抵抗力の設定により行うことができ、従来の密閉電池の如く溶接部の破断強度の設定により行う場合と比べ、その設定及び変更を容易かつ確実に行うことができる上、組付け時の工程管理も容易となって製品に対する信頼性の向上を図り得る。

【0109】請求項2記載の発明によれば、上記請求項1記載の発明による効果に加えて、電池缶内圧に対する安全弁と貫通孔との間の抵抗力を、安全弁と貫通孔との両周面間の摺動摩擦抵抗力によって発揮させるようにしているため、電池缶内圧に応じた上記抵抗力の調整もしくは変更を容易に行うことができる。そして、安全弁を設定内圧値まではその相対変位を阻止した状態に保持し、かつ、設定内圧値を超える電池缶内圧に対し安全弁の導通板側への相対変位を許容するという抵抗力を上記摺動摩擦抵抗力により確実に発揮させることができる。

請求項3記載の発明によれば、上記請求項2記載の発明による効果に加えて、安全弁を、柱状の本体部と、この本体部の基端側から僅かに突出する凸部とから構成しているため、電池缶の内圧の設定内圧値を超えた場合の

上記本体部の相対移動による電流の遮断と、内圧がさらに上昇した場合の上記凸部の貫通孔からの飛び抜けによる電池缶内部の内圧開放とで2段階の安全作動を行なわせることができる。従って、内圧異常時には、まず電流が遮断されて電池の使用者に異常発生を認知せしめ、それでも使用者が放置する場合であっても、内部のガス放出により内圧の異常上昇を回避することができる。

【0110】請求項4記載の発明によれば、上記請求項2記載の発明による効果に加えて、安全弁を柱状の本体部と、その本体部の基端側で拡開したフランジ状の受圧部とから構成する一方、封口蓋に電池缶内部と外部とを互いに連通させるガス抜き孔と、このガス抜き孔の封口蓋内面側で閉止し所定の開放強度を有する閉止体とを設けているため、設定内圧値を超える内圧上昇時に上記本体部の相対移動による電流遮断と、閉止体の開放による内圧開放との2段階の安全作動を行なわせることができる。しかも、上記電流遮断の際、安全弁の受圧部が本体部よりも拡開されて受圧面である安全弁の基端面を請求項4記載の発明の場合と比べ拡大することができるため、内圧の上昇に対し、より応答性よく導通板を押し上げて電流の遮断を行うことができる。

【0111】請求項5記載の発明によれば、上記請求項1記載の発明による効果に加えて、電池缶内圧に対する安全弁と貫通孔との間の抵抗力を、貫通孔の小径孔部に対する安全弁の段差面から大径部にかけての部分の圧入抵抗力によって発揮されるようにしているため、電池缶内圧に応じた上記抵抗力の調整もしくは変更を請求項2記載の発明による場合よりもさらに容易にかつ確実に行うことができる。そして、安全弁を設定内圧値まではその相対変位を阻止した状態に保持し、かつ、設定内圧値を超える電池缶内圧に対し安全弁の導通板側への相対変位を許容するという抵抗力を、安全弁の圧入抵抗力によって確実に発揮させることができる。

【0112】しかも、安全弁を封口蓋に対し組み付ける際に、その安全弁の段差面を貫通孔の当止面に当接させるだけで、安全弁を封口蓋及び導通板に対し確実に所定の内外方向位置に組み付けることができ、その安全弁の封口蓋に対する位置決めを容易に行うことができる。これにより、密閉電池の製造の容易化を図ることができる上、その安全弁を確実に所定位置に位置付けて封口蓋との電氣的接続状態を確実に維持させることができる。

【0113】請求項6記載の発明によれば、上記請求項2または請求項5記載の発明による効果に加えて、安全弁を弾性材料により形成しているため、安全弁と封口蓋の貫通孔との間の摺動摩擦抵抗力または圧入抵抗力の設定をより一層容易に行うことができ、内圧上昇に伴う電氣的接続状態から遮断状態への切換をより確実に行うことができる。

【0114】請求項7記載の発明によれば、上記請求項5記載の発明による効果に加えて、電池缶の内圧の上昇

に伴い、まず、安全弁の段差面から大径部にかけての部分が貫通孔の小径孔部内に圧入し安全弁の先端面が導通板を押し上げて電気的遮断状態に切換えることができ、次に、さらに上昇した内圧を受けて基端側鍔部が大径孔部内に圧入し安全弁全体が貫通孔を導通板側に突き抜けて電池缶の内圧を解放差せることができる。従って、電池缶の内圧の上昇に従って、第1段階として電流が遮断され、第2段階としてその内圧が解放されるという、2段階の安全動作を行なわせることができる。

【0115】請求項8記載の発明によれば、上記請求項5記載の発明による効果に加えて、密閉電池が通常の使用状態にある時には、安全弁のテーパ面が上記貫通孔のテーパ孔面に当接し、これにより、その安全弁が貫通孔内を電池缶の内方側に脱落することを確実に防止することができる。このため、上記安全弁の段差面を貫通孔の当止面に当接した状態に確実に維持することができ、電池缶の内圧が上昇した場合に確実に安全動作を行う態勢に維持することができる。従って、製品の出荷段階から使用段階に至る間、安全弁を封口蓋に対し確実に保持させることができ、電池缶の内圧上昇時の安全動作を確実に行なわせることができる。

【0116】請求項9記載の発明によれば、上記請求項5または請求項8記載の発明による効果に加えて、安全弁を予め封口蓋と一体に形成するようにしているため、封口蓋に対する安全弁の組み付け作業の省力化を図ることができ、密閉電池の製造の容易化、合理化を図ることができる。併せて、予め安全弁を封口蓋及び導通板に対し所定の相対位置に確実に位置付けることができ、通常使用時における封口蓋と導通板との電気的接続状態の維持と、内圧上昇時の安全弁の安全動作とを確実に行なわせることができる。

【0117】請求項10記載の発明によれば、上記請求項1記載の発明による効果に加えて、安全弁の先端面と、可撓性導通板の略中央部内面とを接合するようにしているため、可撓性導通板自体を安全弁と貫通孔との間の摺動摩擦抵抗によって剛性を有する封口蓋により保持された状態にすることができる。このため、上記導通板と封口蓋との当接、つまり、電気的接続状態を安定的に保持することができ、従って、導通板と封口蓋とが溶接されていなくても、上記の電気的接続状態をより確実に保持することができ、通常使用時の製品に対する信頼性についてより一層の向上を図ることができる。

【0118】請求項11記載の発明によれば、上記請求項1記載の発明による効果に加えて、導通板に安全弁の飛び抜けにより押されて曲げ変形した際に破断する脆弱部を設けているため、電池缶内部からのガス抜きをより効果的に行うことができる。

【0119】また、請求項12記載の発明によれば、上記請求項1記載の発明による効果に加えて、リード線と封口体との間の電流経路の途中に、設定温度域で抵抗値

が急増するPTCサーミスタを介装するようにしているため、電池缶の内部温度が上記設定温度より低い適正温度にある間は上記電流経路を通して適正に電流を流すことができる一方、上記内部温度が短絡等の発生に起因して設定温度域以上に上昇した場合、その上昇に対応して上記PTCサーミスタの抵抗値の急増により上記電流経路を通しての通電を実質的に遮断することができ、より安全性を向上させることができる。

【0120】さらに、請求項13記載の発明によれば、上記請求項1記載の発明による効果に加えて、封口体を、安全弁が嵌合された状態の封口蓋と、絶縁部材と、上記封口蓋と電気的に接続された状態の導通板と、正極端子とのそれぞれの外縁部が互いに重合された状態に予め結合一体化されたものとして構成するようにしているため、電池缶の開口部への組み付け作業の容易化を図ることができる上、その組み付け前に上記安全弁の作動を確認するためのテストを行うことができる。

【図面の簡単な説明】

【図1】本発明の第1実施形態を示す上部断面図である。

【図2】封口体の一部を省略した分解斜視図である。

【図3】電気的遮断状態を示す図1対応図である。

【図4】内圧開放状態を示す図1対応図である。

【図5】第2実施形態を示す図1相当図である。

【図6】第3実施形態を示す図1相当図である。

【図7】第4実施形態を示す図1相当図である。

【図8】図7の部分拡大図である。

【図9】第5実施形態を示す図1相当図である。

【図10】図9の部分拡大図である。

【図11】第6実施形態を示す図1相当図である。

【図12】図11の部分拡大図である。

【図13】図8の安全弁についての直径D1、D2と作動圧力との関係図である。

【図14】他の実施形態を示す図1相当図である。

【図15】図14とは異なる他の実施形態を示す図1相当図である。

【図16】図14及び図15とは異なる他の実施形態を示す図1相当図である。

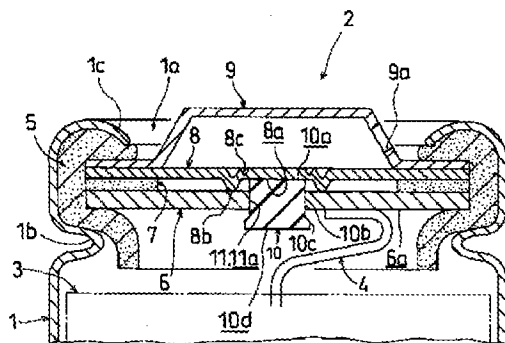
【符号の説明】

1	電池缶
2	封口体
3	発電要素
4	リード線
6, 6'	封口蓋
7, 7', 7''	絶縁リング(絶縁部材)
8, 8', 8''	導通板
9, 9'	正極端子
10, 13, 20, 30, 40	安全弁
10a, 13c, 20f, 30f	上端面(先端

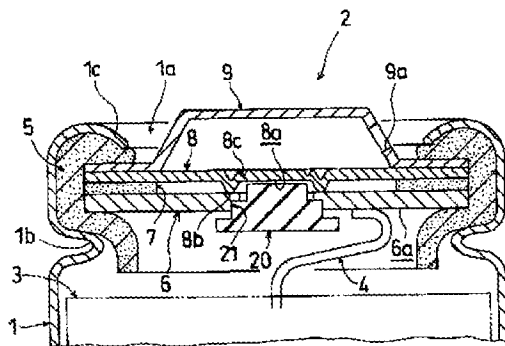
面)	
10d, 13d	下端面(基端
面)	
10b, 13a	本体部
10c	凸部
13b	受圧部
11, 21, 31	貫通孔
12	PTCサーミス
タ	
14	弾性膜(閉止
体)	
15a	ガス抜き孔
20a, 30a, 40a	小径部

20 b, 30 b, 40 c	大径部
20 d, 30 d, 40 e	第1 段差面 (段差面)
20 c, 30 c, 40 d	下端側銹部 (基端側銹部)
21 a, 31 a	大径孔部
21 b, 31 b	小径孔部
21 c, 31 c	当止面
30 a	小径部 (テーパ周面)
31 b	小径孔部 (テーパ孔面)

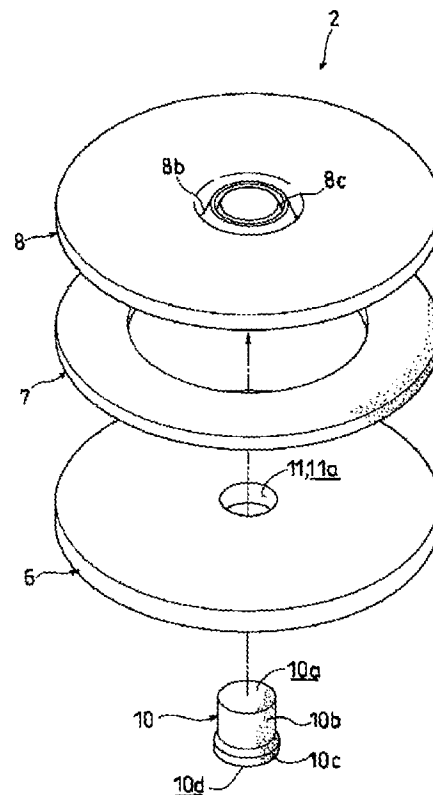
【例 1】



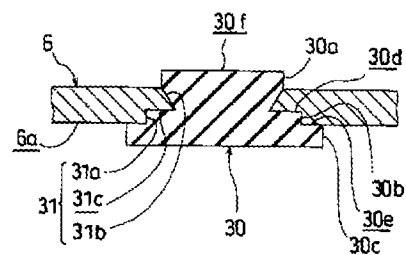
【例7】



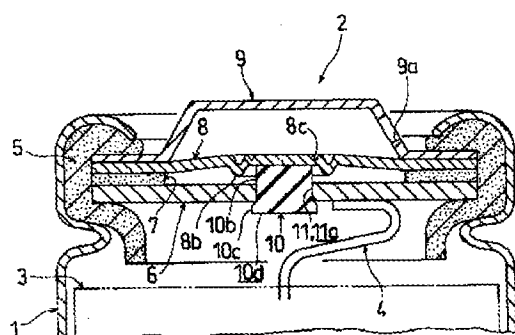
【図2】



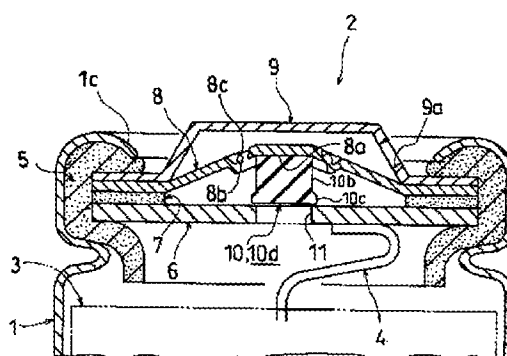
【例 10】



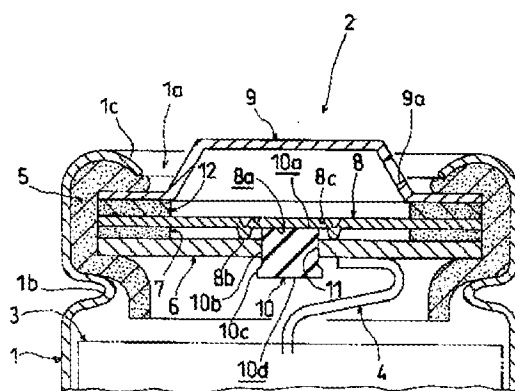
【例3】



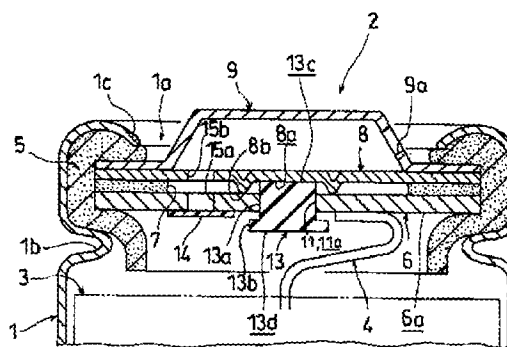
【例4】



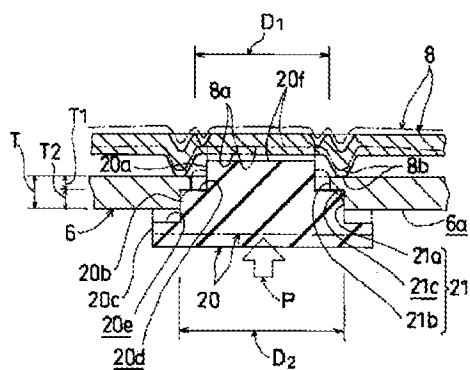
【例5】



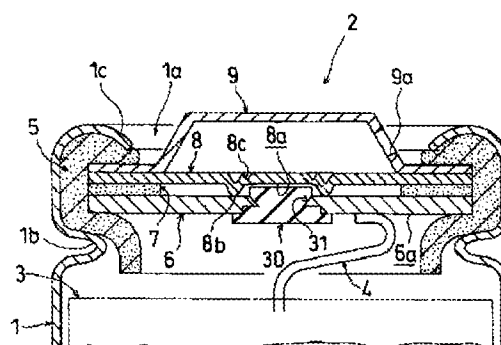
【图6】



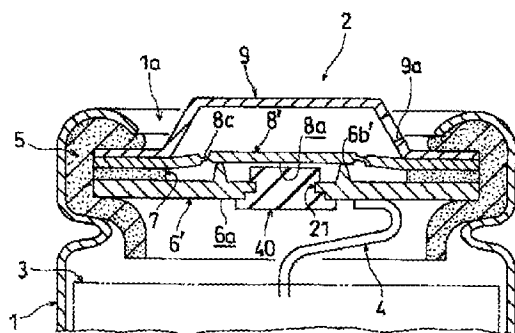
【例8】



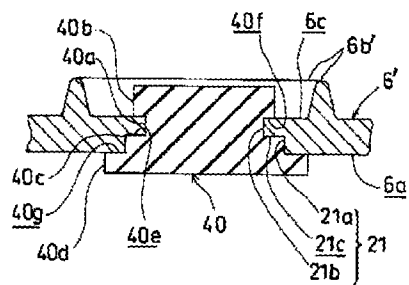
【图9】



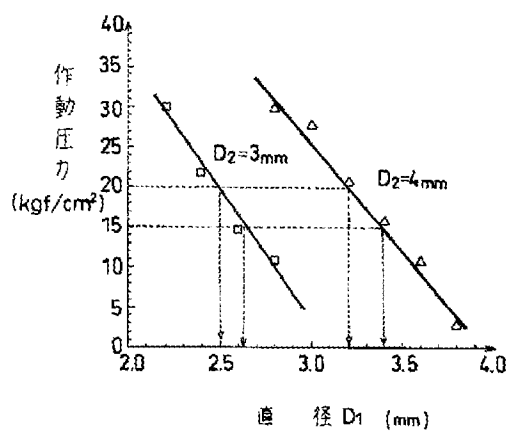
【図11】



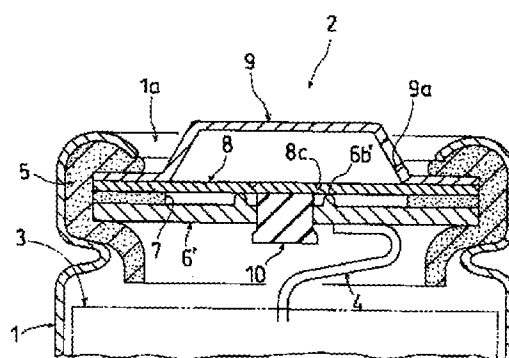
【図12】



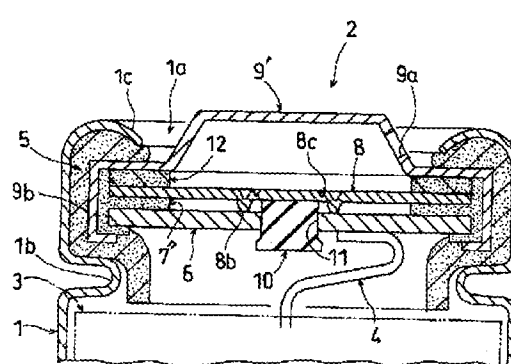
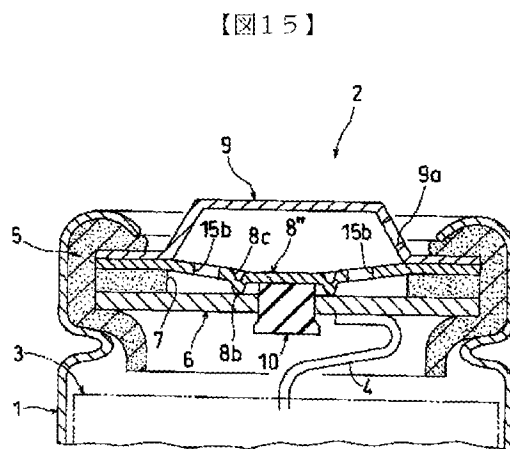
【図13】



【図14】



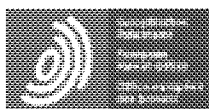
【図15】



フロントページの続き

(72)発明者 柏岡 亨
兵庫県尼崎市東向島西之町8番地 三菱電
線工業株式会社内

(72)発明者 大内 裕
兵庫県尼崎市東向島西之町8番地 三菱電
線工業株式会社内



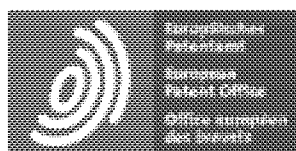
Espacenet

Bibliographic data: KR20150035267 (A) — 2015-04-06

Cap assembly and secondary battery including the same

Inventor(s): 성낙기, ; 류덕현**Applicant(s):** 주식회사 엘지화학**Classification:** - **international:** H01M2/12
- **cooperative:** H01M10/52 (KR); H01M50/147 (KR); H01M50/30 (KR); H01M50/572 (KR); Y02E60/10 (EP)**Application number:** KR20130115616 20130927 Global Dossier**Priority number (s):** KR20130115616 20130927**Also published as:** KR101667966 (B1)**Abstract of KR20150035267 (A)**

The present invention discloses a cap assembly and secondary battery including the same that has an improved structure that allows a quick transformation of safety vent and a prompt emission of pressuring gas to the exterior in the event that the interior pressure of a battery cap increases to cause deformation and rupture of a safety vent. A cap assembly for secondary battery according to the present invention comprises: a top cap that forms anode terminals by being placed on the topmost part in a protruding position and a safety vent, placed at the lower part of the top cap, that is transformed when the interior pressure of the battery can increases and consists of a first notch part that has grooves to facilitate rupture in the event of pressure increase; grooves are formed deeper within a part of the first notch is part than other parts.



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DESCRIPTION KR20150035267A

Cap assembly and secondary battery including the same {Cap assembly and secondary battery including the same}

[0001]

The present invention relates to a technology for manufacturing a secondary battery, and more particularly, to a cap assembly having an improved structure so that internal gas can be easily discharged to the outside when the internal pressure of the battery can is increased, and a secondary battery including the same. .

[0002]

In general, a secondary battery refers to a battery that can be charged and discharged unlike a primary battery that cannot be charged, and is widely used in electronic devices such as mobile phones, notebook computers, camcorders, or electric vehicles. In particular, lithium secondary batteries have an operating voltage of about 3.6V, and have a capacity of about three times that of nickel-cadmium batteries or nickel-hydrogen batteries, which are often used as power sources for electronic equipment, and have a high energy density per unit weight. The degree is on the rise rapidly.

[0003]

These lithium secondary batteries mainly use lithium-based oxides and carbon materials as a positive electrode active material and a negative electrode active material, respectively.

A lithium secondary battery includes an electrode assembly in which a positive electrode plate and a negative electrode plate to which the positive electrode active material and the negative electrode active material are applied, respectively, are disposed with a separator therebetween, and an exterior material for sealing the electrode assembly together with an electrolyte.

[0004]

Meanwhile, the lithium secondary battery may be classified into a can-type secondary battery in which an electrode assembly is embedded in a metal can and a pouch-type secondary battery in which the electrode assembly is embedded in a pouch of an aluminum laminate sheet according to the shape of the battery case.

In addition, can-type secondary batteries can be further classified into cylindrical batteries and prismatic batteries according to the shape of the metal can. The exterior material of such a rectangular or cylindrical secondary battery includes a case having an open end, that is, a battery can and a cap assembly that is sealedly coupled to the open end of the battery can.

[0005]

1 is a cross-sectional view of a cap assembly portion of a cylindrical secondary battery according to the prior art.

[0006]

Referring to FIG. 1, in general, a cylindrical secondary battery is a cylindrical battery can 20, a jelly-roll electrode assembly 30 accommodated in the battery can 20, and is coupled to the upper portion of the battery can 20. A cap assembly 10, a beading portion 40 provided at the front end of the battery can 20 to mount the cap assembly 10, and a clamping portion 50 for sealing the battery are provided.

[0007]

The electrode assembly 30 has a structure wound in a jelly-roll form with a separator interposed between the positive electrode and the negative electrode, and the positive electrode lead 31 is attached to the positive electrode and connected to the cap assembly 10, and the negative electrode has a negative electrode. A lead (not shown) is attached and is connected to the lower end of the battery can 20.

[0008]

The cap assembly 10 includes a top cap 11 forming a positive terminal, a safety element 12 that blocks current by increasing resistance when the temperature inside the battery increases, and blocks current and/or gas when the pressure inside the battery increases. The safety vent 13 that exhausts the gas, an insulating member 14 that electrically separates the safety vent 13 from the current blocking member 15 except for a specific part, and the current to which the positive lead 31 connected to the positive electrode is connected. The blocking members 15 have a structure in which they are sequentially stacked.

In addition, the cap assembly 10 is mounted on the beading portion 40 of the battery can 20 while being mounted on the gasket 16.

Accordingly, under normal operating conditions, the anode of the electrode assembly 30 is connected to the top cap 11 via the anode lead 31, the current blocking member 15, the safety vent 13, and the safety element 12 to conduct electricity. To achieve.

[0009]

However, when gas is generated from the electrode assembly 30 due to a cause such as overcharging and the internal pressure increases, the safety vent 13 is deformed upward while reversing its shape, and at this time, the safety vent 13 is separated from the blocking member 15 to block the current.

Accordingly, charging and discharging may no longer proceed, thereby ensuring the safety of the secondary battery.

Moreover, when the internal pressure of the secondary battery increases above a certain value, the safety vent 13 is ruptured and the pressurized gas is exhausted through the gas hole 11a of the top cap via such a rupture site, thereby preventing the battery from exploding. have.

The safety vent 13 may be provided with a notch 13a having a groove as shown in FIG. 1 in order to facilitate deformation and rupture when the internal pressure of the battery can 20 is increased.

[0010]

However, the safety vent 13 of the cap assembly for a secondary battery according to the prior art has a notch 13a, and the notch 13a has a uniform depth.

According to the prior art, when the internal pressure of the battery can 20 increases, the pressurized gas uniformly applies pressure to the safety vent 13, so that the safety vent 13 cannot be rapidly deformed. In addition, when the internal pressure is further increased and the internal pressure of the secondary battery increases above a certain value, the safety vent 13 is ruptured, and it is possible to predict which point of the notch 13a of the safety vent 13 will rupture. none.

[0011]

2 is a view visually showing a state in which pressurized gas flows in a state in which a safety vent of a cap assembly for a secondary battery according to the prior art is broken.

Referring to FIG. 2, the broken portion of the safety vent shown by F in the drawing is located on the right side of the safety vent 13. Accordingly, the pressurized gas passes through the broken portion F of the safety vent and flows to the right side of the drawing, then changes the direction and flows to the left, and is exhausted to the outside through the gas hole 11a provided in the top cap. In this case, the pressurized gas discharged through the broken portion F of the safety vent may not be properly exhausted through the gas hole 11a of the top cap, and thus a problem may occur in that exhaust is not smoothly performed.

[0012]

Accordingly, the present invention was devised to solve the above problems, and when the safety vent is deformed and ruptured due to an increase in the internal pressure of the battery can, the safety vent is rapidly deformed and pressurized gas is rapidly exhausted to the outside. An object of the present invention is to provide a cap assembly for a secondary battery and a secondary battery including the same.

[0013]

Other objects and advantages of the present invention can be understood by the following description, and will be more clearly understood by examples of the present invention.

In addition, it will be easily understood that the objects and advantages of the present invention can be realized by the means shown in the claims and combinations thereof.

[0014]

The cap assembly for a secondary battery according to the present invention for achieving the above object includes: a top cap disposed in a protruding shape at an uppermost portion to form a positive terminal; And a safety vent that is located under the top cap and changes its shape when the internal pressure of the battery can is increased, wherein the safety vent includes a first notch portion having a groove formed to facilitate fracture when the internal pressure of the battery can is increased, , The first notch portion is characterized in that the groove is formed deeper than the other portion.

[0015]

Preferably, the first notch is characterized in that it is formed in a ring shape.

[0016]

More preferably, the safety vent is characterized in that it further comprises a second notch portion formed in a ring shape outside or inside the first notch portion.

[0017]

Also preferably, the first notch portion is characterized in that it is formed with a plurality of grooves spaced a predetermined distance from each other.

[0018]

Also preferably, the first notch portion is characterized in that one groove is continuously formed.

[0019]

Also preferably, the first notch is characterized in that it is provided in the bent portion of the safety vent.

[0020]

In addition, preferably, the first notch portion is characterized in that the depth of the deepest groove is 2 to 10 times deeper than the depth of the shallowest groove.

[0021]

Also preferably, the first notch portion is characterized in that it is formed so that the length of the deepest grooved portion is 1% to 90%.

[0022]

In addition, preferably, it is disposed between the top cap and the safety vent, characterized in that it further comprises a safety element that blocks the current when the temperature rises.

[0023]

In addition, preferably, an upper portion is connected to the lower end of the safety vent, the lower portion is a current blocking member that can be connected to the electrode assembly; And an insulating member interposed between the safety vent and the current blocking member to electrically insulate the current blocking member from the safety vent except for a portion.

[0024]

In addition, preferably, it is characterized in that it further comprises a gasket surrounding the rims of the top cap and the safety vent.

[0025]

In addition, preferably, the cap assembly for a secondary battery described above is characterized in that it is coupled to a cylindrical secondary battery can.

[0026]

The secondary battery according to the present invention for achieving the above object includes the cap assembly for a secondary battery described above.

[0027]

According to the present invention, the notch portion is configured so that the depth of the groove of the part is different from the depth of the groove of the other part, so that when the internal pressure of the battery can increases, the safety vent is rapidly deformed and the pressurized gas is quickly exhausted to the outside. Can be.

[0028]

Accordingly, even if gas is generated in the electrode assembly due to overcharging and the internal pressure is increased, explosion of the secondary battery can be prevented, and thus a secondary battery with improved safety can be implemented.

[0029]

In addition to the present invention may have a variety of other effects, these other effects of the present invention can be understood by the following description, can be more clearly understood by the embodiments of the present invention.

[0030]

The following drawings attached to the present specification illustrate preferred embodiments of the present invention, and serve to further understand the technical idea of the present invention together with the detailed description of the present invention to be described later, so the present invention is limited to and should not be interpreted.

1 is a cross-sectional view of a cap assembly portion of a cylindrical secondary battery according to the prior art.

2 is a diagram visually showing a state in which pressurized gas flows in a state in which a safety vent of a cap assembly for a secondary battery according to the prior art is broken.

3 is a diagram schematically showing the configuration of a cap assembly portion of a secondary battery according to an embodiment of the present invention.

4 is an enlarged view of portion A of FIG. 3.

5 is a diagram visually showing a state in which pressurized gas flows in a state in which the safety vent of the cap assembly for a secondary battery according to the present invention is broken.

6 is a horizontal cross-sectional view of a cap assembly of a secondary battery according to an embodiment of the present invention.

7 is a horizontal cross-sectional view of a cap assembly of a secondary battery according to another embodiment of the present invention.

[0031]

Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings.

Prior to this, terms or words used in the present specification and claims should not be construed as being limited to their usual or dictionary meanings, and the inventors appropriately explain the concept of terms in order to explain their own invention in the best way. Based on the principle that it can be defined, it should be interpreted as meaning and concept consistent with the technical idea of the present invention.

[0032]

Accordingly, the embodiments described in the present specification and the configurations shown in the drawings are only the most preferred embodiment of the present invention, and do not represent all the technical spirit of the present invention, and thus various alternatives that can be substituted for them at the time of application it should be understood that there may be equivalents and variations.

[0033]

In addition, in describing the present invention, when it is determined that a detailed description of a related known configuration or function may obscure the subject matter of the present invention, a detailed description thereof will be omitted.

[0034]

Since the embodiments of the present invention are provided to more completely describe the present invention to a person skilled in the art, the shapes and sizes of components in the drawings may be exaggerated, omitted, or schematically illustrated for a more clear description.

Therefore, the size or ratio of each component does not entirely reflect the actual size or ratio.

[0035]

3 is a diagram schematically showing the configuration of a cap assembly portion of a secondary battery according to an embodiment of the present invention.

[0036]

Referring to FIG. 3, the secondary battery according to the present invention includes an electrode assembly 300, a battery can 200, and a cap assembly 100.

[0037]

In the electrode assembly 300, a positive electrode plate and a negative electrode plate are disposed with a separator interposed therebetween, and are accommodated in the battery can 200.

At this time, the electrode assembly 300 may be wound and disposed in the form of a jelly roll, and in this case, it is also called a jelly roll.

The electrode plates of the electrode assembly 300 may be formed as a structure in which an active material slurry is applied to a current collector, and the slurry is typically formed by stirring a granular active material, an auxiliary conductor, a binder, and a plasticizer while a solvent is added. I can.

In the direction in which the electrode plates are wound, uncoated portions to which the slurry is not applied may exist at the beginning and end of the current collector, and electrode leads corresponding to respective electrode plates may be attached to the uncoated portions.

In general, in a cylindrical secondary battery, the positive lead is attached to the upper end of the electrode assembly 300 and electrically connected to the cap assembly 100, and the negative lead is attached to the lower end of the electrode assembly 300 to the bottom of the battery can 200. Is connected to

[0038]

Meanwhile, an upper insulating plate may be disposed on the top of the electrode assembly 300.

This upper insulating plate serves to insulate between the electrode assembly 300 and the cap assembly 100.

[0039]

The battery can 200 is made of a lightweight conductive metal material such as aluminum, stainless steel, or an alloy thereof, and may have a cylindrical or prismatic structure having an opening having an open top and a closed bottom facing it. have.

An electrolyte solution is accommodated together with the electrode assembly 300 in the inner space of the battery can 200.

[0040]

The battery can 200 may be formed in a cylindrical shape, but may be formed in various shapes other than a cylindrical shape such as a square.

[0041]

The cap assembly 100 for a secondary battery according to the present invention includes a top cap 110 and a safety vent 130.

The cap assembly 100 is a component of a secondary battery coupled to an upper end of the battery can 200, that is, an open end, and seals the open end of the battery can 200.

The cap assembly 100 may be formed in various shapes such as a circular shape or a square shape according to the shape of the battery can 200.

[0042]

The top cap 110 is disposed on the uppermost portion of the cap assembly 100 to protrude upward to form a positive terminal.

Accordingly, the top cap 110 is electrically connected to an external device, such as a load or a charging device.

In addition, a gas hole 110a through which gas may be discharged may be formed in the top cap 110.

Accordingly, when gas is generated from the electrode assembly 300, gas may be discharged to the outside of the battery can 200 through the gas hole 110a.

The top cap 110 may be formed of, for example, a metal material such as stainless steel or aluminum.

[0043]

The safety vent 130 is positioned under the top cap 110 and is configured to change its shape when the internal pressure of the secondary battery, that is, the internal pressure of the battery can 200 increases to a certain level or more.

For example, the safety vent 130 may be configured to be deformed and burst when the internal pressure of the secondary battery is 12 to 25 kgf/cm².

[0044]

To this end, the safety vent 130 is formed so that the central portion protrudes downward, as shown in the drawing, and a predetermined notch portion may be formed in the vicinity of the central portion.

Therefore, when gas is generated from the inside of the secondary battery, that is, from the side of the electrode assembly 300 and the internal pressure of the battery can 200 increases, the safety vent 130 protrudes upward while its shape is reversed.

At this time, the electrical connection between the safety vent 130 and the electrode assembly 300 is cut off, and when the internal pressure of the secondary battery increases above a predetermined value, the safety vent 130 may rupture around the notch.

Accordingly, the gas filled in the battery can 200 may be discharged to the outside through the ruptured portion of the safety vent 130.

[0045]

In particular, the safety vent 130 according to the present invention may include a first notch portion 131 in which a groove is formed to facilitate fracture when the internal pressure of the battery can 200 is increased.

A portion of the first notch portion 131 may have a deeper groove than another portion.

That is, in the present invention, the first notch portion 131 may have a partially different groove depth.

This will be described in detail with reference to the drawings.

[0046]

4 is an enlarged view of portion A of FIG. 3.

[0047]

Referring to FIG. 4, a portion 131a of the first notch portion has a deeper groove than the other portion 131b of the first notch portion.

That is, as shown in FIG. 4, a portion 131a of the first notch portion has a depth of W1a, and another portion 131b of the first notch portion has a depth of W1b, and W1a > W1b.

Accordingly, when the internal pressure of the battery can 200 increases, the portion 131a in which the groove is formed deeper is more easily deformed than the other portion 131b, and when the internal pressure increases to a certain value or more, the portion 131a in which the groove is formed deeper) Can be broken first.

[0048]

5 is a diagram visually showing a state in which pressurized gas flows in a state in which the safety vent of the cap assembly for a secondary battery according to the present invention is broken.

Referring to FIG. 5, due to structural characteristics, a portion 131a having a deeper groove is more easily deformed and broken than other portions 131b, and thus a broken portion F is formed in the left portion of the drawing.

Therefore, the pressurized gas passes through the broken portion F of the safety vent 130 and flows out, and the gas hole 110a of the top cap 110 is located in the outflow portion, so that the top cap is not required to change direction. It can be exhausted to the outside through the gas hole (110a) of (110).

That is, according to the present invention, when the internal pressure of the battery can 200 increases, the safety vent 130 is rapidly deformed, and a flow path is formed so that gas can be smoothly discharged from the cap assembly 100. Is formed so that the pressurized gas can be quickly exhausted to the outside.

[0049]

Preferably, the first notch portion 131 may be formed in a ring shape.

As shown in the drawing, the cap assembly 100 applied to the cylindrical secondary battery has a cylindrical shape so as to correspond to the cylindrical battery can 200.

Therefore, it is preferable that the first notch portion 131 also has a ring shape so as to be suitable for the cylindrical cap assembly 100.

However, the shape of the first notch portion 131 is not limited to the ring shape, and various shapes such as a square shape may be employed.

[0050]

More preferably, the safety vent 130 may further include a second notch portion 132 outside the first notch portion 131, as shown in FIGS. 3 and 4.

The second notch portion 132 formed on the outside preferably has a shape corresponding to the first notch portion 131 formed on the inside.

That is, when the first notch portion 131 has a ring shape as in the embodiment of the drawing, it is preferable that the second notch portion 132 is also formed in a ring shape.

However, it should not be interpreted to mean that the first notch portion 131 and the second notch portion 132 have different shapes.

[0051]

In addition, third and fourth notches may be further included outside the second notch portion, and the second notch portion may be formed inside the first notch portion, and third and fourth furnaces may be formed inside the second notch portion. It is obvious to those of ordinary skill in the art to which the present invention may be further formed.

[0052]

Also preferably, the first notch portion 131 may be provided in the bent portion (b) of the safety vent 130 as shown in FIGS. 3 and 4.

When the first notch portion 131 is provided on the bent portion b, there is an advantage that the shape deformation can be made more easily when the internal pressure of the battery can 200 is increased as compared to the case where it is not.

[0053]

Also preferably, the first notch portion 131 may be formed with a plurality of grooves spaced apart from each other by a predetermined distance.

[0054]

6 is a horizontal cross-sectional view of a cap assembly of a secondary battery according to an embodiment of the present invention.

[0055]

Referring to FIG. 6, each of the first notch portion 131 and the second notch portion 132 includes a plurality of grooves, and the plurality of grooves are disposed to be spaced apart from each other by a predetermined distance.

Meanwhile, in this embodiment, each of the first notch portion 131 and the second notch portion 132 may have an asymmetric shape around the reference line S1.

That is, in the left portion of the first notch portion 131 and the second notch portion 132, a groove may be formed deeply, like the left first notch portion and the left second notch portion of FIG. 4, and in the right portion of the notch portion 131 and the

second notch portion 132, the first notch portion 131 and the second notch portion 132, like the right first notch portion and the right second notch portion of FIG. 6. The groove may be formed shallower than the left part.

Accordingly, when the internal pressure of the battery can 200 is increased, the first notch portion 131 and the left portion of the second notch portion 132 in which the grooves are formed deeper are the first notch portion 131 and the second notch portion 132) is more easily deformed than the right part, and when the internal pressure increases beyond a certain value, the left part of the notch part where the groove is formed deeper is broken.

Accordingly, the pressurized gas inside the battery can 200 may be exhausted to the outside through a flow path formed in the broken portion on the left side of the notch.

[0056]

Also preferably, the first notch portion 131 may have one groove continuously formed.

[0057]

7 is a horizontal cross-sectional view of a cap assembly of a secondary battery according to another embodiment of the present invention.

[0058]

Referring to FIG. 7, the first notch portion 131 and the second notch portion 132 have one groove continuously formed.

That is, unlike the embodiment shown in FIG. 6, the groove is not intermittently formed but is formed continuously.

[0059]

Meanwhile, in the embodiments of FIGS. 6 and 7, it is preferable that the regions occupied by the deeply formed portions D1 and D2 are formed to be narrower than the regions occupied by the shallowly formed portions.

When the internal pressure of the battery can 200 increases, breakage occurs in the deeply formed portions D1 and D2, so when the areas occupied by the deeply formed portions D1 and D2 are formed to be narrow, the breakable range of the notch portion becomes more it can be narrowly limited.

Accordingly, it becomes easier to guide the pressurized gas inside the battery can 200 to the gas hole 110a of the top cap 110.

[0060]

More preferably, the first notch portion may be formed such that the length of the deepest portion is 1% to 90%.

[0061]

In addition, preferably, the first notch portion may have a depth of the deepest groove being 2 to 10 times deeper than the depth of the shallowest groove.

For example, as in the embodiment shown in FIG. 4, when the first notch portion is formed into a groove having a different depth in two different portions, the depth of the deep groove is 2 to 10 times greater than the depth of the shallow groove. It is better to be formed as possible.

[0062]

On the other hand, the fact that the features applicable to the first notch part 131 can be equally applied to the second notch part 132 is obvious to those of ordinary skill in the art, so it is repetitive. The explanation is omitted.

[0063]

In addition, although the shape of the groove in the embodiment of the drawing is shown as a wedge shape, it is not limited to this embodiment, and that various shapes such as a cylindrical shape and a rod shape may be employed. It is self-evident to those who have it.

[0064]

The cap assembly 100 for a secondary battery according to the present invention may further include a safety element 120.

[0065]

The safety element 120 is disposed between the top cap 110 and the safety vent 130 to electrically connect the top cap 110 and the safety vent 130 in a normal state.

However, the safety element 120 blocks the electrical connection between the top cap 110 and the safety vent 130 in an abnormal state, for example, when the temperature rises.

The safety element 120 is for blocking the flow of current inside the battery due to overheating of the battery, and may be formed of, for example, a PTC device (Positive Temperature Coefficient element).

[0066]

The cap assembly 100 for a secondary battery according to the present invention may further include a current blocking member 150 as shown in FIG. 3.

[0067]

The current blocking member 150, also called CID (Current Interrupt Device), is located between the safety vent 130 and the electrode assembly 300 so that the electrode assembly 300 and the safety vent 130 are electrically connected. do.

That is, at least a portion of the current blocking member 150 is connected to a lower portion of the central protruding portion of the safety vent 130, and at least a portion of the lower portion is connected to an electrode lead of the electrode assembly 300, such as an anode lead.

Accordingly, in a normal state, the current generated from the electrode assembly 300 flows through the positive electrode lead to the current blocking member 150, the safety vent 130, and the top cap 110, thereby discharging the secondary battery.

However, when the internal pressure of the battery increases due to gas generation and the shape of the safety vent 130 is reversed, the contact between the safety vent 130 and the current blocking member 150 is cut off or the current blocking member 150 is damaged. , Electrical connection between the safety vent 130 and the electrode assembly 300 is blocked.

[0068]

As such, the cap assembly 100 may include the current blocking member 150, in this case, as shown in FIG. 3, may further include an insulating member 140.

[0069]

The insulating member 140 is interposed between the safety vent 130 and the current blocking member 150, except for a portion in which the central protruding portion of the safety vent 130 and the current blocking member 150 contact The blocking member 150 and the safety vent 130 are electrically insulated from each other.

[0070]

The cap assembly 100 for a secondary battery according to the present invention may further include a gasket 160 as shown in FIG. 3.

[0071]

The gasket 160 may be configured to surround an edge portion of the top cap 110 and the safety vent 130.

Accordingly, the gasket 160 may be bent in a 'C' shape, as shown in the drawing.

The gasket 160 may be made of a material having electrical insulation, impact resistance, elasticity and durability, for example, polyolefine or polypropylene (PP).

In addition, the gasket 160 is preferably bent by mechanical processing, not by heat treatment, in order to prevent the insulation from deteriorating.

[0072]

The cap assembly 100 for a secondary battery according to the present invention may be coupled to the cylindrical secondary battery can 200.

In this case, the cap assembly 100 may have a cylindrical shape to be coupled to the cylindrical secondary battery can 200.

[0073]

In the above, although the present invention has been described by limited embodiments and drawings, the present invention is not limited thereto, and the technical idea of the present invention and the following will be described by those of ordinary skill in the art to which the present invention pertains. It goes without saying that various modifications and variations are possible within the equal scope of the claims.

[0074]

In the detailed description or drawings of the present invention, the use of terms such as upper, lower, inner, outer, etc. has been used to relatively distinguish one element from another element, and is only a tool concept for increasing the efficiency of description. It should not be construed as a concept used to classify, physical location, relationship between successors and others by absolute criteria.

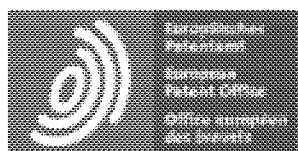
[0075]

Features described in separate embodiments of the present specification may be implemented in combination in a single embodiment.

Conversely, various features described in a single embodiment herein may be individually implemented in various embodiments, or may be implemented in an appropriate subcombination.

[0076]

100: Cap assembly 110: top cap 110a: gas hole 120: safety element 130: safety vent 131: first notch 132: second notch 140: insulating member 150: current blocking member 160: gasket 200: battery can 300: electrode assembly 400: beading portion 500: crimping portion b: bent portion of the safety vent F: fracture portion of the safety vent



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CLAIMS KR20150035267A

1.

A cap assembly for a cylindrical secondary battery, comprising: a top cap which is disposed to protrude from an uppermost portion to form a positive terminal; And a safety vent that is located under the top cap and changes its shape when the internal pressure of the battery can is increased, wherein the safety vent includes a groove having a ring shape to facilitate deformation and breakage when the internal pressure of the battery can is increased. A first notch portion and a second notch portion formed in a ring shape outside the first notch portion is provided with a groove that facilitates deformation and breakage when the internal pressure of the battery can increases, and the first notch portion and the second notch portion are circular A cap assembly for a secondary battery, characterized in that a groove is formed deeper in a circular arc than other portions, and is bent along the first notch portion and the second notch portion to be recessed in a lower direction of the top cap.

2.

delete

3.

delete

4.

delete

5.

delete

6.

delete

7.

The cap assembly of claim 1, wherein the first notch portion has a depth of the deepest groove being 2 to 10 times deeper than a depth of the shallowest groove.

8.

The cap assembly of claim 1, wherein the first notch portion is formed such that a length of a portion in which the deepest groove is formed is 1% to 90%.

9.

The cap assembly of claim 1, further comprising a safety element disposed between the top cap and the safety vent to block a current when a temperature rises.

10.

According to claim 1, An upper portion is connected to the lower end of the safety vent, the lower portion of the current blocking member to be connected to the electrode assembly; And an insulating member interposed between the safety vent and the current blocking member to electrically insulate the current blocking member from the safety vent except for a portion of the cap assembly.

11.

The cap assembly of claim 1, further comprising a gasket surrounding edges of the top cap and the safety vent.

12.

delete

13.

A secondary battery comprising the cap assembly for a secondary battery according to any one of claims 1 and 7 to 11.



Espacenet

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Cap assembly and secondary battery including the same

Inventor(s): SUNG, NAK GI; RYU, DUK HYUN

Applicant(s): LG CHEM. LTD.

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Abstract of KR20150035267 (A)

The present invention discloses a cap assembly and secondary battery including the same that has an improved structure that allows a quick transformation of safety vent and a prompt emission of pressuring gas to the exterior in the event that the interior pressure of a battery cap increases to cause deformation and rupture of a safety vent. A cap assembly for secondary battery according to the present invention comprises: a top cap that forms anode terminals by being placed on the topmost part in a protruding position and a safety vent, placed at the lower part of the top cap, that is transformed when the interior pressure of the battery can increases and consists of a first notch part that has grooves to facilitate rupture in the event of pressure increase; grooves are formed deeper within a part of the first notch is part than other parts.



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주식회사 엘지화학

서울특별시 영등포구 여의대로 128 (여의도동)

(72) 발명자

성낙기

대전광역시 유성구 문지로 188(문지동, LG화학기
술연구원)

류덕현

대전광역시 유성구 문지로 188(문지동, LG화학기
술연구원)

(74) 대리인

특허법인필앤은지

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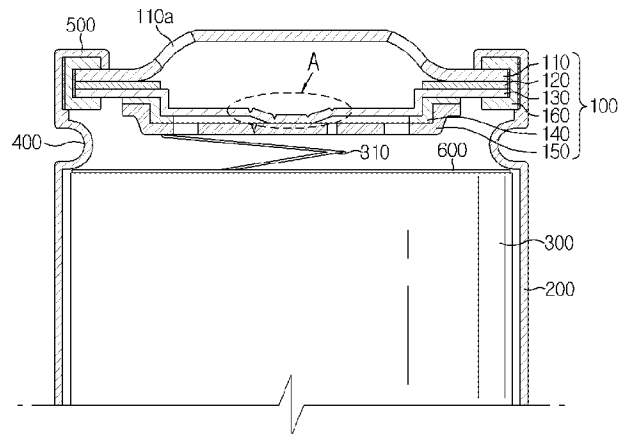
(54) 발명의 명칭 캡 조립체 및 이를 포함하는 이차 전지

(57) 요약

본 발명은 전지 캔의 내압이 증가하여 안전 밸트가 변형 및 파열되는 경우, 안전 밸트의 변형이 신속히 이루어지고, 가압 가스가 외부로 신속하게 배기될 수 있도록 하는 개선된 구조를 갖는 이차 전지용 캡 조립체 및 이를 포함하는 이차 전지를 개시한다.

본 발명에 따른 이차 전지용 캡 조립체는, 최상부에 돌출된 형태로 배치되어 양극 단자를 형성하는 탑 캡; 및 상기 탑 캡의 하부에 위치하여 전지 캔의 내압 증가시 형태가 변형되는 안전 밸트를 포함하되, 상기 안전 밸트는, 상기 전지 캔의 내압 증가시 파단이 용이하도록 홈이 형성된 제1노치부를 구비하고, 상기 제1노치부는 일부분이 다른 부분보다 홈이 더 깊게 형성된 것을 특징으로 한다.

도 3



명세서

청구항의

청구항 1

이차 전지용 캡 조립체에 있어서,

최상부에 돌출된 형태로 배치되어 양극 단자를 형성하는 탑 캡; 및

상기 탑 캡의 하부에 위치하여 전지 캔의 내압 증가시 형태가 변형되는 안전 벤트를 포함하되,

상기 안전 벤트는, 상기 전지 캔의 내압 증가시 파단이 용이하도록 홈이 형성된 제1노치부를 구비하고, 상기 제1노치부는 일부분이 다른 부분보다 홈이 더 깊게 형성된 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 2

제1항에 있어서,

상기 제1노치부는, 링 형상으로 형성된 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 3

제2항에 있어서,

상기 안전 벤트는, 상기 제1노치부의 외측 또는 내측에 링 형상으로 형성된 제2노치부를 더 포함하는 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 4

제1항에 있어서,

상기 제1노치부는, 상호 간 소정 거리 이격된 다수의 홈을 구비하여 형성된 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 5

제1항에 있어서,

상기 제1노치부는, 하나의 홈이 연속하여 형성된 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 6

제1항에 있어서,

상기 제1노치부는, 상기 안전 벤트의 절곡 부분에 구비된 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 7

제1항에 있어서,

상기 제1노치부는, 가장 깊게 형성된 홈의 깊이가 가장 얇게 형성된 홈의 깊이보다 2 배 ~ 10 배 깊은 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 8

제1항에 있어서,

상기 제1노치부는, 가장 깊게 홈이 형성된 부분의 길이가 1 % ~ 90 %가 되도록 형성된 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 9

제1항에 있어서,

상기 탑 캡과 상기 안전 벤트 사이에 배치되어, 온도 상승시 전류를 차단하는 안전 소자를 더 포함하는 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 10

제1항에 있어서,

상부가 상기 안전 벤트의 하단에 연결되고, 하부가 전극 조립체와 연결될 수 있는 전류차단부재; 및

상기 안전 벤트와 상기 전류차단부재 사이에 개재되어 일부분을 제외하고는 상기 전류차단부재를 상기 안전 벤트와 전기적으로 절연시키는 절연부재를 더 포함하는 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 11

제1항에 있어서,

상기 탑 캡 및 상기 안전 벤트의 테두리들을 감싸는 가스켓을 더 포함하는 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 12

제1항 내지 제11항 중 어느 한 항에 있어서,

원통형 이차 전지 캔에 결합되는 것을 특징으로 하는 이차 전지용 캡 조립체.

청구항 13

제1항 내지 제11항 중 어느 한 항에 따른 이차 전지용 캡 조립체를 포함하는 이차 전지.

발명의 설명

기술 분야

[0001] 본 발명은 이차 전지를 제조하는 기술에 관한 것으로서, 보다 상세하게는 진지 캔의 내부 압력증가시 내부의 가스가 외부로 쉽게 배출될 수 있도록 구조가 개선된 캡 조립체 및 이를 포함하는 이차 전지에 관한 것이다.

배경 기술

[0002] 일반적으로, 이차 전지는 충전이 불가능한 일차 전지와 달리, 충전이 가능한 전지를 의미하며, 휴대폰, 노트북 컴퓨터, 캠코더 등의 전자기기 또는 전기 자동차 등에 널리 사용되고 있다. 특히, 리튬 이차 전지는 작동 전압이 3.6V 가량으로서, 전자 장비의 전원으로 많이 사용되는 니켈-카드뮴 전지 또는 니켈-수소 전지보다 약 3배의 용량을 가지며, 단위 중량당 에너지 밀도가 높기 때문에 그 활용 정도가 급속도로 증가되는 추세에 있다.

[0003] 이러한 리튬 이차 전지는 주로 리튬계 산화물과 탄소재를 각각 양극 활물질과 음극 활물질로 사용한다. 리튬 이차 전지는, 이러한 양극 활물질과 음극 활물질이 각각 도포된 양극판과 음극판이 세퍼레이터를 사이에 두고 배치된 전극 조립체와, 전극 조립체를 전해액과 함께 밀봉 수납하는 외장재를 구비한다.

[0004] 한편, 리튬 이차 전지는 전지 케이스의 형상에 따라, 전극 조립체가 금속 캔에 내장되어 있는 캔형 이차 전지와 전극 조립체가 알루미늄 라미네이트 시트의 파우치에 내장되어 있는 파우치형 이차 전지로 분류될 수 있다. 그리고, 캔형 이차 전지는 다시 금속 캔의 형태에 따라 원통형 전지와 각형 전지로 분류될 수 있다. 이러한 각형 또는 원통형 이차 전지의 외장재는 개방단이 형성된 케이스, 즉 전지 캔 및 전지 캔의 개방단에 밀봉 결합되는 캡 조립체를 구비한다.

[0005] 도 1은, 종래기술에 따른 원통형 이차 전지의 캡 조립체 부분의 단면도이다.

[0006] 도 1을 참조하면, 일반적으로 원통형 이차 전지는 원통형 전지 캔(20), 전지 캔(20)의 내부에 수용되는 젤리-롤 형태의 전극 조립체(30), 전지 캔(20)의 상부에 결합되는 캡 조립체(10), 캡 조립체(10)를 장착하기 위해 전지

캔(20)의 선단에 마련된 비딩부(40) 및 전지를 밀봉하기 위한 클립핑 부위(50)를 구비한다.

[0007] 전극 조립체(30)는 양극과 음극 사이에 세퍼레이터가 개재된 상태로 젤리-롤 형태로 권취된 구조이며, 양극에는 양극 리드(31)가 부착되어 캡 조립체(10)에 접속되어 있고, 음극에는 음극 리드(미도시)가 부착되어 전지 캔(20)의 하단에 접속되어 있다.

[0008] 캡 조립체(10)는 양극 단자를 형성하는 탑 캡(11), 전지 내부의 온도 상승시 저항이 증가하여 전류를 차단하는 안전 소자(12), 전지 내부의 압력 상승시 전류를 차단 및/또는 가스를 배기하는 안전 밸브(13), 특정 부분을 제외하고 안전 밸브(13)를 전류차단부재(15)로부터 전기적으로 분리시키는 절연부재(14), 양극에 연결된 양극 리드(31)가 접속되어 있는 전류차단부재(15)가 순차적으로 적층된 구조를 갖는다. 그리고, 이러한 캡 조립체(10)는 가스켓(16)에 장착된 상태로 전지 캔(20)의 비딩부(40)에 장착된다. 따라서, 정상적인 작동 조건에서 전극 조립체(30)의 양극은 양극 리드(31), 전류차단부재(15), 안전 밸브(13) 및 안전 소자(12)를 경유하여 탑 캡(11)에 연결되어 통전을 이룬다.

[0009] 그러나, 과충전 등과 같은 원인에 의해 전극 조립체(30)쪽으로부터 가스가 발생하여 내압이 증가하면, 안전 밸브(13)는 그 형상이 역전되면서 상방으로 변형되게 되고, 이때, 안전 밸브(13)가 전류차단부재(15)로부터 분리되어 전류가 차단되게 된다. 따라서, 충방전이 더 이상 진행되지 않게 되어 이차 전지의 안전성이 확보될 수 있다. 더욱이, 이차 전지의 내압이 일정값 이상으로 증가하면, 안전 밸브(13)가 파열되고 가압 가스는 그러한 파열 부위를 경유하여 탑 캡의 가스 구멍(11a)을 통해 배기됨으로써 전지의 폭발이 방지될 수 있다. 이러한 안전 밸브(13)는 전지 캔(20)의 내압 증가시에 변형 및 파열이 용이하도록 하기 위하여 도 1에 도시된 바와 같이 홈이 형성된 노치부(13a)가 구비될 수 있다.

[0010] 그런데 종래기술에 따른 이차 전지용 캡 조립체의 안전 밸브(13)는 노치부(13a)를 구비하고 있고, 이러한 노치부(13a)는 균일한 깊이를 갖는다. 종래기술에 의하면, 전지 캔(20)의 내압이 증가할 경우, 가압 가스가 균일하게 안전 밸브(13)에 압력을 가하게 되므로 안전 밸브(13)의 변형이 신속히 이루어지지 못하게 된다. 뿐만 아니라, 내압이 더욱 증가하여 이차 전지의 내압이 일정값 이상으로 증가하는 경우 안전 밸브(13)가 파열되는데, 안전 밸브(13)의 노치부(13a)의 어느 지점이 파열될 것인지를 예측할 수 없다.

[0011] 도 2는 종래기술에 따른 이차 전지용 캡 조립체의 안전 밸브가 파단된 상태에서 가압 가스가 흐르는 모습을 시각적으로 나타낸 도면이다. 도 2를 참조하면, 도면에 F로 표시된 안전 밸브의 파단 부분은 안전 밸브(13)의 우측에 위치한다. 따라서, 가압 가스는 안전 밸브의 파단 부분(F)을 통과하여 도면의 우측방향으로 흐른 뒤에 방향을 전환하여 좌측방향으로 흘러 탑 캡에 구비된 가스 구멍(11a)을 통해 외부로 배기된다. 이 경우, 안전 밸브의 파단 부분(F)을 통해 배출된 가압 가스는 탑 캡의 가스 구멍(11a)을 통해 제대로 배기되지 못해 배기가 원활하게 이루어지지 못하는 문제가 발생할 수 있다.

발명의 내용

해결하려는 과제

[0012] 따라서, 본 발명은 상기와 같은 문제점을 해결하기 위해 창안된 것으로서, 전지 캔의 내압이 증가하여 안전 밸브가 변형 및 파열되는 경우, 안전 밸브의 변형이 신속히 이루어지고, 가압 가스가 외부로 신속하게 배기될 수 있도록 하는 이차 전지용 캡 조립체 및 이를 포함하는 이차 전지를 제공하는 것을 목적으로 한다.

[0013] 본 발명의 다른 목적 및 장점들은 하기의 설명에 의해서 이해될 수 있으며, 본 발명의 실시예에 의해 보다 분명하게 알게 될 것이다. 또한, 본 발명의 목적 및 장점들은 특허청구범위에 나타난 수단 및 그 조합에 의해 실현될 수 있음을 쉽게 알 수 있을 것이다.

과제의 해결 수단

[0014] 상기와 같은 목적을 달성하기 위한 본 발명에 따른 이차 전지용 캡 조립체는, 최상부에 돌출된 형태로 배치되어 양극 단자를 형성하는 탑 캡; 및 상기 탑 캡의 하부에 위치하여 전지 캔의 내압 증가시 형태가 변형되는 안전 밸브를 포함하되, 상기 안전 밸브는, 상기 전지 캔의 내압 증가시 파단이 용이하도록 홈이 형성된 제1노치부를 구비하고, 상기 제1노치부는 일부분이 다른 부분보다 홈이 더 깊게 형성된 것을 특징으로 한다.

[0015] 바람직하게는, 상기 제1노치부는, 링 형상으로 형성된 것을 특징으로 한다.

[0016] 더욱 바람직하게는, 상기 안전 밸브는, 상기 제1노치부의 외측 또는 내측에 링 형상으로 형성된 제2노치부를 더

포함하는 것을 특징으로 한다.

- [0017] 또한 바람직하게는, 상기 제1노치부는, 상호 간 소정 거리 이격된 다수의 홈을 구비하여 형성된 것을 특징으로 한다.
- [0018] 또한 바람직하게는, 상기 제1노치부는, 하나의 홈이 연속하여 형성된 것을 특징으로 한다.
- [0019] 또한 바람직하게는, 상기 제1노치부는, 상기 안전 벤트의 절곡 부분에 구비된 것을 특징으로 한다.
- [0020] 또한 바람직하게는, 상기 제1노치부는, 가장 깊게 형성된 홈의 깊이가 가장 얇게 형성된 홈의 깊이보다 2 배 ~ 10 배 깊은 것을 특징으로 한다.
- [0021] 또한 바람직하게는, 상기 제1노치부는, 가장 깊게 홈이 형성된 부분의 길이가 1 % ~ 90 %가 되도록 형성된 것을 특징으로 한다.
- [0022] 또한 바람직하게는, 상기 탑 캡과 상기 안전 벤트 사이에 배치되어, 온도 상승시 전류를 차단하는 안전 소자를 더 포함하는 것을 특징으로 한다.
- [0023] 또한 바람직하게는, 상부가 상기 안전 벤트의 하단에 연결되고, 하부가 전극 조립체와 연결될 수 있는 전류차단 부재; 및 상기 안전 벤트와 상기 전류차단부재 사이에 개재되어 일부분을 제외하고는 상기 전류차단부재를 상기 안전 벤트와 전기적으로 절연시키는 절연부재를 더 포함하는 것을 특징으로 한다.
- [0024] 또한 바람직하게는, 상기 탑 캡 및 상기 안전 벤트의 테두리들을 감싸는 가스켓을 더 포함하는 것을 특징으로 한다.
- [0025] 또한 바람직하게는, 상술한 이차 전지용 캡 조립체는 원통형 이차 전지 캔에 결합되는 것을 특징으로 한다.
- [0026] 상기와 같은 목적을 달성하기 위한 본 발명에 따른 이차 전지는 상술한 이차 전지용 캡 조립체를 포함한다.

발명의 효과

- [0027] 본 발명에 따르면, 노치부는, 일부분의 홈의 깊이가 다른 부분의 홈의 깊이와 다르게 구성되어, 전지 캔의 내압이 증가할 경우, 안전 벤트가 신속하게 변형되고, 가압 가스가 외부로 신속하게 배기될 수 있다.
- [0028] 따라서, 과충전 등에 의해 전극 조립체에서 가스가 발생하여 내압이 증가하더라도, 이차 전지의 폭발 등이 방지될 수 있게 되므로 안전성이 개선된 이차 전지가 구현될 수 있다.
- [0029] 이외에도 본 발명은 다른 다양한 효과를 가질 수 있으며, 이러한 본 발명의 다른 효과들은 하기의 설명에 의해서 이해될 수 있고, 본 발명의 실시예에 의해 보다 분명하게 알 수 있다.

도면의 간단한 설명

- [0030] 본 명세서에 첨부되는 다음의 도면들은 본 발명의 바람직한 실시예를 예시하는 것이며, 후술되는 발명의 상세한 설명과 함께 본 발명의 기술사상을 더욱 이해시키는 역할을 하는 것이므로, 본 발명은 그러한 도면에 기재된 사항에만 한정되어 해석되어서는 아니 된다.

도 1은, 종래기술에 따른 원통형 이차 전지의 캡 조립체 부분의 단면도이다.

도 2는, 종래기술에 따른 이차 전지용 캡 조립체의 안전 벤트가 파단된 상태에서 가압 가스가 흐르는 모습을 시각적으로 나타낸 도면이다.

도 3은, 본 발명의 일 실시예에 따른 이차 전지의 캡 조립체 부분의 구성을 개략적으로 나타낸 도면이다.

도 4는, 도 3의 A부분을 확대하여 나타낸 도면이다.

도 5는, 본 발명에 따른 이차 전지용 캡 조립체의 안전 벤트가 파단된 상태에서 가압 가스가 흐르는 모습을 시각적으로 나타낸 도면이다.

도 6은, 본 발명의 일 실시예에 따른 이차 전지의 캡 조립체의 수평단면도이다.

도 7은, 본 발명의 다른 실시예에 따른 이차 전지의 캡 조립체의 수평단면도이다.

발명을 실시하기 위한 구체적인 내용

- [0031] 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예를 상세히 설명하기로 한다. 이에 앞서, 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야 한다.
- [0032] 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의 가장 바람직한 일 실시예에 불과할 뿐이고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물과 변형예들이 있을 수 있음을 이해하여야 한다.
- [0033] 또한, 본 발명을 설명함에 있어 관련된 공지 구성 또는 기능에 대한 구체적인 설명이 본 발명의 요지를 흐릴 수 있다고 판단되는 경우에는 그 상세한 설명은 생략한다.
- [0034] 본 발명의 실시형태는 통상의 기술자에게 본 발명을 더욱 완전하게 설명하기 위하여 제공되는 것이므로 도면에 서의 구성요소들의 형상 및 크기 등은 보다 명확한 설명을 위해 과장되거나 생략되거나 또는 개략적으로 도시될 수 있다. 따라서, 각 구성요소의 크기나 비율은 실제적인 크기나 비율을 전적으로 반영하는 것은 아니다.
- [0035] 도 3은, 본 발명의 일 실시예에 따른 이차 전지의 캡 조립체 부분의 구성을 개략적으로 나타낸 도면이다.
- [0036] 도 3을 참조하면, 본 발명에 따른 이차 전지는, 전극 조립체(300), 전지 캔(200) 및 캡 조립체(100)를 포함한다.
- [0037] 상기 전극 조립체(300)는, 양극판 및 음극판이 세퍼레이터를 사이에 두고 배치되어 전지 캔(200)에 수납된다. 이때, 전극 조립체(300)는 젤리 롤 형태로 권취되어 배치될 수 있으며, 이 경우 젤리 롤이라고도 불린다. 전극 조립체(300)의 전극판들은 집전체에 활물질 슬러리가 도포된 구조로서 형성될 수 있는데, 슬러리는 통상적으로 입상의 활물질, 보조도체, 바인더 및 가소제 등이 용매가 첨가된 상태에서 교반되어 형성될 수 있다. 전극판들이 감기는 방향으로 집전체의 시작단과 끝단에는 슬러리가 도포되지 않는 무지부가 존재할 수 있는데, 이러한 무지부에는 각각의 전극판에 대응되는 전극 리드가 부착될 수 있다. 일반적으로 원통형 이차 전지에서 양극 리드는 전극 조립체(300)의 상단에 부착되어 캡 조립체(100)에 전기적으로 연결되고, 음극 리드는 전극 조립체(300)의 하단에 부착되어 전지 캔(200)의 바닥에 연결된다.
- [0038] 한편, 전극 조립체(300)의 상단에는 상부 절연판이 배치될 수 있다. 이러한 상부 절연판은 전극 조립체(300)와 캡 조립체(100) 사이를 절연시키는 역할을 한다.
- [0039] 상기 전지 캔(200)은, 알루미늄, 스테인리스 스틸 또는 이들의 합금과 같은 경량의 전도성 금속 재질로 구성되며, 상단이 개방된 개방부와 그와 대향되는 밀폐된 바닥부를 가진 원통형 또는 각형 구조를 가질 수 있다. 이러한 전지 캔(200)의 내부 공간에는 상기 전극 조립체(300)와 함께 전해액이 수납된다.
- [0040] 상기 전지 캔(200)은, 원통형으로 형성될 수 있으나, 각형과 같이 원통형 이외의 다양한 형태로 형성될 수 있다.
- [0041] 본 발명에 따른 이차 전지용 캡 조립체(100)는, 탑 캡(110) 및 안전 벤트(130)를 포함한다. 상기 캡 조립체(100)는 전지 캔(200)의 상단, 즉 개방단에 결합되는 이차 전지의 구성요소로서, 전지 캔(200)의 개방단을 밀폐시킨다. 이러한 캡 조립체(100)는, 전지 캔(200)의 형태에 따라 원형 또는 각형 등 다양한 형태로 형성될 수 있다.
- [0042] 상기 탑 캡(110)은, 캡 조립체(100)의 최상부에 상부 방향으로 돌출된 형태로 배치되어 양극 단자를 형성한다. 따라서, 상기 탑 캡(110)은 외부 장치, 이를테면 부하 또는 충전 장치와 전기적으로 접속된다. 또한, 이러한 탑 캡(110)에는 가스가 배출될 수 있는 가스 구멍(110a)이 형성될 수 있다. 따라서, 전극 조립체(300)로부터 가스 발생시 이러한 가스 구멍(110a)을 통해 전지 캔(200) 외부로 가스가 배출되도록 할 수 있다. 상기 탑 캡(110)은, 예를 들어 스테인리스 스틸이나 알루미늄과 같은 금속 재질로 형성될 수 있다.
- [0043] 상기 안전 벤트(130)는, 상기 탑 캡(110)의 하부에 위치하여, 이차 전지의 내압, 즉 전지 캔(200)의 내압이 일정 수준 이상으로 증가하는 경우 형태가 변형되도록 구성된다. 예를 들어, 상기 안전 벤트(130)는 이차 전지의 내압이 12~25 kgf/cm²일 때 형태가 변형되어 파열되도록 구성될 수 있다.
- [0044] 이를 위해, 상기 안전 벤트(130)는, 도면에 도시된 바와 같이, 중심부가 하부 방향으로 돌출되도록 형성되고, 그러한 중심부 부근에 소정의 노치부가 형성될 수 있다. 따라서, 이차 전지의 내부, 즉 전극 조립체(300) 쪽으로부터 가스가 발생하여 전지 캔(200)의 내압이 증가하게 되면, 안전 벤트(130)는 그것의 형상이 역전되면서 상

향 돌출되게 된다. 이때, 안전 벤트(130)와 전극 조립체(300)의 전기적 연결이 차단되게 되며, 이차 전지의 내압이 일정값 이상으로 증가하면, 안전 벤트(130)는 노치부를 중심으로 파열될 수 있다. 따라서, 이러한 안전 벤트(130)의 파열된 부분을 통해 전지 캔(200)의 내부에 차 있던 가스가 외부로 배출될 수 있게 된다.

[0045] 특히, 본 발명에 따른 안전 벤트(130)는, 전지 캔(200)의 내압 증가시 파단이 용이하도록 홈이 형성된 제1노치부(131)를 구비할 수 있다. 이러한 제1노치부(131)는 일부분이 다른 부분보다 홈이 더 깊게 형성될 수 있다. 즉, 본 발명에 있어서, 제1노치부(131)는 부분적으로 홈의 깊이가 서로 다르게 형성될 수 있다. 이에 대해서는 도면을 참조하여 상세히 설명하도록 한다.

[0046] 도 4는, 도 3의 A부분을 확대하여 나타낸 도면이다.

[0047] 도 4를 참조하면, 제1노치부의 일부분(131a)은 제1노치부의 다른 부분(131b)보다 홈이 더 깊게 형성되어 있다. 즉, 도 4에 도시된 바와 같이, 제1노치부의 일부분(131a)은 W1a의 깊이를 가지고 제1노치부의 다른 부분(131b)은 W1b의 깊이를 가지며, $W1a > W1b$ 이다. 이로 인해, 전지 캔(200)의 내압 증가시 홈이 더 깊게 형성된 부분(131a)이 다른 부분(131b) 보다 쉽게 변형되고, 내압이 일정값 이상으로 증가할 경우, 홈이 더 깊게 형성된 부분(131a)이 먼저 파단될 수 있게 된다.

[0048] 도 5는, 본 발명에 따른 이차 전지용 캡 조립체의 안전 벤트가 파단된 상태에서 가압 가스가 흐르는 모습을 시각적으로 나타낸 도면이다. 도 5를 참조하면, 구조적인 특징으로 인해, 홈이 더 깊게 형성된 부분(131a)이 다른 부분(131b) 보다 쉽게 변형 및 파단 되므로, 도면의 좌측부분에 파단 부분(F)이 형성된다. 따라서, 가압 가스는 안전 벤트(130)의 파단 부분(F)을 통과하여 유출되고, 유출된 부분에 탑 캡(110)의 가스 구멍(110a)이 위치하므로, 방향을 전환할 필요 없이 곧바로 탑 캡(110)의 가스 구멍(110a)을 통해 외부로 배기될 수 있다. 즉, 본 발명에 따르면, 전지 캔(200)의 내압이 증가할 경우, 안전 벤트(130)가 신속하게 변형되고, 캡 조립체(100)에서 가스가 원활하게 배출될 수 있도록 유로(flow path)가 형성되어 가압 가스가 외부로 신속하게 배기될 수 있게 된다.

[0049] 바람직하게는, 상기 제1노치부(131)는 링 형상으로 형성될 수 있다. 도면에 도시된 바와 같이, 원통형 이차 전지에 적용되는 캡 조립체(100)는 원통형 전지 캔(200)에 대응되도록 원통 형상을 갖는다. 따라서, 제1노치부(131)도 원통 형상의 캡 조립체(100)에 적합하도록 링 형상을 갖는 것이 좋다. 다만, 제1노치부(131)의 형상이 링 형상에 제한되는 것은 아니며 각형 등 다양한 형상이 채용될 수도 있음은 물론이다.

[0050] 더욱 바람직하게는, 상기 안전 벤트(130)는, 도 3 및 도 4에 도시된 바와 같이, 제1노치부(131)의 외측에 제2노치부(132)를 더 포함할 수 있다. 외측에 형성된 제2노치부(132)는 내측에 형성된 제1노치부(131)와 대응되는 형상을 갖는 것이 좋다. 즉, 도면의 실시예처럼 제1노치부(131)가 링 형상인 경우에는 제2노치부(132)도 링 형상으로 형성되는 것이 좋다. 다만, 제1노치부(131)와 제2노치부(132)가 서로 다른 형상을 갖는 것을 배제한다는 의미로 해석되어서는 아니된다.

[0051] 또한, 제2노치부 외측에 제3, 제4노치부 등을 더 포함할 수 있고, 제2노치부가 제1노치부의 내측에 형성될 수도 있으며, 제2노치부 내측에 제3, 제4노치부 등이 더 형성될 수도 있음은 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자에게 자명한 사항이다.

[0052] 또한 바람직하게는, 상기 제1노치부(131)는, 도 3 및 도 4에 도시된 바와 같이, 안전 벤트(130)의 절곡 부분(b)에 구비될 수 있다. 제1노치부(131)가 절곡 부분(b) 상에 구비되는 경우에는 그렇지 않은 경우와 비교할 때, 전지 캔(200)의 내압 증가시 형태 변형이 더욱 용이하게 이루어질 수 있는 장점이 있다.

[0053] 또한 바람직하게는, 상기 제1노치부(131)는, 상호 간 소정 거리 이격된 다수의 홈을 구비하여 형성될 수 있다.

[0054] 도 6은, 본 발명의 일 실시예에 따른 이차 전지의 캡 조립체의 수평단면도이다.

[0055] 도 6을 참조하면, 제1노치부(131) 및 제2노치부(132)는 각각 다수의 홈을 구비하고 있고, 이러한 다수의 홈은 상호간 소정의 거리로 이격되어 배치되어 있다. 한편, 이러한 실시예에서, 기준선 S1을 중심으로 제1노치부(131) 및 제2노치부(132)는 각각 비대칭형상을 될 수 있다. 즉, 제1노치부(131) 및 제2노치부(132)의 좌측 부분에는, 도 4의 좌측 제1노치부 및 좌측 제2노치부와 같이, 홈이 깊게 형성될 수 있고, 제1노치부(131) 및 제2노치부(132)의 우측 부분에는, 도 4의 우측 제1노치부 및 우측 제2노치부와 같이, 제1노치부(131) 및 제2노치부(132)의 좌측 부분보다 홈이 얇게 형성될 수 있다. 이로 인해, 전지 캔(200)의 내압 증가시 홈이 더 깊게 형성된 제1노치부(131) 및 제2노치부(132)의 좌측 부분이 제1노치부(131) 및 제2노치부(132)의 우측 부분보다 쉽게 변형되고, 내압이 일정값 이상으로 증가할 경우, 홈이 더 깊게 형성된 노치부의 좌측 부분이 파단되게 된다. 따

로서, 전지 캔(200) 내부의 가압 가스가 노치부의 좌측의 파단 부분에 형성된 유로(flow path)를 통해 외부로 배기될 수 있다.

- [0056] 또한 바람직하게는, 상기 제1노치부(131)는, 하나의 홈이 연속하여 형성될 수 있다.
- [0057] 도 7은, 본 발명의 다른 실시예에 따른 이차 전지의 캡 조립체의 수평단면도이다.
- [0058] 도 7을 참조하면, 제1노치부(131) 및 제2노치부(132)는 하나의 홈이 연속하여 형성되어 있다. 즉, 도 6에 도시된 실시예와 달리 홈이 단속적으로 형성되지 않고 연속적으로 형성되어 있다.
- [0059] 한편, 도 6 및 도 7의 실시예에서, 홈이 깊게 형성된 부분(D1, D2)이 차지하는 영역이 홈이 얇게 형성된 부분이 차지하는 영역보다 좁게 형성되는 것이 좋다. 전지 캔(200)의 내압 증가시 홈이 깊게 형성된 부분(D1, D2)에서 파단이 발생하므로, 홈이 깊게 형성된 부분(D1, D2)이 차지하는 영역이 좁게 형성되면, 노치부의 파단 가능 범위가 더욱 좁게 제한될 수 있다. 따라서, 전지 캔(200) 내부의 가압 가스를 탭 캡(110)의 가스 구멍(110a)으로 유도하기가 더욱 용이해진다.
- [0060] 더욱 바람직하게는, 제1노치부는, 가장 깊게 형성된 부분의 길이가 1 % ~ 90 %가 되도록 형성될 수 있다.
- [0061] 또한 바람직하게는, 제1노치부는, 가장 깊게 형성된 홈의 깊이가 가장 얇게 형성된 홈의 깊이보다 2 배 ~ 10 배 깊을 수 있다. 예를 들어, 도 4에 도시된 실시예와 같이 제1노치부가 서로 다른 2개의 부분에서 각각 다른 깊이를 갖는 홈으로 형성되는 경우, 깊은 홈의 깊이는 얇은 홈의 깊이에 비해 2 배 ~ 10 배가 되도록 형성되는 것이 좋다.
- [0062] 한편, 상기 제1노치부(131)에 적용될 수 있는 특징들은 제2노치부(132)에도 동일하게 적용될 수 있다는 점은 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자에게 자명한 사항이므로 반복적인 설명은 생략한다.
- [0063] 또한, 도면의 실시예에서 홈의 형상은 쐐기 형상으로 도시되었으나, 이러한 실시예에 한정되는 것은 아니며 원통 형상, 타대 형상 등 다양한 형상이 채용될 수 있음은 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자에게 자명한 사항이다.
- [0064] 본 발명에 따른 이차 전지용 캡 조립체(100)는, 안전 소자(120)를 더 포함할 수 있다.
- [0065] 상기 안전 소자(120)는, 탭 캡(110)과 안전 벤트(130) 사이에 배치되어 정상적인 상태에서는 탭 캡(110)과 안전 벤트(130)를 전기적으로 연결시킨다. 다만, 상기 안전 소자(120)는, 비정상 상태, 예를 들어 온도 상승시 탭 캡(110)과 안전 벤트(130) 간의 전기적 연결을 차단시킨다. 상기 안전 소자(120)는 전지의 과열에 의해 전지 내부의 전류의 흐름을 차단하기 위한 것으로서, 이를테면 PTC 소자(Positive Temperature Coefficient element)로 형성될 수 있다.
- [0066] 본 발명에 따른 이차 전지용 캡 조립체(100)는, 도 3에 도시된 바와 같이, 전류차단부재(150)를 더 포함할 수 있다.
- [0067] 상기 전류차단부재(150)는, CID(Current Interrupt Device)라고도 불리는데, 안전 벤트(130)와 전극 조립체(300) 사이에 위치하여, 전극 조립체(300)와 안전 벤트(130)가 전기적으로 접속되도록 한다. 즉, 전류차단부재(150)는, 상부의 적어도 일부분이 안전 벤트(130)의 중앙 돌출 부분 하단에 연결되고, 하부의 적어도 일부분이 전극 조립체(300)의 전극 리드, 이를테면 양극 리드와 연결된다. 따라서, 정상적인 상태에서는 전극 조립체(300)로부터 생성된 전류는 양극 리드를 거쳐 전류차단부재(150), 안전 벤트(130) 및 탭 캡(110)으로 흐름으로써 이차 전지의 방전이 이루어질 수 있다. 그러나, 가스 발생으로 전지의 내압이 증가하여 안전 벤트(130)의 형상이 역전되면, 안전 벤트(130)와 전류차단부재(150) 사이의 접촉이 끊어지거나, 전류차단부재(150)가 파손되어, 안전 벤트(130)와 전극 조립체(300) 사이의 전기적 접속은 차단되게 된다.
- [0068] 이처럼 캡 조립체(100)는 전류차단부재(150)를 포함할 수 있는데, 이 경우, 도 3에 도시된 바와 같이, 절연부재(140)를 더 포함할 수 있다.
- [0069] 상기 절연부재(140)는, 안전 벤트(130)와 전류차단부재(150) 사이에 개재되어, 안전 벤트(130)의 중앙 돌출 부분과 전류차단부재(150)가 접촉되는 부분을 제외하고는 전류차단부재(150)와 안전 벤트(130)가 서로 전기적으로 절연되도록 한다.
- [0070] 본 발명에 따른 이차 전지용 캡 조립체(100)는, 도 3에 도시된 바와 같이, 가스켓(160)을 더 포함할 수 있다.
- [0071] 상기 가스켓(160)은, 탭 캡(110) 및 안전 벤트(130)의 테두리 부위를 감싸도록 구성될 수 있다. 따라서, 상기

가스켓(160)은, 도면에 도시된 바와 같이, 'C'자 형태로 굴곡될 수 있다. 상기 가스켓(160)은 전기 절연성과 내충격성, 탄력성 및 내구성을 가진 소재, 예를 들어 폴리올레핀(polyolefine) 또는 폴리프로필렌(PP)으로 제조될 수 있다. 또한, 상기 가스켓(160)은 절연성이 악화되는 것을 방지하기 위해 열치리에 의하지 않고 기계적 가공에 의해 벤딩되는 것이 좋다.

[0072] 본 발명에 따른 이차 전지용 캡 조립체(100)는, 원통형 이차 전지 캔(200)에 결합될 수 있다. 이 경우, 원통형 이차 전지 캔(200)에 결합되기 위해 캡 조립체(100)는 원통 형상이 될 수 있다.

[0073] 이상에서 본 발명은 비록 한정된 실시예와 도면에 의해 설명되었으나, 본 발명은 이것에 의해 한정되지 않으며 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자에 의해 본 발명의 기술사상과 아래에 기재될 특허청구범위의 균등범위 내에서 다양한 수정 및 변형이 가능함은 물론이다.

[0074] 이상의 본 발명에 대한 상세한 설명 또는 도면에서, 상, 하, 내, 외 등과 같은 용어의 사용은 하나의 요소를 다른 요소와 상대적으로 구분하기 위하여 사용되었으며, 설명의 효율성을 높이기 위한 도구적 개념일 뿐, 물리적인 위치, 선후 관계 등을 절대적인 기준에 의하여 구분하기 위하여 사용된 개념으로 해석되어서는 아니된다.

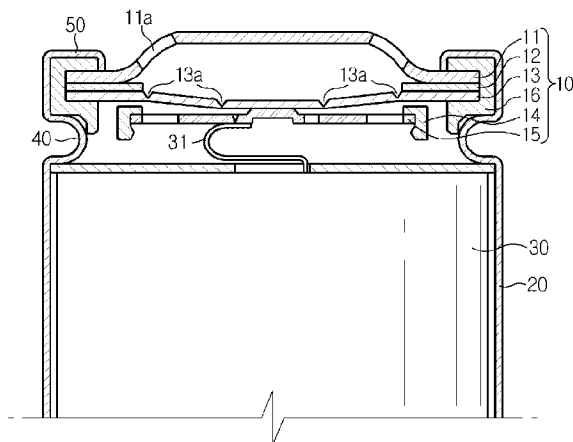
[0075] 본 명세서의 개별적인 실시예에서 설명된 특징들은 단일 실시예에서 결합되어 구현될 수 있다. 반대로, 본 명세서에서 단일 실시예에서 설명된 다양한 특징들은 개별적으로 다양한 실시예에서 구현되거나, 적절한 부결합(subcombination)에서 구현될 수 있다.

부호의 설명

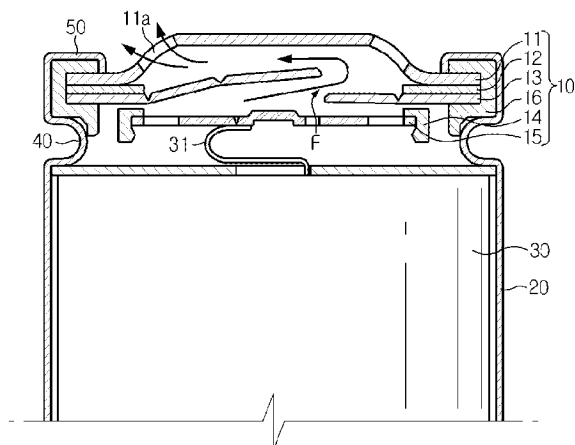
- [0076]
- 100: 캡 조립체
 - 110: 탑 캡
 - 110a: 가스 구멍
 - 120: 안전 소자
 - 130: 안전 벤트
 - 131: 제1노치부
 - 132: 제2노치부
 - 140: 절연부재
 - 150: 전류차단부재
 - 160: 가스켓
 - 200: 전지 캔
 - 300: 전극 조립체
 - 400: 비딩부
 - 500: 클립핑 부위
 - b: 안전 벤트의 절곡 부분
 - F: 안전 벤트의 파단 부분

도면

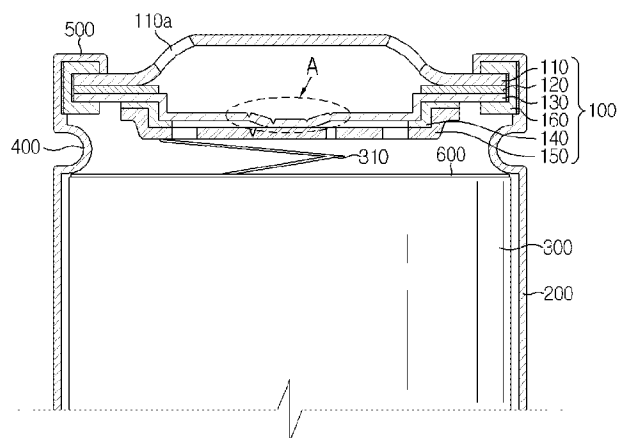
도면1



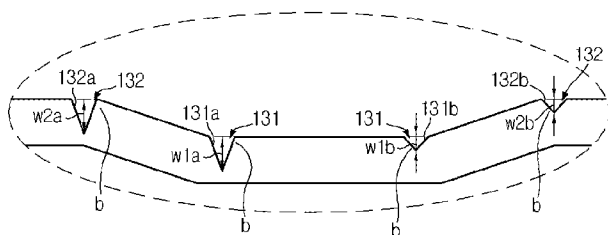
592



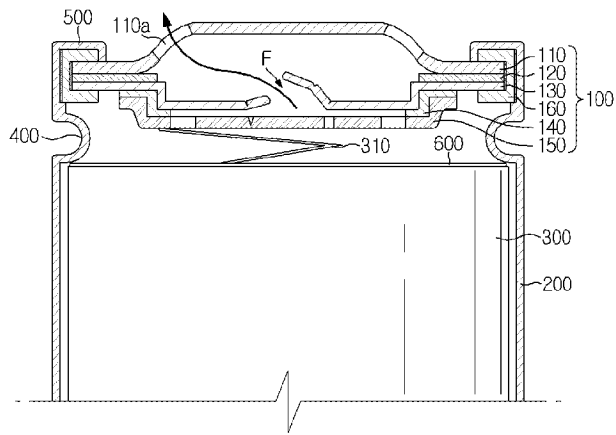
589



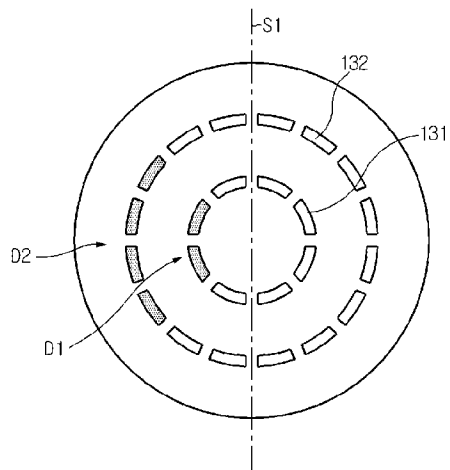
594



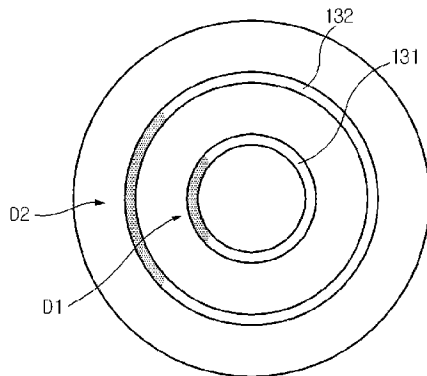
도면5



도면6



도면7



INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2019/015396

A. CLASSIFICATION OF SUBJECT MATTER

H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models: IPC as above

Japanese utility models and applications for utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: gas, notch, bending, fracture

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 05 August 1994 See paragraphs [0019]-[0022], claim 1 and figures 1-4.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 24 December 1996 See paragraph [0043] and figure 11.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 13 April 2001 See paragraphs [0016], [0017] and figures 1-5.	4-9
A	KR 10-2015-0035267 A (LG CHEM, LTD.) 06 April 2015 See the entire document.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 02 October 2018 See the entire document.	1-12



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

“A” document defining the general state of the art which is not considered to be of particular relevance

“E” earlier application or patent but published on or after the international filing date

“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

“O” document referring to an oral disclosure, use, exhibition or other means

“P” document published prior to the international filing date but later than the priority date claimed

“T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

“X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

“Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

“&” document member of the same patent family

Date of the actual completion of the international search

19 FEBRUARY 2020 (19.02.2020)

Date of mailing of the international search report

20 FEBRUARY 2020 (20.02.2020)

Name and mailing address of the ISA/KR

Korean Intellectual Property Office
Government Complex Daejeon Building 4, 189, Cheongsu-ro, Seo-gu,
Daejeon, 35208, Republic of Korea

Facsimile No. +82-42-481-8578

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/KR2019/015396

Patent document cited in search report	Publication date	Patent family member	Publication date
JP 06-215746 A	05/08/1994	None	
JP 08-339793 A	24/12/1996	None	
JP 2001-102038 A	13/04/2001	None	
KR 10-2015-0035267 A	06/04/2015	KR 10-1667966 B1	20/10/2016
US 10090497 B2	02/10/2018	CN 107293687 A	24/10/2017
		KR 10-2017-0116370 A	19/10/2017
		US 2017-0294635 A1	12/10/2017

A. 발명이 속하는 기술분류(국제특허분류(IPC))

H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i

B. 조사된 분야

조사된 최소문헌(국제특허분류를 기재)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

조사된 기술분야에 속하는 최소문헌 이외의 문헌

한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우))

eKOMPASS(특허청 내부 검색시스템) & 키워드: 가스(gas), 노치(notch), 벤딩(bending), 파단(fracture)

C. 관련 문헌

카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 1994.08.05 단락 [0019]~[0022], 청구항 1 및 도면 1-4 참조.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 1996.12.24 단락 [0043] 및 도면 11 참조.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 2001.04.13 단락 [0016],[0017] 및 도면 1-5 참조.	4-9
A	KR 10-2015-0035267 A (주식회사 엘지화학) 2015.04.06 전체 문헌 참조.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 2018.10.02 전체 문헌 참조.	1-12

☐ 추가 문헌이 C(계속)에 기재되어 있습니다.

☒ 대응특허에 관한 별지를 참조하십시오.

* 인용된 문헌의 특별 카테고리:

“A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌

“D” 본 국제출원에서 출원인이 인용한 문헌

“E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후 “X”에 공개된 선출원 또는 특허 문헌

“L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌

“O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌

“P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌

“T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌

“X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다.

“Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다.

“&” 동일한 대응특허문헌에 속하는 문헌

국제조사의 실제 완료일

2020년 02월 19일 (19.02.2020)

국제조사보고서 발송일

2020년 02월 20일 (20.02.2020)

ISA/KR의 명칭 및 우편주소



대한민국 특허청
(35208) 대전광역시 서구 청사로 189,
4동 (둔산동, 정부대전청사)

팩스 번호 +82-42-481-8578

심사관

김연경

전화번호 +82-42-481-3325



국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
JP 06-215746 A	1994/08/05	없음	
JP 08-339793 A	1996/12/24	없음	
JP 2001-102038 A	2001/04/13	없음	
KR 10-2015-0035267 A	2015/04/06	KR 10-1667966 B1	2016/10/20
US 10090497 B2	2018/10/02	CN 107293687 A	2017/10/24
		KR 10-2017-0116370 A	2017/10/19
		US 2017-0294635 A1	2017/10/12

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: January 11, 2021
Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

International Application No.: PCT/KR2019/015396

Confirmation No.: N/A

International Filing Date: November 13, 2019

Art Unit: N/A

For: Secondary Battery

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Dear Sir:

Prior to initiation of the prosecution of the above-identified pending U.S. patent application, the following amendments and remarks are respectfully submitted.

IN THE SPECIFICATION

Please amend the specification as originally filed and substitute therefore the enclosed revised specification which is presented in both mark-up and clean copy versions.

IN THE CLAIMS

1. (currently amended) A secondary battery comprising:
an electrode assembly-(1);
a can member-(3) configured to accommodate the electrode assembly (1)-therein; and
a top cap assembly-(100) configured to cover an opening of the can member-(3),
wherein the top cap assembly-(100) comprises a top cap (101) positioned at ~~the~~an uppermost portion and a safety vent part (100)-positioned below the top cap-(101),
wherein the safety vent part (110)-comprises:
a rupture portion configured to be (111)-ruptured when an internal pressure within the can member-(3) increases to equal or exceed a predetermined pressure ~~or more~~;
a rupture notch portion-(113) that defines a boundary of the rupture portion-(111); and
a bending notch portion-(115) which is disposed at a position spaced apart from the rupture notch portion-(113) toward a center of the rupture portion-(111) on a bottom surface of the rupture portion-(111) facing ~~the~~an inside of the can member-(3),
wherein the bending notch portion-(115) is bent when the internal pressure within the can member-(3) increases.
2. (currently amended) The secondary battery of claim 1, wherein the rupture notch portion (113)-is formed as a recessed ~~from within~~ a lower surface of the safety vent part-(110), and
the bending notch portion is formed as a (115)-~~recessed from~~ within the lower surface of the safety vent part-(110),
wherein a recessed depth of the bending notch portion (115)-is less than a recessed depth ~~that~~ of the rupture notch portion-(113).
3. (currently amended) The secondary battery of claim 1, wherein the rupture notch portion (113)-is provided in the bottom surface of the rupture portion-(111).
4. (currently amended) The secondary battery of claim 1, wherein the top cap assembly (100)-further comprises a current interrupt device (120)-provided with an interrupt portion (121)

which is attached to the rupture portion (111) and which is configured to be ruptured to interrupt current when the internal pressure within the can member (3) increases.

5. (currently amended) The secondary battery of claim 4, wherein the interrupt portion (121) of the current interrupt device (120) is attached to the rupture portion (111) inside a circumference of the bending notch portion (115).

6. (currently amended) The secondary battery of claim 1, wherein the bending notch portion (115) has a continuous circular or elliptical shape.

7. (currently amended) The secondary battery of claim 1, wherein the bending notch portion (115) has a discontinuous circular or elliptical shape.

8. (currently amended) The secondary battery of claim 1, wherein the bending notch portion (115) ~~is provided in~~ comprises a plurality of bending notches, and the plurality of bending notches ~~portions~~ have different lengths.

9. (currently amended) The secondary battery of claim 1, wherein the rupture notch portion (113) has a discontinuous circular or elliptical shape.

10. (currently amended) The secondary battery of claim 1, wherein a connection portion (119) between an edge portion (117) and the rupture portion (111) of the safety vent part (110) is bent upward.

11. (currently amended) The secondary battery of claim 10, wherein the rupture portion (111) comprises:

an inclined section (111a) inclined so that a height thereof gradually decreases from the connection portion (119) toward the bending notch portion (115); and

a horizontal section (111b) having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion (111),

wherein the bending notch portion-~~(115)~~ is positioned on a boundary between the inclined section-~~(111a)~~ and the horizontal section-~~(111b)~~.

12. (currently amended) The secondary battery of claim 1, wherein a spaced distance between an edge end of the safety vent part ~~(110)~~ and the rupture notch portion ~~(113)~~ is equal to a spaced distance between the rupture notch portion (113) and the bending notch portion-~~(115)~~.

IN THE ABSTRACT

Please amend the abstract as shown below:

The present invention relates to a secondary battery capable of improving venting pressure distribution. ~~Provided is a~~A secondary battery including an electrode assembly, a can member that accommodates the electrode assembly ~~therein~~, and a top cap assembly that covers an opening of the can member. The top cap assembly includes a top cap positioned at the uppermost portion and a safety vent part positioned below the top cap. The safety vent part includes a rupture portion that is ruptured when an internal pressure within the can member equals or exceeds~~increases to~~ a predetermined pressure ~~or more~~, a rupture notch portion that defines a boundary of the rupture portion, and a bending notch portion which is ~~disposed at a position~~ spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion ~~facing the inside of the can member~~, and the bending notch portion is bent when the internal pressure within the can member increases.

REMARKS

The above-noted amendment of claims 1-12, as well as the submission of a substitute Specification and revised Abstract, are respectfully submitted prior to initiation of the prosecution of this application in the U.S. Patent and Trademark Office. These amendments are respectfully submitted in order to more clearly and appropriately claim at least one aspect of the subject matter that Applicant considers to constitute the inventive contribution and/or in order to clarify and correct the application and to conform it to the requirements of U.S. practice. No new matter is included in these amendments.

In view of the above, it is respectfully requested that these amendments now be entered, and that prosecution on the merits of this application now be initiated. If, however, for any reason the Examiner does not believe such action can be taken, it is respectfully requested that the Examiner telephone Applicant's attorney at (908) 654-5000 in order to overcome any objections that the Examiner may have.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: January 11, 2021

Respectfully submitted,
Electronic signature:
/Jordan T. Riviello/
Jordan T. Riviello
Registration No.: 72,515
LERNER, DAVID, LITTENBERG,
KRUMHOLZ & MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

DESCRIPTION**SECONDARY BATTERY****CROSS-REFERENCE TO RELATED APPLICATION**

[0001] The present application claims the benefit of the priority of Korean Patent Application No. 10-2018-0149369, filed on November 28, 2018, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery, and more particularly, to a secondary battery capable of improving venting pressure distribution.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction of materials to supply power to the outside are used when alternate current (AC) power to be supplied to the building is not obtained, or direct current (DC) power is required according to the living environments surrounded by various electrical/electronic devices.

[0004] Among such batteries, primary batteries and secondary batteries, which are chemical cells using chemical reaction, are being generally used. The primary batteries are consumable cells which are collectively referred to as dry cells. Also, the secondary batteries are rechargeable batteries manufactured by using a material in which oxidation and reduction processes between current and a material are capable of being repeated several times. That is, power is recharged when the reduction reaction to the material is performed by

the current, and power is discharged when the oxidation reaction to the material is performed. The recharging and discharging are repeatedly performed to generate electricity.

[0005] A lithium ion battery among the secondary batteries is manufactured in such a manner in which: an active material is applied, with a predetermined thickness, to each of a positive electrode conductive foil and a negative electrode conductive foil; a separator is disposed between both the conductive foils so as to be wound several times in an approximate jelly roll or cylindrical shape and thereby to manufacture an electrode assembly; and the electrode assembly is accommodated in a cylindrical or prismatic can, a pouch, or the like and is sealed.

[0006] A secondary battery according to the related art is disclosed in Korean Patent Publication No. 10-2011-0095118.

[0007] In the secondary battery, an internal gas is vented to the outside through a top cap assembly when an internal pressure increases, and the top cap assembly includes a safety vent for venting the internal gas.

[0008] Such a safety vent has a structure in which a notch defined in a vent flange part disposed between the top cap assembly and a can is cut when the internal pressure within the can increases, and thus a passage for venting the gas is provided.

[0009] Particularly, a certain component, such as the safety vent, in a circular battery induces venting to prevent the battery from being exploded and ignited when an abnormal accident such as overcharging occurs.

[0010] Such a safety vent may cause a safety problem when the venting is not induced at a pre-designed venting pressure.

[0011] Also, a top cap interferes with the safety vent according to the related art during measurement of a venting pressure before the notch of the safety vent is cut, and thus the venting may be delayed.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0012] Accordingly, the present invention has been made to solve the above problem, and an object of the present invention is to provide a secondary battery in which the interference with a top cap is prevented to be able to improve the venting pressure distribution.

TECHNICAL SOLUTION

[0013] A secondary battery according to an embodiment of the present invention includes: an electrode assembly; a can member that accommodates the electrode assembly therein; and a top cap assembly that covers an opening of the can member, wherein the top cap assembly includes a top cap positioned at the uppermost portion and a safety vent part positioned below the top cap, wherein the safety vent part includes: a rupture portion ruptured when an internal pressure within the can member increases to a predetermined pressure or more; a rupture notch portion that defines a boundary of the rupture portion; and a bending notch portion which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing the inside of the can member, wherein the bending notch portion is bent when the internal pressure within the can member increases.

[0014] The rupture notch portion may be recessed from a lower surface of the safety vent part, and the bending notch portion may be recessed from the lower surface of the safety vent part,

wherein a recessed depth of the bending notch portion is less than that of the rupture notch portion.

[0015] The rupture notch portion may be provided in the bottom surface of the rupture portion.

[0016] The top cap assembly may further include a current interrupt device provided with an interrupt portion which is attached to the rupture portion and ruptured to interrupt current when the internal pressure within the can member increases.

[0017] The interrupt portion of the current interrupt device may be attached to the rupture portion inside a circumference of the bending notch portion.

[0018] The bending notch portion may have a continuous circular or elliptical shape.

[0019] The bending notch portion may have a discontinuous circular or elliptical shape.

[0020] The bending notch portion may be provided in plurality, and the plurality of bending notch portions have different lengths.

[0021] The rupture notch portion may have a discontinuous circular or elliptical shape.

[0022] A connection portion between an edge portion and the rupture portion of the safety vent part may be bent upward.

[0023] The rupture portion may include: an inclined section inclined so that a height thereof gradually decreases from the connection portion toward the bending notch portion; and a horizontal section having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion, wherein the bending notch portion is positioned on a boundary between the inclined section and the

horizontal section.

[0024] A spaced distance between an edge end of the safety vent part and the rupture notch portion may be equal to a spaced distance between the rupture notch portion and the bending notch portion.

ADVANTAGEOUS EFFECTS

[0025] According to the present invention, the bending notch portion as well as the rupture notch portion is further provided in the bottom surface of the safety vent part, and thus the safety vent part is not interfered with the top cap when the internal pressure within the can member increases.

[0026] Also, since the safety vent part is not interfered with the top cap, the rupture notch portion stably performs the venting at the preset venting pressure, and thus the venting pressure distribution may be improved.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a perspective view of a secondary battery according to a first embodiment of the present invention.

[0028] FIG. 2 is a partial cross-sectional view taken along line A-A and illustrating main parts in an upper inner portion of the secondary battery.

[0029] FIG. 3 is a view illustrating a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in the internal pressure of a can member of FIG. 2.

[0030] FIG. 4 is a view illustrating a state of use in which a rupture notch portion of a safety vent part of FIG. 3 is cut to provide venting.

[0031] FIG. 5 is a bottom view illustrating only the safety vent part according to the first embodiment of the present

invention.

[0032] FIG. 6 is a cross-sectional view taken along line B-B of FIG. 5.

[0033] FIG. 7 is a bottom view illustrating only a safety vent part according to a second embodiment of the present invention.

[0034] FIG. 8 is a bottom view illustrating only a safety vent part according to a third embodiment of the present invention.

[0035] FIG. 9 is a cross-sectional view taken along line C-C of FIG. 8.

[0036] FIG. 10 is a view of a secondary battery in which a safety vent part according to the related art is installed, illustrating, on the same cross-section of FIG. 3, a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in an internal pressure of a can member.

[0037] FIG. 11 shows the degrees of distribution for comparing the venting pressure distribution in the secondary battery in which the safety vent part according to the related art is installed and the venting pressure distribution in the secondary battery in which the safety vent part according to the present invention is installed.

MODE FOR CARRYING OUT THE INVENTION

[0038] Hereinafter, preferred embodiments of the present invention will now be described in detail with reference to the accompanying drawings.

[0039] Terms or words used in this specification and claims should not be restrictively interpreted as ordinary meanings or dictionary-based meanings, but should be interpreted as

meanings and concepts conforming to the scope of the present invention on the basis of the principle that an inventor can properly define the concept of a term to describe and explain his or her invention in the best ways. Therefore, the embodiments described in this specification and the configurations illustrated in the drawings are only preferred embodiments of the present invention, and may not describe the technical idea thoroughly. Accordingly, it should be understood that various equivalents and modifications that can be substituted for the embodiments might be provided at the filing date of the present application.

[0040] In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience and clarity of illustration. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

[0041] FIG. 1 is a perspective view of a secondary battery according to a first embodiment of the present invention, and FIG. 2 is a partial cross-sectional view taken along line A-A and illustrating main parts in an upper inner portion of the secondary battery.

[0042] As illustrated in FIGS. 1 and 2, a secondary battery according to the embodiment includes an electrode assembly 1, a can member 3 for accommodating the electrode assembly 1 therein, and a top cap assembly 100 for covering an opening of the can member 3. The top cap assembly 100 includes a top cap 101 positioned at the uppermost portion and a safety vent part

100 positioned below the top cap 101. The safety vent part 110 includes a rupture portion 111 ruptured when an internal pressure within the can member 3 increases to a predetermined pressure or more, a rupture notch portion 113 that defines a boundary of the rupture portion 111, and a bending notch portion 115 which is disposed at a position spaced apart from the rupture notch portion 113 toward a center of the rupture portion 111 on a bottom surface of the rupture portion 111 facing the inside of the can member 3. The bending notch portion 115 is bent when the internal pressure within the can member 3 increases.

[0043] The electrode assembly 1 may be manufactured by stacking, multiple times, a positive electrode coated with a positive electrode active material, a negative electrode coated with a negative electrode active material, and a separator interposed between the negative electrode and the positive electrode.

[0044] Also, the electrode assembly 1 may be manufactured by winding, in a jelly roll structure, a stack body in which the positive electrodes, the separators, and the negative electrodes are stacked on each other.

[0045] The can member 3 is a metal container having an open top surface of an approximately cylindrical structure in a cylindrical lithium ion secondary battery or a metal container having an open top surface of an approximately prismatic structure in a prismatic lithium ion secondary battery. The can member 3 is made of aluminum or an aluminum alloy that is generally lightweight and easy to manage corrosion.

[0046] The can member 3 serves as a container for the electrode assembly 1 and an electrolyte (not shown). The

electrode assembly 1 is inserted into the can member 3 via an upper open end, i.e., an upper end that is open in the can member 3, and then the upper open end of the can member 3 is sealed by the top cap assembly 100.

[0047] The top cap assembly 100 may further include a current interrupt device 120 provided with an interrupt portion 121 which is ruptured to interrupt current when an internal pressure within the can member 3 increases.

[0048] FIG. 3 is a view illustrating a state of use in which the interrupt portion is ruptured from the current interrupt device due to an increase in the internal pressure of the can member of FIG. 2.

[0049] Referring to FIG. 3, the interrupt portion 121 of the current interrupt device 120 is attached to the rupture portion 111 of the safety vent part 110. The rupture portion 121 is ruptured from the current interrupt device 120 and moves upward toward the top cap 101 of the top cap assembly 100, together with the rupture portion 111 of the safety vent part 110.

[0050] According to the present invention, the rupture portion 121 of the current interrupt device 120 may be attached to the center in the rupture portion 111 inside a circumference of the bending notch portion 115.

[0051] The rupture notch portion 113 of the safety vent part 110 is recessed from the surface of the safety vent part 110 so as to be cut when the internal pressure within the can member 3 increases.

[0052] The rupture notch portion 113 may be defined in one of a top surface and a bottom surface of the safety vent part 110. In a case in which the rupture notch portion 113 is provided in the bottom surface of the safety vent part 110 facing the

inside of the can member 3, it may be advantageous to secure a spaced distance d to the top cap 101 when the safety vent part 110 rises upward due to the internal pressure within the can member 3.

[0053] Similar to the rupture notch portion 113, the bending notch portion 115 is concavely indented (or grooved) from the surface of the safety vent part 110. The recessed depth of the bending notch portion 115 is less than the recessed depth of the rupture notch portion 113.

[0054] This may be done for maintaining a thickness of a notch to a thickness not allowing the notch to be cut, and thus the bending notch portion 115 is not cut but bent due to the internal pressure within the can member 3.

[0055] Also, the bending notch portion 115 is provided in the bottom surface of the safety vent part 110, and thus the spaced distance d to the top cap 101 may be secured when the safety vent part 110 is moved upward due to the internal pressure within the can member 3.

[0056] FIG. 4 is a view illustrating a state of use in which the rupture notch portion of the safety vent part of FIG. 3 is cut to provide venting.

[0057] As illustrated in FIG. 4, in a case in which the spaced distance d to the top cap 101 is secured by the bending of the bending notch portion 115 provided in the bottom surface of the safety vent part 110 when the safety vent part 110 is moved upward due to the internal pressure inside the can member 3, the rupture notch portion 113 is cut at a preset venting pressure, and thus the gas within the can member 3 may be safely vented to the outside.

[0058] FIG. 5 is a bottom view illustrating only the safety

vent part according to the first embodiment of the present invention.

[0059] According to the embodiment of the present invention as illustrated in FIG. 5, the rupture notch portion 113 may have a discontinuous circular or elliptical shape in the bottom surface of the safety vent part 110.

[0060] As described above, the rupture notch portion 113 may be provided corresponding to the bottom surface of the safety vent part 110. That is, when the bottom surface of the safety vent part 110 has a circular shape, the rupture notch portion 113 may also have the circular shape, and when the bottom surface of the safety vent part 110 has an elliptical shape, the rupture notch portion 113 may also have the elliptical shape.

[0061] The bending notch portion 115 may have a continuous circular or elliptical shape in the bottom surface of the safety vent part 110.

[0062] Similar to the rupture notch portion 113, the bending notch portion 115 may also be provided corresponding to the bottom surface of the safety vent part 110.

[0063] Also, the bending notch portion 115 has a smaller circumferential size than the rupture notch portion 113, and thus may be provided in the bottom surface of the safety vent part 110 inside a circumference of the rupture notch portion 113.

[0064] Here, if a spaced distance between an edge end on the bottom surface of the safety vent part 110 and the rupture notch portion 113 is equal, without being biased to the one side, to a spaced distance between the rupture notch portion 113 and the bending notch portion 115 when viewed from the

bottom surface of the safety vent part 110, the spaced gap between the safety vent part 110 and the top cap 101 may be most effectively secured when the safety vent part 110 is moved upward due to the internal pressure within the can member 3.

[0065] FIG. 6 is a cross-sectional view taken along line B-B of FIG. 5.

[0066] As illustrated in FIG. 6, in the secondary battery according to the embodiment of the present invention, a connection portion 119 between an edge portion 117 and the rupture portion 111 of the safety vent part 110 may be bent upward.

[0067] That is, since the connection portion 119 is bent upward, although the safety vent part 110 is moved upward due to the internal pressure within the can member 3, the rupture notch portion 113 is not easily cut, and the safety vent part 110 may be moved upward.

[0068] The rupture portion 111 may include an inclined section 111a which is inclined so that a height thereof gradually decreases from the connection portion 119 toward the bending notch portion 115.

[0069] Also, the rupture portion 111 may include a horizontal section 111b having a horizontal shape and extending from an end of the inclined section 111a toward the center of the rupture portion 111.

[0070] Also, the bending notch portion 115 may be positioned on a boundary between the inclined section 111a and the horizontal section 111b.

[0071] That is, as the inclined section 111a is inclined so that a height thereof gradually decreases toward the horizontal section 111b, an increase in the height of the connection

portion 119 by being bent upward is compensated. Thus, the horizontal section 111b of the rupture portion 111 is allowed to be in a lowermost position so as to maintain the horizontal level.

[0072] As described above, the horizontal section 111b of the rupture portion 111 may be positioned at the lowermost portion of the safety vent part 110 so as to be coupled to the interrupt portion 121 of the current interrupt device 120 installed at the lower side.

[0073] Also, when the bending notch portion 115 is positioned on the boundary between the inclined section 111a and the horizontal section 111b, it may be significantly effective in bending the bending notch portion 115 to secure an optimal spaced distance d to the top cap 101.

[0074] FIG. 7 is a bottom view illustrating only a safety vent part according to a second embodiment of the present invention.

[0075] As illustrated in FIG. 7, in a secondary battery according to the second embodiment of the present invention, a bending notch portion 115 of a safety vent part 110 may have a discontinuous circular or elliptical shape.

[0076] When the bending notch portion 115 has the discontinuous circular or elliptical shape, a recessed region of the bending notch portion 115 may be minimized, and thus the bending notch portion 115 may be effectively prevented from being cut when an internal pressure inside a can member 3 increases.

[0077] FIG. 8 is a bottom view illustrating only a safety vent part according to a third embodiment of the present invention, and FIG. 9 is a cross-sectional view taken along

line C-C of FIG. 8.

[0078] As illustrated in FIGS. 8 and 9, in a secondary battery according to a third embodiment of the present invention, a plurality of bending notch portions 115 of the safety vent part 110 are provided having different lengths.

[0079] In a case in which the plurality of bending notch portions 115 are provided having different lengths, even when the safety vent part 110 rapidly rises due to an instantaneous increase in an internal pressure within the can member 3, the plurality of bending notch portions 115 may be bent to mitigate a rising impact.

[0080] That is, the rupture notch portions 113 may be prevented from being cut, at a pressure before reaching a preset venting pressure, due to the instant and sudden rising impact.

[0081] FIG. 10 is provided as Comparative example. FIG. 10 is a view of a secondary battery in which a safety vent part according to the related art is installed, illustrating, on the same cross-section of FIG. 3, a state of use in which an interrupt portion is ruptured from a current interrupt device due to an increase in an internal pressure of a can member.

[0082] In the secondary battery according to the related art as illustrated in FIG. 10, only a rupture notch portion 113' is provided in a safety vent part 110' without the bending notch portion of the present invention.

[0083] In such a secondary battery according to the related art, the safety vent part 110' moves toward a top cap 101' when an internal pressure within a can member 3 increases. Here, a central portion of the safety vent part 110' farthest away from an edge of the safety vent part 110' coupled to the

can member 3 may rise to the uppermost side due to the internal pressure of a gas.

[0084] Thus, the central portion of the safety vent part 110' soars to the vicinity of the top cap 101', and approaches the top cap 101' until a spaced distance d_2 therebetween is close to approximately 123 μm . Furthermore, the central portion of the safety vent part 110' may come into contact with the top cap 101'.

[0085] A rupture notch is cut when the spaced distance d_2 between the safety vent part 110' and the top cap 101' is very small, or the safety vent part 110' comes into contact with the top cap 101' as described above. That is, since the rupture notch is not cut at a preset venting pressure at which an internal gas should be vented, the venting may not be performed stably.

[0086] In this case, safety problems such as explosion, heat generation, and ignition of a secondary battery may occur.

[0087] In the secondary battery according to the embodiment of the present invention as illustrated in FIG. 3, when the internal pressure within the can member 3 increases, the safety vent part 110 moves toward the top cap 101 while the inclined section 111a of the rupture portion 111 is bent upward. When the safety vent part 110 moves toward top cap 101, the bending notch portion 115 is bent and absorbs a portion of the internal pressure within the can member 3, and thus the horizontal section 111b of the rupture portion 111 may be prevented from soaring toward top cap 101.

[0088] Thus, a spaced distance d_1 between the safety vent part 110 and the top cap 101 may be sufficiently secured to about 186 μm .

[0089] As the spaced distance d_1 between the safety vent part 110 and the top cap 101 is secured as described above, the rupture notch portion 113 is cut at a preset vent pressure, which may prevent interference with the top cap 101 and stably vent the gas within the can member 3.

[0090] FIG. 11 shows the degrees of distribution for comparing the venting pressure distribution in the secondary battery in which the safety vent part according to the related art is installed and the venting pressure distribution in the secondary battery in which the safety vent part according to the present invention is installed.

[0091] In FIG. 11, V_0 is a distribution group of venting pressures, M_x is the maximum value of the venting pressures, M_n is the minimum value of the venting pressures, and each of V_1 and V_2 is a portion of the venting pressure distribution.

[0092] When comparing the venting pressures of the safety vent part according to the related art in which the bending notch portion is not provided and the venting pressures of the improved safety vent part according to the present invention in which the bending notch portion is provided with reference to FIG. 11, it may be found that the venting pressure distribution of the safety vent part according to the present invention is improved as compared to the venting pressure distribution of the safety vent part according to the related art. The difference therebetween may be about 1.5 times.

[0093] Particularly, during the venting of gas within the can member in the safety vent part according to the related art, the central portion of the safety vent part comes into contact with the top cap in a state in which the rupture notch portion is not cut, and thus the venting is not properly performed at

the preset venting pressure. The distribution in which the venting pressures greatly change during abnormal venting described above may be indicated as V_1 and V_2 .

[0094] According to the present invention as described above, the bending notch portion as well as the rupture notch portion is further provided in the bottom surface of the safety vent, and thus the safety vent is not interfered with the top cap when the internal pressure within the can member increases.

[0095] Also, since the safety vent is not interfered with the top cap, the rupture notch portion performs the venting at the preset venting pressure, and thus the venting pressure distribution may be improved.

[0096] Although the secondary battery according to the present invention has been described with reference to the exemplary drawings as described above, the present invention is not limited to the embodiments and drawings described and illustrated above, and various embodiments may be made within the scope of the appended claims by those skilled in the art pertaining to the present invention.

CLAIMS

1. A secondary battery comprising:
 - an electrode assembly (1);
 - a can member (3) configured to accommodate the electrode assembly (1) therein; and
 - a top cap assembly (100) configured to cover an opening of the can member (3),
 - wherein the top cap assembly (100) comprises a top cap (101) positioned at the uppermost portion and a safety vent part (100) positioned below the top cap (101),
 - wherein the safety vent part (110) comprises:
 - a rupture portion (111) ruptured when an internal pressure within the can member (3) increases to a predetermined pressure or more;
 - a rupture notch portion (113) that defines a boundary of the rupture portion (111); and
 - a bending notch portion (115) which is disposed at a position spaced apart from the rupture notch portion (113) toward a center of the rupture portion (111) on a bottom surface of the rupture portion (111) facing the inside of the can member (3),
 - wherein the bending notch portion (115) is bent when the internal pressure within the can member (3) increases.
2. The secondary battery of claim 1, wherein the rupture notch portion (113) is recessed from a lower surface of the safety vent part (110), and
 - the bending notch portion (115) is recessed from the lower surface of the safety vent part (110),

wherein a recessed depth of the bending notch portion (115) is less than that of the rupture notch portion (113).

3. The secondary battery of claim 1, wherein the rupture notch portion (113) is provided in the bottom surface of the rupture portion (111).

4. The secondary battery of claim 1, wherein the top cap assembly (100) further comprises a current interrupt device (120) provided with an interrupt portion (121) which is attached to the rupture portion (111) and ruptured to interrupt current when the internal pressure within the can member (3) increases.

5. The secondary battery of claim 4, wherein the interrupt portion (121) of the current interrupt device (120) is attached to the rupture portion (111) inside a circumference of the bending notch portion (115).

6. The secondary battery of claim 1, wherein the bending notch portion (115) has a continuous circular or elliptical shape.

7. The secondary battery of claim 1, wherein the bending notch portion (115) has a discontinuous circular or elliptical shape.

8. The secondary battery of claim 1, wherein the bending notch portion (115) is provided in plurality, and the plurality of bending notch portions have different lengths.

9. The secondary battery of claim 1, wherein the rupture notch portion (113) has a discontinuous circular or elliptical shape.

10. The secondary battery of claim 1, wherein a connection portion (119) between an edge portion (117) and the rupture portion (111) of the safety vent part (110) is bent upward.

11. The secondary battery of claim 10, wherein the rupture portion (111) comprises:

an inclined section (111a) inclined so that a height thereof gradually decreases from the connection portion (119) toward the bending notch portion (115); and

a horizontal section (111b) having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion (111),

wherein the bending notch portion (115) is positioned on a boundary between the inclined section (111a) and the horizontal section (111b).

12. The secondary battery of claim 1, wherein a spaced distance between an edge end of the safety vent part (110) and the rupture notch portion (113) is equal to a spaced distance between the rupture notch portion (113) and the bending notch portion (115).

ABSTRACT

The present invention relates to a secondary battery capable of improving venting pressure distribution. Provided is a secondary battery including an electrode assembly, a can member that accommodates the electrode assembly therein, and a top cap assembly that covers an opening of the can member. The top cap assembly includes a top cap positioned at the uppermost portion and a safety vent part positioned below the top cap. The safety vent part includes a rupture portion ruptured when an internal pressure within the can member increases to a predetermined pressure or more, a rupture notch portion that defines a boundary of the rupture portion, and a bending notch portion which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing the inside of the can member, and the bending notch portion is bent when the internal pressure within the can member increases.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2019/015396**A. CLASSIFICATION OF SUBJECT MATTER*****H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i***

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models: IPC as above

Japanese utility models and applications for utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: gas, notch, bending, fracture

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 05 August 1994 See paragraphs [0019]-[0022], claim 1 and figures 1-4.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 24 December 1996 See paragraph [0043] and figure 11.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 13 April 2001 See paragraphs [0016], [0017] and figures 1-5.	4-9
A	KR 10-2015-0035267 A (LG CHEM, LTD.) 06 April 2015 See the entire document.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 02 October 2018 See the entire document.	1-12



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

“A” document defining the general state of the art which is not considered to be of particular relevance

“E” earlier application or patent but published on or after the international filing date

“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

“O” document referring to an oral disclosure, use, exhibition or other means

“P” document published prior to the international filing date but later than the priority date claimed

“T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

“X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

“Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

“&” document member of the same patent family

Date of the actual completion of the international search

19 FEBRUARY 2020 (19.02.2020)

Date of mailing of the international search report

20 FEBRUARY 2020 (20.02.2020)

Name and mailing address of the ISA/KR

Korean Intellectual Property Office
Government Complex Daejeon Building 4, 189, Cheongsu-ro, Seo-gu,
Daejeon, 35298, Republic of Korea

Facsimile No. +82-42-481-8578

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.

PCT/KR2019/015396

Patent document cited in search report	Publication date	Patent family member	Publication date
JP 06-215746 A	05/08/1994	None	
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KR 10-2015-0035267 A	06/04/2015	KR 10-1667966 B1	20/10/2016
US 10090497 B2	02/10/2018	CN 107293687 A	24/10/2017
		KR 10-2017-0116370 A	19/10/2017
		US 2017-0294635 A1	12/10/2017

A. 발명이 속하는 기술분류(국제특허분류(IPC))

H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/02(2006.01)i

B. 조사된 분야

조사된 최소문헌(국제특허분류를 기재)

H01M 2/12; H01M 2/30; H01M 2/34; H01M 2/02

조사된 기술분야에 속하는 최소문헌 이외의 문헌

한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우))

eKOMPASS(특허청 내부 검색시스템) & 키워드: 가스(gas), 노치(notch), 벤딩(bending), 파단(fracture)

C. 관련 문헌

카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
Y	JP 06-215746 A (MATSUSHITA ELECTRIC IND CO., LTD.) 1994.08.05 단락 [0019]-[0022], 청구항 1 및 도면 1-4 참조.	1-12
Y	JP 08-339793 A (MITSUBISHI CABLE IND LTD.) 1996.12.24 단락 [0043] 및 도면 11 참조.	1-12
Y	JP 2001-102038 A (TOYO KOHAN CO., LTD.) 2001.04.13 단락 [0016],[0017] 및 도면 1-5 참조.	4-9
A	KR 10-2015-0035267 A (주식회사 엘지화학) 2015.04.06 전체 문헌 참조.	1-12
A	US 10090497 B2 (SAMSUNG SDI CO., LTD.) 2018.10.02 전체 문헌 참조.	1-12

☐ 추가 문헌이 C(계속)에 기재되어 있습니다.

☒ 대응특허에 관한 별지를 참조하십시오.

* 인용된 문헌의 특별 카테고리:

“A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌

“D” 본 국제출원에서 출원인이 인용한 문헌

“E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후 “X”에 공개된 선출원 또는 특허 문헌

“L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌

“O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌

“P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌

“T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌

“X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다.

“Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다.

“&” 동일한 대응특허문헌에 속하는 문헌

국제조사의 실제 완료일

2020년 02월 19일 (19.02.2020)

국제조사보고서 발송일

2020년 02월 20일 (20.02.2020)

ISA/KR의 명칭 및 우편주소



대한민국 특허청
(35208) 대전광역시 서구 청사로 189,
4동 (둔산동, 정부대전청사)

팩스 번호 +82-42-481-8578

심사관

김연경

전화번호 +82-42-481-3325



국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
JP 06-215746 A	1994/08/05	없음	
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KR 10-2015-0035267 A	2015/04/06	KR 10-1667966 B1	2016/10/20
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		KR 10-2017-0116370 A	2017/10/19
		US 2017-0294635 A1	2017/10/12



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U.S. APPLICATION NO.	FIRST NAMED INVENTOR	ATTY. DOCKET NO.
17/259,395	Geon Woo Min	LGCHEM 3.3F-3059

86765
LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

INTERNATIONAL APPLICATION NO.	
PCT/KR2019/015396	
I.A. FILING DATE	PRIORITY DATE
11/13/2019	11/28/2018

CONFIRMATION NO. 1017
371 ACCEPTANCE LETTER



Date Mailed: 05/11/2021

NOTICE OF ACCEPTANCE OF APPLICATION UNDER 35 U.S.C 371 AND 37 CFR 1.495

The applicant is hereby advised that the United States Patent and Trademark Office, in its capacity as a Designated / Elected Office (37 CFR 1.495), has ACCEPTED the above identified international application for national patentability examination in the United States Patent and Trademark Office.

The United States Application Number assigned to the application is shown above. A Filing Receipt will be issued for the present application in due course. **THE DATE APPEARING ON THE FILING RECEIPT AS THE "FILING DATE or 371(c) DATE" IS THE DATE ON WHICH THE LAST OF THE 35 U.S.C. 371 (c)(1) and (c)(2) REQUIREMENTS HAS BEEN RECEIVED IN THE OFFICE. THIS DATE IS SHOWN BELOW.** The filing date of the above identified application is the international filing date of the international application (Article 11(3) and 35 U.S.C. 363)

01/11/2021
DATE OF RECEIPT OF 35 U.S.C.
371(c)(1) and (c)(2) REQUIREMENTS

The following items have been received:

- Copy of the International Application filed on 01/11/2021
- English Translation of the IA filed on 01/11/2021
- Copy of the International Search Report filed on 01/11/2021
- Preliminary Amendments filed on 01/11/2021
- Information Disclosure Statements filed on 01/11/2021
- Inventor's Oath or Declaration filed on 01/11/2021
- Request for Immediate Examination filed on 01/11/2021
- U.S. Basic National Fees filed on 01/11/2021
- Substitute Specification filed on 01/11/2021
- Assignment filed on 01/11/2021
- Assignee Statement for PGPUB filed on 01/11/2021
- Authorize Access to Search Results filed on 01/11/2021
- Priority Documents filed on 01/11/2021
- Power of Attorney filed on 01/11/2021
- Authorization to Permit Access filed on 01/11/2021
- Application Data Sheet (37 CFR 1.76) filed on 01/11/2021

Applicant is reminded that any communications to the United States Patent and Trademark Office must be mailed to the address given in the heading and include the U.S. application no. shown above (37 CFR 1.5)

TRAYNA M LAWRENCE

Telephone: (703) 756-1210



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/259,395	01/11/2021		1660	LGCHEM 3.3F-3059	12	1

CONFIRMATION NO. 1017

FILING RECEIPT

86765

LGCHEM

Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016



0000000125461674

Date Mailed: 05/11/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Geon Woo Min, Daejeon, KOREA, REPUBLIC OF;
Je Jun Lee, Daejeon, KOREA, REPUBLIC OF;
Sang Suk Jung, Daejeon, KOREA, REPUBLIC OF;

Applicant(s)

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF;

Assignment For Published Patent Application

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF

Power of Attorney: The patent practitioners associated with Customer Number 86765

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/KR2019/015396 11/13/2019

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

REPUBLIC OF KOREA 10-2018-0149369 11/28/2018

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 05/08/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/259,395**

Projected Publication Date: 08/19/2021

Non-Publication Request: No

Early Publication Request: No

Title

Secondary Battery

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/259,395

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	12 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	320
N/A	540
N/A	800
x 100 =	0.00
x 480 =	0.00
	0.00
	0.00
TOTAL	1660

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

MULTIPLE DEPENDENT CLAIM FEE CALCULATION SHEET Substitute for Form PTO-1360 (For use with Form PTO/SB/06)							Application Number		Filing Date				
							17259395						
							Applicant(s) Geon Woo Min						
							* May be used for additional claims or amendments						
CLAIMS	AS FILED		AFTER FIRST AMENDMENT		AFTER SECOND AMENDMENT			*		*		*	
	Indep	Depend	Indep	Depend	Indep	Depend		Indep	Depend	Indep	Depend	Indep	Depend
1	1		1										
2		1		1									
3		1		1									
4		1		1									
5		1		1									
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Total Indep	1		1		0								
Total Depend	11		11		0								
Total Claims	12		12		0								

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To: eOfficeAction@lernerdavid.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 86765

May 11, 2021 04:33:49 AM

Dear PAIR Customer:

LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 86765 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

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The list of documents shown below is provided as a courtesy and is not part of the official file wrapper. The content of the images shown in PAIR is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
17259395	M903	05/11/2021	LGCHEM 3.3F-3059
	APP.FILE.REC	05/11/2021	LGCHEM 3.3F-3059

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/259,395

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	540
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(j))	12	minus 20 = *			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1	minus 3 = *				x 480 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
			TOTAL			TOTAL	1660

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

U.S. APPLICATION NO.	FIRST NAMED INVENTOR	ATTY. DOCKET NO.
17/259,395	Geon Woo Min	LGCHEM 3.3F-3059

86765
LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

INTERNATIONAL APPLICATION NO.	
PCT/KR2019/015396	
I.A. FILING DATE	PRIORITY DATE
11/13/2019	11/28/2018

CONFIRMATION NO. 1017
371 FORMALITIES LETTER



OC000000127003052

Date Mailed: 07/15/2021

NOTIFICATION OF MISSING REQUIREMENTS UNDER 35 U.S.C. 371 IN THE UNITED STATES DESIGNATED/ELECTED OFFICE (DO/EO/US)

The following items have been submitted by the applicant or the International Bureau to the United States Patent and Trademark Office as a Designated / Elected Office (37 CFR 1.495).

- Priority Document
- Copy of the International Application filed on 01/11/2021
- English Translation of the IA filed on 01/11/2021
- Copy of the International Search Report filed on 01/11/2021
- Preliminary Amendments filed on 01/11/2021
- Information Disclosure Statements filed on 01/11/2021
- Inventor's Oath or Declaration filed on 01/11/2021
- Request for Immediate Examination filed on 01/11/2021
- U.S. Basic National Fees filed on 01/11/2021
- Substitute Specification filed on 01/11/2021
- Assignment filed on 01/11/2021
- Assignee Statement for PGPUB filed on 01/11/2021
- Authorize Access to Search Results filed on 01/11/2021
- Priority Documents filed on 01/11/2021
- Power of Attorney filed on 01/11/2021
- Authorization to Permit Access filed on 01/11/2021
- Application Data Sheet (37 CFR 1.76) filed on 01/11/2021

The following items **MUST** be furnished within the period set forth below:

- Translation of the application into English. The current translation of the application into English is defective as described below. Note a processing fee will be required if submitted later than 30 months from the priority date.
 - The text in the drawings has not been properly translated.
- Processing fee of \$140 for providing the translation of the application and/or the Annexes later than 30 months from the priority date (37 CFR 1.492(i)).

SUMMARY OF FEES DUE:

Total additional fees required for this application is **\$140** for an Undiscounted Entity:

- **\$140** for English translation surcharge required.

ALL OF THE ITEMS SET FORTH ABOVE MUST BE SUBMITTED WITHIN TWO (2) MONTHS FROM THE DATE OF THIS NOTICE OR BY 32 MONTHS FROM THE PRIORITY DATE FOR THE APPLICATION, WHICHEVER IS LATER. FAILURE TO PROPERLY RESPOND WILL RESULT IN ABANDONMENT.

The time period set above may be extended by filing a petition and fee for extension of time under the provisions of 37 CFR 1.136(a).

Applicant is reminded that any communications to the United States Patent and Trademark Office must be mailed to the address given in the heading and include the U.S. application no. shown above (37 CFR 1.5)

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.

<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

TRAYNA M LAWRENCE

Telephone: (703) 756-1210



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

U.S. APPLICATION NO.	FIRST NAMED APPLICANT	ATTY. DOCKET NO.
17/259,395	Geon Woo Min	LGCHEM 3.3F-3059

86765
LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

INTERNATIONAL APPLICATION NO.	
PCT/KR2019/015396	
I.A. FILING DATE	PRIORITY DATE
11/13/2019	11/28/2018

CONFIRMATION NO. 1017
371 WITHDRAWAL NOTICE



Date Mailed: 07/15/2021

Letter Regarding a New Notice and/or the Status of the Application

If a new notice or Filing Receipt is enclosed, applicant may disregard the previous notice mailed on 05/11/2021. The time period for reply runs from the mail date of the new notice. Within the time period for reply, applicant is required to file a reply in compliance with the requirements set forth in the new notice to avoid abandonment of the application.

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at
1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If the reply is not filed electronically via EFS-Web, the reply must be accompanied by a copy of the new notice.

If the Office previously granted a petition to withdraw the holding of abandonment or a petition to revive under 37 CFR 1.137, the status of the application has been returned to pending status.

TRAYNA M LAWRENCE

Telephone: (703) 756-1210

To: eOfficeAction@lernerdavid.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 86765

Jul 15, 2021 04:08:55 AM

Dear PAIR Customer:

LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 86765 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

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Application	Document	Mailroom Date	Attorney Docket No.
17259395	M905	07/15/2021	LGCHEM 3.3F-3059
	PEFN	07/15/2021	LGCHEM 3.3F-3059

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If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: July 16, 2021
Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: January 11, 2021

Art Unit: N/A

For: Secondary Battery

Examiner: Not Yet Assigned

RESPONSE TO NOTICE TO FILE MISSING PARTS

MAIL STOP MISSING PARTS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

In response to the Notice to File Missing Parts, attached please find an English translation of the drawings, which was inadvertently omitted at the time the above-referenced application was filed.

If there are any questions concerning this application, the Examiner is encouraged to call Applicant's attorney at (908) 654-5000. All required fees, if any, are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments, to our Deposit Account No. 12-1095.

Dated: July 16, 2021

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER, DAVID, LITTENBERG,
KRUMHOLZ & MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant

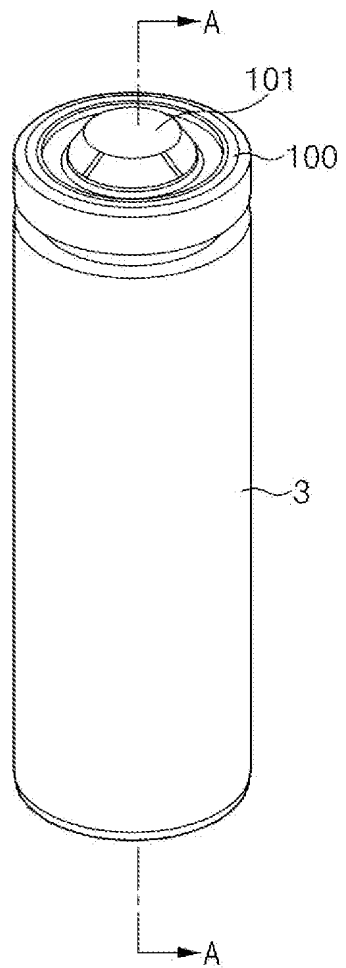


FIG. 1

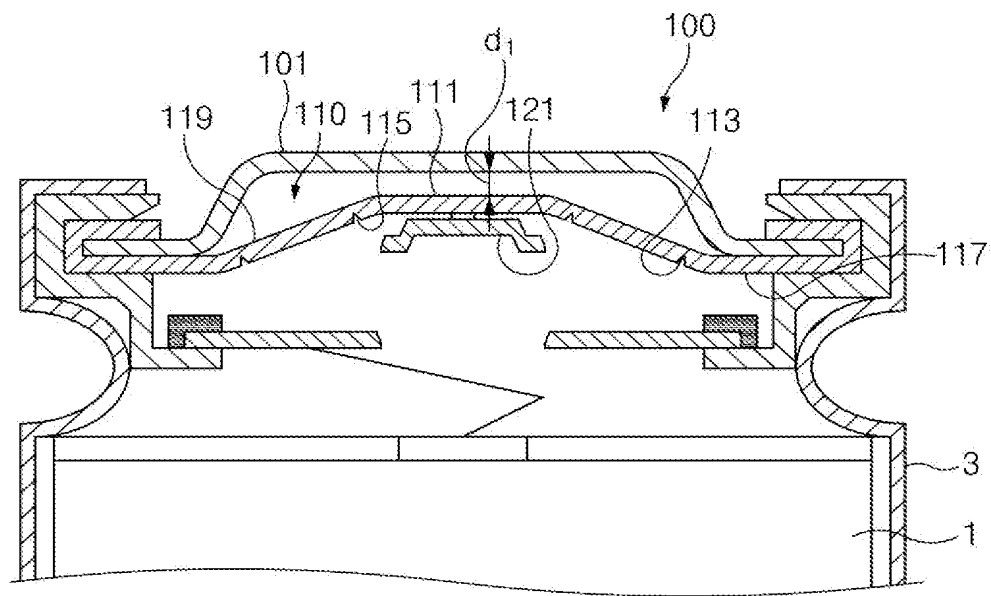


FIG. 3

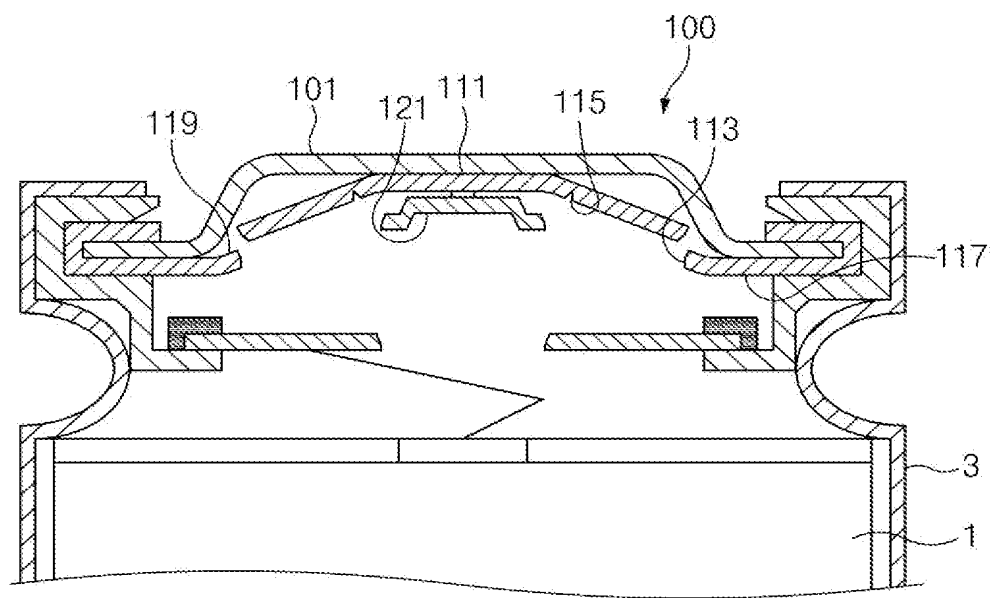


FIG. 4

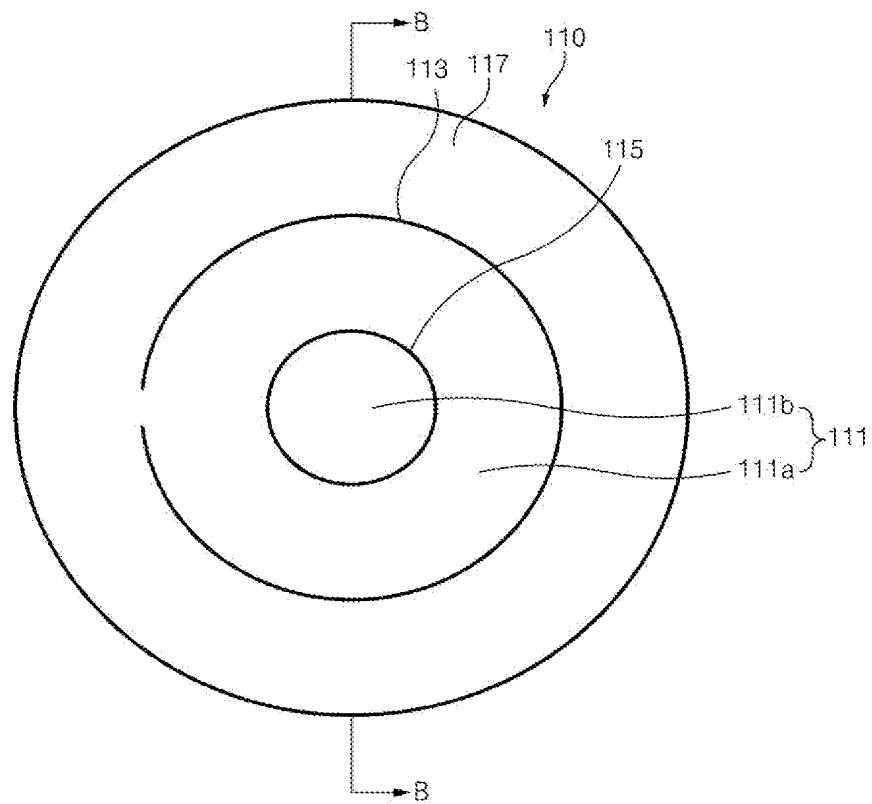


FIG. 5

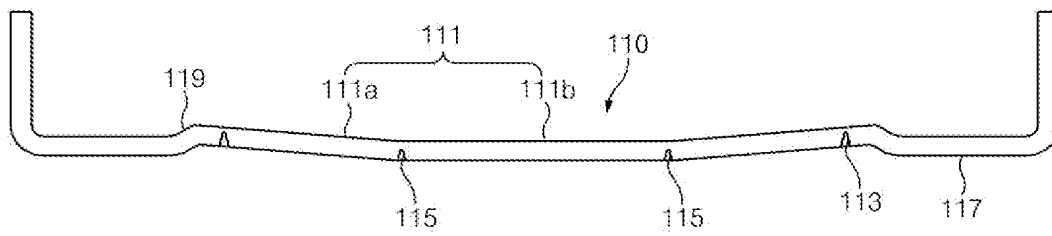


FIG. 6

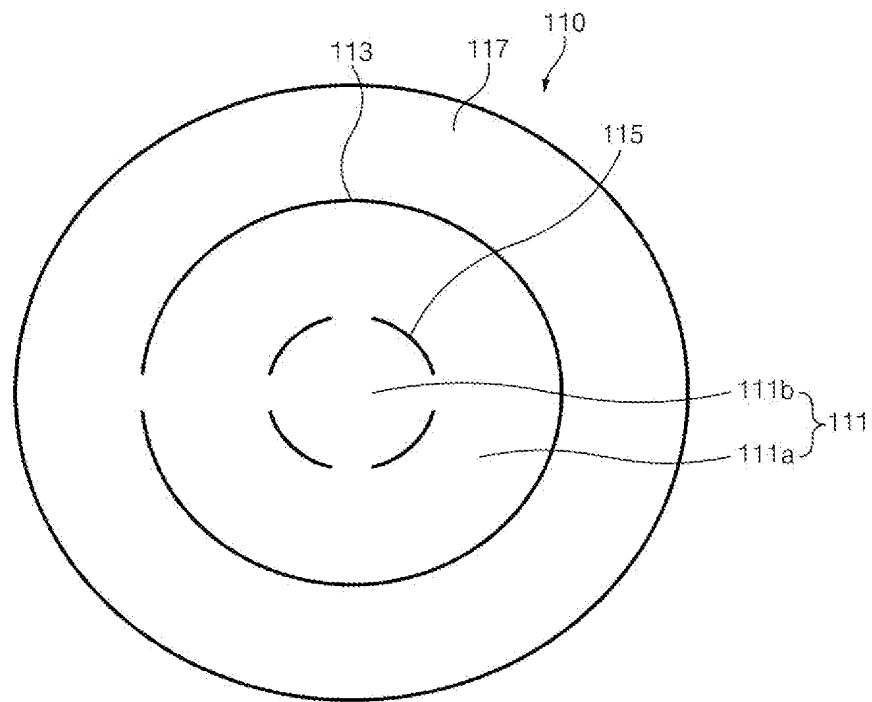


FIG. 7

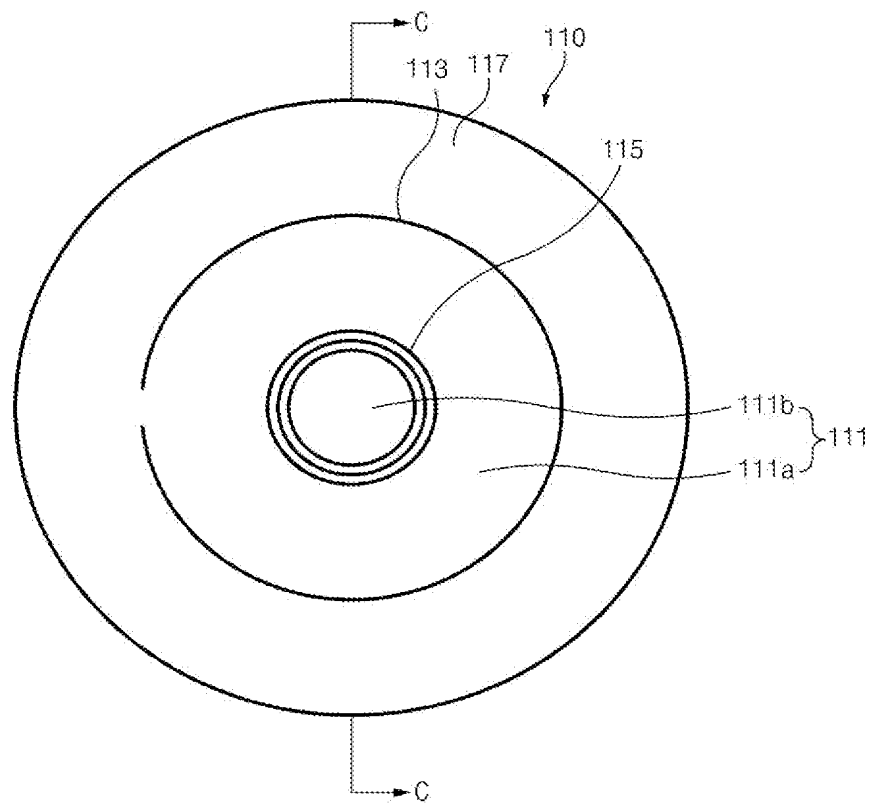


FIG. 8

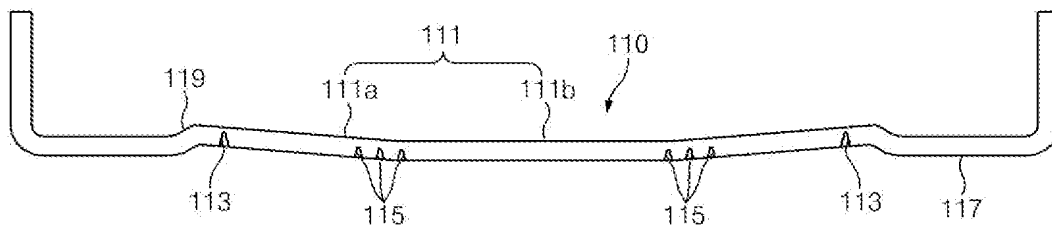


FIG. 9

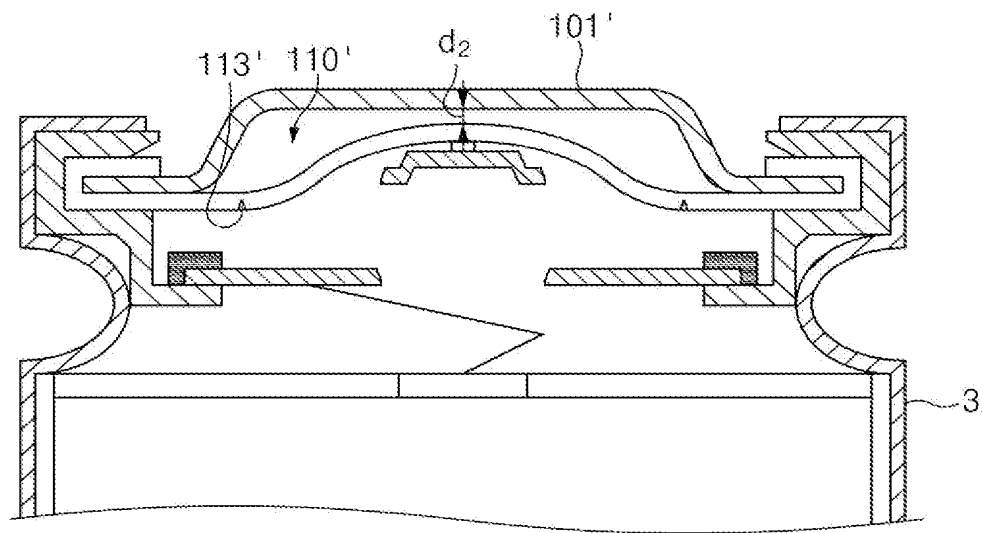


FIG. 10

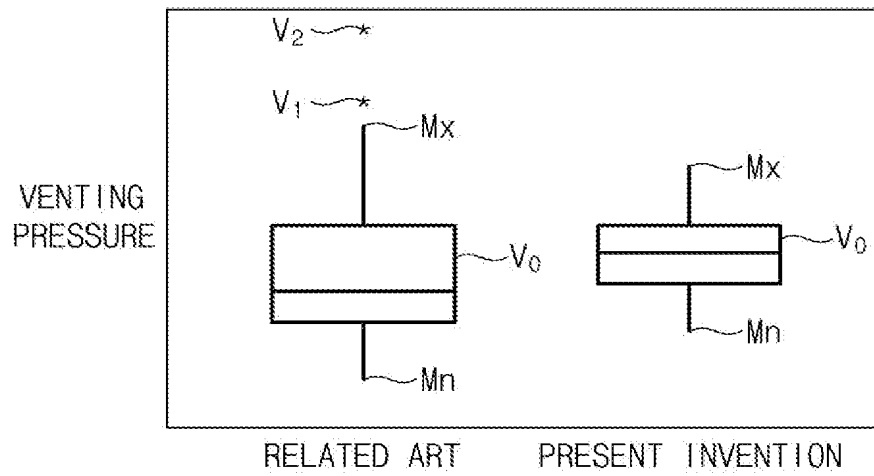


FIG. 11



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United States Patent and Trademark Office
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P.O. Box 1450
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U.S. APPLICATION NO.	FIRST NAMED INVENTOR	ATTY. DOCKET NO.
17/259,395	Geon Woo Min	LGCHEM 3.3F-3059

86765

LGCHEM

Lerner, David, Littenberg, Krumholz & Mentlik, LLP

20 Commerce Drive

Cranford, NJ 07016

INTERNATIONAL APPLICATION NO.

PCT/KR2019/015396

I.A. FILING DATE

11/13/2019

PRIORITY DATE

11/28/2018

CONFIRMATION NO. 1017
371 FORMALITIES LETTER



OC000000127003052

Date Mailed: 07/15/2021

NOTIFICATION OF MISSING REQUIREMENTS UNDER 35 U.S.C. 371 IN THE UNITED STATES DESIGNATED/ELECTED OFFICE (DO/EO/US)

The following items have been submitted by the applicant or the International Bureau to the United States Patent and Trademark Office as a Designated / Elected Office (37 CFR 1.495).

- Priority Document
- Copy of the International Application filed on 01/11/2021
- English Translation of the IA filed on 01/11/2021
- Copy of the International Search Report filed on 01/11/2021
- Preliminary Amendments filed on 01/11/2021
- Information Disclosure Statements filed on 01/11/2021
- Inventor's Oath or Declaration filed on 01/11/2021
- Request for Immediate Examination filed on 01/11/2021
- U.S. Basic National Fees filed on 01/11/2021
- Substitute Specification filed on 01/11/2021
- Assignment filed on 01/11/2021
- Assignee Statement for PGPUB filed on 01/11/2021
- Authorize Access to Search Results filed on 01/11/2021
- Priority Documents filed on 01/11/2021
- Power of Attorney filed on 01/11/2021
- Authorization to Permit Access filed on 01/11/2021
- Application Data Sheet (37 CFR 1.76) filed on 01/11/2021

The following items **MUST** be furnished within the period set forth below:

- Translation of the application into English. The current translation of the application into English is defective as described below. Note a processing fee will be required if submitted later than 30 months from the priority date.
 - The text in the drawings has not been properly translated.
- Processing fee of \$140 for providing the translation of the application and/or the Annexes later than 30 months from the priority date (37 CFR 1.492(i)).

SUMMARY OF FEES DUE:

Total additional fees required for this application is **\$140** for an Undiscounted Entity:

- **\$140** for English translation surcharge required.

ALL OF THE ITEMS SET FORTH ABOVE MUST BE SUBMITTED WITHIN TWO (2) MONTHS FROM THE DATE OF THIS NOTICE OR BY 32 MONTHS FROM THE PRIORITY DATE FOR THE APPLICATION, WHICHEVER IS LATER. FAILURE TO PROPERLY RESPOND WILL RESULT IN ABANDONMENT.

The time period set above may be extended by filing a petition and fee for extension of time under the provisions of 37 CFR 1.136(a).

Applicant is reminded that any communications to the United States Patent and Trademark Office must be mailed to the address given in the heading and include the U.S. application no. shown above (37 CFR 1.5)

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TRAYNA M LAWRENCE

Telephone: (703) 756-1210

Electronic Patent Application Fee Transmittal

Application Number:	17259395			
Filing Date:				
Title of Invention:	Secondary Battery			
First Named Inventor/Applicant Name:	Geon Woo Min			
Filer:	Stephen M. Lund/Cindy Corb			
Attorney Docket Number:	LGCHEM 3.3F-3059			
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
ENG. TRANSL. > 30 MO. FROM PRIORITY DATE	1618	1	140	140
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				140

Electronic Acknowledgement Receipt

EFS ID:	43269197
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Stephen M. Lund/Cindy Corb
Filer Authorized By:	Stephen M. Lund
Attorney Docket Number:	LGCHEM 3.3F-3059
Receipt Date:	16-JUL-2021
Filing Date:	
Time Stamp:	11:41:36
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 140
RAM confirmation Number	E20217FB42102947
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	Response_to_Notice_to_File_Missing_Parts_Dec_and_Fees_LD-468b.pdf	21113	no	1
			1c67c1cecf2f3c1d31aa46cd73ea8f492168489		
Warnings:					
Information:					
2	Drawings-only black and white line drawings	Formal_Drawings.PDF	2461833	no	11
			e8981155c5ae590aa0c128894999313943e60249		
Warnings:					
Information:					
3	Miscellaneous Incoming Letter	Notice_of_DO_EO_Missing_Requirements_Mailed-_mailed_7_15_2021.PDF	71422	no	2
			ba1937eed3e534da531bcb129297cc577f406664		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	30607	no	2
			59b1230756e618af2ec2c61a1536875904eb78b6		
Warnings:					
Information:					
Total Files Size (in bytes):			2584975		

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/259,395	07/16/2021	1725	1800	LGCHEM 3.3F-3059	12	1

CONFIRMATION NO. 1017

FILING RECEIPT

86765

LGCHEM

Lerner, David, Littenberg, Krumholz & Mentlik, LLP

20 Commerce Drive

Cranford, NJ 07016



0000000127054624

Date Mailed: 07/20/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Geon Woo Min, Daejeon, KOREA, REPUBLIC OF;
Je Jun Lee, Daejeon, KOREA, REPUBLIC OF;
Sang Suk Jung, Daejeon, KOREA, REPUBLIC OF;

Applicant(s)

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF;

Assignment For Published Patent Application

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF

Power of Attorney: The patent practitioners associated with Customer Number 86765

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/KR2019/015396 11/13/2019

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

REPUBLIC OF KOREA 10-2018-0149369 11/28/2018

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 05/08/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/259,395**

Projected Publication Date: 10/28/2021

Non-Publication Request: No

Early Publication Request: No
Title

Secondary Battery

Preliminary Class

429

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 17/259,395				
APPLICATION AS FILED - PART I										
(Column 1)		(Column 2)		SMALL ENTITY		OTHER THAN SMALL ENTITY				
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)			
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320			
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	540			
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800			
TOTAL CLAIMS (37 CFR 1.16(j))	12	minus 20 = *			OR	x 100 =	0.00			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1	minus 3 = *				x 480 =	0.00			
APPLICATION SIZE FEE (37 CFR 1.16(s)) If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							0.00			
							0.00			
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00			
			TOTAL			TOTAL	1660			
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED - PART II										
(Column 1)		(Column 2)		(Column 3)		SMALL ENTITY		OTHER THAN SMALL ENTITY		
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =	
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =	
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.										



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U.S. APPLICATION NO.	FIRST NAMED INVENTOR	ATTY. DOCKET NO.
17/259,395	Geon Woo Min	LGCHEM 3.3F-3059

86765
LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

INTERNATIONAL APPLICATION NO.	
PCT/KR2019/015396	
I.A. FILING DATE	PRIORITY DATE
11/13/2019	11/28/2018

CONFIRMATION NO. 1017
371 ACCEPTANCE LETTER



Date Mailed: 07/20/2021

NOTICE OF ACCEPTANCE OF APPLICATION UNDER 35 U.S.C 371 AND 37 CFR 1.495

The applicant is hereby advised that the United States Patent and Trademark Office, in its capacity as a Designated / Elected Office (37 CFR 1.495), has ACCEPTED the above identified international application for national patentability examination in the United States Patent and Trademark Office.

The United States Application Number assigned to the application is shown above. A Filing Receipt will be issued for the present application in due course. **THE DATE APPEARING ON THE FILING RECEIPT AS THE "FILING DATE or 371(c) DATE" IS THE DATE ON WHICH THE LAST OF THE 35 U.S.C. 371 (c)(1) and (c)(2) REQUIREMENTS HAS BEEN RECEIVED IN THE OFFICE. THIS DATE IS SHOWN BELOW.** The filing date of the above identified application is the international filing date of the international application (Article 11(3) and 35 U.S.C. 363)

07/16/2021
DATE OF RECEIPT OF 35 U.S.C.
371(c)(1) and (c)(2) REQUIREMENTS

The following items have been received:

- Copy of the International Application filed on 01/11/2021
- English Translation of the IA filed on 07/16/2021
- Copy of the International Search Report filed on 01/11/2021
- Preliminary Amendments filed on 01/11/2021
- Information Disclosure Statements filed on 01/11/2021
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- Power of Attorney filed on 01/11/2021
- Authorization to Permit Access filed on 01/11/2021
- Application Data Sheet (37 CFR 1.76) filed on 01/11/2021

Applicant is reminded that any communications to the United States Patent and Trademark Office must be mailed to the address given in the heading and include the U.S. application no. shown above (37 CFR 1.5)

TRAYNA M LAWRENCE

Telephone: (703) 756-1210

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From: PAIR_eOfficeAction@uspto.gov
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Jul 20, 2021 03:42:31 AM

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Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
17259395	APP.FILE.REC	07/20/2021	LGCHEM 3.3F-3059
	M903	07/20/2021	LGCHEM 3.3F-3059

To view your correspondence online or update your email addresses, please visit us anytime at <https://portal.uspto.gov/secure/myportal/privatepair>.

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/259,395	07/16/2021	Geon Woo Min	LGCHEM 3.3F-3059

CONFIRMATION NO. 1017

PUBLICATION NOTICE



OC000000129432001

86765
LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

Title:Secondary Battery

Publication No.US-2021-0336309-A1

Publication Date:10/28/2021

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

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To: eOfficeAction@lernerdavid.com,,
From: PAIR_eOfficeAction@uspto.gov
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Oct 29, 2021 04:56:22 AM

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LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016
UNITED STATES

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The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

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Application	Document	Mailroom Date	Attorney Docket No.
17259395	NTC.PUB	10/28/2021	LGCHEM 3.3F-3059

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	Filing Date	2021-07-16
	First Named Inventor	Geon Woo Min
	Art Unit	
	Examiner Name	Not Yet Assigned
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	2	20170124343	KR	A	2017-11-10	Lg Chemical Ltd		×
	3	2018030836	WO	A1	2018-02-15	Shin Heung Energy & Electronic Co.,Ltd		□

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17259395
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**INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: July 16, 2021

Art Unit: Not Yet Assigned

For: Secondary Battery

Examiner: Not Yet Assigned

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Dated: November 1, 2021

Respectfully submitted,

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/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant(s)

LD-458A

Electronic Acknowledgement Receipt

EFS ID:	44171975
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Stephen M. Lund/Kathleen Reynolds
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Application Type:	U.S. National Stage under 35 USC 371

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(72) Inventor: Kim, Jin-Hee
Gyeonggi-do (KR)

(30) Priority: 02.10.2008 US 102275 P

(74) Representative: Gulde Hengelhaupt Ziebig &
Schneider
Patentanwälte - Rechtsanwälte
Wallstraße 58/59
10179 Berlin (DE)

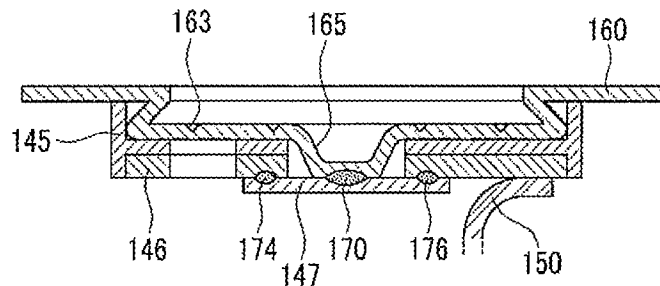
(71) Applicant: Samsung SDI Co., Ltd.
Gyeonggi-do (KR)

(54) Rechargeable battery

(57) A rechargeable battery according to the present invention includes an electrode assembly that includes a positive electrode, a negative electrode, and a separator between the positive electrode and the negative electrode, a case that contains the electrode assembly and a cap assembly that is connected to the case. The positive electrode includes a current collector and a positive active material layer that includes an NCM-based

positive active material disposed at the collector. The cap assembly includes a vent plate having a notch, and a current interrupt portion that interrupts electrical connection in a case of an increase in pressure inside the case. When a current interrupt pressure of the current interrupt portion is "A," a vent fracture pressure at which the vent plate fractures at the notch is "B," and the capacity of the rechargeable battery is "X," they satisfy a formula of $A/B \approx 1.22 - 0.39X + D$ ($-0.08 \leq D \leq 0.08$).

FIG.2A



EP 2 175 507 A1

FIG.2B

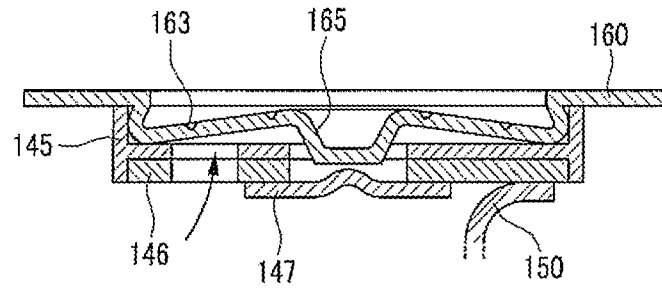
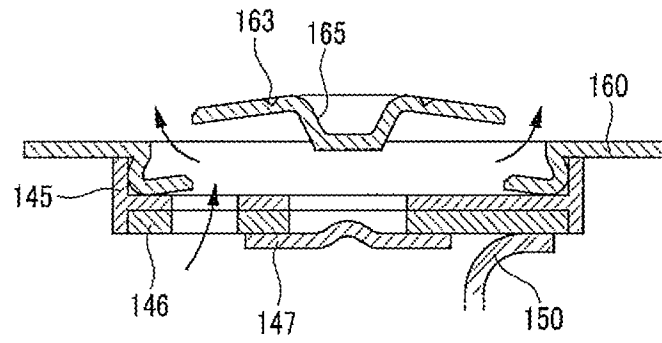


FIG.2C



Description

BACKGROUND OF THE INVENTION

1. Field of the invention

[0001] The present invention relates to a rechargeable battery. More particularly, the present invention relates to a rechargeable battery that includes a cap assembly.

2. Description of the Related Art

[0002] Unlike a primary battery, which cannot be recharged, a rechargeable battery can be repeatedly recharged and discharged. A low capacity rechargeable battery (e.g., a rechargeable battery composed of one cell) can be utilized as a power source for driving a small and portable electronic device, such as a mobile phone, a laptop computer and a camcorder. A large capacity rechargeable battery (e.g., a rechargeable battery composed of a plurality of cells connected to form a pack) can be utilized as a power source for driving a motor of a hybrid vehicle.

[0003] Rechargeable batteries are produced in various suitable shapes, and typically in a shape of a cylinder or polygon.

[0004] Further, a plurality of rechargeable batteries (or battery cells) can be serially connected and included as part of a high capacity rechargeable battery module such that the module can be used for driving a motor of an electric vehicle that needs a large amount of electrical power.

[0005] A rechargeable battery is composed of an electrode assembly having a positive electrode, a negative electrode, and a separator between the positive electrode and the negative electrode; a case containing the electrode assembly; and a cap assembly for closing an opening side of the case.

[0006] In the case of a cylindrical rechargeable battery, the positive electrode and the negative electrode include parts in the electrode assembly that are not coated with an active material. The positive electrode and negative electrode uncoated parts face opposite directions.

[0007] A negative current collector plate is attached to the negative uncoated part, and a positive current collector plate is attached to the positive uncoated part. The negative current collector plate is electrically connected to the case, and the positive electrode current collector plate is electrically connected to the cap assembly and is for supplying the current out of the electrode assembly. Therefore, the case operates as a negative terminal, and a cap-up disposed at the cap assembly operates as a positive terminal.

[0008] While a rechargeable battery is repeatedly recharged and discharged, a gas builds up inside the rechargeable battery and internal pressure rises. Thus, if unchecked, the increased internal pressure may cause an explosion of the rechargeable battery. In order to protect from (or prevent) the explosion, a vent plate is provided under the cap-up, having a notch, such that the vent plate fractures at a certain level of pressure.

[0009] The vent plate includes a convex portion facing downward, and the convex portion is welded to a sub-plate that is electrically connected to the electrode assembly. When the pressure rises inside the rechargeable battery, the convex portion is first disengaged from the sub-plate, interrupting electric current flow between the vent plate and the sub-plate. If the pressure further increases, the vent plate fractures at the notch, thereby discharging gas to the outside.

[0010] A current interrupt pressure (at which electric current is interrupted) and a vent fracture pressure (at which the notch at the vent plate) is fractured are important elements in the aspects of reliability and safety of the rechargeable battery.

[0011] That is, if the current interrupt pressure and the vent fracture pressure are set to be low, the safety of the battery increases but this leads to a problem that the battery does not operate even with a small increase in internal pressure.

[0012] Particularly, a rechargeable battery having a lithium-nickel-cobalt-manganese oxide (LiNiCoMnO_2 , hereinafter "NCM")-based positive active material, shows a different pressure-behavior than a rechargeable battery with a lithium cobalt oxide (LiCoO_2 , hereinafter "LCO")-based positive active material. As such, the rechargeable battery with an NCM-based positive active material should have different current interrupt pressure and vent fracture pressure than those of a rechargeable battery with an LCO-based positive active material.

[0013] If the current interrupt pressure and vent fracture pressure derived for an LCO-based positive active material are directly applied to a battery with an NCM-based positive active material, there will be a problem of reduced reliability and safety of the rechargeable battery.

SUMMARY OF THE INVENTION

[0014] An aspect of an embodiment of the present invention is directed toward a rechargeable battery with improved reliability and safety.

[0015] According to the present invention there is provided a rechargeable battery comprising:

an electrode assembly comprising a positive electrode, a negative electrode, and a separator between the positive electrode and the negative electrode, the positive electrode comprising a current collector and a positive active material coupled to the current collector, wherein the positive active material is a lithium-nickel-cobalt-manganese oxide (NCM)-based positive active material;

a case containing the electrode assembly; and

a cap assembly coupled to the case,

the cap assembly comprising a vent plate having a notch configured to fracture the vent plate due to a predefined vent fracture pressure B inside the case and a current interrupt portion configured to interrupt electrical connection due to a predetermined current interrupt pressure A inside the case,

characterized in that

a ratio of the current interrupt pressure A and the vent fracture pressure B satisfies the following equation:

$$A/B = 1.22 - 0.39X + D,$$

wherein

A represents the current interrupt pressure measured in kgf/cm² of the current interrupt portion;

B represents the vent fracture pressure measured in kgf/cm² of the notch,

X represents a capacity of the rechargeable battery measured in Ah, and

D is a factor within the range of $-0.08 \leq D \leq 0.08$.

[0016] Preferred embodiments of the invention are subject matter of the dependent claims.

[0017] In one embodiment, the cap assembly further includes a cap-up, a sub-plate, a cap-down, and a protrusion. The cap-up is coupled to the case. The vent plate is between the cap-up and the sub-plate and includes a convex portion facing a bottom end of the case. The sub-plate is welded to the convex portion, and electrically connected to the electrode assembly. The cap-down is between the vent plate and the sub-plate, and the protrusion is between the cap-down and the sub-plate. Here, the protrusion may include a welded portion for welding the sub-plate and the cap-down together. The rechargeable battery may further include a lead member electrically connected to the electrode assembly, and the cap-down may be electrically connected to the electrode assembly via the lead member. The sub-plate may be electrically connected to the electrode assembly via the cap-down. In addition, the rechargeable battery may further include an insulating plate between the cap-down and the vent plate. The current interrupt portion may include a welded portion for welding the convex portion and the sub-plate together.

[0018] In one embodiment, the rechargeable battery has a cylindrical shape.

[0019] In one embodiment, the active material has an overcharge additive in an amount between 0.5 and 2.0 wt% of the positive active material. The overcharge additive may include a carbon compound. The carbon compound may include lithium carbonate.

[0020] In one embodiment, the current interrupt pressure of the current interrupt portion is between 8 kgf/cm² and 9 kgf/cm².

[0021] In one embodiment, the vent fracture pressure of the notch is between 20 kgf/cm² and 25 kgf/cm².

[0022] In one embodiment, the current interrupt pressure of the current interrupt portion is between 8 kgf/cm² and 9 kgf/cm², and the vent fracture pressure of the notch is between 20 kgf/cm² and 25 kgf/cm².

[0023] In one embodiment, the positive active material is coated on the current collector. The current collector may be made of aluminum.

[0024] As such, according to an embodiment of the present invention, a rechargeable battery with an NCM-based positive active material has improved safety because the electric current is interrupted and the vent is fractured at the correct moment.

[0025] In addition, according to an embodiment of the present invention, reliability is enhanced because early interruption of electric current and fracture of the vent are reduced (or prevented).

[0026] Moreover, according to an embodiment of the present invention, reliability and safety are further improved due to an overcharge additive in the positive active material.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027]

FIG. 1 is a schematic exploded cross-sectional perspective view of a rechargeable battery according to an embodiment of the present invention.

FIG. 2A to FIG. 2C are schematic cross-sectional views of processes of interruption of electric current and fracture of a vent.

FIG. 3 is a graph showing changes in pressure inside a rechargeable battery with an LCO-based positive active material and a rechargeable battery with an NCM-based positive active material with time when the batteries are fully charged and kept at 90°C.

FIG. 4 is a graph illustrating the relationship between current interrupt pressure and vent fracture pressure according to capacity of a battery with an LCO-based positive active material.

FIG. 5 is a graph illustrating the relationship between current interrupt pressure and vent fracture pressure according to capacity of a battery with an NCM-based positive active material.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0028] Hereinafter, the present invention will be described more fully with reference to the accompanying drawings, in which exemplary embodiments of the invention are shown.

[0029] FIG. 1 is a schematic exploded cross-sectional perspective view of a rechargeable battery 100 according to an exemplary embodiment of the present invention.

[0030] Referring to FIG. 1, the rechargeable battery includes an electrode assembly 110 that includes a positive electrode 112, a negative electrode 113, and a separator 114 therebetween, and a case 120 that has an opening at one end thereof such that the case 120 can contain the electrode assembly 110 and an electrolyte solution. Further, a cap assembly 140 for closing the case 120 via a gasket 144 is disposed at the opening of the case 120.

[0031] More particularly, the case 120 is composed of a conductive metal such as aluminum, an aluminum alloy, or nickel-plated steel.

[0032] In addition, the case 120 according to the present exemplary embodiment has a cylindrical shape with an internal space in which the electrode assembly 110 is located. The cap assembly 140 is inserted into the case 120 and then clamped thereto. In this process, a crimped portion 125 and a beaded portion 123 are formed at the case 120.

[0033] The electrode assembly 110 is shaped into a cylindrical shape (or into a jelly roll shape) after the positive electrode 112, the separator 114, and the negative electrode 113 are stacked together. However, the structure of the electrode assembly 110 is not limited to this cylindrical shape, and can have various different suitable shapes.

[0034] A positive electrode uncoated part 112a, over which a positive active material is not coated, is disposed at an upper end of the positive electrode 112 and is electrically connected to a positive current collector plate 138. In addition, a negative electrode uncoated part 113a, over which a negative active material is not coated, is disposed at the lower end of the negative electrode 113 and is electrically connected to a negative current collector plate 132.

[0035] In one embodiment, the negative electrode 113 includes a current collector that is made up of copper or other suitable metal and over which a carbon-based active material is coated. The positive electrode 112 includes a current collector that is made up of aluminum or other suitable metal and over which an NCM-based positive active material is coated.

[0036] The cap assembly 140 includes a cap-up 143 that includes an external terminal 143a that protrudes outward and a gas discharge outlet 143b, and a vent plate 160 that is disposed below the cap-up 143 and includes a notch 163 that fractures at a set (or predetermined) pressure and discharges gas. The vent plate 160 interrupts electrical connection between the electrode assembly 110 and the cap-up 143 at a set (or predetermined) pressure.

[0037] A positive temperature coefficient element 141 is disposed between the cap-up 143 and the vent plate 160. The positive temperature coefficient element 141 has electrical resistance that infinitely increases when it exceeds a set (or predetermined) temperature, and interrupts electric current flow when the temperature of the rechargeable battery 100 rises above a set (or predetermined) value.

[0038] A convex portion 165 that protrudes downward is disposed at the center of the vent plate 160, and a sub-plate 147 is welded to the bottom surface of the convex portion 165.

[0039] A cap-down 146 is disposed between the vent plate 160 and the sub-plate 147, and the cap-down 146 is shaped as a disk that has a hole in the middle such that the convex portion 165 can be inserted into it.

[0040] An insulating member 145 is disposed between the cap-down 146 and the vent plate 160, and electrically insulates the cap-down 146 and the vent plate 160. The insulating member 145 has a hole in the middle such that the convex portion 165 of the cap-up 143 can be inserted into it.

[0041] Here, the convex portion 165 of the vent plate 160 can pass through the holes and be easily connected to the sub-plate 147.

[0042] The sub-plate 147 is welded to the convex portion 165 and the cap-down 146, respectively. The cap-down 146 is electrically connected to the electrode assembly 110 via a lead member 150. Consequently, electric current flows easily into the vent plate 160 and the vent plate 160 is connected to the cap-up 143 and carries the electric current to

the external terminal 143a of the cap-up 143.

[0043] FIG. 2A to FIG. 2C are schematic cross-sectional views of processes of interruption of electric current and fracture of a vent.

[0044] Referring to FIG. 2A to FIG. 2C, the sub-plate 147 and the convex portion 165 are connected by ultrasonic welding such that a current interrupt portion 170 is formed between the sub-plate 147 and the convex portion 165. Further, the sub-plate 147 and the cap-down 146 are connected by laser welding such that welded portions 174 and 176 are formed between the sub-plate 147 and the cap-down 16.

[0045] If internal pressure increases inside the rechargeable battery due to swelling during repeated recharge and discharge, the convex portion 165 of the current interrupt portion 170 becomes disengaged from the sub-plate 147 due to the increased pressure, as shown in FIG. 2B. When the convex portion 165 is disengaged from the sub-plate 147, electric current flow is interrupted between the vent plate 160 and the sub-plate 147. The pressure at which the convex portion 165 is disengaged from the sub-plate 147 is referred to as "current interrupt pressure."

[0046] If, as shown in FIG. 2C, the internal pressure of the battery further increases, the vent plate 160 fractures at the notch 163 and exhausts the gas inside the battery to the outside. The pressure at which the vent plate 160 fractures at the notch 163 is referred to as "vent fracture pressure".

[0047] Positive active materials can be divided into LCO (LiCoO_2)-based and NCM (LiNiCoMnO_2)-based positive active materials. While an LCO-based positive active material enhances safety, the price of cobalt has increased, leading to a discrepancy between supply and demand and reducing profitability. Therefore, research on NCM-based positive active materials is being carried out.

[0048] The NCM-based positive active material has a different (or completely different) profile of gas generation caused by a reaction with an electrolyte solution than the LCO-based positive active material. Therefore, it is important to set (or configure) an adequate interrupt pressure and an adequate vent fracture pressure for a battery made with the NCM-based positive active material.

[0049] FIG. 3 is a graph showing changes in pressure inside the rechargeable battery with time when it is fully charged and kept at 90°C. As shown in FIG. 3, the LCO-based positive active material shows a small amount of gas generation during the initial period, but it shows a rapid increase of the amount of the gas generated after a certain (or predetermined) time period. By contrast, the NCM-based positive active material shows a large amount of gas generated during the initial time period but shows almost no further increase in the amount of the gas over time.

[0050] Both reliability and safety must be ensured with regard to a rechargeable battery, but the reliability is inversely related to safety.

[0051] Enhanced safety leads to reduced reliability and vice versa. For instance, it is desired to set a high current interrupt pressure in order to satisfy a criterion of reliability in stable battery operation (i.e., no interruption of battery operation) of the CID (current interrupt device) for 8 hours while a fully-charged rechargeable battery remains at 90 °C. The category of 90 °C is used for determining reliability such that mobile devices are not turned off even at an extremely high temperature. By contrast, it is desired to set a low current interrupt pressure because the current interrupt pressure is related to the amount of gas generated during the overcharge. That is, because overcharge current must be interrupted before combustion and explosion in order to satisfy a criterion of safety that there must be no combustion and explosion while a rechargeable battery is adiabatically overcharged at a 2C rate.

[0052] The biggest cause of generation of gas inside the battery is that electrolyte solution decomposes through a reaction of the electrolyte solution and the positive active material.

[0053] The present invention will be described in more detail with reference to the following examples.

[Table 1]

	Positive electrode active material	LiCoO_3 (wt%)	Current interrupt pressure (kgf/cm ²)	Vent fracture pressure (kgf/cm ²)	First criterion	Second criterion	Third criterion
					90°C (current interrupt time, hr)	2C adiabatic overcharge	150°C exposure to heat (combustion time, min)
Comparative Example 1	LCO	0	6	26	OK (15.0)	OK	OK (15)
Comparative Example 2	LCO	0	7	30	OK (16.1)	OK	OK (16)

(continued)

		Positive electrode active material	LiCO ₃ (wt%)	Current interrupt pressure (kgf/cm ²)	Vent fracture pressure (kgf/ cm ²)	First criterion 90°C (current interrupt time, hr)	Second criterion 2C adiabatic overcharge	Third criterion 150°C exposure to heat (combustion time, min)
5								
10	Comparative Example 3	NCM	0	6	26	NG (6.3)	NG	NG (7)
	Comparative Example 4	NCM	0	7	30	NG (7.2)	NG	NG (8)
15	Comparative Example 5	NCM	0	6	20	NG (6.4)	NG	OK (20)
	Comparative Example 6	NCM	0	7	25	NG (6.5)	NG	OK (22)
20	Comparative Example 7	NCM	0	8	26	OK (10.2)	NG	NG (5)
	Comparative Example 8	NCM	0	9	30	OK (9.3)	NG	NG (6)
25	Comparative Example 9	NCM	0.2	8	20	OK 8.9)	NG	OK (21)
	Comparative Example 10	NCM	0.2	9	25	OK (9.5)	NG	OK (19)
30	Comparative Example 11	NCM	0.4	8	20	OK (10.5)	NG	OK (17)
	Comparative Example 12	NCM	0.4	9	25	OK (11)	NG	OK (18)
35	Comparative Example 13	NCM	2.2	8	20	NG (5.6)	OK	OK (22)
	Comparative Example 14	NCM	2.2	9	25	NG (5.8)	OK	OK (23)
40	Comparative Example 15	NCM	1.0	11	20	OK (10.2)	NG	OK (23)
	Comparative Example 16	NCM	1.0	9	18	NG (4.6)	OK	OK (20)
45	Exemplary Embodiment 1	NCM	0.5	8	20	OK (12.5)	OK	OK (30)
	Exemplary Embodiment 2	NCM	0.5	9	25	OK (13.6)	OK	OK (31)
50	Exemplary Embodiment 3	NCM	1.0	8	20	OK (15.7)	OK	OK (33)
55	Exemplary Embodiment 4	NCM	1.0	9	25	OK (16.5)	OK	OK (29)

(continued)

	Positive electrode active material	LiCO ₃ (wt%)	Current interrupt pressure (kgf/cm ²)	Vent fracture pressure (kgf/cm ²)	First criterion 90°C (current interrupt time, hr)	Second criterion 2C adiabatic overcharge	Third criterion 150°C exposure to heat (combustion time, min)
Exemplary Embodiment 5	NCM	2.0	8	20	OK (17.1)	OK	OK (30)
Exemplary Embodiment 6	NCM	2.0	9	25	OK (19.9)	OK	OK (32)

[0054] Table 1 shows results of experiments regarding the three criteria according to the positive active material, the current interrupt pressure, and the vent fracture pressure. One kilogram-force (kgf) is by definition equal to 9.80665 Newtons.

[0055] The first criterion is directed toward the change in voltage after ten batteries fully charged at 1C (2.4Ah) rate and 4.2V are put into an oven heated to 90 °C. Here, if the current is not interrupted for 8 hours and the voltage is kept above 4.0V, the criterion is categorized as "OK" or otherwise as "NG".

[0056] The second criterion is directed toward ten batteries fully charged at a 1C (2.4Ah) rate and 4.2V are surrounded by an insulating material and then overcharged at a 2C rate and 18V for 3 hours. Here, if the current is interrupted and there is no combustion or rupture, the criterion is categorized as "OK" or otherwise as "NG".

[0057] The third criterion is directed toward the change in voltage and temperature after ten batteries that are fully charged at a 1C (2.4Ah) rate and 4.2V are put into an oven. The temperature of the oven is increased at a rate of 5°C per minute and reaches 150 °C after 30 minutes. If the vent does not fracture and there is no combustion and rupture within 10 minutes after the temperature of the battery reaches 150°C, the criterion is categorized as "OK" or otherwise as "NG".

[0058] The first criterion is about reliability and the second and third criteria are about safety.

<Comparative Example>

[0059] Comparative Example 1 is directed toward a battery with an LCO-based positive active material, wherein the current interrupt pressure was set at 6 Kgf/cm² and the vent fracture pressure was set at 26 Kgf/cm². Comparative Example 2 is directed toward a battery with an LCO-based positive active material, wherein the current interrupt pressure was set at 7 Kgf/cm² and the vent fracture pressure was set at 30 Kgf/cm². Referring to Table 1, Comparative Examples 1 and 2 satisfy the first, second, and third criteria.

[0060] The LCO-based positive active material, with a small amount of gas generation, satisfied all of the above criteria.

[0061] Comparative Example 3 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 6 Kgf/cm² and the vent fracture pressure was set at 26 Kgf/cm². Comparative Example 4 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 7 Kgf/cm² and the vent fracture pressure was set at 30 Kgf/cm². Referring to Table 1, Comparative Examples 3 and 4 satisfied none of the criteria.

[0062] The NCM-based positive active material does not ensure reliability and safety under the above conditions since its gas generation pattern differs from that of an LCO-based positive active material.

[0063] Comparative Example 5 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 6 Kgf/cm² and the vent fracture pressure was set at 20 Kgf/cm². Comparative Example 6 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 6 Kgf/cm² and the vent fracture pressure was set at 25 Kgf/cm². It was regarded that the NCM-based positive active material was not able to satisfy the safety criterion in the Comparative Examples 3 and 4; the amount of generated gas was not sufficient to fracture the vent even in the case of exposure to heat since it generates less gas over time. Therefore, the vent fracture pressure was lowered.

[0064] Referring to Table 1, Comparative Examples 5 and 6 satisfied the third criterion but did not satisfy the first and the second criteria.

[0065] An NCM-base positive active material generates a lot of gas initially and reaches the current interrupt pressure earlier than an LCO-based positive active material when kept at 90°C. Accordingly, a battery with an NCM-based positive active material is able to satisfy the first criterion in which the current interrupt device does not interrupt battery operation for 8 hours when the battery is kept at 90°C, only if its current interrupt pressure is set to be higher than that of an LCO-based positive active material.

[0066] Comparative Example 7 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 8 Kg/cm² and the vent fracture pressure was set at 26 Kg/cm². Comparative Example 8 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9 Kg/cm² and the vent fracture pressure was set at 30 Kg/cm².

[0067] As shown in Table 1, with high current interrupt pressure Comparative Examples 7 and 8 satisfied the first criterion but did not satisfy the second and the third criteria.

[0068] From the above comparative examples, it was determined that the first and third criteria could be satisfied with control of the current interrupt pressure and the vent fracture pressure. However, the second criteria could not be easily satisfied with the control of the current interrupt pressure and the vent fracture pressure.

[0069] An overcharge additive was introduced in order to satisfy the second criterion. LiCO₃, an overcharge additive, reacts with an electrolyte solution and generates gas in case of overcharge. With a controlled amount of LiCO₃, a gas that can reach the current interrupt pressure only in case of overcharge can be generated.

[0070] Comparative Example 9 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 8Kg/cm², the vent fracture pressure was set at 20Kg/cm², and 0.2 wt% of LiCO₃ was added to the positive active material. Comparative Example 10 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9Kg/cm², the vent fracture pressure was set at 25Kg/cm², and 0.2 wt% of LiCO₃ was added to the positive active material.

[0071] As shown in Table 1, with high current interrupt pressure and vent fracture pressure, Comparative Examples 9 and 10 satisfied the first and third criteria but did not satisfy the second criterion due to the small amount of LiCO₃.

[0072] Comparative Example 11 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 8 Kg/cm², the vent fracture pressure was set at 20 Kg/cm², and 0.4 wt% of LiCO₃ was added to the positive active material. Comparative Example 12 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9 Kg/cm², the vent fracture pressure was set at 25 Kg/cm², and 0.4 wt% of LiCO₃ was added to the positive active material.

[0073] As shown in Table 1, with high current interrupt pressure and vent fracture pressure, Comparative Examples 11 and 12 satisfied the first and third criteria, but did not satisfy the second criterion due to the small amount of LiCO₃.

[0074] Comparative Example 13 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 8 Kg/cm², the vent fracture pressure was set at 20 Kg/cm², and 2.2wt% of LiCO₃ was added to the positive active material. Comparative Example 14 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9 Kg/cm², the vent fracture pressure was set at 25 Kg/cm², and 2.2wt% of LiCO₃ was added to the positive active material.

[0075] As shown in Table 1, with a high vent fracture pressure and a large amount of LiCO₃, Comparative Examples 13 and 14 satisfied the second and third criteria, but did not satisfy the first criterion since the electric current was interrupted early due to too much LiCO₃.

[0076] Comparative Example 15 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 11 Kg/cm², the vent fracture pressure was set at 20 Kg/cm², and 1.0 wt% of LiCO₃ was added to the positive active material. As shown in Table 1, with the current interrupt pressure of 11 Kg/cm² higher than 9 Kg/cm², Comparative Example 15 did not satisfy the second criterion; combustion occurred as the current was not interrupted at the right moment.

[0077] Comparative Example 16 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9 Kg/cm², the vent fracture pressure was set at 18 Kg/cm², and 1.0 wt% of LiCO₃ was added to the positive active material. As shown in Table 1, with the current interrupt pressure of 18 Kg/cm² lower than 20 Kg/cm², Comparative Example 16 was faced with a problem that the vent fractured too early.

<Exemplary Embodiments>

[0078] Exemplary Embodiment 1 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 8 Kg/cm², the vent fracture pressure was set at 20 Kg/cm², and 0.5 wt% of LiCO₃ was added to the positive active material.

[0079] As shown in Table 1, with a high current interrupt pressure and vent fracture pressure as well as an appropriate amount of LiCO₃, Exemplary Embodiment 1 satisfied the first, second, and third criteria.

[0080] Exemplary Embodiment 2 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9 Kg/cm², the vent fracture pressure was set at 25 Kg/cm², and 0.5 wt% of

LiCO₃ was added to the positive active material.

[0081] As shown in Table 1, with a high current interrupt pressure and vent fracture pressure as well as an appropriate amount of LiCO₃, Exemplary Embodiment 2 satisfied the first, second, and third criteria.

[0082] Exemplary Embodiment 3 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 8 Kg/cm², the vent fracture pressure was set at 20 Kg/cm², and 1.0 wt% of LiCO₃ was added to the positive active material.

[0083] As shown in Table 1, with a high current interrupt pressure and vent fracture pressure as well as an appropriate amount of LiCO₃, Exemplary Embodiment 3 satisfied the first, second, and third criteria.

[0084] Exemplary Embodiment 4 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9 Kg/cm², the vent fracture pressure was set at 25 Kg/cm², and 1.0 wt% of LiCO₃ was added to the positive active material.

[0085] As shown in Table 1, with a high current interrupt pressure and vent fracture pressure as well as an appropriate amount of LiCO₃, Exemplary Embodiment 4 satisfied the first, second, and third criteria.

[0086] Exemplary Embodiment 5 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 8 Kg/cm², the vent fracture pressure was set at 20 Kg/cm², and 2.0 wt% of LiCO₃ was added to the positive active material.

[0087] As shown in Table 1, with a high current interrupt pressure and vent fracture pressure as well as an appropriate amount of LiCO₃, Exemplary Embodiment 5 satisfied the first, second, and third criteria.

[0088] Exemplary Embodiment 6 is directed toward a battery with an NCM-based positive active material, wherein the current interrupt pressure was set at 9 Kg/cm², the vent fracture pressure was set at 25 Kg/cm², and 2.0 wt% of LiCO₃ was added to the positive active material.

[0089] As shown in Table 1, with a high current interrupt pressure and vent fracture pressure as well as an appropriate amount of LiCO₃, Exemplary Embodiment 6 satisfied the first, second, and third criteria.

[0090] As described above, a battery with an NCM-based positive active material shows the desired effect when the amount of LiCO₃, an overcharge additive added to the positive active material, is between 0.5 wt% and 2.0 wt%. That is, as shown in Comparative Examples 11 and 12, when less than 0.5 wt% of the overcharge additive was added, the amount of generated gas was small, causing a problem that the electric current was not interrupted at the right moment and combustion occurred. In addition, as shown in Comparative Examples 13 and 14, when more than 2.0 wt% of the overcharge additive was added, so much gas was generated that reliability was reduced.

[0091] Further, a battery with an NCM-based positive active material showed the desired effect when the current interrupt pressure was set between 8 Kg/cm² and 9 Kg/cm² and the vent fracture pressure was set between 20 Kg/cm² and 25 Kg/cm².

[0092] That is, as shown in Comparative Example 6, if the current interrupt pressure was lower than 8 Kg/cm², the electric current was interrupted too early even without the danger of combustion, thereby deteriorating reliability. In addition, as shown in Comparative Example 15, if the current interrupt pressure was higher than 9 Kg/cm², the electric current was not interrupted at the right moment, causing a problem of combustion or explosion.

[0093] Further, as shown in Comparative Example 16, if the vent fracture pressure was lower than 20 Kg/cm², the vent fractured too early; and as shown in Comparative Example 7, if the vent fracture pressure was higher than 25 Kg/cm², the vent did not fracture at the right moment, triggering combustion or explosion of the battery.

[0094] However, the current interrupt pressure and the vent fracture pressure vary with battery capacity. Their relationship is shown in FIG. 4 and FIG. 5.

[0095] FIG. 4 is a graph illustrating the relationship between current interrupt pressure and vent fracture pressure according to capacity of a battery with an LCO-based positive active material.

[Table 2]

Capacity (Ah) = X	2.2	2.4	2.6	2.8	3.0
Current interrupt pressure (Kg/cm ²) = A	8	8	9.5	15	15
Vent fracture pressure (Kg/cm ²) = B	20	20	20	30	30
A / B = Y	0.4	0.4	0.48	0.5	0.5

[0096] Table 2 shows a relationship of current interrupt pressure and vent fracture pressure with battery capacity, and the graph in FIG. 4 was drawn based upon Table 2.

[0097] As shown in FIG. 4, the battery with an LCO-based positive active material shows an increase in the value of Y as the capacity increases. Y is the ratio of current interrupt pressure to vent fracture pressure.

[0098] Based upon the above data, the relationship between the capacity (X) and the ratio (Y) of current interrupt

pressure to the vent fracture pressure can be shown via the following Relationship 1 below, which is also graphically shown in FIG. 4.

Relationship 1

$$Y = 0.06500 + 0.1500 \cdot X$$

[0099] Referring to Relationship 1, an LCO-based positive active material shows an increase in the ratio (Y) of the current interrupt pressure and the vent fracture pressure according to an increase in the capacity (X).

[0100] FIG. 5 is a graph illustrating the relationship between current interrupt pressure and vent fracture pressure according to capacity of a battery with an NCM-based positive active material.

[Table 3]

Capacity (Ah) = X	2.0	2.0	2.0	2.2	2.2	2.2	2.4	2.4	2.4
Current interrupt pressure (Kgf/cm ²) = A	9.0	9.5	9.5	8	8.5	9.0	8	8	8
Vent fracture pressure (Kgf/cm ²) = B	21.2	18.1	20	25	18	24.3	21.6	25	30
A / B = Y	0.425	0.525	0.475	0.32	0.47	0.37	0.37	0.32	0.27

[0101] Table 3 shows a relationship of current interrupt pressure and vent fracture pressure with battery capacity, and the graph in FIG. 5 was drawn based upon Table 3.

[0102] The data in Table 3 is represented with circular dots in FIG. 5.

[0103] As shown in FIG. 5, the battery with an NCM-based positive active material shows a gradual decrease in the value of Y as the capacity increases. Y is the ratio of current interrupt pressure to vent fracture pressure.

[0104] Based upon the above data, the relationship between the capacity (X) and the ratio (Y) of current interrupt pressure to the vent fracture pressure can be shown via the following Relationship 2, which is also graphically shown in FIG. 5.

Relationship 2

$$Y = 1.22 - 0.39 \cdot X + D \quad (-0.08 \leq D \leq 0.08) \quad \text{or} \quad Y = 1.224 - 0.3875 \cdot X + D \quad (-0.08 \leq$$

$$D \leq 0.08)$$

[0105] In FIG. 5, the solid line represents $Y = 1.22 - 0.39X$, and the dotted lines represent $Y = 1.22 - 0.39X + 0.08$ and $Y = 1.22 - 0.39X - 0.08$, respectively.

[0106] Referring to Relationship 2, an NCM-based positive active material shows a decrease in the ratio (Y) of the current interrupt pressure to the vent fracture pressure according to an increase in the capacity (X).

[Table 4]

Capacity (Ah) = X	2.0	2.2	2.4
Current interrupt pressure (Kgf/cm ²) = A	9.0	7.0	8
Vent fracture pressure (Kgf/cm ²) = B	26	14	19
A / B = Y	0.346	0.5	0.42

[0107] Table 4 shows a comparative example, which is represented with triangular dots in FIG. 5. When the capacity was 2.0 Ah, the vent did not fracture, causing an explosion of the battery; and, when the capacity was 2.2 Ah, the electric current was interrupted too early and the vent fractured too early, failing to ensure reliability. Further, when the capacity was 2.4 Ah, the vent fractured too early, deteriorating reliability.

[0108] A rechargeable battery with an NCM-based positive active material is able to ensure reliability and safety as long as the relationship between the current interrupt pressure and the vent fracture pressure according to the capacity follows the above Relationship 2.

[0109] To put it another way, a rechargeable battery needs to ensure both its reliability and safety. However, reliability of the battery is inversely related to its safety. In addition, an LCO-based positive active material generates a small amount of gas during its initial operation time period, but it generates a large amount of gas after a certain time period. By contrast, an NCM-based positive active material generates a large amount of gas during its initial operation time period, but it generates no further increase in the amount of the gas over time.

[0110] As such, in view of the above, an embodiment of the present invention includes a rechargeable battery that has an electrode assembly including a positive electrode, a case containing the electrode assembly, and a cap assembly coupled to the case. Here, the positive electrode includes a current collector and an NCM-based positive active material coupled to the current collector, and the cap assembly includes a current interrupt portion and a vent plate having a notch. The current interrupt portion is configured to interrupt electrical connection due to a current interrupt pressure (kgf/cm²) of A inside the case. The notch is configured to fracture the vent plate due to a vent fracture pressure (kgf/cm²) of B inside the case. The rechargeable battery is configured to have a capacity (Ah) of X, and the rechargeable battery is configured to satisfy $A/B = 1.22 - 0.39X + D$ ($-0.08 \leq D \leq 0.08$).

[0111] In one embodiment, the current interrupt pressure of the current interrupt portion is between 8 kgf/cm² and 9 kgf/cm², and the vent fracture pressure of the notch of the vent plate is between 20 kgf/cm² and 25 kgf/cm². In addition, the NCM-based positive active material may include an overcharge additive in an amount between 0.5 and 2.0 wt% of the positive active material. The overcharge additive is LiCO₃. As such, the rechargeable battery with the NCM-based positive active material has improved safety and reliability because the electric current is interrupted and the vent is fractured at the proper moment. In addition, reliability and safety are further improved due to the overcharge additive in the positive active material.

Claims

1. A rechargeable battery (100) comprising:

an electrode assembly (110) comprising a positive electrode, a negative electrode, and a separator between the positive electrode and the negative electrode, the positive electrode comprising a current collector and a positive active material coupled to the current collector, wherein the positive active material is a lithium-nickel-cobalt-manganese oxide (NCM)-based positive active material;
a case (120) containing the electrode assembly (110); and
a cap assembly (140) coupled to the case (120),
the cap assembly (140) comprising a vent plate (160) having a notch (163) configured to fracture the vent plate (160) due to a predefined vent fracture pressure B inside the case (120) and a current interrupt portion (170) configured to interrupt electrical connection due to a predetermined current interrupt pressure A inside the case (120), **characterized in that**
a ratio of the current interrupt pressure A and the vent fracture pressure B satisfies the following equation:

$$A/B = 1.22 - 0.39X + D,$$

wherein

A represents the current interrupt pressure measured in kgf/cm² of the current interrupt portion (170);

B represents the vent fracture pressure measured in kgf/cm² of the notch (163),

X represents a capacity of the rechargeable battery measured in Ah, and

D is a factor within the range of $-0.08 \leq D \leq 0.08$.

2. The rechargeable battery of claim 1, wherein the cap assembly (110) comprises:

a cap-up (143) coupled to the case (120);

a sub-plate (147),

the vent plate (160) being between the cap-up (143) and the sub-plate (147) and comprising a convex portion (165) facing a bottom end of the case (120),

the sub-plate (147) being welded to the convex portion (165), and electrically connected to the electrode assembly

(110);
a cap-down (146) between the vent plate (160) and the sub-plate (147); and
a protrusion between the cap-down (146) and the sub-plate (147).

- 5 **3.** The rechargeable battery of claim 2, wherein the protrusion comprises a welded portion (174, 176) for welding the sub-plate (147) and the cap-down (146) together.
- 10 **4.** The rechargeable battery of claim 2 or 3, further comprising a lead member (150) electrically connected to the electrode assembly (110), wherein the cap-down (146) is electrically connected to the electrode assembly (110) via the lead member (150).
- 15 **5.** The rechargeable battery of any of claims 2 through 4, wherein the sub-plate (147) is electrically connected to the electrode assembly (110) via the cap-down (146).
- 20 **6.** The rechargeable battery of any of claims 2 through 5, further comprising an insulating plate (145) between the cap-down (146) and the vent plate (160).
- 25 **7.** The rechargeable battery of any of claims 2 through 6, wherein the current interrupt portion (170) comprises a welded portion for welding the convex portion (165) and the sub-plate (147) together.
- 30 **8.** The rechargeable battery of any of the preceding claims, wherein the rechargeable battery (100) has a cylindrical shape.
- 35 **9.** The rechargeable battery of any of the preceding claims, wherein the positive active material comprises an overcharge additive in an amount between 0.5 and 2.0 wt% of the positive active material.
- 40 **10.** The rechargeable battery of Claim 9, wherein the overcharge additive comprises a carbon compound.
- 45 **11.** The rechargeable battery of Claim 10, wherein the carbon compound is lithium carbonate.
- 50 **12.** The rechargeable battery of any of the preceding claims, wherein the current interrupt pressure of the current interrupt portion (170) is between 8 kgf/cm² and 9 kgf/cm² with a capacity of the rechargeable battery of 2.4 Ah.
- 55 **13.** The rechargeable battery of any of the preceding claims, wherein the vent fracture pressure of the notch (163) is between 20 kgf/cm² and 25 kgf/cm² with a capacity of the rechargeable battery of 2.4 Ah.
- 60 **14.** The rechargeable battery of any of the preceding claims, wherein the positive active material is coated on the current collector.
- 65 **15.** The rechargeable battery of claim 14, wherein the current collector is made of aluminum.

FIG.1

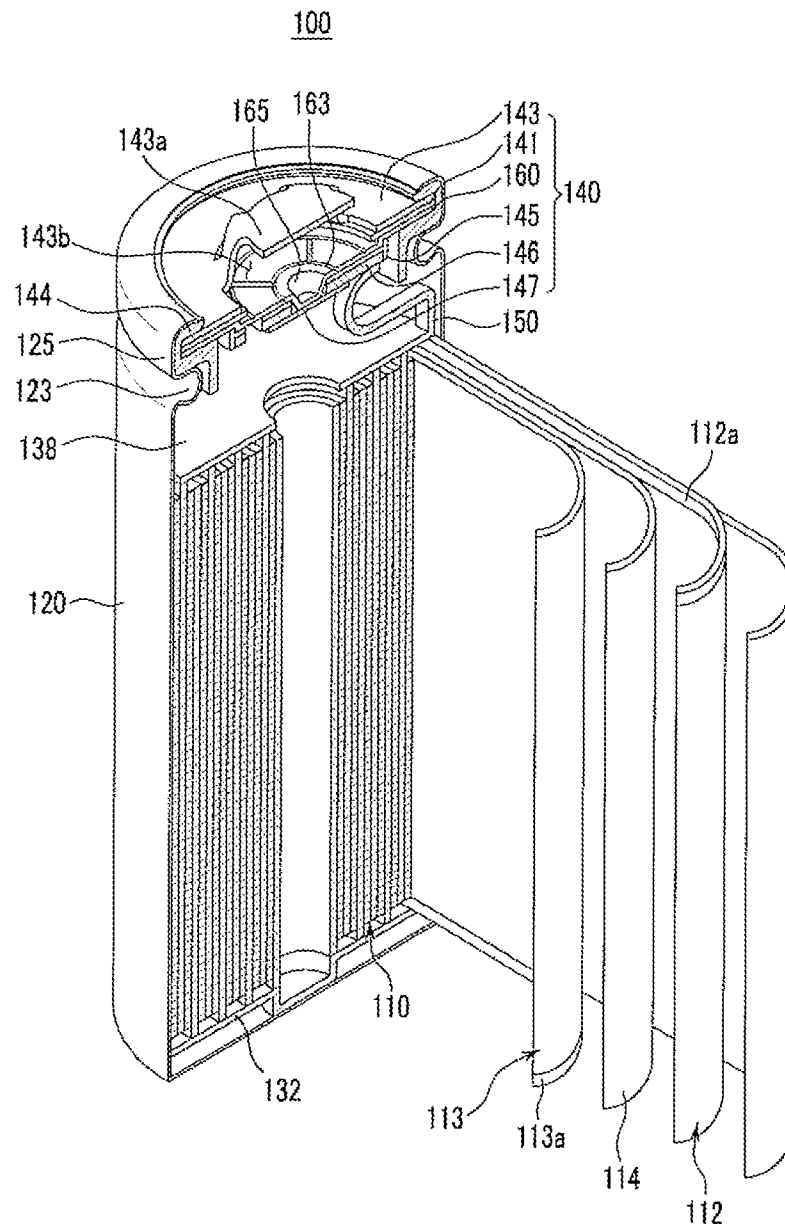


FIG.2A

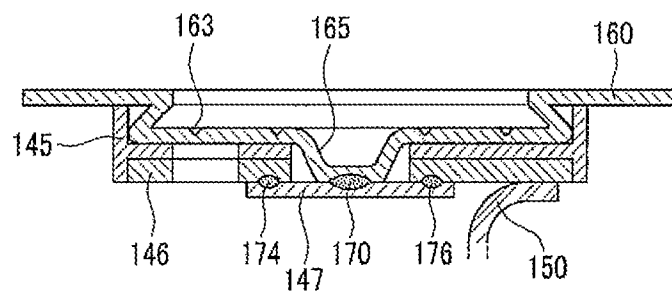


FIG.2B

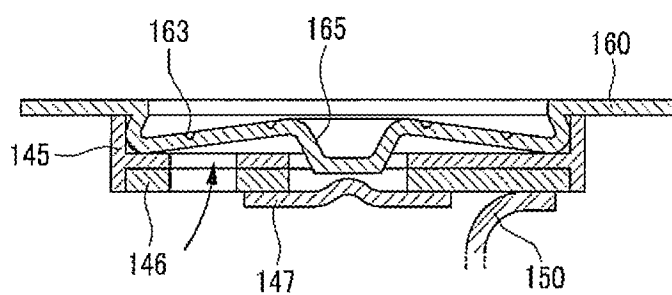


FIG.2C

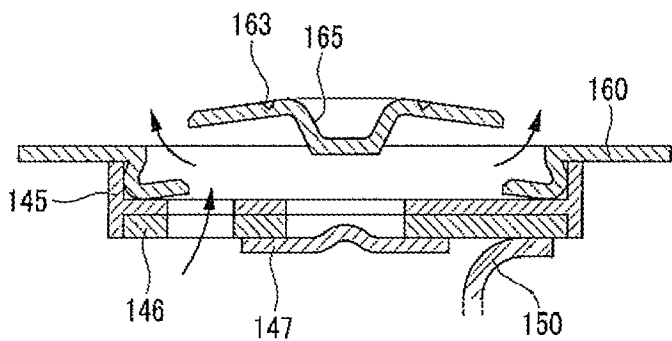


FIG.3

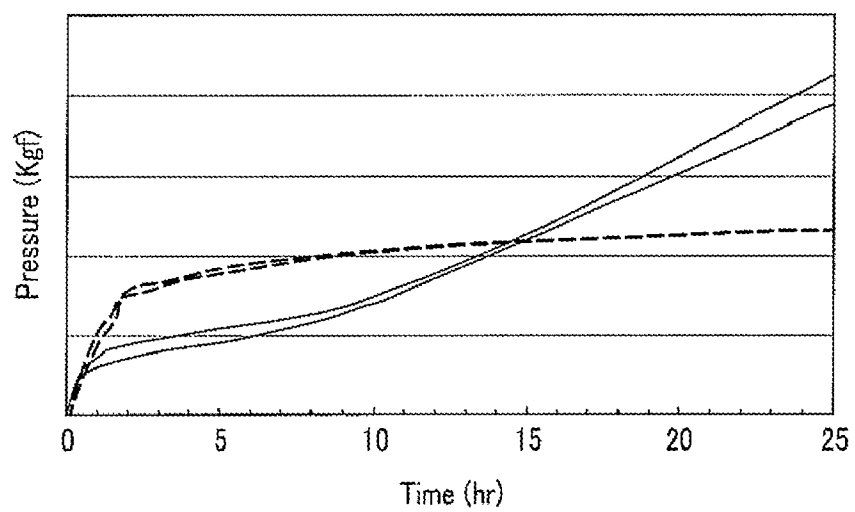


FIG.4

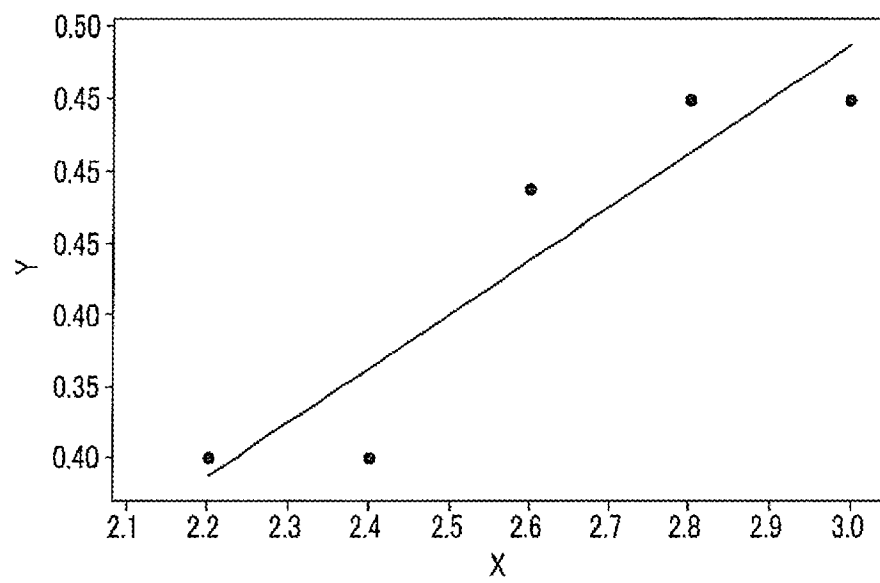
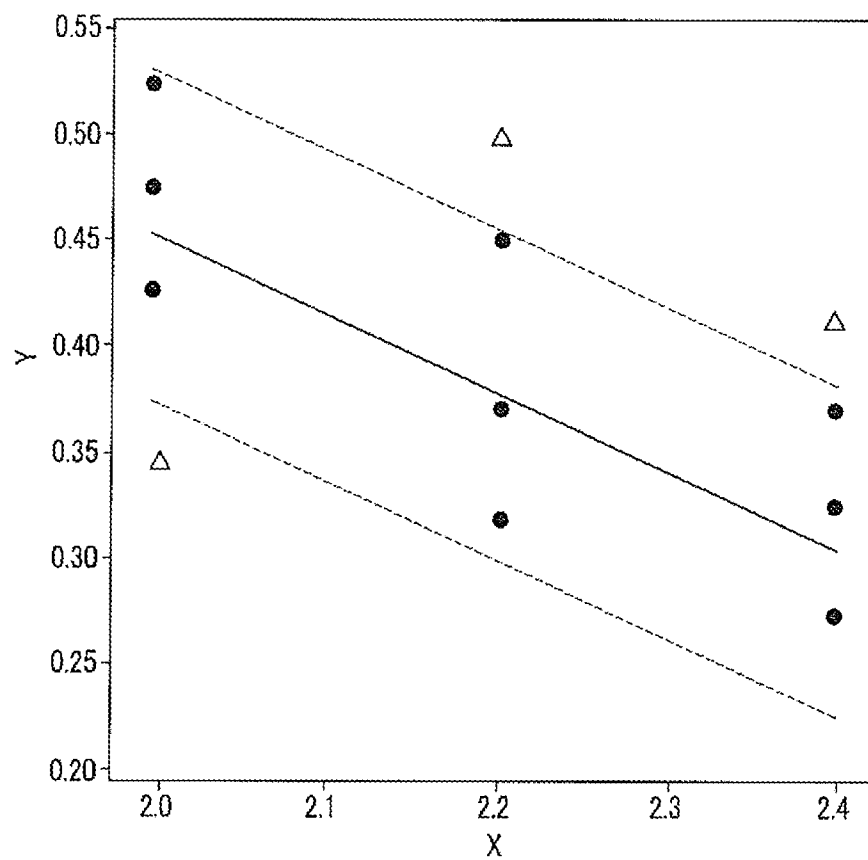


FIG.5





EUROPEAN SEARCH REPORT

 Application Number
EP 09 17 1983

DOCUMENTS CONSIDERED TO BE RELEVANT			
Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X	EP 1 049 189 A (SHIN KOBE ELECTRIC MACHINERY [JP]) 2 November 2000 (2000-11-02) * paragraph [0003] * * paragraph [0044] - paragraph [0048] * -----	1-15	INV. H01M2/12 H01M6/50 H01M10/052 H01M10/058 H01M2/34 H01M10/42
A	US 2004/157127 A1 (CHOI YOUNG-MIN [KR] ET AL) 12 August 2004 (2004-08-12) * paragraphs [0012] - [0014], [0045] * * table 2 * * claims 1,2 * -----	9-11	
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The present search report has been drawn up for all claims			
Place of search The Hague		Date of completion of the search 2 November 2009	Examiner Standaert, Frans
CATEGORY OF CITED DOCUMENTS X : particularly relevant if taken alone Y : particularly relevant if combined with another document of the same category A : technological background O : non-written disclosure P : intermediate document T : theory or principle underlying the invention E : earlier patent document, but published on, or after the filing date D : document cited in the application L : document cited for other reasons & : member of the same patent family, corresponding document			

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EPO FORM 1503 03 02 (P04001)

**ANNEX TO THE EUROPEAN SEARCH REPORT
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This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report.
The members are as contained in the European Patent Office EDP file on
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02-11-2009

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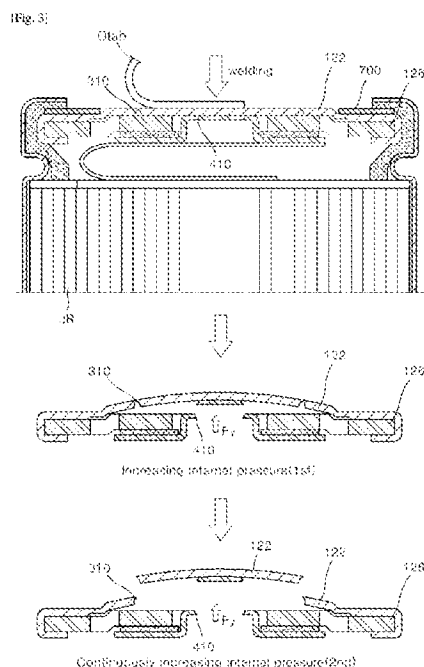
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CO., LTD. [KR/KR]; 48, Yangsan-ro, Osan-si, Gyeong-
gi-do 18100 (KR).(72) Inventors: KIM, Gi Rin; 303-1604, 111, Yeong-
tong-ro, Yeongtong-gu, Suwon-si, Gyeonggi-do 16687
(KR). HWANG, Man Yong; 423-702, 20, Jungang-gong-
won-ro, Bundang-gu, Seongnam-si, Gyeonggi-do 13589
(KR). ABE, Takan; 48, Yangsan-ro, Osan-si, Gyeonggi-do
18100 (KR). KIM, Jong Ho; 316-201, 16, Ogeum-ro, Gun-
po-si, Gyeonggi-do 15863 (KR).(74) Agent: JI, Jeong Hun; 4F, 40-4, Teheran-ro 14-gil, Gang-
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KM, ML, MR, NE, SN, TD, TG).(54) Title: CAP ASSEMBLY FOR SECONDARY BATTERY AND SECONDARY BATTERY INCLUDING THE CAP ASSEM-
BLY(57) Abstract: Disclosed are a cap assembly for a secondary battery and a sec-
ondary battery including the cap assembly. The cap assembly includes: a safety vent
made of a circular metal plate and consisting of a central portion corresponding to
the central area of the metal plate and a peripheral surrounding the central portion,
the central portion including a current-carrying contact forming a first conductive
path through which an electric current generated from a power generating device
flows to the outside and a breakable junction forming a second conductive path at
the center of the central portion; a cap-down opposite to the safety vent, having a
central hole through which the breakable junction is exposed, and including a sup-
port member in close contact with the peripheral portion; and a sub-plate joined to
the breakable junction and having a notch formed around the breakable junction,
the outer circumferential surface of the sub-plate and the cap-down interposing an
insulating layer therebetween. The cap assembly can be used to increase the capaci-
ty of the power generating device in the secondary battery compared to when used
in other secondary batteries of the same size, minimizes the contact resistance be-
tween terminals, is constructed at reduced cost, and can protect a gasket disposed
between a battery can and the safety vent from being melted or lost by energy or
force generated upon external electrical shorting to prevent electrical shorting of
the secondary battery.

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Description

Title of Invention: CAP ASSEMBLY FOR SECONDARY BATTERY AND SECONDARY BATTERY INCLUDING THE CAP ASSEMBLY

Technical Field

- [1] The present invention relates to a cap assembly for a secondary battery and a secondary battery including the cap assembly. More specifically, the present invention relates to a cap assembly for a secondary battery that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery, and a secondary battery including the cap assembly.

[2]

Background Art

- [3] Secondary batteries are classified into cylindrical, prismatic, and pouch types depending on the shape of battery cases they employ. The cylindrical batteries include an electrode assembly accommodated in a cylindrical metal can. The prismatic batteries include an electrode assembly accommodated in a prismatic metal can. The pouch type batteries include an electrode assembly accommodated in a pouch type case made of an aluminum laminate sheet.
- [4] Electrode assemblies accommodated in battery cases are power generating devices capable of repeated charge/discharge cycles that have a cathode/separator/anode laminate structure. Electrode assemblies are classified into jelly-roll and stack types by their structure. The jelly-roll type structure is constructed by interposing a separator between an anode and a cathode, each of which is in the form of a long sheet coated with an active material, and winding the laminate in a roll form. The stack type structure is constructed by sequentially stacking a plurality of electrode units, each of which includes a cathode having a predetermined size, an anode having a predetermined size, and a separator interposed therebetween. The jelly-roll type electrode assemblies are most widely used in secondary batteries due to their ease of construction and high energy density per unit weight. The jelly-roll type electrode assemblies are usually employed in cylindrical batteries.
- [5] Secondary batteries employing jelly-roll type electrode assemblies are likely to

deform because the jelly-roll type electrode assemblies undergo repeated swelling and shrinkage during charge/discharge. This deformation may cause internal short circuits. Heat generated by internal short circuits decomposes organic solvents to produce gases. The gases increase the internal gas pressure of batteries to burst the batteries.

[6] An increase in the internal gas pressure of batteries may take place also when internal short circuits are caused by an external impact.

[7] Many attempts have been made to solve the safety problems of batteries. For example, known is a cap assembly for a cylindrical battery having a structure in which a safety vent, safety devices, and a top cap are fixed by a gasket. High-pressure gases are released through the safety vent. The safety devices include a PTC device adapted to interrupt the flow of electric current at high temperature and a current interrupt device (CID) adapted to interrupt the flow of electric current when the internal pressure of the battery increases. The top cap forms a protruding terminal adapted to protect the safety devices. Examples of such attempts are disclosed in many patent publications, for example, Japanese Patent Publication Nos. 2006-286561, 2005-100927, and 2002-373711 and Korean Patent Publication Nos. 2012-0103394 and 1999-67165. In each of the cap assemblies disclosed in these publications, the gasket serves to seal the interface between the can and the cap assembly while surrounding the outer circumferences of the safety devices, but the structure of the top cap limits the capacity of the power generating device, permits the release of a limited amount of gases when the safety vent ruptures, causes an increase in contact resistance, and increases in the construction cost of the cap assembly.

[8] In a cylindrical secondary battery 100 illustrated in Fig. 9, a power generating jelly roll JR is accommodated in a battery can 110 such that a tab is connected to a sub-plate 128. This connection enables power supply through the battery can and a terminal of a cap-up 124. An open end (e) of the battery can is clamped to fix a cap assembly 120. A safety vent 122 of the cap assembly 120 is bent to surround the cap-up 124. Alternatively, the cap assembly may have a structure in which the safety vent is not bent and the cap-up is exposed to the top so as to be in direct contact with a gasket 130. When a module consisting of the secondary battery and one or more other secondary batteries is used in an electric vehicle or an energy storage system (ESS), energy or force generated by electrically shorting of the other secondary batteries or a power grid can melt or structurally weaken the gasket 130 interposed between the clamped battery can 110 and the cap-up 124 or the safety vent 122, and as a result, the battery can 110 comes into contact with the safety vent, causing short circuits. The short circuits increase the risk of leakage of an electrolyte from the secondary battery or safety accidents, such as explosion.

[9]

Disclosure of Invention

Technical Problem

- [10] A first object of the present invention is to provide a cap assembly for a secondary battery that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.
- [11] A second object of the present invention is to provide a secondary battery including a cap assembly that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.

[12]

Solution to Problem

- [13] A first aspect of the present invention provides a cap assembly for a secondary battery including: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion including a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and including a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween.
- [14] According to one embodiment of the present invention, the safety vent may be bent downwardly and extend to form a bent portion.
- [15] According to a further embodiment of the present invention, the breakable junction may be welded to a protrusion of the sub-plate entering the central hole formed in the support member.
- [16] According to another embodiment of the present invention, the protrusion may

include a portion that protrudes upwardly opposite to the central portion and is opposite and melt-bonded to the breakable junction.

- [17] According to another embodiment of the present invention, the notch may be formed in the shape of a long home on the outer circumference of the portion opposite to the breakable junction.
- [18] According to another embodiment of the present invention, the protrusion may include a portion that protrudes upwardly opposite to the central portion and is opposite to and in close contact with the current-carrying contact.
- [19] According to another embodiment of the present invention, the portion opposite to the current-carrying contact may be formed on the outer circumference of the notch.
- [20] According to another embodiment of the present invention, a breakable notch may be formed at the bottom of the peripheral portion such that the safety vent is bent by a rupture pressure.
- [21] According to another embodiment of the present invention, the cap assembly may further include a bending-relieving portion adjacent to the breakable notch to space the current-carrying contact from the cap-down and forming a space where a rupture pressure is applied to the breakable junction.
- [22] According to another embodiment of the present invention, the bottom width of the bending-relieving portion may be larger than that of the breakable notch.
- [23] According to another embodiment of the present invention, the cap-down may include a through-hole that is spaced apart from and formed around the central hole and through which internal gases and liquids pass.
- [24] According to another embodiment of the present invention, the insulating layer may extend beyond the end of the sub-plate and may reach the end of the cap-down.
- [25] According to another embodiment of the present invention, the insulating layer may have a through-hole opposite to the through-hole of the cap-down.
- [26] According to another embodiment of the present invention, spacer bridges may be formed across the opposite through-hole.
- [27] According to another embodiment of the present invention, a short-circuit preventing layer may be formed at a position opposite to the end of the battery can through a gasket of the safety vent.
- [28] According to another embodiment of the present invention, the short-circuit preventing layer may include a polymeric or organic material that is resistant to heat at a temperature of 300 °C or more.
- [29] According to another embodiment of the present invention, the short-circuit preventing layer may be an oxide or nitride film formed by surface treatment with high energy, such as a laser, after removal of a plating layer.
- [30] According to another embodiment of the present invention, the short-circuit

preventing layer may be made of an anodized ring-shaped aluminum material.

[31] According to another embodiment of the present invention, the aluminum material may be melt-bonded to an adherend by soldering.

[32] According to another embodiment of the present invention, the cap-down and the sub-plate between which the insulating layer is interposed may have rough sides opposite to each other.

[33] A second aspect of the present invention provides a secondary battery including the cap assembly.

[34]

Advantageous Effects of Invention

[35] The cap assembly of the present invention can be used to increase the capacity of a power generating device in a secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.

[36]

Brief Description of Drawings

[37] Fig. 1 is a plan view illustrating a cap assembly of the present invention.

[38] Fig. 2 is a cross-sectional view of the cap assembly of Fig. 1.

[39] Fig. 3 is a cross-sectional view illustrating a state in which a pack-tab (Otab) is welded to a secondary battery fabricated by assembly of a cap assembly according to the present invention and a battery can.

[40] Fig. 4 is a cross-sectional view illustrating states in which a safety vent of a cap assembly according to the present invention is deformed at operating and rupture pressures.

[41] Fig. 5 is a cross-sectional view illustrating a state in which an electrode tab attached to a sub-plate of a cap assembly according to the present invention is deformed in a battery can and is in close contact with a cap-down through an insulating layer interposed therebetween.

[42] Fig. 6 is a detailed plan view of the insulating layer of Fig. 5.

[43] Fig. 7 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together without the need for forging.

[44] Fig. 8 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together

without the need for forging.

- [45] Fig. 9 is a cross-sectional view of a secondary battery mounted with a conventional cap assembly and illustrates a state in which the end of a battery is in contact with a safety vent as a result of melting or damage of a gasket by thermal energy generated in an external circuit or load and a state in which a cap-down is in contact with the safety vent as a result of melt-bonding of an insulating layer.

[46]

Best Mode for Carrying out the Invention

- [47] The present invention will now be described in more detail.

- [48] Technical terms used in this specification are used to merely illustrate specific embodiments, and should be understood that they are not intended to limit the present disclosure.

- [49] As far as not being defined differently, all terms used herein including technical or scientific terms may have the same meaning as those generally understood by an ordinary person skilled in the art to which the present disclosure belongs to, and should not be construed in an excessively comprehensive meaning or an excessively restricted meaning.

- [50] In addition, if a technical term used in the description of the present disclosure is an erroneous term that fails to clearly express the idea of the present disclosure, it should be replaced by a technical term that can be properly understood by the skilled person in the art. In addition, general terms used in the description of the present disclosure should be construed according to definitions in dictionaries or according to its front or rear context, and should not be construed to have an excessively restrained meaning. A singular representation may include a plural representation as far as it represents a definitely different meaning from the context. Terms "include" or "has" used herein should be understood that they are intended to indicate an existence of several components or several steps, disclosed in the specification, and it may also be understood that part of the components or steps may not be included or additional components or steps may further be included. In the description of the present invention, detailed explanations of related art are omitted when it is deemed that they may unnecessarily obscure the essence of the invention. In the accompanying drawings, those components are rendered the same reference numeral that are the same or are in correspondence, regardless of the figure number, and redundant explanations are omitted.

- [51] Fig. 1 is a plan view illustrating a cap assembly of the present invention, Fig. 2 is a cross-sectional view of the cap assembly of Fig. 1, Fig. 3 is a cross-sectional view illustrating a state in which a pack-tab (Otab) is welded to a secondary battery fabricated

by assembly of a cap assembly according to the present invention and a battery can, Fig. 4 is a cross-sectional view illustrating states in which a safety vent of a cap assembly according to the present invention is deformed at operating and rupture pressures, Fig. 5 is a cross-sectional view illustrating a state in which an electrode tab attached to a sub-plate of a cap assembly according to the present invention is deformed in a battery can and is in close contact with a cap-down through an insulating layer interposed therebetween, Fig. 6 is a detailed plan view of the insulating layer of Fig. 5, and Fig. 7 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together without the need for forging.

- [52] A cap assembly 120 for a secondary battery according to the present invention includes: a safety vent 122 made of a circular metal plate and consisting of a central portion 200 corresponding to the central area of the metal plate and a peripheral portion 300 surrounding the central portion, the central portion including a current-carrying contact 210 forming a first conductive path through which an electric current generated from a power generating device JR flows to the outside and a breakable junction 220 forming a second conductive path at the center of the central portion; a cap-down 126 opposite to the safety vent, having a central hole 510 through which the breakable junction is exposed, and including a support member 500 in close contact with the peripheral portion; and a sub-plate 400 joined to the breakable junction and having a notch 410 formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer 600 therebetween.
- [53] The current-carrying contact 210 in the central portion 200 functions to reduce the contact resistance due to the formation of the conductive path and imparts a support force against a pressure applied when a pack-tab is attached to the cap assembly. The breakable junction 220 included in the central portion 200 may be welded to a protrusion 420 of the sub-plate entering the central hole 510 formed in the central portion surrounded by the support member. The welding may be performed by various bonding processes. Preferably, laser welding or melt-bonding is used such that the welded portion is not detached even under high internal pressures (for example, 1st pressure and 2nd pressure) generated upon abnormal operation of the secondary battery.
- [54] That is to say, the protrusion 420 of the sub-plate 400 protrudes upward opposite to the central portion 200. Specifically, the protrusion 420 enters the central hole 510 and is welded to the breakable junction 220. For the prevention of short circuits, the protrusion 420 and the central hole should be spaced apart from each other such that they do not come into contact and in electrical communication with each other.

- [55] Furthermore, the insulating layer 600 has a central hole 620 opposite to the central hole 510 and through which the central portion 200 is exposed and a standing portion 625 in the form of a barrier protruding along the circumference of the central hole 620. Due to this structure, the risk of short circuits between the protrusion 420 of the sub-plate and the central hole 510 during assembly can be completely avoided.
- [56] The support member 500 of the cap-down 126 is opposite to and in close contact with the peripheral portion 300 of the safety vent 122. This arrangement reduces the contact resistance when power is applied from the power generating device JR to the outside, ensuring good current-carrying properties. In addition, the support member 500 serves to prevent the cap assembly from being deformed caused by an excessive load or pressure during welding of a pack-tab (Otab) to a package module in which the battery is connected in series or parallel with one or more other batteries.
- [57] Furthermore, the protrusion 420 is in close contact with and supports the central portion 200 over a large area. This arrangement is advantageous in preventing the cap assembly from being deformed when a pack-tab is welded to the secondary battery. Thus, the protrusion 420 is designed to have a portion 220' opposite and welded to the breakable junction 220 and a portion 210' opposite to the current-carrying contact around the circumference of the portion 220'. The portion 210' is formed in close contact with the current-carrying contact other than the notch 410 in the form of a recessed long home that is broken by high pressures upon abnormal operation of the secondary battery and permits the release of internal gases into the outside. Due to this design, a support force by the portion 210' is ensured and the conductive paths can be maintained, which is beneficial in terms of current-carrying properties.
- [58] The cap-down 126 includes a through-hole 530 spaced apart from but adjacent to the central hole 510 and through which internal gases and liquids are released into the outside. The through-hole 530 is filled with high-pressure gases, liquids, and a flowable electrolyte upon abnormal operation of the secondary battery, and as a result, operating and rupture pressures are applied to the safety vent 122 therethrough.
- [59] Here, the operating pressure refers to a 1_{st} pressure at which the contact and junction between the current-carrying contact 210 of the safety vent and the portion 210' opposite thereto and between the breakable junction 220 of the safety vent and the portion 220' opposite thereto are released to block the application of power to the outside. The rupture pressure refers to a 2_{nd} pressure at which the safety vent 122 ruptures so that the internal gases leak out before explosion of the secondary battery.
- [60] Accordingly, it is necessary for the operating pressure to be smaller than the rupture pressure. The operating pressure is designed to be much larger than the rupture pressure. For example, the operating pressure and the rupture pressure may be adjusted to about 16 kgf/cm² and about 30 kgf/cm², respectively, to clearly distinguish the

operation and rupture of the cap assembly. The structural behaviors of such operation and rupture will be discussed in detail. The internal pressure of the secondary battery increases continuously upon internal abnormal operation of the secondary battery. By the operating pressure, the current-carrying contact 210 is detached from the portion 210' opposite thereto and the breakable junction 220 is detached from the portion 220' opposite thereto. A flowable electrolyte, liquids, and gases passes through the through-hole 530 of the support member to increase the internal pressure and directly lifts the safety vent 122 because the breakable junction is welded to the protrusion 420. A vertical force (F_y) to detach the breakable junction in the upward and downward directions and a horizontal force (F_x) in the horizontal direction are effective in providing a force (F) to break the portion 220' opposite to the breakable junction. By the rupture force (F), the notch 410 is broken to interrupt the flow of electric current in the secondary battery.

- [61] That is, the detachment of the current-carrying contact and the breakage of the breakable junction can be accomplished by a combination of the vertical force (F_y) and the horizontal force (F_x).
- [62] Even after the notch 410 is broken by the operating pressure to interrupt the flow of electric current, the 2nd pressure of the secondary battery may increase continuously. Since the increasing 2nd pressure increases the danger of explosion, it is necessary to reduce the internal pressure beforehand by the removal of internal gases and liquids (outgasing).
- [63] To this end, a breakable notch 310 in the form of a long home is formed in the peripheral portion of the safety vent. The breakable notch 310 is designed to be broken by the internal pressure of the secondary battery. The safety vent is in close contact with the cap-down, making it difficult to bend or break the breakable notch 310. Thus, a bending-relieving portion 320 in the shape of a long home or groove is formed adjacent to the breakable notch 310. Due to its shape, the bending-relieving portion 320 can vary the bending angle (θ) of the breakable notch, enabling efficient breakage of the breakable notch. The bending-relieving portion is designed to have a bottom area (G) larger than that (N) of the breakable notch. This design is advantageous in varying the bending angle (θ) of the breakable notch to freely form a space between the bending-relieving portion and the cap-down opposite thereto, thus being effective in deforming or breaking the breakable notch.
- [64] The breakable notch 310 and the bending-relieving portion 320 may be arranged concentrically around the center of the central portion on the bottom of the peripheral portion of the safety vent. The internal pressure increases continuously even after interruption of the flow of electric current by the 1st pressure and reaches the 2nd pressure at which the breakable notch is broken, and as a result, the secondary battery can be

protected from explosion.

[65] The safety vent 122 may be bent downwardly and extend to form a bent portion 1221 that surrounds the cap-down 126. The formation of the bent portion is beneficial in ensuring adhesion of the safety vent 122 to the cap-down 126. The extension length of the bent portion on the bottom of the cap-down is limited to increase the power generation capacity of the secondary battery.

[66] The power generating device JR is a device that produces electrical energy through chemical reactions. For example, in the case of a secondary lithium-ion battery, the power generating device JR may be a jelly-roll type electrode assembly including an anode material, a cathode material, an electrolyte, and a center pin. One of the electrodes is in electrical communication with the cap assembly 120 through an electrode tab. Power generated in the electrode assembly is supplied to an external circuit or load through the electrode tab.

[67] The insulating layer 600 extends beyond the end of the sub-plate 400 and reaches the end of the cap-down. Thus, the insulating layer 600 precludes an extension of the electrode tab other than the portion welded to the sub-plate 400 from being bent up toward the cap assembly 120 and coming into contact with the cap-down 126 to prevent the occurrence of electrical shorting during fabrication of the secondary battery and use in an application.

[68] The insulating layer 600 is interposed between and insulates the cap-down and the sub-plate and can prevent the electrode tab from being in electrical communication with the cap-down. However, the insulating layer 600 may impede the flow of gases through the through-hole 530, which is solved by the arrangement of a through-hole 610 at a position opposite to the through-hole 530. Spacer bridges 612 extend across the opposite through-hole to space the electrode tab from the cap-down. The spacer bridges can be understood as bar-type bridges traversing the opposite through-hole.

[69] As explained previously, the central hole 620 of the insulating layer 600 is formed opposite to the central hole 510 such that the central portion 200 is exposed. The sub-plate 400 is insertable into the insulating layer 600 through the central hole 620. The standing portion 625 in the form of a barrier protrudes along the circumference of the central hole 620 to prevent short circuits.

[70] For better attachment and adhesion of the insulating layer 600 between the cap-down 126 and the sub-plate 400, the cap-down and the sub-plate have long homes 1265 and 4005 in the form of depressions or grooves at sides opposite to each other, respectively. The insulating layer is interposed between the cap-down and the sub-plate, followed by pressing or thermal pressing to form a filling portion 615 in which the insulating layer is filled in the long homes.

[71] A gasket 130 is disposed between the battery can 110 and the safety vent 122. The

gasket 130 should be protected from being melted or lost by the application of energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery. To this end, a short-circuit preventing layer 700 can be interposed between the safety vent and a gasket 130 at a position opposite to the end of the battery can.

- [72] Any insulating material may be used without particular limitation for the short-circuit preventing layer 700. For example, the short-circuit preventing layer may be made of a metallic, nonferrous metallic or non-metallic material. The surface of the short-circuit preventing layer made of a metallic or nonferrous metallic material is preferably insulated.
- [73] The material for the insulating layer is insulated by wet or dry physical or chemical process in such a manner that the surface chemical composition is changed to increase insulating properties. For example, aluminum is a current-carrying material but its insulating properties increase when an oxide layer is formed on the surface. An oxide film is formed on the surface of a metallic or nonferrous metallic material, for example, by dipping in an acidic or basic solution, electrolysis, anodizing or laser irradiation.
- [74] Particularly, a dry process by laser irradiation may have difficulty in processing the insulating layer or film whose surface is plated. In this case, after removal of the plating layer, high-energy laser beams are irradiated to form an oxide or nitride film on the surface of the insulating layer or film, achieving good insulating properties.
- [75] Alternatively, the short-circuit preventing layer 700 may be formed using a polymeric material. Since the short-circuit preventing layer 700 is required to have a sufficient physical support force in a high-temperature environment where the gasket 130 is melted or structurally weakened by thermal energy or force generated upon external short circuits, the material for the short-circuit preventing layer 700 should be resistant to heat at a temperature of 300 °C or more. However, the material is not particularly limited so long as its physical properties are maintained at a temperature of 300 °C or more.
- [76] The short-circuit preventing layer 700 may include an inorganic material. For example, the short-circuit preventing layer may be formed using an inorganic material or may be an insulating layer whose surface is coated with an inorganic material, for example, a powder of inorganic particles, to form an inorganic layer (not shown).
- [77] The short-circuit preventing layer 700 may take the form of a ring made of an oxidized or anodized metallic or nonferrous metallic material, for example, an aluminum material. The insulating layer acts as a kind of spacer that maintains the interval between the end (e) of the battery can and the safety vent 122 to prevent short circuits. The short-circuit preventing layer 700 may be provided in plurality. In this

case, the number of the short-circuit preventing layers is not limited so long as they are arranged between the end (e) of the battery can and the safety vent 122. It is beneficial from the viewpoint of production processability that the short-circuit preventing layer 700 is in the form of a ring. The short-circuit preventing layer 700 may be fixed to the safety vent 122 by soldering.

- [78] On the other hand, the present invention provides a secondary battery including the cap assembly. The secondary battery of the present invention may be fabricated by assembling an anode material, a cathode material, an electrolyte, a center pin, a jelly roll, etc. together in a battery can and aligning the cap assembly at the open end of the battery can (clamping). The fabrication method of the secondary battery is generally known in the art and detailed description thereof is thus omitted.

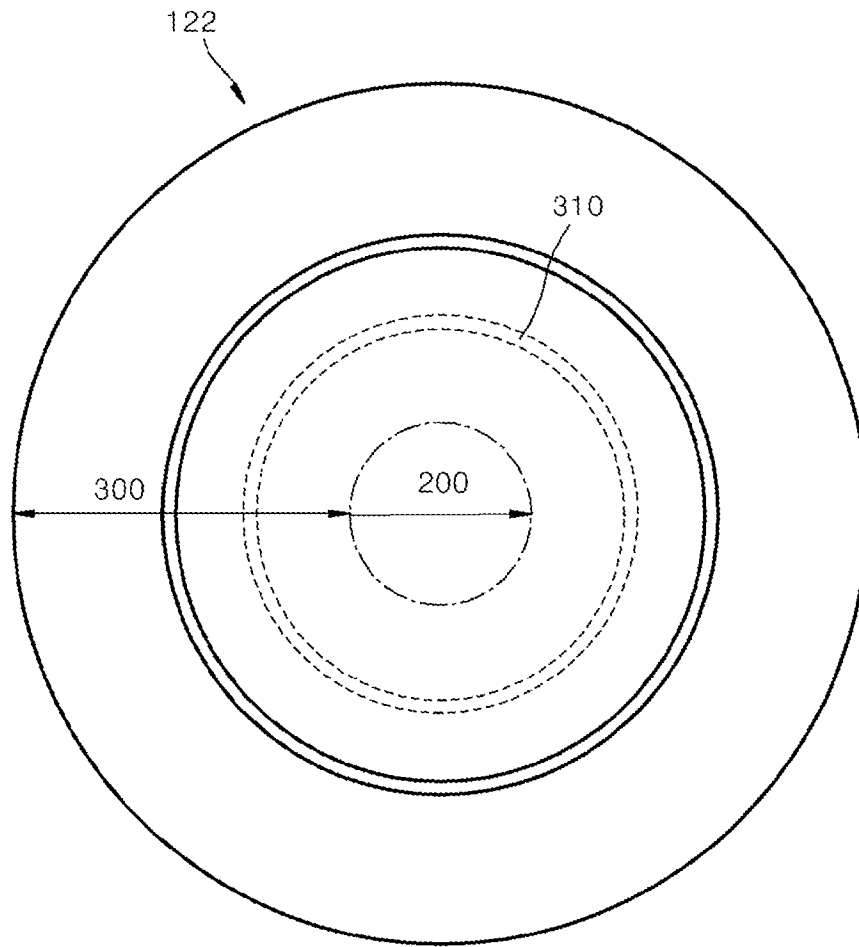
Claims

- [Claim 1] A cap assembly for a secondary battery comprising: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion comprising a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and comprising a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween.
- [Claim 2] The cap assembly according to claim 1, wherein the safety vent is bent downwardly and extends to form a bent portion.
- [Claim 3] The cap assembly according to claim 1, wherein the breakable junction is welded to a protrusion of the sub-plate entering the central hole formed in the support member.
- [Claim 4] The cap assembly according to claim 3, wherein the protrusion comprises a portion that protrudes upwardly opposite to the central portion and is opposite and melt-bonded to the breakable junction.
- [Claim 5] The cap assembly according to claim 4, wherein the notch is formed in the shape of a long home on the outer circumference of the portion opposite to the breakable junction.
- [Claim 6] The cap assembly according to claim 3, wherein the protrusion comprises a portion that protrudes upwardly opposite to the central portion and is opposite to and in close contact with the current-carrying contact.
- [Claim 7] The cap assembly according to claim 6, wherein the portion opposite to the current-carrying contact is formed on the outer circumference of the notch.
- [Claim 8] The cap assembly according to claim 1, wherein a breakable notch is formed at the bottom of the peripheral portion such that the safety vent is bent by a rupture pressure.
- [Claim 9] The cap assembly according to claim 8, further comprising a bending-

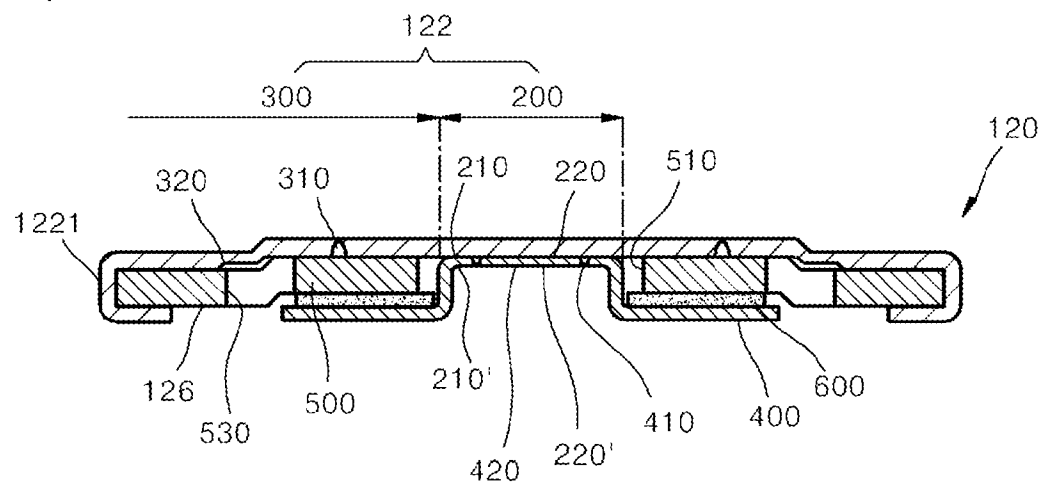
relieving portion adjacent to the breakable notch to space the current-carrying contact from the cap-down and forming a space where a rupture pressure is applied to the breakable junction.

- [Claim 10] The cap assembly according to claim 9, wherein the bottom width of the bending-relieving portion is larger than that of the breakable notch.
- [Claim 11] The cap assembly according to claim 1, wherein the cap-down comprises a through-hole that is spaced apart from and formed around the central hole and through which internal gases and liquids pass.
- [Claim 12] The cap assembly according to claim 1, wherein the insulating layer extends beyond the end of the sub-plate and reaches the end of the cap-down.
- [Claim 13] The cap assembly according to claim 12, wherein the insulating layer has a through-hole opposite to the through-hole of the cap-down.
- [Claim 14] The cap assembly according to claim 13, wherein spacer bridges are formed across the opposite through-hole.
- [Claim 15] The cap assembly according to claim 1, wherein a short-circuit preventing layer is formed at a position opposite to the end of the battery can through a gasket of the safety vent.
- [Claim 16] A secondary battery comprising the cap assembly according to any one of claims 1 to 15.

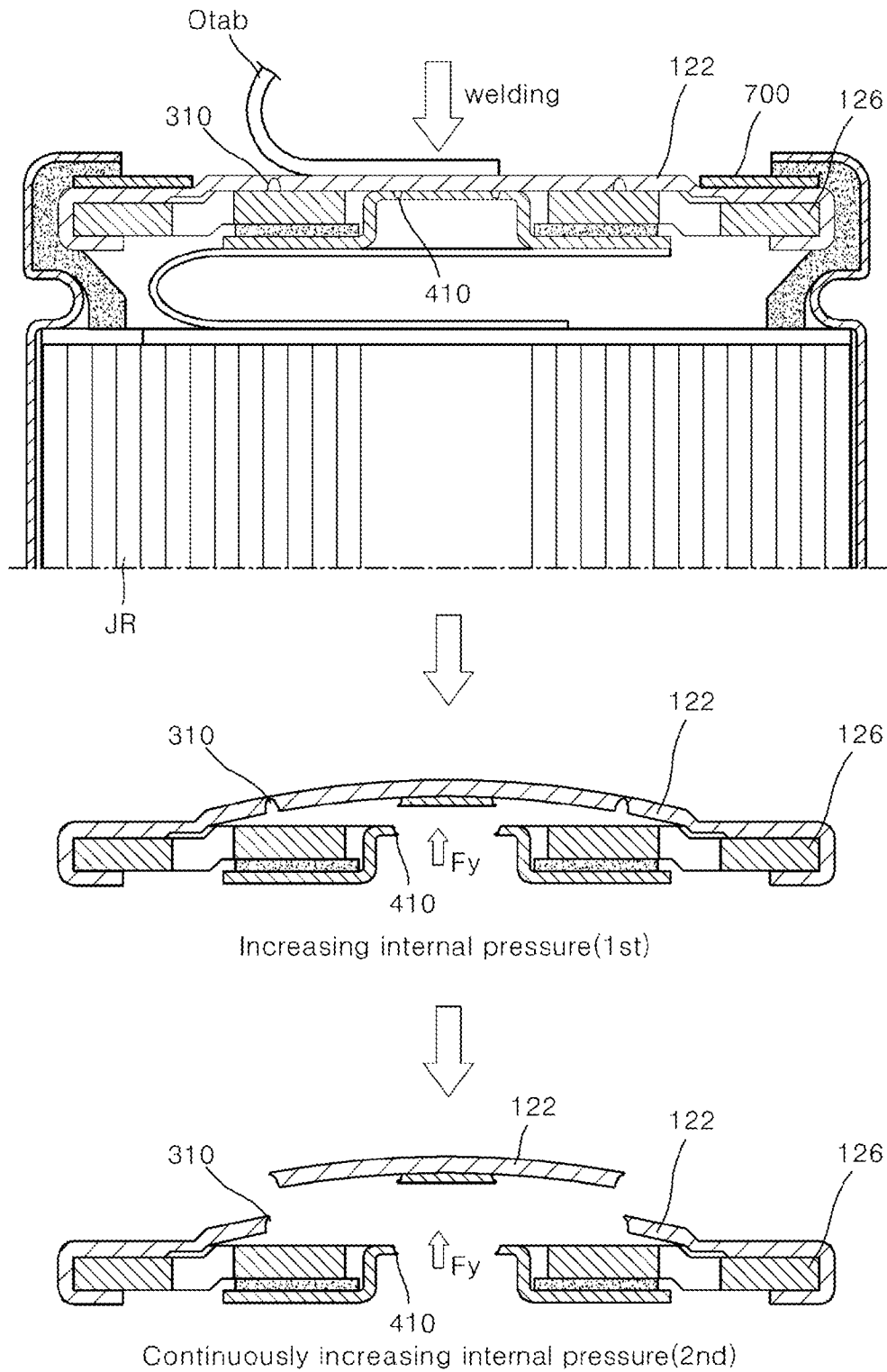
[Fig. 1]



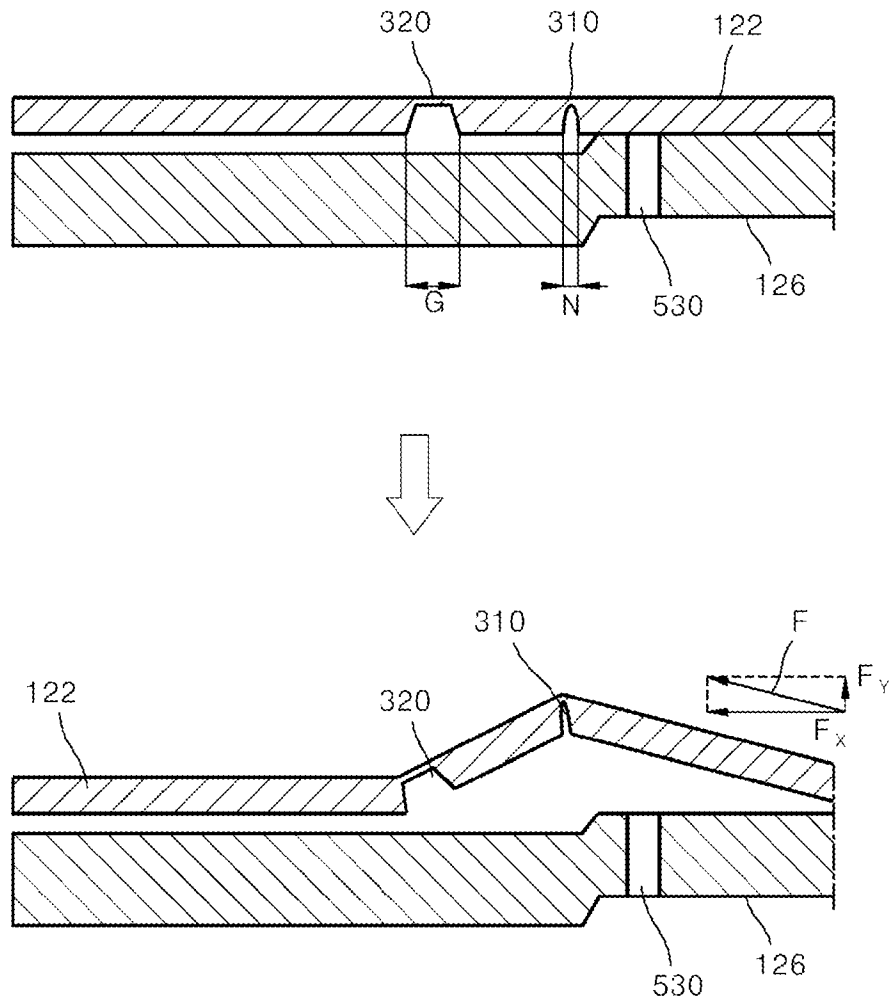
[Fig. 2]



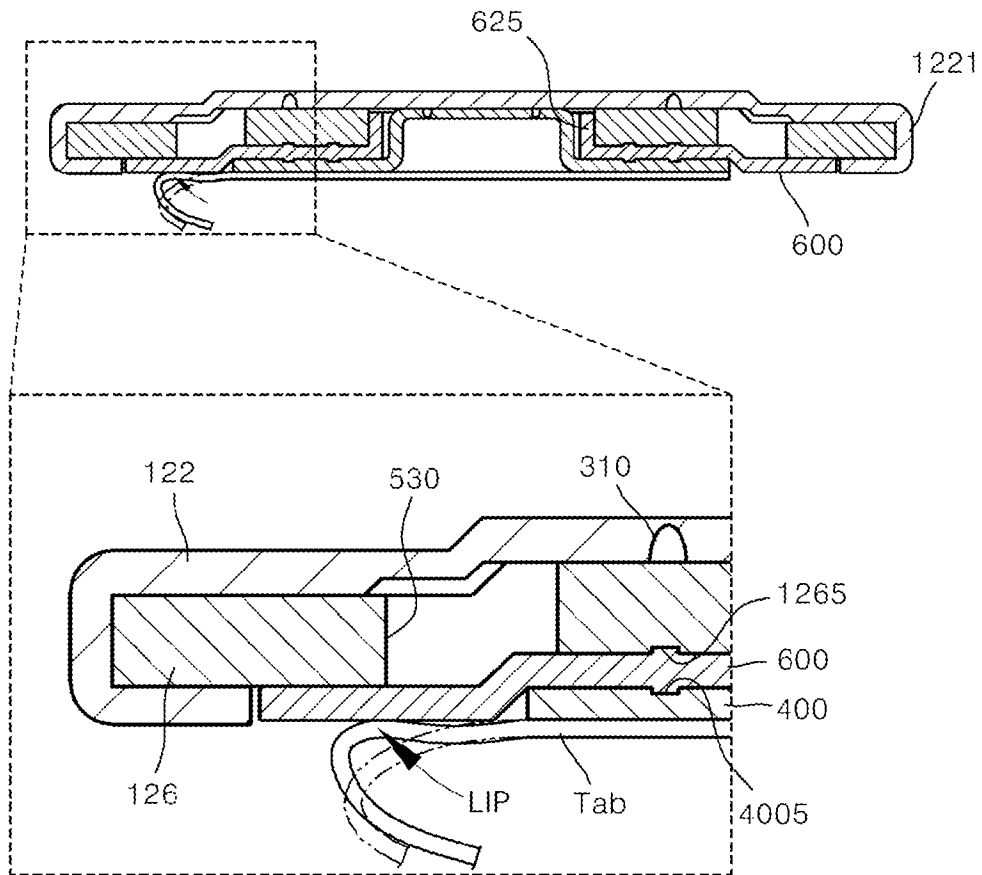
[Fig. 3]



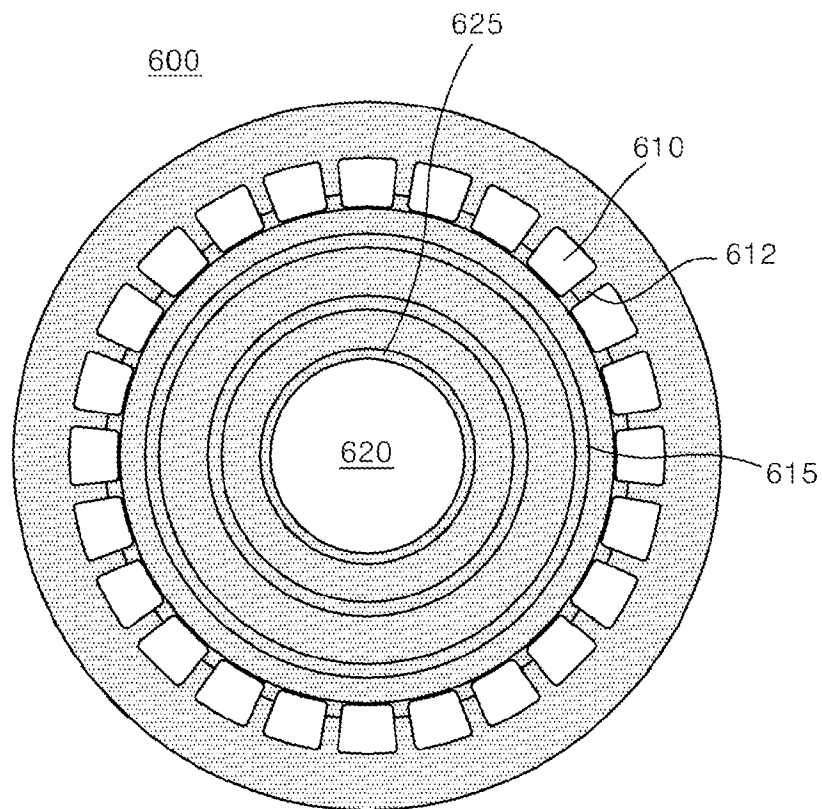
[Fig. 4]



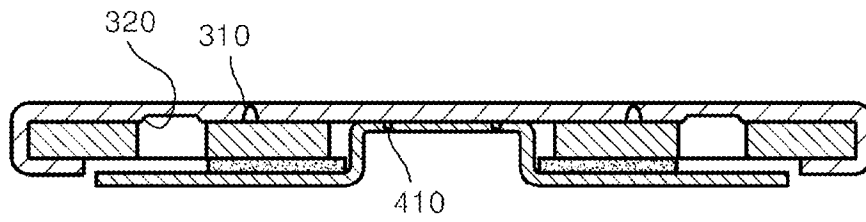
[Fig. 5]



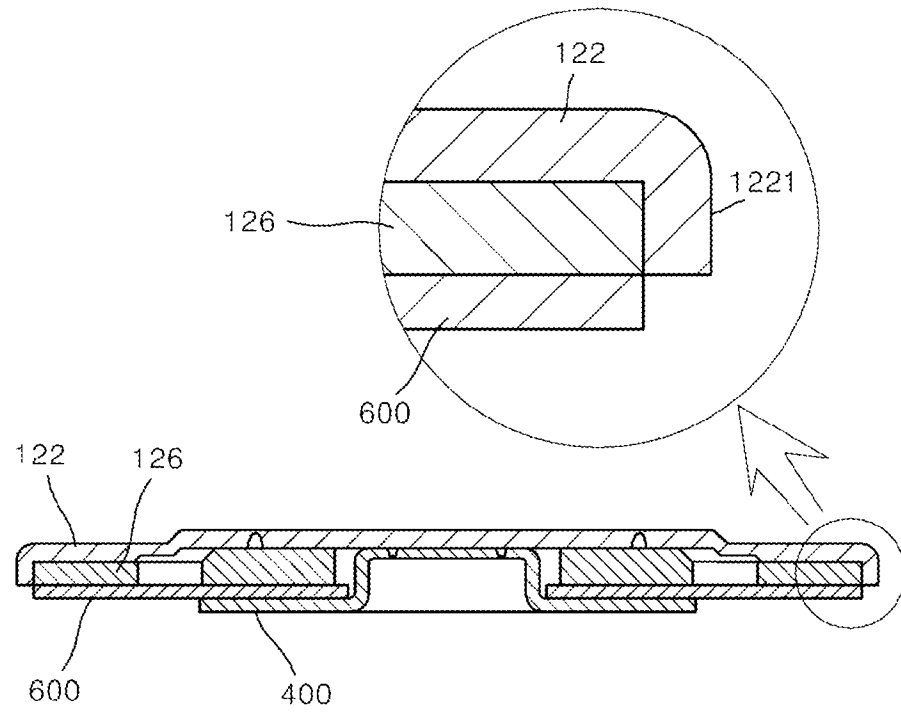
[Fig. 6]



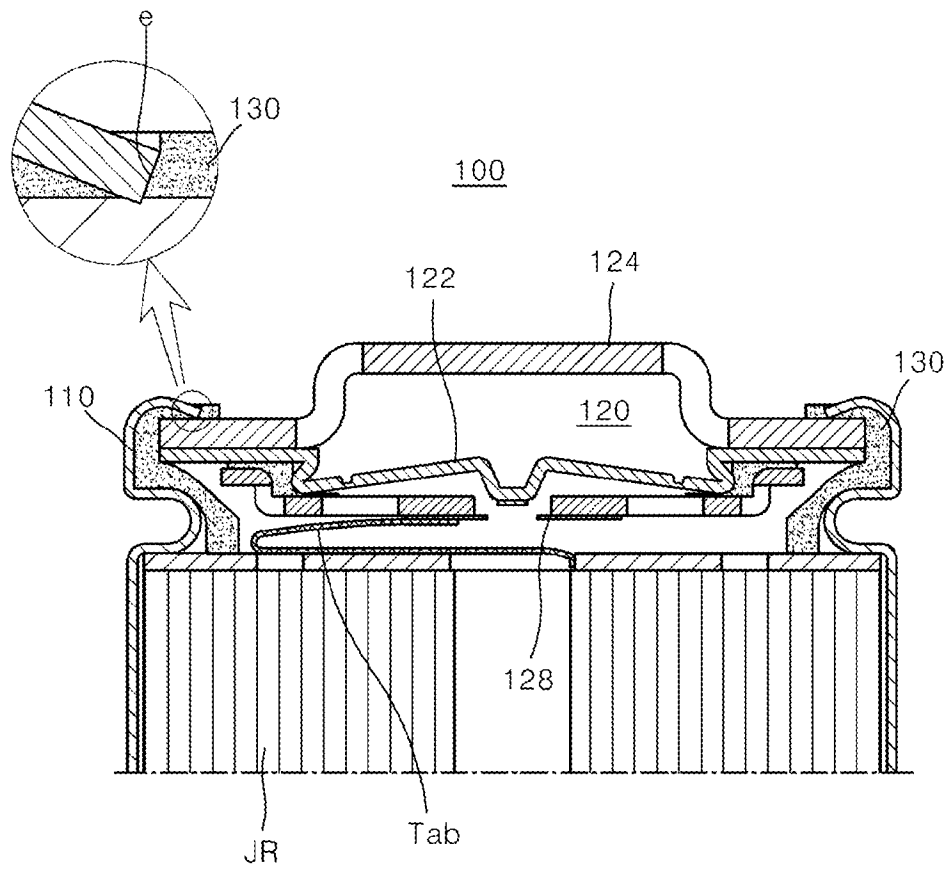
[Fig. 7]



[Fig. 8]



[Fig. 9]



INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2017/008736**A. CLASSIFICATION OF SUBJECT MATTER****H01M 2/04(2006.01)i, H01M 2/12(2006.01)i, H01M 10/04(2006.01)i**

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/04; H01M 2/12; H01M 2/08; H01M 2/34; H01M 2/20; H01M 2/26; H01M 2/10; H01M 2/02; H01M 10/04

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models

Japanese utility models and applications for utility models

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS(KIPO internal) & keywords: cap assembly, secondary battery, safety bent, shorting, notch

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KR 10-2016-0034794 A (SHIN HEUNG ENERGY & ELECTRONIC CO., LTD.) 30 March 2016 See figures 1-7; and claims 1-2.	1-16
A	US 2011-0206958 A1 (KIYAMA, A. et al.) 25 August 2011 See the entire document.	1-16
A	JP 2005-100927 A (MATSUSHITA ELECTRIC INDUSTRY CO., LTD.) 14 April 2005 See the entire document.	1-16
A	KR 10-2010-0062886 A (SAMSUNG SDI CO., LTD.) 10 June 2010 See the entire document.	1-16
A	KR 10-2006-0065378 A (LG CHEM, LTD.) 14 June 2006 See the entire document.	1-16



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

06 December 2017 (06.12.2017)

Date of mailing of the international search report

06 December 2017 (06.12.2017)

Name and mailing address of the ISA/KR

International Application Division
Korean Intellectual Property Office
189 Cheongsu-ro, Seo-gu, Daejeon, 35208, Republic of Korea

Facsimile No. +82-42-481-8578

Authorized officer

LEE, Dong Wook

Telephone No. +82-42-481-8163



INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No.

PCT/KR2017/008736

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
KR 10-2016-0034794 A	30/03/2016	KR 10-1772266 B1	30/08/2017
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		JP 4596289 B2	08/12/2010
		KR 10-1310015 B1	24/09/2013
		KR 10-2011-0082187 A	18/07/2011
		US 8936861 B2	20/01/2015
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		JP 5191473 B2	08/05/2013
		US 2010-0136388 A1	03/06/2010
		US 8486546 B2	16/07/2013
KR 10-2006-0065378 A	14/06/2006	KR 10-0858414 B1	11/09/2008

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Plasseraud IP
66, rue de la Chaussée d'Antin
75440 Paris Cedex 09
FRANCE

Date	17.08.21
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Reference MRA/201420/LLA	Application No./Patent No. 19889598.9 - 1108 / 3800697 PCT/KR2019015396
Applicant/Proprietor LG CHEM, LTD.	

Communication

The extended European search report is enclosed.

The extended European search report includes, pursuant to Rule 62 EPC, the supplementary European search report (Art. 153(7) EPC) and the European search opinion.

Copies of documents cited in the European search report are attached.

☒ 0 additional set(s) of copies of such documents is (are) enclosed as well.

Refund of the search fee

If applicable under Article 9 Rules relating to fees, a separate communication from the Receiving Section on the refund of the search fee will be sent later.

Should you wish to further prosecute this application in the examination phase, your attention is drawn to the provisions of Rule 70a EPC. An invitation to respond to the extended European search report will be issued shortly (R. 70(2) EPC).



**SUPPLEMENTARY
EUROPEAN SEARCH REPORT**

Application Number
EP 19 88 9598

DOCUMENTS CONSIDERED TO BE RELEVANT			
Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
Y	US 2012/114979 A1 (KIM JI YOUNG [KR] ET AL) 10 May 2012 (2012-05-10) * the whole document *	1-12	INV. H01M50/152
Y	US 2011/008654 A1 (KIM SUNGJONG [KR] ET AL) 13 January 2011 (2011-01-13) * the whole document *	1-12	
Y	EP 2 175 507 A1 (SAMSUNG SDI CO LTD [KR]) 14 April 2010 (2010-04-14) * the whole document *	1,4,11	
Y	WO 2018/030836 A1 (SHIN HEUNG ENERGY & ELECTRONIC CO LTD [KR]) 15 February 2018 (2018-02-15) * figures 1-7 *	1,2,4	
Y	KR 2017 0124343 A (LG CHEMICAL LTD [KR]) 10 November 2017 (2017-11-10) * figures 1-5 *	2,8	
The supplementary search report has been based on the last set of claims valid and available at the start of the search.			TECHNICAL FIELDS SEARCHED (IPC) H01M
Place of search		Date of completion of the search	Examiner
Munich		2 August 2021	Cappadonia, Marcella
CATEGORY OF CITED DOCUMENTS X : particularly relevant if taken alone Y : particularly relevant if combined with another document of the same category A : technological background O : non-written disclosure P : intermediate document T : theory or principle underlying the invention E : earlier patent document, but published on, or after the filing date D : document cited in the application L : document cited for other reasons & : member of the same patent family, corresponding document			

**ANNEX TO THE EUROPEAN SEARCH REPORT
ON EUROPEAN PATENT APPLICATION NO.**

EP 19 88 9598

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report.
The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

02-08-2021

Patent document cited in search report		Publication date	Patent family member(s)	Publication date
US 2012114979	A1	10-05-2012	CN 102473883 A	23-05-2012
			EP 2466668 A2	20-06-2012
			JP 5577405 B2	20-08-2014
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			CN 101714650 A	26-05-2010
			EP 2175507 A1	14-04-2010
			JP 5227285 B2	03-07-2013
			JP 2010086968 A	15-04-2010
			KR 20100038054 A	12-04-2010
			US 2010086835 A1	08-04-2010

WO 2018030836	A1	15-02-2018	NONE	

KR 20170124343	A	10-11-2017	NONE	

Information on Search Strategy - Pilot phase (see OJ 2015, A86)

The type of information contained in this sheet may change during the pilot for improving the usefulness of this new service.

Application Number

EP 19 88 9598

TITLE: SECONDARY BATTERY

APPLICANT: LG CHEM, LTD.

IPC CLASSIFICATION: H01M50/152

EXAMINER: Cappadonia, Marcella

CONSULTED DATABASES: WPI

CLASSIFICATION SYMBOLS DEFINING EXTENT OF THE SEARCH:

IPC:

CPC:

FI/F-TERMS:

KEYWORDS OR OTHER ELEMENTS FEATURING THE INVENTION:

The examination is being carried out on the **following application documents**

Description, Pages

1-17 filed with entry into the regional phase before the EPO

Claims, Numbers

1-12 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/11-11/11 filed with entry into the regional phase before the EPO

1. Art. 84 EPC

Claim 1 does not meet the requirements of Art. 84 EPC since the notches (113) and (115) are defined by the result to be achieved rather than by their physical characteristics, the notches are distinguishable only after explosion of the safety vent.

Claims 1, 2 and 3 are unclear since it is not evident the difference between "bottom surface" and "lower surface". It is also noted that said terms relate to the placement of the battery in the space and therefore are relative terms not limiting the subject-matter.

Claim 8 is not clear since not supported by the description contrary to art. 84 EPC: the plurality of notch portions with different lengths can be arranged along a single circle forming discontinuous notch.

Further, a relative terms "upward", "horizontal" or "height...decreases" are present in claims 10 and 11, which are not considered meeting the requirements of Art. 84 EPC.

2.

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2012/114979 A1 (KIM JI YOUNG [KR] ET AL) 10 May 2012
(2012-05-10)

D2 US 2011/008654 A1 (KIM SUNGJONG [KR] ET AL) 13 January 2011
(2011-01-13)

- D3 EP 2 175 507 A1 (SAMSUNG SDI CO LTD [KR]) 14 April 2010
(2010-04-14)
- D4 WO 2018/030836 A1 (SHIN HEUNG ENERGY & ELECTRONIC CO LTD
[KR]) 15 February 2018 (2018-02-15)
- D5 KR 2017 0124343 A (LG CHEMICAL LTD [KR]) 10 November 2017
(2017-11-10)

The following document has been cited in the international search report; the numbering will be adhered to in the rest of the procedure.

- D6 JP H06 215746 A (MATSUSHITA ELECTRIC IND CO., LTD.)

3.

D6 discloses a secondary battery comprising:

an electrode assembly;

a can member configured to accommodate the electrode assembly therein; and

a top cap assembly configured to cover an opening of the can member ,

wherein the top cap assembly comprises a top cap (5) positioned at the uppermost portion and a safety vent part (1) positioned below the top cap ,

wherein the safety vent part comprises:

a rupture portion ruptured when an internal pressure within the can member increases to a predetermined pressure or more;

a rupture notch portion (1a,1c) that defines a boundary of the rupture portion; and

a bending notch portion (1b) which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing the inside of the can member,

wherein the bending notch portion is bent when the internal pressure within the can member increases.

D6 is novelty destroying for the subject-matter of claims 1,2,3,4,5,6 (Art. 54(1) and (2) EPC), considering that the meaning of notch according to the Oxford Lexicon is “a hollow in a surface” and the folding elements 1a and 1b form two depression in the surface of the vent facing the inside of the can.

Further, the objection of lack of inventive step raised in the PCT-phase is fully maintained under the provisions of the EPC.

4.

D1, D2 and D3 disclose a secondary battery comprising:
an electrode assembly;
a can member configured to accommodate the electrode assembly therein; and
a top cap assembly configured to cover an opening of the can member,
wherein the top cap assembly comprises a top cap positioned at the uppermost portion and a safety vent part positioned below the top cap,
wherein the safety vent part comprises:
a rupture portion ruptured when an internal pressure within the can member increases to a predetermined pressure or more;
a rupture notch portion (D1 and D2, 324; D3, outer 163 as shown in fig 2A) that defines a boundary of the rupture portion; and
a bending notch portion (D1 and D2, 326; D3, inner 163 as shown in fig 2C) which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion,
wherein the bending notch portion is bent when the internal pressure within the can member increases.

The subject-matter of claim 1 of the present application differs from the battery of the prior art in that the bending notch is disposed on a surface of the safety vent facing the inside of the can member.

Said feature is just a design modification common in the field as shown in the comparative example of fig. 10 or, for example, in D4.

Therefore, the subject-matter of claim 1 does not involve an inventive step (art. 56 EPC).

Dependent claims 2-12 do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to inventive step.

The features of claims 2 and 8 are known in the art see D5.

The feature of claim 3 are disclosed in D4.

The feature of claims 4,5,7,9,11 are disclosed in D1 and D2 or D3.

The feature of claims 6 and 10 are common in the field.

The feature is just a design modification of the vent according to D1, D2 and D3, moreover no effect can be attributed to the claimed feature, also considering that none of the embodiment presented in the figures and in corresponding description do not fall within the scope of this claim, it is noted that the outer circle in fig .5, 7 and 8 does not represent the edge end of the safety vent but the position of the folding as shown in fig. 6 and 9.

Finally, it is noted that the comparative battery shown in fig 10 does not represent the closest prior art.



IRISH PATENTS OFFICE

Espacenet

Bibliographic data: KR20170124343 (A) — 2017-11-10

CAP ASSEMBLY FOR RECHARGEABLE BATTERY

Inventor(s): KWON TAE HYUN [KR]; SUNG JOO HWAN [KR]; KIM HYOUNG KWON [KR] ± (권태현, ; 성주환, ; 김형권)

Applicant(s): LG CHEMICAL LTD [KR] ± (주식회사 엘지화학)

Classification: - international: H01M2/04; H01M2/12
- cooperative: H01M50/152 (KR); H01M50/3425 (KR); H01M50/383 (KR); H01M2200/20 (KR); Y02E60/10 (EP, KR)

Application number: KR20160054122 20160502 Global Dossier

Priority number(s): KR20160054122 20160502

Also published as: KR102155706 (B1)

Abstract of KR20170124343 (A)

The present invention relates to a cap assembly for a secondary battery. According to the present invention, the cap assembly for a secondary battery comprises: a top cap located in an outer side of a battery wherein the top cap has a discharging hole formed therein; and a safe vent having a rupture notch ruptured in case of an internal pressure increase to discharge an internal gas to the discharging hole and having a bending notch formed on the rupture unit of which a location or a shape is changed due to ruptures wherein the bending notch is bent in case of ruptures of the rupture notch.



(19) 대한민국특허청(KR)
(12) 공개특허공보(A)

(11) 공개번호 10-2017-0124343
(43) 공개일자 2017년11월10일

(51) 국제특허분류(Int. Cl.)
H01M 2/12 (2006.01) H01M 2/04 (2006.01)
(52) CPC특허분류
H01M 2/1247 (2013.01)
H01M 2/046 (2013.01)
(21) 출원번호 10-2016-0054122
(22) 출원일자 2016년05월02일
심사청구일자 없음

(71) 출원인
주식회사 엘지화학
서울특별시 영등포구 여의대로 128 (여의도동)
(72) 발명자
권태현
대전광역시 유성구 문지로 188 LG화학 기술연구원 내
성주환
대전광역시 유성구 문지로 188 LG화학 기술연구원 내
김형권
대전광역시 유성구 문지로 188 LG화학 기술연구원 내
(74) 대리인
특허법인태평양

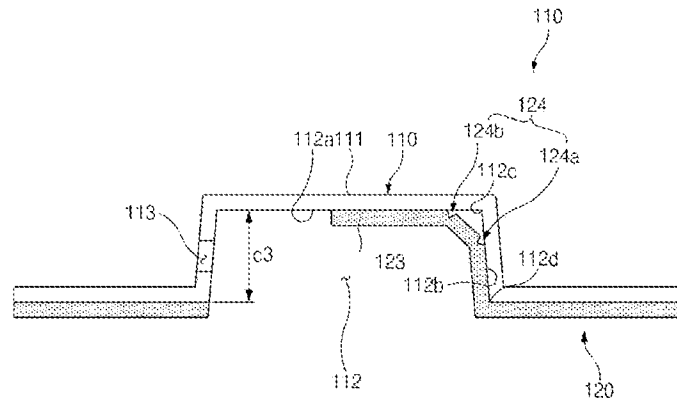
전체 청구항 수 : 총 26 항

(54) 발명의 명칭 이차전지용 캡 조립체

(57) 요약

본 발명은 이차전지용 캡 조립체에 관한 것으로, 본 발명에 따른 이차전지용 캡 조립체는, 전지의 외각에 위치되어 배출구가 형성된 탈 캡 및 상기 배출구로 내부 가스의 배출이 가능하도록 내압 상승시 파열되는 파열 노치가 형성되고, 상기 파열 노치의 파열시 절곡되는 절곡 노치가 상기 파열로 인해 위치 또는 형태가 변하는 파열부 상에 형성되는 안전 밴드를 포함한다.

도 1은 도 3



(52) CPC특허분류

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명세서

청구범위

청구항 1

전지의 외각에 위치되어 배출구가 형성된 탭 캡; 및

상기 배출구로 내부 가스의 배출이 가능하도록 내압 상승시 파열되는 파열 노치가 형성되고, 상기 파열 노치의 파열시 절곡되는 절곡 노치가 상기 파열로 인해 위치 또는 형태가 변하는 파열부 상에 형성되는 안전 밴트를 포함하는 이차전지용 캡 조립체.

청구항 2

청구항 1에 있어서,

상기 탭 캡 보다 상기 전지의 내측에 위치된 상기 안전 밴트에서 상기 파열 노치의 파열시 상기 파열부가 상기 탭 캡과 충돌되며 상기 절곡 노치가 절곡되는 이차전지용 캡 조립체.

청구항 3

청구항 2에 있어서,

상기 파열 노치의 파열 전에는 상기 탭 캡의 상기 배출구는 상기 안전 밴트에 의해 모두 폐쇄되고,

상기 파열 노치의 파열 및 상기 절곡 노치의 절곡 후에는, 상기 배출구가 모두 개방되는 이차전지용 캡 조립체.

청구항 4

청구항 2에 있어서,

상기 파열 노치의 파열시 상기 파열부가 상기 탭 캡의 내측면과 대응되는 형태로 절곡되도록 상기 절곡 노치가 절곡되는 이차전지용 캡 조립체.

청구항 5

청구항 2에 있어서,

상기 파열 노치는 상기 안전 밴트의 평면도 상으로 일측이 개방된 원호 형태로 형성되는 이차전지용 캡 조립체.

청구항 6

청구항 5에 있어서,

상기 절곡 노치는 상기 평면도 상에서 상기 파열 노치의 원호 내측부에 형성되는 이차전지용 캡 조립체.

청구항 7

청구항 6에 있어서,

상기 절곡 노치는 상기 파열 노치의 개방 부위와 상기 파열 노치의 원호 중심 사이에 위치되는 이차전지용 캡 조립체.

청구항 8

청구항 6에 있어서,

상기 절곡 노치는 상기 평면도 상에서 직선 형태로 형성되는 이차전지용 캡 조립체.

청구항 9

청구항 6에 있어서,

상기 절곡 노치는 상기 평면도 상에 형성되는 상기 파열 노치의 원호 중심에서 상기 파열 노치의 개방 방향으로 연장된 선에 대해 수직으로 라인을 형성하는 이차전지용 캡 조립체.

청구항 10

청구항 5에 있어서,

상기 절곡 노치는 평면도 상에 형성되는 상기 파열 노치의 원호 중심에서 상기 파열 노치의 개방 부위로 향하는 방향에 대해 기울어진 방향으로 연장되며 형성되는 이차전지용 캡 조립체.

청구항 11

청구항 10에 있어서,

상기 절곡 노치의 양단은 상기 파열 노치의 원호 상에 위치되는 이차전지용 캡 조립체.

청구항 12

청구항 1 내지 청구항 11 중 어느 한 항에 있어서,

상기 파열 노치가 상기 절곡 노치 보다 먼저 파열되도록, 상기 파열 노치가 상기 절곡 노치 보다 크게 형성되는 이차전지용 캡 조립체.

청구항 13

청구항 12에 있어서,

상기 파열 노치의 깊이는 상기 절곡 노치의 깊이 보다 깊게 형성되는 이차전지용 캡 조립체.

청구항 14

청구항 12에 있어서,

상기 파열 노치의 폭은 상기 절곡 노치의 폭 보다 넓게 형성되는 이차전지용 캡 조립체.

청구항 15

청구항 1 내지 청구항 11 중 어느 한 항에 있어서,

상기 절곡 노치는 복수개로 형성되는 이차전지용 캡 조립체.

청구항 16

청구항 5 내지 청구항 10 중 어느 한 항에 있어서,

상기 탭 캡은 상기 안전 밸브의 상기 파열부가 수용되는 수용부를 포함하는 이차전지용 캡 조립체.

청구항 17

청구항 16에 있어서,

상기 배출구는 상기 수용부의 측면에 형성되어 상기 전지의 내부에서 발생하는 가스가 배출되게 하는 이차전지용 캡 조립체.

청구항 18

청구항 17에 있어서,

상기 탭 캡에서 수용부의 천장면과 측면 사이의 각도는 수직으로 형성되는 이차전지용 캡 조립체.

청구항 19

청구항 17에 있어서,

상기 탭 캡에서 수용부의 천장면과 측면 사이의 각도는 85~100° 로 형성되는 이차전지용 캡 조립체.

청구항 20

청구항 17에 있어서,

상기 안전 밴트에서 상기 파열 노치의 파열시 평면상으로 상기 파열 노치의 원호 상 노치가 형성되지 않은 부분인 비노치부가 상기 탭 캡 방향으로 절곡되도록 형성되는 이차전지용 캡 조립체.

청구항 21

청구항 20에 있어서,

상기 안전 밴트의 상기 비노치부는 상기 파열 노치의 파열시 상기 수용부의 가장자리에 의해 절곡되는 이차전지용 캡 조립체.

청구항 22

청구항 21에 있어서,

상기 절곡 노치는 상기 파열 노치의 파열시 상기 수용부의 천장 모서리부에 대응되는 위치에 위치되도록 형성되는 이차전지용 캡 조립체.

청구항 23

청구항 21에 있어서,

상기 절곡 노치는 제1 절곡 노치 및 제2 절곡 노치를 포함하고,

상기 제1 절곡 노치 및 상기 제2 절곡 노치는 상호 일정거리 이격되어 나란하게 형성되는 이차전지용 캡 조립체.

청구항 24

청구항 23에 있어서,

상기 비노치부에서 상기 제1 절곡 노치 까지의 거리는 상기 탭 캡의 상기 수용부 길이 보다 작게 형성되고,

상기 비노치부에서 상기 제2 절곡 노치 까지의 거리는 상기 탭 캡의 수용부 길이 보다 크게 형성되는 이차전지용 캡 조립체.

청구항 25

청구항 20에 있어서,

상기 수용부를 중심으로 4방향 중 어느 한 방향에 상기 안전 밴트의 상기 비노치부가 위치되고, 나머지 하나 이상의 방향에 상기 배출구가 위치되는 이차전지용 캡 조립체.

청구항 26

청구항 20에 있어서,

상기 수용부를 중심으로 4 방향 중 어느 한 방향에 상기 안전 밴트의 상기 비노치부가 위치되고, 나머지 3 방향에 상기 배출구가 위치되는 이차전지용 캡 조립체.

발명의 설명

기술분야

[0001] 본 발명은 이차전지용 캡 조립체에 관한 것이다.

배경기술

[0003] 이차 전지는 일차 전지와는 달리 재충전이 가능하고, 또 소형 및 대용량화 가능성으로 인해 근래에 많이 연구

개발되고 있다. 모바일 기기에 대한 기술 개발과 수요가 증가함에 따라 에너지원으로서의 이차 전지의 수요가 급격하게 증가하고 있다.

[0004] 특히, 리튬 이차 전지는 작동 전압이 3.6V로서, 전자 장비의 전원으로 많이 사용되는 니켈-카드뮴 전지 또는 니켈-수소 전지보다 약3배의 용량을 가지며, 단위 중량당 에너지 밀도가 높기 때문에 그 활용 정도가 급속도로 증가되는 추세에 있다.

[0005] 이러한 리튬 이차 전지는 주로 리튬계 산화물과 탄소재를 각각 양극 활물질과 음극 활물질로 사용한다. 또한, 리튬 이차 전지는 각형 전지, 원통형 전지, 파우치형 전지로 구분될 수 있다.

[0006] 리튬 이온 이차 전지는 양극/세퍼레이터/음극이 순차적으로 배치된 전극 조립체와, 전극 조립체를 전해액과 함께 밀봉 수납하는 외장재를 구비한다. 특히, 각형 또는 원통형 이차 전지의 외장재는 개방단이 형성된 케이스 및 케이스의 개방단에 밀봉 결합되는 캡 조립체를 구비한다.

선행기술문헌

특허문헌

[0008] (특허문헌 0001) 한국 공개특허 제10-2009-0105547호

발명의 내용

해결하려는 과제

[0009] 본 발명의 하나의 관점은 전지 내부의 가스 발생으로 전지의 내압 상승시 가스를 전지 외부로 용이하게 배출할 수 있는 이차전지용 캡 조립체를 제공하기 위한 것이다.

과제의 해결 수단

[0011] 본 발명의 일 실시예에 따른 이차전지용 캡 조립체는, 전지의 외각에 위치되어 배출구가 형성된 탑 캡 및 상기 배출구로 내부 가스의 배출이 가능하도록 내압 상승시 파열되는 파열 노치가 형성되고, 상기 파열 노치의 파열 시 절곡되는 절곡 노치가 상기 파열로 인해 위치 또는 형태가 변하는 파열부 상에 형성되는 안전 벤트를 포함할 수 있다.

발명의 효과

[0013] 본 발명에 따른 이차전지용 캡 조립체는 전지 내부의 가스 발생으로 전지의 내압 상승시 가스 배출이 용이한 형태로 탑 캡 및 안전 벤트가 형성되어 가스 배출의 용이성을 확보할 수 있다.

[0014] 특히, 본 발명에 따른 이차전지용 캡 조립체는 내압 상승으로 인해 파열되는 안전 벤트의 파열부가 가스 배출이 용이한 형태로 파열 및 절곡되어 가스 배출에 필요한 통로를 용이하게 확보할 수 있다.

도면의 간단한 설명

[0016] 도 1은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체가 설치되는 이차전지를 도시한 절개 사시도이다.

도 2는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체를 나타낸 단면도이다.

도 3은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 벤트가 파열된 후 상태를 나타낸 단면도이다.

도 4는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 벤트가 파열된 후 상태를 나타낸 개념도이다.

다.

도 5는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 밴트를 나타낸 평면도이다.

도 6는 도 5에서 A-A'선을 따라 절개한 단면이다.

발명을 실시하기 위한 구체적인 내용

- [0017] 본 발명의 목적, 특정한 장점들 및 신규한 특징들은 첨부된 도면들과 연관되어지는 이하의 상세한 설명과 바람직한 실시예들로부터 더욱 명백해질 것이다. 본 명세서에서 각 도면의 구성요소들에 참조번호를 부가함에 있어서, 동일한 구성 요소들에 한해서는 비록 다른 도면상에 표시되더라도 가능한 한 동일한 번호를 가지도록 하고 있음에 유의하여야 한다. 또한, 본 발명은 여러 가지 상이한 형태로 구현될 수 있으며 여기에서 설명하는 실시예에 한정되지 않는다. 그리고, 본 발명을 설명함에 있어서, 본 발명의 요지를 불필요하게 흐릴 수 있는 관련된 공지 기술에 대한 상세한 설명은 생략하도록 한다.
- [0019] 도 1은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체가 설치되는 이차전지를 도시한 절개 사시도이고, 도 2는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체를 나타낸 단면도이다.
- [0020] 도 1 및 도 2를 참고하면, 본 발명의 일 실시예에 따른 이차전지용 캡 조립체(100)는, 배출구(113)가 형성된 탑 캡(Top cap)(110) 및 이차 전지(10)의 내압 상승시 파열되는 파열 노치(Notch)(121) 및 절곡되는 절곡 노치(124)가 형성된 안전 밴트(bent)(120)를 포함한다.
- [0022] 도 3은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 밴트가 파열된 후 상태를 나타낸 단면도이고, 도 4는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 밴트가 파열된 후 상태를 나타낸 사시도이다.
- [0023] 이하에서, 도 1 내지 도 6을 참조하여, 본 발명의 일 실시예인 이차전지용 캡 조립체에 대해 보다 상세히 설명하기로 한다.
- [0024] 도 1 및 도 2를 참고하면, 탑 캡(110)은 이차 전지(10)의 측외각에 위치되어, 외부 단자(111)와, 수용부(112) 및 배출구(113)를 포함하여 형성될 수 있다.
- [0025] 외부 단자(111)는 탑 캡(110)의 상단에 돌출 형성되어 이차 전지(10)의 내부에 수용된 전극 적층체(12)와 전기적으로 연결되어 양극 단자 또는 음극 단자를 형성할 수 있다.
- [0026] 전극 적층체(12)는 케이스(11)에 수용되고, 케이스(11)의 내부에 전해액이 주입된다. 또한, 전극 적층체(12)는 충방전이 가능한 발전소자로서, 양극(12a)과 음극(12b) 및 분리막(12c)이 결합되어 적층된 구조를 형성한다. 여기서, 전극 적층체(12)의 양극(12a)은 탑 캡(110)의 외부 단자(111)와 전기적으로 연결되어 외부 단자(111)가 양극 단자의 기능을 할 수 있다.
- [0027] 분리막(12c)은 양극(12a)과 음극(12b)을 분리하여 전기적으로 절연시킨다. 이때, 양극(12a) 및 음극(12b)은 분리막(12c)과 함께 권취되어 젤리 롤(Jelly roll) 형태로 형성될 수 있다. 아울러, 분리막(12c)은 절연 재질로 이루어져 양극(12a)과 음극(12b) 사이를 전기적으로 절연한다. 여기서, 분리막(12c)은 예를 들어 미다공성을 가지는 폴리에틸렌, 폴리프로필렌 등 폴리올레핀계 수지막으로 형성될 수 있다.
- [0028] 한편, 전극 적층체(12)의 중심부에는 원통형의 센터 핀(13)이 구비될 수 있다. 센터 핀(13)은 소정의 강도를 부여하기 위해 금속 소재로 이루어질 수 있으며, 관재를 통괄해 절곡한 중공형의 원통형 구조로 이루어질 수 있다. 이러한 센터 핀(13)은 전극 적층체(12)를 고정 및 지지하는 작용과 충방전 및 작동시 내부 반응에 의해 발생하는 가스를 방출하는 통로로서 작용할 수 있다.
- [0029] 도 2 및 도 3을 참고하면, 수용부(112)는 외부 단자(111)의 하부에 안전 밴트(120)의 파열부(123)가 수용되도록 안전 밴트(120) 방면으로 개방된 형태로 형성될 수 있다.
- [0030] 또한, 탑 캡(110)에서 수용부(112)의 전장면(112a)과 측면(112b) 사이의 각도(α)는 85° ~ 100° 로 형성될 수 있다. 이때, 탑 캡(110)의 수용부(112)에서 전장면(112a)과 측면(112b) 사이의 각도는 예를 들어 수직으로 형성될 수 있다.

- [0031] 도 3 및 도 4를 참고하면, 배출구(113)는 수용부(112)의 측면(112b)에 형성되어 전지 내부의 가스를 배출할 수 있다. 또한, 배출구(113)는 수용부(112)를 중심으로 4 방향 중에서 어느 한 방향을 제외한 방향에 형성될 수 있다. 이때, 배출구(113)는 예를 들어 수용부(112)를 중심으로 4 방향 중에서 3 방향에 형성될 수 있다.
- [0033] 도 5는 본 발명의 일 실시예에 따른 이차전지용 랩 조립체에서 안전 밸트를 나타낸 평면도이고, 도 6은 도 5에서 A-A'선을 따라 절개한 단면이다.
- [0034] 도 3 및 도 5를 참고하면, 안전 밸트(120)는 랩 캡(110) 보다 전지의 내측에 위치되어, 파열 노치(121) 및 절곡 노치(124)가 형성될 수 있다.
- [0035] 또한, 안전 밸트(120)는 랩 캡(110)의 배출구(113)를 개방 및 폐쇄할 수 있다. 여기서, 랩 캡(110)의 배출구(113)와 전기 내부를 연결하는 통로 상에 안전 밸트(120)가 위치된다. 이에 따라, 파열 노치(121)의 파열 전에는 랩 캡(110)의 배출구(113)가 폐쇄되고, 전지의 내압 상승으로 안전 밸트(120)의 파열 노치(121)가 파열되면 랩 캡(110)의 배출구(113)가 개방될 수 있다.
- [0036] 파열 노치(121)는 전지 내압의 상승시 랩 캡(110) 방향으로 파열될 수 있다. 이때, 파열 노치(121)의 파열에 따라 위치 또는 형태가 변하는 파열부(123)는 랩 캡(110)의 수용부(112) 방향으로 파열되어 수용부(112)에 수용될 수 있다.
- [0037] 아울러, 파열 노치(121)는 평면도 상으로 일측이 개방된 원호 형태로 형성될 수 있다. 여기서, 파열 노치(121)의 개방 부위는 원호 중심으로 소정 각도로 형성될 수 있다. 이때, 파열 노치(121)의 개방 부위 각도(β)는 예를 들어 60° 이내로 형성될 수 있다.
- [0038] 절곡 노치(124)는 파열 노치(121)의 파열시 절곡되도록 파열 노치(121)의 파열에 따라 위치 또는 형태가 변하는 파열부(123) 상에 형성된다. 여기서, 파열부(123)는 파열시 랩 캡(110)과 충돌하게 되고, 랩 캡(110)과의 충돌력에 의해 절곡 노치(124)가 절곡되게 된다. 이때, 파열 노치(121)의 파열시 파열부(123)가 랩 캡(110)의 내측면과 대응되는 형태로 절곡되도록 절곡 노치(124)가 절곡될 수 있다. 따라서, 파열 노치(121)의 파열시 파열부(123)와 랩 캡(110)의 내측면 사이에 공간 발생을 줄여 랩 캡(110)의 배출구(113)가 모두 개방될 수 있다. 이에 따라, 전지의 내부 가스 배출에 필요한 통로를 최대한 확보할 수 있다. 특히, 랩 캡(110)에서 수용부(112)의 천장면(112a)과 측면(112b) 사이의 각도(α)가 $85^\circ\sim 100^\circ$ 또는 수직으로 형성되면, 파열부(123)가 랩 캡(110)의 수용부(112) 내측면에 대응되도록 절곡시 배출구(113)의 개방이 더욱 용이해짐에 따라 가스 배출의 용이성을 훨씬 더 효과적으로 확보할 수 있다. 여기서, 랩 캡(110)의 수용부(112) 내측면은 천장면(112a) 및 측면(112b)을 포함하여 구성될 수 있다.
- [0039] 또한, 절곡 노치(124)는 평면도 상으로 파열 노치(121)의 원호 내측부에 형성될 수 있다. 이때, 절곡 노치(124)는 파열 노치(121)의 개방 부위와 파열 노치(121)의 원호 중심(C) 사이에 위치될 수 있다. 그리고, 절곡 노치(124)는 평면도 상으로 직선 형태로 형성될 수 있다.
- [0040] 아울러, 절곡 노치(124)는 평면상으로 파열 노치(121)의 원호 중심에서 파열 노치(121)의 개방 부위로 향하는 방향에 대해 기울어진 방향으로 연장되며 형성될 수 있으며, 또한, 절곡 노치(124)는 평면상으로 파열 노치(121)의 원호 중심에서 파열 노치(121)의 개방 방향으로 연장되는 선(L)에 대해 수직으로 라인을 형성할 수 있다. 여기서, 절곡 노치(124)의 양단은 파열 노치(121)의 원호 상에 위치될 수 있다.
- [0041] 한편, 안전 밸트(120)에서 파열 노치(121)의 파열시 평면상으로 파열 노치(121)의 원호 상 노치가 형성되지 않은 부분인 비노치부(122)가 랩 캡(110) 방향으로 절곡되도록 형성될 수 있다. 여기서, 안전 밸트(120)의 비노치부(122)는 수용부(112)의 가장자리에 위치한 모서리(112d)에 대응되도록 위치되어, 파열 노치(121)의 파열시 수용부(112)의 가장자리에 위치한 모서리(112d)에 의해 절곡될 수 있다. 이때, 안전 밸트(120)에서 랩 캡(110)의 하면과 접촉되지 않는 파열부(123)가 비노치부(122)를 기준으로 랩 캡(110)의 수용부(112)에 수용되는 방향으로 절곡될 수 있다.
- [0042] 그리고, 절곡 노치(124)는 파열 노치(121)의 파열시 수용부(112)의 천장면(112a)과 측면(112b) 사이에 형성된 천장 모서리부(112c)에 대응되는 위치에 위치되도록 형성될 수 있다. 즉, 안전 밸트(120)에서 절곡되는 비노치부(122)에서 절곡 노치(124)까지의 거리는 수용부(112)의 깊이에 대응되도록 형성될 수 있다.
- [0043] 또한, 절곡 노치(124)는 복수개로 형성될 수 있다. 이때, 절곡 노치(124)는 제1 절곡 노치(124a) 및 제2 절곡 노치(124b)를 포함하고, 제1 절곡 노치(124a) 및 제2 절곡 노치(124b)는 상호 일정거리 이격되어 나란하게 형성

될 수 있다.

[0044] 여기서, 비노치부(122)에서 제1 절곡 노치(124a) 까지의 거리(c1)는 탑 캡(110)의 수용부(112) 깊이(c3) 보다 작게 형성되고, 비노치부(122)에서 제2 절곡 노치(124b)까지의 거리(c2)는 탑 캡(110)의 수용부(112) 깊이(c3) 보다 크게 형성될 수 있다.

[0045] 한편, 도 6을 참고하면, 파열 노치(121)가 절곡 노치(124) 보다 먼저 파열되도록, 파열 노치(121)가 절곡 노치(124) 보다 크게 형성되거나 깊게 형성될 수 있다. 여기서, 파열 노치(121)의 폭(a1)은 절곡 노치(124)의 폭(a2) 보다 넓게 형성될 수 있다. 또한, 파열 노치(121)의 길이(b1)는 절곡 노치(124)의 길이(b2) 보다 깊게 형성될 수 있다.

[0046] 한편, 도 4 및 도 5를 참고하면, 수용부(112)를 중심으로 4방향 중 어느 한 방향에 안전 벨트(120)의 비노치부(122)가 위치되고, 나머지 하나 이상의 방향에 상기 배출구(113)가 위치될 수 있다. 여기서, 수용부(112)를 중심으로 4 방향 중 어느 한 방향에 안전 벨트(120)의 비노치부(122)가 위치되고, 나머지 3 방향에 상기 배출구(113)가 위치 될 수 있다.

[0048] 이상 본 발명을 구체적인 실시예를 통하여 상세히 설명하였으나, 이는 본 발명을 구체적으로 설명하기 위한 것으로, 본 발명에 따른 이차전지용 캡 조립체는 이에 한정되지 않는다. 본 발명의 기술적 사상 내에서 당해 분야의 통상의 지식을 가진 자에 의해 다양한 실시가 가능하다고 할 것이다.

[0049] 또한, 발명의 구체적인 보호 범위는 첨부된 특허청구범위에 의하여 명확해질 것이다.

부호의 설명

- [0051]
- 10: 이차 전지
 - 11: 케이스
 - 12: 전극 적층체
 - 12a: 양극
 - 12b: 음극
 - 12c: 분리막
 - 13: 셀터면
 - 100: 캡 조립체
 - 110: 탑 캡
 - 111: 외부 단자
 - 112: 수용부
 - 112a: 전장면
 - 112b: 측면
 - 112c: 전장 모서리부
 - 112d: 모서리
 - 113: 배출구
 - 120: 안전 벨트
 - 121: 파열 노치
 - 122: 비노치부

123: 화열부

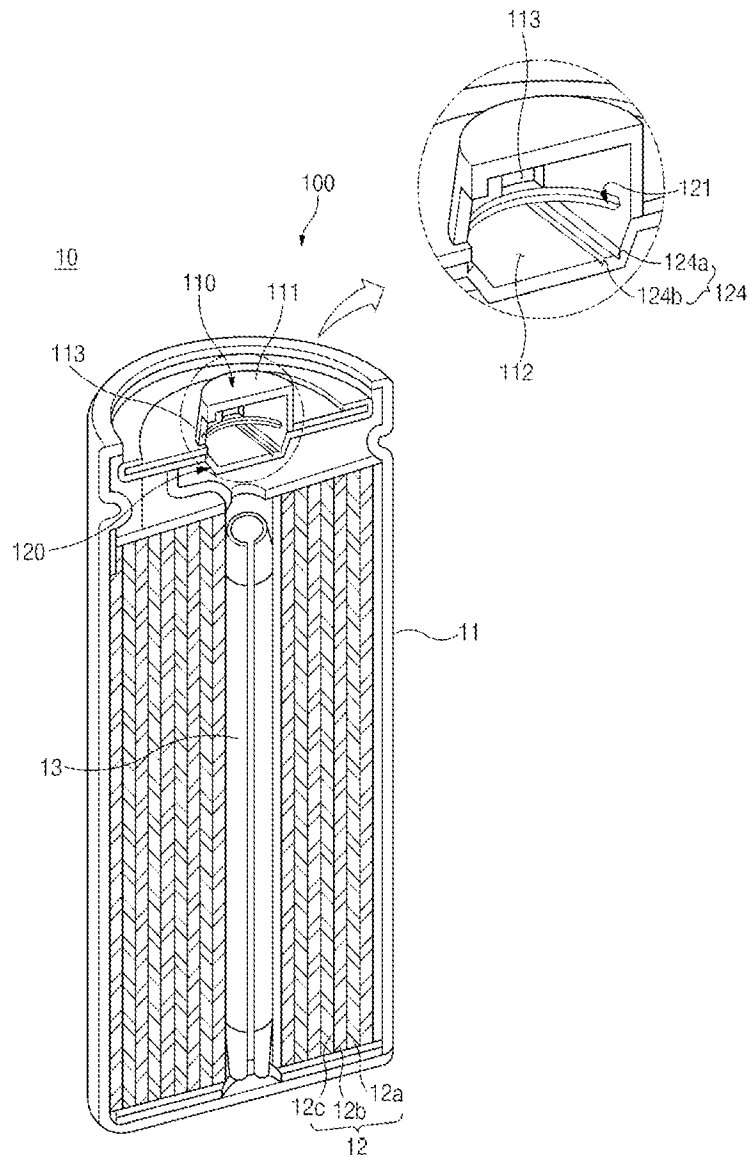
124: 절곡 노치

124a: 제1 절곡 노치

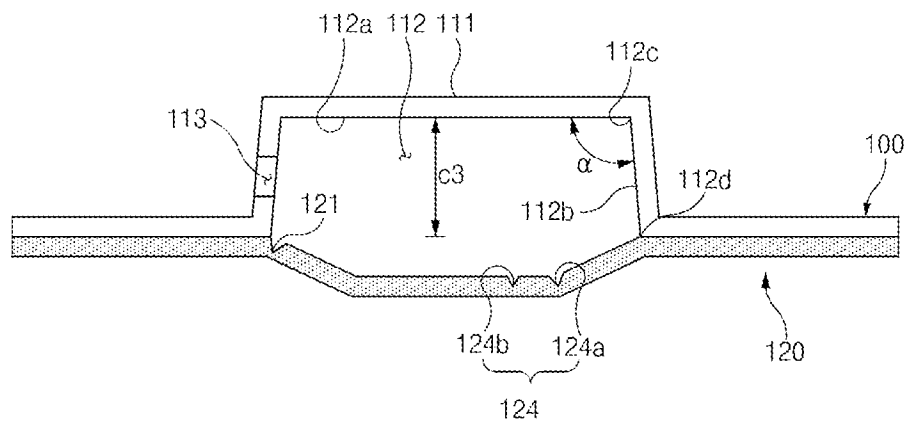
124b: 제2 절곡 노치

도면

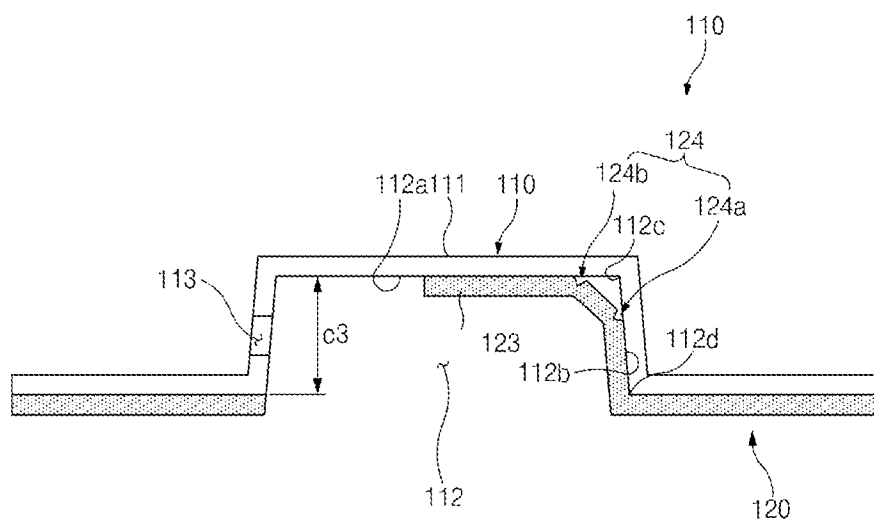
도면



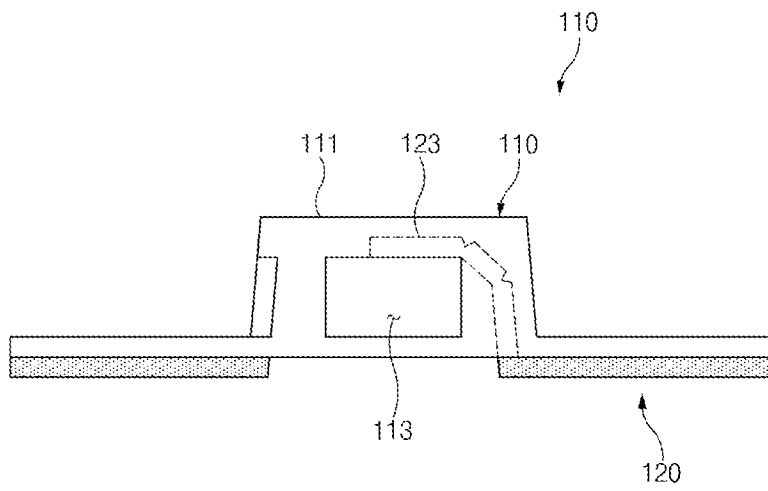
도면2



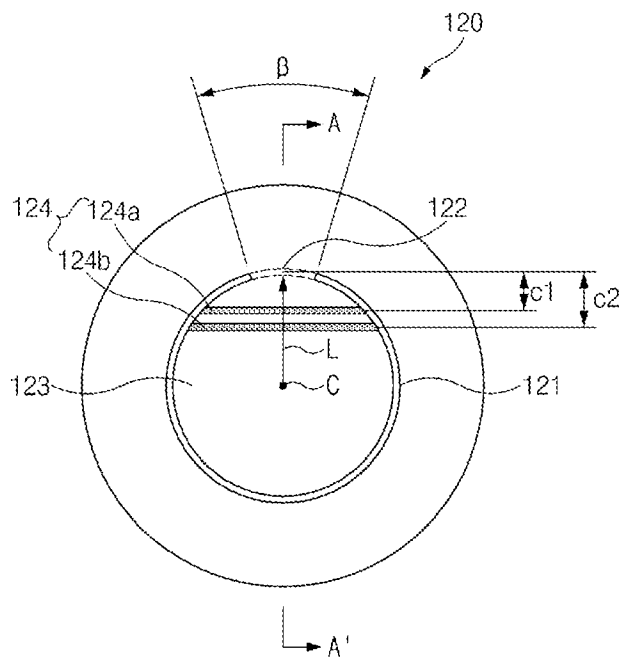
도면3



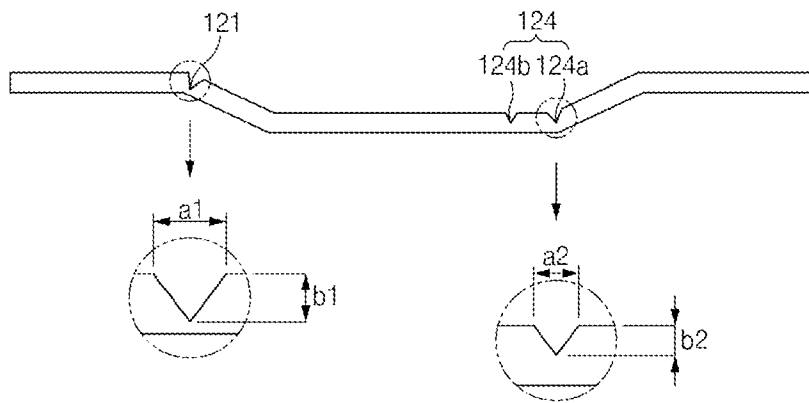
도면4

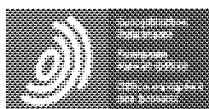


도면5



도면6





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CAP ASSEMBLY FOR RECHARGEABLE BATTERY

Inventor(s): KWON TAE HYUN [KR]; SUNG JOO HWAN [KR]; KIM HYOUNG KWON [KR] ± (권태현, ; 성주환, ; 김형권)

Applicant(s): LG CHEMICAL LTD [KR] ± (주식회사 엘지화학)

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(EP, KR)

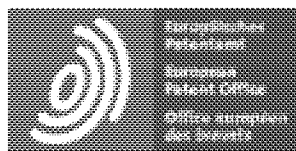
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Abstract of KR20170124343 (A)

The present invention relates to a cap assembly for a secondary battery. According to the present invention, the cap assembly for a secondary battery comprises: a top cap located in an outer side of a battery wherein the top cap has a discharging hole formed therein; and a safe vent having a rupture notch ruptured in case of an internal pressure increase to discharge an internal gas to the discharging hole and having a bending notch formed on the rupture unit of which a location or a shape is changed due to ruptures wherein the bending notch is bent in case of ruptures of the rupture notch.



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DESCRIPTION KR20170124343A

Cap assembly for secondary battery {CAP ASSEMBLY FOR RECHARGEABLE BATTERY}

[0001]

The present invention relates to a cap assembly for a secondary battery.

[0003]

Secondary batteries, unlike primary batteries, are rechargeable, and have been widely researched and developed in recent years due to their small size and large capacity. As technology development and demand for mobile devices increase, the demand for secondary batteries as an energy source is rapidly increasing.

[0004]

In particular, the lithium secondary battery has an operating voltage of 3.6V, which has about three times the capacity of a nickel-cadmium battery or a nickel-hydrogen battery, which are often used as power sources for electronic equipment, and has a high energy density per unit weight, so the degree of utilization is in a rapidly increasing trend.

[0005]

Such a lithium secondary battery mainly uses a lithium-based oxide and a carbon material as a positive electrode active material and a negative electrode active material, respectively.

In addition, the lithium secondary battery may be classified into a prismatic battery, a cylindrical battery, and a pouch-type battery.

[0006]

A lithium ion secondary battery includes an electrode assembly in which a positive electrode/a separator/a negative electrode is sequentially disposed, and a packaging material for sealingly housing the electrode assembly together with an electrolyte.

In particular, the casing of a prismatic or cylindrical secondary battery includes a case having an open end and a cap assembly sealingly coupled to the open end of the case.

[0008]

Korean Patent Publication No. 10-2009-0105547

[0009]

One aspect of the present invention is to provide a cap assembly for a secondary battery capable of easily discharging gas to the outside of the battery when the internal pressure of the battery is increased due to the generation of gas inside the battery.

[0011]

The cap assembly for a secondary battery according to an embodiment of the present invention includes a top cap positioned on the outer shell of the battery and having an outlet formed therein, and a rupture notch that ruptures when the internal pressure rises to allow the internal gas to be discharged to the outlet, and the rupture notch and a safety vent in which a bending notch that is bent upon rupture of the rupture is formed on the rupture portion whose position or shape is changed due to the rupture.

[0013]

In the cap assembly for a secondary battery according to the present invention, a top cap and a safety vent are formed in a form in which gas is easily discharged when the internal pressure of the battery is increased due to the generation of gas inside the battery, so that it is possible to secure the ease of gas discharge.

[0014]

In particular, in the cap assembly for a secondary battery according to the present invention, the rupture portion of the safety vent that ruptures due to an increase in internal pressure is ruptured and bent in a form for easy gas discharge, so that a passage necessary for gas discharge can be easily secured.

[0016]

1 is a cutaway perspective view illustrating a secondary battery in which a cap assembly for a secondary battery according to an embodiment of the present invention is installed.

2 is a cross-sectional view illustrating a cap assembly for a secondary battery according to an embodiment of the present invention.

3 is a cross-sectional view illustrating a state after a safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention.

4 is a conceptual diagram illustrating a state after a safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention.

5 is a plan view illustrating a safety vent in the cap assembly for a secondary battery according to an embodiment of the present invention.

FIG. 6 is a cross-sectional view taken along line A-A' in FIG. 5 .

[0017]

The objects, specific advantages and novel features of the present invention will become more apparent from the following detailed description and preferred embodiments taken in conjunction with the accompanying drawings.

In the present specification, in adding reference numbers to the components of each drawing, it should be noted that only the same components are given the same number as possible even though they are indicated on different drawings.

In addition, the present invention may be embodied in several different forms and is not limited to the embodiments described herein.

And, in describing the present invention, detailed descriptions of related known technologies that may unnecessarily obscure the gist of the present invention will be omitted.

[0019]

1 is a cutaway perspective view illustrating a secondary battery in which a cap assembly for a secondary battery according to an embodiment of the present invention is installed, and FIG. 2 is a cross-sectional view showing a cap assembly for a secondary battery according to an embodiment of the present invention.

[0020]

1 and 2, in the cap assembly 100 for a secondary battery according to an embodiment of the present invention, a top cap 110 having an outlet 113 is formed and the internal pressure of the secondary battery 10 is increased, and a safety vent 120 formed with a rupture notch 121 and a bending notch 124 to be bent.

[0022]

3 is a cross-sectional view illustrating a state after the safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention, and FIG. 4 is after the safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention. It is a perspective view showing the state.

[0023]

Hereinafter, a cap assembly for a secondary battery according to an embodiment of the present invention will be described in more detail with reference to FIGS. 1 to 6.

[0024]

1 and 2, the top cap 110 is positioned at the outermost portion of the secondary battery 10, and may be formed to include an external terminal 111, an accommodating part 112, and an outlet 113. have.

[0025]

The external terminal 111 may protrude from the top of the top cap 110 and may be electrically connected to the electrode stack 12 accommodated in the secondary battery 10 to form a positive terminal or a negative terminal.

[0026]

The electrode stack 12 is accommodated in the case 11, and an electrolyte is injected into the case 11.

In addition, the electrode stack 12 is a power generating element capable of charging and discharging, and the anode 12a, the cathode 12b, and the separator 12c are assembled to form a stacked structure.

Here, the positive electrode 12a of the electrode stack 12 is electrically connected to the external terminal 111 of the top cap 110 so that the external terminal 111 may function as a positive electrode terminal.

[0027]

The separator 12c separates and electrically insulates the anode 12a and the cathode 12b.

At this time, the positive electrode 12a and the negative electrode 12b may be wound together with the separator 12c to form a jelly roll type.

In addition, the separator 12c is made of an insulating material to electrically insulate between the anode 12a and the cathode 12b.

Here, the separator 12c may be formed of, for example, a polyolefin-based resin film such as polyethylene or polypropylene having microporosity.

[0028]

Meanwhile, a cylindrical center pin 13 may be provided at the center of the electrode stack 12.

The center pin 13 may be made of a metal material to impart a predetermined strength, and may have a hollow cylindrical structure obtained by bending a plate material in a round shape.

The center pin 13 may act as a passage for fixing and supporting the electrode stack 12 and for discharging gas generated by an internal reaction during charging, discharging, and operation.

[0029]

2 and 3, the receiving part 112 may be formed in an open shape toward the safety vent 120 so that the rupture part 123 of the safety vent 120 is accommodated in the lower part of the external terminal 111. can

[0030]

In addition, the angle α between the ceiling surface 112a and the side surface 112b of the receiving part 112 in the top cap 110 may be formed in a range of 85 to 100°.

In this case, the angle between the ceiling surface 112a and the side surface 112b in the receiving part 112 of the top cap 110 may be formed vertically, for example.

[0031]

Referring to FIGS. 3 and 4, the outlet 113 is formed on the side 112b of the receiving part 112 to discharge gas inside the battery.

In addition, the outlet 113 may be formed in a direction except for any one of the four directions with respect to the receiving portion 112.

In this case, the outlet 113 may be formed in, for example, 3 directions among 4 directions with respect to the receiving part 112.

[0033]

5 is a plan view illustrating a safety vent in the cap assembly for a secondary battery according to an embodiment of the present invention, and FIG. 6 is a cross-sectional view taken along line A-A' in FIG. 5.

[0034]

3 and 5, the safety vent 120 is located inside the battery rather than the top cap 110, so that a rupture notch 121 and a bending notch 124 may be formed.

[0035]

In addition, the safety vent 120 may open and close the outlet 113 of the top cap 110.

Here, the safety vent 120 is positioned on the passage connecting the outlet 113 of the top cap 110 and the inside of the battery.

Accordingly, before the rupture of the rupture notch 121, the outlet 113 of the top cap 110 is closed, and when the rupture notch 121 of the safety vent 120 ruptures due to an increase in internal pressure of the battery, the outlet 113 may be opened.

[0036]

The rupture notch 121 may rupture in the direction of the top cap 110 when the internal pressure of the battery increases.

At this time, the rupture part 123 whose position or shape is changed according to the rupture of the rupture notch 121 may be ruptured in the direction of the receiving part 112 of the top cap 110 to be accommodated in the receiving part 112.

[0037]

In addition, the rupture notch 121 may be formed in a circular arc shape with one side open in a plan view.

Here, the opening portion of the rupture notch 121 may be formed at a predetermined angle with respect to the center of the arc. In this case, the opening angle β of the rupture notch 121 may be formed within, for example, 60° .

[0038]

The bending notch 124 is formed on the tearing portion 123 whose position or shape changes according to the tearing of the tearing notch 121 so as to be bent upon the tearing of the tearing notch 121.

Here, the rupture part 123 collides with the top cap 110 when ruptured, and the bending notch 124 is bent by the collision force with the top cap 110.

In this case, the bending notch 124 may be bent so that the rupture part 123 is bent in a shape corresponding to the inner surface of the top cap 110 when the rupture notch 121 is ruptured.

Accordingly, when the rupture notch 121 is ruptured, space generation between the rupture part 123 and the inner surface of the top cap 110 is reduced, so that all of the outlets 113 of the top cap 110 can be opened.

Accordingly, a passage necessary for discharging the internal gas of the battery can be maximally secured.

In particular, when the angle α between the ceiling surface 112a and the side surface 112b of the receiving part 112 in the top cap 110 is 85 to 100° or is formed vertically, the rupture part 123 is formed in the top cap (As the opening of the outlet 113 becomes easier when bending to correspond to the inner surface of the receiving portion 112 of the 110), the ease of gas discharge can be secured much more effectively.

Here, the inner surface of the receiving part 112 of the top cap 110 may be configured to include a ceiling surface 112a and a side surface 112b.

[0039]

In addition, the bending notch 124 may be formed on the inner side of the arc of the tear notch 121 in a plan view.

In this case, the bending notch 124 may be located between the opening portion of the tear notch 121 and the arc center C of the tear notch 121.

In addition, the bending notch 124 may be formed in a straight shape in a plan view.

[0040]

In addition, the bending notch 124 may be formed to extend in a direction inclined with respect to the direction from the center of the arc of the tear notch 121 toward the opening portion of the tear notch 121 in plan view, and also the bending notch 124 () may form a line perpendicular to the line L extending in the opening direction of the tear notch 121 from the center of the circular arc of the tear notch 121 in plan view.

Here, both ends of the bending notch 124 may be positioned on the arc of the rupture notch 121.

[0041]

On the other hand, when the rupture notch 121 is ruptured in the safety vent 120, the non-notch portion 122, which is a portion where the notch is not formed on the arc of the rupture notch 121, is bent in the direction of the top cap 110 in a plan view. can

Here, the non-notch part 122 of the safety vent 120 is positioned to correspond to the edge 112d positioned at the edge of the receiving part 112, and is positioned at the edge of the receiving part 112 when the tear notch 121 is ruptured. It can be bent by the corner edge (112d).

At this time, in the safety vent 120, the rupture part 123 that does not come into contact with the lower surface of the top cap 110 is bent in a direction to be accommodated in the receiving part 112 of the top cap 110 with respect to the non-notch part 122. can be

[0042]

In addition, the bending notch 124 may be formed to be positioned at a position corresponding to the ceiling edge portion 112c formed between the ceiling surface 112a and the side surface 112b of the receiving part 112 when the tear notch 121 is ruptured, have.

That is, the distance from the non-notch part 122 bent in the safety vent 120 to the bent notch 124 may be formed to correspond to the depth of the receiving part 112 .

[0043]

In addition, the bending notches 124 may be formed in plurality.

At this time, the bending notch 124 includes a first bending notch 124a and a second bending notch 124b, and the first bending notch 124a and the second bending notch 124b are spaced apart from each other by a predetermined distance and arranged side by side, can be formed.

[0044]

Here, the distance c1 from the non-notch part 122 to the first bending notch 124a is formed to be smaller than the depth c3 of the receiving part 112 of the top cap 110 , and from the non-notch part 122 to the second The distance c2 to the bending notch 124b may be greater than the depth c3 of the receiving portion 112 of the top cap 110 .

[0045]

Meanwhile, referring to FIG. 6 , the tear notch 121 may be formed larger or deeper than the bending notch 124 so that the tear notch 121 ruptures before the bending notch 124 .

Here, the width a1 of the rupture notch 121 may be wider than the width a2 of the bending notch 124 .

In addition, the depth b1 of the rupture notch 121 may be formed to be deeper than the depth b2 of the bending notch 124 .

[0046]

Meanwhile, referring to FIGS. 4 and 5 , the non-notch part 122 of the safety vent 120 is positioned in any one direction of the four directions around the receiving part 112 , and the outlet 113 is located in the other one or more directions.) can be located.

Here, the non-notch part 122 of the safety vent 120 may be positioned in any one of four directions around the receiving part 112 , and the outlet 113 may be positioned in the other three directions.

[0048]

Although the present invention has been described in detail through specific embodiments, this is intended to describe the present invention in detail, and the cap assembly for a secondary battery according to the present invention is not limited thereto.

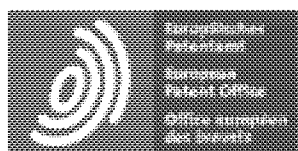
It will be said that various implementations are possible by those of ordinary skill in the art within the technical spirit of the present invention.

[0049]

In addition, the specific protection scope of the invention will become clear by the appended claims.

[0051]

10: Secondary Battery 11: Case 12: Electrode stack 12a: positive electrode 12b: negative electrode 12c: separator 13: center pin 100: cap assembly 110: top cap 111: external terminal 112: receiving part 112a: ceiling surface 112b: side 112c: ceiling edge Part 112d: edge 113: outlet 120: safety vent 121: tear notch 122: unnotched portion 123: burst 124: bend notch 124a: first bend notch 124b: second bend notch



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CLAIMS KR20170124343A

1.

a top cap positioned on the outer shell of the battery and formed with an outlet; and a safety vent in which a rupture notch that ruptures when the internal pressure rises is formed so as to allow the internal gas to be discharged to the outlet, and a bending notch that is bent when the rupture notch is ruptured is formed on the rupture part whose position or shape is changed due to the rupture. A cap assembly for a secondary battery comprising.

2.

The cap assembly for a secondary battery according to claim 1, wherein the rupture part collides with the top cap and the bent notch is bent when the rupture notch is ruptured in the safety vent located inside the battery than the top cap.

3.

The cap for a rechargeable battery of claim 2, wherein all of the outlets of the top cap are closed by the safety vent before the rupture of the rupture notch, and all of the outlets are opened after the rupture of the rupture notch and the bending of the bending notch assembly.

4.

The cap assembly for a rechargeable battery of claim 2, wherein the bending notch is bent so that the rupture part is bent in a shape corresponding to the inner surface of the top cap when the rupture notch is ruptured.

5.

The cap assembly for a secondary battery according to claim 2, wherein the rupture notch is formed in an arc shape with one side open in a plan view of the safety vent.

6.

The cap assembly for a secondary battery according to claim 5, wherein the bending notch is formed on an inner side of an arc of the rupture notch in the plan view.

7.

The cap assembly for a secondary battery according to claim 6, wherein the bending notch is positioned between an open portion of the rupture notch and an arc center of the rupture notch.

8.

The cap assembly for a secondary battery according to claim 6, wherein the bent notch is formed in a straight shape in the plan view.

9.

The cap assembly for a secondary battery according to claim 6, wherein the bent notch forms a line perpendicular to a line extending in an opening direction of the rupture notch from the center of an arc of the rupture notch formed in the plan view.

10.

The cap assembly for a secondary battery according to claim 5, wherein the bending notch extends in a direction inclined with respect to a direction toward an opening portion of the rupture notch from the center of a circular arc of the rupture notch formed in a plan view.

11.

The cap assembly for a secondary battery according to claim 10, wherein both ends of the bending notch are positioned on an arc of the rupture notch.

12.

The cap assembly for a secondary battery according to any one of claims 1 to 11, wherein the rupture notch is formed to be larger than the bending notch so that the rupture notch is ruptured before the bending notch.

13.

The cap assembly for a secondary battery of claim 12, wherein a depth of the rupture notch is greater than a depth of the bending notch.

14.

The cap assembly of claim 12, wherein a width of the rupture notch is wider than a width of the bending notch.

15.

The cap assembly for a secondary battery according to any one of claims 1 to 11, wherein the bent notches are formed in plurality.

16.

The cap assembly for a secondary battery according to any one of claims 5 to 10, wherein the top cap includes a receiving part in which the rupture part of the safety vent is accommodated.

17.

The cap assembly for a secondary battery according to claim 16, wherein the outlet is formed on a side surface of the accommodating part to discharge gas generated inside the battery.

18.

The cap assembly for a secondary battery according to claim 17, wherein an angle between a ceiling surface and a side surface of the receiving part in the top cap is formed vertically.

19.

The cap assembly for a secondary battery according to claim 17, wherein an angle between the ceiling surface and the side surface of the receiving part in the top cap is 85 to 100°.

20.

The cap assembly for a secondary battery according to claim 17, wherein when the tear notch is ruptured in the safety vent, a non-notch portion, which is a portion on an arc of the tear notch in which a notch is not formed, is formed to be bent in the direction of the top cap.

21.

The cap assembly for a secondary battery according to claim 20, wherein the non-notch portion of the safety vent is bent by an edge of the receiving portion when the tear notch is ruptured.

22.

The cap assembly for a secondary battery according to claim 21, wherein the bending notch is formed to be positioned at a position corresponding to a ceiling edge of the receiving part when the rupture notch is ruptured.

23.

The cap assembly for a secondary battery of claim 21, wherein the bending notch includes a first bending notch and a second bending notch, and the first bending notch and the second bending notch are spaced apart from each other by a predetermined distance and formed in parallel.

24.

The method according to claim 23, The distance from the non-notch portion to the first bending notch is formed to be smaller than the depth of the receiving portion of the top cap, and the distance from the non-notch portion to the second bending notch is greater than the depth of the receiving portion of the top cap. A cap assembly for a secondary battery that is formed large.

25.

The cap assembly for a secondary battery according to claim 20, wherein the non-notch portion of the safety vent is positioned in any one direction among four directions with respect to the receiving part, and the outlet is positioned in one or more other directions.

26.

The cap assembly for a secondary battery according to claim 20, wherein the non-notch part of the safety vent is positioned in any one of four directions around the receiving part and the outlet is positioned in the other three directions.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	17259395
	Filing Date	2021-07-16
	First Named Inventor	Geon Woo Min
	Art Unit	OPAP
	Examiner Name	Not Yet Assigned
	Attorney Docket Number	LGCHEM 3.3F-3059

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17259395
	Filing Date		2021-07-16
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	Attorney Docket Number	LGCHEM 3.3F-3059	

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	OPAP
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2022-03-04
Name/Print	Jordan T. Riviello	Registration Number	72,515

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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Dated: March 4, 2022

Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: July 16, 2021

Art Unit: OPAP

For: Secondary Battery

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

It is respectfully requested that the reference listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of the reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited reference is legally available prior art or that the same is pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-3059, from which the undersigned is authorized to draw.

Dated: March 4, 2022

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant(s)

LD-458A

Electronic Acknowledgement Receipt

EFS ID:	45149846
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Stephen M. Lund/Kathleen Reynolds
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Attorney Docket Number:	LGCHEM 3.3F-3059
Receipt Date:	04-MAR-2022
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Application Type:	U.S. National Stage under 35 USC 371

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Foreign Reference	LGCHEM3430_- _JP2013-502035A.pdf	777926	no	29
			25fbac7597f8c1eb8689096068427e3bcb3 d7d50		

Warnings:

Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM3430_-_Information_Disclosure_Statement_Fillable_PDF.pdf	1263203 e3139975834a34c1dc17fe8711c705833091eb1d	no	4
Warnings:					
Information:					
A U.S. Patent Number Citation or a U.S. Publication Number Citation is required in the Information Disclosure Statement (IDS) form for autoloading of data into USPTO systems. You may remove the form to add the required data in order to correct the Informational Message if you are citing U.S. References. If you chose not to include U.S. References, the image of the form will be processed and be made available within the Image File Wrapper (IFW) system. However, no data will be extracted from this form. Any additional data such as Foreign Patent Documents or Non Patent Literature will be manually reviewed and keyed into USPTO systems.					
3	Transmittal Letter	IDS_-_Information_Disclosure_Statement_-_Reference_Enclosed_LD-458.pdf	21375 3154c5fcf5f1938b6a0b11de850db4c2704a7d47	no	1
Warnings:					
Information:					
Total Files Size (in bytes):			2062504		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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Bibliographic data: JP2013502035 (A) — 2013-01-17

CYLINDRICAL SECONDARY BATTERY OF IMPROVED SAFETY

Inventor(s):

Applicant(s):

Classification: - international: H01M10/0567; H01M2/02; H01M2/04; H01M2/06;
H01M2/08; H01M2/12; H01M2/26; H01M2/34
- cooperative: H01M10/0431 (EP, US); H01M10/0587 (KR);
H01M50/3425 (EP, US); H01M50/578 (EP, US);
H01M10/0525 (KR); Y02E60/10 (EP); Y02P70/50 (EP)

Application number: JP20120524651 20100813 Global Dossier

Priority number(s): KR20090075156 20090814 ; WO2010KR05338 20100813

Also published as: CN102473883 (A) CN102473883 (B) EP2466668 (A2)
EP2466668 (A4) EP2466668 (B1) JP5577405 (B2)
KR101058386 (B1) KR20110017835 (A) US2012114979 (A1)
US9112234 (B2) WO2011019237 (A2) WO2011019237 (A3) less

Abstract not available for JP2013502035 (A)

Abstract of corresponding document: US2012114979 (A1)

Provided is a cylindrical battery in which an electrode assembly fabricated by rolling a cathode/separator/anode and an electrolyte are provided in a cylindrical can, wherein a cap assembly mounted on the opening top of the cylindrical can comprises: a safety vent provided with a predetermined notch, to allow breakage due to high-pressure gas of the battery, a current interruptive device to interrupt current, welded to the bottom of the safety vent, and a gasket for the current interruptive device to surround the periphery of the current interruptive device, wherein the current interruptive device comprises two or more through holes to allow exhaustion of gas, wherein the through holes have a size of 20 to 50% with respect to the total area of the current interruptive device.

(19)

(12) 公表特許公報 (A)

JP 2013-502035 A 2013.1.17

(11)

特表2013-502035

(P 2 0 1 3 - 5 0 2 0 3 5 A)

(43)

25 1 17 (2013.1.17)

(51)Int.Cl.

<i>H O 1 M</i>	<i>2/34</i>	<i>(2006. 01)</i>	<i>2/34</i>
<i>H O 1 M</i>	<i>10/0567</i>	<i>(2010. 01)</i>	<i>10/00</i>
<i>H O 1 M</i>	<i>2/02</i>	<i>(2006. 01)</i>	<i>2/02</i>
<i>H O 1 M</i>	<i>2/04</i>	<i>(2006. 01)</i>	<i>2/04</i>
<i>H O 1 M</i>	<i>2/06</i>	<i>(2006. 01)</i>	<i>2/06</i>

28

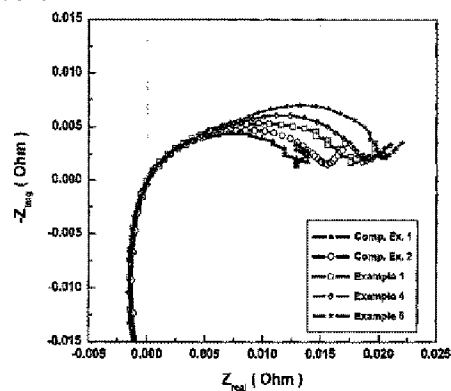
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(86)(22)	22 8 13 (2010.8.13)
(85)	24 3 23 (2012.3.23)
(86)	PCT/KR2010/005338
(87)	WO2011/019237
(87)	23 2 17 (2011.2.17)
(31)	10-2009-0075156
(32)	21 8 14 (2009.8.14)
(33)	(KR)

(71)	500239823
(74)	100110364
(74)	100122161
(72)	

(54)

(57)

[Fig. 16]



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coefficient

positive temperature

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I LiClO₄ LiBF₄ LiB₁₀Cl₁₀ LiPF₆ LiCF₃SO₃ LiCF₃CO₂ LiAsF₆ LiSbF₆ LiAlCl₄
 CH₃SO₃Li CF₃SO₃Li (CF₃SO₂)₂NLi

30

N -

hexaphosphoric triamide

-

-

-

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beading member

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the anode collector

10

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μ

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beading

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10

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30

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frequency

50

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<表 1> (単位：電流 (A))

	1	2	平均値
実施例 1	46.8	50	48.4
実施例 4	44.5	45.8	45.2
実施例 5	40.9	43.2	42.0
実施例 6	45.3	45.6	45.4
実施例 7	43.9	42.1	43.0
比較例 1	51.2	53.6	52.4
比較例 2	51	52	51.5

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<表 2>

	開始温度 (°C)	熱容量(J/g)			
		1ピーク	2ピーク	3ピーク	合計
実施例 1	206±1	71±4	47±4	-	118±6
実施例 4	202±1	55±3	45±5	-	100±6
実施例 5	204±1	50±4	40±4	-	90±4
実施例 6	203±1	53±2	50±4	-	103±4
実施例 7	205±1	48±5	43±5	-	91±6
比較例 1	198±1	223±6			223±6
比較例 2	204±1	98±11	56±5	49±6	203±6

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specific heat capacity

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nail test

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<表 3>

	貫通孔の面積比	安全ベントの切断領域	爆発	電池缶の変形
実施例 1	30%	第一ノッチ	×	×
実施例 2	30%	第二ノッチ	×	△
実施例 3	30%	内側のノッチ (増加した直径)	×	△
比較例 3	18%	第一ノッチ	○	○

10

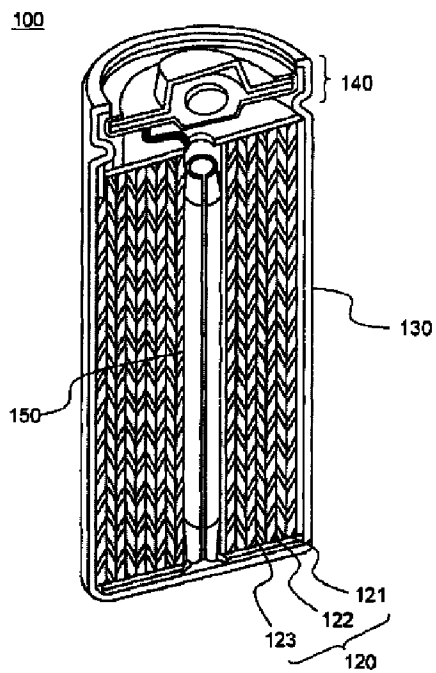
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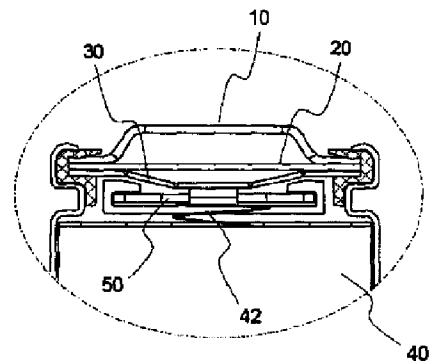
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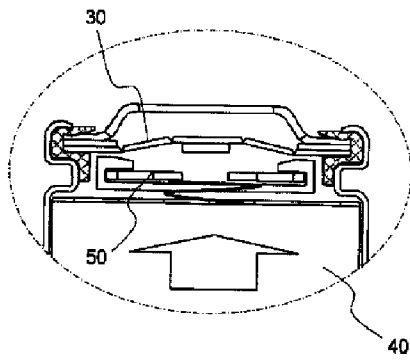
[Fig. 1]



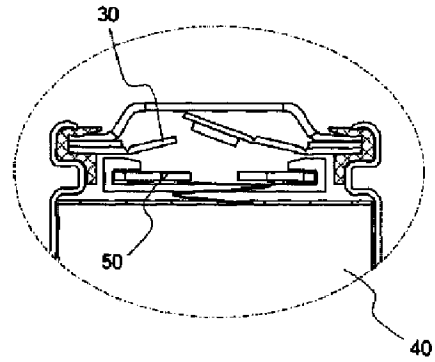
[Fig. 2]



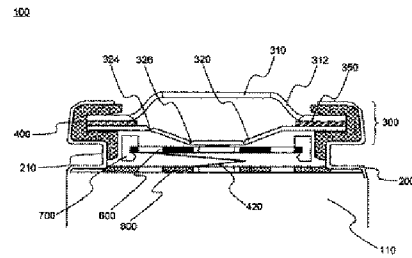
[Fig. 3]



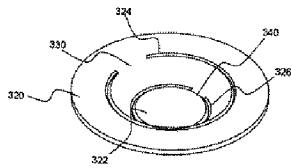
[Fig. 4]



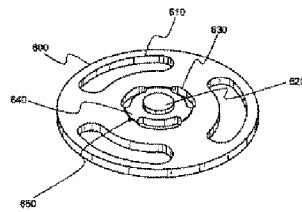
[Fig. 5]



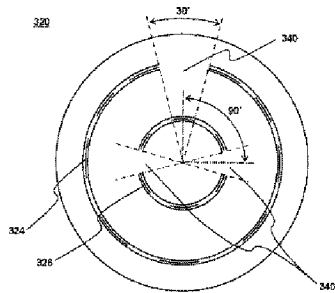
[Fig. 6]



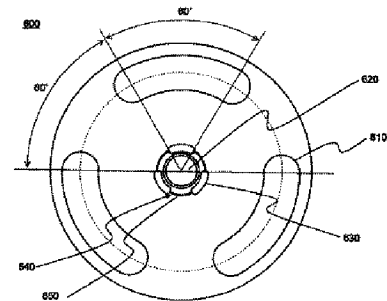
[Fig. 6]



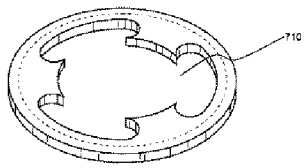
[Fig. 7]



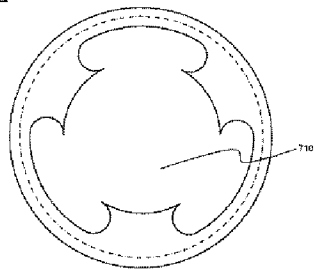
[Fig. 9]



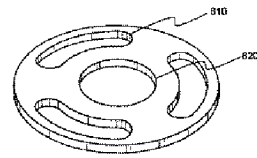
[Fig. 10]
290



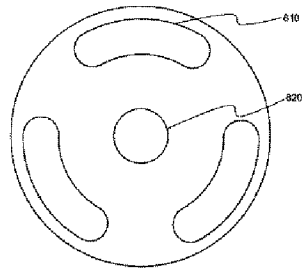
[Fig. 11]
700



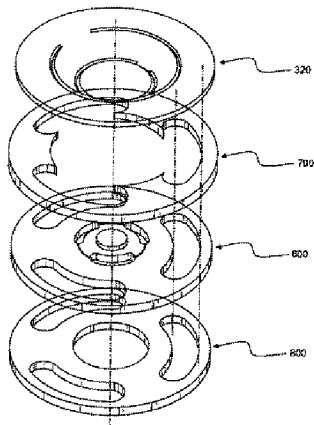
[Fig. 12]
800



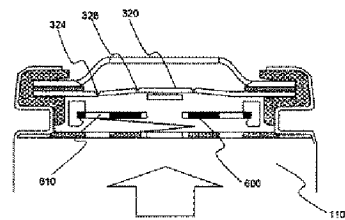
[Fig. 13]
800



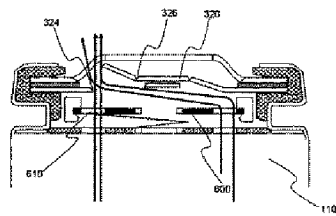
[Fig. 14]



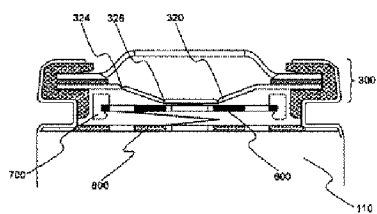
[Fig. 16]

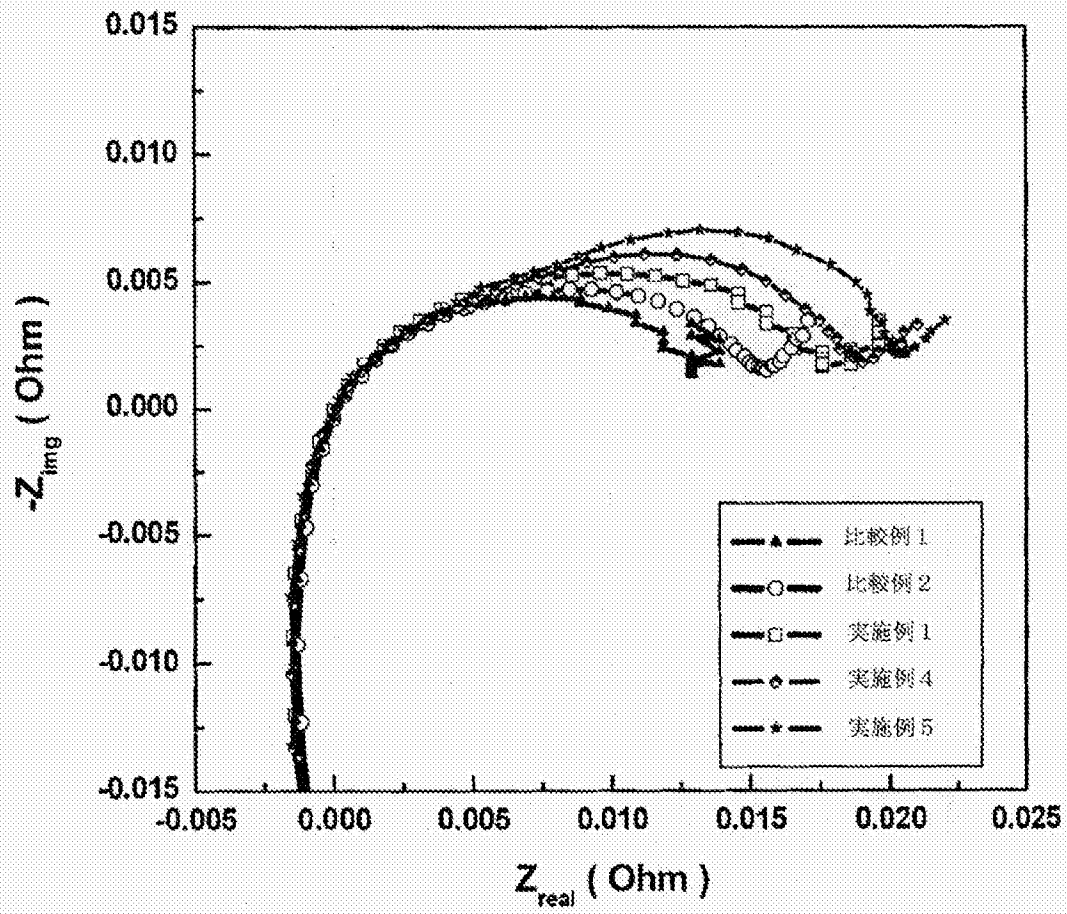


[Fig. 17]



[Fig. 15]





INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2010/005338**A. CLASSIFICATION OF SUBJECT MATTER****H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/08(2006.01)i**

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/12; H01M 10/058; H01M 10/0525; H01M 2/26; H01M 2/10; H01M 10/36; H01M 10/50

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean Utility models and applications for Utility models: IPC as above

Japanese Utility models and applications for Utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: cap assembly, safety belt, current interrupt, gasket, through-hole

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KR 10-0855181 B1 (LG CHEM. LTD.) 29 August 2008 See abstract, claims 1-6 and figures 1-6b	1-22
A	KR 10-0619631 B1 (LG CHEM. LTD.) 12 September 2006 See abstract, claims 1-6 and figures 1-4c	1-22
A	KR 10-2004-0037579 A (SAMSUNG SDI CO., LTD.) 07 May 2004 See abstract, claims 1-23 and figures 1-4	1-22
A	JP 2008-262832 A (MATSUSHITA ELECTRIC IND CO LTD) 30 October 2008 See abstract, claims 1-8 and figure 1	1-22

☐ Further documents are listed in the continuation of Box C.☒ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"G" document member of the same patent family

Date of the actual completion of the international search

08 MARCH 2011 (08.03.2011)

Date of mailing of the international search report

08 MARCH 2011 (08.03.2011)

Name and mailing address of the ISA/KR


 Korean Intellectual Property Office
 Government Complex-Daejeon, 139 Seonsa-ro, Daejeon 302-701,
 Republic of Korea

Facsimile No. 82-42-472-7140

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.

PCT/KR2010/005338

Patent document cited in search report	Publication date	Patent family member	Publication date
KR 10-0855181 B1	29.08.2008	NONE	
KR 10-0619631 B1	12.09.2006	NONE	
KR 10-2004-0037579 A	07.05.2004	NONE	
JP 2008-262832 A	30.10.2008	CN 101542820 A	23.09.2009
		JP 2008-262832 A	30.10.2008
		KR 20090028729A	19.03.2009
		US 2009-311585 A1	17.12.2009
		WO 2008-136177 A1	13.11.2008

국제조사보고서

국제출원번호
PCT/KR2010/005338

A. 발명이 속하는 기술분류(국제특허분류(IPC))		
H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/08(2006.01)i		
B. 조사된 분야		
조사된 최소문헌(국제특허분류상 기재) H01M 2/12; H01M 10/058; H01M 10/0525; H01M 2/26; H01M 2/10; H01M 10/36; H01M 10/50		
조사된 기술분야에 속하는 최소문헌 이외의 문헌 한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC		
국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우)) eCOMPASS(특허청 내부 검색시스템) & 키워드: 캡 어셈블리, 안전펜트, 전류차단, 가스켓, 판봉구,		
C. 관련 문헌		
카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
A	KR 10-0855181 B1 (주식회사 엘지화학) 2008.08.29 요약, 청구항 1-6항 및 도면 1-6a참조	1-22
A	KR 10-0619631 B1 (주식회사 엘지화학) 2006.09.12 요약, 청구항 1-6항 및 도면 1-4c참조	1-22
A	KR 10-2004-0037579 A (삼성세스티아이 주식회사) 2004.05.07 요약, 청구항 1-23항 및 도면 1-4참조	1-22
A	JP 2008-262832 A (MATSUSHITA ELECTRIC IND CO LTD) 2008.10.30 요약, 청구항 1-8항 및 도면 1참조	1-22
☐ 추가 문헌이 C(계속)에 기재되어 있습니다. ☒ 대응특허에 관한 별지를 참조하십시오.		
* 인용된 문헌의 특별 카테고리: "A" 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌 "B" 국제출원일보다 빠른 출원일 또는 우선일을 가져나 국제출원일 이후에 공개된 선출원 또는 특허 문헌 "L" 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌 "O" 구조 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌 "P" 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌 "T" 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌 "X" 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다. "Y" 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다. "&" 동일한 대응특허문헌에 속하는 문헌		
국제조사의 실제 완료일 2011년 03월 08일 (08.03.2011)	국제조사보고서 발송일 2011년 03월 08일 (08.03.2011)	
ISA/KR의 명칭 및 우편주소 대한민국 특허청 (302-701) 대전광역시 서구 선사로 139, 정부대전청사 팩스 번호 82-42-472-7140	심사관 이창희 전화번호 82-42-481-5644	

서식 PCT/ISA/210 (두 번째 용지) (2009년 7월)

국제조사보고서
대응특허에 관한 정보

국제출원번호

PCT/KR2010/005338

국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
KR 10-0855181 B1	2008.08.29	없음	
KR 10-0619631 B1	2006.09.12	없음	
KR 10-2004-0037579 A	2004.06.07	없음	
JP 2008-262832 A	2008.10.30	CN 101542820 A	2009.09.23
		JP 2008-262832 A	2008.10.30
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17259395
	Filing Date		2021-07-16
	First Named Inventor	Geon Woo Min	
	Art Unit		
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGCHEM 3.3F-3059	

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	1	2572931	CA	A1	2008-06-21	Wu, Donald P. H		
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	3	103155225	CN	A	2013-06-12	Lg Chemical Ltd	U.S. Publication No. 20130216871 A1 is an English language counterpart.	☒

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		17259395
Filing Date		2021-07-16
First Named Inventor	Geon Woo Min	
Art Unit		
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Search Report dated June 22, 2022 from the Office Action for Chinese Application No. 201980046222.8 issued June 30, 2022, 3 pages.	×

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2022-09-14
Name/Print	Jordan T. Riviello	Registration Number	72,515

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Dated: September 14, 2022
Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: July 16, 2021

Art Unit: Not Yet Assigned

For: Secondary Battery

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

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Dear Commissioner:

It is respectfully requested that the references listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited references are legally available prior art or that the same are pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-3059, from which the undersigned is authorized to draw.

Dated: September 14, 2022

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant(s)

LD-458

Electronic Acknowledgement Receipt

EFS ID:	46613849
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Kevin M. Kocun/Sharon McGee
Filer Authorized By:	Kevin M. Kocun
Attorney Docket Number:	LGCHEM 3.3F-3059
Receipt Date:	14-SEP-2022
Filing Date:	16-JUL-2021
Time Stamp:	12:17:51
Application Type:	U.S. National Stage under 35 USC 371

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Espacenet

Bibliographic data: CA2572931 (A1) — 2008-06-21

SAFETY STRUCTURE FOR A PLASTIC BATTERY CASE

Inventor(s): WU DONALD P H [TW] ± (WU, DONALD P. H)

Applicant(s): WU DONALD P H [TW] ± (WU, DONALD P. H)

Classification: - **international:** H01M2/02
- **cooperative:** H01M10/0413 (EP); H01M10/0431 (EP);
H01M10/0463 (EP); H01M10/052 (EP); H01M10/0587
(EP); H01M50/121 (EP); H01M50/15 (EP);
H01M50/152 (EP); H01M50/3425 (EP); H01M50/528
(EP); Y02E60/10 (EP)

Application number: CA20062572931 20061221

Priority number(s): CA20062572931 20061221

Abstract of CA2572931 (A1)

A safety structure for a plastic battery case, the battery case is a plastic unitary structure, in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces the interior of the battery case, so as to facilitate the collection of pressure, and in the bottom of the groove is formed a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the closest grooves will bulge instantly, and the bulged working portion will not be ruptured right away. The working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure safely.



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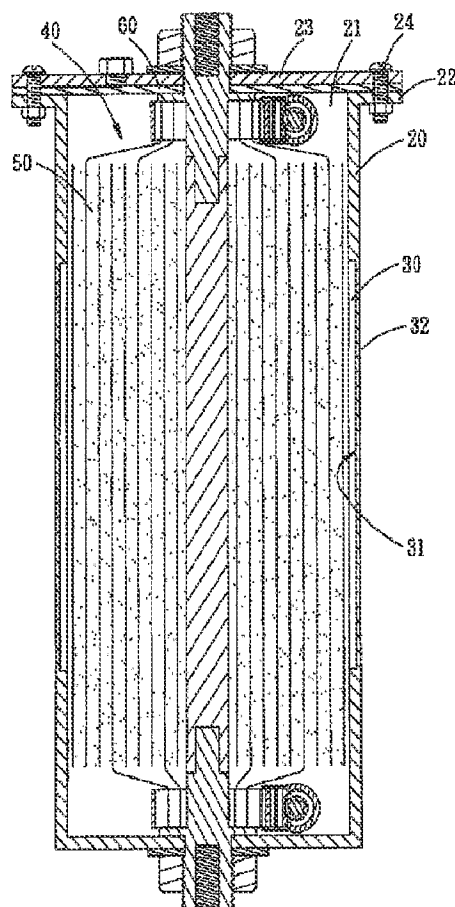
(71) Demandeur/Applicant:
WU, DONALD P. H., TW

(72) Inventeur/Inventor:
WU, DONALD P. H., TW

(74) Agent: BATTISON WILLIAMS DUPUIS

(54) Titre : STRUCTURE DE SECURITE POUR BOITIER DE PILE EN PLASTIQUE

(54) Title: SAFETY STRUCTURE FOR A PLASTIC BATTERY CASE



(57) Abrégé/Abstract:

A safety structure for a plastic battery case, the battery case is a plastic unitary structure, in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces the interior of the battery case, so as to facilitate the

(57) **Abrégé(suite)/Abstract(continued):**

collection of pressure, and in the bottom of the groove is formed a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the closest grooves will bulge instantly, and the bulged working portion will not be ruptured right away. The working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure safely.

ABSTRACT OF THE DISCLOSURE

A safety structure for a plastic battery case, the battery case is a plastic unitary structure, in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces the interior of the battery case, so as to
5 facilitate the collection of pressure, and in the bottom of the groove is formed a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the closest grooves will bulge instantly, and the bulged working portion will not be ruptured right away. The working portion of the groove will bulge outward and become thinner and
10 consequently form a slit to release the pressure safely.

SAFETY STRUCTURE FOR A PLASTIC BATTERY CASE

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a safety structure for a plastic battery case, and
5 more particularly to a safety design for a plastic battery case for safely releasing the
internal pressure thereof.

Description of the Prior Art

Due to the great break-through in material technique, lithium secondary battery,
as a high capacity power supply (such as LiFePO_4 type lithium secondary battery),
10 has been used on power devices, such as electric bicycle, electric wheelchair, and etc.
The power storage and supply capacity of such a high capacity non-aqueous
electrolyte type lithium secondary battery are greater than that of a conventional
lithium battery. The battery case of the abovementioned lithium secondary battery is
usually a metal case made of aluminum. Since the battery case is filled with electrode
15 board and non-aqueous electrolyte, and in order to ensure a long term and stable
sealing performance, the open end of the battery case should be sealed with a sealing
plate by laser welding.

If the abovementioned lithium secondary battery is overcharged and
consequently the interior current value exceeds the normal value, or the battery is
20 misused and cause short circuit, the non-aqueous electrolyte in the battery case will be
fully decomposed and produce a great deal of gas. When the gas pressure inside the

battery case exceeds the maximum pressure limit, the battery case will be ruptured and gas will be gushed out of the battery case instantly, causing damage to the powered device the battery and causing electrolyte contamination.

Therefore, the aluminum sealing plate of the conventional non-aqueous
5 electrolyte lithium secondary battery is usually defined with an air hole sealed with a thin aluminum safety valve. When the internal gas pressure of the battery case exceeds a predetermined value, the gas will break through the aluminum valve, thus releasing the pressure from the battery case.

It is to be noted that the safety valve is located on the surface of the sealing
10 plate, considering the battery as a whole, the safety valve is located comparatively far away from mid portion of the electrode board, and since the air hole is limited by the size of the sealing plate, the inner diameter of the air hole will not be large. The gas will take a long time to break through the safety valve. The normal condition is that, when the internal pressure of the battery case increases, the mid portion of the battery
15 case will be inflated and ruptured instantly, however, the safety valve doesn't work. Obviously, such a conventional safety valve is unable to solve the safety problem of the battery case.

Therefore, as shown in Fig. 1, a safety mechanism for an aluminum rectangular battery case was made and was disclosed in US Pat. No. 6,964,690, in the elongated
20 side surface of the battery case 10 is formed a plurality of cutting grooves 11. Between a groove bottom surface of the cutting groove 11 and an inner surface of the

battery case 10 is provided a thin-walled easily-rupturable portion designed to rupture at an instant when an internal pressure of the battery case reaches a predetermined value, thus releasing the gas pressure from the battery case.

For such a safety mechanism design of cutting the cutting grooves in the metal
5 battery case, when the internal pressure of the battery case increases, the easily-rupturable portion will be ruptured. It is to be noted that a great deal of gas pressure impacts the small easily-rupturable portion in a very short time, the resultant ruptured grooves in the battery case will be very large, and as a result, the gas and the non-aqueous electrolyte will be gushed out of the battery case via the ruptured
10 grooves, and the electrode board inside the battery case will be exposed out of the ruptured portion. This is a very large threat to a powered device.

The safety structure of the present invention is such that, when the internal pressure of the battery case increases, the strong impact force caused by the unusual gas pressure increase is buffered and then is released through small slits.

15

SUMMARY OF THE INVENTION

The primary objective of the present invention is to provide a plastic integrally formed battery case, in the inner surface of the battery case is formed a plurality of grooves whose open side face the interior of the battery case. When the internal pressure of the battery case is too high, the closest grooves will bulge instantly, and
20 the bulged working portion will not be ruptured right away. When the thickness of the working portion is thinned to zero, a small slit will appear. At this moment, the gas

can be released from the slit, thus preventing a great deal of gas from gushing out of the battery case in an instant and contaminating the powered device.

The secondary objective of the present invention is to provide a plastic integrally formed battery case, in the inner surface of the battery case is formed a plurality of grooves whose open side face the interior of the battery case, the open
5 side of the groove is opposite the direction of the gas pressure, thus facilitating the collection of pressure, and making it easier for the thinnest working portion to bulge outward.

BRIEF DESCRIPTION OF THE DRAWINGS

10 Fig. 1 is a perspective view of showing a conventional safety structure for an aluminum rectangular battery disclosed in US Pat. No. 6,964,690;

Fig. 2 is an exploded view of a circular plastic battery case and a core assembly in accordance with the present invention;

Fig. 3 is a longitudinal cross sectional view in accordance with the present
15 invention of showing a circular plastic battery case;

Fig. 4 is a transverse cross sectional view in accordance with the present invention of showing a circular plastic battery case;

Fig. 5 is a longitudinal cross sectional view in accordance with the present invention of showing that the working portion in Fig. 4 is being bulged;

20 Fig. 6 is a longitudinal cross sectional view in accordance with the present invention of showing that the working portion in Fig. 4 is bulged and ruptured to

release the gas;

Fig. 7 is an exploded view of a circular plastic battery case with longitudinal and transverse grooves and a core assembly in accordance with the present invention;

Fig. 8 is a longitudinal cross sectional view in accordance with the present invention of showing that the longitudinal and transverse working portions in Fig. 7 is bulged and ruptured to release the gas;

Fig. 9 is an exploded view of a rectangular plastic battery case and a core assembly in accordance with the present invention; and

Fig. 10 is a longitudinal cross sectional view in accordance with the present invention of showing that the working portion in Fig. 9 is bulged and ruptured to release the gas.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention will be more clear from the following description when viewed together with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiment in accordance with the present invention.

Referring to Figs. 2, 3 and 4, a safety structure for a plastic battery case in accordance with a preferred embodiment of the present invention comprises at least one groove formed in a battery case, wherein:

The battery case 20 is a circular plastic unitary structure in which is defined a receiving space 21 with an open end for accommodation of a core assembly 40, a leakage-proof washer 60, and non-aqueous electrolyte 50. A connecting flange 22 is

formed around the peripheral edge of open end of the receiving space, a cover 23 is fixed to the connecting flange 22 by screws 24 to seal the open end, thus creating an airtight receiving space 21.

In the inner surface of the receiving space 21 of the battery case 20 is formed at least one groove 30 whose open side 31 faces the interior of the battery case 20, and between the bottom of the groove 30 and the outer surface of the battery case 20 is a working portion 32 which is the thinnest portion of the wall of the battery case 20.

When unusual factors (including: overcharge, short circuit of the electrode plate, overload) occur and consequently the non-aqueous electrolyte is decomposed into high pressure gas, the working portion can substantially bulge outward and then become thinner and consequently form a slit to release the gas from the slit.

The battery case of the present invention is made of plastic material, in the inner surface of the battery case is formed at least one groove, and when the internal pressure of the battery case exceeds a predetermined value, the working portion of the groove will bulge outward and become thinner and consequently form a slit for releasing the pressure. And such a pressure-releasing process can prevent the battery from explosion, and ensure that the internal pressure can be released safely. The operation of the embodiment is explained as follows:

Referring to Figs. 2, 3 and 4 again, the battery case 20 is a unitary structure made of plastic material, in the inner surface of the battery case 20 is formed a plurality of grooves 30, and between the bottom of the grooves 30 and the outer wall

of the battery case 20 is a working portion 32 which is the thinnest portion of the wall of the battery case 20. The number of the grooves 30 is based on the size of the battery case 20. For example, a 20 cm diameter circular battery case 20 can be axially formed in its inner surface with four grooves 30, and the grooves 30 are equidistantly
5 arranged.

Referring then to Figs. 4, 5 and 6, when unusual factors occur and consequently the non-aqueous electrolyte is decomposed into high pressure gas 70, the gas pressure will make the battery case 20 bulge outward. It is to be noted that the location where the unusual factors take place may be located at one of the inner sides of the battery
10 case 20 only, and the pressure on this inner side of the battery case 20 is the greatest. The working portion 32 of the grooves 30 located closest to the high pressure point will bulge instantly. The battery case 20 is made of plastic material, and the plastic surface of the battery case 20 has certain flexibility, so that the bulged working portion 32 will not be ruptured right away. The rupture time is prolonged, and then a
15 small slit 33 will appear when the thickness of the working portion 32 is thinned to zero. At this moment, the gas 70 will be released from the battery via the slit 33.

It has to be mentioned that the battery case of the present invention is made of plastic material, and as compared with the conventional metal battery case, the plastic battery case makes it easier for the battery to pass the free-fall test.

20 It is to be noted that, as shown in Fig. 4, each of the grooves 30 is V-shaped in cross section, and the open side 31 of the grooves 30 faces the interior of the battery

case 20. When the unusual gas pressure is applied to the inner surface of the battery case 20, the open side 31 of the grooves 30 is opposite the direction of the gas pressure for facilitating the collection of pressure, and making it easier for the thinnest working portion 32 to bulge toward the radial direction of the battery and consequently form a slit 33. And thus the pressure can be released safely. Therefore, it is very important that the open side 31 of the grooves 30 opens toward the interior of the battery case 20.

In addition, the location of the grooves 30 is the position where the gas 70 inside the battery case 20 is to be released. With the structural design, the grooves 30 can be arranged at a predetermined position, so that the position where the gas 70 to be released can be anticipated, thus preventing the powered device from being contaminated and damaged. In addition to the longitudinal direction as shown in Fig. 7, the direction in which the grooves 30 are arranged can also be arranged in transverse direction. For example, as shown in Fig. 8, the battery case can be formed with longitudinal grooves 30 and transverse grooves 30a, and it has the same pressure releasing effect.

In addition to being used on circular battery case, the safety structure design of the present invention is also suitable for use in a rectangular battery case. With reference to Fig. 9, the grooves 30 and the working portion 32 are formed in the rectangular battery case. And as shown in Fig. 10, which is an illustrative view of showing that the working portion 32 of the rectangular battery 20a is bulged and

ruptured.

It can be seen from the above description that the battery case of the present invention is formed in its inner surface thereof with at least one groove, and since the plastic battery case is flexible, when the internal pressure of the battery case exceeds a
5 predetermined value, the working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure. In other words, the safety pressure release is achieved by the flexible inflation of the plastic material, so that the strong impact force caused by the unusual gas pressure increase is buffered and then is released.

10 While we have shown and described various embodiments in accordance with the present invention, it is clear to those skilled in the art that further embodiments may be made without departing from the scope of the present invention.

WHAT IS CLAIMED IS:

1. A safety structure for a plastic battery case, the battery case being a plastic unitary structure in which being defined a receiving space with an open end for accommodation of electrode plate and electrolyte, the open end being sealed with a cover, thus creating an airtight receiving space; the safety structure is characterized in that:

in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces an interior of the battery case, and between a bottom of the groove and an outer surface of the battery case is a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure.

2. The safety structure for a plastic battery case as claimed in claim 1, wherein the groove is V-shaped in cross section, and the open side of the groove faces the interior of the battery case.

3. The safety structure for a plastic battery case as claimed in claim 1, wherein the groove is arranged in a longitudinal direction.

4. The safety structure for a plastic battery case as claimed in claim 1, wherein the groove is arranged both in longitudinal and transverse directions.

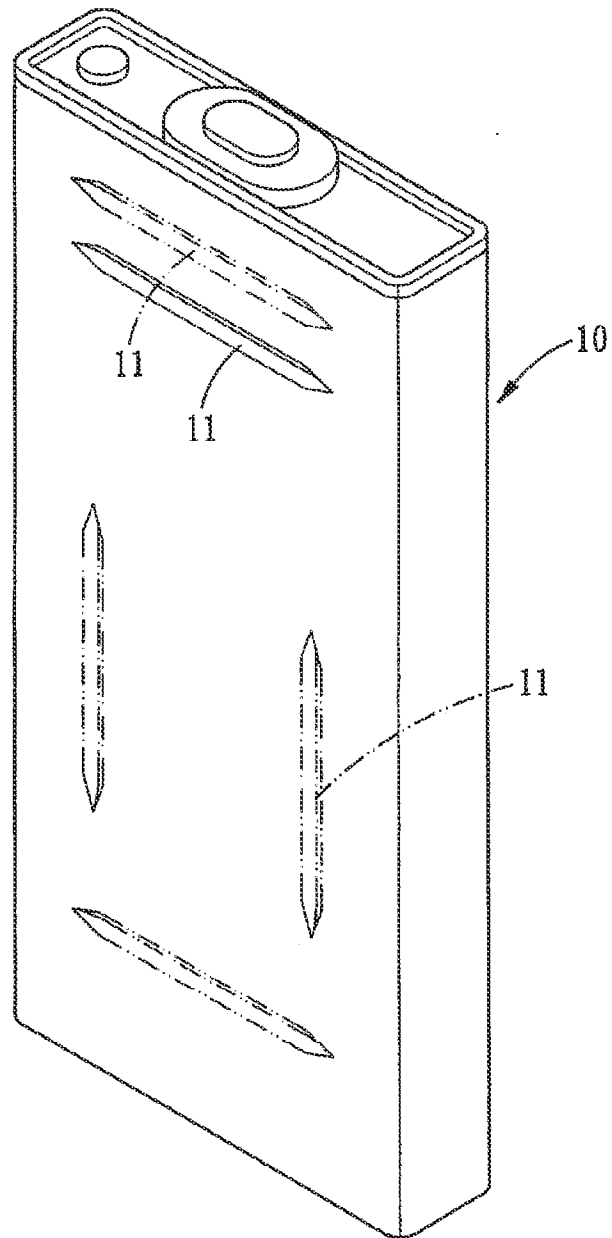
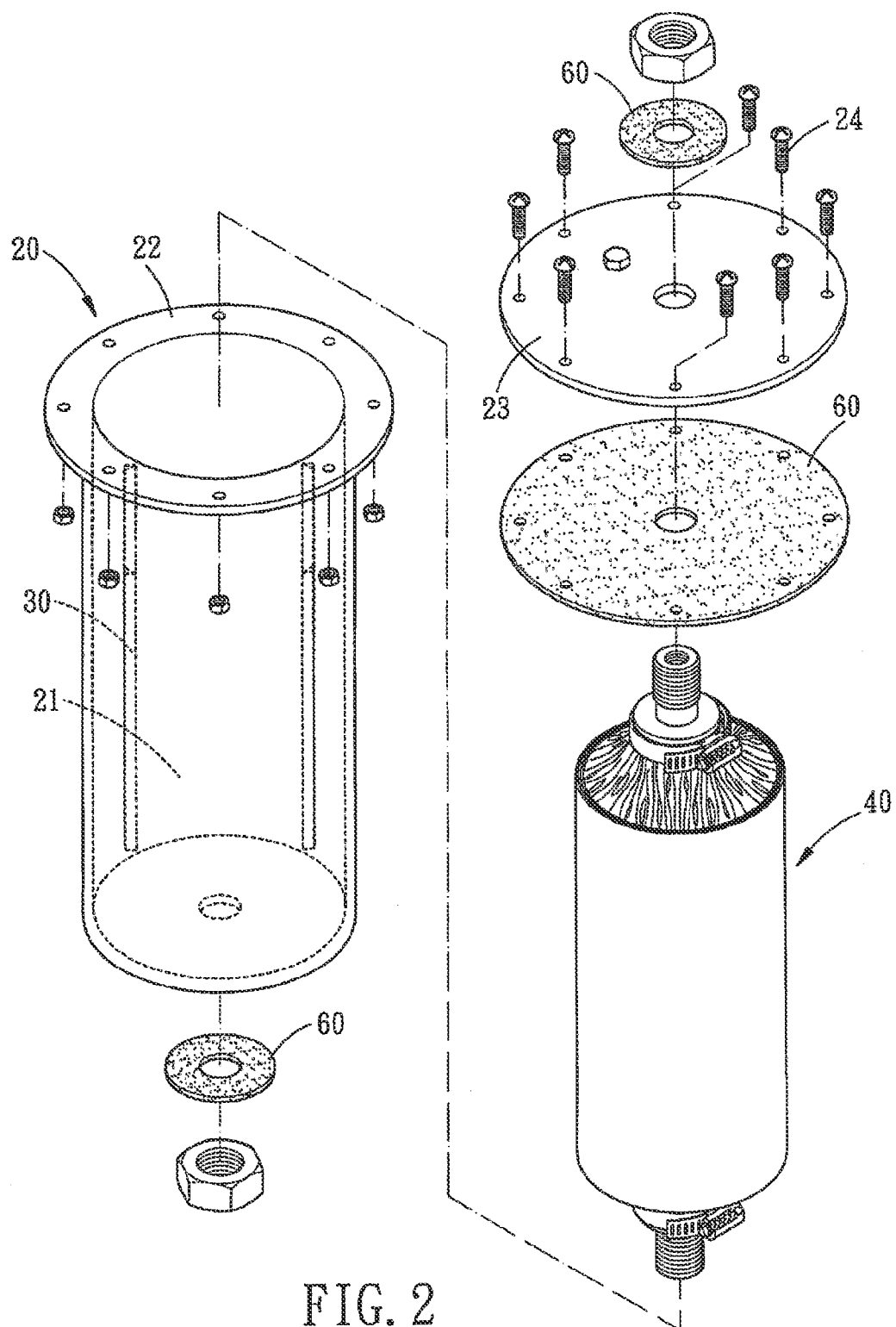


FIG. 1
PRIOR ART



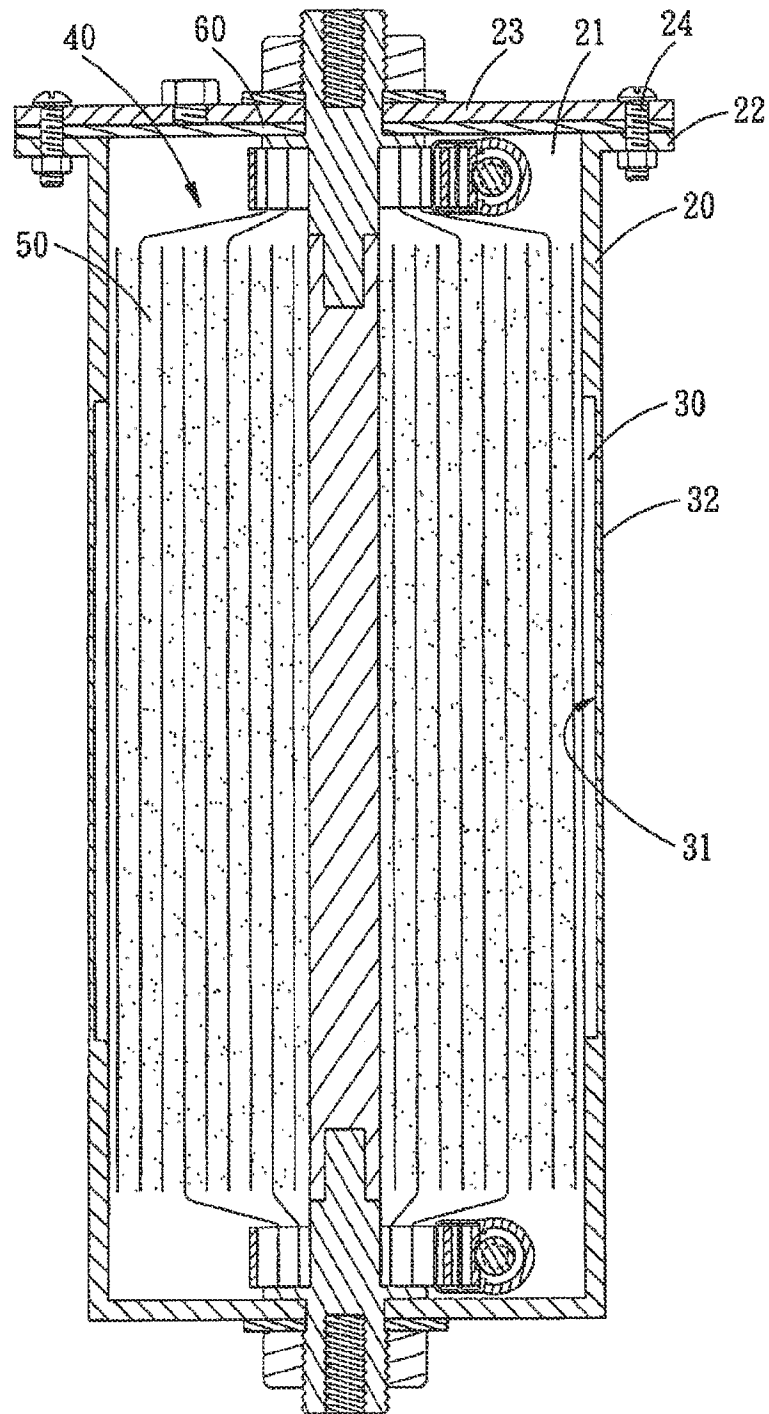


FIG. 3

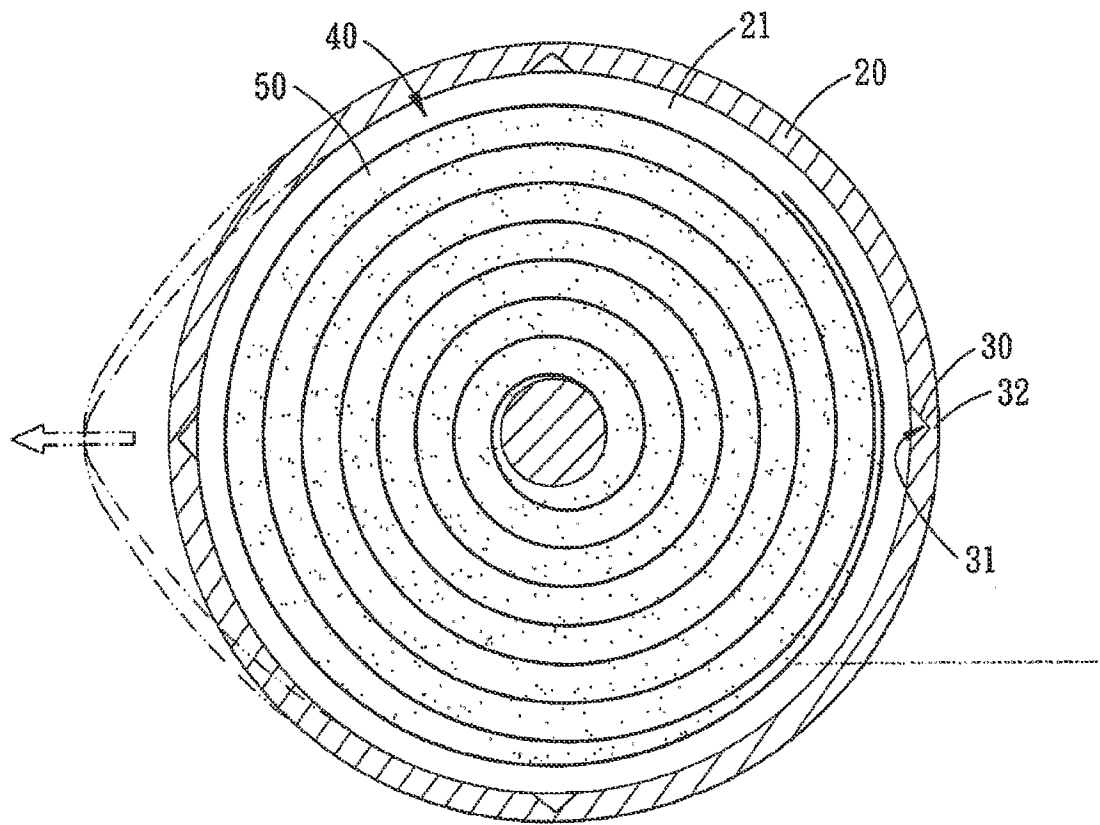


FIG. 4

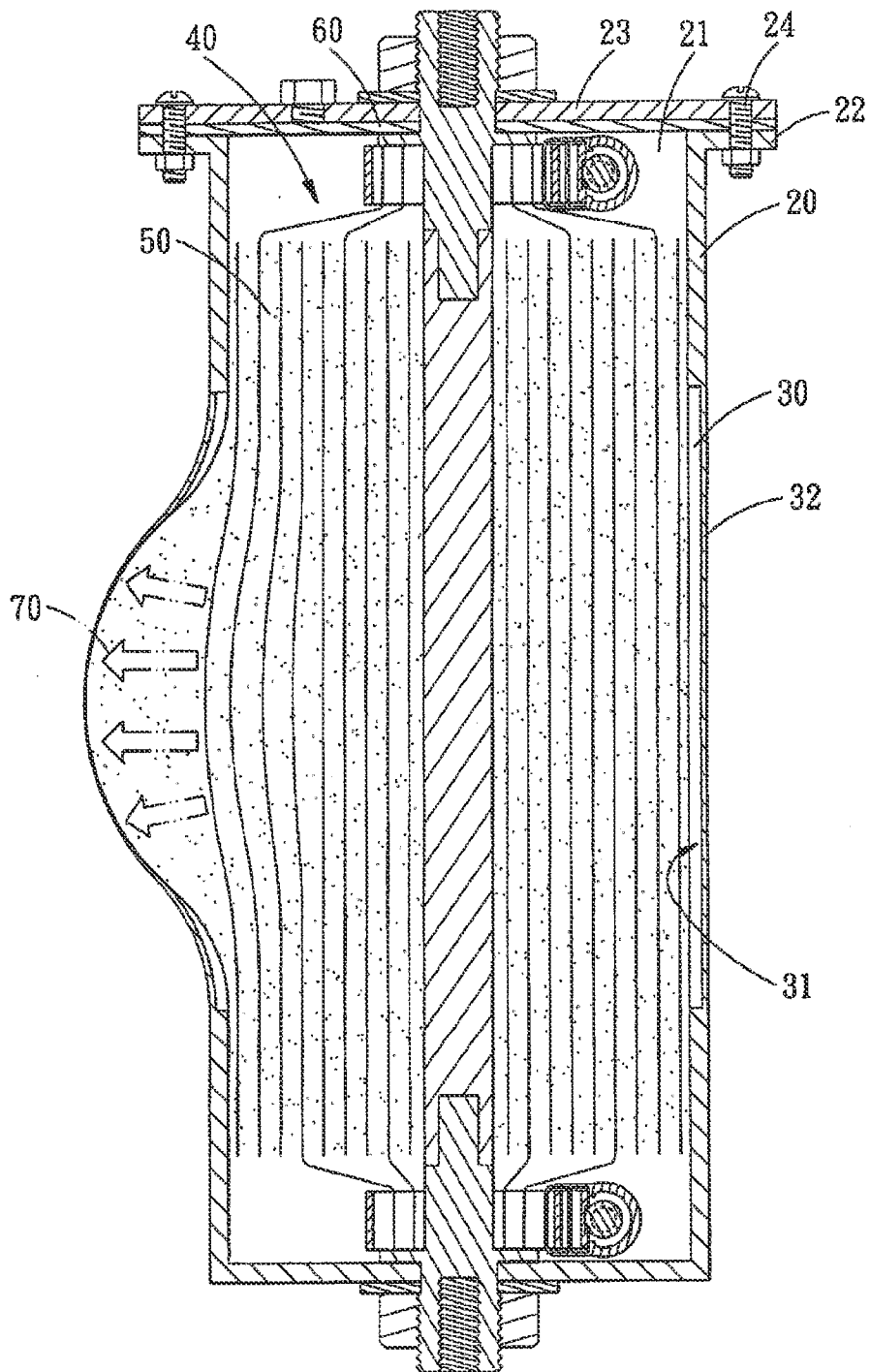


FIG. 5

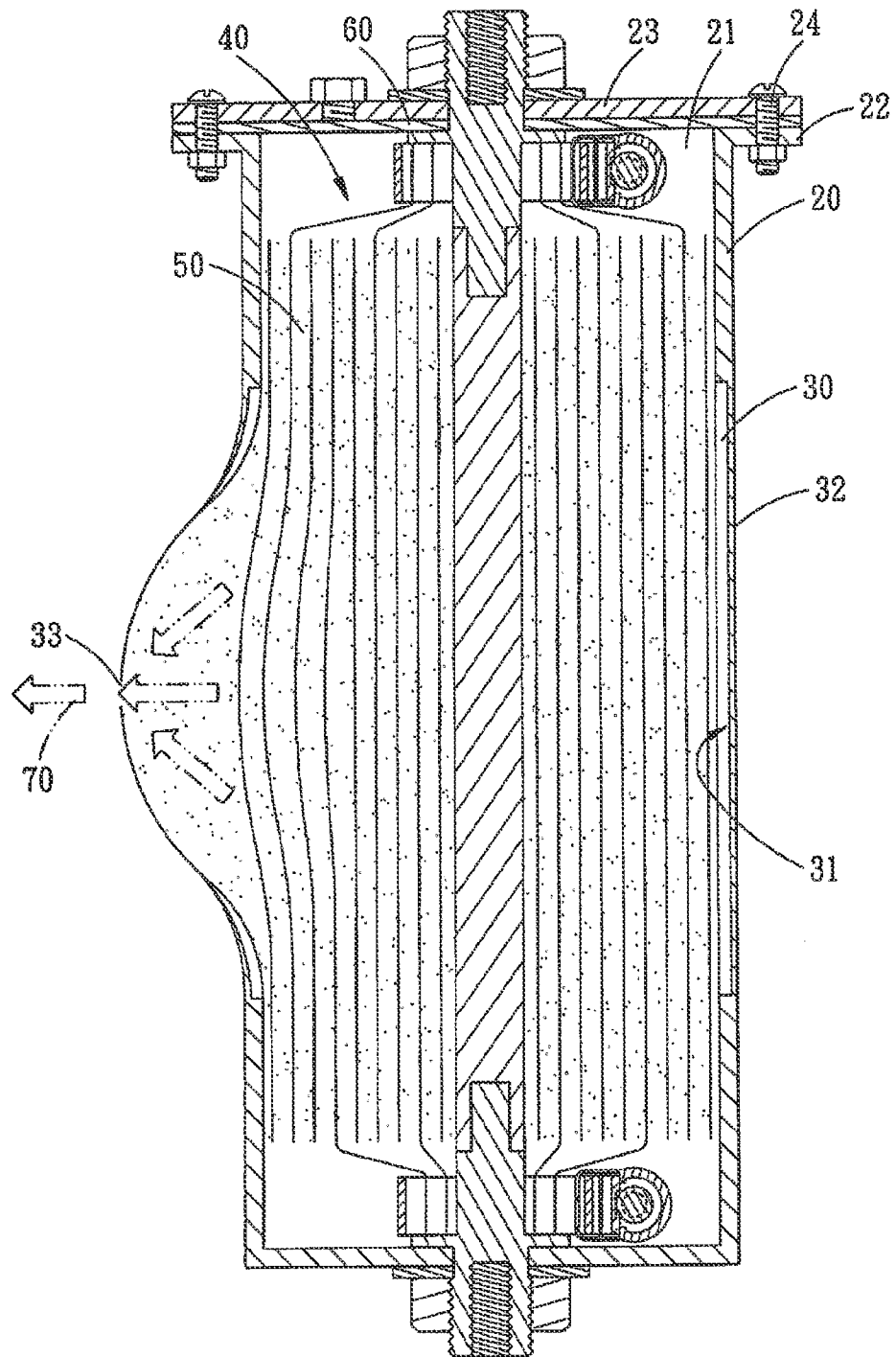


FIG. 6

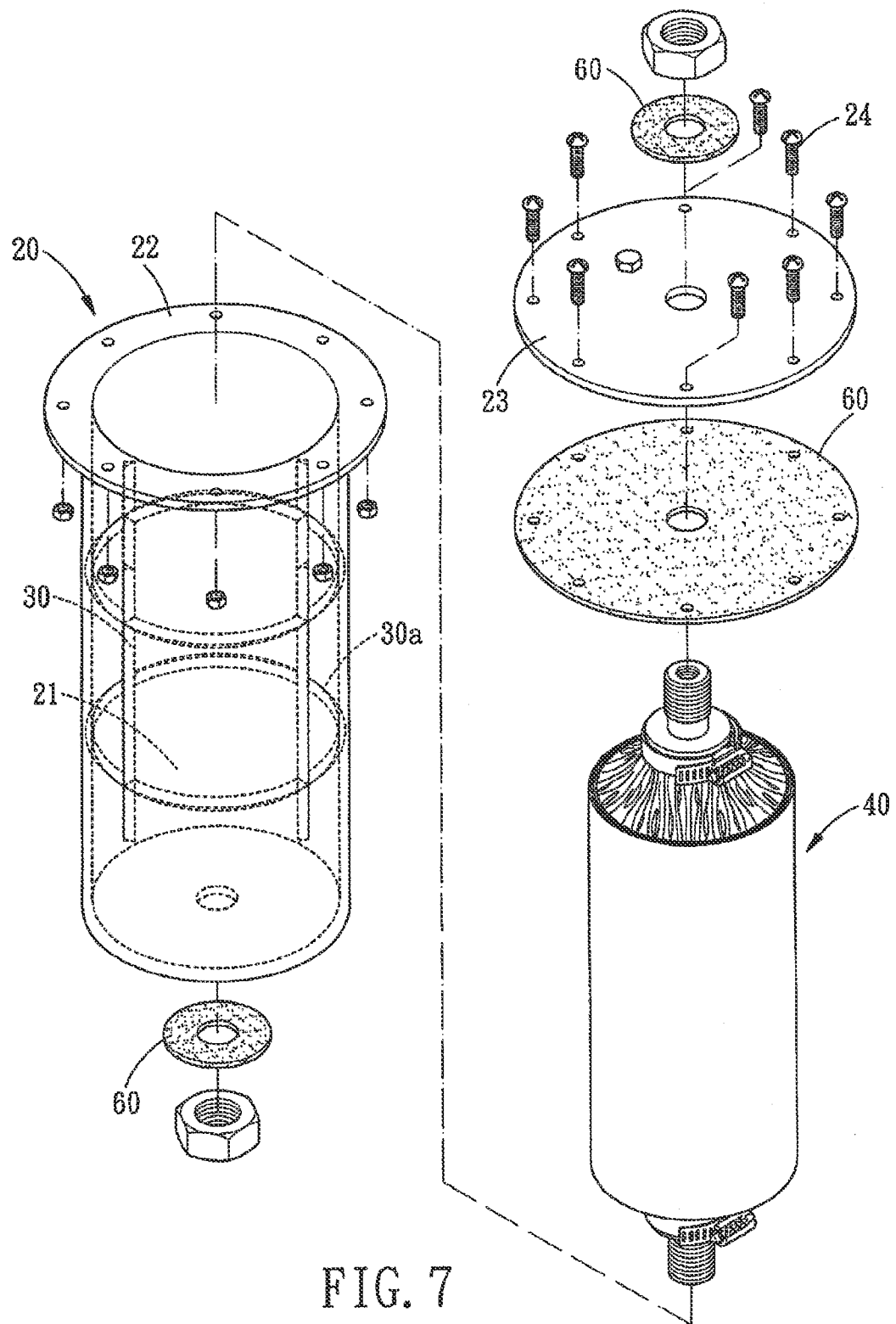


FIG. 7

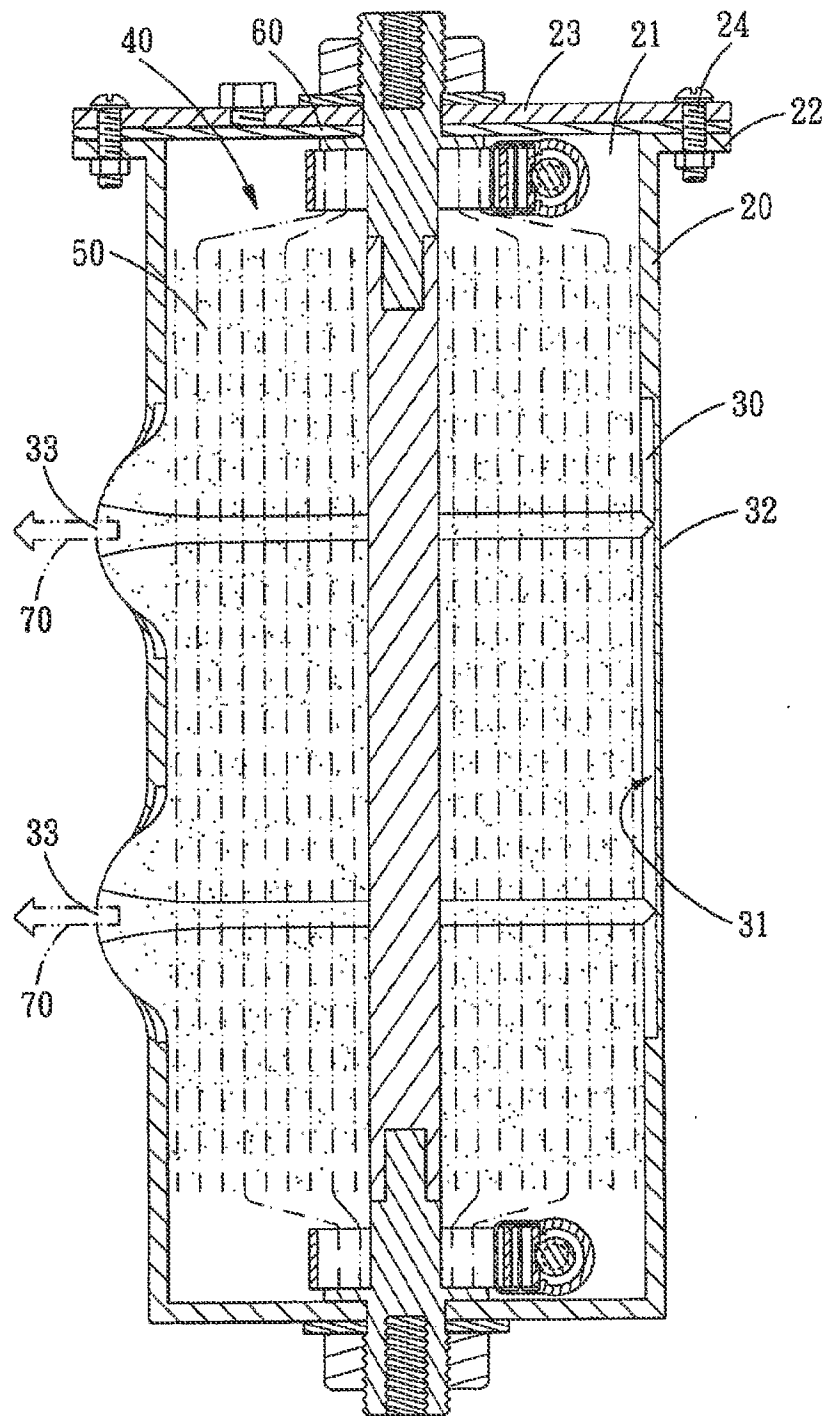


FIG. 8

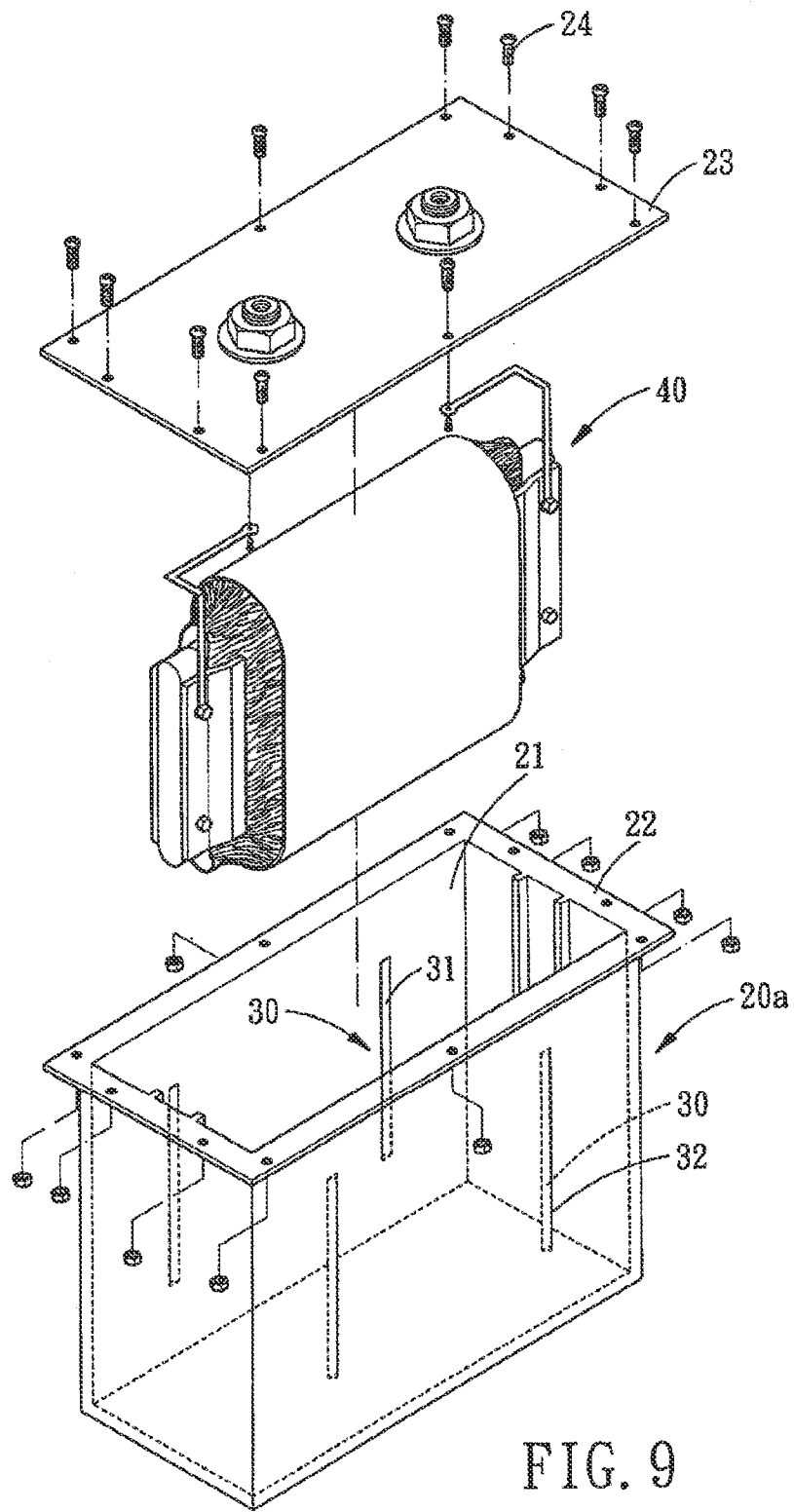


FIG. 9

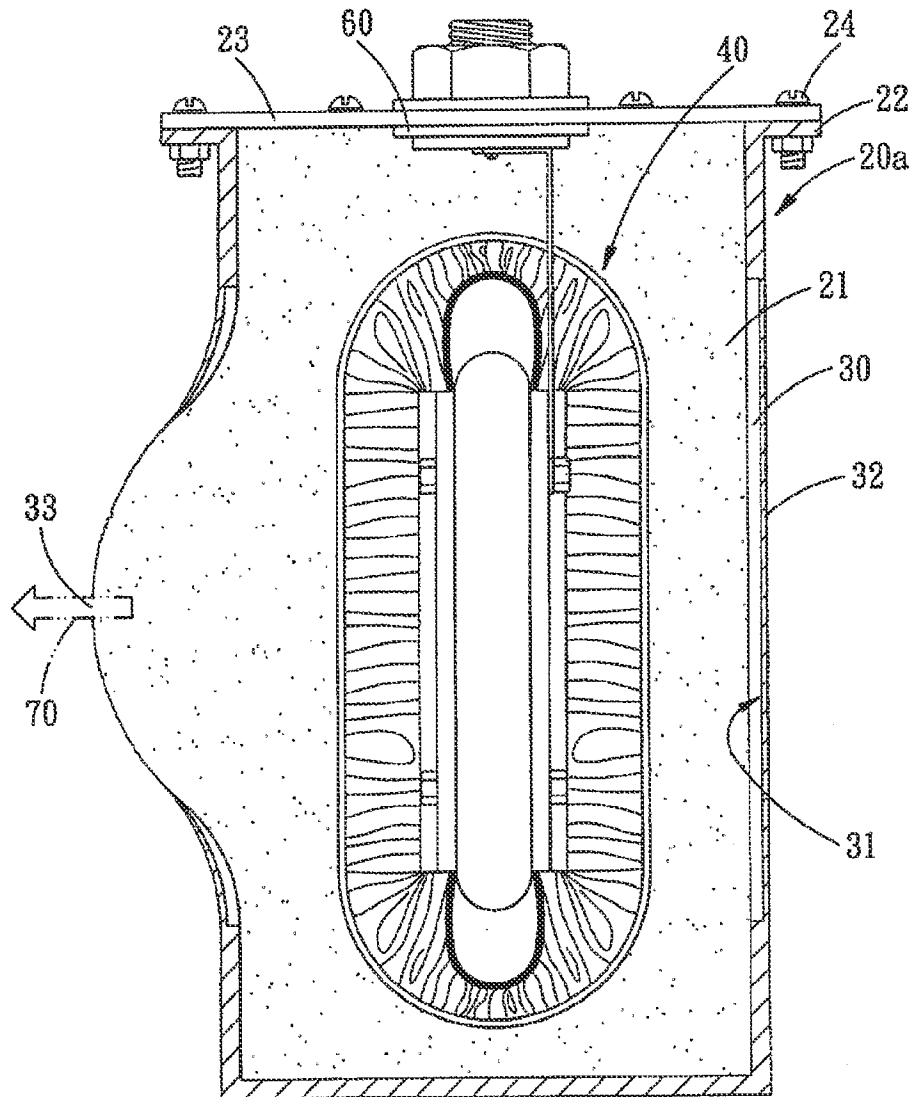
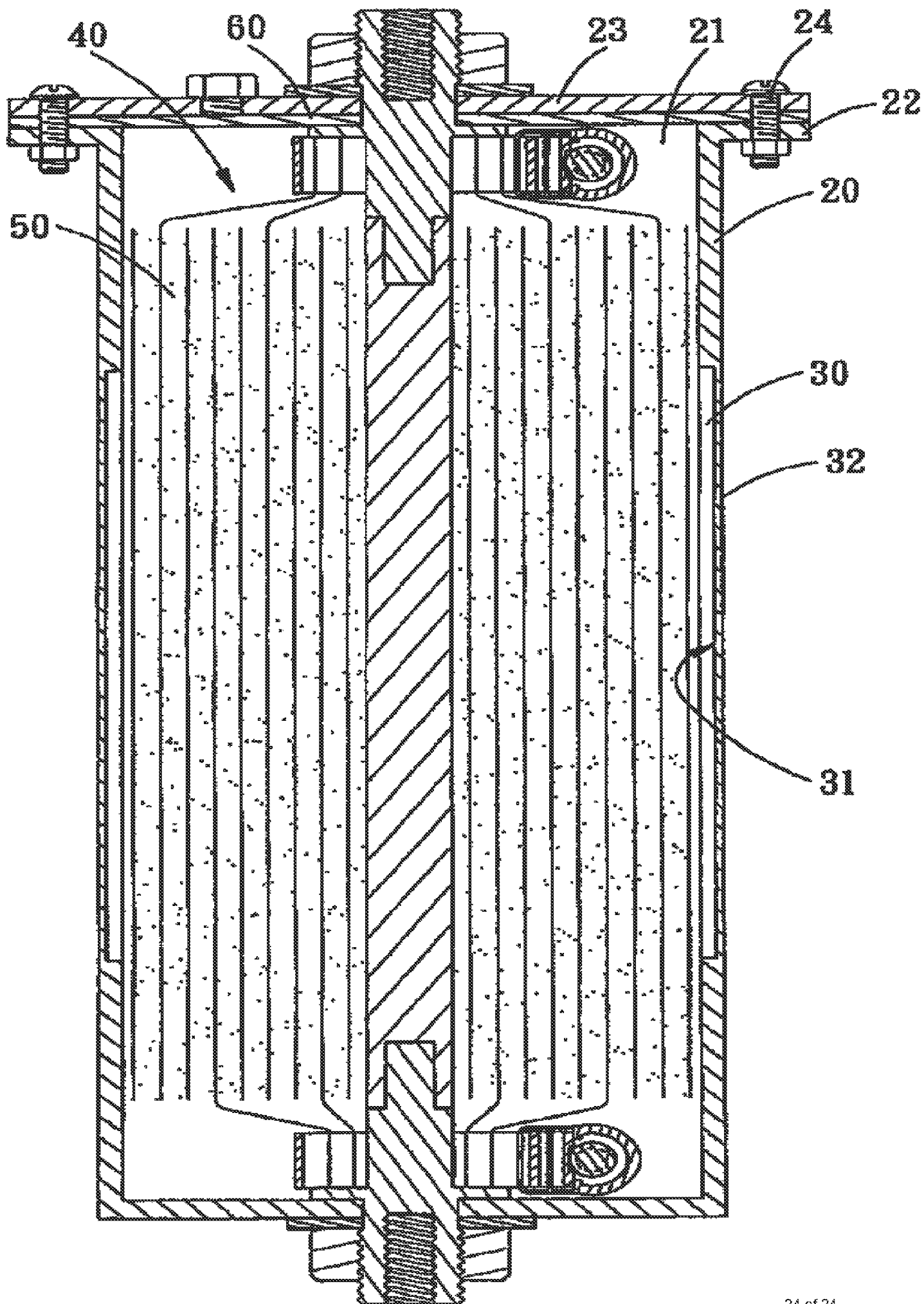


FIG. 10





Espacenet

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Cylindrical rechargeable battery with improved stability

Inventor(s): JI-YOUNG KIM; JUNG-JIN KIM; KWAN-SOO LEE; SUNG-JONG KIM; SOO-RYOUNG KIM; CHAHUN KU; DUKHYUN RYU; BYUNG-KYU JUNG ± (KIM JI-YOUNG, ; KIM JUNG-JIN, ; LEE KWAN-SOO, ; KIM SUNG-JONG, ; KIM SOO-RYOUNG, ; KU CHAHUN, ; RYU DUKHYUN, ; JUNG BYUNG-KYU, ; KU CHA-HUN, ; RYU DUK-HYUN)

Applicant(s): LG CHEMICAL LTD ± (LG CHEMICAL LTD, ; LG CHEM LTD)

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H01M10/0525 (KR); Y02E60/10 (EP); Y02P70/50 (EP)

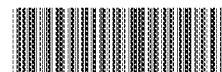
Application number: CN20108034848 20100813 Global Dossier

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KR101058386 (B1) KR20110017835 (A) US2012114979 (A1)
US9112234 (B2) WO2011019237 (A2) WO2011019237 (A3) less

Abstract of CN102473883 (A)

Provided is a cylindrical rechargeable battery in which an electrode assembly produced by the winding of an anode/separating film/cathode and an electrolytic solution are housed inside a cylindrical can, wherein a cap assembly mounted on the open upper part of the cylindrical can comprises: a safety vent formed with a predetermined notch so as to be ruptured by high-pressure gas in the battery; a current-blocking member which is coupled by welding to the lower edge of the safety vent and blocks current when the internal pressure of the battery rises; and a gasket for the current-blocking member surrounding the outer circumferential surface of the current-blocking member, and wherein the current-blocking member comprises two or more through holes for discharging gas, and the through holes are formed with a size such that their overall surface area is from 20 to 50% of the overall surface area of the current-blocking member.



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(71) 申请人 株式会社 LG 化学

地址 韩国首尔

(72) 发明人 金志映 金正镇 李营秀 金成钟

金秀铃 具滋训 柳德铉 郑炳圭

(74) 专利代理机构 中原信达知识产权代理有限

责任公司 11219

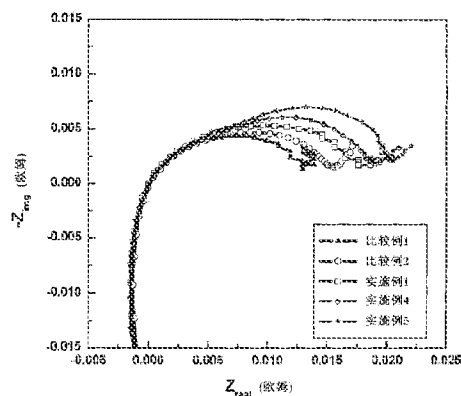
权利要求书 2 页 说明书 13 页 附图 10 页

(54) 发明名称

具有改善的安全性的圆筒形二次电池

(57) 摘要

本发明提供了一种其中通过卷绕正极 / 隔膜 / 负极而制造的凝胶卷型电极组件以及电解质设置在圆筒形罐中的圆筒形二次电池, 其中安装在所述圆筒形罐的开放顶部上的盖组件包含: 安全排气口, 所述安全排气口具有预定凹口以允许由于电池的高压气体而造成的破裂; 用于中断电流的电流中断装置, 所述电流中断装置焊接至安全排气口的底部; 和用于所述电流中断装置的衬垫, 所述衬垫用于围绕所述电流中断装置的周边, 其中所述电流中断装置包含两个以上允许排放气体的通孔, 其中所述通孔具有相对于所述电流中断装置的总面积为 20% 至 50% 的尺寸。



1. 一种圆筒形电池,其中通过卷绕正极/隔膜/负极而制造的电极组件(凝胶卷)和电解质设置在圆筒形罐中,

其中安装在所述圆筒形罐的开放顶部上的盖组件包含:

安全排气口,所述安全排气口具有预定凹口以允许由于所述电池的高压气体而造成的破裂;

用于中断电流的电流中断装置,所述电流中断装置焊接至所述安全排气口的底部;和用于所述电流中断装置的衬垫,所述衬垫用于围绕所述电流中断装置的周边,

其中所述电流中断装置包含两个以上允许排放气体的通孔,其中所述通孔具有相对于所述电流中断装置的总面积为 20%至 50%的尺寸。

2. 根据权利要求 1 所述的圆筒形电池,其中所述电流中断装置具有如下结构:在其中心处具有向上突出且焊接至所述安全排气口的凹陷部的底部的突出部,并且连接至所述凝胶卷的正极的正极引线通过所述突出部之外的区域的所述底部发生电连接;和桥,所述桥具有三至五个通孔和用于连接相邻通孔的凹口,所述通孔基于所述突出部的中心同心地形成。

3. 根据权利要求 1 所述的圆筒形电池,其中所述通孔具有相对于所述电流中断装置的总面积为 30%至 40%的面积。

4. 根据权利要求 1 所述的圆筒形电池,其中所述通孔在基于所述电流中断装置的中心径向对称的位置处以圆弧形式形成。

5. 根据权利要求 2 所述的圆筒形电池,其中所述三个通孔被布置成使得它们基于所述电流中断装置的中心以圆弧形式彼此间隔预定距离,所述圆弧具有与 50 度至 80 度的角度相对应的长度。

6. 根据权利要求 2 所述的圆筒形电池,其中所述通孔被布置成使得它们基于所述电流中断装置的中心以圆弧形式彼此间隔预定距离,所述圆弧具有与 40 度至 55 度的角度相对应的长度。

7. 根据权利要求 1 所述的圆筒形电池,其中所述安全排气口具有向下凹陷的中心,并且构成所述凹陷部的上弯曲部和下弯曲部分别具有第一凹口和第二凹口,

其中设置在上部区域上的所述第一凹口具有其中一侧敞开的开口曲线。

8. 根据权利要求 7 所述的圆筒形电池,其中所述第二凹口具有其中彼此面对的两端都敞开的开口弯曲结构。

9. 根据权利要求 7 所述的圆筒形电池,其中所述第一凹口具有与假想线的尺寸相等的尺寸,所述假想线用于连接所述通孔的中心线。

10. 根据权利要求 7 所述的圆筒形电池,其中相对于所述安全排气口的中心,所述第一凹口的开口区域具有与 20 度至 50 度的角度相对应的圆弧长度,从而在所述第一凹口被切开时防止所述凹陷部被完全切开。

11. 根据权利要求 8 所述的圆筒形电池,其中在所述第二凹口中敞开的所述两端形成于不与所述第一凹口的开口区域相对应的区域中。

12. 根据权利要求 1 所述的圆筒形电池,其中所述盖组件具有电流中断装置、用于电流中断装置的衬垫、安全排气口、PTC 元件和其中形成一个或多个气体出口的顶盖的层压结构,其中在所述层压结构的外围上安装有衬垫。

13. 根据权利要求 1 所述的圆筒形电池,其中所述衬垫在其中心处具有开口,并且所述开口与所述电流中断装置的所述通孔连通。

14. 根据权利要求 13 所述的圆筒形电池,其中所述电流中断装置的所述通孔、用于电流中断装置的衬垫的所述中心开口和所述安全排气口的所述凹口以使得它们在直线上彼此连通的方式形成。

15. 根据权利要求 1 所述的圆筒形电池,其中所述圆筒形电池包含含添加剂(电解质添加剂)的电解质以提高热稳定性。

16. 根据权利要求 15 所述的圆筒形电池,其中所述电解质添加剂在 150℃ 以上的温度下随着温度增加而引起连续的电解质分解,从而产生气体。

17. 根据权利要求 16 所述的圆筒形电池,其中所述电解质添加剂是在 150℃ 至 250℃ 下引起一次电解质分解和在 250℃ 至 300℃ 下引起二次电解质分解的材料。

18. 根据权利要求 16 所述的圆筒形电池,其中所述电解质添加剂为氟代碳酸亚乙酯(FEC)。

19. 根据权利要求 15 所述的圆筒形电池,其中基于电解质的总重量,所述电解质添加剂的含量为 11 重量%至 40 重量%。

20. 根据权利要求 15 所述的圆筒形电池,其中基于电解质的总重量,所述电解质添加剂的含量为 15 重量%至 35 重量%。

21. 一种安装在电池中的圆筒形罐的开放顶部上的盖组件,在所述电池中电极组件和电解质设置在圆筒形罐中,

其中,基于电解质的重量,电池的电解质包含 11 重量%至 40 重量%的添加剂,从而在 150℃ 以上的温度下随着温度增加通过连续的电解质分解而产生气体,

其中,在凝胶卷的顶部上安装有板型绝缘构件,所述板型绝缘构件在其中心处具有允许排放气体以及连接电极端子的穿孔开口,并具有布置在所述开口周围的一个或多个通孔,所述板型绝缘构件的所述通孔具有相对于所述板型绝缘构件的总面积为 20%至 50% 的尺寸。

22. 根据权利要求 21 所述的盖组件,其中所述板型绝缘构件的所述通孔的位置使得所述通孔在直线上与用于电流中断装置的衬垫的中心开口和所述电流中断装置的通孔连通。

具有改善的安全性的圆筒形二次电池

技术领域

[0001] 本发明涉及圆筒形二次电池。更具体而言,本发明涉及其中通过卷绕正极/隔膜/负极而制造的电极组件和电解质设置在圆筒形罐中的圆筒形电池,其中安装在圆筒形罐的开放顶部上的盖组件可包含:安全排气口,所述安全排气口具有预定凹口以允许因电池的高压气体而造成的破裂;电流中断装置,所述电流中断装置焊接至安全排气口的底部并用于中断电流;和用于所述电流中断装置的衬垫,所述衬垫用于围绕所述电流中断装置的周边,其中所述电流中断装置包含两个以上允许排放气体的通孔,其中所述通孔具有相对于所述电流中断装置的总面积为 20%至 50%的尺寸。

背景技术

[0002] 技术发展和对移动设备增加的需求已经导致对作为能源的二次电池的需求迅速增加。在这些二次电池中,正在进行与具有高能量密度和放电电压的锂二次电池相关的大量研究,并且这种锂二次电池可商购获得并被广泛使用。

[0003] 根据电池壳体的形状,二次电池分为其中电极组件安装在圆筒形或角形金属容器内的圆筒形电池和角形电池以及其中电极组件安装在由铝层压片制成的袋型壳体內的袋型电池。其中,圆筒形电池具有容量和结构稳定性相对高的优点。

[0004] 安装在电池壳体中的电极组件是用于产生电的可再充电且可放电的装置,并且具有正极/隔膜/负极的层压结构。电极组件分为:凝胶卷型(jelly-roll type)电极组件,所述电极组件通过将隔膜夹在负极和涂布了活性材料的具有长片形状的正极之间,然后进行卷绕而制成;和堆叠型电极组件,所述电极组件通过依次层压具有预定尺寸的多个正极和具有预定尺寸的多个负极,使得将隔膜夹在每个正极和每个负极之间而制成。其中,凝胶卷型电极组件具有易于制造且单位重量的能量密度高的优点。

[0005] 在这一点上,图 1 是示意性地示出一般圆筒形电池的横截面透视图。

[0006] 参考图 1,通过将凝胶卷型电极组件 120 放置在圆筒形壳体 130 中;将电解质注入到圆筒形壳体 130 内,并且将具有电极端子(例如,正极端子,未示出)的顶盖 140 连接到壳体 130 的开放顶部,制造了圆筒形二次电池 100。

[0007] 电极组件 120 具有如下结构,其中正极 121、负极 122 和夹在两者之间的隔膜 123 被卷绕,并且圆筒形中心销 150 插入在电极组件 120 的卷中心(凝胶卷中心)。中心销 150 通常由金属材料制成以提供预定强度并具有其中片材料被圆形弯曲的中空圆筒形结构。中心销 150 固定和支撑电极组件并充当通道,从而使得可排放由于在充电/放电时的内部反应而产生的气体。

[0008] 同时,锂二次电池具有低稳定性的缺点。例如,在其中将电池过充电至约 4.5V 以上的情况中,正极活性材料发生分解,在负极上生长锂枝晶,并且电解质发生分解。这些过程涉及热反应和分解反应并且因此迅速进行多个副反应。最终,可能发生电池的燃烧和爆炸。

[0009] 因此,为了解决这些问题,一般的圆筒形二次电池具有电流中断装置(CID)和安

全排气口以在电池异常运行时中断电流并减小设置在电极组件和顶盖之间的空间中的内部压力。

[0010] 将参考图 2 至 4 对该机构进行描述。

[0011] 参考附图,顶盖 10 形成突出形式的正极端子并具有穿孔排气口。在顶盖 10 下方依次设置有:正温度系数 (PTC) 元件 20,所述正温度系数 (PTC) 元件 20 在电池的内部温度增大时大大增加电池的电阻并从而中断电流;安全排气口 30,所述安全排气口 30 在正常状态下向下突出,但在电池的内部压力增大时突出并同时发生破裂,从而导致排放气体;和连接板 50,所述连接板 50 的顶部一侧连接至安全排气口 30 且其底部的另一侧连接至电极组件 40 的正极。

[0012] 因此,在正常运行条件下,电极组件 40 的正极依次通过引线 42、连接板 50、安全排气口 30 和 PTC 元件 20 连接至顶盖 10 以施加电。

[0013] 然而,如图 3 中所示,当由于例如过充电而从电极组件 40 产生气体并且内部压力增大时,安全排气口 30 的形状反转,由此向上突出。此时,安全排气口 30 与连接板 50 分离,并从而中断电流。因此,确保了安全性,以便防止进一步过充电。然而,如图 4 中所示,当内部压力继续增大时,安全排气口 30 发生破裂,并且压力气体穿过破裂的区域,然后经顶盖 10 的排气口排出,从而防止电池爆炸。

[0014] 这种运行过程取决于在电极组件中产生的气体的量和气体到安全排气口的输送效率。例如,虽然产生大量气体,但在其中气体未有效输送至安全排气口的情况中,不能进行期望的安全运行过程。此外,除非在短时间内产生大量气体,并且气体到达安全排气口并诱发预定的运行过程,否则电池的内部压力将迅速增大,从而引起爆炸。

[0015] 此外,由电解质经由高温加热的分解而引起气体的产生。当从电池产生的热量在短时间内迅速增加时,可能发生热失控。热失控现象发生在电池处于连续加电状态时。当该现象进一步加速时,电池可能燃烧或爆炸的风险大大增加,从而不利地引起严重的安全问题。

[0016] 因此,对于开发如下圆筒形二次电池具有日益增加的需求,所述圆筒形二次电池能够在电池异常运行状态下更稳定地诱导电池的内部气体并将所述气体迅速排放至电池外部。

发明内容

[0017] 【技术问题】

[0018] 因此,完成了本发明以解决上述问题以及其他尚待解决的技术问题。

[0019] 本发明的一个目的是提供一种二次电池,所述二次电池使得所产生的气体可以经由具有特定结构的盖组件而有效地通过安全构件,并且使得预定的安全过程能够进行,从而最终提高电池安全性。

[0020] 本发明的另一个目的是提供一种二次电池,所述二次电池随着温度的增加而诱发电解质的连续分解反应,并且抑制电池内部压力的迅速增加,从而使得安全构件能够有效运行。

[0021] 本发明的又一个目的是提供一种二次电池,所述二次电池诱发电解质的分解反应,同时使产生的热最小化并抑制热失控的发生。

【0022】 【技术解决方案】

[0023] 因此,根据本发明的圆筒形二次电池是其中通过卷绕正极/隔膜/负极而制造的凝胶卷型电极组件以及电解质设置在圆筒形罐中的圆筒形二次电池,其中安装在所述圆筒形罐的开放顶部上的盖组件包含:安全排气口,所述安全排气口具有预定凹口以允许由于电池的高压气体而造成的破裂;电流中断装置,所述电流中断装置焊接至安全排气口的底部并用于中断电流;和用于所述电流中断装置的衬垫,所述衬垫用于围绕所述电流中断装置的周边,其中所述电流中断装置包含两个以上允许排放气体的通孔,其中所述通孔具有相对于所述电流中断装置的总面积为 20%至 50%的尺寸。

[0024] 因此,圆筒形电池使得产生的气体可以通过具有特定结构的盖组件而直接作用于安全构件,诱发期望的断电并且通过使安全排气口破裂而使得能够将气体迅速排放至外部,从而确保电池的安全性。

[0025] 电流中断装置是当电池的内部压力增大时发生破裂并从而中断电流的构件,并且可具有如下结构:在其中心处具有向上突出且焊接至安全排气口的凹陷部的底部的突出部,并且连接至所述凝胶卷的正极的正极引线通过所述突出部之外的区域的底部发生电连接。

[0026] 照这样,由于将电流中断装置连接至安全构件,所以在电流中断装置和正极之间的连接区根据安全排气口的形状反转而发生破裂,从而导致电流中断。在具体实例中,可在突出部的周边表面处形成凹口,所述凹口根据安全排气口的形状反转而发生破裂,并且所述突出部可容易地与电流中断装置和正极之间的连接区分离,同时突出部仍然连接至安全排气口。

[0027] 此外,电流中断装置具有两个以上通孔以使得气体可向上移动,从而导致电池的内部气体增加,并由此使安全排气口反转。

[0028] 然而,本发明人确认了,在设置在电流中断装置处的通孔的面积小于 20%的情况下,当在电池中产生高压时,电池的燃烧或爆炸率增加。认为这是由于以下事实所造成的:由于穿过通孔的气体的量少而导致有效断电失败,或者由于气体的向上移动通道发生阻塞或变化而导致中断电流需要花费长时间。同时,从难以确保元件的预定强度和扭转(torsion)并且在加工时发生部件的扭转的观点来看,通孔具有过大的尺寸是不优选的。

[0029] 因此,在本发明的圆筒形电池中,电流中断装置包含两个以上允许排放气体的通孔,并且所述通孔具有相对于电流中断装置的总面积为 20%至 50%、更优选 30%至 40%的面积,从而提高向上移动(排放)的气体的量并诱导气体的快速向上移动(排放)。

[0030] 优选地,在基于电流中断装置的中心径向对称的位置处以圆弧形形成通孔。在其中在这种径向对称位置处形成通孔的情况下,有利之处在于,可以防止气体排放集中在一个方向上并且可以确保电流中断装置的强度。

[0031] 此时,通孔的数量没有特别限制,并且优选为 2 至 4 个。

[0032] 在一个优选实施方式中,可以设置三个通孔,使得它们基于电流中断装置的中心以圆弧形彼此间隔预定距离,所述圆弧具有与 50 至 80 度的角度相对应的长度。

[0033] 同时,所述突出部具有桥,所述桥具有三个通孔和用于连接相邻通孔的凹口,所述通孔基于所述突出部的中心而同心地形成。可以对通孔进行设置,使得它们基于电流中断装置的中心以圆弧形彼此间隔预定距离,所述圆弧在与通孔相对应的位置处且具有与 40

至 55 度的角度相对应的长度。

[0034] 安全排气口是在电池的內部压力增大时排放气体从而确保电池安全性的元件。例如,当在电池内部产生气体并且内部压力由此增大至临界水平以上时,从破裂区域排放的气体可通过设置在顶盖上的一个或多个气体出口排放。

[0035] 优选地,安全排气口具有向下凹陷的中心。在这种结构中,当电池的內部压力增大时,向下的凹陷部反转成向上的凹陷部,与电流中断装置分离且中断电连接。

[0036] 考虑到具有这种结构的安全排气口的运行过程,在电池内部产生的气体穿过通孔并对安全排气口施加向上的压力,安全排气口的向下凹陷部在预定压力(简称为“第一临界压力”)以上的压力下发生反转,附着至凹陷部的突出部与电流中断装置分离并且中断从电流中断装置到安全排气口的电施加。

[0037] 凹陷部包含例如上弯曲部和下弯曲部,并且所述上弯曲部优选具有第一凹口。在常规方法中,提出了其中仅有下弯曲部被切开的结构。然而,下弯曲部具有狭窄的区域并因此不适合有效地排放气体。另一方面,根据本发明,在上弯曲部中设置有用于破裂的凹口,从而增加切口面积并使得能够将大量气体迅速排放至外部。

[0038] 当压力继续增大并达到预定水平(简称为“第二临界压力”)时,尽管安全排气口中断电流,但是第一凹口可被切开。为此,受压的内部气体通过设置在顶盖上的气体出口而平稳地排放至外部。

[0039] 同时,为了防止整个上弯曲部与安全排气口分离,第一凹口可具有其中一侧敞开的开口曲线。即,一侧的开口充当用于将安全排气口的凹陷部连接至其剩余区域的桥。

[0040] 第一凹口的尺寸没有特别限制,并且可以优选地对应于假想线的尺寸,所述假想线用于连接通孔的中心线以使得气体可以从电流中断装置向上移动而对第一凹口直接施加压力。

[0041] 此外,第一凹口的开口区域具有如上所述当第一凹口被切开时防止凹陷部被完全切开的尺寸,并且相对于安全排气口的中心,优选具有与 20 度至 50 度、更优选 25 至 40 度的角度相对应的圆弧长度。

[0042] 在一个优选实施方式中,在下弯曲部中形成第二凹口,但具有其中彼此面对的两端敞开的开口弯曲结构,以便在内部压力增大时使得能够仅将上弯曲部切开。其中两端敞开的这种结构防止下弯曲部被切开,并且通过切开后弯曲部而更有效地引起打开。

[0043] 优选地,在不与第一凹口的开口区域相对应的区域中形成在第二凹口中开口的两端。在一个更优选的实施方式中,以约 ± 90 度的角度形成在第二凹口中开口的两端。

[0044] 在一个优选实施方式中,将第一凹口形成为大于第二凹口的深度,从而使得第一凹口在第二凹口之前被切开。

[0045] 安全排气口没有特别限制,并且可以由具有 0.15 至 0.4mm 厚度的铝板材料制成以提供强度,从而使得在内部压力增大至临界水平以上时可以将凹口切开。

[0046] 根据本发明的圆筒形电池的盖组件还可包含诸如正温度系数(PTC)元件的构件,从而在电池的內部温度增大时大大增加电池的电阻并由此中断电流。

[0047] 盖组件具有电流中断装置、电流中断装置用衬垫、安全排气口、PTC 元件、以及其中形成一个或多个气体出口的顶盖的层压结构。可以将衬垫进一步添加至层压结构的外围。可以将 PTC 元件夹在顶盖和安全排气口之间,使得其附着至设置在两者之间的区域。

[0048] 此外,电流中断装置用衬垫围绕电流中断装置的周边。在一个优选实施方式中,衬垫在其中心处具有开口。受压气体优选以直线移动,以便使得内部气体可有效地向上移动。因此,中心开口具有不遮挡电流中断装置的通孔的形状。

[0049] 在常规技术中,由于衬垫遮挡电流中断装置的通孔的一部分,所以穿过通孔的受压气体被衬垫干扰,结果,排气量减小并且通道发生变化,从而使得难以诱发安全排气口的即时运行。然而,如上所述,其中衬垫不遮挡通孔的结构可确保电池的稳定性,因为高压气体完全直接到达安全排气口。

[0050] 在一个优选实施方式中,以使得它们在直线上彼此连通的方式形成电流中断装置的通孔、电流中断装置用衬垫的中心开口和安全排气口的凹口。

[0051] 因此,尽管在电池中产生的高压气体经由中心开口移动穿过电流中断装置的通孔而到达安全排气口的凹口,但是气体通道可保持直线,由此诱发安全排气口的即时运行并确保电池稳定性。

[0052] 在一个优选实施方式中,圆筒形电池可包含含添加剂(电解质添加剂)以提高热稳定性的电解质。

[0053] 电解质添加剂可随着温度增加而通过电解质的连续分解反应产生气体,并且例如可在 150℃ 以上的温度下发生反应。

[0054] 因此,如在下面的实验例中所证明的,根据本发明的圆筒形电池随着温度增加而诱发电解质的连续分解反应,并由此抑制热失控和电池内部压力的迅速增加。这种电解质连续分解反应的诱发防止了在将大量气体输送至安全元件时发生一种瓶颈现象。因此,电解质添加剂发挥与特定盖组件结构配合的协同效应。

[0055] 在一个优选实施方式中,电解质添加剂是在 150℃ 至 250℃ 下引起一次电解质分解和在 250℃ 至 300℃ 下引起二次电解质分解的材料。例如,电解质添加剂在一次电解质分解反应中引起约 2/3 的总分解反应,然后在高于一次电解质分解温度的温度下的二次电解质分解反应中引起约 1/3 的总分解反应。

[0056] 因此,气体随着温度的增加而连续地产生,从而使得电流中断装置和安全排气口可以运行。为此,能够解决与在短时间内产生大量气体相关的问题,并且如上所述,尽管连续产生气体,但所产生的气体经由特有的盖组件结构而直接起作用,从而诱发期望的断电。通过安全排气口的破裂可以将气体迅速排放至电池的外部,并且能最终有效地防止电池的燃烧和爆炸。

[0057] 电解质添加剂的优选实例是氟代碳酸亚乙酯(FEC),但本发明不限于此。

[0058] 本发明的发明人通过实验证明了,如上所述,电解质添加剂随着温度的增加而诱发连续的电解质分解,并且在正极的表面上形成保护膜而增加正极界面处的电阻;并且这种电阻的增加减少了短路时施加的电量,从而导致改善的电池稳定性。

[0059] 基于电解质的总重量,电解质添加剂优选以 11 重量%至 40 重量%、更优选 15 重量%至 35 重量%的量存在。

[0060] 当电解质添加剂的含量过小时,不能实现高温下的连续气体产生和在短路时的电流降低效果,另一方面,当电解质添加剂的含量过大时,不利地减小了电池容量。

[0061] 优选地,将电解质添加剂用于诸如在 4.3V 至 4.35V 的充电/放电电压下的高压电池。在这种情况下,电解质添加剂的含量优选为 20 重量%至 35 重量%。

[0062] 电解质通常包含在非水有机溶剂中的锂盐,这在本领域中是已知的。

[0063] 作为可用于本发明的非水电解液,例如可以提及的有非质子有机溶剂,如 N- 甲基 -2- 吡咯烷酮、碳酸亚丙酯、碳酸亚乙酯、碳酸亚丁酯、碳酸二甲酯、碳酸二乙酯、 γ - 丁内酯、1,2- 二甲氧基乙烷、四羟基 Franc、2- 甲基四氢呋喃、二甲基亚砜、1,3- 二氧戊环、甲酰胺、二甲基甲酰胺、二氧戊环、乙腈、硝基甲烷、甲酸甲酯、乙酸甲酯、磷酸三酯、三甲氧基甲烷、二氧戊环衍生物、环丁砜、甲基环丁砜、1,3- 二甲基 -2- 咪唑啉酮、碳酸亚丙酯衍生物、四氢呋喃衍生物、醚、丙酸甲酯以及丙酸乙酯。其中,优选的是由线性碳酸酯化合物、环状碳酸酯化合物或其组合构成的碳酸酯基溶剂。

[0064] 锂盐是易溶于上述非水电解质的材料,且可以包括例如 LiCl、LiBr、LiI、LiClO₄、LiBF₄、LiB₁₀Cl₁₀、LiPF₆、LiCF₃SO₃、LiCF₃CO₂、LiAsF₆、LiSbF₆、LiAlCl₄、CH₃SO₃Li、CF₃SO₃Li、(CF₃SO₂)₂NLi、氯硼烷锂、低级脂族羧酸锂、四苯基硼酸锂和酰亚胺。

[0065] 另外,为了改善充放电特性和阻燃性,可以向非水电解质内添加例如吡啶、亚磷酸三乙酯、三乙醇胺、环醚、乙二胺、正甘醇二甲醚、六磷酸三酰胺(hexaphosphoric triamide)、硝基苯衍生物、硫、酞亚胺染料、N- 取代的咪唑烷酮、N, N- 取代的咪唑烷、乙二醇二烷基醚、铵盐、吡咯、2- 甲氧基乙醇、三氯化铝等。如果需要,为了赋予不燃性,非水电解质还可包含含卤素的溶剂如四氯化碳和三氟乙烯。此外,为了改善高温贮藏特性,非水电解质可另外包含二氧化碳气体和丙烯磺酸内酯(PRS)。

[0066] 本发明还提供了安装在电池中的圆筒形罐的开放顶部上的盖组件,在所述电池中电极组件和电解质设置在圆筒形罐中。具体地,基于电解质的重量,电池的电解质包含 11 重量%至 40 重量%的添加剂以在 150℃ 以上的温度下随着温度的增加而通过连续的电解质分解产生气体。对于盖组件,可在凝胶卷的顶部上安装板型绝缘构件,所述构件在中心处具有穿孔开口以使得可排放气体并连接电极端子,并且具有设置在开口周围的一个或多个通孔。

[0067] 在一个优选实施方式中,板型绝缘构件的通孔可具有这样的位置,使得其在直线上与电流中断装置用衬垫的中心开口和电流中断装置的通孔连通。板型绝缘构件的通孔具有相对于板型绝缘构件的总面积为 20%至 50%的尺寸。

[0068] 可以使用任何绝缘构件而没有任何限制,只要其由绝缘材料制成即可,并且例如可以是聚丙烯树脂。其中,特别优选的是高强度聚丙烯树脂,其能够在恒定的厚度条件下发挥优异的机械强度并且在施加强度的外部冲击时防止凝胶卷移动和分离。

[0069] 【有利效果】

[0070] 如从上文显而易见的,本发明提供了一种圆筒形电池,其包含具有多个具有相对大面积的通孔的电流中断装置,从而有效地排放内部高压气体。此外,在形成在安全排气口中发生破裂的凹口以对应于通孔的情况中,当产生高压气体时,高压气体可以直线排放,且可以诱发快速断电,气体排放可迅速进行,并且可以显著地提高电池安全性。

[0071] 此外,在向电解质中添加特定电解质添加剂的情况中,添加剂随着温度的增加而引起连续的电解质分解,从而稳定地产生气体并减小短路电流,由此大大降低由发热而引起的燃烧或爆炸的可能性。

附图说明

[0072] 根据结合附图的以下详细描述,将更清楚地理解本发明的上述和其他目的、特征及其它优点,在附图中:

[0073] 图 1 是示意性地示出一般圆筒形电池的横截面透视图;

[0074] 图 2 至 4 是示出其中通过常规圆筒形电池中的安全元件的运行而中断电流并排放高压气体的一系列过程的横截面图;

[0075] 图 5 是示出根据本发明一个实施方式的圆筒形电池的示意性截面图;

[0076] 图 6 和 7 示出了用于图 5 的盖组件结构的安全排气口的透视图和平面图;

[0077] 图 8 和 9 是示出用于图 5 的圆筒形电池的电流中断装置的透视图和平面图;

[0078] 图 10 和 11 是示出用于图 5 的圆筒形电池的电流中断装置用衬垫的透视图和平面图;

[0079] 图 12 和 13 是示出用于图 5 的圆筒形电池的板型绝缘构件的透视图和平面图;

[0080] 图 14 是示出用于层压根据本发明一个实施方式的板型绝缘构件、电流中断装置用衬垫和电流中断装置的过程的示意图;

[0081] 图 15 至 17 是示出其中通过图 5 的圆筒形电池中的安全元件的运行而中断电流并排放高压气体的一系列过程的横截面图;以及

[0082] 图 18 是示出在实施例 1、4 和 5 以及比较例 1 和 2 中制造的电池的正极界面电阻结果的图。

具体实施方式

[0083] 现在,将参考附图更详细地描述本发明,并且不应将其理解为限制本发明的范围和主旨。

[0084] 图 5 示出了根据本发明一个实施方式的圆筒形电池的盖组件结构。图 6 和 7 是用于图 5 的盖组件结构的安全排气口的透视图和平面图。

[0085] 参考图 5,通过下列方式来制造圆筒形电池 100:将凝胶卷 110 插入到圆筒形罐 200 中;将电解质注入到其中;将板型绝缘构件 800 安装在凝胶卷 110 的顶部上;并且将盖组件 300 安装在圆筒形罐 200 的开放顶部上。

[0086] 电解质包含添加剂以使热产生最小化并诱发电解质的连续分解反应。

[0087] 盖组件 300 具有如下结构,其中顶盖 310、PTC 元件 350 和用于减小内部压力的安全排气口 320 附着至用于气密性的衬垫 400 的内部,所述衬垫 400 安装在圆筒形罐 200 的上部卷边构件 210 上。

[0088] 电流中断装置 600 通过焊接连接至安全排气口 320 的底部,所述电流中断装置 600 在电池内部压力增大时发生破裂并从而中断电流,且用于所述电流中断装置的衬垫 700 围绕电流中断装置的周边。

[0089] PTC 元件 350 夹在顶盖 310 和安全排气口 320 之间,使得其附着至设置在其间的电流电路,并且在电池内部温度增大时大大增加电池电阻,从而中断电流。顶盖 310 具有向上突出的中心,由此通过连接至外部电路而充当正极端子。沿着突出部的周边形成用于排放在圆筒形罐 200 中存在的受压气体的多个气体出口 312。

[0090] 将参考图 6 和 7 更详细地描述具有这种结构的盖组件 300 中的安全排气口 320 的结构。

[0091] 安全排气口 320 具有导电薄膜结构,所述结构具有在其中心处的向下凹陷的凹陷部 322、在其上弯曲部处的第一凹口 324 和在其下弯曲部处的第二凹口 326。

[0092] 如示出安全排气口 320 的平面图的图 7 中所示,设置在上弯曲部上的第一凹口 324 具有其中一侧敞开的开口曲线。该开口区域为非凹口区域 330,这防止了整个第一凹口的分开,尽管将其中由高压气体形成凹口的区域切开。

[0093] 此外,第二凹口 326 形成在下弯曲部中并具有其中彼此面对的两端敞开的开口弯曲结构。两个非凹口区域 340 设置在第二凹口 326 的两个开口端上。非凹口区域 340 防止通过切开面将第二凹口 326 分开。为了更精确地防止第二凹口 326 分开,非凹口区域 340 优选设置在不与第一凹口 324 中形成的非凹口区域 330 相对应的区域中,更优选地,如图中所示,基于第一凹口 324 的非凹口区域 330 的中心线,以约 ± 90 度的角度形成非凹口区域 340,并且可以以彼此面对的方式设置两个非凹口区域 340。

[0094] 图 8 和 9 示出了用于图 5 的圆筒形电池的电流中断装置 600。

[0095] 参考图 8 和 9 以及图 5,在电流中断装置 600 的中心处形成向上突出的突出部 620,所述突出部 620 通过焊接而附着至安全排气口 320 的凹陷部底部 (322,参考图 6),并且将连接至凝胶卷 110 的正极的正极引线 420 电连接至电流中断装置 600 底部上的突出部 620 之外的区域。

[0096] 突出部 620 具有桥 640,所述桥 640 具有三个通孔 630 和用于连接相邻通孔 630 的凹口 650,所述通孔基于突出部 620 的中心同心地形成。

[0097] 此外,在电流中断装置 600 的周边上,在基于中心轴径向对称的位置处,以圆弧形式形成用于气体排放的三个通孔 610。通孔 610 具有相对于电流中断装置 600 的总面积为 30% 的尺寸,并且增加了在电池内部存在的高压气体的排放量,从而能发挥可靠的电流中断。通孔 610 以约 120° 的角度彼此隔开且具有相同的形状和尺寸,并且其间的距离基本均匀。基于这种结构,电流中断装置 600 使气体排放最大化并保持高机械强度。

[0098] 此时,优选地,用于连接通孔 610 的中心线的假想线 (参见虚线) 的尺寸基本等于安全排气口 (320,参见图 6 和 7) 的第一凹口 324 的尺寸。因此,当产生预定水平以上的内部压力时,穿过通孔 610 的高压气体经由直线通道压制第一凹口 324,并且促进切开第一凹口 324,从而使得能够快速排放气体。

[0099] 图 10 和 11 示出了围绕电流中断装置 600 的电流中断装置用衬垫 700。为了便于说明,简单地示出电流中断装置用衬垫以仅表示开口的形状。

[0100] 参考图 10 和 11 以及图 5,电流中断装置用衬垫 700 在其中心处具有开口 710,并且中心开口 710 具有不遮挡电流中断装置 600 的通孔 610 的形状。即,电流中断装置用衬垫 700 的形状基本对应于其中在电流中断装置 600 中未形成开口的区域的周边的形状,由此围绕电流中断装置 600 的周边且不遮挡通孔 610,从而确保通孔 610 起到排气口的作用。

[0101] 同时,图 12 和 13 示出了安装在根据图 5 的对称电池 100 中的凝胶卷 110 的顶部上的板型绝缘构件 800 的实例。参考这些附图以及图 2,板型绝缘构件 800 在其中心处具有穿孔开口 820 以使得可排放气体并连接电极端子,并且具有设置在开口 820 周围的一个或多个通孔 810。板型绝缘构件 800 的通孔 810 可以以与电流中断装置 600 的通孔 610 相对应的形状、尺寸和位置形成,并且具有相对于板型绝缘构件 800 的总面积为 30% 的尺寸。

[0102] 图 14 示意性地示出了其中在圆筒形罐上依次安装有板型绝缘构件 800、电流中断

装置用衬垫 700、电流中断装置 600 和安全排气口 320 的层压体。

[0103] 参考图 14 以及以上所示的图 2 至 6, 形成板型绝缘构件 800 的通孔 810、电流中断装置用衬垫 700 的中心开口 710、电流中断装置 600 的通孔 610 和安全排气口 320 的第一凹口 324, 使得它们在直线上彼此连通, 以便使得可以将电池内部的受压气体有效地排放到外部。

[0104] 图 15 至 17 示意性地示出了其中在电池的内部压力异常增大的情况中排放气体的过程。

[0105] 参考这些附图, 当电池的内部压力异常增大时, 受压气体穿过通孔 630 以及电流中断装置 600 的通孔 610, 并对安全排气口 320 施加向上的压力。

[0106] 这种压力导致安全排气口的向下凹陷部 322 升高, 而同时导致焊接至向下凹陷部 322 的突出部 620 在第一临界压力以上的压力下与电流中断装置 600 分离, 从而中断从电流中断装置 600 到安全排气口 320 的电施加, 如图 16 中所示。

[0107] 尽管具有这种电流阻断, 但是如果压力继续增大并随后达到第二临界压力以上, 则安全排气口 320 的第一凹口 324 发生破裂, 并且同时, 通过气体出口 312 将内部受压气体排放到外部, 如图 17 中所示。

[0108] 与其中将具有小直径的第二凹口切开的情况相比, 根据本发明的其中将具有大直径的第一凹口 324 切开的情况显示更大的气体排放量, 从而诱导高压气体的快速排放。

[0109] 下面将参考以下实施例更详细地描述本发明。提供这些实施例仅用于说明本发明, 且不应被理解为限制本发明的主旨和范围。

[0110] [实施例 1]

[0111] 以下列方式制造圆筒形二次电池。

[0112] 1-1 正极和负极的制造

[0113] 将包含 LiCoO_2 的正极活性材料涂布到铝集电器上, 并且将正极抽头 (tap) 附着至集电器的端部, 使得正极抽头向上突出, 从而制造正极。将包含人造石墨的负极活性材料涂布到铜集电器上, 并且将负极抽头附着至负极集电器的端部, 使得负极抽头向上突出, 从而制造负极。

[0114] 1-2. 电解质的制备

[0115] 通过基于电解质的总重量, 将 11 重量% 的作为添加剂的氟代碳酸亚乙酯添加至非水溶剂中来制备用于锂二次电池的电解质, 所述非水溶剂包含 1.0M LiPF_6 作为电解质并由 30/5/65 (体积比) 的碳酸亚乙酯 (EC) / 碳酸亚丙酯 (PC) / 碳酸二甲酯 (DMC) 组成。

[0116] 1-3 使用中心销卷绕层压体

[0117] 在 1-1 部分中获得的其中隔膜夹在正极和负极之间的片型层压体中插入中心销, 然后卷绕至 $350\text{g}/\text{cm}^2$ 的应力以制造凝胶卷。

[0118] 将由此制造的凝胶卷置于连接至引线的圆筒形壳体中, 并且将 1-2 部分中制备的电解质插入到其中。

[0119] 1-4 圆筒形二次电池的制造

[0120] 在具有 16mm 的外径和 0.3mm 的厚度的铝板材料上的具有 30 度角度的圆弧之外的区域中, 形成了具有 8.0mm 的直径和 0.06mm 的厚度的第一凹口, 如图 7 中所示, 并且制造了具有凹陷部的安全排气口, 所述凹陷部向下突出至 0.65mm 的深度。

[0121] 此外,在具有 11mm 的外径和 0.5mm 的厚度的铝板材料上径向地形成三个通孔,使得通孔的总面积相对于板材料的总面积为约 30%,如图 9 中所示。然后,在其中心处形成具有 1.53mm 的直径和 0.20mm 的高度的突出部,在距离中心突出部的中心为 1.5mm 的距离处打出具有 0.6mm 的宽度和 2.61mm 的周长的三个通孔,并且在用于连接通孔的各桥处形成具有约 70 μ m 的厚度的凹口以制造电流中断装置。

[0122] 此外,在具有 12mm 的外径和 0.5mm 的厚度的聚丙烯板材料上制造了不遮挡电流中断装置的通孔的电流中断装置用衬垫,如图 11 中所示。

[0123] 将电流中断装置的外围插入到电流中断装置用衬垫中,并且通过激光焊接将安全排气口的弯曲的底部附着至电流中断装置的突出部的顶部。

[0124] 将板型绝缘构件安装在 1-3 的圆筒形壳体的开口顶部上,通过卷边来固定容器的顶部,将具有电流中断装置以及其中插入了 PTC 元件和顶盖的衬垫的安全排气口插入到卷边部分中,并且通过向内压制容器顶部而夹紧衬垫以制造电池。

[0125] [实施例 2]

[0126] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制造安全排气口的过程中第一凹口的厚度为 0.1mm,并且第二凹口的厚度为 0.06mm。

[0127] [实施例 3]

[0128] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制造安全排气口的过程中第一凹口的厚度为 7mm,并且第二凹口的厚度为 0.06mm。

[0129] [实施例 4]

[0130] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制备电解质的过程中,基于电解质的总重量,氟代碳酸亚乙酯的含量为 20 重量%。

[0131] [实施例 5]

[0132] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制备电解质的过程中,基于电解质的总重量,氟代碳酸亚乙酯的含量为 30 重量%。

[0133] [实施例 6]

[0134] 以与实施例 4 中相同的方式制造电池,不同之处在于,在电解质的制备中使用由体积比为 8 : 92 的碳酸亚丙酯 (PC) 和丙酸甲酯 (MP) 构成的非水溶剂。

[0135] [实施例 7]

[0136] 以与实施例 4 中相同的方式制造电池,不同之处在于,在制备电解质的制备中使用由体积比为 8 : 92 的碳酸亚丙酯 (PC) 和丙酸甲酯 (MP) 构成的非水溶剂。

[0137] [比较例 1]

[0138] 以与实施例 1 中相同的方式制造袋型单电池,不同之处在于,在电解质的制备中不添加添加剂。

[0139] [比较例 2]

[0140] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制备电解质的过程中,基于电解质的总重量,氟代碳酸亚乙酯的含量为 5 重量%。

[0141] [比较例 3]

[0142] 以与实施例 1 中相同的方式制造电池,不同之处在于,相对于电流中断装置的总面积,电流中断装置的总面积为 18%。

[0143] [实验例 1]

[0144] 制备了在实施例 1、4 和 5 以及比较例 1 和 2 中制造的电池，并且测量了其正极界面电阻。将结果示于图 18 中。

[0145] 参考图 18，使用电化学阻抗谱 (EIS) 设备测量单位频率的电阻，并且将所得特性转化为半圆并示于图中。可以看出，与其中未添加添加剂的电池（比较例 1）和包含 5 重量%的添加剂的电池（比较例 2）相比，其中将氟代碳酸亚乙酯添加至电解质的实施例 1 的电池显示电阻的相对增加。特别地，可以看出，包含 20 重量%的添加剂的电池（实施例 4）和包含 30 重量%的添加剂的电池（实施例 5）显示正极界面电阻的显著增加。

[0146] [实验例 2]

[0147] 制备两个实施例 1、4 至 7 以及比较例 1 和 2 中制造的各电池，在充满电的电池中诱发外部短路并测量其电路电流。将由此获得的结果示于表 1 中。

[0148] <表 1> (单位：电流 (A))

[0149]

	1	2	平均值
实施例 1	46.8	50	48.4
实施例 4	44.5	45.8	45.2
实施例 5	40.9	43.2	42.0
实施例 6	45.3	45.6	45.4
实施例 7	43.9	42.1	43.0
比较例 1	51.2	53.6	52.4
比较例 2	51	52	51.5

[0150] 从上表 1 中可以看出，与其中未添加添加剂的比较例 1 的电池和其中将 5 重量%的添加剂添加至电解质的比较例 2 的电池相比，其中将氟代碳酸亚乙酯添加至电解质的实施例 1 的电池显示在短路时的电流低。

[0151] 此外，与包含 11 重量%添加剂的实施例 1 的电池相比，包含 20 重量%添加剂的实施例 4 的电池和包含 30 重量%添加剂的实施例 5 的电池显示相当低的短路电流。在具有与实施例 4 和 5 不同的非水电解质组合物的实施例 6 和 7 中类似地显示了这种特性。

[0152] 因此，考虑到短路时产生的热与电流量成比例地增加的事实，根据本发明的电池因产生的热量少而相对稳定。

[0153] [实验例 3]

[0154] 在热稳定性方面，将实施例 1 和 4 至 7 中制造的电池与比较例 1 和 2 中的电池进行比较。将由此获得的结果示于下表 2 中。

[0155] <表 2>

[0156]

	开始温度 (°C)	热容量(J/g)			
		1峰值	2峰值	3峰值	总计
实施例1	206±1	71±4	47±4	-	118±6
实施例4	202±1	55±3	45±5	-	100±6
实施例5	204±1	50±4	40±4	-	90±4
实施例6	203±1	53±2	50±4	-	103±4
实施例7	205±1	48±5	43±5	-	91±6
比较例1	198±1	223±6			223±6
比较例2	204±1	98±11	56±5	49±6	203±6

[0157] 上表 2 中的“开始温度”表示热分解的初始温度,“热容量”表示在热分析时的峰面积。通过比较这些值来评价热稳定性。

[0158] 从实验中可以看出,在不包含氟代碳酸亚乙酯的比较例 1 的电池中,在约 210°C 下诱发一次电解质分解,而在包含氟代碳酸亚乙酯的实施例 1 的电池中,在约 225°C 和 275°C 下诱发两次电解质分解。因此可以看出,抑制了一次产生气体的浓度。

[0159] 此外,从表 2 中可以看出,实施例 1 的电池显示等于比较例 1 的电池的约 50% 的比热容,这表明由产生高热量而导致的安全问题较不严重。特别地,与实施例 1 的电池相比,分别包含 20 重量%和 30 重量%的电解质添加剂的实施例 4 和 5 的电池显示相当低的比热容,这表明实施例 4 和 5 的电池显示优异的稳定性。在使用具有 PC/MP 的组合物的非水电解质的实施例 6 和 7 中显示了这种所产生的热显著减少的现象。

[0160] 同时,与包含 11 重量%电解质添加剂的实施例 1 的电池相比,包含 5 重量%电解质添加剂的比较例 2 的电池显示具有 70% 以上的比热容增加。这意味着,当氟代碳酸亚乙酯以小于 10 重量%的量存在时,尽管添加了其,但是通过添加而获得的效果也相当低。

[0161] [实验例 4]

[0162] 对实施例 1 和 3 及比较例 3 中制造的电池进行针刺试验。将由此获得的结果示于下表 3 中。

[0163] <表 3>

[0164]

	通孔面积比	安全排气口的 切口区域	爆炸	电池容器 的变形
实施例1	30%	第一凹口	X	X
实施例2	30%	第二凹口	X	Δ
实施例3	30%	内部凹口 (增加的直径)	X	Δ
比较例3	18%	第一凹口	O	O

[0165] 从上表 3 中可以看出,当使用具有 30%的通孔面积比的电流中断装置时,电池爆炸和变形都未发生。

[0166] 另一方面,使用具有 18%的通孔面积比的电流中断装置的比较例 3 的电池在针刺试验期间发生爆炸。此外,如实施例 2 和 3 中所示,在其中将第二凹口切开并且气体未从安全排气口平稳地排出的情况中,电池未发生爆炸,但是电池容器发生变形。

[0167] 可以看到,在高容量电池中,在安全排气口的切口部分和电流中断装置的通孔的尺寸不足的情况中,气体不能充分地通过这些元件。此外,可以看到,在其中安全排气口的切口部分设置在电池内部的第二凹口的情况中,尽管使凹口的直径变宽,但是通过在切开之后安全排气口倾斜的内部凹口可能不利地产生对气体排放的阻力。

[0168] 如从上文中显而易见的,电流中断装置的通孔的面积增加,并且使用安全排气口的切口部分作为第一凹口的结构产生电池稳定性提高的效果,以便有效地从电池中排出气体。

[0169] 尽管为了进行说明性目的公开了本发明的优选实施方式,但是本领域的技术人员应理解,在不背离所附权利要求书中公开的本发明的范围和主旨的情况下,可以进行各种修改、增加和替代。

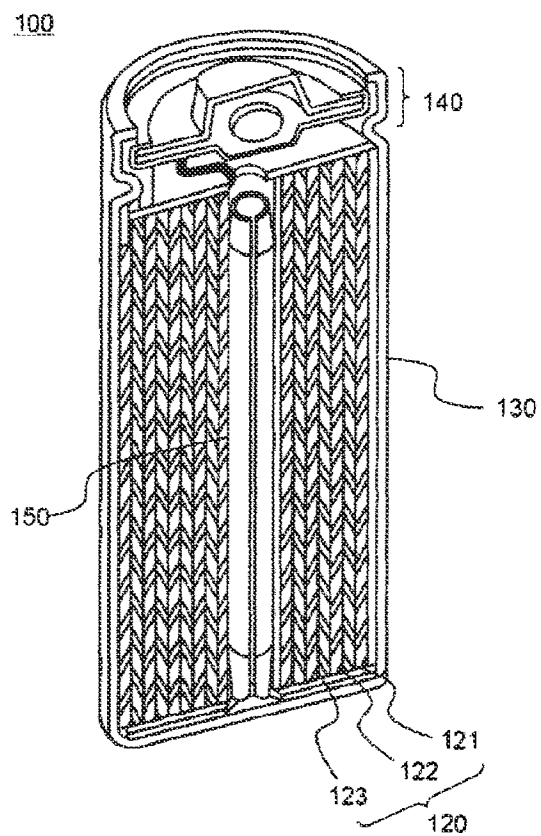


图 1

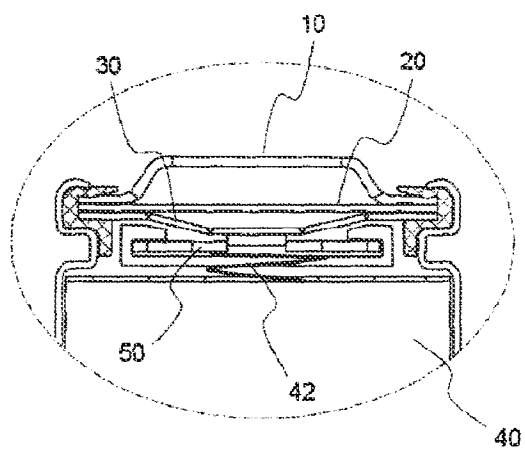


图 2

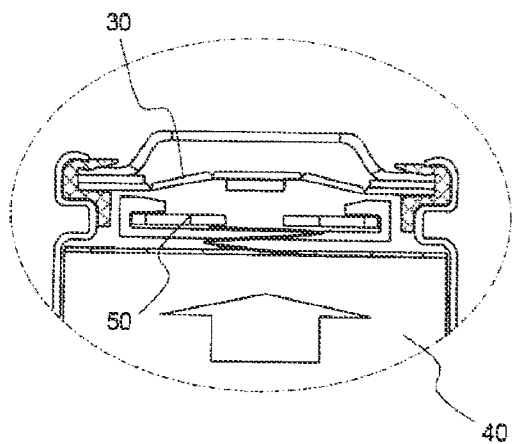


图 3

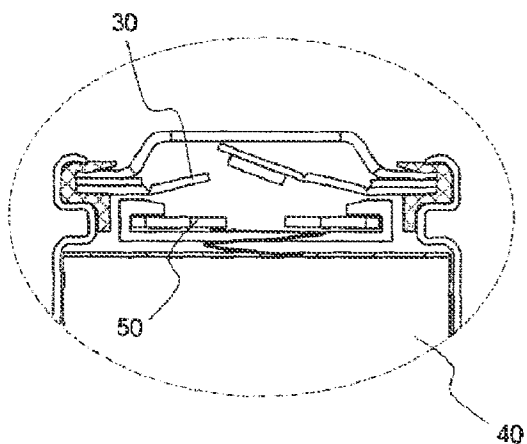


图 4

100

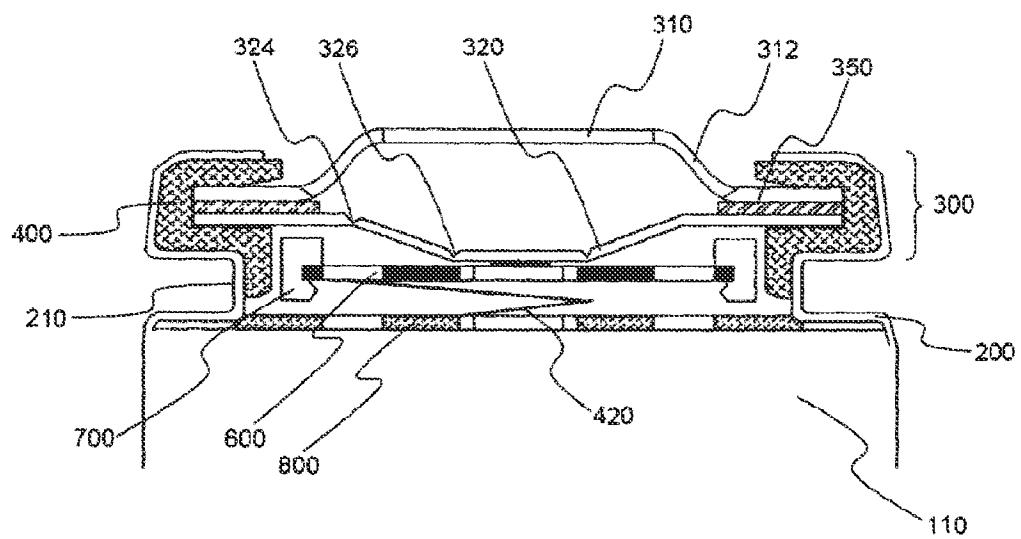


图 5

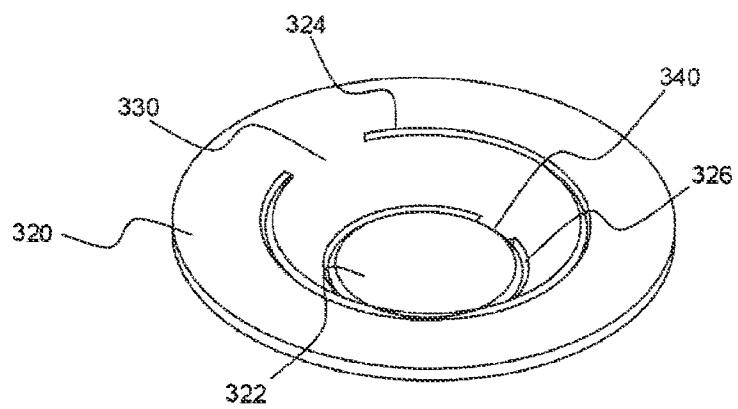


图 6

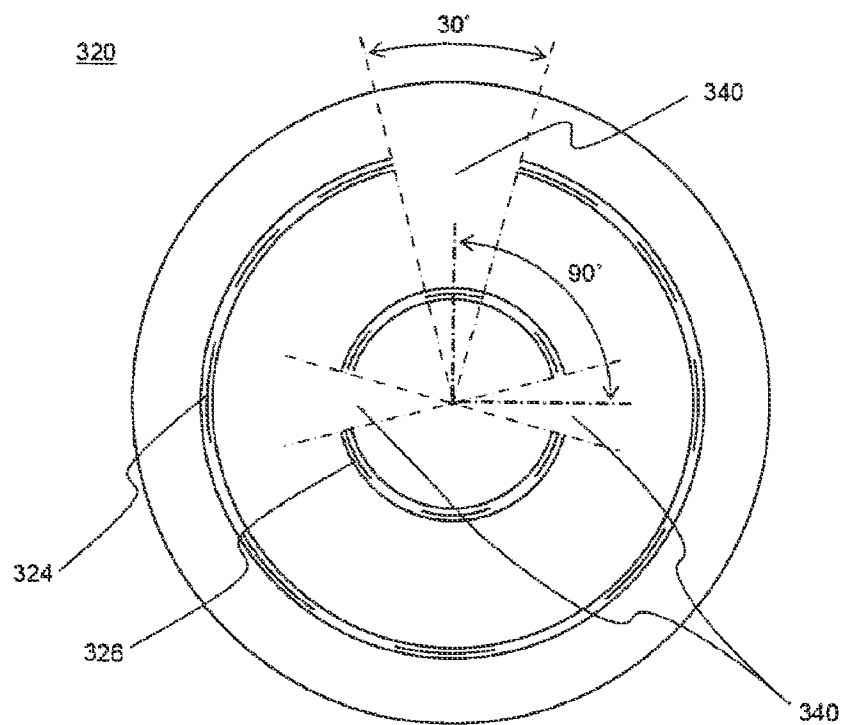


图 7

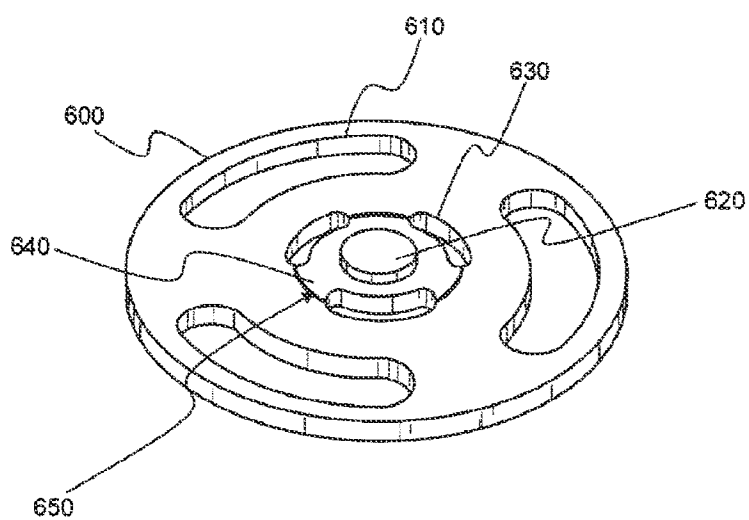


图 8

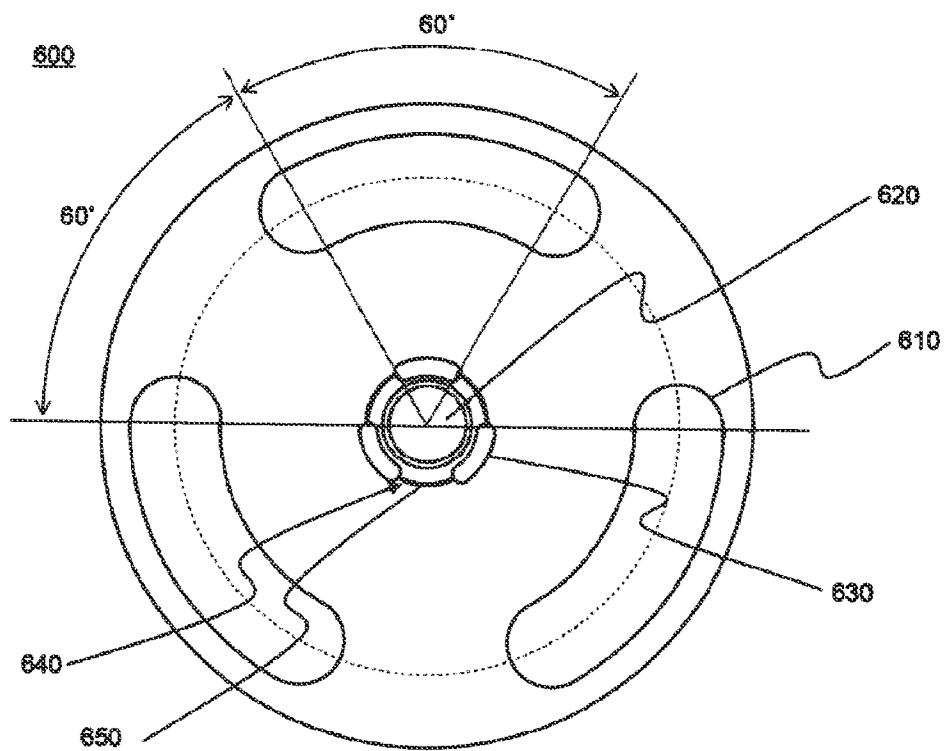


图 9

700

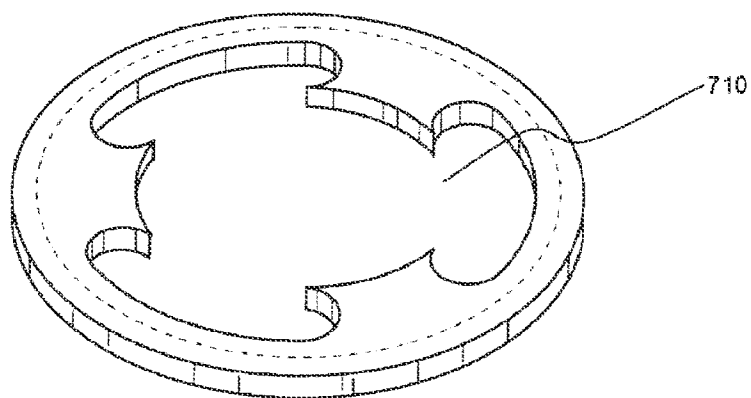


图 10

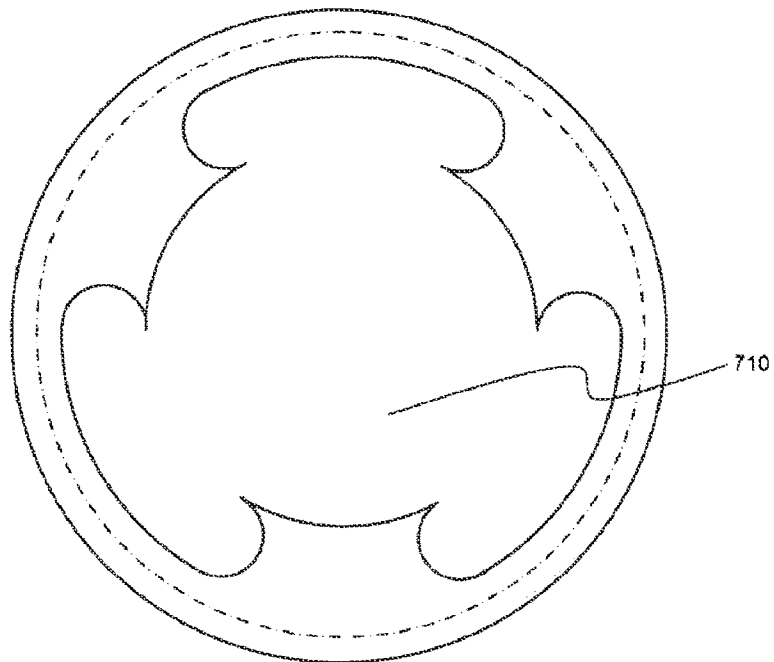
700

图 11

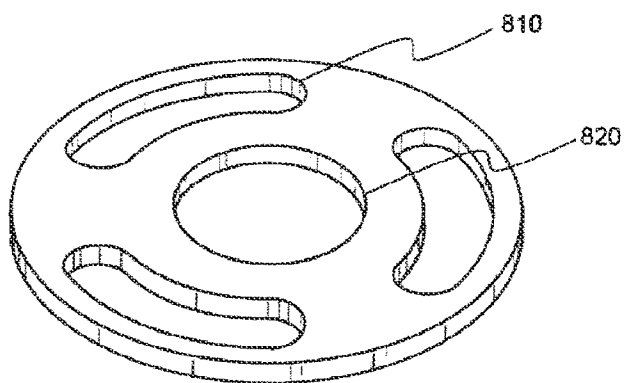
800

图 12

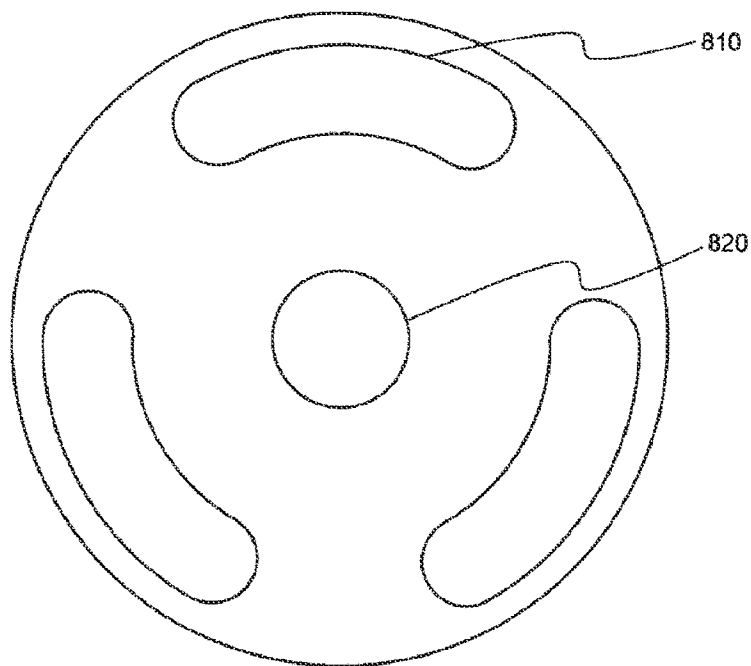
800

图 13

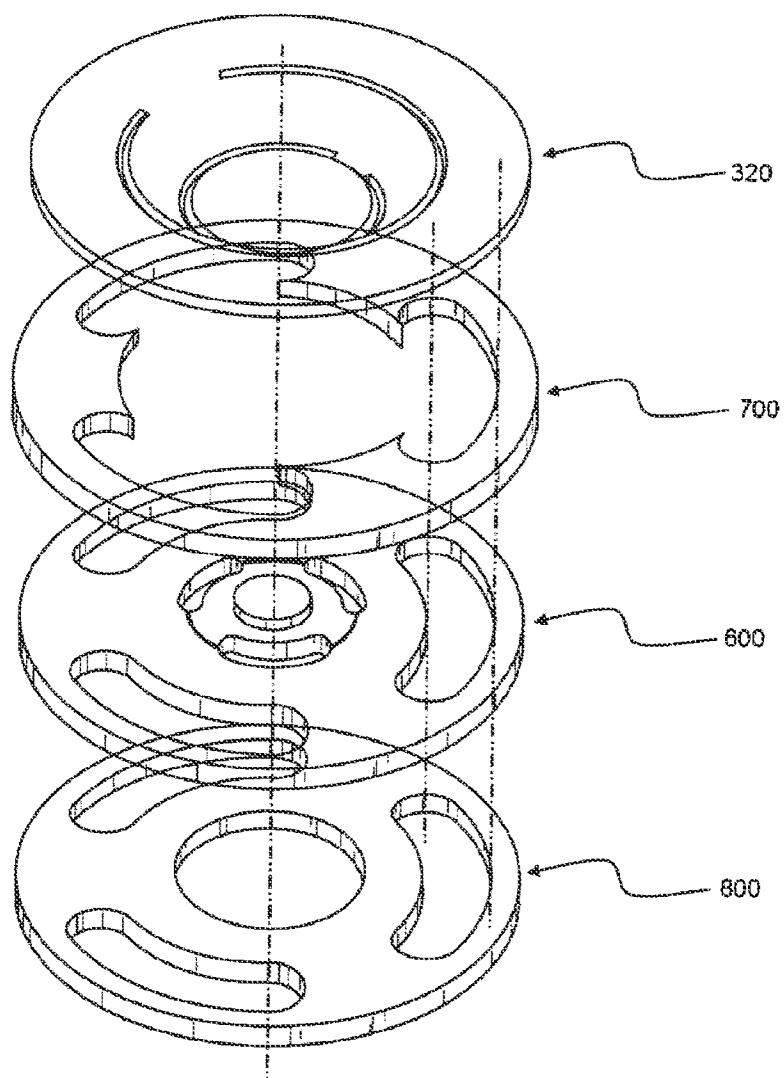


图 14

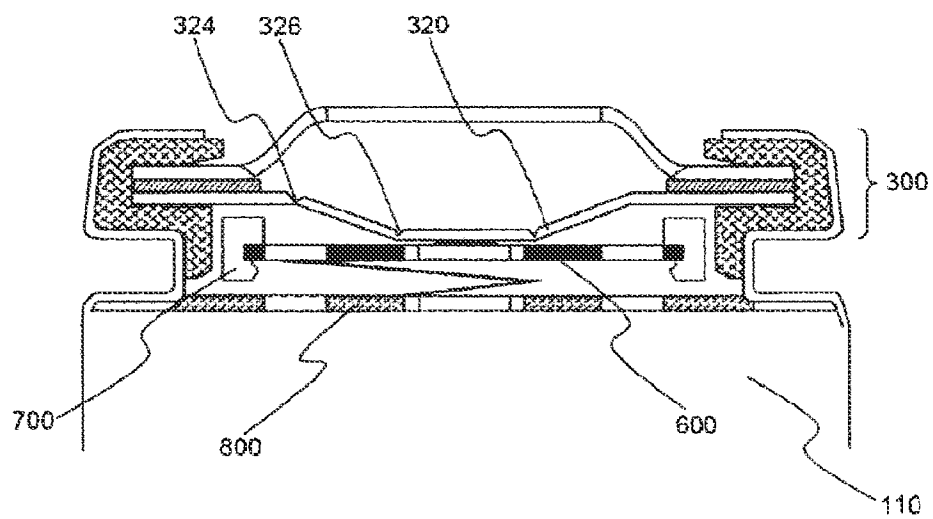


图 15

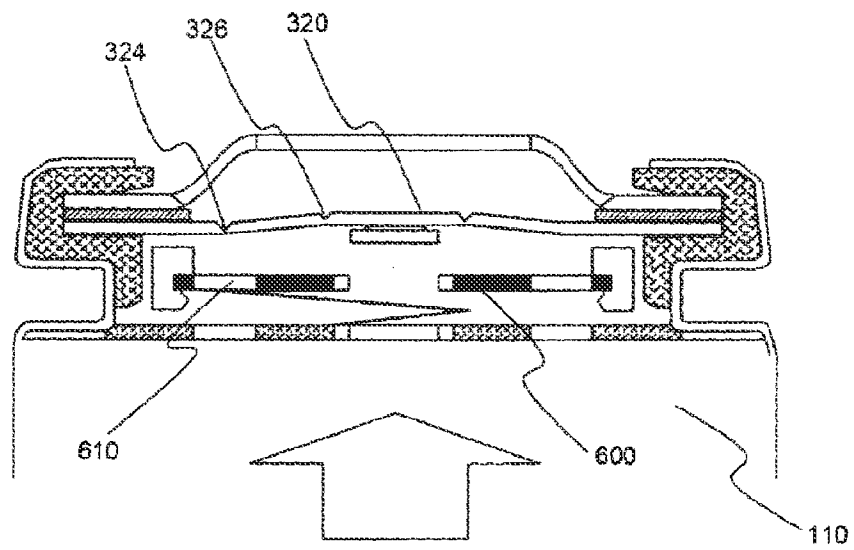


图 16

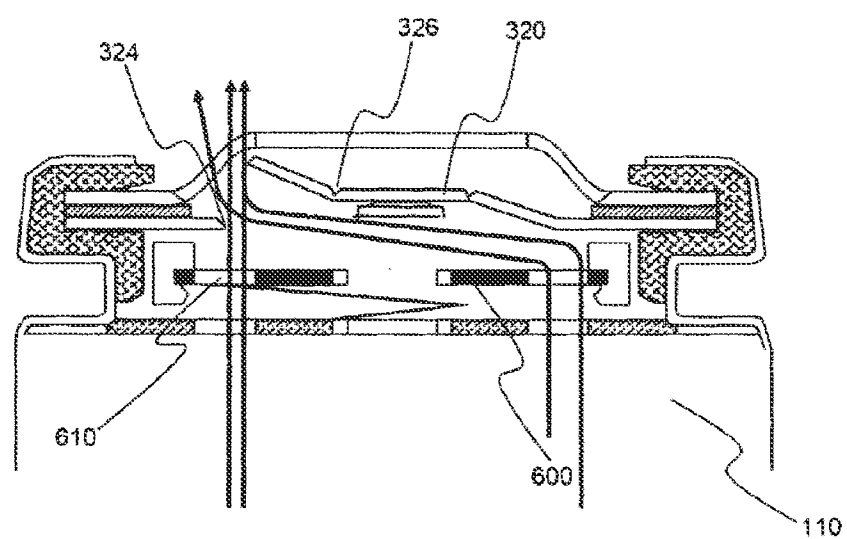


图 17

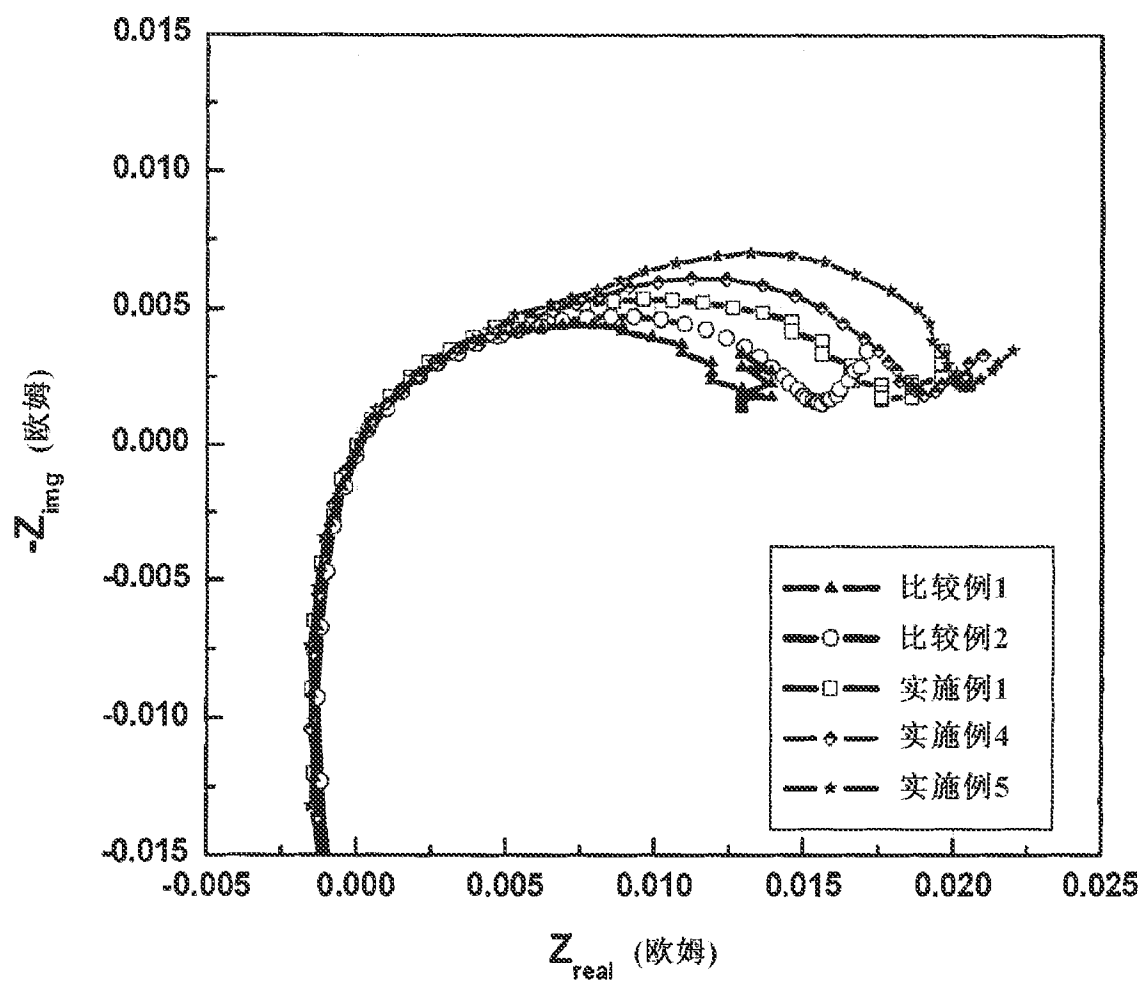


图 18



Espacenet

Bibliographic data: CN103155225 (A) — 2013-06-12

Cap assembly having novel structure and cylindrical battery containing same

Inventor(s): LEE DONG-SUB; KIM DONG-MYUNG; CHANG JAE-DONG; MOON JUN-HO; JUNG SANG-SOK ± (LEE DONG-SUB, ; KIM DONG-MYUNG, ; CHANG JAE-DONG, ; MOON JUN-HO, ; JUNG SANG-SOK)

Applicant(s): LG CHEMICAL LTD ± (LG CHEM. LTD)

Classification: - **international:** H01M50/528
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Abstract of CN103155225 (A)

The present invention provides a cap assembly mounted on an open upper end of a can in a jelly-roll type electrode assembly having a cathode/separator/anode structure, wherein an electrode lead of the jelly-roll is electrically connected to the lower end of a safety vent having notches which are broken to discharge high pressure gas in case high pressure is generated inside of a battery, and a safety member for intercepting current at a predetermined temperature is placed between the safety vent and the electrode lead.



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(71) 申请人 株式会社 LG 化学

地址 韩国首尔

(72) 发明人 李东燮 金东明 张在栋 文俊皓

郑湘锡

(74) 专利代理机构 中原信达知识产权代理有限
责任公司 11219

代理人 高伟 陆弋

权利要求书1页 说明书6页 附图3页

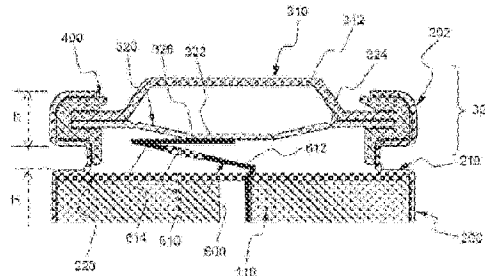
(54) 发明名称

新型结构的帽组件与利用该帽组件的圆筒形
电池

(57) 摘要

本发明提供了一种帽组件,该帽组件安装在具有阴极/隔板/阳极结构的果冻卷式电极组件中的容器的开放上端上,其中,果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端,在电池的內部产生高压的情况下,该切口破裂以排出高压气体,并且用于在预定的温度下中断电流的安全元件被设置在安全排放孔与电极引线之间。

100a



1. 一种布置在电池的圆筒形容器的开放的上端处的帽组件, 在所述电池中, 阴极 / 隔板 / 阳极结构的电极组件 (果冻卷状物) 被安装在所述圆筒形容器中, 其中

所述果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端, 并且用于在预定的温度下中断电流的安全元件设置在所述安全排放孔与所述电极引线之间, 其中, 当在所述电池中产生高压时, 所述切口破裂以排出高压气体。

2. 根据权利要求 1 所述的帽组件, 其中, 用于防止所述果冻卷状物与所述安全元件之间接触的绝缘板被安装在所述果冻卷状物的顶部。

3. 根据权利要求 1 所述的帽组件, 其中, 所述安全元件包括热熔丝。

4. 根据权利要求 1 所述的帽组件, 其中, 所述预定的温度是 80℃ 至 110℃。

5. 根据权利要求 1 所述的帽组件, 其中, 所述预定的温度是 90℃ 至 100℃。

6. 根据权利要求 1 所述的帽组件, 其中, 所述电极引线被从所述电极引线的联接有所述安全元件的区域的前端向外弯曲, 使得所述电极引线从所述果冻卷状物的中央被向外引导, 并且然后所述电极引线被从所述电极引线的联接有所述安全元件的所述区域的后端向内弯曲。

7. 根据权利要求 6 所述的帽组件, 其中, 所述安全元件形成在所述电极引线的上端的一部分处。

8. 根据权利要求 1 所述的帽组件, 其中, 所述安全排放孔由具有 0.15mm 至 0.4mm 的厚度的铝片形成。

9. 根据权利要求 1 所述的帽组件, 其中, 在所述安全排放孔的中央处设有被构造成向下凹入的凹陷, 第一切口和第二切口分别形成在限定所述凹陷的上弯曲区域和下弯曲区域处, 所述第一切口形成闭合曲线, 形成在所述第一切口的下方的所述第二切口被构造成一侧开放的开放曲线的形状, 并且所述第二切口形成成为比所述第一切口深。

10. 根据权利要求 1 所述的帽组件, 其中, 所述帽组件被构造成具有如下结构, 在该结构中, 顶帽和所述安全排放孔被堆叠, 至少一个气体排出口形成在所述顶帽处, 并且垫圈被安装在所述帽组件的外周处。

11. 一种包括根据权利要求 1 至 10 中的任一项所述的帽组件的圆筒形电池。

12. 根据权利要求 11 所述的圆筒形电池, 其中, 所述电池包括锂二次电池。

新型结构的帽组件与利用该帽组件的圆筒形电池

技术领域

[0001] 本发明涉及一种新型结构的帽组件,并且,尤其涉及一种布置在电池的圆筒形容器的开放的上端处的帽组件,在该电池中,阴极/隔板/阳极结构的电极组件(果冻卷状物)被安装在圆筒形容器中,其中,果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端,当在电池中产生高压时,该切口破裂以排出高压气体,并且用于在预定的温度下中断电流的安全元件设置在安全排放孔与电极引线之间。

背景技术

[0002] 随着移动装置已经日益改进并且对该移动装置的需求已经增加,对作为用于移动装置的能源的二次电池的需求也已经大大增加。在该二次电池之中的是具有高能量密度和高放电电压的锂二次电池,在该锂二次电池中已经进行了许多研究并且现在该锂二次电池被商业化且广泛地使用。

[0003] 取决于电池壳体的形状,二次电池可以被划分为具有安装在圆筒形金属容器中的电极组件的圆筒形电池、具有安装在棱柱形金属容器中的电极组件的棱柱形电池或具有安装在由铝层压片形成的袋形壳体中的电极组件的袋形电池。圆筒形电池的优点在于,圆筒形电池具有相对大的容量并且在结构上是稳定的。

[0004] 另外,安装在电池壳体中的电极组件充当具有阴极/隔板/阳极堆结构的功率产生元件,该功率产生元件能够被充电和放电。电极组件可以被划分为:果冻卷式电极组件,该果冻卷式电极组件被构造成具有如下结构,被涂敷活性材料的长片式阴极和长片式阳极被卷绕,而隔板被布置在阴极与阳极之间;或堆叠式电极组件,该堆叠式电极组件被构造成具有如下结构,具有预定尺寸的多个阴极和阳极被依次堆叠,而隔板被分别布置在阴极与阳极之间。果冻卷式电极组件的优点在于,果冻卷式电极组件易于制造并且具有高的能量密度每单位质量。

[0005] 图1是典型地示出一般圆筒形电池的竖直截面透视图。

[0006] 参考图1,通过如下过程制造圆筒形电池100:将果冻卷式(卷绕式)电极组件120插入圆筒形壳体130中,将电解质注入圆筒形壳体130中,并且将具有电极端子(未示出)例如阴极端子的顶帽140联接到圆筒形壳体130的开放的上端。

[0007] 果冻卷式电极组件120被构造成具有如下结构,在该结构中,在隔板123被布置在阴极121与阳极122之间的状态下,阴极121和阳极122被卷绕成圆形。圆筒形中心销150被布置在卷的中央,即,电极组件120的中央。中心销150由金属材料制成以呈现预定的强度。中心销150被构造成具有通过卷起金属片形成的中空圆筒形结构。中心销150用于固定并且支撑电极组件120。此外,当对二次电池充电和放电时或当操作二次电池时,中心销150充当通过二次电池的内部反应产生的排出的气体的通道。

[0008] 同时,锂二次电池的缺点在于,锂二次电池具有低安全性。例如,当电池被过度充电至约4.5V或以上时,阴极活性材料被分解,在阳极发生锂金属的树枝状生长,并且电解质被分解。这时,从电池产生热,结果是,上文提及的分解和若干子分解迅速地进行,并且最

后,电池可能着火并且爆炸。

[0009] 为了解决上文提及的问题,因此,一般的圆筒形二次电池包括电流中断装置(CID),该电流中断装置被安装在限定在果冻卷式电极组件与顶帽之间的空间中以当二次电池发生故障时中断电流并且释放内部压力。

[0010] 图2至图4中示出由该CID执行的一系列操作。

[0011] 参考该附图,顶帽10突出以形成阴极端子。顶帽10具有排气口。在顶帽10的下方依次布置正温度系数(PTC)元件20,该正温度系数元件20用于当电池的内部温度升高时通过电池电阻的显著增加来中断电流;安全排放孔30,该安全排放孔30被构造成在正常状态下具有向下凹陷的形状以及用于当电池的内部压力增加时突出并且破裂以用于排出气体;以及电流中断装置50,该电流中断装置50在其上端的一侧处被联接到安全排放孔30并且在其下端的一侧处被连接到电极组件40的阴极。另外,电流中断装置50的外周被电流中断装置垫圈52包围以固定电流中断装置50。

[0012] 因此,在正常操作条件下,电极组件40的阴极经由电极引线42、电流中断装置50、安全排放孔30以及PTC元件20连接到顶帽10以实现电气传导。

[0013] 然而,当由于各种原因诸如过度充电从电极组件40产生气体时,电池的内部压力增加,并且安全排放孔30的形状被反向,如图3中所示。即,安全排放孔30向上突出。这时,安全排放孔30从电流中断装置50分离以中断电流。因此,防止过度充电进一步进行,从而实现安全性。然而,当电池的内部压力继续增加时,安全排放孔30破裂,如图4中所示,结果是,加压气体通过顶帽10的排气口经由破裂的安全排放孔30被排出,从而防止电池爆炸。

[0014] 同时,除上述的过度充电特性之外,圆筒形二次电池必须呈现高温存储特性。

[0015] 高温存储特性需要电池的如下独特特性,在该电池中,当电池被维持在预定的温度下(例如,75℃)持续长时间段时,电池的电压不会下降。具体地,甚至在测试之后高温存储特性需要电池在正常电压和电流下的操作。

[0016] 当电池被充电到比正常电压高的水平时,过度充电特性需要CID用于有效地中断电压和电流的正常运行,从而防止电池在由于电池的内部压力和温度增加使电池爆炸之前爆炸。

[0017] 在8至14kgf/cm²的短路压力范围内调节注入的电解质的量的方法主要被用于满足高温存储特性和过度充电特性。

[0018] 如果将少量电解质注入圆筒形电池容器中,则尽管高温存储特性变得有利,但过度充电特性变得不利。即,如果电解质的量是小的,则电池单元的内部空间增加。因此,如果在电池单元被过度充电的状态下在电池单元中产生气体,则在用于使CID破裂的充足的气体产生之前,电池单元的内部温度增加,结果使电池单元爆炸。

[0019] 另一方面,如果将大量的电解质注入圆筒形电池容器中,则尽管过度充电特性变得有利,但高温存储特性变得不利。即,如果电解质的量是大的,则电池单元的内部空间减小。因此,如果在电池单元在高温下的状态下在电池单元中产生气体,则CID破裂,结果电池单元不能使用。

[0020] 因此,非常需要新型结构的帽组件以及包括诸如帽组件的圆筒形电池,该帽组件能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。

发明内容

[0021] 技术问题

[0022] 因此,已经完成本发明来解决以上问题以及尚未解决的其它的技术问题。

[0023] 为了解决如上文所述的问题,作为各种广泛和深入研究的结果,本申请的发明人已经发现,当用于在预定的温度下中断电流的安全元件被联接在帽组件的安全排放孔与电极引线之间时,能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。基于该发现已经完成了本发明。

[0024] 技术方案

[0025] 根据本发明的一个方面,以上的和其它的目的可以通过提供一种布置在电池的圆筒形容器的开放的上端处的帽组件来实现,在该电池中,阴极/隔板/阳极结构的电极组件(果冻卷状物)被安装在圆筒形容器中,其中,果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端,当在电池中产生高压时,该切口破裂以排出高压气体,并且用于在预定的温度下中断电流的安全元件设置在安全排放孔与电极引线之间。

[0026] 即,在根据本发明的帽组件中,帽组件的安全排放孔和果冻卷状物的电极引线经由用于在预定的温度下中断电流的安全元件被电连接到彼此。因此,能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。

[0027] 具体地,在根据本发明的帽组件中,安全元件用于在预定的温度下中断电流,并且因此,虽然电池单元的内部压力稍微增加,但是电流不中断,而不像常规帽组件那样,在常规帽组件中,当电解质的量是大的时,电池单元的内部空间减小,结果当产生气体时使电流中断装置(CID)破裂。因此,根据本发明的帽组件有助于高温存储特性。

[0028] 另外,甚至在电解质的量较小的情况下,当电池单元的内部温度由于过度充电而连续地增加时,电流被安全元件中断,并且因此,电池单元的温度在爆炸之前达到预定的温度。因此,能够抑制温度的进一步上升,从而有效地防止电池单元的爆炸。

[0029] 此外,根据本发明的帽组件不需要电流中断装置以及用于电流中断装置的垫圈,并且因此,能够降低制造成本。另外,充分的是,注入适当的量的电解质到电池单元被正常操作的程度,并且因此,能够提高注入效率。

[0030] 此外,安全元件位于在安全排放孔与电极引线之间使得安全元件被布置成与果冻卷状物相邻。因此,安全元件能够较迅速地反应以改变果冻卷状物的温度以在预定的温度下中断电流。

[0031] 优选地,用于防止在果冻卷状物与安全元件之间的接触的绝缘板被安装在果冻卷状物的顶部以防止由于在安全元件与果冻卷状物之间的接触而发生短路。

[0032] 在优选的示例中,安全元件可以包括热熔丝。

[0033] 当电池被热冲击、过度充电或电压下降时,热熔丝中断电流,从而提高电池的安全性。另外,热熔丝容易地解决基于过程的问题,诸如短路压力的增加和减小。

[0034] 预定的温度可以是 80℃至 110℃,优选是 80℃至 100℃。如果温度太低,则在高温存储条件下可能使电流中断。另一方面,如果温度太高,则在过度充电条件下可能使电流不中断。

[0035] 电极引线可以被从电极引线的安全元件联接到的区域的前端向外弯曲,使得电极

引线被从果冻卷状物的中央向外引导；并且然后电极引线被从电极引线的安全元件联接到的区域的后端向内弯曲。因此，防止安全元件弯曲和损坏。

[0036] 在以上结构中，安全元件可以被联接至电极引线的上端的一部分。例如，安全元件可以通过焊接被一体地联接至电极引线的上端的一部分。

[0037] 安全排放孔是一种安全装置，当由于电池的异常操作或组成电池的部件的劣化而使电池的内部压力增加时，该安全装置排出来自电池的气体，从而确保电池的安全。例如，当在电池中产生气体并且电池的内部压力超过临界值时，安全排放孔破裂，并且因此，气体穿过破裂的安全排放孔并随后被从电池通过形成在顶帽处的一个或多个气体排出口排出。

[0038] 安全排放孔的材料不被特别地限制。优选地，安全排放孔由具有 0.15mm 至 0.4mm 的厚度的铝片形成，使得当电池的内部压力超过临界值时安全排放孔破裂。

[0039] 在优选的示例中，安全排放孔在其中央处可以设有被构造成向下凹入的凹陷，第一切口和第二切口可以分别形成在限定凹陷的上弯曲区域和下弯曲区域处，第一切口可以形成闭合曲线，在第一切口的下方形成的第二切口可以被构造成一侧开放的开放曲线的形状，并且第二切口可以形成为比第一切口深。

[0040] 由于第二切口形成为比第一切口深，所以当其压力超过临界压力的加压气体被施加到安全排放孔时，第二切口被切断。另一方面，当其压力小于临界压力的加压气体被施加到安全排放孔时，第一切口与第二切口协作以使凹陷向上升高。

[0041] 在优选的示例中，帽组件可以被构造成具有如下结构，在该结构中，顶帽和安全排放孔被堆叠，至少一个气体排出口形成在该顶帽处，并且垫圈可以被安装在帽组件的外周。

[0042] 因此，电极引线的顶部通过焊接被联接至安全排放孔，安全元件被联接至该电极引线，并且因此，能够通过连续的过程制造帽组件。

[0043] 另外，与包括如图 2 中所示的顶帽、安全排放孔、PTC 元件、CID 和电流中断装置垫圈的常规帽组件相比，被构造成具有上文的结构帽组件不需要正温度系数 (PTC) 元件、电流中断装置 (CID) 和电流中断装置垫圈。

[0044] 由于根据本发明的帽组件不需要如上文所述的电流中断装置和电流中断装置垫圈，所以能够降低帽组件的制造成本。另外，由于根据本发明的帽组件不需要 PTC 元件，所以能够减小与 PTC 元件的高度成正比的压褶部分 (crimping part) 的高度，结果卷边部分 (beaded part) 的高度增加，从而增加果冻卷状物的电池容量。

[0045] 根据本发明的另一个方面，提供了一种包括帽组件的圆筒形电池。

[0046] 根据本发明的圆筒形二次电池通过如下过程制造：在果冻卷式电极组件被安装在圆筒形金属容器中的状态下，将帽组件联接至圆筒形金属容器的开放的上端，将电极组件的阳极焊接到容器的下端，并且在电极组件和电解质被包含在容器中的状态下，将电极组件的阴极焊接到联接至容器的上端的帽组件以密闭地密封电池。

[0047] 根据本发明的电池可以是具有高能量密度、高放电电压以及高输出稳定性的锂二次电池。以下将详细描述根据本发明的锂二次电池的其它部件。

[0048] 一般地，锂二次电池包括阴极、阳极、隔板以及含有锂盐的非水电解质。

[0049] 例如通过将阴极活性材料、导电材料以及粘合剂的混合物涂敷到阴极集电器并且对所涂敷的混合物进行干燥可以制造阴极。根据需要，可以进一步添加填料。通过将阳极材料涂敷到阳极集电器并且对所涂敷的阳极材料进行干燥可以制造阳极。根据需要，可以

进一步添加上文提及的成分。

[0050] 隔板被布置在阳极与阴极之间。隔板可以由呈现高离子渗透性和高机械强度的绝缘薄膜形成。

[0051] 含有锂盐的非水电解质由非水电解质和锂盐组成。液体非水电解质、固体非水电解质或无机固体非水电解质可以被用作非水电解质。

[0052] 在本发明所属的领域中,集电器、电极活性材料、导电材料、粘合剂、填料、隔板、电解质以及锂盐是公知的,因此,将省略其详细描述。

[0053] 可以使用本发明所属的领域中公知的一般方法来制造根据本发明的锂二次电池。即,通过将多孔隔板布置在阴极与阳极之间并且将电解质注入多孔隔板中可以制造锂二次电池。

[0054] 例如,通过将由先前描述的锂过渡金属氧化物活性材料、导电材料以及粘合剂组成的浆料涂敷到集电器并且对浆料进行干燥可以制造阴极。以相同的方式,例如,通过将由先前描述的碳活性材料、导电材料以及粘合剂组成的浆料涂敷到薄的集电器并且对浆料进行干燥可以制造阳极。

[0055] 有利的效果

[0056] 如从上文的描述明显的是,在根据本发明的帽组件中,用于在在预定的温度下中断电流的安全元件被连接在安全排放孔与电极引线之间。因此,能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。

附图说明

[0057] 根据结合附图进行的下列详细描述,将更清楚地理解本发明的上述和其它目的、特征以及其它优点,其中:

[0058] 图 1 是示出一般圆筒形电池的竖直截面透视图;

[0059] 图 2 至图 4 是示出通过电流中断装置(CID)的操作来中断电流并且对来自圆筒形电池的高压气体进行排放的一系列过程的竖直截面图;

[0060] 图 5 是典型地示出根据本发明的实施例的圆筒形电池的局部截面图;

[0061] 图 6 是示出在图 5 的圆筒形电池中使用的安全排放孔的透视图;并且

[0062] 图 7 是典型地示出图 5 的安全元件的截面图。

具体实施方式

[0063] 现在,将参考附图详细地描述本发明的优选实施例。然而,应注意,本发明的范围并不限于示出的实施例。

[0064] 图 5 是典型地示出根据本发明的实施例的圆筒形电池的局部截面图,并且图 6 是典型地示出在图 5 的圆筒形电池中使用的安全排放孔的透视图。

[0065] 参考这些附图,根据本发明的实施例的圆筒形电池 100a 通过如下过程制造:将果冻卷状物 110 插入容器 200 中,将电解质注入容器 200 中并且将帽组件 300 安装到容器 200 的开放的上端。

[0066] 帽组件 300 被构造具有如下结构,在该结构中,顶帽 310 和用于减小电池的内部压力的安全排放孔 320 在相互紧密接触的同时被布置在气密的垫圈 400 中,该气密的垫圈

400 被安装在容器的压褶部分 202 的内侧并且在容器 200 的卷边部分 210 的上部分的内侧。顶帽 310 的中央向上突出以充当用于与外部电路连接的阴极端子。多个通孔 312 围绕顶帽 310 的突起形成,使得压缩气体通过通孔 312 被从容器 200 排出。

[0067] 另外,帽组件 300 被构造成具有如下结构,在该结构中,顶帽 310 和安全排放孔 320 被堆叠,充当气体排出口的通孔 312 形成在顶帽 310 处。垫圈 400 被安装在帽组件 300 的外周处。

[0068] 因此,与图 2 的帽组件相比,图 5 的帽组件 300 不需要正温度系数(PTC)元件 200。因此,与 PTC 元件 200 的高度成正比例的压褶部分 202 的高度减小,结果卷边部分 210 的高度 H 增加,从而增加果冻卷状物 110 的电池容量。

[0069] 安全排放孔 320 是传输电流的薄膜结构。使安全排放孔 320 的中央凹入以形成凹入的中央部分 322。具有不同的深度的两个切口 324 和 326 分别形成在中央部分 322 的上弯曲区域和下弯曲区域处。

[0070] 另外,果冻卷状物 110 的电极引线 600 被连接到安全排放孔 320 的下端。用于在 80℃至 110℃的温度下中断电流的安全元件 610 被联接到电极引线 600。

[0071] 由于安全元件 610 被布置成与果冻卷状物 100 相邻,所以安全元件 610 能够较迅速地反应以改变果冻卷状物 100 的温度来中断电流。

[0072] 用于防止在果冻卷状物 110 与安全元件 610 之间接触的绝缘板 220 被安装在果冻卷状物 110 的顶部,从而防止由于在果冻卷状物 110 与安全元件 610 之间接触而发生短路。

[0073] 另外,电极引线 600 被从电极引线 600 的联接有安全元件 610 的区域的前端向外弯曲,使得电极引线 600 被从果冻卷状物的中央向外引导;并且然后电极引线 600 被从电极引线 600 的联接有安全元件 610 的区域的后端向内弯曲。

[0074] 同时,形成在安全排放孔 320 的上部分处的第一切口 324 形成闭合曲线,并且形成在安全排放孔 320 的下部分处的第二切口 326 被构造成其一侧是开放的开放曲线的形状。另外,第二切口 326 形成为比第一切口 324 深。因此,第二切口 326 的联接力小于第一切口 324 的联接力。

[0075] 因此,当容器 200 的内部压力超过临界压力时,安全排放孔 320 的第二切口 326 破裂,结果加压气体被从容器 200 通过顶帽 310 的通孔 312 排出。

[0076] 图 7 是典型地示出图 5 的安全元件的截面图。

[0077] 参考图 7,作为安全元件的热熔丝 610 被构造成使得热熔丝 610 的中间区域 A 形成薄片形状。因此,例如,当电池的内部温度是 80℃或以下时,电流不被中断,从而满足高温存储特性。另一方面,当电池单元的内部温度超过 80℃时,热熔丝 610 的相对薄的中间区域 A 破裂以使从果冻卷状物 110 流到安全排放孔 320 的电流中断,从而防止电池由于过度充电而爆炸。

[0078] 除了上文提及的结构之外,可以使用其它不同的结构来在预定的温度下中断电流。

[0079] 虽然出于说明性目的已经公开了本发明的优选实施例,但是本领域的技术人员应理解,在不脱离如在所附权利要求中公开的本发明的范围和精神的情况下,能够进行各种变形、添加和替换。

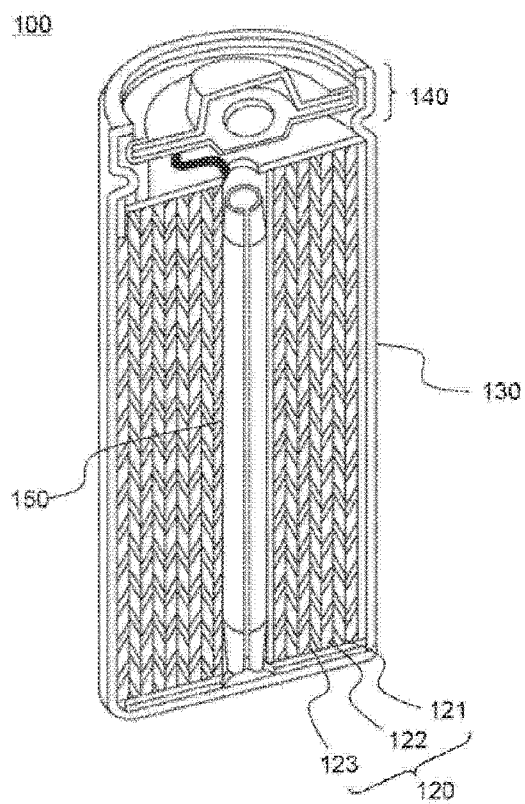


图 1

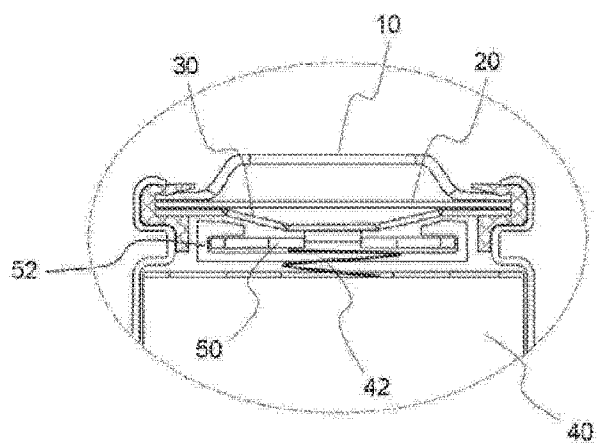


图 2

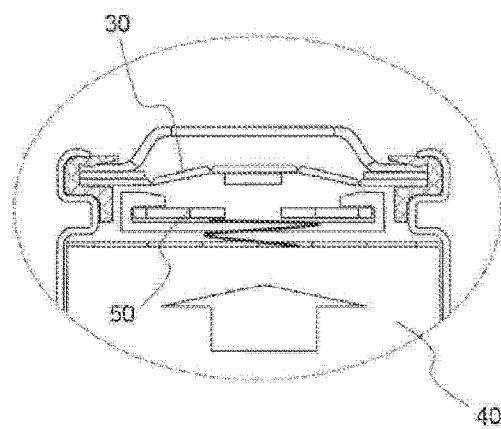


图 3

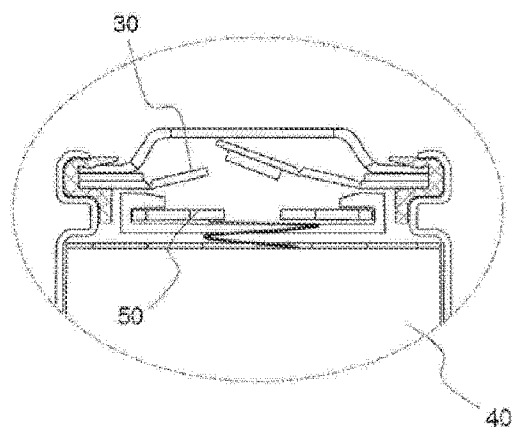


图 4

100a

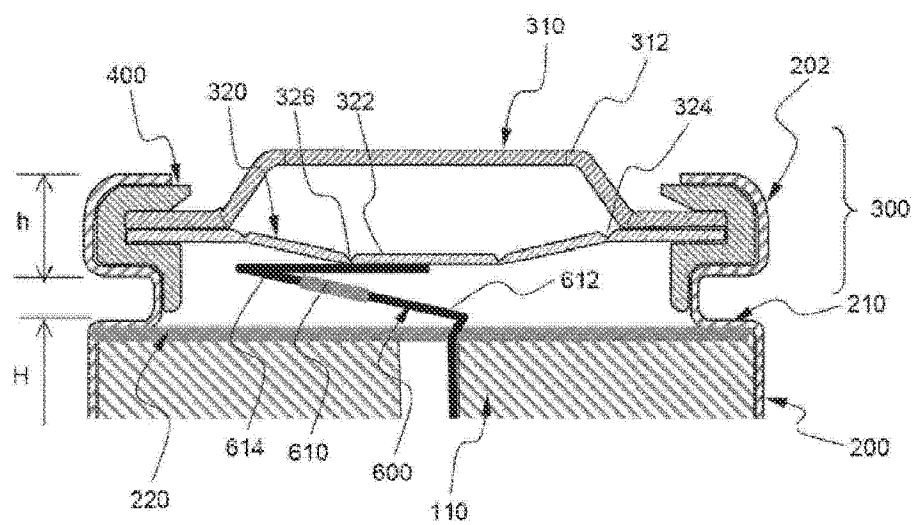


图 5

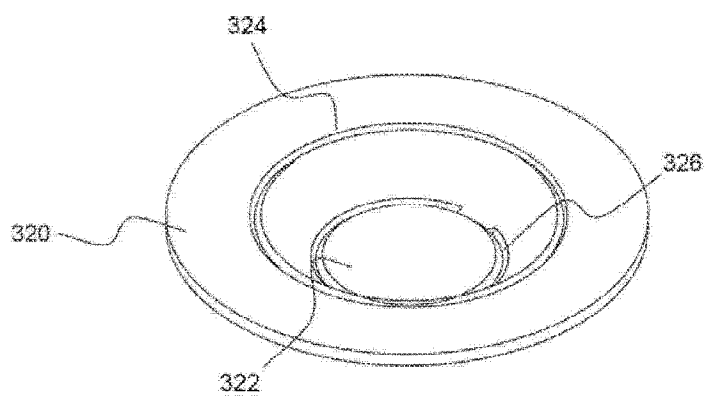


图 6

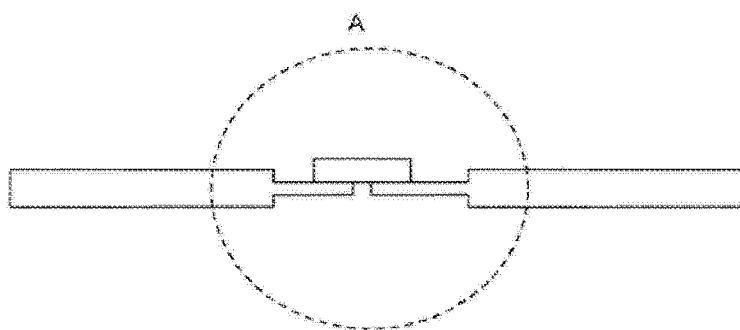


图 7



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检索报告

申请号: 2019800462228	申请日: 20191113	首次检索			
申请人: 株式会社 LG 新能源	最早的优先权日: 20181128				
权利要求项数: 12	说明书段数: 100+11				
审查员确定的 IPC 分类号: H01M50/103,H01M50/107,H01M50/15,H01M50/152,H01M50/342,H01M50/578					
检索记录信息: CN102473883A: 获自本申请同族 EP19889598.9 检索报告对比文件的中文同族					
WO2018/030836A1: 获自本申请同族 JP2021-500123 检索报告					
JP 特开平 6-215746A: 获自本申请同族 KR10-2018-0149369 检索报告					
JP 特开平 8-339793A: 获自本申请同族 PCT/KR2019/015396 检索报告					
CN103155225A: 112 CNTXT, (/ic H01M50/103 or H01M50/107 or H01M50/15 or H01M50/152 or H01M50/342 or H01M50/578 or H01M2/0? or H01M2/2? or H01M2/3?) and ((盖 or 顶 or 罩) s (排气 or 泄压 or 卸压) s ((断 or 裂) 3d (口 or 切 or 凹))); 语义排序,语义基准:2019800462228					
CA2572931A1: 429 VEN, (/cpc H01M50/103 or H01M50/107 or H01M50/15 or H01M50/152 or H01M50/342 or H01M50/578) and (notch??? or ruptur???); 语义排序,语义基准:2019800462228					
相 关 专 利 文 献					
类型	国别以及代码[11] 给出的文献号	代码[43]或[45] 给出的日期	IPC 分类号	相关的段落 和 / 或图号	涉及的权 利要求
Y	CN102473883A	20120523	H01M2/12	说明书第 [0069]-[0109]]段, 图 1-18	1-12
Y	WO2018/030836A1	20180215	H01M2/04	说明书第 [0051]-[0065]]段, 图 1-9	1-12
A	JP 特开平 6-215746A	19940805	H01M2/12	全文	1-12
A	JP 特开平 8-339793A	19961224	H01M2/12	全文	1-12



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A	CN103155225A	20130612	H01M2/34	全文	1-12
A	CA2572931A1	20080621	H01M2/02	全文	1-12



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相 关 非 专 利 文 献					
类型	书名(包括版本号和卷号)	出版日期	作者姓名和出版者名称	相关页数	涉及的权利要求
类型	期刊或文摘名称 (包括卷号和期号)	发行日期	作者姓名和文章标题	相关页数	涉及的权利要求

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 - R: 任何单位或个人在申请日向专利局提交的、属于同样的发明创造的专利或专利申请文件。
 - P: 中间文件, 其公开日在申请的申请日与所要求的优先权日之间的文件;
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审 查 员: 续绍聘
2022 年 06 月 22 日

审查部门: 专利审查协作河南中心电学发明审查部

Application Number: 2019800462228	Date of filing: 20210108	Type of Search: 首次检索
Applicant(s): 株式会社 LG 新能源	Earliest Priority Date: 20181128	
Number of Claims:12	Number of Paragraphs of Description: 100+11	
IPC H01M50/103, H01M50/107, H01M50/15, H01M50/152, H01M50/342, H01M50/578	Number Certified	by Examiner:
Patent search record information: CN102473883A: 获自本申请同族 EP19889598.9 检索报告对比文件的中文同族		
W02018/030836A1: 获自本申请同族 JP2021-500123 检索报告		
JP 特开平 6-215746A: 获自本申请同族 KR10-2018-0149369 检索报告		
JP 特开平 8-339793A: 获自本申请同族 PCT/KR2019/015396 检索报告		
CN103155225A: 112 CNTXT, (/ic H01M50/103 or H01M50/107 or H01M50/15 or H01M50/152 or H01M50/342 or H01M50/578 or H01M2/0? or H01M2/2? or H01M2/3?) and ((盖 or 顶 or 罩) s (排气 or 泄压 or 卸压) s ((断 or 裂) 3d (口 or 切 or 凹))); 语义排序, 语义基准:2019800462228		
CA2572931A1: 429 VEN, (/cpc H01M50/103 or H01M50/107 or H01M50/15 or H01M50/152 or H01M50/342 or H01M50/578) and (notch??? or ruptur???); 语义排序, 语义基准: 2019800462228		

Cited Patent Document(s)

Type	Nationality	Document Number	Publication Date	IPC Number	Involved Claims	Relate Page	Type of Document
A	CN	103155225	20130612	H01M2/34	1-12	全文	A
A	CA	2572931	20080621	H01M2/02	1-12	全文	A1
A	JP	特开平 8-339793	19961224	H01M2/12	1-12	全文	A
Y	CN	102473883	20120523	H01M2/12	1-12	说明书 第 [0069] - [0109] 段, 图 1-18	A
A	JP	特开平 6-215746	19940805	H01M2/12	1-12	全文	A
Y	WO	2018/030	20180215	H01M2/	1-12	说明书	A1

		836		04		第 [0051] - [0065] 段, 图 1-9	
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Cited Periodical(s)

Type	Name of Periodical or Abstract	Volume Number	Issue Number	Release Date	Author(s)	Title	Involved Claims	Related Page
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Cited Book

Type	Title	Volume Number	Version	Publication Date	Author(s)	Title	Involved Claims	Reference Page
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017
86765	7590	02/08/2023	EXAMINER	
LG Lerner, David, Littenberg, Krumholz & Mentlik, LLP 20 Commerce Drive Cranford, NJ 07016			PARK, LISA S	
			ART UNIT	PAPER NUMBER
			1729	
			NOTIFICATION DATE	DELIVERY MODE
			02/08/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOfficeAction@lernerdavid.com

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, or 365(c) is acknowledged. Receipt is acknowledged of papers submitted under 35 U.S.C. 119(a)-(d) or (f), which papers have been placed of record in the file.

Information Disclosure Statement

3. Information disclosure statements (IDS), submitted November 1, 2021, March 4, 2022, and September 14, 2022, have been received and considered by the examiner.

Claim Interpretation

4. All "wherein" clauses are given patentable weight unless otherwise noted. Please see MPEP 2111.04 regarding optional claim language.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

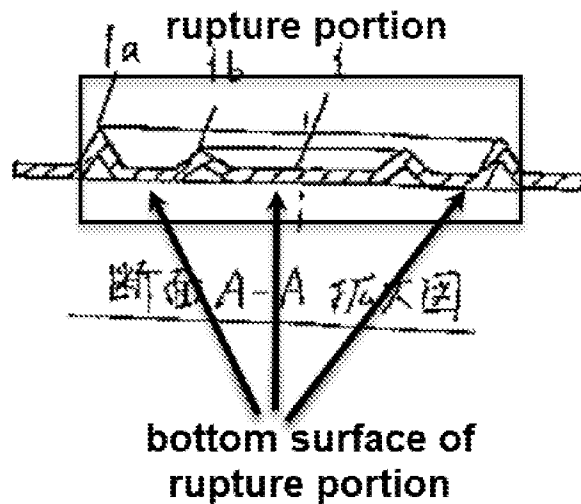
(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

5. Claims 1-3 and 6 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Mitamura JPH06215746.

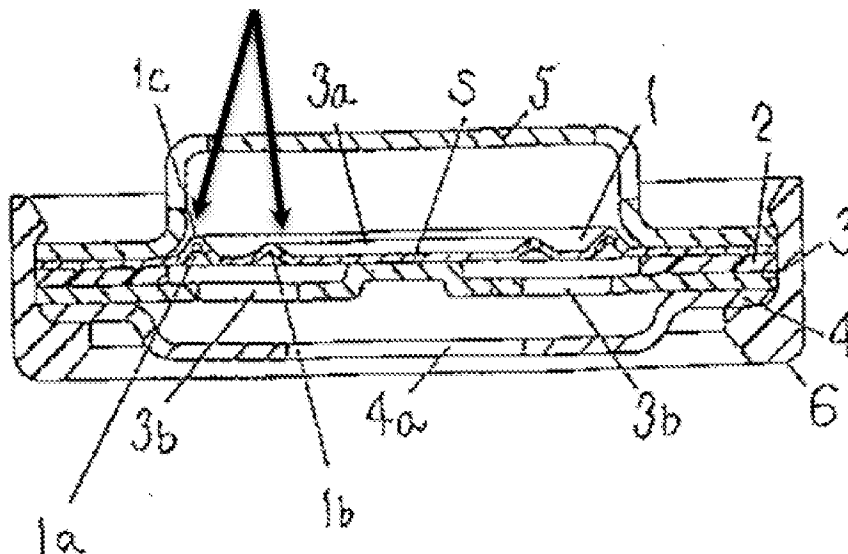
Regarding Claim 1, Mitamura discloses a secondary battery comprising an electrode assembly (electrode plate group), a can member (battery case) **8** configured to accommodate the electrode assembly therein, and a top cap assembly (sealing plate) configured to cover an opening of the can member **8**, wherein the top cap assembly comprises a top cap (external terminal plate) **5** positioned at an uppermost portion and a safety vent part (safety valve) **1** positioned below the top cap **5**, wherein the safety vent part **1** comprises a rupture portion (the circle having **1a** as its boundary) designed to be ruptured when the internal pressure within the can member **8** increases to equal or exceed a predetermined pressure, a rupture notch **1a** that defines a boundary of the rupture portion, and a bending notch portion **1b** which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing an inside of the can member **8**, wherein the bending notch portion **1b** is bent when the internal pressure within the can member **8** increases (see at least Figs 1-4b; paras 0019-0024). The Office notes that Fig. 3 shows the original (pre-venting) state where **1b** is in its original position, Fig. 4a shows a first elevated pressure situation where **1b** bends initially, and Fig. 4b shows a second (higher) elevated pressure situation where the rupture takes place at rupture notch portion **1a**.⁹⁷ Since the outer periphery of the metal safety valve **1** is crimped and fixed by the battery case **8**, (para 0021: “when the battery internal pressure reaches a predetermined value, the central portion is lifted upward and bent as shown in FIG. 4A. The portion **1b** is reversely deformed, and the welded portions are peeled off by this stress, and the connection with the perforated metal terminal plate **3** is broken, thereby interrupting the current”).

Regarding Claims 2 and 3, Mitamura discloses wherein the rupture notch **1a** is formed as a recess within a lower surface of the safety vent part and the bending notch portion **1b** is formed as a recess within the lower surface of the safety vent part and of the rupture portion (meeting **Claim 3**), wherein a

recessed depth of the bending notch part is less than a recessed depth of the rupture notch portion (see annotated figure below, as well as Fig. 2 showing A-A perspective). This is especially true when considering the thinned portion 1c, which increases the recessed depth of 1a even farther.



1a and 1b are clearly designed with different heights (1b being shorter, i.e. having a less recessed depth) as indicated by the parallel lines connecting respective 1a notches and 1b notches.



The portion below of the Fig 2 of Mitamura shows the rupture portion, and it is clear that the rupture notch **1a** is in the bottom surface of the rupture portion.



Regarding Claim 6, Fig. 2 (stack of top cap assembly) shows that the rupture notch portion **1a** and the bending notch portion **1b** have continuous circular shapes.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

6. Claims 4-5, 7-9, and 12 are rejected under 35 U.S.C. 103 as being unpatentable over Mitamura JPH06215746, as applied to Claim 1, and further in view of Kim US PG Publication 2012/0114979 (hereinafter “Kim ‘979”).

Regarding Claims 4 and 5, Mitamura discloses the claimed secondary battery as described in the rejection of Claim 1, which is incorporated herein in its entirety. Mitamura fails to specifically disclose wherein the top cap assembly further comprises a current interrupt device provided with an interrupt portion which is attached to the rupture portion which is configured to be ruptured to interrupt current when the internal pressure within the can member increases and wherein the interrupt portion of the current interrupt portion is attached to the rupture portion inside a circumference of the bending notch portion. However, in the same field of endeavor of cylindrical secondary batteries having safety vents in the top cap, Kim ‘979 discloses a top cap assembly **300** comprising a current interrupt device **600** which is attached to a rupture portion inside a circumference of a bending notch portion **324** and inside a rupture notch portion **326** (or the reverse, see Fig. 17 where **324** is ruptured and is therefore rupture notch portion) and which is configured to be ruptured to interrupt current when the internal pressure within the can member **200** increases, which improves safety, in conjunction with discharging pressurized gas as necessary via the ruptured safety vent (see at least Figs.5-6, 17; paras 0085, 0101-0104). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the battery of Mitamura such that the top cap assembly further comprises a current interrupt device provided with an interrupt portion which is attached to the rupture portion which is configured to be ruptured to interrupt current when the internal pressure within the can member increases and wherein the interrupt portion of the current interrupt portion is attached to the rupture portion inside a circumference of the bending notch portion because Kim ‘979 teaches that this feature provides additional safety via interrupting current.

Regarding Claims 7 and 9, Mitamura fails to specifically disclose wherein the rupture notch portion and the bending notch portion have a discontinuous circular or elliptical shape. However, Kim '979 teaches that a rupture notch portion (second notch) **326** and bending notch portion (first notch) **324** both have an open curve structure to allow for rupture without full separation for either notch portion (Figs 6-7; paras 0089-0090). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the rupture notch portion or the bending notch portion of Mitamura with a discontinuous circular or elliptical shape because Kim '979 teaches that this structure is known for a safety vent notch portion of a safety vent in a cylindrical secondary battery to provide for bending or rupturing without full separation of the parts, and the use of a known technique to improve similar devices (methods or products) in the same way is likely to be obvious. See *KSR International Co. v. Teleflex Inc.*, 550 U.S. ___, ___, 82 USPQ2d 1385, 1395 – 97 (2007) (see MPEP § 2143, C.). Further, the change in form or shape, without any new or unexpected results, is an obvious engineering design. See *In re Dailey*, 149 USPQ 47 (CCPA 1976) (see MPEP § 2144.04).

Regarding Claim 8, Kim '979 teaches that the discontinuous shape of the rupture portion (see e.g. Figs. 7, 17) results in more than one segment of the second notch **326**, and in the case where this notch only bends and does not rupture, then there are multiple bending notch portions (each section of the discontinuous circle is a bending notch portion). IT would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the multiple bending notch portions of Mitamura and Kim '979 having different lengths in order to more finely control the rupture or bending action since the change in form or shape, without any new or unexpected results, is an obvious engineering design. See *In re Dailey*, 149 USPQ 47 (CCPA 1976) (see MPEP § 2144.04). A change in proportion or relative dimension is obvious in the absence of unexpected results. In *Gardner v. TEC Systems, Inc.*, 725 F.2d 1338, 220 USPQ 777 (Fed. Cir. 1984), *cert. denied*, 469 U.S. 830, 225 USPQ 232 (1984), the Federal Circuit held that, where the only difference between the prior art and the claims was a recitation of relative dimensions of the claimed device and a device having the claimed relative

dimensions would not perform differently than the prior art device, the claimed device was not patentably distinct from the prior art device.

Regarding Claim 12, Mitamura modified by Kim ‘979 does not specifically disclose wherein a spaced distance between an edge end of the safety vent part and the rupture notch portion is equal to a spaced distance between the rupture notch portion and the vending notch portion. However, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to space these parts evenly from one another such that the spaced distance between an edge end of the safety vent part and the rupture notch portion is equal to a spaced distance between the rupture notch portion and the vending notch portion of Mitamura modified by Kim ‘979 in order to best fit the notches with the inner cell structure and to ensure even response to pressure changes since a change in proportion or relative dimension is obvious in the absence of unexpected results. In *Gardner v. TEC Systems, Inc.*, 725 F.2d 1338, 220 USPQ 777 (Fed. Cir. 1984), *cert. denied*, 469 U.S. 830, 225 USPQ 232 (1984), the Federal Circuit held that, where the only difference between the prior art and the claims was a recitation of relative dimensions of the claimed device and a device having the claimed relative dimensions would not perform differently than the prior art device, the claimed device was not patentably distinct from the prior art device.

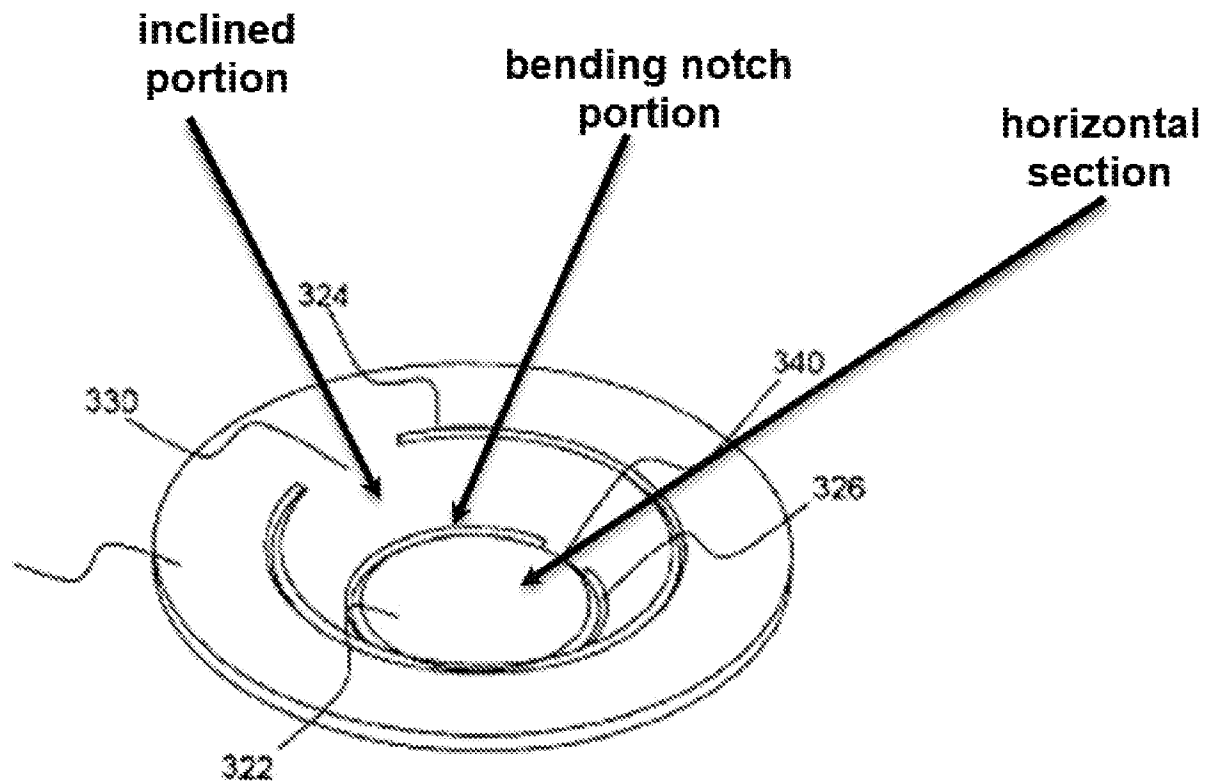
7. Claim 10 is rejected under 35 U.S.C. 103 as being unpatentable over Mitamura JPH06215746, as applied to Claim 1, and further in view of Kim WO2018/030836 (hereinafter “Kim ‘836”).

Regarding Claim 10, Mitamura discloses the claimed secondary battery as described in the rejection of Claim 1, which is incorporated herein in its entirety. Mitamura fails to specifically disclose wherein a connection portion between an edge portion and the rupture portion of the safety vent part is bent upward. However, in the same field of endeavor of cylindrical secondary batteries having safety vents in the top cap, Kim ‘836 teaches that the safety vent includes a connection portion between an edge portion and the

[illegible]

Regarding Claim 11, Mitamura modified by Kim ‘836 discloses the claimed secondary battery as described in the rejection of Claim 10, which is incorporated herein in its entirety. Mitamura modified by

Kim '836 fails to specifically disclose wherein the rupture portion comprises an inclined section inclined so that a height thereof gradually decreases from the connection portion toward the bending notch portion and a horizontal section having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion, wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section. However, previously cited Kim '979 teaches a safety vent having a rupture portion wherein an inclined portion is inclined so that a height thereof gradually decreases from a connection portion toward a bending notch portion **326** (Fig 17 shows an embodiment wherein **326** is a bending notch portion while **324** which is a rupture notch portion) and a horizontal section **322** having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion (see Figs 5-7) wherein the bending notch portion **324** is positioned on a boundary between the inclined portion and the horizontal section.



Kim '979 teaches that this design allows the safety vent to function by providing the lower portion **322** (horizontal section) that is thrust upward when the pressure in the battery rises and allows for current interruption via CID **600** thereby improving safety (see paras 0084-0090, Figs 5-7 and 14). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the battery of Mitamura modified by Kim '836 such that the rupture portion comprises an inclined section inclined so that a height thereof gradually decreases from the connection portion toward the bending notch portion and a horizontal section having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion, wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section because Kim '979 teaches that this design allows the safety vent to function by providing the horizontal section that is thrust upward when the pressure in the battery rises and allows for current interruption via CID. Applying a known technique to a known device (method or product) ready for improvement to yield predictable results is likely to be obvious. See *KSR International Co. v. Teleflex Inc.*, 550 U.S. ___, ___, 82 USPQ2d 1385, 1395 – 97 (2007) (see MPEP § 2143, D.).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached on M-Th 6 or 7 to 3 or 4; Alt Fri afternoons.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

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Primary Examiner, Art Unit 1729

Notice of References Cited	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.	
	Examiner LISA S PARK	Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20120114979-A1	05-2012	KIM; Ji Young	H01M50/3425	429/7
	B					
	C					
	D					
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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
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	O	WO2018/030836	02-2018	WIPO	Kim	
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	Q					
	R					
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CPC - Searched*		
Symbol	Date	Examiner
H01M50/3425 or H01M50/107 or H01M50/578 or H01M50/574 or H01M2200/20 or H01M50/56 or H01M50/152 or Y02E60/10 or H01M50/116 (text-limited)	01/31/2023	/L.S.P./

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Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search history)	01/31/2023	/L.S.P./
Analysis of Global Dossier file	02/02/2023	/L.S.P./
PE2E Applicant/Assignee search	02/02/2023	/L.S.P./
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Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	OPAP
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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First Named Inventor	Geon Woo Min
Art Unit	OPAP
Examiner Name	Not Yet Assigned
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(71) Applicant: **SHIN HEUNG ENERGY & ELECTRONIC CO., LTD.** [KR/KR]; 48, Yangsan-ro, Osan-si, Gyeonggi-do 18100 (KR).

(72) Inventors: **KIM, Gi Rin**; 303-1604, 111, Yeongtong-ro, Yeongtong-gu, Suwon-si, Gyeonggi-do 16687 (KR). **HWANG, Man Yong**; 423-702, 20, Jungang-gongwon-ro, Bundang-gu, Seongnam-si, Gyeonggi-do 13589 (KR). **ABE, Takao**; 48, Yangsan-ro, Osan-si, Gyeonggi-do 18100 (KR). **KIM, Jong Ho**; 316-201, 16, Ogeum-ro, Gunpo-si, Gyeonggi-do 15863 (KR).

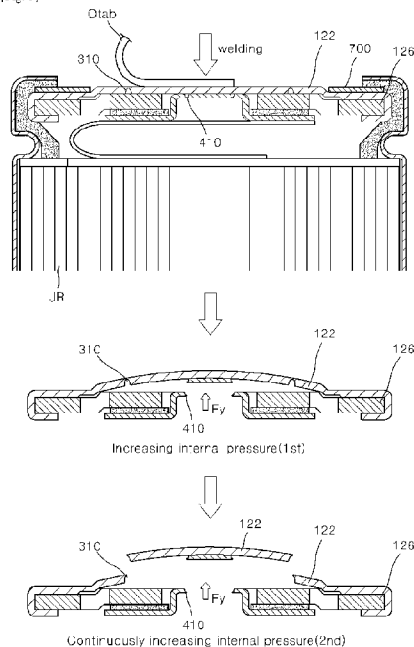
(74) Agent: **Ji, Jeong Hun**; 4F, 40-4, Teheran-ro 14-gil, Gangnam-gu, Seoul 06239 (KR).

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(54) Title: CAP ASSEMBLY FOR SECONDARY BATTERY AND SECONDARY BATTERY INCLUDING THE CAP ASSEMBLY

[Fig. 3]



(57) Abstract: Disclosed are a cap assembly for a secondary battery and a secondary battery including the cap assembly. The cap assembly includes: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion including a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and including a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween. The cap assembly can be used to increase the capacity of the power generating device in the secondary battery compared to when used in other secondary batteries of the same size, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and the safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.

Published:

- *with international search report (Art. 21(3))*
- *before the expiration of the time limit for amending the claims and to be republished in the event of receipt of amendments (Rule 48.2(h))*

Description

Title of Invention: CAP ASSEMBLY FOR SECONDARY BATTERY AND SECONDARY BATTERY INCLUDING THE CAP ASSEMBLY

Technical Field

- [1] The present invention relates to a cap assembly for a secondary battery and a secondary battery including the cap assembly. More specifically, the present invention relates to a cap assembly for a secondary battery that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery, and a secondary battery including the cap assembly.

[2]

Background Art

- [3] Secondary batteries are classified into cylindrical, prismatic, and pouch types depending on the shape of battery cases they employ. The cylindrical batteries include an electrode assembly accommodated in a cylindrical metal can. The prismatic batteries include an electrode assembly accommodated in a prismatic metal can. The pouch type batteries include an electrode assembly accommodated in a pouch type case made of an aluminum laminate sheet.
- [4] Electrode assemblies accommodated in battery cases are power generating devices capable of repeated charge/discharge cycles that have a cathode/separator/anode laminate structure. Electrode assemblies are classified into jelly-roll and stack types by their structure. The jelly-roll type structure is constructed by interposing a separator between an anode and a cathode, each of which is in the form of a long sheet coated with an active material, and winding the laminate in a roll form. The stack type structure is constructed by sequentially stacking a plurality of electrode units, each of which includes a cathode having a predetermined size, an anode having a predetermined size, and a separator interposed therebetween. The jelly-roll type electrode assemblies are most widely used in secondary batteries due to their ease of construction and high energy density per unit weight. The jelly-roll type electrode assemblies are usually employed in cylindrical batteries.
- [5] Secondary batteries employing jelly-roll type electrode assemblies are likely to

deform because the jelly-roll type electrode assemblies undergo repeated swelling and shrinkage during charge/discharge. This deformation may cause internal short circuits. Heat generated by internal short circuits decomposes organic solvents to produce gases. The gases increase the internal gas pressure of batteries to burst the batteries.

- [6] An increase in the internal gas pressure of batteries may take place also when internal short circuits are caused by an external impact.
- [7] Many attempts have been made to solve the safety problems of batteries. For example, known is a cap assembly for a cylindrical battery having a structure in which a safety vent, safety devices, and a top cap are fixed by a gasket. High-pressure gases are released through the safety vent. The safety devices include a PTC device adapted to interrupt the flow of electric current at high temperature and a current interrupt device (CID) adapted to interrupt the flow of electric current when the internal pressure of the battery increases. The top cap forms a protruding terminal adapted to protect the safety devices. Examples of such attempts are disclosed in many patent publications, for example, Japanese Patent Publication Nos. 2006-286561, 2005-100927, and 2002-373711 and Korean Patent Publication Nos. 2012-0103394 and 1999-67165. In each of the cap assemblies disclosed in these publications, the gasket serves to seal the interface between the can and the cap assembly while surrounding the outer circumferences of the safety devices, but the structure of the top cap limits the capacity of the power generating device, permits the release of a limited amount of gases when the safety vent ruptures, causes an increase in contact resistance, and increases in the construction cost of the cap assembly.
- [8] In a cylindrical secondary battery 100 illustrated in Fig. 9, a power generating jelly roll JR is accommodated in a battery can 110 such that a tab is connected to a sub-plate 128. This connection enables power supply through the battery can and a terminal of a cap-up 124. An open end (e) of the battery can is clamped to fix a cap assembly 120. A safety vent 122 of the cap assembly 120 is bent to surround the cap-up 124. Alternatively, the cap assembly may have a structure in which the safety vent is not bent and the cap-up is exposed to the top so as to be in direct contact with a gasket 130. When a module consisting of the secondary battery and one or more other secondary batteries is used in an electric vehicle or an energy storage system (ESS), energy or force generated by electrically shorting of the other secondary batteries or a power grid can melt or structurally weaken the gasket 130 interposed between the clamped battery can 110 and the cap-up 124 or the safety vent 122, and as a result, the battery can 110 comes into contact with the safety vent, causing short circuits. The short circuits increase the risk of leakage of an electrolyte from the secondary battery or safety accidents, such as explosion.

[9]

Disclosure of Invention

Technical Problem

- [10] A first object of the present invention is to provide a cap assembly for a secondary battery that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.
- [11] A second object of the present invention is to provide a secondary battery including a cap assembly that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.
- [12]

Solution to Problem

- [13] A first aspect of the present invention provides a cap assembly for a secondary battery including: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion including a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and including a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween.
- [14] According to one embodiment of the present invention, the safety vent may be bent downwardly and extend to form a bent portion.
- [15] According to a further embodiment of the present invention, the breakable junction may be welded to a protrusion of the sub-plate entering the central hole formed in the support member.
- [16] According to another embodiment of the present invention, the protrusion may

include a portion that protrudes upwardly opposite to the central portion and is opposite and melt-bonded to the breakable junction.

- [17] According to another embodiment of the present invention, the notch may be formed in the shape of a long home on the outer circumference of the portion opposite to the breakable junction.
- [18] According to another embodiment of the present invention, the protrusion may include a portion that protrudes upwardly opposite to the central portion and is opposite to and in close contact with the current-carrying contact.
- [19] According to another embodiment of the present invention, the portion opposite to the current-carrying contact may be formed on the outer circumference of the notch.
- [20] According to another embodiment of the present invention, a breakable notch may be formed at the bottom of the peripheral portion such that the safety vent is bent by a rupture pressure.
- [21] According to another embodiment of the present invention, the cap assembly may further include a bending-relieving portion adjacent to the breakable notch to space the current-carrying contact from the cap-down and forming a space where a rupture pressure is applied to the breakable junction.
- [22] According to another embodiment of the present invention, the bottom width of the bending-relieving portion may be larger than that of the breakable notch.
- [23] According to another embodiment of the present invention, the cap-down may include a through-hole that is spaced apart from and formed around the central hole and through which internal gases and liquids pass.
- [24] According to another embodiment of the present invention, the insulating layer may extend beyond the end of the sub-plate and may reach the end of the cap-down.
- [25] According to another embodiment of the present invention, the insulating layer may have a through-hole opposite to the through-hole of the cap-down.
- [26] According to another embodiment of the present invention, spacer bridges may be formed across the opposite through-hole.
- [27] According to another embodiment of the present invention, a short-circuit preventing layer may be formed at a position opposite to the end of the battery can through a gasket of the safety vent.
- [28] According to another embodiment of the present invention, the short-circuit preventing layer may include a polymeric or organic material that is resistant to heat at a temperature of 300 °C or more.
- [29] According to another embodiment of the present invention, the short-circuit preventing layer may be an oxide or nitride film formed by surface treatment with high energy, such as a laser, after removal of a plating layer.
- [30] According to another embodiment of the present invention, the short-circuit

preventing layer may be made of an anodized ring-shaped aluminum material.

[31] According to another embodiment of the present invention, the aluminum material may be melt-bonded to an adherend by soldering.

[32] According to another embodiment of the present invention, the cap-down and the sub-plate between which the insulating layer is interposed may have rough sides opposite to each other.

[33] A second aspect of the present invention provides a secondary battery including the cap assembly.

[34]

Advantageous Effects of Invention

[35] The cap assembly of the present invention can be used to increase the capacity of a power generating device in a secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.

[36]

Brief Description of Drawings

[37] Fig. 1 is a plan view illustrating a cap assembly of the present invention.

[38] Fig. 2 is a cross-sectional view of the cap assembly of Fig. 1.

[39] Fig. 3 is a cross-sectional view illustrating a state in which a pack-tab (Otab) is welded to a secondary battery fabricated by assembly of a cap assembly according to the present invention and a battery can.

[40] Fig. 4 is a cross-sectional view illustrating states in which a safety vent of a cap assembly according to the present invention is deformed at operating and rupture pressures.

[41] Fig. 5 is a cross-sectional view illustrating a state in which an electrode tab attached to a sub-plate of a cap assembly according to the present invention is deformed in a battery can and is in close contact with a cap-down through an insulating layer interposed therebetween.

[42] Fig. 6 is a detailed plan view of the insulating layer of Fig. 5.

[43] Fig. 7 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together without the need for forging.

[44] Fig. 8 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together

without the need for forging.

- [45] Fig. 9 is a cross-sectional view of a secondary battery mounted with a conventional cap assembly and illustrates a state in which the end of a battery is in contact with a safety vent as a result of melting or damage of a gasket by thermal energy generated in an external circuit or load and a state in which a cap-down is in contact with the safety vent as a result of melt-bonding of an insulating layer.

[46]

Best Mode for Carrying out the Invention

- [47] The present invention will now be described in more detail.

- [48] Technical terms used in this specification are used to merely illustrate specific embodiments, and should be understood that they are not intended to limit the present disclosure.

- [49] As far as not being defined differently, all terms used herein including technical or scientific terms may have the same meaning as those generally understood by an ordinary person skilled in the art to which the present disclosure belongs to, and should not be construed in an excessively comprehensive meaning or an excessively restricted meaning.

- [50] In addition, if a technical term used in the description of the present disclosure is an erroneous term that fails to clearly express the idea of the present disclosure, it should be replaced by a technical term that can be properly understood by the skilled person in the art. In addition, general terms used in the description of the present disclosure should be construed according to definitions in dictionaries or according to its front or rear context, and should not be construed to have an excessively restrained meaning. A singular representation may include a plural representation as far as it represents a definitely different meaning from the context. Terms "include" or "has" used herein should be understood that they are intended to indicate an existence of several components or several steps, disclosed in the specification, and it may also be understood that part of the components or steps may not be included or additional components or steps may further be included. In the description of the present invention, detailed explanations of related art are omitted when it is deemed that they may unnecessarily obscure the essence of the invention. In the accompanying drawings, those components are rendered the same reference numeral that are the same or are in correspondence, regardless of the figure number, and redundant explanations are omitted.

- [51] Fig. 1 is a plan view illustrating a cap assembly of the present invention, Fig. 2 is a cross-sectional view of the cap assembly of Fig. 1, Fig. 3 is a cross-sectional view illustrating a state in which a pack-tab (Otab) is welded to a secondary battery fabricated

by assembly of a cap assembly according to the present invention and a battery can, Fig. 4 is a cross-sectional view illustrating states in which a safety vent of a cap assembly according to the present invention is deformed at operating and rupture pressures, Fig. 5 is a cross-sectional view illustrating a state in which an electrode tab attached to a sub-plate of a cap assembly according to the present invention is deformed in a battery can and is in close contact with a cap-down through an insulating layer interposed therebetween, Fig. 6 is a detailed plan view of the insulating layer of Fig. 5, and Fig. 7 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together without the need for forging.

- [52] A cap assembly 120 for a secondary battery according to the present invention includes: a safety vent 122 made of a circular metal plate and consisting of a central portion 200 corresponding to the central area of the metal plate and a peripheral portion 300 surrounding the central portion, the central portion including a current-carrying contact 210 forming a first conductive path through which an electric current generated from a power generating device JR flows to the outside and a breakable junction 220 forming a second conductive path at the center of the central portion; a cap-down 126 opposite to the safety vent, having a central hole 510 through which the breakable junction is exposed, and including a support member 500 in close contact with the peripheral portion; and a sub-plate 400 joined to the breakable junction and having a notch 410 formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer 600 therebetween.
- [53] The current-carrying contact 210 in the central portion 200 functions to reduce the contact resistance due to the formation of the conductive path and imparts a support force against a pressure applied when a pack-tab is attached to the cap assembly. The breakable junction 220 included in the central portion 200 may be welded to a protrusion 420 of the sub-plate entering the central hole 510 formed in the central portion surrounded by the support member. The welding may be performed by various bonding processes. Preferably, laser welding or melt-bonding is used such that the welded portion is not detached even under high internal pressures (for example, 1st pressure and 2nd pressure) generated upon abnormal operation of the secondary battery.
- [54] That is to say, the protrusion 420 of the sub-plate 400 protrudes upward opposite to the central portion 200. Specifically, the protrusion 420 enters the central hole 510 and is welded to the breakable junction 220. For the prevention of short circuits, the protrusion 420 and the central hole should be spaced apart from each other such that they do not come into contact and in electrical communication with each other.

- [55] Furthermore, the insulating layer 600 has a central hole 620 opposite to the central hole 510 and through which the central portion 200 is exposed and a standing portion 625 in the form of a barrier protruding along the circumference of the central hole 620. Due to this structure, the risk of short circuits between the protrusion 420 of the sub-plate and the central hole 510 during assembly can be completely avoided.
- [56] The support member 500 of the cap-down 126 is opposite to and in close contact with the peripheral portion 300 of the safety vent 122. This arrangement reduces the contact resistance when power is applied from the power generating device JR to the outside, ensuring good current-carrying properties. In addition, the support member 500 serves to prevent the cap assembly from being deformed caused by an excessive load or pressure during welding of a pack-tab (Otab) to a package module in which the battery is connected in series or parallel with one or more other batteries.
- [57] Furthermore, the protrusion 420 is in close contact with and supports the central portion 200 over a large area. This arrangement is advantageous in preventing the cap assembly from being deformed when a pack-tab is welded to the secondary battery. Thus, the protrusion 420 is designed to have a portion 220' opposite and welded to the breakable junction 220 and a portion 210' opposite to the current-carrying contact around the circumference of the portion 220'. The portion 210' is formed in close contact with the current-carrying contact other than the notch 410 in the form of a recessed long home that is broken by high pressures upon abnormal operation of the secondary battery and permits the release of internal gases into the outside. Due to this design, a support force by the portion 210' is ensured and the conductive paths can be maintained, which is beneficial in terms of current-carrying properties.
- [58] The cap-down 126 includes a through-hole 530 spaced apart from but adjacent to the central hole 510 and through which internal gases and liquids are released into the outside. The through-hole 530 is filled with high-pressure gases, liquids, and a flowable electrolyte upon abnormal operation of the secondary battery, and as a result, operating and rupture pressures are applied to the safety vent 122 therethrough.
- [59] Here, the operating pressure refers to a 1_{st} pressure at which the contact and junction between the current-carrying contact 210 of the safety vent and the portion 210' opposite thereto and between the breakable junction 220 of the safety vent and the portion 220' opposite thereto are released to block the application of power to the outside. The rupture pressure refers to a 2_{nd} pressure at which the safety vent 122 ruptures so that the internal gases leak out before explosion of the secondary battery.
- [60] Accordingly, it is necessary for the operating pressure to be smaller than the rupture pressure. The operating pressure is designed to be much larger than the rupture pressure. For example, the operating pressure and the rupture pressure may be adjusted to about 16 kgf/cm² and about 30 kgf/cm², respectively, to clearly distinguish the

operation and rupture of the cap assembly. The structural behaviors of such operation and rupture will be discussed in detail. The internal pressure of the secondary battery increases continuously upon internal abnormal operation of the secondary battery. By the operating pressure, the current-carrying contact 210 is detached from the portion 210' opposite thereto and the breakable junction 220 is detached from the portion 220' opposite thereto. A flowable electrolyte, liquids, and gases passes through the through-hole 530 of the support member to increase the internal pressure and directly lifts the safety vent 122 because the breakable junction is welded to the protrusion 420. A vertical force (F_y) to detach the breakable junction in the upward and downward directions and a horizontal force (F_x) in the horizontal direction are effective in providing a force (F) to break the portion 220' opposite to the breakable junction. By the rupture force (F), the notch 410 is broken to interrupt the flow of electric current in the secondary battery.

[61] That is, the detachment of the current-carrying contact and the breakage of the breakable junction can be accomplished by a combination of the vertical force (F_y) and the horizontal force (F_x).

[62] Even after the notch 410 is broken by the operating pressure to interrupt the flow of electric current, the 2nd pressure of the secondary battery may increase continuously. Since the increasing 2nd pressure increases the danger of explosion, it is necessary to reduce the internal pressure beforehand by the removal of internal gases and liquids (outgasing).

[63] To this end, a breakable notch 310 in the form of a long home is formed in the peripheral portion of the safety vent. The breakable notch 310 is designed to be broken by the internal pressure of the secondary battery. The safety vent is in close contact with the cap-down, making it difficult to bend or break the breakable notch 310. Thus, a bending-relieving portion 320 in the shape of a long home or groove is formed adjacent to the breakable notch 310. Due to its shape, the bending-relieving portion 320 can vary the bending angle (θ) of the breakable notch, enabling efficient breakage of the breakable notch. The bending-relieving portion is designed to have a bottom area (G) larger than that (N) of the breakable notch. This design is advantageous in varying the bending angle (θ) of the breakable notch to freely form a space between the bending-relieving portion and the cap-down opposite thereto, thus being effective in deforming or breaking the breakable notch.

[64] The breakable notch 310 and the bending-relieving portion 320 may be arranged concentrically around the center of the central portion on the bottom of the peripheral portion of the safety vent. The internal pressure increases continuously even after interruption of the flow of electric current by the 1st pressure and reaches the 2nd pressure at which the breakable notch is broken, and as a result, the secondary battery can be

protected from explosion.

- [65] The safety vent 122 may be bent downwardly and extend to form a bent portion 1221 that surrounds the cap-down 126. The formation of the bent portion is beneficial in ensuring adhesion of the safety vent 122 to the cap-down 126. The extension length of the bent portion on the bottom of the cap-down is limited to increase the power generation capacity of the secondary battery.
- [66] The power generating device JR is a device that produces electrical energy through chemical reactions. For example, in the case of a secondary lithium-ion battery, the power generating device JR may be a jelly-roll type electrode assembly including an anode material, a cathode material, an electrolyte, and a center pin. One of the electrodes is in electrical communication with the cap assembly 120 through an electrode tab. Power generated in the electrode assembly is supplied to an external circuit or load through the electrode tab.
- [67] The insulating layer 600 extends beyond the end of the sub-plate 400 and reaches the end of the cap-down. Thus, the insulating layer 600 precludes an extension of the electrode tab other than the portion welded to the sub-plate 400 from being bent up toward the cap assembly 120 and coming into contact with the cap-down 126 to prevent the occurrence of electrical shorting during fabrication of the secondary battery and use in an application.
- [68] The insulating layer 600 is interposed between and insulates the cap-down and the sub-plate and can prevent the electrode tab from being in electrical communication with the cap-down. However, the insulating layer 600 may impede the flow of gases through the through-hole 530, which is solved by the arrangement of a through-hole 610 at a position opposite to the through-hole 530. Spacer bridges 612 extend across the opposite through-hole to space the electrode tab from the cap-down. The spacer bridges can be understood as bar-type bridges traversing the opposite through-hole.
- [69] As explained previously, the central hole 620 of the insulating layer 600 is formed opposite to the central hole 510 such that the central portion 200 is exposed. The sub-plate 400 is insertable into the insulating layer 600 through the central hole 620. The standing portion 625 in the form of a barrier protrudes along the circumference of the central hole 620 to prevent short circuits.
- [70] For better attachment and adhesion of the insulating layer 600 between the cap-down 126 and the sub-plate 400, the cap-down and the sub-plate have long homes 1265 and 4005 in the form of depressions or grooves at sides opposite to each other, respectively. The insulating layer is interposed between the cap-down and the sub-plate, followed by pressing or thermal pressing to form a filling portion 615 in which the insulating layer is filled in the long homes.
- [71] A gasket 130 is disposed between the battery can 110 and the safety vent 122. The

gasket 130 should be protected from being melted or lost by the application of energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery. To this end, a short-circuit preventing layer 700 can be interposed between the safety vent and a gasket 130 at a position opposite to the end of the battery can.

- [72] Any insulating material may be used without particular limitation for the short-circuit preventing layer 700. For example, the short-circuit preventing layer may be made of a metallic, nonferrous metallic or non-metallic material. The surface of the short-circuit preventing layer made of a metallic or nonferrous metallic material is preferably insulated.
- [73] The material for the insulating layer is insulated by wet or dry physical or chemical process in such a manner that the surface chemical composition is changed to increase insulating properties. For example, aluminum is a current-carrying material but its insulating properties increase when an oxide layer is formed on the surface. An oxide film is formed on the surface of a metallic or nonferrous metallic material, for example, by dipping in an acidic or basic solution, electrolysis, anodizing or laser irradiation.
- [74] Particularly, a dry process by laser irradiation may have difficulty in processing the insulating layer or film whose surface is plated. In this case, after removal of the plating layer, high-energy laser beams are irradiated to form an oxide or nitride film on the surface of the insulating layer or film, achieving good insulating properties.
- [75] Alternatively, the short-circuit preventing layer 700 may be formed using a polymeric material. Since the short-circuit preventing layer 700 is required to have a sufficient physical support force in a high-temperature environment where the gasket 130 is melted or structurally weakened by thermal energy or force generated upon external short circuits, the material for the short-circuit preventing layer 700 should be resistant to heat at a temperature of 300 °C or more. However, the material is not particularly limited so long as its physical properties are maintained at a temperature of 300 °C or more.
- [76] The short-circuit preventing layer 700 may include an inorganic material. For example, the short-circuit preventing layer may be formed using an inorganic material or may be an insulating layer whose surface is coated with an inorganic material, for example, a powder of inorganic particles, to form an inorganic layer (not shown).
- [77] The short-circuit preventing layer 700 may take the form of a ring made of an oxidized or anodized metallic or nonferrous metallic material, for example, an aluminum material. The insulating layer acts as a kind of spacer that maintains the interval between the end (e) of the battery can and the safety vent 122 to prevent short circuits. The short-circuit preventing layer 700 may be provided in plurality. In this

case, the number of the short-circuit preventing layers is not limited so long as they are arranged between the end (c) of the battery can and the safety vent 122. It is beneficial from the viewpoint of production processability that the short-circuit preventing layer 700 is in the form of a ring. The short-circuit preventing layer 700 may be fixed to the safety vent 122 by soldering.

- [78] On the other hand, the present invention provides a secondary battery including the cap assembly. The secondary battery of the present invention may be fabricated by assembling an anode material, a cathode material, an electrolyte, a center pin, a jelly roll, etc. together in a battery can and aligning the cap assembly at the open end of the battery can (clamping). The fabrication method of the secondary battery is generally known in the art and detailed description thereof is thus omitted.

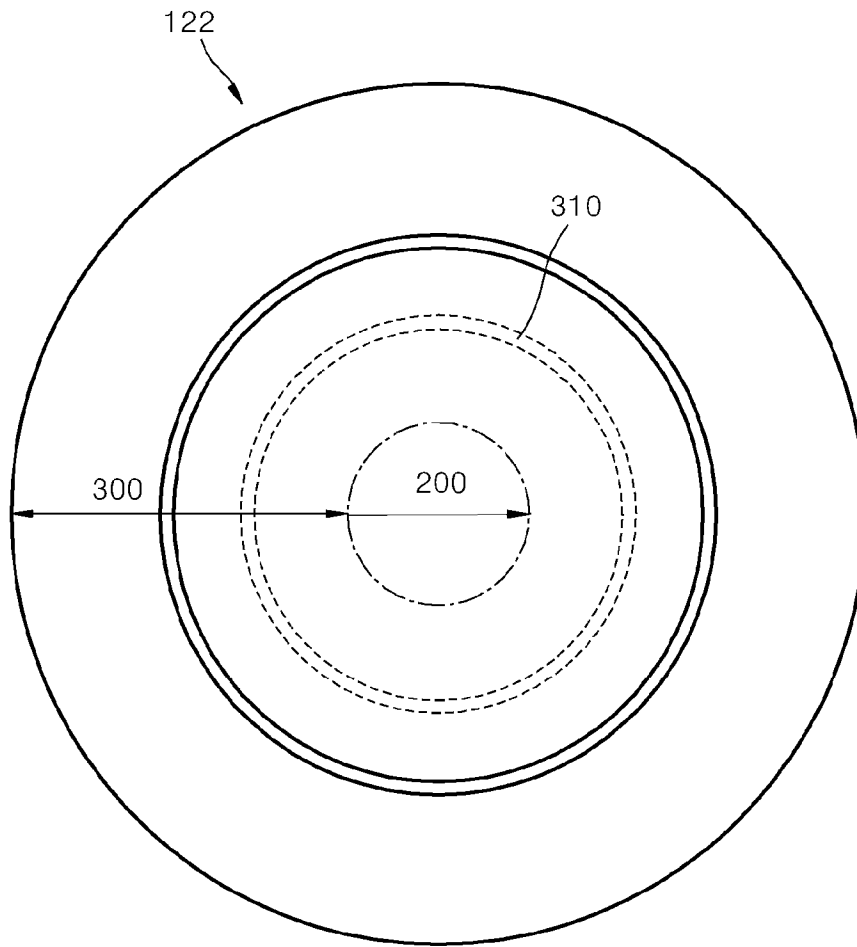
Claims

- [Claim 1] A cap assembly for a secondary battery comprising: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion comprising a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and comprising a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween.
- [Claim 2] The cap assembly according to claim 1, wherein the safety vent is bent downwardly and extends to form a bent portion.
- [Claim 3] The cap assembly according to claim 1, wherein the breakable junction is welded to a protrusion of the sub-plate entering the central hole formed in the support member.
- [Claim 4] The cap assembly according to claim 3, wherein the protrusion comprises a portion that protrudes upwardly opposite to the central portion and is opposite and melt-bonded to the breakable junction.
- [Claim 5] The cap assembly according to claim 4, wherein the notch is formed in the shape of a long home on the outer circumference of the portion opposite to the breakable junction.
- [Claim 6] The cap assembly according to claim 3, wherein the protrusion comprises a portion that protrudes upwardly opposite to the central portion and is opposite to and in close contact with the current-carrying contact.
- [Claim 7] The cap assembly according to claim 6, wherein the portion opposite to the current-carrying contact is formed on the outer circumference of the notch.
- [Claim 8] The cap assembly according to claim 1, wherein a breakable notch is formed at the bottom of the peripheral portion such that the safety vent is bent by a rupture pressure.
- [Claim 9] The cap assembly according to claim 8, further comprising a bending-

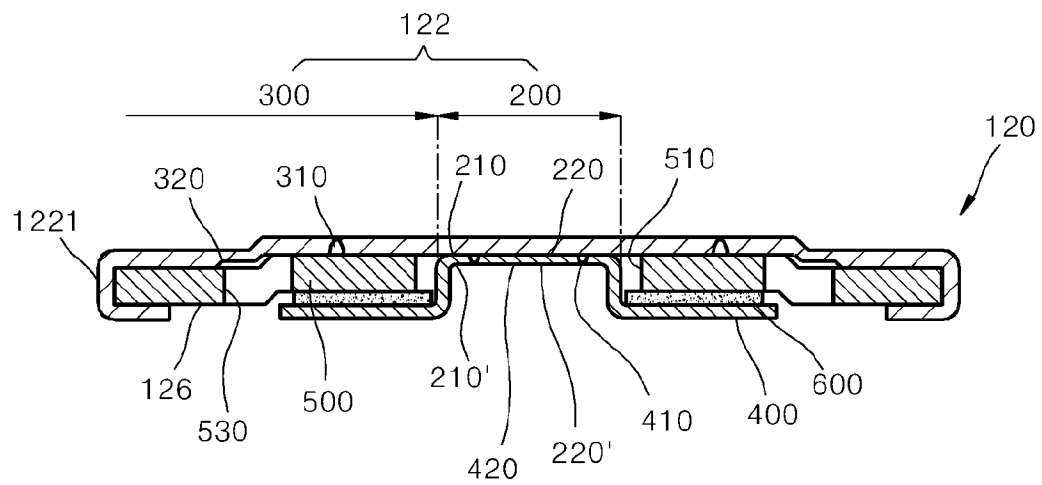
relieving portion adjacent to the breakable notch to space the current-carrying contact from the cap-down and forming a space where a rupture pressure is applied to the breakable junction.

- [Claim 10] The cap assembly according to claim 9, wherein the bottom width of the bending-relieving portion is larger than that of the breakable notch.
- [Claim 11] The cap assembly according to claim 1, wherein the cap-down comprises a through-hole that is spaced apart from and formed around the central hole and through which internal gases and liquids pass.
- [Claim 12] The cap assembly according to claim 1, wherein the insulating layer extends beyond the end of the sub-plate and reaches the end of the cap-down.
- [Claim 13] The cap assembly according to claim 12, wherein the insulating layer has a through-hole opposite to the through-hole of the cap-down.
- [Claim 14] The cap assembly according to claim 13, wherein spacer bridges are formed across the opposite through-hole.
- [Claim 15] The cap assembly according to claim 1, wherein a short-circuit preventing layer is formed at a position opposite to the end of the battery can through a gasket of the safety vent.
- [Claim 16] A secondary battery comprising the cap assembly according to any one of claims 1 to 15.

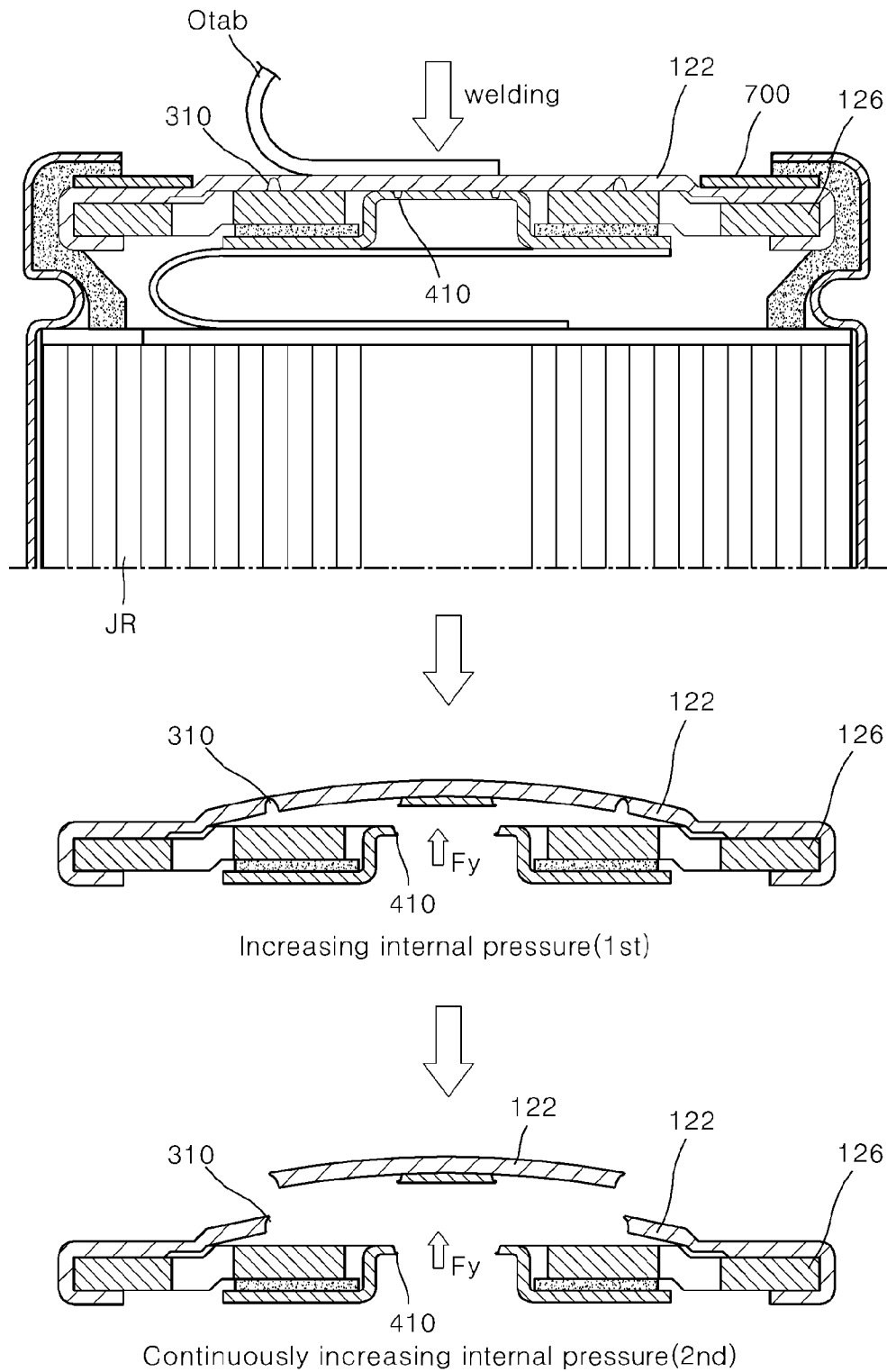
[Fig. 1]



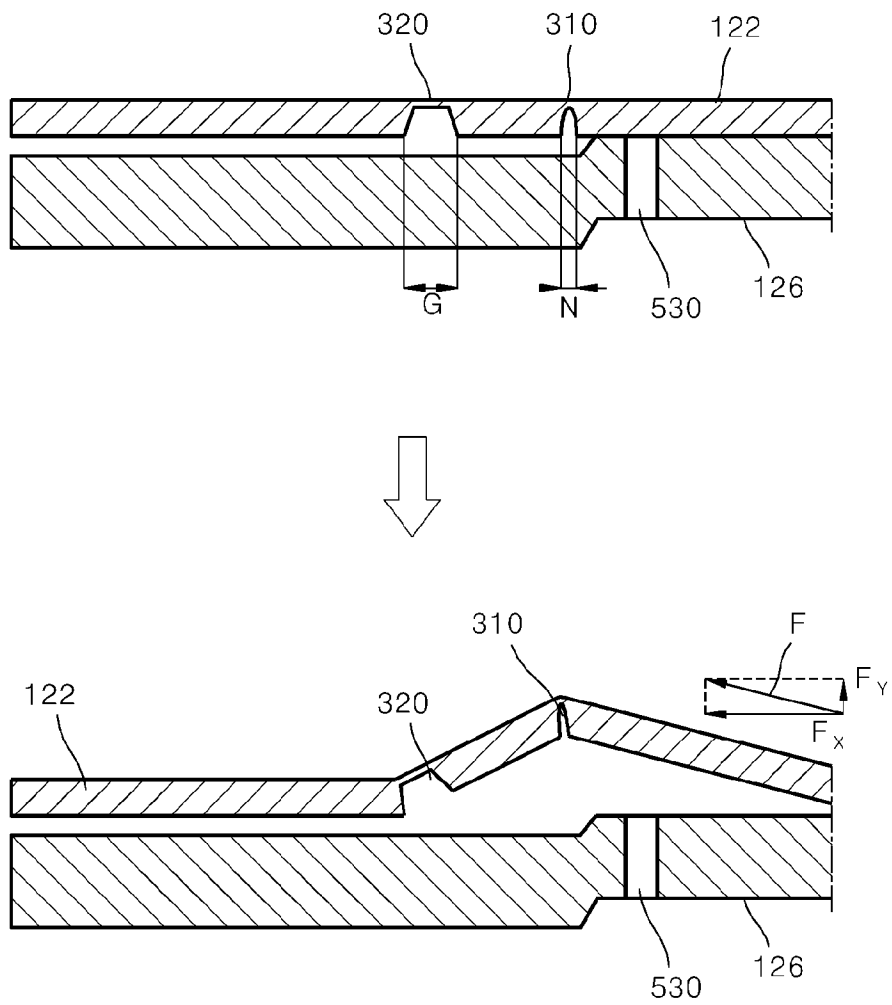
[Fig. 2]



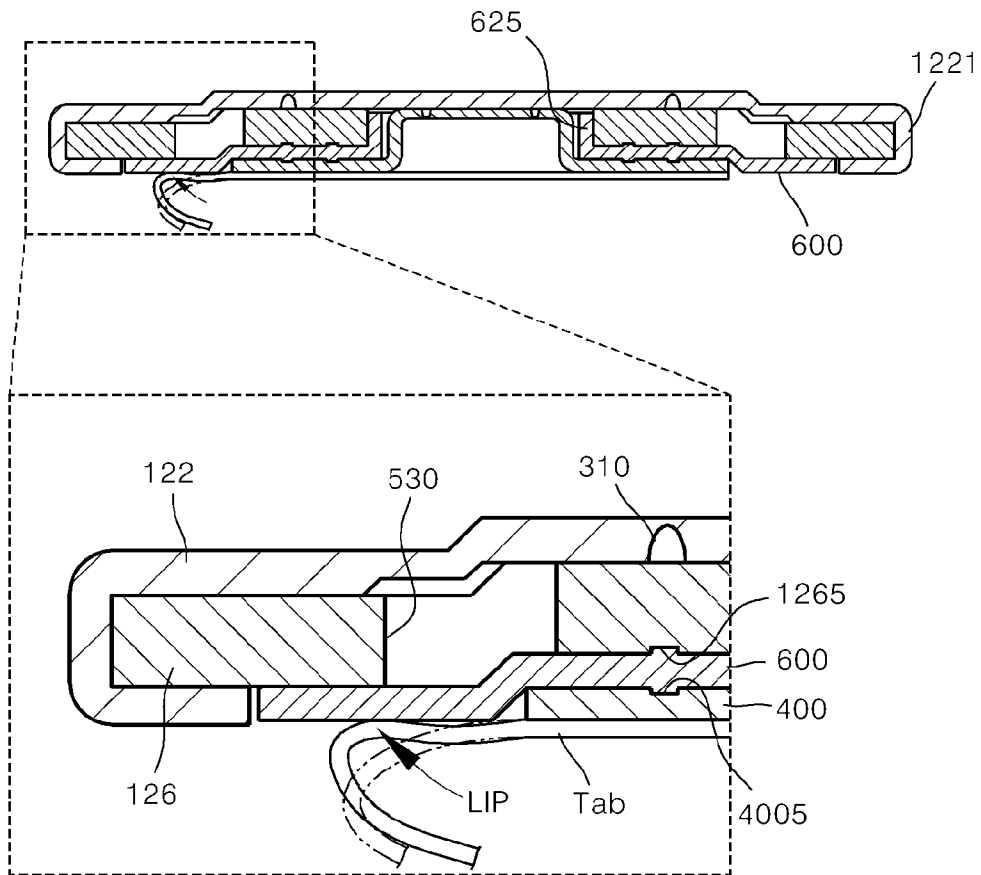
[Fig. 3]



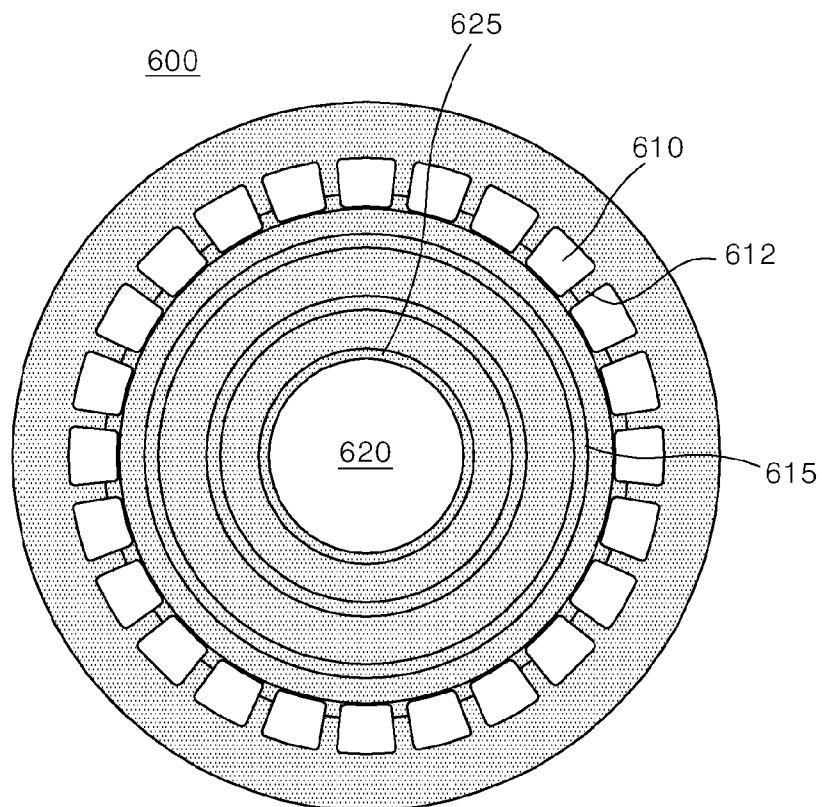
[Fig. 4]



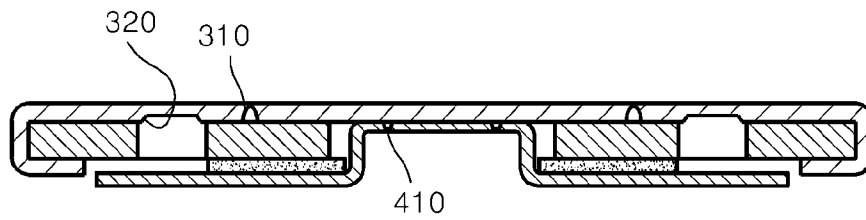
[Fig. 5]



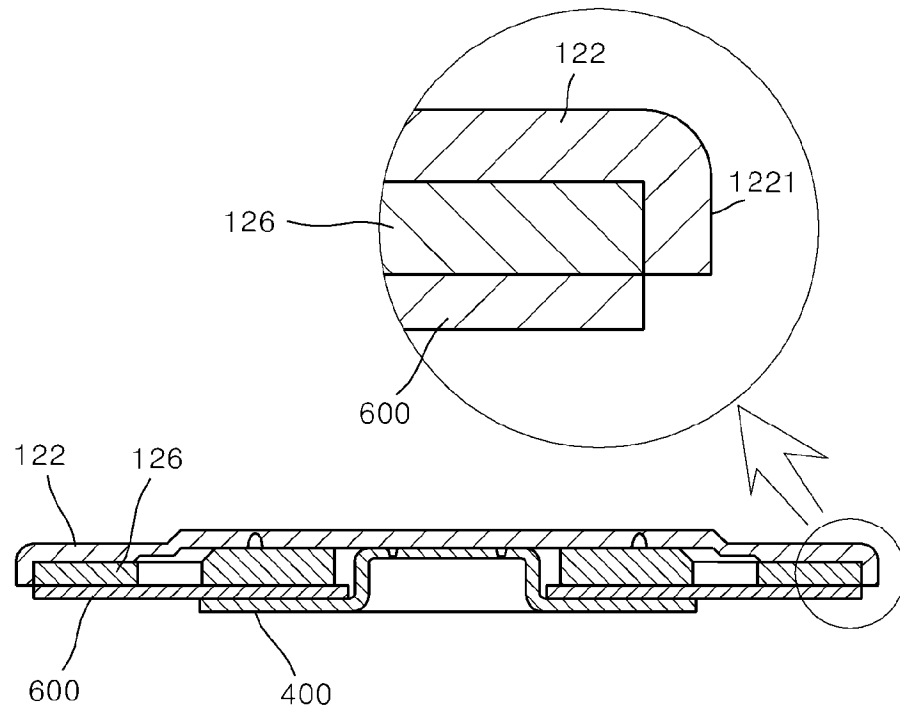
[Fig. 6]



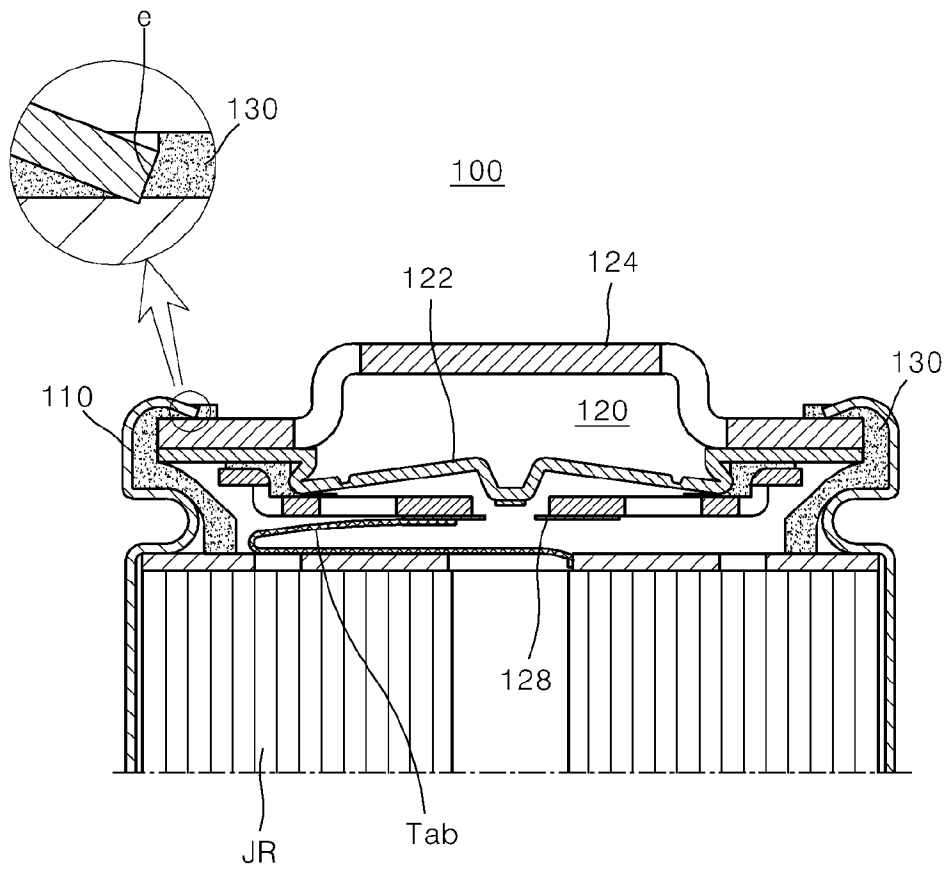
[Fig. 7]



[Fig. 8]



[Fig. 9]



INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2017/008736**A. CLASSIFICATION OF SUBJECT MATTER****H01M 2/04(2006.01)i, H01M 2/12(2006.01)i, H01M 10/04(2006.01)i**

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/04; H01M 2/12; H01M 2/08; H01M 2/34; H01M 2/20; H01M 2/26; H01M 2/10; H01M 2/02; H01M 10/04

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models

Japanese utility models and applications for utility models

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS(KIPO internal) & keywords: cap assembly, secondary battery, safety bent, shorting, notch

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KR 10-2016-0034794 A (SHIN HEUNG ENERGY & ELECTRONIC CO., LTD.) 30 March 2016 See figures 1-7; and claims 1-2.	1-16
A	US 2011 0206958 A1 (KIYAMA, A. et al.) 25 August 2011 See the entire document.	1 16
A	JP 2005 100927 A (MATSUSHITA ELECTRIC INDUSTRY CO., LTD.) 14 April 2005 See the entire document.	1 16
A	KR 10-2010-0062886 A (SAMSUNG SDI CO., LTD.) 10 June 2010 See the entire document.	1-16
A	KR 10-2006-0065378 A (LG CHEM, LTD.) 14 June 2006 See the entire document.	1-16



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Date of the actual completion of the international search

06 December 2017 (06.12.2017)

Date of mailing of the international search report

06 December 2017 (06.12.2017)

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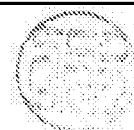
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LEE, Dong Wook

Telephone No. +82-42-481-8163



INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No.

PCT/KR2017/008736

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
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		US 8486546 B2	16/07/2013
KR 10-2006-0065378 A	14/06/2006	KR 10-0858414 B1	11/09/2008

Doc code: IDS

PTO/SB/08a (02-18)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2020. OMB 0651-0031

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Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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	1	2175507	EP	A1	2010-04-14	Samsung Sdi Co., Ltd	U.S. Publication No. 20100086835 is the English language counterpart which is previously cited.	
	2	20170124343	KR	A	2017-11-10	Lg Chemical Ltd		×
	3	2018030836	WO	A1	2018-02-15	Shin Heung Energy & Electronic Co.,Ltd		□

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
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Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Extended European Search Report for Application No. EP 19889598.9, dated August 17, 2021, 8 pages.	

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EXAMINER SIGNATURE

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

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Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

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Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2021-11-01
Name/Print	Jordan T. Riviello	Registration Number	72,515

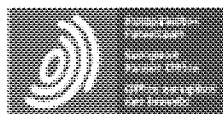
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**Espacenet****Bibliographic data: JPH06215746 (A) — 1994-08-05**

EXPLOSION-PROOF SAFETY DEVICE AND BATTERY HAVING IT

Inventor(s): MITAMURA TOMOKAZU; KONISHI AKIYA; TSURUTA KUNIO;
MURAKAMI TETSUYA ± (MITAMURA TOMOKAZU, ; KONISHI
AKIYA, ; TSURUTA KUNIO, ; MURAKAMI TETSUYA)

Applicant(s): MATSUSHITA ELECTRIC IND CO LTD ± (MATSUSHITA ELECTRIC
IND CO LTD)

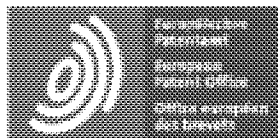
Classification: - **international:** *H01M2/12; H01M2/34*; (IPC1-7): H01M2/12
- **cooperative:** H01M50/3425 (EP); H01M50/574 (EP, US);
H01M50/578 (EP); Y02E60/10 (EP)

**Application
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**Priority
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Abstract of JPH06215746 (A)

PURPOSE: To prevent the rupture of a battery, by interrupting an electric path between a terminal board and a power generating element, by a first inversion deformation when internal pressure in the battery is increased, and breaking a safety valve, concurrently with a second inversion deformation, to exhaust gas. **CONSTITUTION:** When battery internal pressure is increased by a short circuit, overcharge, and reverse charge, the battery internal pressure is transmitted to a metal-made safety valve 1 through an air vent hole provided on a metallic lead fitting plate 4, a metallic perforated terminal board 3, and an insulating ring 2. In the valve 1, since its outer peripheral part is calked by a battery case, when the battery internal pressure is reached to a given value, a center part is risen upward, a folded part 1b is reversely deformed, a fused part S is separated, and the connection with the terminal board 3 is released to interrupt current. Next, when battery internal pressure is suddenly increased, the valve 1 is ruptured concurrently with the reverse deformation of the folded part 1a of the valve 1 to exhaust gas to outside a battery through an air vent hole provided on a terminal board 5. Consequently, the rupture of the battery can be prevented.



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DESCRIPTION JPH06215746A

[0001]

¹³ [Industrial fields of application]

[0002]

¹⁷ The present invention relates to an explosion-proof safety device for use in batteries.

[0003]

²¹ [Prior art]

[0004]

²⁵ 2. Description of the Related Art In recent years, electronic devices such as AV devices and personal computers are becoming more portable and cordless.

²⁷ Lithium secondary batteries, in particular, have high expectations as batteries having a particularly high voltage and high energy density.

[0005]

³² However, sealed batteries, such as lithium secondary batteries that use lithium metal or carbon material as the negative electrode, generate gas inside the battery and increase internal pressure when misused such as short circuit, overcharge, or reverse charge. may explode and lose its function or damage peripheral devices.

[0006]

³⁹ In particular, if the overcharged state continues, the decomposition of the electrolyte and the decomposition of the active material may lead to a rapid temperature rise of the battery or an explosion.

[0007]

⁴⁵ In order to prevent the battery from exploding in such a short-circuit, overcharge, or reverse charge state, conventionally, as disclosed, for example, in Japanese Utility Model Publication No. 54-137734 and Japanese Utility Model Publication No. 59-15398, , an explosion-proof safety device has been proposed which has a flexible thin plate (diaphragm) that breaks as the internal pressure of the battery rises and discharges the internal pressure to the outside of the battery.

⁵¹ As shown in FIGS. 5(A) and 5(B), as the internal pressure rises, this safety device deforms and breaks the flexible thin plate 42 (diaphragm) outward to release gas. Although it is discharged outside the battery to prevent the battery from exploding, short-circuiting, overcharging, and reverse charging of the battery continue.

⁵⁵ Therefore, when this safety device is used in a battery with a high energy density (the amount of electrical energy that can be extracted from the battery per unit volume or unit weight), such as a lithium secondary battery, the internal pressure of the battery rises sharply especially if the overcharged state continues. However, there were cases in which gas was not exhausted due to the destruction of the flexible thin plate, leading to an explosion.

[0008]

⁶³ Japanese Patent Laid-Open Nos. 112151/1990 and 288063/1990 propose an explosion-proof safety device in which a lead and an explosion-proof valve are separated or cut off as the internal pressure of the battery rises to cut off current.

⁶⁶ As shown in FIGS. 6A and 6B, this explosion-proof safety device has an explosion-proof valve 21 that deforms outward as the internal pressure rises, and a protrusion 22a of the explosion-proof valve 21 contacts a lead blocking stripper 22. When the internal pressure reaches a predetermined value, the explosion-proof valve 21 ruptures the lead plate 23 to cut off the current and prevent the explosion.

[0009]

⁷⁴ However, in the case of the above conventional explosion-proof safety device, as can be seen in FIG. Since the lead plate 23 led out from one electrode plate is welded to the tip of the electrode, the connection strength tends to be unstable.

⁷⁷ If the connection strength varies, the battery internal pressure (operating pressure) when the current is interrupted will vary significantly.

[0010]

⁸² The connection strength can be stabilized by selecting suitable welding conditions for laser welding, ultrasonic welding, or the like.

[0011]

⁸⁷ However, lithium secondary batteries, which have a relatively high energy density among sealed batteries, use a non-aqueous electrolyte made of an organic solvent with a low flash point. The connection with the valve must be made in advance before filling the battery with the electrolyte.

⁹¹ That is, when assembling a battery, it is common to insert the electrode plate group into the battery case, connect the sealing plate to the lead plate, inject the electrolyte, and then seal.

⁹³ Therefore, even if the lead plate and the explosion-proof valve are connected with a certain connection strength, mechanical stress is applied to the connection during the assembly process before sealing, which causes variations in connection strength. Occurred.

⁹⁶ Reducing the mechanical stress applied to the connection between the lead plate and the explosion-proof valve as much as possible is likely to become an obstacle in mass production using an automatic assembly machine or the like.

⁹⁹ In addition, when the battery is subjected to mechanical vibration or external impact such as being dropped, the electrode plate group moves within the battery case, and the lead plate separates from the explosion-proof valve, resulting in loss of battery function. .

[0012]

¹⁰⁵ Therefore, the present invention facilitates mass production of batteries, and ensures the safety and reliability of batteries in short-circuit, overcharge, and reverse charge states even after shocks such as vibrations and drops are applied to the batteries. An object of the present invention is to provide an explosion-proof safety device capable of

[0013]

¹¹² The explosion-proof safety device of the present invention is electrically connected to the external terminal plate at the peripheral portion and to the internal perforated terminal plate at the central portion, respectively, so that the electricity from the power generation element to the external terminal plate is connected. a first reversal portion having a safety valve which also acts as an electrical conduction path, the safety valve being disposed centrally thereof and electrically connected to the internal perforated terminal plate in a normal state; and a second reversing portion disposed outside the portion, wherein the first reversing portion reversely deforms at a relatively low battery internal pressure to cut off the electrical connection between the internal perforated terminal plate and the safety valve. In addition, the second reversing portion is reversely deformed by a relatively high battery internal

pressure, and at least a part of the second reversing portion is broken, and the gas in the battery is discharged. It is.

[0014]

¹²⁷ It is desirable that the reversing portion is formed by at least two folds, and that a thin portion is formed in the bending portion that forms the second reversing portion.

[0015]

¹³² A battery of the present invention is characterized by comprising the above explosion-proof safety device.

[0016]

¹³⁷ [Operation] With the above configuration, when the internal pressure of the battery rises due to a short circuit, overcharge, or reverse charge, when the internal pressure of the battery reaches the lower set value, the bent portion corresponding to the lower set value causes reversal deformation, which cuts off the electrical conduction path and prevents rapid temperature rise of the battery. , the bent portion corresponding to the higher setting is reversed and destroyed at the same time, allowing the gas inside the battery to dissipate and prevent an explosion.

[0017]

¹⁴⁷ In addition, since the explosion-proof safety device itself has the function of interrupting electrical connection, it is possible to strengthen the electrical connection between the lead plate and the sealing plate, which facilitates the mass production of batteries, and further reduces the risk of vibration, dropping, etc. of batteries. Even if the electrode plate group moves within the battery case due to the impact of , the lead plate does not separate from the sealing plate, and a battery with high safety and reliability can be obtained.

[0018]

¹⁵⁶ BEST MODE FOR CARRYING OUT THE INVENTION The present invention will now be described with reference to specific embodiments together with the drawings.

[0019]

¹⁶¹ 1, 2 and 3 are diagrams showing an assembled sealing plate including an explosion-proof safety device according to an embodiment of the present invention.

¹⁶³ This assembly sealing plate consists of a flexible metal safety valve 1, an insulating ring 2,

and a metal internal perforated terminal plate 3 having a projection 3a provided in the center and a ventilation hole 3b formed around the projection 3a. , a metal lead mounting plate 4 , an external terminal plate 5 and an insulating gasket 6 .

- 167 The metal safety valve 1 is integrally formed with annular bent portions 1a and 1b, and the central portion of the metal safety valve 1 is a protrusion of a perforated metal terminal plate 3 projecting through an insulating ring 2. It is welded to the upper surface of the portion 3a, and the metal safety valve 1 and the terminal plate 3 are electrically connected by forming the welded portion s.
- 172 The metal safety valve 1 and the projecting portion 3a of the perforated metal terminal plate 3 are preferably welded together by ultrasonic welding, resistance welding, laser welding, or the like.
- 175 The bent portion 1a of the metal safety valve 1 may be provided with a thin portion 1c as shown in FIG.

[0020]

- 180 The metal lead mounting plate 4 is provided with a vent hole 4a, and the internal pressure of the battery is released through the vent hole 4a, the inner opening of the insulating ring 2, the vent hole 3b of the perforated metal terminal plate 3, and the metal terminal plate 3. It is configured to be fully added to the safety valve 1 .

[0021]

- 187 Next, the operation of the assembled sealing plate according to the present embodiment to cut off current in response to an increase in the internal pressure of the battery will be described with reference to FIGS. 4(A) and 4(B).
- 190 The assembly sealing plate is welded to the lead plate 7 attached to the electrode plate, and after the electrolyte is injected, it is attached by crimping the battery case 8, which also serves as the terminal plate, to seal the battery.
- 193 When the internal pressure of the battery rises due to a short circuit, overcharging, or reverse charging, the internal pressure of the battery passes through the metal lead mounting plate 4, the perforated metal terminal plate 3, and the metal safety valve 1 through the ventilation holes provided in the insulating ring 2. be communicated to
- 197 Since the outer periphery of the metal safety valve 1 is crimped and fixed by the battery case 8, when the battery internal pressure reaches a predetermined value, the central portion is lifted upward and bent as shown in FIG. 4A. The portion 1b is reversely deformed, and the welded portion s is peeled off by this stress, and the connection with the perforated metal terminal plate 3 is broken, thereby interrupting the current.

[0022]

- 205 It should be noted that current cut-off between metal safety valve 1 and metal perforated

terminal plate 3 due to an increase in internal pressure of the battery described above is caused by short circuit, overcharge, or breakage of metal safety valve 1 due to internal pressure in order to release the battery from a reverse charged state. For this reason, the welding strength between the metal safety valve 1 and the projecting portion 3a of the perforated metal terminal plate 3 should be made smaller than the breaking strength of the metal safety valve 1. It is configured.

212 In this embodiment, the bent portion 1b of the metal safety valve 1 is bent at an obtuse angle and the bent portion 1a is bent at an acute angle so that the bent portion 1b is reversely deformed at a lower pressure. ing.

[0023]

218 In addition, since the metal safety valve 1 and the perforated metal terminal plate 3 are sufficiently separated by the reversal deformation, the metal safety valve 1 and the perforated metal terminal plate 3 come into contact with each other due to fluctuations in the internal pressure of the battery after the current is cut off, causing a chattering phenomenon. I can't wake you up.

[0024]

226 3, that is, if the bent portion 1a of the metal safety valve 1 is provided with the thin portion 1c, as described above, in the short-circuit, overcharge, and reverse charge states, some other factor 4B, the thin portion 1c provided at the bent portion 1a of the metal safety valve 1 breaks simultaneously with the reverse deformation of the bent portion 1a. Thus, the gas can be reliably discharged to the outside of the battery through the ventilation holes 5a provided in the terminal plate 5.

[0025]

235 When the explosion-proof safety device of the present invention is used, since the explosion-proof safety device itself has a current interrupting mechanism, the welding strength to the lead plate can be made relatively high, so troubles such as the dislodgment of the assembly sealing plate during the battery assembly process can be prevented. It is possible to avoid this and facilitate the mass production of batteries.

240 Also, even if the battery is subjected to shock such as vibration or drop, it is possible to avoid troubles such as battery operation failure due to detachment of leads from the assembly sealing plate.

[0026]

246 Effect of the Invention According to the explosion-proof safety device of the present invention, it is possible to effectively prevent heat generation and explosion of the battery

even in the case of short circuit, overcharge, and reverse charge. Therefore, it is possible to obtain a battery equipped with a highly reliable explosion-proof safety device in which malfunction is unlikely to occur.

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(71)出願人 000005321

松下電器産業株式会社

大阪府門真市大字門真1006番地

(72)発明者 三田村 知一

大阪府門真市大字門真1006番地 松下電器
産業株式会社内

(72)発明者 小西 昭治

大阪府門真市大字門真1006番地 松下電器
産業株式会社内

(72)発明者 鶴田 邦夫

大阪府門真市大字門真1006番地 松下電器
産業株式会社内

(74)代理人 弁理士 小鍛冶 明 (外 2 名)

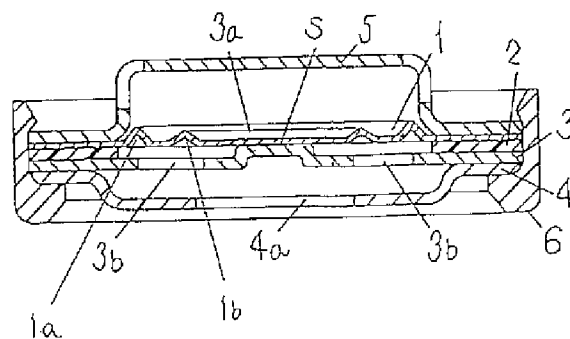
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(54)【発明の名称】 防爆安全装置および防爆安全装置を備えた電池

(57)【要約】

【目的】 電池が短絡、過充電、逆充電状態にあっても安全性と信頼性を確保できる防爆安全装置を提供することを目的とする。

【構成】 二重の反転変形構造を有し、電池内の内圧が上昇した時に、まず第一の反転変形によって端子板と発電要素間の電氣的導通経路を遮断し、次に第二の反転変形と同時に安全弁を破壊して電池内部の発生ガスを円滑に外部に排出し、電池の破裂を未然に防止する。



【特許請求の範囲】

【請求項1】 周縁部において外部端子板と、中央部において内部有孔端子板とそれぞれ電氣的に接続され、発電要素から外部端子板への電氣的導通経路としても作用する安全弁を備え、この安全弁が、その中央側に配置され、前記内部有孔端子板と通常状態においては電氣的に接続されている第1反転部と、該第1反転部の外方に配置された第2反転部とを備え、前記第1反転部が、比較的低い設定電池内部圧力で反転変形して、前記内部有孔端子板と安全弁との電氣的接続を遮断するようになっているとともに、前記第2反転部が、比較的高い設定電池内部圧力で反転変形するとともに、少なくとも一部が破断し、電池内ガスを排出するようになっている防爆安全装置。

【請求項2】 前記反転部が、少なくとも2重の折り曲がり部によって形成されている請求項1の防爆安全装置。

【請求項3】 前記第2反転部を形成する折れ曲がり部は、薄肉部が形成されている請求項2の防爆安全装置。

【請求項4】 請求項1ないし3のいずれかの防爆安全装置を備えた電池。

【発明の詳細な説明】

【0001】

【産業上の利用分野】

【0002】本発明は、電池に用いる防爆安全装置に関するものである。

【0003】

【従来の技術】

【0004】近年、AV機器あるいはパソコン等の電子機器のポータブル化、コードレス化が急速に進んでおり、これらの駆動用電源として小型、軽量で高エネルギー密度を有する密閉型電池への要望が高い。特にリチウム二次電池は、とりわけ高電圧、高エネルギー密度を有する電池として期待が大きい。

【0005】ところが、密閉型電池、例えばリチウム金属やカーボン材料を負極として使用するリチウム二次電池は、短絡、過充電、逆充電等の誤使用された時に、電池内にガスが発生し、内圧が上昇して破裂しその機能を失ったり、周辺機器に損傷を与えることがあった。

【0006】特に、過充電状態が継続すると、電解液の分解および活物質の分解により、電池が急激に温度上昇したり、爆発を招く恐れがあるという問題があった。

【0007】このような、短絡、過充電、逆充電状態における電池の爆発を未然に防止するため、従来は、例えば実開昭54-137734号公報、実公昭59-15398号公報に示されているように、電池内圧上昇に伴い破壊して内圧を電池外に排出する可撓性薄板（ダイヤフラム）を有する防爆安全装置が提案されている。この安全装置は、図5の（A）および（B）に示したように内圧の上昇に伴い、この内圧により可撓性薄板42（ダ

イアフラム）が外方に変形を生じて破壊してガスを電池外に排出し電池の爆発を未然に防止するものであるが、電池の短絡、過充電、逆充電状態は継続される。したがって、この安全装置をエネルギー密度（単位体積もしくは単位重量当りの電池から取り出せる電気エネルギー量）の高い電池、例えばリチウム二次電池に用いた場合、特に過充電状態が継続すると電池内圧が急激に上昇し、可撓性薄板の破壊によるガスの排出が追いつかず、爆発にいたる場合があった。

【0008】その改良案として特開平2-112151号、特開平2-288063号には、電池内圧上昇に伴いリードと防爆弁を剥離もしくは切断して電流を遮断する防爆安全装置を提案している。この防爆安全装置は、図6の（A）および（B）に示したように内圧の上昇に伴い外方に変形を生じる防爆弁21を、その突起部22aがリード遮断用ストリッパ22に接触するようにして取り付け、所定の内圧に達したとき、防爆弁21の作用によりリード板23を破断するようにして電流を遮断し、爆発を防止するようにしたものである。

【0009】

【発明が解決しようとする課題】しかしながら、上記の従来の防爆安全装置の場合、図6で解るようにリード遮断用ストリッパ22の穴を介して突出した金属薄板製の防爆弁21の突起部22aの先端に一方の極板から導出したリード板23を溶接しているため、接続強度が不安定となりやすい。接続強度にバラツキが発生すると、電流遮断されるときに電池内圧（作動圧）の著しいバラツキとなる。

【0010】接続強度を安定化することは、レーザー溶接、超音波溶着等で適切な溶接条件を選定することで可能である。

【0011】ところが、密閉型電池のなかでも比較的エネルギー密度の高いリチウム二次電池の場合、電解液として、低引火点の有機溶媒からなる非水電解液を用いるため、火花の発生を伴うリード板と防爆弁との接続は、電池に電解液を注液する前にあらかじめ実施する必要がある。つまり、電池を組み立てる場合、極板群を電池ケースに挿入した後に、封口板をリード板に接続し、電解液を注液し、しかる後に封口することが一般的である。そこで、一定の接続強度でリード板と防爆弁を接続しても、その後封口するまでに、注液工程を経るため組み立て工程において接続部に機械的ストレスが加わり、結果的に接続強度のバラツキが発生した。リード板と防爆弁との接続部に加わる機械的ストレスを可能な限り少なくすることは自動組立機などで大量生産する際に障害となりやすい。また、電池に機械的振動もしくは、落下等外部から衝撃を加えた場合、電池ケース内において極板群が移動し、リード板が防爆弁より剥離し、電池としての機能を失うという問題があった。

【0012】そこで、本発明は、電池の大量生産が容易

で、電池に振動、落下等の衝撃が加わった後においても、短絡、過充電、逆充電状態における電池の安全性と信頼性を確保することのできる防爆安全装置を提供することを目的とするものである。

【0013】

【課題を解決するための手段】本発明の防爆安全装置は、周縁部において外部端子板と、中央部において内部有孔端子板とそれぞれ電氣的に接続され、発電要素から外部端子板への電氣的導通経路としても作用する安全弁を備え、この安全弁が、その中央側に配置され、前記内部有孔端子板と通常状態においては電氣的に接続されている第1反転部と、該第1反転部の外方に配置された第2反転部とを備え、前記第1反転部が、比較的低い電池内部圧力で反転変形して、前記内部有孔端子板と安全弁との電氣的接続を遮断するようになっているとともに、前記第2反転部が、比較的高い電池内部圧力で反転変形するとともに、少なくとも一部が破断し、電池内ガスを排出するようになっていることを特徴とするものである。

【0014】上記反転部は、少なくとも2重の折れ曲がり部によって形成され、前記第2反転部を形成する折れ曲がり部に、薄肉部が形成されていることが望ましい。

【0015】本発明の電池は、上記の防爆安全装置を備えていることを特徴とするものである。

【0016】

【作用】以上の構成により、短絡、過充電、逆充電状態により、電池内圧が上昇した場合、まず低い方の設定値に電池の内部圧力が達した時に、低い方の設定に該当する折れ曲がり部が反転変形を生じ、このことにより電氣的導通経路を遮断し電池の急激な温度上昇を防止することができ、また、急激なガス発生によって高い方の設定値に電池の内部圧力が達した時には、高い方の設定に該当する折れ曲がり部が反転変形を生じると同時に破壊され、電池内部のガスを逸散させ爆発を未然に防ぐことが可能である。

【0017】また、防爆安全装置自体が電氣的接続遮断の機能を有するため、リード板と封口板との電氣的接続を強くすることが可能となり、電池の量産が容易であり、さらに電池に振動、落下等の衝撃が加わり極板群が電池ケース内を移動しても、リード板が封口板から剥離することもなく、安全性、信頼性の高い電池を得ることができる。

【0018】

【実施例】以下、図面とともに本発明を具体的な実施例によって説明する。

【0019】図1、図2および図3は、本発明の実施例における防爆安全装置を含む組立封口板を示す図である。この組立封口板は、可撓性の金属製安全弁1、絶縁リング2、中央に突起部3aが設けられ、この突起部3aの周囲に通気孔3bが形成された金属製内部有孔端子

板3、金属製リード取りつけ板4、外部端子板5および絶縁ガasket6より構成されている。金属製安全弁1は、環状の折り曲がり部1aおよび1bが一体に形成されており、また金属製安全弁1の中央部は、絶縁リング2を介して突出する金属製有孔端子板3の突起部3aの上面に溶着しており、溶着部sの形成により、金属製安全弁1と端子板3とは電氣的に接続されている。なお、金属製安全弁1と金属製有孔端子板3の突起部3aとの溶着は、超音波溶着もしくは抵抗溶接、レーザー溶接等により行うことが適切である。金属製安全弁1の折り曲がり部1aには図3に示したように薄肉部1cが設けられていてもよい。

【0020】また、金属製リード取りつけ板4には、通気孔4aが設けられており、電池内圧は、この通気孔4a、絶縁リング2の内開口および金属製有孔端子板3の通気孔3b、金属製安全弁1に充分加えられるよう構成されている。

【0021】次に、上記本実施例による組立封口板が、電池内圧の上昇に応じて電流を遮断する動作について、図4の(A)および(B)を参照して説明する。組立封口板は、極板に付属したリード板7と溶接され、電解液注液後に、端子板を兼ねる電池ケース8をかしめることにより取り付けられ、電池を密閉している。短絡、過充電もしくは、逆充電等により電池内圧が上昇した場合、この電池内圧が金属製リード取りつけ板4、金属製有孔端子板3、絶縁リング2に設けられた通気孔を通して金属製安全弁1に伝えられる。金属製安全弁1はその外周部が電池ケース8によりかしめられ固定されているため、電池内圧が所定の値に達すると、図4の(A)に示すように中心部が上方へ持ち上げられ折り曲がり部1bが反転変形し、この応力によって前記溶着部sが剥離され金属製有孔端子板3との接続が解かれて電流が遮断される。

【0022】なお、上述の電池内圧の上昇による金属製安全弁1と金属製有孔端子板3との電流遮断は、電池の短絡、過充電もしくは、逆充電状態を解くため内圧による金属製安全弁1の破壊される以前に確実に行う必要があるが、そのために、金属製安全弁1と金属製有孔端子板3の突起部3aとの溶着はその溶着強度を金属製安全弁1の破壊強度よりも小さくするよう構成されている。本実施例においては、金属製安全弁1の折り曲がり部1bの折り曲がり角度を鈍角に折れ曲がり部1aの折り曲がり角度を鋭角に構成し、折り曲がり部1bの方が低圧で反転変形を生じるようにしている。

【0023】さらに、反転変形によって金属製安全弁1と金属製有孔端子板3が充分隔離されるため、電流遮断後に電池内圧の変動によって金属製安全弁1と金属製有孔端子板3が接触しチャタリング現象を起こすこともない。

【0024】また、図3に示すように構成すれば、すな

わち、金属製安全弁1の折れ曲がり部1aに薄肉部1cを設ければ、上述のように、短絡、過充電、逆充電状態において、他のなんらかの要因で電池の内圧が急激に上昇した場合においても、図4の(B)に示すように前記金属製安全弁1の折れ曲がり部1aに設けた薄肉部1cが折れ曲がり部1aの反転変形と同時に破壊され、端子板5に設けた通気孔5aを通してガスを電池外部に確実に排出することができる。

【0025】本発明の防爆安全装置を用いれば、防爆安全装置自体が電流遮断機構を有するため、リード板との溶接強度を比較的高くすることができるので、電池組み立て工程における組立封口板はずれのトラブル発生を回避でき電池の量産化を容易にすることができる。また、電池に振動あるいは落下等の衝撃を加えても組立封口板からのリードはずれによる電池動作不良といったトラブルも回避できる。

【0026】

【発明の効果】本発明における防爆安全装置によれば、短絡、過充電、逆充電時においても、電池の発熱、爆発を効果的に防止することができ、振動、落下等の衝撃を加えても動作不良が発生しにくい信頼性の高い防爆安全装置を備えた電池を得ることが可能である。

【図面の簡単な説明】

【図1】本発明の実施例の防爆安全装置を含む組立封口板の断面図である。

【図2】本発明の実施例の防爆安全装置を含む組立封口板の分解斜視図である。

【図3】本発明の実施例の防爆安全装置を含む組立封口板の断面図である。

【図4】上記防爆安全装置を備えた電池の作動状態を説明するための説明図である。

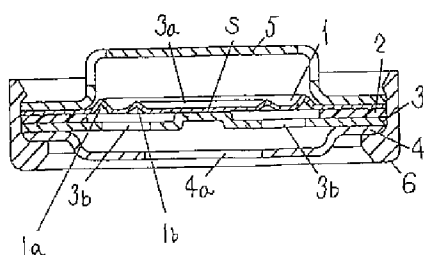
【図5】従来の密閉型電池断面図である。

【図6】従来の密閉型電池断面図である。

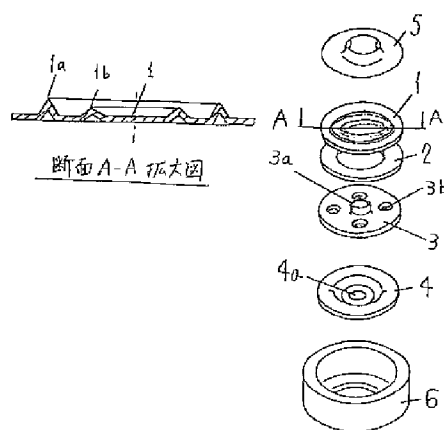
【符号の説明】

- 1 金属製安全弁
- 2 絶縁リング
- 3 金属製有孔端子板
- 4 金属製リード取り付け板
- 5 端子板
- 6 絶縁ガスカート
- 7 リード板
- 8 電池ケース
- 1a 折れ曲がり部
- 1b 折れ曲がり部
- 1c 薄肉部
- 3a 突起部
- 3b 通気孔
- 4a 通気孔
- 5a 通気孔

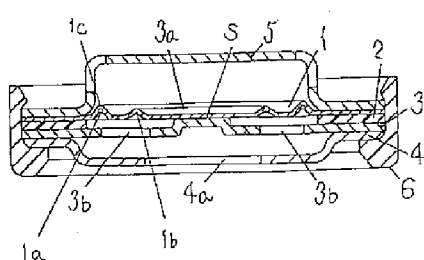
【図1】



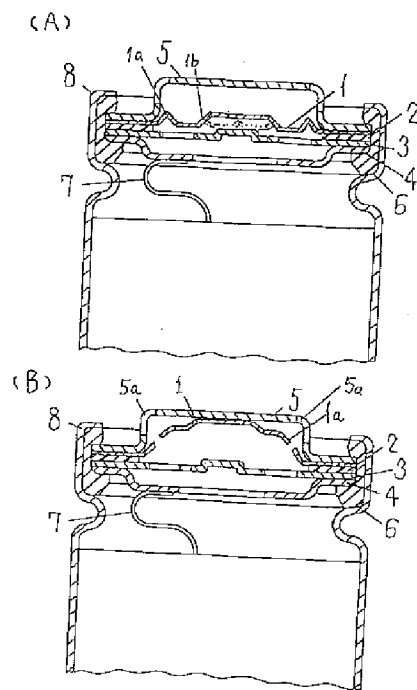
【図2】



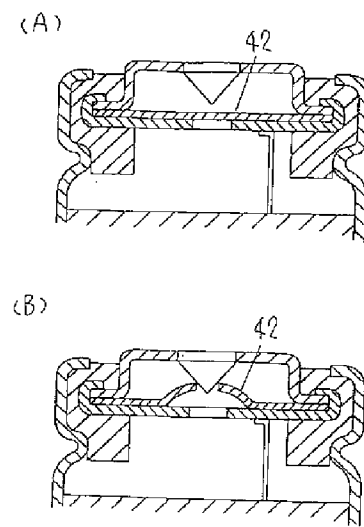
【図3】



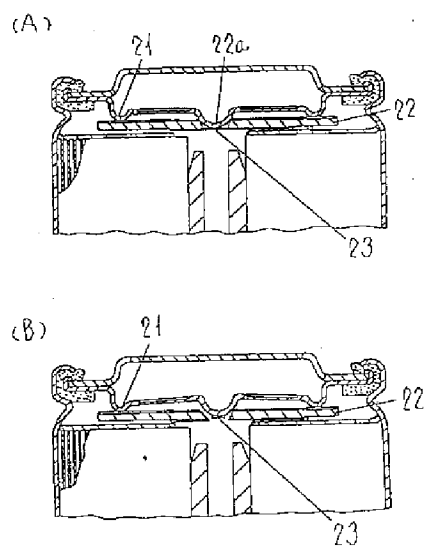
【図4】



【図5】



【図6】



フロントページの続き

(72)発明者 村上 哲哉
大阪府門真市大字門真1006番地 松下電器
産業株式会社内

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	Filing Date		2021-07-16
	First Named Inventor	Geon Woo Min	
	Art Unit		
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGCHEM 3.3F-3059	

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	1	2572931	CA	A1	2008-06-21	Wu, Donald P. H		
	2	102473883	CN	A	2012-05-23	Lg Chemical Ltd	U.S. Publication No. 20120114979 A1 is an English language counterpart, which has been previously cited.	×
	3	103155225	CN	A	2013-06-12	Lg Chemical Ltd	U.S. Publication No. 20130216871 A1 is an English language counterpart.	☒

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Search Report dated June 22, 2022 from the Office Action for Chinese Application No. 201980046222.8 issued June 30, 2022, 3 pages.	×

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Name/Print	Jordan T. Riviello	Registration Number	72,515

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L2	1547994	(H01M50/3425 or H01M50/107 or H01M50/578 or H01M50/574 or H01M2200/20 or H01M50/56 or H01M50/152 or Y02E60/10 or H01M50/116).cpc.	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:17 PM
L3	3881	2 AND ruptur\$4 WITH (lid cap top cover)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:17 PM
L4	285	2 AND ruptur\$4 WITH (lid cap top cover) WITH notch	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:18 PM
L5	26	2 AND ruptur\$4 WITH (lid cap top cover) WITH (weak\$5 notch\$2) WITH bend\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:18 PM
L6	7	("20130216871" OR "20110008654" OR "10090497" OR "20050214641" OR "20100086835" OR "20110200854" OR "20120114979").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/01/31 02:42 PM
L7	23	2 AND ruptur\$4 WITH (lid cap top cover) WITH (weak\$5 notch\$2) WITH deform\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:46 PM
L8	155	2 AND ruptur\$4 WITH (lid cap top cover) WITH predetermin\$4 ADJ pressure	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:56 PM
L9	63	8 AND notch\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:56 PM
L10	1	("20170294641").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/01/31 03:04 PM

L11	8	("20130273401" "6509114" "20100209765" "20140199567" "6258477" "20090148766" "20130230749" "20080314482").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 07:05 AM
L12	7596	((("LG") near3 ("CHEM") near3 ("LTD"))).AS,AANM.	(USPAT)	OR	ON	ON	2023/02/02 07:13 AM
L13	33	12 AND vent\$4 AND ruptur\$4 AND (bend\$4 bent)	(USPAT)	OR	ON	ON	2023/02/02 07:14 AM
L14	119	((("MIN") near3 ("Geon") near3 ("Woo")) OR (("LEE") near3 ("Je") near3 ("Jun")) OR (("JUNG") near3 ("Sang") near3 ("Suk"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/02/02 02:18 PM
L15	33	14 AND vent\$4 AND ruptur\$4 AND (bend\$4 bent)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/02/02 02:18 PM
L16	26	("20090311583" "20050026033" "20110171508" "20130309529" "20020122974" "20110200854" "20170309881" "20100203367" "20180062122" "20060024576" "20170309881" "20050026033" "20150004446" "20100159310" "20150236317").PN.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/02/02 02:31 PM
L17	13	("20090311583" "20050026033" "20110171508" "20130309529" "20020122974" "20110200854" "20170309881" "20100203367" "20180062122" "20060024576" "20170309881" "20050026033" "20150004446" "20100159310" "20150236317").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 02:51 PM
L18	1	("20110076527").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 03:15 PM
L19	36	2 AND bending ADJ (part portion) WITH (c-shaped (open discontinuous) ADJ (shape circle))	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:21 PM
L20	36	2 AND bending ADJ (part portion) WITH (c-shaped ((open discontinuous) ADJ	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT;	OR	ON	ON	2023/02/02 03:31 PM

L21	974	(shape circle))) 2 AND bending ADJ (part portion notch) WITH (plurality first second) WITH (length width)	IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L22	991	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) WITH (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L23	9	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) WITH (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L24	3	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) WITH (longer bigger) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L25	0	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) near4 (longer bigger) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:41 PM
L26	4	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) near4 (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:41 PM
L27	2	2 AND safety ADJ vent WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4) WITH (plurality first second) near4 (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:41 PM
L28	2	2 AND safety ADJ vent WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4 bent) WITH (plurality first second) near4 (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:55 PM
L29	18	2 AND safety ADJ vent WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4 bent) near4 (multiple plurality first)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 04:16 PM
L30	3	("20120114979").pn.	(US-PGPUB; USPAT;	OR	ON	ON	2023/02/02

L31	1	("20120114979").pn.	USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	04:19 PM
L32	46	(h01m50/342.ipc. h01m50/107.ipc. h01m50/578.ipc. h01m50/574.ipc. h01m50/56.ipc. h01m50/152.ipc. h01m50/116.ipc.)	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 04:20 PM
L33	3	32 AND vent	(JPO)	OR	ON	ON	2023/02/02 05:24 PM
							2023/02/02 05:25 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Bibliographic Data

Application No: 17/259,395

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged:

/LISA S PARK/

Examiner's Signature

Initials

Title:

Secondary Battery

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
07/16/2021	429	1729	LGES 3.3F-3059
RULE			

APPLICANTS

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF

INVENTORS

Geon Woo Min, Daejeon, KOREA, REPUBLIC OF

Je Jun Lee, Daejeon, KOREA, REPUBLIC OF

Sang Suk Jung, Daejeon, KOREA, REPUBLIC OF

CONTINUING DATA

This application is a 371 of PCT/KR2019/015396 11/13/2019

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF KR10-2018-0149369 11/28/2018

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05/08/2021

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Lerner, David, Littenberg, Krumholz & Mentlik, LLP

20 Commerce Drive

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Application	Document	Mailroom Date	Attorney Docket No.
17259395	CTNF	02/08/2023	LGES 3.3F-3059
	892	02/08/2023	LGES 3.3F-3059
	1449	02/08/2023	LGES 3.3F-3059
	1449	02/08/2023	LGES 3.3F-3059
	1449	02/08/2023	LGES 3.3F-3059

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USPTO Automated Interview Request (AIR)

Mar 8 2023

This paper requesting to schedule and/or conduct an interview is appropriate because:

This submission is requested to be accepted as an authorization for this interview to communicate via the internet. Recognizing that Internet communications are not secure, I hereby authorize the USPTO to communicate with the undersigned concerning scheduling of the interview via video conference, instant messaging, or electronic mail, and to conduct the interview in accordance with office practice including video conferencing.

Name(s) :

Jordan T Riviello

S-signature:

/Jordan T Riviello/

Registration Number:

72515

U.S. Application Number:

17259395

Confirmation Number:

1017

E-mail Address:

jriviello@lerner david.com

Phone Number:

+1 9085186373

Proposed Time of Interview:

3-21-2023 1:00 PM ET

Alternative Proposed Time(s) of Interview:

3-22-2023 11:30 AM ET

Preferred Interview Type:

Telephonic

I am the applicant or applicant's representative for this application.

Topic for Discussion:

prior art rejections



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017
86765	7590	03/24/2023	EXAMINER	
LG			PARK, LISA S	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1729	
			NOTIFICATION DATE	
			DELIVERY MODE	
			03/24/2023	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

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<i>Applicant-Initiated Interview Summary</i>	Application No. 17/259,395	Applicant(s) Min et al.		
	Examiner LISA S PARK	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
LISA S PARK	Primary Examiner	Telephonic
Jordan Riviello	Attorney of Record	

Date of Interview: 21 March 2023

Issues Discussed:

Proposed Amendment(s)

Proposed amendments were discussed to overcome the rejection of record. No specific agreement was reached.

☒ Attachment

/LISA S PARK/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

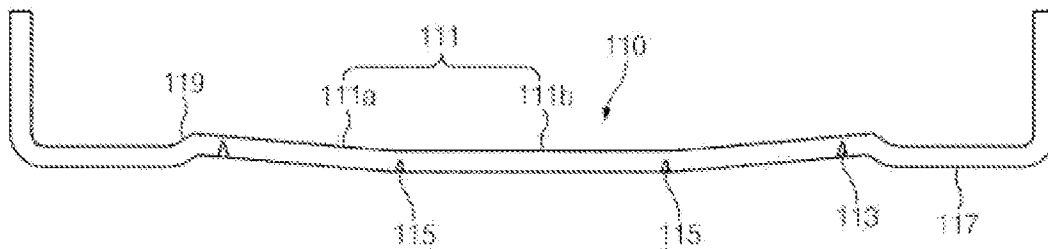
From: Riviello, Jordan T.
To: Park, Lisa
Cc: "LGES 2 OF 1736 - 4188 Correspondence"
Subject: [interview agenda] US 17/259,395 (LGES.4118)
Date: Monday, March 20, 2023 6:48:13 PM
Attachments: [image001.png](#)
[image002.png](#)

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Examiner Park,

During our interview scheduled for Tuesday, March 20, 2023 at 1:00 pm EST, I would like to discuss the following information.

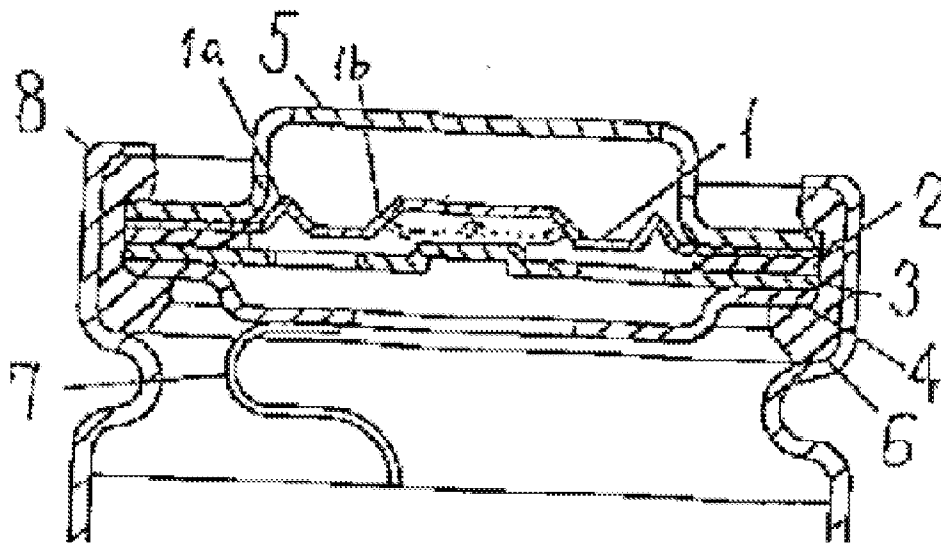
First, the rupture and bending notches are notches or grooves defined into the rupture portion. See for example FIG. 6 of the present application (reproduced below).



This mechanically changes the strength of the rupture portion and allows it to be cut at a preset venting pressure. See paragraph [0057]. In contrast, the prior art merely teaches a “rupture portion” that bends based on pressure but there is no “notch” defined into the structure. See for example FIG. 6 of the cited art where the structure is bent differently such that there is no bending notch present. With respect to the prior art, the rupturing is a result of the welding strength between the safety valve (1) and the projecting portion (3a). See page 6 of the English-language translation.

【 図 4 】

(A)



Moreover, we note that that this configuration allows the horizontal part (111b) to move from a lowermost position beneath rupture notch 113 (see fig. 2) where it can be coupled to the interrupt portion (121) and rise when the internal pressure increases to a location above the rupture notches (see FIG. 3). The prior art does not teach such movement.

I look forward to discussing the above distinctions with you in further detail during our telephone call tomorrow.

Kindly acknowledge receipt of this email.

Best regards,
Jordan

Jordan T. Riviello
jriviello@lernerjacobson.com
908-518-6373 Direct


Lerner David Liftenberg Krumholz & Mentlik LLP
20 Commerce Drive, Cranford, NJ 07016
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Application	Document	Mailroom Date	Attorney Docket No.
17259395	EXIN	03/24/2023	LGES 3.3F-3059
	OA.APPENDIX	03/24/2023	LGES 3.3F-3059

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Electronic Patent Application Fee Transmittal

Application Number:	17259395			
Filing Date:	16-Jul-2021			
Title of Invention:	Secondary Battery			
First Named Inventor/Applicant Name:	Geon Woo Min			
Filer:	James A. Vezeris/Kathleen Reynolds			
Attorney Docket Number:	LGES 3.3F-3059			
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	1251	1	220	220
Miscellaneous:				
Total in USD (\$)				220

Electronic Acknowledgement Receipt

EFS ID:	47974631
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	James A. Vezeris/Kathleen Reynolds
Filer Authorized By:	James A. Vezeris
Attorney Docket Number:	LGES 3.3F-3059
Receipt Date:	09-MAY-2023
Filing Date:	16-JUL-2021
Time Stamp:	15:13:09
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

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Payment Type	CARD
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RAM confirmation Number	E202359F13577437
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1	Extension of Time	One_Month_Request_for_Extension_of_Time_Under_37_CFR_1136a.pdf	24417	no	1
			8e106ff24a704533636322310fc0ad5b6c5012ca		
Warnings:					
Information:					
2	Miscellaneous Incoming Letter	Amendment_Transmittal.pdf	22637	no	1
			fa627fd8e10184329de5cc02540096642c9dfa42		
Warnings:					
Information:					
3	Amendment/Request for Reconsideration-After Non-Final Rejection	Amendment.pdf	45023	no	7
			ecd9b4591226bdb3f75eb9605ca49125bed9e35e		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	38046	no	2
			b1edb3e253596326c609f8924ee203b93317d58a		
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Information:					
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If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

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PETITION FOR EXTENSION OF TIME UNDER 37 CFR 1.136(a)		Docket Number (Optional) LGES 3.3F-3059																															
Application Number 17/259,395-Conf. #1017		Filed July 16, 2021																															
For Secondary Battery																																	
Art Unit 1729		Examiner L. S. Park																															
This is a request under the provisions of 37 CFR 1.136(a) to extend the period for filing a reply in the above-identified application. The requested extension and fee are as follows (check time period desired and enter the appropriate fee below): <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 40%;"></th> <th style="width: 15%; text-align: center;">Fee</th> <th style="width: 15%; text-align: center;">Small Entity Fee</th> <th style="width: 15%; text-align: center;">Micro Entity Fee</th> <th style="width: 15%;"></th> </tr> </thead> <tbody> <tr> <td>☒ One month (37 CFR 1.17(a)(1))</td> <td style="text-align: center;">\$220</td> <td style="text-align: center;">\$88</td> <td style="text-align: center;">\$44</td> <td style="text-align: right;">\$ <u>220.00</u></td> </tr> <tr> <td>☐ Two months (37 CFR 1.17(a)(2))</td> <td style="text-align: center;">\$640</td> <td style="text-align: center;">\$256</td> <td style="text-align: center;">\$128</td> <td style="text-align: right;">\$ _____</td> </tr> <tr> <td>☐ Three months (37 CFR 1.17(a)(3))</td> <td style="text-align: center;">\$1,480</td> <td style="text-align: center;">\$592</td> <td style="text-align: center;">\$296</td> <td style="text-align: right;">\$ _____</td> </tr> <tr> <td>☐ Four months (37 CFR 1.17(a)(4))</td> <td style="text-align: center;">\$2,320</td> <td style="text-align: center;">\$928</td> <td style="text-align: center;">\$464</td> <td style="text-align: right;">\$ _____</td> </tr> <tr> <td>☐ Five months (37 CFR 1.17(a)(5))</td> <td style="text-align: center;">\$3,160</td> <td style="text-align: center;">\$1,264</td> <td style="text-align: center;">\$632</td> <td style="text-align: right;">\$ _____</td> </tr> </tbody> </table> ☐ Applicant asserts small entity status. See 37 CFR 1.27. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously. ☐ A check in the amount of the fee is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The Director has already been authorized to charge fees in this application to a Deposit Account. ☒ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment, to Deposit Account Number <u>12-1095</u>. ☒ Payment made via USPTO's patent electronic filing system (Patent Center or EFS-Web). WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. I am the ☐ applicant. ☒ attorney or agent of record. Registration number <u>72,515</u>. ☐ attorney or agent acting under 37 CFR 1.34. Registration number _____. _____ /Jordan T. Riviello/ Signature _____ Jordan T. Riviello Typed or printed name _____ May 9, 2023 Date _____ (908) 654-5000 Telephone Number NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.					Fee	Small Entity Fee	Micro Entity Fee		☒ One month (37 CFR 1.17(a)(1))	\$220	\$88	\$44	\$ <u>220.00</u>	☐ Two months (37 CFR 1.17(a)(2))	\$640	\$256	\$128	\$ _____	☐ Three months (37 CFR 1.17(a)(3))	\$1,480	\$592	\$296	\$ _____	☐ Four months (37 CFR 1.17(a)(4))	\$2,320	\$928	\$464	\$ _____	☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,264	\$632	\$ _____
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☐ * Total of 1 forms are submitted.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 9, 2023

Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

AMENDMENT TRANSMITTAL LETTER			Docket No. LGES 3.3F-3059	
Application No. 17/259,395-Conf. #1017	Filing Date July 16, 2021	Examiner L. S. Park	Art Unit 1729	
Applicant(s): LG Chem, Ltd.				
Invention: Secondary Battery				
TO THE COMMISSIONER FOR PATENTS Transmitted herewith is an amendment in the above-identified application. The fee has been calculated and is transmitted as shown below.				
CLAIMS AS AMENDED				
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate
Total Claims	12	- 20 =	0	x
Independent Claims	1	- 3 =	0	x
Multiple Dependent Claims (check if applicable) ☐				
Other fee (please specify):				
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:				0.00
☒ Large Entity ☐ Small Entity				
☒ No additional fee is required for this amendment.				
☐ Please charge Deposit Account No. _____ in the amount of \$ _____.				
☐ A check in the amount of \$ _____ to cover the filing fee is enclosed.				
☐ Payment by credit card. Form PTO-2038 is attached.				
☒ The Director is hereby authorized to charge and credit Deposit Account No. <u>12-1095</u> as described below. A duplicate copy of this sheet is enclosed.				
☒ Credit any overpayment.				
☒ Charge any additional filing or application processing fees required under 37 C.F.R. § 1.16 and 1.17.				
<u>/Jordan T. Riviello/</u> Jordan T. Riviello (Atty/Agent Reg. No.: 72,515) LERNER DAVID LLP 20 Commerce Dr. Cranford, New Jersey 07016 (908) 654-5000		Dated: <u>May 9, 2023</u>		

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 9, 2023

Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 9, 2023
Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: July 16, 2021

Art Unit: 1729

For: Secondary Battery

Examiner: L. S. Park

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Dear Commissioner:

In response to the official action mailed February 8, 2023, Applicant submits the following amendments and remarks, along with a one-month extension of time and the requisite fee.

IN THE CLAIMS

1. (currently amended) A secondary battery comprising:
an electrode assembly;
a can member configured to accommodate the electrode assembly therein; and
a top cap assembly configured to cover an opening of the can member,
wherein the top cap assembly comprises a top cap positioned at an uppermost portion and a safety vent part positioned below the top cap,
wherein the safety vent part comprises:
a rupture portion configured to be ruptured when an internal pressure within the can member ~~increases to equal or exceeds~~ a predetermined pressure, the rupture portion including a first section and a second section;
a rupture notch portion that defines a boundary of the rupture portion; and
a bending notch portion which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing an inside of the can member, the bending notch portion separating the first section and the second section,
wherein the bending notch portion is bent when the internal pressure within the can member increases such that the first section is moved at least partially above the bending notch portion when the internal pressure is equal to the predetermined pressure.
2. (previously presented) The secondary battery of claim 1, wherein the rupture notch portion is formed as a recess within a lower surface of the safety vent part, and
the bending notch portion is formed as a recess within the lower surface of the safety vent part,
wherein a recessed depth of the bending notch portion is less than a recessed depth of the rupture notch portion.
3. (previously presented) The secondary battery of claim 1, wherein the rupture notch portion is provided in the bottom surface of the rupture portion.

4. (previously presented) The secondary battery of claim 1, wherein the top cap assembly further comprises a current interrupt device provided with an interrupt portion which is attached to the rupture portion and which is configured to be ruptured to interrupt current when the internal pressure within the can member increases.

5. (previously presented) The secondary battery of claim 4, wherein the interrupt portion of the current interrupt device is attached to the rupture portion inside a circumference of the bending notch portion.

6. (previously presented) The secondary battery of claim 1, wherein the bending notch portion has a continuous circular or elliptical shape.

7. (previously presented) The secondary battery of claim 1, wherein the bending notch portion has a discontinuous circular or elliptical shape.

8. (previously presented) The secondary battery of claim 1, wherein the bending notch portion comprises a plurality of bending notches, and the plurality of bending notches have different lengths.

9. (previously presented) The secondary battery of claim 1, wherein the rupture notch portion has a discontinuous circular or elliptical shape.

10. (currently amended) The secondary battery of claim 1, wherein a connection portion between an edge portion (417) and the rupture portion of the safety vent part is bent upward.

11. (currently amended) The secondary battery of claim 10, wherein ~~the rupture portion comprises:~~

the second section of the rupture portion is declined ~~an inclined section inclined~~ so that a height thereof gradually decreases from the connection portion toward the bending notch portion; and

~~the first section a horizontal section having~~has a horizontal shape and extending from an end of the ~~inclined declined~~ section toward the center of the rupture portion;

~~wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section.~~

12. (currently amended) The secondary battery of claim 1, wherein a spaced distance between an edge end of the safety vent part and the rupture notch portion is equal to a spaced distance between the rupture notch portion (113) and the bending notch portion.

REMARKS

This Amendment is responsive to the February 8, 2023 non-final Office Action (“the Action”). No new claims have been added by way of this Amendment and no claims have been cancelled herein. Consequently, claims 1-12 remain pending in this application.

Initially, Applicant thanks the Examiner of the courtesies extended to Applicant’s representative during the telephonic interview of March 21, 2023 (“the Interview”) during which the prior art rejections and potential claim amendments for overcoming same were discussed. The Interview is evidenced by the Applicant-Initiated Interview Summary mailed on March 24, 2023 (“Interview Summary”). While no specific claim language was agreed upon during Interview, it is believed that a general agreement for advancing prosecution was reached. Independent claim 1 has been amended in view of this agreement. Support for the amendments to independent claim 1 may be found at least in FIGS. 2 and 3 of the original application. Claims 10-12 have been amended for clarity.

Claim Rejections under 35 U.S.C. §§ 102 and 103

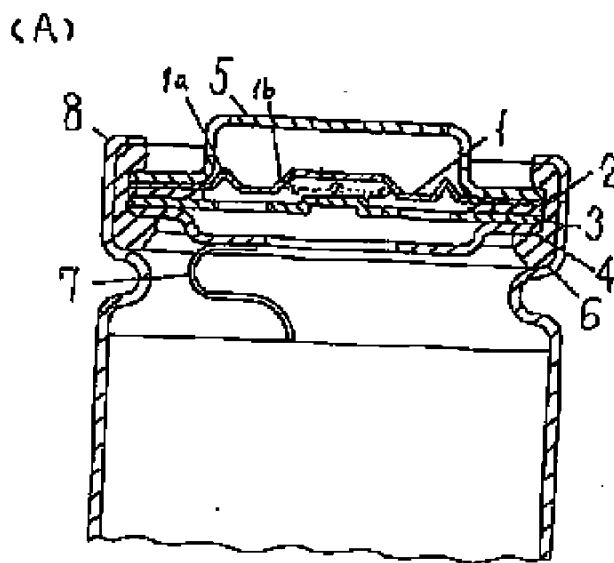
Claims 1-3 and 6 are rejected under 35 USC §102 as being anticipated by JPH 06215746 (“Mitamura”). Claims 4, 5, 7-9 and 12 are rejected under 35 USC §103 as obvious over Mitamura in view of US 2012/0114979 (“Kim ‘979”). Claim 10 is rejected under 35 USC §103 as obvious over Mitamura in view of WO 2018/030836 (“Kim ‘836”). Claim 11 is rejected under 35 USC §103 as obvious over Mitamura in view of Kim ‘979 and further in view of Kim ‘836. In view of the appended claim amendments and the following remarks, reconsideration and withdrawal of the rejections is requested.

In the Action, the bent portions (1a, 1b) of Mitamura are interpreted as the claimed “rupture notch portion” and the claimed “bending notch portion,” respectively. Although Applicant maintains that the bends are not “notches,” independent claim 1 has nevertheless been amended to expedite prosecution.

Amended independent claim 1 recites “the bending notch portion is bent when the internal pressure within the can member increases such that the first section is moved at least partially above the bending notch portion when the internal pressure is equal to the predetermined pressure.” In contrast thereto, as internal pressure within the device of Mitamura increases to the claimed predetermined pressure, the safety valve deforms. More specifically, the bent portion (1b) (equated

to the bending notch) deforms such that it no longer forms a “notch.” Consequently, in this state, the first section of Mitamura cannot be located at least partially above the bending notch portion as required by amended independent claim 1. *See* FIG. 4 of Mitamura reproduced below.

【 図 4 】



Kim '979 and Kim '836 are cited in the Action merely in support of the assertion that the references teach one or more features recited in dependent claims 4, 5, 7-9, 11 and 12. Nevertheless, neither Kim '979 nor Kim '836 cure the deficiencies of Mitamura with respect to amended claim 1.

For these reasons, independent claim 1 is not anticipated or rendered obvious by the cited prior art and is in condition for allowance. Claims 2-12, depending directly or indirectly therefrom, are in condition for allowance for at least the same reasons.

Reconsideration of the rejections and allowance of the claims are earnestly solicited. If, however, for any reason the Examiner does not believe that such action can be taken at this time, the Examiner is encouraged to telephone Applicant's undersigned representative at (908) 654-5000 to discuss any additional objections which the Examiner might have.

All required fees, if any, are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments, to our Deposit Account No. 12-1095.

Dated: May 9, 2023

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER DAVID LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/259,395	Filing Date 07/16/2021	☐ To be Mailed
---	--	---------------------------	---------------------------------------

ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	05/09/2023		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	* 12	Minus	** 20	= 0	x \$100 = 0
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$480 = 0
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						0
AMENDMENT			CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/PAMELA V THERATT/
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".						
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

86765 7590 06/01/2023
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EXAMINER	
PARK, LISA S	
ART UNIT	PAPER NUMBER
1729	
DATE MAILED: 06/01/2023	

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017

TITLE OF INVENTION: Secondary Battery

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	09/01/2023

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

86765 7590 06/01/2023
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017

TITLE OF INVENTION: Secondary Battery

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	09/01/2023

EXAMINER	ART UNIT	CLASS-SUBCLASS
PARK, LISA S	1729	429-056000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via Patent Center or EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017
86765	7590	06/01/2023	EXAMINER	
LG			PARK, LISA S	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1729	
DATE MAILED: 06/01/2023				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 17/259,395	Applicant(s) Min et al.	
	Examiner LISA S PARK	Art Unit 1729	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed 5/8/2023.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-12. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☒ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>20210111</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/LISA S PARK/
Primary Examiner, Art Unit 1729

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. In response to the amendment received on 5/9/2023:
 - a. Claims 1-12 are pending in the current application. Claims 1 and 10-12 are amended.
 - b. The previous prior art rejections are overcome in light of the amendment.

Allowable Subject Matter

3. **Claims 1-12 are allowed.**

The following is an examiner's statement of reasons for allowance: none of the prior art of record, alone or in combination, appears to teach, suggest, or render obvious the invention of Claims 1-12.

Independent Claim 1 recites a very specific structure for a safety vent of a battery top cap assembly, said vent comprising a rupture portion and a bending notch portion that have structural relationships relative to one another (e.g. the first section being moved partially above the bending notch portion and the bending notch portion separating first and second sections) that are not taught by the prior art, and there is no prior art available to teach or make obvious such specific structure. Therefore, the reference fails to teach or suggest the particulars of the independent Claim 1, and it's not obvious to modify these teachings to give the instant claimed

invention. Thus none of the prior art of the record, alone or in combination, appears to teach, suggest, or render obvious the invention of independent Claim 1. Since Claims 2-12 depend on Claim 1, they are allowable for the same reason.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached on M-Th 6 or 7 to 3 or 4; Alt Fri afternoons.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LISA S PARK/
Primary Examiner, Art Unit 1729

Issue Classification 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

CPC						
Symbol					Type	Version
H01M	/	50	/	3425	F	2021-01-01
H01M	/	50	/	107	I	2021-01-01
H01M	/	50	/	578	I	2021-01-01
H01M	/	2200	/	20	A	2013-01-01
H01M	/	50	/	574	A	2021-01-01

CPC Combination Sets							
Symbol				Type	Set	Ranking	Version
	/		/				

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	12	
/LISA S PARK/ Primary Examiner, Art Unit 1729	20 May 2023	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	4

Issue Classification 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

INTERNATIONAL CLASSIFICATION				
CLAIMED				
H01M	/	50	/	342
H01M	/	50	/	107
H01M	/	50	/	578
H01M	/	50	/	574
H01M	/	50	/	56
H01M	/	50	/	152
H01M	/	50	/	116
NON-CLAIMED				
	/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	12	
/LISA S PARK/ Primary Examiner, Art Unit 1729	20 May 2023	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	4

<i>Issue Classification</i> 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10												
2	2	11	11												
3	3	12	12												
4	4														
5	5														
6	6														
7	7														
8	8														
9	9														

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 12	
/LISA S PARK/ Primary Examiner, Art Unit 1729 (Primary Examiner) _____ (Date) 20 May 2023		O.G. Print Claim(s) 1	O.G. Print Figure 4

<i>Search Notes</i> 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M50/3425 or H01M50/107 or H01M50/578 or H01M50/574 or H01M2200/20 or H01M50/56 or H01M50/152 or Y02E60/10 or H01M50/116 (text-limited)	01/31/2023, 5/20/2023	/L.S.P./

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search history)	02/02/2023, 5/20/2023	/L.S.P./
Analysis of Global Dossier file	02/02/2023	/L.S.P./
PE2E Applicant/Assignee search	02/02/2023	/L.S.P./
PE2E foreign database search	02/02/2023	/L.S.P./

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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<i>Search Notes</i> 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H01M50	3425, 107, 578, 574, 56, 152, 116	05/20/2023	/L.S.P./
H01M2200	20	05/20/2023	/L.S.P./
Y02E60	10	05/20/2023	/L.S.P./

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L34	50	2 AND vent\$4 WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4 bent) near4 (multiple plurality first) AND pressure	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/05/20 03:17 PM

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	83	(H01M50/3425 OR H01M50/107 OR H01M50/578 OR H01M50/574 OR H01M2200/20 OR H01M50/56 OR H01M50/152 OR Y02E60/10 OR H01M50/116).cpc. AND ((break\$4 ruptur\$4) AND (bend\$4 bent) AND (section portion part) near3 (first second)).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2023/05/20 03:19 PM
N2	5	(H01M50/3425 OR H01M50/107 OR H01M50/578 OR H01M50/574 OR H01M2200/20 OR H01M50/56 OR H01M50/152 OR Y02E60/10 OR H01M50/116).cpc. AND ((break\$4 ruptur\$4) AND (bend\$4 bent) AND (section portion part) near3 (first second) AND predetermined ADJ pressure).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2023/05/20 03:19 PM

Doc code: IDS

PTO/SB/08a (02-18)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2020. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number

Filing Date

First Named Inventor

Geon Woo Min

Art Unit

N/A

Examiner Name

Not Yet Assigned

Attorney Docket Number

LGCHEM 3.3F-3059

U.S.PATENTS

Remove

Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	10090497	B2	2018-10-02	Kim et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

Add

U.S.PATENT APPLICATION PUBLICATIONS

Remove

Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20050214641	A1	2005-09-29	Kim	
	2	20100086835	A1	2010-04-08	KIM	
	3	20110200854	A1	2011-08-18	Bak et al.	
	4	20120114979	A1	2012-05-10	KIM et al.	

If you wish to add additional U.S. Published Application citation information please click the Add button.

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FOREIGN PATENT DOCUMENTS

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²ⁱ	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Geon Woo Min	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

1	2001102038	JP	A	2001-04-13	Toyo Kohan Co Ltd	Machine generated English translation attached.	×
2	H06215746	JP	A	1994-08-05	Matsushita Electric Ind Co Ltd	Machine generated English translation attached.	☒
3	H08339793	JP	A	1996-12-24	Mitsubishi Cable Ind Ltd	Machine generated English translation attached.	☒
4	100578805	KR	B1	2006-05-11	Samsung Sdi Co., Ltd	U.S. Publication No. 20050214641 A1 is an English language counterpart.	☒
5	101058386	KR	B1	2011-08-22	Lg Chem. Ltd	U.S. Publication No. 20120114979 A1 is an English language counterpart.	☒
6	20100038054	KR	A	2010-04-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20100086835 A1 is an English language counterpart.	☒
7	20110095118	KR	A	2011-08-24	Samsung Sdi Co., Ltd	U.S. Publication No. 20110200854 A1 is an English language counterpart.	☒
8	20150035267	KR	A	2015-04-06	Lg Chem. Ltd	Machine generated English translation attached.	☒

If you wish to add additional Foreign Patent Document citation information please click the Add button

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NON-PATENT LITERATURE DOCUMENTS

Remove

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report for Application No. PCT/KR2019/015396, mailed February 20, 2020, pages 1-2.	×

If you wish to add additional non-patent literature document citation information please click the Add button

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Geon Woo Min	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

EXAMINER SIGNATURE

Examiner Signature	/LISA S PARK/	Date Considered	05/20/2023
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Geon Woo Min	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2021-01-11
Name/Print	Jordan T. Riviello	Registration Number	72,515

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Thu Jun 01 05:04:38 EDT 2023

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Correspondence Address:

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Cranford, NEW JERSEY, 07016

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Application	Document	Mailroom Date	Attorney Docket No.
17259395	NOA	06/01/2023	LGES 3.3F-3059
17259395	1449	06/01/2023	LGES 3.3F-3059

To view your correspondence online, please sign in to [Patent Center](#) and then select Workbench/View correspondence. To update your email address(es), select Manage/Manage customer numbers.

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The Patent Center Team

PART B-FEE(S) TRANSMITTAL

Complete and send this form, together with the applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

By fax, send to: (571) 273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

Jordan T. Riviello	(Typed or printed name)
/Jordan T. Riviello/	(Signature)
July 25, 2023	(Date)

APPLICATION NO	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO	CONFIRMATION NO
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017

TITLE OF INVENTION: Secondary Battery

APPLN: TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1,200.00			\$1,200.00	09/01/2023

EXAMINER	ART UNIT	CLASS-SUBCLASS
L. S. Park	1729	429-056000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363)

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1. Lerner David LLP

2.

3.

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE or COUNTRY)

LG Energy Solution, Ltd.

Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent):

☐ Individual ☒ Corporation or other private group entity ☐ Government
4a. Fees Submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment (Please first reapply any previously paid fee shown above):

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5. Change of Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29.
- ☐ Applicant asserting small entity status. See 37 CFR 1.27.
- ☐ Applicant changing to regular undiscounted fee status.

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Authorized Signature

/Jordan T. Riviello/

Date July 25, 2023

Typed or printed name

Jordan T. Riviello

Registration No.

72,515

Electronic Patent Application Fee Transmittal

Application Number:	17259395			
Filing Date:	16-Jul-2021			
Title of Invention:	Secondary Battery			
First Named Inventor/Applicant Name:	Geon Woo Min			
Filer:	Keith McWha/Melinda Cormier			
Attorney Docket Number:	LGES 3.3F-3059			
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	48346471
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Keith McWha/Melinda Cormier
Filer Authorized By:	Keith McWha
Attorney Docket Number:	LGES 3.3F-3059
Receipt Date:	25-JUL-2023
Filing Date:	16-JUL-2021
Time Stamp:	15:26:51
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E20237OF27428204
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	Issue_Fee_Transmittal.pdf	37048	no	1
			c55f4cee834c11874aad868670ad994e5cc60b6d		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	37667	no	2
			c74f59021a213d817a28c1d45c9eb42ce4fc36ab		

Warnings:

Information:

Total Files Size (in bytes):			74715
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

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New International Application Filed with the USPTO as a Receiving Office

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	09/05/2023	11749866	LGES 3.3F-3059	1017

86765 7590 08/16/2023

LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 194 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Geon Woo Min, Daejeon, KOREA, REPUBLIC OF;

Je Jun Lee, Daejeon, KOREA, REPUBLIC OF;

Sang Suk Jung, Daejeon, KOREA, REPUBLIC OF;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF;

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CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Fri Aug 18 05:00:57 EDT 2023

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Application	Document	Mailroom Date	Attorney Docket No.
17259395	ISSUE.NTF	08/16/2023	LGES 3.3F-3059

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017
86765	7590	09/05/2023	EXAMINER	
LG			PARK, LISA S	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1729	
			NOTIFICATION DATE	DELIVERY MODE
			09/05/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

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APPLICATION NO.	ISSUE DATE	PATENT NO.
17/259,395	05-Sep-2023	11749866

LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EGRANT NOTIFICATION

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The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>

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Tue Sep 05 06:01:13 EDT 2023

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Application	Patent	Document	Mailroom Date	Attorney Docket No.
17259395	11749866	EGRANT.NTF	09/05/2023	LGES 3.3F-3059

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